

5. Enzyme sequences and DNA codons

Codon-optimized NHI 1 (CitB) sequence:

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ATGCCTATTAGCACTAAGTCGAGTTTCTACCTTCCCGCCGTTGATATCAGTCCATATT
TACAAGACCCCAACTCTGACGCAGCAAGAAAGGTTATAGACGACGTGCGGGCCGC
CTGCACTAGCACGGGTTTCTTCCAATTACTTGGACATGGGATTTTCGCCTGCACTTCA
ACAGTCGGTATTCGCGGCAGCTGCCAAGTTCTTTGCCTTGCCGTCAGATGTTAAGT
CACGTTGCCGCAACGTAGGTTTCCGCGGTTACGACCCAATGGCAAGCCAGTCTTAC
GAGTTAGGAGTGCTGCCCCGATCTGAAAGAGGGGTTTCATCGCCGGCAAAGACATCC
CTCTTGACGATCCGCGGGTTCGCATCTCAGCGGTTCTTCATGGGTCAGAACGCTTG
GCCGCCTTCCGAGCTGCTCCCCGAGGCGAACTTTGCGCGACCTATTGAGGAGTAT
TACCAGGCCATGCTCAAACGTGTGTTGGGTAGTCCTGGACTTAGTAGCAGCAACTTTA
CCTTACGGTCCGCATGTTTTCGACGAGTTCAAAGAGAACGACCCTGCTTGCCCCTT
ACGGCTTCTGCACTACCCACCTGCTCCAGCGCCTGATGTGGCTAAGGGGCGTCAA
CTCGGGTCCTCAGCCCATACAGACTTTGGCGCTATTACACTTTTGCTTCAGGACGA
CCATAGCGGACTTGAGGTGCAAGACTGCGAAACCGGTGAATGGATCGGCGTGCCA
CCTAACAAGGATGCGTACGTGGTGAATTTAGGCGACATGATGTCGCGTATCACCAG
AGGCCATTATAAGTCTTCCATTCATCGTGTGATCAACCAAACTTGACAGACCGTTAT
AGCGTCGTGTTCTTCTTCGATGGGAATCTTGACTACCGGCTTCGACCCTTGATAG
AGTTGGTCAAACTGGGACGAAGAGGACACACTTACTGTGGAAGAACACATGCTC
GAACGTACAACCACGACATATAACCTGAAGGTGAAGTAG
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NHI 1 (CitB) protein sequence:

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MPISTKSSFYLPVDISPYLQDPNSDAARKVIDDVRAACTSTGFFQLLGHGISPALQQSV
FAAAAKFFALPSDVKSRCRNVGFRGYDPMASQSYELGVLPDLKEGFIAGKDIPLDDPR
VASQRFFMGQNAWPPSELLPEANFRRPIEEYYQAMLKLCWVVLDLVAATLPYGPHVFD
EFKENDPACPLRLLHYPPAPAPDVAKGRQLGSSAHTDFGAILLLLQDDHSGLEVQDCE
TGEWIGVPPNKDAYVVNLGDMMSRITRGHYKSSIHRVINQNLTDYRSVFFFDGNLDY
RLRPLDRVGQNWDEEDTLTVEEHMLERTTTTTYNLKVK
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Codon-optimized NHI 2 (ClaD) sequence:

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ATGGGCAGCAGCCATCATCATCATCACAGCAGCGGCCTGGTGCCGCGCGGCA
GCCATATGATGCCGGTGCTGTGCAATCCGAGCTTTTATCTGCCACTGGTGGATATCA
CCCCGTTCTTGAGAAACCCGCATGGCGCAGCGGCGCAGGATGTCATCGAAAGCG
TGCGTACCGCCTGCAAAAGCACCGGTTTTTTTCAGATCAAAGGTCATCAGGTGCCG
CTGCGTTTGCAAAAATCGGTGTTTCGAAGCCAGCGCGCGTTTCTTTGCGCTGCCTCT
CAAAAACAAACTCGAACTCGACAGCCGCAAAACCGTTGGCTTCCGCGGTTACGATG
TCATGGAAACGCAAAGCTACGAACCTGGAATTTGGCGCGGTGCAGGAAGCGGATGC
GTTGCGCGATATCAAAGAAGGCTTTTTTATTGCCACCGACCTGCCGCCGGATCATC
CGCATGTCGCCAACGGTCGTTTTTTGCAAGGGCCCGAACGTCTGGCCGAAACCAGA
ACAGCTGGCGCCGGAAGATTTCCAGAGCGTACTGGAAGAATATTACACCGAAATGC
AGCGTCTGTCTCATGTGGTGTTGTCGCTGCTGGCAGCAACGCTGCCGTATGGTCC
GCACGTTTTTCGACGAGCTGGAAACCTGCGATCCGATGTCGTTGCTGCGCCTGCTG
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CATTATCCGCGCGGCTTGGAAAAACAGGATGGCAAAAAACTGCAACTGGGTGCCG
GGGAACATACCGATTTTGGTACCTTTACCCTGTTACTGCAGGACGAGCATCCTGGC
CTGGAAGTGCAAGACAGTGTCAACCGGCGAATGGCACGGCGTGCCACCGCAGGAG
GATGTTTATATCGTCAACGTTGCCGATATTCTGTCCACCATGACCGAGGGCGATTAC
AAGTCTTCTGTGCATCGCGTGTGGAACATCAAGAGCAACGATCGCTACAGCGTCGT
GTTTTTCTACGACGGCAACCTGGATTACAAAGTCAAACCGTTACGCAGCAGCGGTC
AGGACGAAAACGAAGAAATCGATGCACCGACCATCGAAGAGCACGTCCGTTTCGCG
TCTCACTGCCAGTTATGCCATTTAA

NHI 2 (ClaD) protein sequence:

MPVLSNPSFYLPVLDITPFLNPHGAAAQDVIESVRTACKSTGFFQIKGHQVPLRLQKS
VFEASARFFALPLKNKLELDSRKTGVFRGYDVMETQSYELEFGAVQEADALRDIKEGFF
IATDLPPDHPHVANGRFLQGPVWPKPEQLAPEDFQSVLEEYYTEMQRLSHVLSLLA
ATLPYGPHVFDELETCDPMSLLRLLHYPRGLEKQDGKKLQLGAGEHTDFGTFTLLLQD
EHPGLEVQDSVTGEWHGVPPQEDVYIVNVADILSTMTEGDYKSSVHRVWNIKSNDRY
SVVFFYDGNLDYKVKPLRSSGQDENEEIDAPTIEEHVRSRLTASYAI

Codon-optimized NHI 3 (TropC) sequence:

ATGTCCATTGGTGACGAAGTCATACCGACGGTTGATATAAGTGCGTGGCTCTCGTC
CACCGCGAGCCCGGAGTCAAAGAATAAGGTTGTGCGAGGAAGTTCGTTCCGGCGTGT
AATAAATATGGATTCTTCAATCTTGTAGGTACGGTATCCCGGCTGAGGCCCGCGAG
AAAATTTTCGGGTGCACTAAGAAATTCTTCGACCTGCCCCCTGGAAGAAAAGATGAA
GATCTCCGTTGACAAGTCACTTGGAAGTCTTCCGCGGGTATGAGCCTTCCCTGA
TCCAAACTCACCAAGACGGTTTGTACCAGACACGAAGGAGTGCTTCATTACTGGG
GCAGAAATTCCGGCTGACCACCCTGATGCTGGTAAATTCTCAACTGGGCCCAATCT
GTGGCCAGAGGGCCTGTCAGACAAGGAATTCCGTCAACCCGTGATGGAGTATAGA
GCGCTGATGCTGGACCTCGTGTCAACCATTGTGCGCATCTTAGGTCAGGGCATTCA
TAAGGCTTTCGGCCATCCTAGTGATGTACTTAATGACATCCTCATTAACTAGCATC
CCCATGCGACTGCTTCACTATGCTCCTCAAGAGAACCCGGATCCACGACAATTTGG
AGTTGGCGACCACACTGATTTTGGCTGTGTCTCTATACTGCTGCAGCAAAAGGGTA
CGAAAGGCCTGGAAGTGTGGTACCCACCCAAAGAGACGTGGATACCAGTACCTGT
CATCGAAGACGCCTTCGTTATTAACATGGGCGATACTATGCATCGTTGGACCGGCG
GTTACTACAGAAGTGCCCGCCACCGTGTATACATAACGGGTGAGCGTCGGTACAGC
GTTGCATTCTTCTTGAACGGTAATCTGAATCTGAAGATAAAGCCTTTGGATGGAAGT
GGCGGAGAAGCTTCAGTAGGAGAACACATCAATAGTAGACTCGCACATACTTTGGG
AGACAATGCTAAATACCTGCGTTGA

NHI 3 (TropC) protein sequence:

MSIGDEVIPTVDISAWLSSTASPESKNKVVEEVRSAKNKYGFFNLVGHGIPAEAREKIFG
CTKKFFDLPLEEKMKISVDKSLGKSFRGYEPSLIQTHQDGLLPDTKECFITGAEIPADHP
DAGKFSTGPNLWPEGLSDKEFRQPVMEYRALMLDLVSTIVRILGQGIHKAFGHPSDVL
NDILINPSIPMROLLHYAPQENPDPRQFGVGDHTDFGCVSILLQQKGTKGLEVWYPPKET

WIPVPVIEDAFVINMGDTMHRWTGGYYRSARHRVYITGERRYVVAFFLNGNLNLKIKPL
DGSGGEASVGEHINSRLAHTLGDNAKYLR

Codon-optimized NHI 4 (VioC) sequence:

ATGACTGAATCTCCTACAACGCATCATGGAGCGGCACCGCCTGACTCTGTAGCTAC
TCCAGTGCGCCCGTGGAGTGAGTTCCGTCTGACCCCTGCAGAAGCAGCAGCTGCT
GCTGCGCTGGCCGCACGTTGCGCACAAAGATACGATGAGACTGACGGACCCGAGT
TCTTGCTGGACGCTCCGGTAATCGCACATGAGTTGCCTCGGCGACTCAGAACGTTCC
ATGGCGCGGGCCCGGCTTGACGCATGGCCGCACGCATTAGTTGTTCTGTGGGAACC
CAGTGGATGACGCCGCGTTGGGGTCTACACCTGTTCACTGGCGCACAGCCAGAAC
GCCTGGATCCCGGGCCCCTGTCTTTCTTACTGATGCTGTATGCCGGCTTATTGGGAG
ACGTTTTTCGGCTGGGCAACGCAGCAGGACGGTCGGGTCTGTTACAGATGTGCTTCC
CATTAGGGGCGGGGAGCATAACCCTTGTGAGTTCTGCTTCGAGACAAGAGTTGGGAT
GGCACACAGAGGATGCGTTCAGCCCTTATCGAGCGGATTACGTGGGCCTGCTGTC
TCTTCGTAATCCCGACGGCGTGCGGACCACTTTAGCAGGGGTCCCGTTAGACGAC
TTAGATGAGCGGACTCTGGACGTTCTGTTTCAAGAGCGGTTCCCTCATTAGACCCGA
CGACAGCCACCTTCAGGTAAATAACTCAACGGCCCAACAAGGGAGAGTGGAGTTC
GAAGGTATCGCTCAAGCGGCGGATCGGCCTGAGCCAGTGGCAATCTTGACCGGTC
ATAGAGCAGCTCCGCACCTTCGAGTTGACGGTGACTTTTCTGCCCCCGCCGAGGG
CGACGAGGAAGCTGCCGCTGCGTTGGGAACACTGCGCAAGCTTATCGATGCATCC
CTCTACGAATTAGTATTAGATCAAGGAGATGTCGCTTTCATCGATAATCGACGTGCTG
TACACGGGAGACGTGCGTTCCAACCGCGATATGACGGCCGTGACCGGTGGCTGAA
ACGTATTAATATTACTCGTGACCTGCACCGCTCCCGGAAGGCATGGGCAGGCGACT
CGCGAGTGCTTGGCCAGCGT

NHI 4 (VioC) protein sequence:

MTESPTTHHGAAPPDSVATPVRPWSEFRLTPAEAAAAAALARCAQRYDETGDPEFLL
DAPVIAHELPRRLRTFMARARLDAWPHALVVRGNPVDDAALGSTPVHWRTARTPGSR
PLSFLMLYAGLLGDVFGWATQQDGRVVTDVLPKGGEHTLVSSSSRQELGWHTEDAF
SPYRADYVGLLSLRNPDGVATTLAGVPLDDLDERTLDFQERFLIRPDDSHLQVNNST
AQQGRVEFEGIAQAADRPEPVAILTGHRAAPHLRVDGDFSAPAEGDEEAAAALGTLRK
LIDASLYELVLDQGDVAFIDNRRRAVHGRRAFQPRYDGRDRWLKRINITRDLHRSRKAWA
GDSRVLGQR

Codon-optimized NHI 5 (SxtS) sequence:

ATGGTCGCGGCTGTTCTCACCTGTTATGGCGGGATGATCCATTGAACCCTTTAGA
CTGCGAGGGAATTAAGGAAATGTTGACAGTCGAACAGAAACAAGCCTACGTGAACG
ATGGGTATTTACGGTTGAGGAGACGGTACCGAAGGCGCTTATAGAGGAGATTCGG
CATGAGGTAGAATTAATTACGGAGCAAAAGCGCGGTGGTGTGCTTGCGGGCGACTA
CGAGTGGTGGTCAGACCACACGATTCCCGACCCGATCCGCTATCAGAAAATCATAC
AACGCCTTCTGGAGCTCCCAACCGTTATGGAGCCAGTCCAAGCCCTGATCGGTTCA
GATATATTCCTCTTGATCACTGACCTCGCGATTATTCGCGCAGGGACCGGTACATT
GCTTGGCACCAAGATCACGGCTACTTGGTCGAGGTCCTTATAGCCCCCGAGTCAAA

GAGCACAAATGTGCTGAACGACGACGCCTTCCGGTTGGTGTACCCGTTGTACAAC
AAACGACCGTCTTCATAACCATCTACCTGCAGGACACTGATAACACTATGGGAGCTA
TGCGGGTGATTCCGTCAAGTCACCAAGTGGGAACATTATTGGACCCTTCGTCTGCG
AACTCTTTAGATGCCGAAATTTGCCTTCCTTTGCCTGGTGGCGCGGCGATGTTCTAC
ACACCAACTGTGTGGCACACGGCCGCCGCCAACACCAGCATCACCGATTACCGTAT
GTTAACTCTTATCTTTACGAAGAACAACATCAAGCCGTTACTGGTGGATGCTCTTAA
GGGCGTATC

NHI 5 (SxtS) protein sequence:

MVA AVL TLLWRDDPLNPLDCEGIKEMLTVEQKQAYVNDGYFTVEETVPKALIEEIRHEVE
LITEQKRGGVLAGDYEWWS DHTIPDPIRYQKIIQRLLLEPTVMEPVQALIGSDIFLLITDL
AIRAGTGYIAWHQDHGYLVEVLIAPESKSTNVLNDDAFRLVLPVVQQTTFITIIYLQD TD
NTMGAMRVIPSSHQWEHSLDPSSANS LDAEICLPLPGGAAMFYTPTVWHTAAANTSIT
DYRMLTLIFTKNNIKPLLVDALKGRI

Codon-optimized NHI 6 sequence:

ATGCCAGTGCCGACCAAGTCATCTTTCTACCTTCCCGCTGTAGATATCAGCCCTTAC
TTGCAAGACCCGAAC TCGGACGGTGCGCGGAAGGTTATAAACGACGTGCGCGCCG
CATGTACCTCAACGGGATTCTTTCAATTACTTGGCCATGGAATATCTCCTGAGTTGCA
ACGGTCCGTGTTTCGCAGCCGCGGCCAAAATTCTTTGCATTACCTACGGATGTAAAATC
ATCCTGCCGAAACGTGCGCTTCCGTGGCTACGACCCCATGGCTTCCAGTCTTACG
AACCGGGCGTTCTGCCCCGACCTCAAGGAATCTTTCATTTCCGGGTACGGACATTCCT
CTTGACGACCCCCGCGTTATGAGTCGGCGTTTCTTCATGGGTCAAATGCCTGGCC
ACCCGGCGAACTCCTGCCAGAGACGGACTTTCGTGCCCCCGTCGAAGAATACTAC
CAAGCCATGATGAAGCTTTGCTGGGCTGTATTGGACTTAGTTGCGGCCACCTTGCC
TTACGGGCCGCATGTCTTCGATGAGTTCAAGGAGAATGACCCCGCGTGTCCTTTAC
GCTTGTTACTACTACCCACCGACCCCCGCCCCGACGTGGTCAAGGGTCGTCAGCT
GGGGTCTTCAGCTCACACTGACTTTGGCGCAATAACTCTCTTACTGCAGGATGACC
ACCCTGGGTTAGAGGTGCAAGATCGCGAAACCGGAGATTGGATTGGCGTGCCCCC
AAAGTCAGATGCGTACGTTGTTAATTTGGGCGATATGATGAGCAGAATCACTAGAGG
CCACTACAAGTCGTCAATACATCGTGTGATTAACAAGAACCTGACAGACCGTTATTC
TG TAGTATTCTTCTTTGACGGAATCTCGATTATAAGTTACGCCCACTGGACCGTGTA
GAGCAAGATCAGGACGAGGAGAACGTGCTGACCGTAGAAGAGCACATGTTAGAGC
GGACGACAACAACCTATAACACCAAAGCTAAATGA

NHI 6 protein sequence:

MPVPTKSSFYLPV DISPYLQDPNSDGARKVINDVRAACTSTGFFQLLGHGISP ELQRS
VF AAAAKFFALPTDVKSSCRNVGFRGYDPMASQSYEPGVLPDLKESFISGTDIPLDDPR
VMSRRFFMGQNAWPPGELLPETDFRRPVEEYYQAMMKLCWAVLDLVAATLPYGP HVF
DEFKENDPACPLRLLHYPPTPAPDVVKGRQLGSSAHTDFGAILLLLQDDHPGLEVQDR
ETGDWIGVPPKSDAYVVNLGDMMSRITRGHYKSSIHRVINKNLTD RYSVVFFFDGNLDY
KLRPLDRVEQDQDEENVLTVEEHMLERTTTTTYNTKAK

Codon-optimized NHI 7 sequence:

ATGCCTTCCTCAACAAAGTCAAGCTTCTACCTGCCAAGTGTTAACATAACACCGTAT
CTGCAAGACCCTGATAGCAGTGAGGCCCAAAGGTGATAGATGACATCCGTACCGC
GTGCCGGTCAACAGGTTTCTTTCAACTGCTTGGCCACGGGATCCCGCGTACCCTG
CAGCAATCCGTGTTTCGACGCGGGGAAACGTTTCTTTGCACTCCCGACCGATGTGA
AGCTCGGTATGGATTGTCGTCGGAATGTAGGCTTCCGGGGCTACGACGCGATGGC
TAGCCAGTCTTACGAGTCTGGGGTACTCCCCGACTTAAAGGAAGGCTTTATGATGG
GCACTGACATGAGCGAAAACGACTTCCGAGTGATCAATAGAAGATTCTTTATGGGG
CCCAACGTTTGGCCCCCAGAGAGCTTACTGAAGAAGGAAGACTTCCGTCAACCGC
TGGAACGTACTTCGAAGCGATGATGAAGTTATGTTGGACCGTTATGGACTTAGTTG
CTGCCACCCTGCCGTACGGGCCCAATATATTCGATGAGTGGAAGGCGAACGACCCA
GCATGTCCATTCCGGTTATTACATTACCCACCTACCCCCGAGCCTGTTGAGGGACAA
AAGCGTCAACTTGGGTCAAGCGCACACACAGATTCGGCGCCATCACATTATTGCT
GCAGGACGAGCATGCAGGATTAGAGATCCAAGACCGTGAAACCGGAGAGTGATT
CCGGTCCCGCCGAAGGAAGATGCGTACGTAATTAATATGGGAGACATGATTTACG
CATAACAGATGGCCACTACAAATCGTCAATCCATAGAGTCATAAACCGGAACCTCGAA
CGACCGGTACTCCATTGTGTTCTTCTTCGATGGCAATATCGATTACAAGCTGCGACG
GCTTGACTCGAAGGTCCAGTCAGAGGACGAGGCACTGACTGTTGAGGAGCACATG
GAAGAACGTCGTAATACGACGTACGGCCTCAAGAAGAAGTAA

NHI 7 protein sequence:

MPSSTKSSFYLPSVNITPYLQDPDSSEAQKVIDDIRTACRSTGFFQLLGHGIPRTLQQSV
FDAGKRFFALPTDVKLGMDCRRNVGFRGYDAMASQSYESGVLPDLKEGFMMGTDMS
ENDFRVINRRFFMGPNVWPPESLLKKEDFRQPLETYFEAMMKLCWTVMDLVAATLPY
GPNIFDEWKANDPACPFRLHYPPTPEPVEGQKRQLGSSAHTDFGAILLLQDEHAGL
EIQDRETGEWIPVPPKEDAYVINMGDMISRITDGHYKSSIHRVINRNSNDYRSIVFFFDG
NIDYKLRLRLDSKVQSEDEALTVEEHMEERRNTTYGLKKK

Codon-optimized NHI 8 sequence:

ATGCCTTCCTCTACTAAGAACTCATCATTTTATCTCCCATTAGTGAACATAACACCATA
TCTCCAGGACCCGAGCAGTATAGCTGGCCAACAAGTACTTGACGACGTTTCGACGG
GCGTGCACTTCAACAGGTTTCTTCCAGTTAGTTGGTCATGGCGTCCCTGCTACCTTA
CAGAAGAATTTGTACAACGCCGCGGCAAAGTTCTTCGCGCTGCCTATGGACGTCAA
GCTTCCTTTGGACGTACGAAAGAACATCGGGTTCGTTGGGTATGACGTGATGAGCA
GTCAATCATATGAGAAAGACGTTTTGCCGGACCTTAAAGAAGGTTACATTGTTGGTA
CGGATATCCCACTGGACGATCCCCGAGTAATGAAACGTCGCTTCTTCATGGGACGA
AACATATGGCCGCCAACCAGATTCTTGCCTTACGAGGACTTCCAAAAGGTTGTTGA
GGAGTACAACGACGCAATGTTGCGTCTTTCGTGGGTGTGGTCTGACTTAATAGCAG
CTACCCTGCCGTACGGTCCGCACGTCTTCGATGAATTAAGGAAAACGATCCGGCA
TGTCCACTTCGCTTGCTCCATTATCCGCCCACACCAAAGCAAGTTGGGAATGAGCG
CCGCCAATTCGGCTCATCAGCACACACTGACTTTGGGTTCATAACACTCTTGTTGCA
AGACGAGAATCCCGGTCTTGAAGTACAAGACCGGGAGACTGGTGAGTGATTGGG

GTACCGCCTAACGAGGAAGCTTATGTGGTAAATATGGGAGATATGATATCTAGAGTG
ACTGGCAACCTCTATAAGTCATCACTTCACCGCGTACTTAACAAGGCTCCAACCGAT
CGGTATTCAAGTCGTGTTCTTCGTGGATGGTAATTTAGACTACAAGCTTCGTGCTTG
GATAAAATTGGTCAACCTGGCGAGGACGAAGAGGCCTTAACAGTGGAAGAGCACAT
GGAAGAGCGCCGCCGAACACTACATATGGACTTCGTAAGAAGTAA

NHI 8 protein sequence:

MPSSTKNSSFYLPLVNITPYLQDPSSIAGQQVLDDVRRACTSTGFFQLVGHGVPATLQK
NLYNAAAKFFALPMDVKLPLDVRKNIGFRGYDVMSSQSSEKDVLPDLKEGYIVGTDIPL
DDPRVMKRRFFMGRNIWPPTEFLPYEDFQKVVEEYNDAMLRLSWVVDLIAATLPYGP
HVFDELKENDPACPLRLLHYPPTPKQVGNERRQFGSSAHTDFGFITLLLQDENPGLEV
QDRETGEWIGVPPNEEAYVVMGMISRVGTGNLYKSSLHRVLNKAPTDRYSVFFVD
GNLDYKLRRLDKIGQPGEDEEALVVEEHMEERRRTTYGLRKK

Codon-optimized NHI 9 sequence:

ATGGCGAGTCCAATCCCGTTTAGCGTGCCTACCATTGACTTCACAGCATACCTTAGT
GACCCGACGTCAGCAGAGTCTTCGAAGATTGTATCAGATATTAAACAGCTTGTACC
ACCACAGGCTTCTTTCAGCTGATAGGTCATGGTATCCCGGCGAGCTTACAGGAAGA
CGTGTTCAAGGGTTCCAAAGCTCTTTTCAAGCTGCCTATGGAAGAGAAGGTCAAGC
TTGACCGCGGTAAATCAGTAGGTGCTAGCAACCGAGGGTATGAGCTCATTGGCGGT
CAAGGACTGCAAGAGGGAACCTTACCGGATCTTAAAGAGGGATTCTACGTCGGACA
GGACATTCCTGCGACAGACCCACGTGTACAAAACAACGCGTTCCTGATGGGACCA
AACTTATGGCCAAGCAGTGATTTAGTGTCTGAGGCGGTCTTTAAGAAGCCAATGGA
AAAGTACTTTAGTAAGATGTTACCTTAACGTTGAAGGTCATGGATGCGATAGCCGC
TGGGATGCCATATGGATCGTCTGTTTTCGACGAGTTCACGAGTAACGACGCCGTTT
CCTCGATCCGCCTGCTGCACTACCCGCCAGACAAGAGCAATAACGAGCTGCAGCT
CGGGGCAGGCGCTCACACGGACTTTGGGGCAATTACTCTGCTGTTGCAAGACCAA
ATCGGCGGACTTCAGGTTTGGGATTATGGTAACAAGGAGTGGAAGGACGTAACCTC
GGTAAAGGATGCGTACGTCGTTAACGTTGGCGACATGCTGCAAATGTGGACTTCGG
GTGTGTATAAAAGTTCACTGCACCGAGTTCTCAATCGCTCCGGGAAAGACCGTTAC
TCGGTTCGGTTCTTCTTCGACGGAAACGTCGATTGCGTACTGACACCCCTGGGTGG
CGGGGAAGAGGGTAAGAAGTACCCTACCGTCGAAGAGCATATGAAGGAGCGTTTT
GGGCAAACCTATGGTCGTGGTAAGAAGAGAGAAGACGGCGGCGAGAACGTTAAGT
TCTGA

NHI 9 protein sequence:

MASPIFVSPTIDFTAYLSDP TSAESSKIVSDIKTACTTTGFFQLIGHGIPASLQEDVFKGS
KALFKLPMEEEKVKLDRGKSVGASNRGYELIGGQGLQEGTLPDLKEGFYVGQDIPATDP
RVQNN AFLMGP NLWPSSDLVSEAVFKKPMKEYFSKMFTLTLKVM DAIAAGMPY GSSVF
DEFTSND AVSSIRLLHYPPDKSNNELQLGAGAHTDFGAITLLLQDQIGGLQVWDYGNK
EWKDVTPVKDAYVVNVGDMLQMWTSGVYKSSLHRVLNRSGKDRYSVPFFFDGNVDC
VLTPLGGGEEGKKYPTVEEHMKERFGQTYGRGKKREDGGENVKF

Codon-optimized NHI 10 sequence:

ATGAAAACTATGTTTCGTGAGCTATCGGCTTAATATTTGTGTTCCCTACAGTAGACATATC
ACCATATTTAGCAGATCCAAAGTCGGATGCTGCCCAAAGGTAATCGAGGACGTTTC
GTGAGGCCTGCATATCAACCGGCTTCTTTTCAGATAACAGGGCACGGAATTTCTAAGA
GTCTTCAACAAAACATATTTGACGCGGCTCATTCTTTGCCTTGCCGCTTGAGG
AGAAGAAGAACTGAACGCTGCAAATTTTCATCGGACATCGTGGCTACGACGTGCTG
GCTAGTCAGAGTTACGAGGAAGGTGTATTACCGGACCTGAAGGAAGGTTATTACGT
TGGGTCGGATGTTTTACCTAGTGACCCACTTTATGGACGCTTCTTCATGGGGCAAAA
CGTATGGCCAAGTACCGAACTTTTATCTATCATTCAAGGAGCCTTGTGAGCG
CTACCACGAATCCGTACAACATCTCGCTTTCAAGGTCCTGCAATTGGTAGGAGATAC
GATGATACCTCACGGTCATTCAACCGTTGGAATGACCGAGAACATGCCGACAGTCT
TACATAACCTCATCCGCCTGGATAAAATACCAGCTTGCCCATTAAGACTTCTTCACTA
CCCCTCAGCAACAAGAAACGGTGGGTCTGTACAGGTAAACCTCAATACGGTGCG
TCGGCGCATACGGACTTCGGAGTTATCACTCTTCTGCTGCAAGACGACAACCCTGG
GCTGGAAGTGCTGGTTGAGAAGGATGGTAAGCAAGTTTGGTGGCCGATTGACCCC
AATCCATCTGCATATGTTGTAAATATCGGTGATATGATCAGTATGGTGACCAATGGCAT
CTATAAGTCCAGTATGCACCGCGTTGTGTGCAAACGGCCTGAGCGTGAGCGTTACA
GCATAGTGTTCTTCTTAGATGGATGCTTGAATGCTAATTTAGAGCCGGTCGAGGGTT
TCGGCCATCCGGAAGAAGGCGTCGGGTTTTCTCACAAGACCGTGGAGCAACACAT
GACCGAAAGACTGAGCATGTCTTATGCGAAAGAGGGTAAGGACCCTCGTCATAACG
AAGACACATTCTCTGCCTGA

NHI 10 protein sequence:

MKTMFVSYRLNICVPTVDISPYLADPKSDAAQKVIEDVREACISTGFFQITGHGISKSLQ
QNIFDAAHSFFALPLEEKKKLNAANFIGHRGYDVLASQSYEEGVLPLDLKEGYVVGSDVL
PSDPLYGRFFMGQNVWPSTELLSIIQFQEPCEHYHESVQHLAFKVLQLVGD TMIPHGH
STVGMTENMPTVLHNLIRLDKIPACPLRLLHYP SATRNGGSVTGKPQYGASAHTDFGVI
TLLLQDDNPGLEVLVEKD GKQVWWPIDPNPSAYVVNIGDMISMV TNGIYKSSMHRVVC
KRPERERYSIVFFLDGCLNANLEPVEGF GHPEEGVGFSHKTVEQHMTERLSMSYAKE
GKDPRHNEDTFSA

Codon-optimized NHI 11 sequence:

ATGGCCATAGCCAGTTCATTTTCCGTGCCGACGATTGACATTTCTCCATATCTGAGT
GACGCGACCTCAGAGGCGGGCTTAAAGGTCATCGAGGACGTTAAGGCTGCCTGTA
CGGGCACGGGATTCTTCCAATTGATCGGACACGGCATCTCTCCGGAACCTCAAGAG
AAAGTCTTCAAAGGCGCAGAGCAGCTCTTTGCGCTTCCCATGGAAGAGAAGGTAAA
ACTTGATAAGTCACAATCGGTAGGTGCTAGTAATCGTGGGTACGAACTGATCGGTGG
CCAGGGGTTACAAGAGGGGACTCTGCCCGACTTAAAGGAAGGTTTCTACGTTGGT
CAGGACATTCCGGCTAGTGATCCGCGCGTACAGAAGAAGGCATTCTTAATGGGTCC
GAATCTTTGGCCCCCGAGCGACCTGGTCCCTGAGGACGTTTTCAAGGAACCCATG
ACCAGCTATTATGCGCAAATCTTCGCTCTCTCGTTGAAAGTATTAGAAGTTATTGCGG
CCGGCATGGAGTATCCCGGTGTGAAAACAGTCTTTGAGGAGTTCACTGCCAACGA

CGCCGTCGCTTCTATTCTGCTTACTTCACTACCCGCCTGACAAGGGTAACGATGCAC
GACAATTGGGAGCTGGGGCACATACCGATTTCCGGTGCTATCACCTTATTACTGCAAG
ACGACATAGGTGGGCTGCAAGTGTGGGACTACGTCGGTAAGGAATGGCGGGACGT
TAAGCCTGTAAAGAAGCGTACGTAGTCAATGTGGGTGACATGTTACAAATGTGGAC
CTCTGGCGTCTACAAAAGCTCGCTGCACCGAGTTGTAAACCGTTCGGGGCGTGAT
CGGTATAGCGTTCCTTTCTTCTTCGACGGTAATGTCGACTTCGAGCTGCGTCCACTT
GACGGGCGTGAGCGCGAAGGTGGTGTTCCTTACGGTTGAAGCCCACATGAAGGAG
CGCTTCGCGTCGACGTATGGCCGCGGCGCTCAAAGGTGCGCCGGTCTCTGA

NHI 11 protein sequence:

MAIASSFSVPTIDISPYLSDATSEAGLKVIEDVKAACGTGFFQLIGHGISPELQEKVFKG
AEQLFALPMEEEKVKLDKSQSVGASNRGYELIGGQGLQEGTLPDLKEGFYVGQDIPASD
PRVQKKAFLMGPNLWPPSDLVPEDVFKEPMTSYYAQIFALSLKVLEVIAAGMEYPGVKT
VFEEFTANDAVASIRLLHYPPDKGNDARQLGAGAHTDFGAILLLLQDDIGGLQVWDYVG
KEWRDVKPVKEAYVVNVGDMLQMWTSGVYKSSLHRVVNRSGRDRYSVPFFFDGNV
DFELRPLDGREREGGVLTV EAHMKERFASTYGRGAQKVAGL

Codon-optimized NHI 12 sequence:

ATGCCAGCGGCCAGCGCTACGACTAGCAGCTTCTATTTACCGGATATCAACATAGCC
CCATGGCTTGAGGATCCTGGTAGTCCATCAGGCCAACACGTCGTTAATGAGGTACG
TAGAGCATGTACATCGACAGGGTTCTTCCAATTAACAGGTCACGGTTTCCCGCCTG
CGGTAATTAAAGACATATTCGCCGCAGCAGCTAAGTTCTTCGCGTTGCCGAAGGAG
ATTAAGCAAGACCTGAACTTCACGAAGAACGCCGGTTTCAGAGGCTACGAGCTCGT
AGGAGCCCAAGTGTACGAGACGGATGTGCTCCCCGACCTCAAGGAAGGCTTTCTC
GCGGGCATTGACCTTCCTCGAGACGATGTCCGTGTTCAAATAAACGGTTCTTCGC
AGGACAAAACGTTTGGCCTAGCCCCGACGTTCTGCCATATTCGGAATTTGCACCC
CTGTCGAGACGTACTACAAGTCTATAATGCAACTGTGTTGGACCGTTATGGACCTGA
TAGCCTCGACGCTTCCTTACGGTCCTAATGTGTTGACGAATGGAAAGCTCACGAG
CCTGCGTGTCTCTGCGTCTCTTGCACTATCCGCCAACCCCGAAGCACGTCGCGG
GAAAGACTCGCCAACTGGGAAGCAGTGCACACACTGACTTCGGCGGCGTTACACT
TCTGCTGCAGGACAGTCATGACGGTCTGGAAATACTTGATAGAGAGACTAGATCCT
GGGTACCGGTACCACCAAAGCCGGGTGCTTTCGTCATCAATGTTGCCGATATGATG
ACAATGATTACCGGCGGCGAGTACAAGAGCAGCATGCATCGAGTGATTAACCAAAA
CGAGACCGACCGTTACTCCGTTGTATTCTTCATGGATGGCAACGTCGATTACAAGTT
AAGACGTTTAGACAAAATAGGGCAACCGGTTGGCGAGGACGAAGAGCTCATTACAG
TCGAAGAGTACATGCTTGGTAAGCGCAATACTACGTACGTTAAGTGA

NHI 12 protein sequence:

MPAASATTSSFYLPDINIAPWLEDPGSPSGQHVVNEVRRACTSTGFFQLTGHGFPPAVI
KDIFAAAKFFALPKEIKQDLNFTKNAGFRGYELVGAQVYETDVLPLDKEGFLAGIDLPR
DDVRVQNKRRFFAGQNVWPSDVLPHYSEFRTPVETYYKSIMQLCWTVMDLIASTLPYGP
NVFDEWKAHEPACPLRLLHYPPTPKHVAGKTRQLGSSAHTDFGGVTLLLQDSHDGLEI

LDRETRSWVPVPPKPGAFVINVADMMTMITGGEYKSSMHRVINQNETDRYSVFFMD
GNVDYKLRRLDKIGQPVGEDEELITVEEYMLGKRNTTYVK

Codon-optimized NHI 13 sequence:

ATGACCAAAGGCAGCAGCCCAAATGTGCCAACGGTGGACATCTCACCATATCTGGA
AGATCCAAAAAGCAAAGCCGCCCGTAAAGTGATCGATGATGTTTCGCGAAGCGTGCA
CGTCAACCGGCTTTTTCCAAATCACGGGTCACGGCATTAGCAAAGAGCAACAAAAA
TCGGTGTTTGATGCGGCGCACGCCTTCTTCGCGTTACCATTAGAAGAAAAAAAAAA
ACTGAATGCCGCGCACTTTATCGGCCATCGTGGGTATGATGTTTTAGCGTCGCAATC
ATACGAGGAAGGTGTATTACCGGATTTAAAGAGGGTTATTATGTAGGCTCGGATATT
CTGCCGAGCGATCCACTGTATGGCCGTTTCTTTATGGGTAGCAACGTGTGGCCGCC
GGCGACCTTACTGAGCGCCTCACAGTTTCAGGAACCGTGCGAACGCTATCATAAAA
ACGTGCAGCAACTGGCGTTTAAAGTGCTGCAATTAATTGGTGATACGATGATCCCAT
GTGAGCGTAGCACGATCTCGATGACGGGTGGCGTACCGACGGTTCTGCACGATTTA
ATTCGCTTAGATGAAACGCCGGCGTGCCCGCTGCGTCTGTTACACTATCCGTCAGC
GGCGCGTCACGGTGATGTTGCCGGTAAACCGCAATATGGTGCGAGCGCGCACACG
GATTTTCGGCGTAATCACGTTACTGCTGCAAGATGATAACCCAGGCCTGGAAGTTCT
GGTTGAGAAAGATGGGAAACAGGTTTGGCTGCCAATTAACCCGAACCCAGATGCCT
ATGTGGTGAATATCGGGGATATGGCGTCGATGGTGACGAATGGGGTATATAAATCGA
GCATCCACCGTGTGGTGTGCAAACGCCCGGAACGCGAGCGTTATAGCATCGTATTT
TTCCTGGATGGCTGTTTAGATGCGTGTCTGCGTCCGATCGAGGAGTTCGGGCATCC
GGAGGGTGGGATCGCGGATAGCCACAAAACCGTTGAGCAGCACATGATCGAACGT
TTATCAATGAGCTATGCGAAAGAGGGCAAAGATCCACGTCATAACGAGGATACGTTT
TCGGTG

NHI 13 protein sequence:

MTKGSSPNVPTVDISPYLEDPKSKAARKVIDDVREACTSTGFFQITGHGISKEQQKSVF
DAAHAFFALPLEEKKKLNAAHFIGHRGYDVLASQSYEEGVLPDLKEGYVVGSDILPSDP
LYGRFFMGSNVWPPATLLSASQFQEPCCERYHKNVQQLAFKVLQLIGDTMIPCERSTIS
MTGGVPTVLHDLIRLDETPACPLRLLHYPsAARHGDVAGKPQYGASAHTDFGVITLLLQ
DDNPGLEVLVEKD GKQVWLPINPNPDAYV VNI GDMASMTNGVYKSSIHRV VCKRPER
ERY SIVFFLDGCLDA CLRPIEEFGHPEGGIADSHKTVEQHMIERLSMSYAKEGKDPRHN
EDTFSV

Codon-optimized NHI 14 sequence:

ATGAAAACGATGTTTCGTTTCGTATCGTCTGAACATTTGTGAGTTATATTTTGTGATTCT
GCAACAACCAACGCACAAAGTTAAAGATCGCTCACCAACGATCCATCCGTCGCGCG
ATATAGCCTGCATCGTATCACAATTTATTCTGAATAGCGATAAAATGACGAAAGGTAG
CTTCTCAGGTGTGCCAACGGTGGACATCTCGCCATATCTGGCGGATCCAAAAAGCG
ATGCCAGCCGCAAAGTGATCGAAGATGTTTCGTGAAGCGTGATCTCAACGGGGTTT
TTCCAGATTACGGGTCATGGCATTAGCAAATCGCTGCAACAAAACATTTTTTGATGCG
GCGCATAGCTTCTTCGCCTTACCATTAGAAGAGAAAAAAACTGAATGCCGCGAAT
TTTATCGGTCATCGTGGGTATGATGTTTTAGCCAGCCAAAGCTATGAAGAAGGCGTA

TTACCAGATTTAAAAGAGGGTTATTATGTAGGCAGCAACGTGCTGCCAAGCGATCCA
TTATATGGTCGTTTTTTTTATGGGCCAGAACGTGTGGCCATCAACGGAATTACTGAGC
ACGATCCGCTTTCAAGAACCATGCGAGCGCTATCACGAAAGCGTGACGACACCTGG
CCTTTAAAGTGCTGCAATTAGTAGGTGATACGATGATCCACACGGGCATAGCACGG
TGGGTATGACGGAATCGATGCCAACGGTGCTGCACAATTTAATTCGTTTAGATAAAAT
CCCGGCGTGTCGCTGCGTCTGCTGCACTATCCAAGCGCCACGCGTAACGGTGGG
TCGGTTACGGGCAAACCGCAATATGGTGCGTCAGCGCACACGGATTTTCGGTGTAAT
TACGTTACTGCTGCAAGATGATAATCCAGGCCTGGAAGTGCTGGTGGAAAAAGATG
GTAAACAAGTTTGGTGGCCAATCGATCCGAATCCAAGCGCCTATGTTGTGAATATCG
GTGATATGATCAGCATGGTGACGAACGGCATCTATAAAAGCAGCATGCATCGTGTGG
TATGCAAACGTCCAGAACGCGAGCGCTATAGCATCGTATTTTTCTGGATGGCTGTT
TAAACGCGAATCTGGAGCCAGTAGAGGGTTTCGGGCACCCAGAAGAAGGCGTGGG
TTTTTCGCACAAAACCGTAGAACAACACATGACGGAACGTTTATCAATGAGCTATGC
GAAAGAGGGCAAAGATCCACGTCATAACGAGGATACGTTTTTCGGCG

NHI 14 protein sequence:

MKTMFVSYRLNICELYFVILQQPTHKVKDRSPTIHPSRDIACIVSQFILNSDKMTKGSFS
GVPTVDISPYLADPKSDASRKVIEDVREACISTGFFQITGHGISKSLQQNIFDAAHSFFAL
PLEEKKKLNAANFIGHRGYDVLASQSYEEGVLPDLKEGYVVGSNVLPSPPLYGRFFMG
QNVWPSTELLSTIRFQEPCEYHESVQHFLAFKVLQLVGDTMIPHGHSTVGMTESMPTV
LHNLIRLDKIPACPLRLLHYPSTRNGGSVTGKPQYGASAHTDFGVITLLQDDNPGLEV
LVEKDQKQVWWPIDPNPSAYVVNIGDMISMVTNGIYKSSMHRVVCKRPERERYSIVFFL
DGCLNANLEPVEGFHPEEGVGFSHKTVEQHMTERLSMSYAKEGKDPRHNEDTFSA

Codon-optimized NHI 15 sequence:

ATGAGCTCAACGACGGATCGGAGCCACATCAGAAAAAATTTTATATTCCACAGGTT
AATATTGAACCATTCCTGAAAGATCCAGATAGCGCCACGAGCAAAACGATTATTGAA
GAAGTGCGTGCCGCCTGCCGTTCAACCGGTTTTTTTTGAAATCACGGGCCACGGCA
TCCCAAAAAGCCTTCAGAAAGAAGTTTTTCGGTGCCGCGTCGCAGTTTTTTAACTG
AGCTTCGATGAAAAAAAACGTTTAGATGCGTCAAAACAGATTGGTTATCGTGGTTAT
GATATGCTTGCGTCACAGGCCTATGGCAAAGATCTGCTTCGGATTAAAAGAAGGT
TTTTATATTGGCATAGATATACCAAAAACGGATCCTCGTGCGGGTCGTTTTTTTACGG
GCCCAAATGTGTGGCCAGATGAAAGCCTGCTGCCAGCCAGCCAGTTTAAAGATCCA
TGCACCATGTATTTTTTTCAGCCATGCTTCAGCTGAGCTTCGGTATAATGGCCCTGCTG
GCGCGTACGCTGCCATACGGTCCACATATTTTCGATGAATTTATCGCCAACGATCCA
GCCGCCCCACTTCGTCTGTTACACTATCCGCCACCACCAATGGCCTCAGTTGGCCC
AGAACAGCAGAAAAAACAGTTCGGCGCCTCGGCGCATACCGATTTTCGGTGCCATAA
CCATATTACTTCAGGATGAAAATCCGGGTCTGGAAGTTCTGGATCAGGATGCCGATA
GCTGGGTTCCAATCGATCCGAATCCAGGTGCCTATGTTGTGAATGTTGGTGATATGC
TGTCGTTATGGACGTCGGGTGAATATAAATCGAGCGTTCATCGCGTTATCAGCAAAA
ATCCGACGGATCGGTATAGCGTTGTTTTTTTCTTAGATGGGAATTTAGATTGTTCAATTA
AACCCATTAGATGGCTCGGAAAACCGGGGTCCGAAAACGGTTGAAGAACACATGAC
CCGGCGGTTAACGGCCTCGTATGGGGTTGAAAAAGAAAAAGAT

NHI 15 protein sequence:

MSSTTDRSPHQKKFYIPQVNI EPFLKDPDSATSKTIIIEVRAACRSTGFFEITGHGIPKSL
QKEVFGAASQFFKLSFDEKKRLDASKQIGYRGYDMLASQAYGKD LLPDLKEGFYIGIDI
PKTDPRAGRFFTGP NVWPDESLLPASQFKDPCTMYFSAMLQLSFRIMALLARTLPYGP
HIFDEFIANDPAAPLRLLHYPPPPMASVGPEQQKKQFGASAHTDFGAITILLQDENPGLE
VLDQDADSWVPIDPNPGAYVVNVGDMLSLWTSGEYKSSVHRVISKNPTDRYSVVFFLD
GNLDCSLNPLDGSEN RGPKTVEEHMTRRLTASYGVEKEKD

Codon-optimized NHI 16 sequence:

ATGGGAAGCGTGGCTCCTCCTCTGGTTCCAACGATTGATATTGCGGCGT TTTCTGGC
GGATTCAACCTCAGAAGAGGCGGAAGATATAGTAGATGCGGTTTCGTCAGGCGTGCA
CGAAAACCGGCTTTTTTTCAGATTACCGGTCATGGATTTCCAGAAGCGGTT CAGGAT
GCGATGTTCCAGGCGAGCAAAAAATTTTTCGCGCTGCCAATGGAAGAAAAAGTGAA
ACTGGATGCTAAAACGACGATTGGCTATCGGGGCTATGATGTGCTGGCGTCACAGG
TGTATGATACCTCAGGAGAAGTATTACCTGACCTGAAAGAAGGCTTTTATGTAGGCC
ATGATCTGCCGCTGTCAGATCCAGTGGTAGTAGCGCGCCGCTTTTTTATGGGCCCA
AACGTGTGGCCACCAGTGAGCTCACTGGCGGCTGCGGAATTT CAGCATCCAGTGG
AGGCGTATCATGGCGCGGTGAACAGCCTGGCGCTGAAAATGCTGGATCTGGTGGC
GCGCACCTGCCATACGATACGAAACAAATTTTCGCGGAATTTGCGACCGGCTATG
TTATTGCGCCGCTGCGTCTGCTGCATTATCCGCCGGCGGATACCGATGCTGCGACC
CAGACCCGTCCGCGCACCGGGGCGGGCGCGCATACGGATATGGGCGCGATTACC
CTGCTGCTGCAAAACGATCAGCCAGGGCTGGAAGTTCTGCATCCAGGCACCGGCG
AATTTGTGGCGGTGGATCCAACCCAGGCGCGTTTCGTGGTGAACGTGGGCGATAT
GCTTAGCGCGTGGACGGGCGGGCTGTATAAAAGCAGCGTTCATCGCGTGATTAACC
GTAACCCAGTGGATCGCTATTCAGCTGCGTTCTTTTTTGATGGCTCACTGGATTGCC
CGCTGGACCCGCTGGATGGGAGCGGCACCGGACCAAAAGGGGA ACTGACCGTGG
AAAAACACGTTATAAATCGTATTATGCAGAGCTAT

NHI 16 protein sequence:

MGSVAPPLVPTIDIAAFLADSTSEEAE DIVDAVRQACTKTGFFQITGHGFPEAVQDAMF
QASKKFFALPME EKVKLDAKTTIGYRGYDV LASQVYDTSGEVLPDLKEGFYVG HDLPL
SDPVVVARRFFMGP NVWPPVSSLAAAEFQHPVEAYHGAVNSLALKMLDLVARTLPYDT
KQIFAEFATGYVIAPLRLLHYPPADTDAATQTRPRTGAGAHTDMGAITLLLQNDQP GLEV
LHPGTGEFVAVDPTPGAFVVNVGDMLSAWTGGLYKSSVHRVINRNPVD RYSAAFFFD
GSLDCPLDPLDGSGTGPKGELTVEKHVINRIMQSY

Codon-optimized NHI 17 sequence:

ATGCCATCAGCGATTGAACAACCAGGATTT CATATACCAACGGTCGATATAGCGAGC
TATCTTGCTGATCCGAGCAGCGCTGATGCTCAACAGGTGGTAGACGCTGTACGGCT
GGCTTGCAACGAGCGGCTTTTTTCAACTGGTGGGCCACGGCATACTATAACAC
TGCGGGCTACGATGTTTGCGGGCTCAAAGCTCTGTTTGAGTTACCGTTTGACCAG
AAAAAGCGCTGAAACGGGCTGGAAATCGGGGCTATGAGGTGATGGGCTCACAGG

CGCTGCAAGATGGCACCCCTGCCAGATATGAAGGAAGGCTATTATATTGGCGAGGAA
ATACCATTTGAAGAAGGCCAAAGCGTACCGCCCGTTTATGGAACCAAACATTTGGCCA
TCAGAAGAACAGATTCCACATAGCATTTTTTCGGCTGCCAATGAACGAATATTATAAAC
GGGTGTCAGCGCTGGGCGTGGTGGTGGTGGTATTTTTTGC GGCGGGCGCTTCCTTAT
GGCCCAAACGTGTTTGATAAATTAACAAGAACCCTTCAGCGGCGAGCATAACGCCT
GCTTCATTATCCGCCTCAGCGGACGACGGATGAAAAGCAGCTGGGCGCGGGGGGC
GCATACGGACTTTGGCGTGCTGACCCTGCTGCTGACCGATGGCCATCCAGGCCTG
CAAGTCCTGAACGCTGCGACCGACACCTGGGTGCCTGTACCACCGGATGCGGAC
GCGTATGTGGTGAATGTAGGAGATATGTTACAGCAGATAACGGACGGATATTTTAAAA
GCAACACGCATCGGGTGGTGAACCTAGGAGACGACGACCGGTATAGCGTACCGTAT
TTTTTCGATGGGTGTCTGGA CTGTCGGTTAGAACGGTTAGATGGAAAGCACGAAGA
GAAAGTATTAACGGTCGAGGAACACATGAACGAACGGTTCGCTACCACCTACGGAC
GGGGCCAGCAAAAAACCGCT

NHI 17 protein sequence:

MPSAIEQPGFHIPTVDIASYLADPSSADAQQVVDAVRLACTTSGFFQLVGHGIPITLRAT
MFAGSKALFELPFDQKKALKRAGNRGYEVMGSQALQDGLPDMKEGYIIGEEIPFEE
GKAYRPFMEPNIW PSEEQIPHSIFRLPMNEYKRV SALGVVMDIFAAALPYGPNVFDK
LKQEPSAASIRLLHYPPQRTTDEKQLGAGAHTDFGVLTLLTLDGHPGLQVLNAATDTWV
PVPPDADAYVVNVGDMLQQITDGYFKSNTHRVVNLGDDDRYSVPYFFDGLDCRLER
LDGKHEEKVLTVEEHMNERFATTYGRGQKTA

Codon-optimized NHI 18 sequence:

ATGGCGAGCTCAGCGACGAGCAGCCCATATATTCCGGTTGTTGATATTACCCCATTT
TTAGAGAATCCACAGGGGACGGCCGCGCAACAGGTTGTGGATAACGTTCCGGCGTG
CCTGCATGTCAACGGGGTTTTTTCTGATGAAAGGCCACGGCATCAGCCCAAACTG
CAAACTCAGTGTTTCAGGCGTCAGCCAAATTTTTTTCGTTACCGCACGAGGCCAA
ACTGAGCTTAGATGCGCGGAAAAATGTGGGCTTTCGTGGCTATGATGTTATGGAGT
CACAATCATACGAGCTGGAAACCCGTAGCGGCCCAACAAGATGGGTCAGAAGTAATG
CGCGATACGAAAGAAGGCTTTTTT GCCAGCACCGATGTGCCACTGGATCACCCACG
CGTTGCCGCCGGCCGTTTCCTGGTTGGTCCAAATGTTTGGCCGGATGCCGATGTG
CTGAGCCACGAAGATTTCCGTGCCCACTGGAAGAATATTTAGGGGAGATGAAACG
TCTGAGCCACGTTGTGCTGAATCTGATTGCCGCGACGCTGCCATACGGGCCGCAC
GTATTCGATGAACTGGAAGCCGATGATCCAATGTGCCTGCTGCGTCTGCTGCACTAT
CCGCCGACCCACACAGCGTAGATGGCATGAAATTAGGCGCCGGCGCCACACGG
ATTTCCGGTGCCATTACCCTGCTGTTACAGGATGAGCATCCAGGCCTGGAGGTGCTG
GATGAGAAAACCGGCGAGTGGCAGGGCGTTCCGCCACAAAAAGATGTGTATATCGT
TAACATGGCCGATATTATGAGCATTCTGACCAAAGGTGCCTATAAAAGCAGCGTTCAT
CGGGTTTGAACAAAAATACGGAAGATCGCTATAGCGTGGTTTTTTCTTTGATGGC
AACCTGGATCTGAACTGAAACCGCTGGATGGGACGGAACAACGCCAGGATAATAA
CCAGGAGATTCTGACGCTGGAGGAGCATGTGCGCGGCCGTATGACCGGTAGCTAT
AGCATCCCAAAAAAT

NHI 18 protein sequence:

MASSATSSPYIPVVDITPFLENPQGTAAQQVVDNVRRACMSTGFFLMKGHGISPQLQN
SVFQASAKFFALPHEAKLSLDARKNVGFRGYDVMESQSYELETRSGPQDGSEVMRDT
KEGFFASTDVPLDHPRVAAGRFLVGPVWPADVLSHEDFRAPLEEYLGEMKRLSHV
VLNLIATLPYGPHVFDELEADDPMCLLRLLHYPPTPHSV DGMKLGAGAHTDFGAILLL
LQDEHPGLEVLDEKTGEWQGVPPQKDVYIVNMADIMSILTKGAYKSSVHRVWNKNT
DRYSVVFFFDGNLDLKLKPLDGTQRQDNNQEILTLEEHVRGRMTGSYSIPKN

Codon-optimized NHI 19 sequence:

ATGGCGAAAGGTAACCTCTCGGGGGTGCCAACCGTGATATAAGCCCATATCTGAC
GGATCCAAAAAGCGATGCCGCCCGTCGTGTGATCGAAGATGTCCGTGAAGCGTGC
GTGTCAACGGGGTTTTTCCAGATTACGGGTCATGGCATTAGCGAATCGCTGCAACA
AAAAATTTTTGATGCGGCGCATGCCTTCTTCGCGCTGCCATTAGAAGAGAAAAAAA
ACTGAATGCCGCGAACTTTATCGGTCATCGTGGGTATGATGTCTTAGCCAGCCAAAG
CTATGAAGAAGGTGTATTACCAGATTTAAAGAGGGTTATTATGTAGGCAGCGATGTG
CTGCCAAGCGATCCATTATATGGTCGTTTTTTTTATGGGCCAGAACGTGTGGCCATCA
ACGGGTTTACTGAGCCCAATCCAGTTTCAAGAACCATGCGAGCGTTATCACGAGAG
CGTGCGAGCACCTGGCCTTTAAAGTGCTGCAATTAATCGGTGATACGATGATCCAC
CGGGCATAGCACGGTGGGTATGACGGAAAATATGCCAACGGTGCTGCACAATTTAA
TTCGTTTAGATAAAATTCCGGCGTGTCCGCTGCGCCTGCTGCACTATCCAAGCGCC
ACGCGCAACGGTGGTTCGGTCGCCGGCAAACCACAATATGGTGCGTCAGCGCACA
CGGATTTTCGGTGTAATTACGCTGCTGCTGCAAGATGATAATCCAGGCCTGGAAGTG
CTGGTGGA AAAAGATGGTAAACAAGTCTGGTGGCCAATCGATCCGAATCCAAGCGC
CTATGTCGTGAATATCGGTGATATGATCAGCATGGTGACGAACAGCATCTATAAAGC
AGCATGCATCGCGTGGTATGCAAACGTCCAGAACGTGAACGTTATAGCGTGGTATTT
TTTCTGGATGGCTGTTTAAACGCGAATTTAGAGCCAATCGAGGGCTTTGGTCACCC
AGAGGAAGGGTTCGGGTTTTTACACAAAACCGTAGAACACACATGACGGAACGTT
TATCAATGAGCTATGCGAAAGAGGGTAAAGATCCACGTCATAACGAGGATACGTTTT
CGGCG

NHI 19 protein sequence:

MAKGNFSGVPTVDISPYLTDPKSDAARRVIEDVREACVSTGFFQITGHGISESLQQKIFD
AAHAFFALPLEEKKKLNAANFIGHRGYDVLASQSYEEGVLPDLKEGYVVGSDVLPSPD
LYGRFFMGQNVWPSTGLLSPIQFQEP CERYHESVQHLAFKVLQLIGDTMIPHGHSTVG
MTENMPTVLHNLIRLDKIPACPLRLLHYP SATRNNGGSVAGKPQYGASAHTDFGVITLLL
QDDNPGLEVLVEKD GKQVWWPIDPNPSAYV VNI GDMISMV TNSIYKSSMHRV VCKRP
ERERYSVVFFLDGCLNANLEPIEGFGHP EEGFGF SHKTVEQH MTERLSMSYAKEGKD
PRHNEDTFS A

Codon-optimized NHI 20 sequence:

ATGGCGTCGCCAGCCAGCACGCCAGCCAGTAGCTCGTCACTGGTGCCAACTATCG
ATATTAGCCCATCTTATTAAACCCAACGAGCGATGCAGCAAACGACGTATGCCAGG

CGATCCGGGCGGCCTGTCTGCATACGGGCTTTTTTCAGGTGATCGGCCACGCCGT
GAGCACCAGCACGCTGACTGGTGGCTTTGAAGCATCACGGCGTTTTTTTACTTTAC
CACTGGAACAAAAACAGAAATTAGACATTACCAAACAACTGGGTTTTTCGGGGCTATG
ACGGCCTTGGCACTCAGTCATACAGCGGTGACACGCTTCCAGACCTGAAAGAATC
GTTCTTTCTGGGCCGTGATATACCACCAAGCCACCCAGACTATGGTCGCATACTGAC
GGGCCCCAACATTTGGCCTAGCTTCGCGGTGCTGCCGGCAACGACGTTTAAAGAA
CCAGTGGAAGCACTTTATTAGCACTGATGCAACTGGCATGCAAAATTCTGGAAATT
CTGGCCCCGCACCCTTCCACACGGTGCAGGCCGAAGGCATTTTTTGATCGGTTCTGTGC
GCGACCCGGCCGCCCAATGCGGCTGCTTCATTACCCACCAATGGATGTGATTGAT
GAACGCCAGCTGGGTGCCTCAGCGCATACCGATTTTGGCGCAATTACTCTGCTTCT
GCAAGACCAGGTCCCGGGTTTACAGGTGCACGATTGAGACACCGGTGCCTGGCTG
GATGTGCCACCATGTCAGGACTCGATTGTGGTGAATGTCGGTGACATGATCACTCG
GTGGACTGCGGGCGAATACAAATCGAGCGTGCACCGTGTGATTAACCGGTCAGGT
ACCCACCGGTATTCAATCGCGTTCTTCTTTGACGGCAATGTGGATTGTCTGTAGAC
CCGTTAGACGGTACCACTGCCCCGGGTGGTTCAATCACTGTGGAAAAACACATGAT
CGAATGTATACGTTTCATCATACGCGTCGAAA

NHI 20 protein sequence:

MASPASTPASSSSLVPTIDISPFLNPTSDAANDVCQAIRAACLHTGFFQVIGHAVSTSTL
TGGFEASRRFFTLPLEQKQKLDITKQLGFRGYDGLGTQSYSGDTLPDLKESFFLGRDIP
PSHPDYGRILTGPNIWPSFAVLPAITTFKEPVEALYSALMQLACKILEILARTLPHGAGEGI
FDRFVRDPAAPMRLLYPPMDVIDERQLGASAHTDFGAILLLQDQVPGLQVHDSDTG
AWLDVPPCQDSIVNVGDMITRWAGEYKSSVHRVINRSGTHRYSIAFFFDGNVDCLL
DPLDGTAPGGSITVEKHMIECIRSSYASK

Codon-optimized NHI 21 sequence:

ATGGTTAAAGGTAACCTTCTGTGGTGTGCCAACGGTGGACATCTCACCATATCTGGAA
GATCCAGGGAGCAACGCCGCCCGCAAAGTGATCGAAGATGTTGCGGAAGCGTGCA
TCTCAACCGGCTTTTTTCAAATCACGGGTCATGGCATTCCAAAAAATCTGCAACAGA
ATGTGTTTGATGCGGGCGCATGCCTTTTTTCGCCTTACCATTAGAAGAGAAAAAAAAC
TGAACGCCGCGAACTTTATCGGTCATCGCGGGTATGATGTTTTAGCCAGCCAAAGC
TATGAGGAAGGTGTATTACCAGATTTAAAGAGGGTTATTATGTAGGCAGCGATGTG
CTGCCATCAGATCCATTATATGGCCGTTTTTTTTATGGGCCGTAACGTGTGGCCATCA
ACGAAATTTTTAAGCACGGCGCAATTTCAAGAACCATGCGAGCGTTATCACGAATCG
GTGCAGCAACTGGCCTTTAAAGTGTTACAATTAGTGGGTGATACGATGATCCACGC
GGTCATAGCAGCATCGGTATGACGGATAATATGCCAACGGTGTGTCATGATTTAATT
CGTTTAGATAACATCCCGGCGTGCCCGCTGCGCCTGCTGCACTATCCATCGGCCGT
GCGTAACCGCGGGGATGTTGCCGGTAAACCGCAATATGGTGCGTCAGCGCATACG
GATTTTGGTGTAAATTACGTTACTGCTGCAAGATGATAATCCAGGGCTGGAAGTTCTG
GTAGAAATCGATGGTAAACAAGTTTGGTGGCCAGTGGATCCGAATCCAGGCGCCTA
TGTGGTGAATATCGGGGATATGACGAGCATGGTGACGAATGGCACGTATAAAAGCA
GCATGCATCGTGTGGTTTGCAAACAACCAGAACGCGAGCGTTATAGCATCGTATTTT
TTCTGGATGGTTGTTTAAATGCGAAATTAACCAATCGAGGGCTTCGGGCATCCGG

AAGAAGATATTGAAACGTGCGACAAAACCGTAGAGCAACACATGACGGAACGTTTAT
CAATGAGCTATGCGAAAGAGGGTAAAGATCCGCGTCACAACGAGGATGCCTTTAAC
GCG

NHI 21 protein sequence:

MVKGNFCGVPTVDISPYLEDPGSNAARKVIEDVREACISTGFFQITGHGIPKNLQQNVF
DAAHAFFALPLEEKKKLNAANFIGHRGYDVLASQSYEEGVLPDLKEGYVVGSDVLP
PLYGRFFMGRNVWPSTKFLSTAQFQEP CERYHESVQQLAFKVLQLVGDTMIPRGHSSI
GMTDNMPTVLHDLIRLDNIPACPLRLLHYPSAVRNRGDVAGKPQYGASAHTDFGVITLL
LQDDNPGLEVLVEIDGKQVWWPVDPNPGAYVVNIGDMTSMVTNGTYKSSMHRVVCK
QPERERYSIVFFLDGCLNAKLKPIEGFGHPEEDIETSHKTVEQHMTERLSMSYAKEGKD
PRHNEDAFNA

Codon-optimized NHI 22 sequence:

ATGCATAGCAGCTTTTATCTGCCACTGGTGGACATTACGCCATTTCTGGATGATCCA
AGCAGCCCAGCGGCCCCAGCAGGTGATTGATGACGTGCGTCGGGCCTGCCGTTTCG
ACCGGCTTTTTTCAGATGAAAGGTCATGGCGTTCCACTGAAACTGCAAAAAAGCATT
CTGGCCGCCGCCGCCAAATTTTTTCCCCTGCCGACCGAAACGAAACTGGCGCTGG
ATGCGCGCAAAAATGTGGGGTTTCGCGGGTATGATGTAATGGAAACGCAGAGCTAT
GAACTGGAAATTGGCGTACCACAGGGGCCAGGAAGCCCAGGTGGTGC GCGATATTA
AAGAAGGCTTTTTTATTAGCAGCGATCTGCCATTAGATCATCCGCGTGTTTCGCGACG
GCCGTTTTTTACAGGGGCCAAACGTGTGGCCGGATAAAGAAGTTCTGGCCCCGCA
GGATTTTCGTGTCGTTCTGGAAGAATATCTGAGCGAAATGCAGCGTCTGTCACAGG
TTGTTCTGTCACTGGTTGCCGCGACGCTGCCGTATGGCCCCGCGTGTTTTTCGATGAA
CTGGACTCAGATGATCCAATGTGTCTGCTGCGTCTGCTGCATTACCCGCCGGATGG
GGAACAGGCCCGGCGTGAACGGGGGCAGCTGGGGAGCGGTGAACATACGGACTT
TGGTACCCTGACCCTGCTGCTGCAAGACGAACATGCGGGGCTGGAAGTTCAGGAC
CATGAAACCGGCCGAATGGCATGGTGTTCACCGCAGGAAGATGTTTATATCGTGAA
CATGGCGGATCTGATGGAAATTATGACCAGCGGGGAATACCGGTCGTCGTGGCATC
GTGTCCTGAATCGTAACCCGGAAGACCGCTATTTCGGTTGTCTTTTTCTATGACGGCA
ACCTGGACTACAAATTACAGCCGCTGGACCCGGCGAAAAAAGTCGAAGAAGAAAAT
GTTCCAACGATCGAAGAGCATGTGCGTCTGCGCCTTACCAGTAGCTATAGCATCGC
GCAGAAA

NHI 22 protein sequence:

MHSSFYLPVDITPFLDDPSSPAAQQVIDDVRRACRSTGFFQMKGHGVPLKLQKSILAA
AAKFFALPTETKLALDARKNVGFRGYDVMETQSYELEIGVPQGQEAQVVRDIKEGFFIS
SDLPLDHPRVRDGRFLQGPNVWPDKEVLAPQDFRVVLEEYLSEMQRLSQVVL SLVAA
TLPYGPRVFDELDSDDP MCLLRLLHYPPDGEQARRERGQLGSGEHTDFGTLTLLQD
EHAGLEVQDHETGEWHGVPPQEDVYIVNMADLMEIMTSGEYRSSWHRVLNRNPEDR
YSVFFFYDGNLDYKLQPLDPAKKVEEENVPTIEEHVRLRLTSSYSIAQK

Codon-optimized NHI 23 sequence:

ATGACGAAAGGTAGCTTCTCAGGTGTGCCAACGGTGGACATCTCGCCATATCTGGC
GGATCCAAAAAGCGATGCCGCCCAGAAAGTGATCGAAGATGTTTCGTGAAGCGTGC
ATCTCAACGGGGTTTTTCCAGATTACGGGTCATGGCATTAGCAAATCGCTGCAACAA
AACATTTTTGATGCGGCGCATAGCTTCTTCGCGTTACCATTAGAAGAGAAAAAAA
CTGAATGCGGCGAACTTTATCGGTCATCGTGGGTATGATGTTTTAGCCAGCCAAAGC
TATGAAGAAGGCGTATTACCAGATTTAAAAGAGGGTTATTATGTAGGCAGCGATGTG
CTGCCAAGCGATCCATTATATGGTCGTTTTTTCATGGGCCAGAACGTGTGGCCATCA
ACGGAATTACTGAGCATTATCCAGTTTCAAGAACCATGCGAGCGCTATCACGAAAGC
GTGCAGCACCTGGCCTTTAAAGTGCTGCAATTAGTAGGTGATACGATGATCCCACAC
GGGCATAGCACGGTGGGTATGACGGAAAATATGCCAACGGTGCTGCACAATTTAATT
CGTTTAGATAAAATCCCGGCGTGTCGCTGCGCCTGCTGCACTATCCAAGCGCCAC
GCGCAACGGTGGGTGCGTTACGGGCAAACCACAATATGGTGCGTCAGCGCACACG
GATTTTCGGTGTAATTACGCTGCTGCTGCAAGATGATAATCCAGGCCTGGAAGTGCT
GGTGGA AAAAGATGGTAAACAAGTTTGGTGGCCAATCGATCCGAATCCAAGCGCCT
ATGTTGTGAATATCGGTGATATGATCAGCATGGTGACGAACGGCATCTATAAAAGCA
GCATGCATCGCGTGGTATGCAAACGTCCAGAACGTGAGCGTTATAGCATCGTATTTT
TTCTGGATGGCTGTTTAAATGCGAATCTGGAGCCAGTAGAGGGTTTCGGGCACCCA
GAAGGCGTGGGTTTTTCGCACAAAACCGTAGAACACACATGACGGAACGTTTATC
AATGAGCTATGCGAAAGAGGGTAAAGATCCACGTCATAACGAGGATACGTTTTTCGG
CG

NHI 23 protein sequence:

MTKGSFSGVPTVDISPYLADPKSDAAQKVIEDVREACISTGFFQITGHGISKSLQQNIFD
AAHSFFALPLEEKKKLNAANFIGHRGYDVLASQSYEEGVLPDLKEGYVVGSDVLPSPD
LYGRFFMGQNVWPSTELLSIIQFQPCERYHESVQHLAFKVLQLVGDTMIPHGHSTVG
MTENMPTVLHNLIRLDKIPACPLRLLHYPSTRNGGSVTGKPQYGASAHTDFGVITLLL
QDDNPGLEVLVEKDQKQVWWPIDPNPSAYVVNIGDMISMVTNGIYKSSMHRVVKRP
ERERYSIVFFLDGCLNANLEPVEGFHPEGVGFVSHKTVEQHMTERLSMSYAKEGKDP
RHNEDTFSA

Codon-optimized NHI 24 sequence:

ATGAGCCCAAACGTGATTGTGGATGAAAACATTACCAGCAAAACCACCACGGTGGA
TAGCTTTCATATTCCAACGGTTGATATTTACAGCGTTTCTGGCGGATCCGGATTCACCA
GAAGCGCAGAGCATTATTCCAATTGTTTCGGAGCGCGTGCCGCTCAACCGGTTTTTT
TCAGATTACCGGTCATGGCATCAGCAAAGATCTGCAACAGGCGGTTTTTTCGGGGCG
CGAAACAGTTTTTTTAGCCTGCCATTTGAAACGAAAAAAGCGCTGGATGCGCATTTAA
ATCAGGGCCTGCGGGGTTATGATGTATTAAGCTCACAGAGCTATGAAGATGGGATTC
TGCCGGATCTGAAAGAAGGCTTTTTTACCGGTGTTGATCTGCCGGCGGATGATCCT
CGTGTTTCAGGCGCGCAAATTTCTGATGGGTAGCAATGTGTGGCCAGATGCGAGCCT
GCTGCCAGAAGATGAATTTTCGCAAACCAATTAAAGAATATCATGATGTTCTGATGAAA
CTGAGCATTAACTGCTGAACTGATCGAACGCACCCTGCCATACGGTCCGGGTAT
TTTTGATGAATTTATTGGCAACGATCCAATTATGCCGATGCGCATGCTGCATTATCCG
CCGGCGCGCCCAGAACATGAACGGAACGGTAAAACCCAGTATGGTGCGAGCGCG

CATACCGATTTTGGCGCGATCACGCTGCTGCTGCAAGAAAAAATCCAGGGCTGGA
AGTTTTTGGATTACGTACCGATGAATGGAAACCAATCCCGCCGAATCCGGATGTTTA
TGTTGTGAATATCGGCGATATGCTGTCGTTTTGGACCAAAAACGAATATAAAAGCTC
GGTTCATCGCGTTATTAATCGCGAACCAGGCGATCGGTATTGATTGTTTTTTTTGTT
GATGGGAATGGGGATGCGAAACTGTCACCGCTGGATGGGTCAGAAGAAGATAATCC
GCTGACGGTGGAACAGCATCTGATCCGTCGGCTGGCGGAAAGCTATGGCAAAAAA

NHI 24 protein sequence:

MSPNVIVDENITSKTTTVDSFHIPTVDISAFLADPDSPEAQSIPIVRSACRSTGFFQITGH
GISKDLQQAVFAGAKQFFSLPFETKKALDAHLNQGLRGYDVLSSQSYEDGILPDLKEGF
FTGVDLPADDPVQARKFLMGSNVWPDASLLPEDEFKPKIKEYHDVLMKLSIKLLKLIE
RTLPGYGPFIGDEFIGNDPIMPMRMLHYPPARPEHERNGKTQYGASAHTDFGAILLLLQE
KNPGLVFDSTDEWKPIPPNPDVYVNVNIGDMLSFWTKNEYKSSVHRVINREPGDRYS
IVFFVDGNGDAKLSPLDGSEEDNPLTVEQHLIRRLAESYGKK

Codon-optimized NHI 25 sequence:

ATGCCGCCCTTACAGTACACCTGGAAAATTCTCCGCGGCGCGGCCCTTCAGAACG
CTTTGTCACCGTCTAGTACGACCAGCCAGCTGCGTGTGACGCGGCCGATTGCTGC
ATTTAACCCTTTTAGCCTAGCCACGTACGCACCTTCAAAGCACCTGCTCGCCGTC
GTCAATCATTGCGGGCAACGGAAACGACATCTACCGCGACCCCGTCATCGAGTGTG
CCGCCGACCCTGCCCACGCCGCGCGCTGCGACCACCACCCACCTAGCTTCACC
ATTCCGACTGTAGACATCGCCCCCTATTTGAGCAATCCAAGCTCGCCGGCAGCTCA
GCGCGTAATTGAGCAGATTAACCAAGCATGCCGTAGCACGGGCTTCTTCCAAATAG
TTAATCATGGTGTCCCTGCGGCCCTGCAATCAGAGGTGTTTAGAGCTGCCGAACGC
TTCTTTTCGCTTACCGCTGGAAGAAAAGAAGAACTGGATGCGAAAACCACCGTGGG
TCATCGTGGCTACGATGTGCTGGAATCCCAGAGCTATGAAGAAGGTGTGTTGCCTG
ACTTAAAAGAAGGCTTTTATGTGGGTAACGACATTCCAGCGAGCGACAGCCGTGCC
CAAGCGCGCCGCTTCTTTATGGGCCCGAATGTGTGGCCGTCGTCACTGCAACCTC
ATGAACTGCAACAACCGGCAGAAGCGTATTTTCGCAGCAGTGTATACCCTGTCATTAC
GCATACTTGACATGATTGCGGCATCACTCCCGTACGATGTCTCGCCTATGTTGCGAC
GTTTCACCCAGCCAGCTGACCTGGTTGCCGCGCCACTACGTCTGCTTCACTATCCG
CCTGCCCGACGCTCTAAACAGAGGCAGCTGGGGGGCGTCTGCGCATACTGATTTTG
GCGCTATCACCTGTCTGCTACAGGATGGTAATCCGGGCCTGCAAGTGCTTGATCAG
CATAGCGAAAGCTGGGTGGACGTGCCCTCGACGCGTAATACGTATGTCGTGAATGT
TGGTGACATGCTTGAGAAATGGACGGGTGGCTACTACCGCAGTAGCGTTCATCGTG
TGCTGAATCCGCAGCCAAAAGATCGCTATTGAGTCGTTTTCTTCTTTGATGGTAACTT
GGACTGTCCCTTAGTTCCGCTCAATCCGAATGATGCCGCGAAATCGTCAGCCCTGC
CGGGTGGCGAGGGCGGTAAAGAAGATGGTGTGCTGACTGTGAGAAACACATGAT
TAACCGCATGACAACTAGCTATGGTAAGGGCTCGAAGAAGTAA

NHI 25 protein sequence:

MPPLQYTWKILRGAALQNALSPSSTTSQLRVTRPIAAFNPFLATSRTFKAPARRRQSF
AATETTSTATPSSSVPTLPTPRAATTTTPPSFTIPTVDIAPYLSNPSSPAAQRVIEQINQA

CRSTGFFQIVNHGVPAALQSEVFRAAERFFRLPLEEKKKLDKTTVGHRGYDVLESQS
YEEGVLPDLKEGFYVGNIPASDSRAQARRFFMGPVWPSSLQPHELQQPAEAYFAA
VYTLSLRILDMIAASLPYDVSPMFARFTQPADLVAAPLRLLHYPPARRRSKQRQLGASAHT
DFGAITLLLQDGNPGLQVLDQHSSESWVDVPSTRNTYVVNVGDMLEKWTGGYYRSSV
HRVLNPQPKDRYSVVFVFDGNLDCPLVPLNPNDAAKSSALPGGEGGKEDGVLTVEKH
MINRMTTSYGKGSKK

Codon-optimized NHI 26 sequence:

ATGCCTAGCGCGATCGAACAACCACAGTTTCACATTCCGACGGTGGACATCGGTCC
ATATTTGGCTGACCCGTCTAGTGCGGAAGCACATCATGTTGTCGATGCGGTTCCGCC
TGGCGTGTACGACTTCGGGCTTCTTCCAATTAACGGGTCATGGCATTGAGCCGGAG
ATTCGCGATGCAATGTTTAGCGGTTTCGAAAGCCTTATTTCGATCTGCCGTTTCGATCAG
AAGAAAGCGCTGCGGCGCGGGAAGAACC GCGGTTATGAAGTCATTGGCTCGCAG
GCCTTACAAGATGGGACCTTGCCTGATCTGAAAGAAGGGTATTATATCGGTGAAGAT
ACCACAGATTTTGAAGAGGGGAAAACCTACCGCCCGTTTATGGATCCTAACGTTTG
GCCGTCAGAGGATCAGGTCCCGAGTAGTGTGTTTCGGAAACCGATGAATCAATATT
ATGAACGTGTAGCTGCGCTCAGCAAAACCGTAATGGACATTATTGCGGCAGCGCTC
CCGTATGGGCCCACGGTGTTTGATCAATACAAGCTGGAACCAATTGCCGCGAGCAT
CCGCCTGCTGCATTATCCGCCGCAGCGAACCAACGACGGTAAACAGCTGGGTGCC
GGCGCGCATACCGACTTCGGCTGGATCACTCTGCTGCTCACCGACGGACATCCAG
GCCTGGAAGTGTTAAATCAGAGCACCGGCGAATGGGTACAAGTCCCGCCGAACAA
GGATGCCTACATTGTGAATGTCGGCGATATGCTGCAACAGATTACCGGGGGCTATTT
TAAATCGAACTTTTCATCGTGTTTTGAACTTAGGTGAATCAGATCGTTATAGCGTTCCG
TACTTCTTTGATGGTTGCCTCGACGCGAAACTCGCGCGCCTTGACGGTCGCGACG
AAGGTAAAGTACTGACAGTGGAAGAACACATGCTGGAGCGTTTTGCGACGACTTAT
GGTCGCGGCCAGGCCAAAAGATTAA

NHI 26 protein sequence:

MPSAIEQPQFHIPTVDIGPYLADPSSAEAHHVVDVRLACTTSGFFQLTGHGIEPEIRDA
MFSGSKALFDLPFDQKKALRRGKNRGYEVIGSQALQDGTLPDLKEGYIYIGEDTTDFEE
GKTYRPFMDPNVWPSEDQVPSSVFRKPMNQYYERVAALSKTVMIDIIAALPYGPTVFD
QYKLEPIAASIRLLHYPPQRTNDGKQLGAGAHTDFGWITLLLTDGHPGLEVLNQSTGE
WVQVPPNKDAYIVNVGDMQLQKITGGYFKNFHRVLNLGESDRYSVPYFFDGCLDAKL
ARLDGRDEGKVLTVEEHMLERFATTYGRGQAKD

Codon-optimized NHI 27 sequence:

ATGAGTCCAAGCAAGACCCCGAGCTTTACCATTCCGACGGTAGACATTAGCAGCTAT
GTCGCCGACCCGACCTCGCCCGCTGCGAAATCCGTTATTGAGGATGTTAGAAACG
CATGCATCACTAGCGGTTTCTTTTACGCTGGTGGGTCATGGAGTTCCACGCGAACTG
CAAGAAGCCGTGCTGAACGGCGCTAAAGCATTCTTTAATTTACCGCAAGACGAAAA
GCTGAAGTATGAGAAAAGCACAGTCGGCAAAGGGAATCGTGGCTACGAGATCATG
GGGAAACAGATCTTACAGAAAGGAGCGCTGCCAGATCTAAAAGAAGGTTTCTATGT
GGGCGAAGATTTTGAAGCAAGCCCGGATCCCAACCATAAGTATAGCTGGTTCAATC

CGCCCAATGTGTGGCCACGTCGTTGGCTGACAGCGAAATCGTATTTGCAAACCT
ATCAACGAATACTATTCCCACATGTTCCCATTAAGCACGACGATTCTGGACATTATGG
CCGCAGGTATGCCGTATGGTCCTTCCGTTTTTCGATGAGTTCAAAAGCGATCCAGTG
GTAGCAGTGATGCGCCTGTTACATTATCCACCGCAGCCTGAATCTTCTGACGCCTTG
CAGTTCGGTGTGGCGCACATACCGATTTTGGCTCTATCACACTGCTCCTACAGGAT
ATGGCTGGCGGTTTAGAAGTGTTGAACCACGACACGAACGAATGGGTCCCTGTTAC
CCCTAATCCGGACGCGTATGTAGTGAACCTCGGTGACATGATGCAGCGTTGGACGC
GCGGTGCTTACCGCTCTAACTTGCATCGTGTAAGAATATGGGTGGGGGGGATCGC
TACAGCGTACCGTTCTTCTTCGATGGCAACATGAAAACAAAGCTCGCCCCGTTCTGA
CGGCGGAGAAACCGATGAGGAAGTAATTACCGCCGAAGATCACATGATGGAACGAT
TTAGAACCACCGCAGGCTTTTAA

NHI 27 protein sequence:

MSPSKTPSFTIPTVDISSYVADPTSPAASKSVIEDVRNACITSGFFQLVGHGVPRELQEAV
LNGAKAFFNLQPDEKLKYEKSTVGKGNRGYEIMGKQILQKGALPDLKEGFYVGEDFEP
SPDPNHKYSWFNPPNVWPTSLADSEIVFRKPINEYYSHMFPLSTTILDIMAAGMPYGPS
VFDEFKSDPVVAVMRLHYPPQPESSDALQFGVGAHTDFGSITLLLQDMAGGLEVLNH
DTNEWVPVTPNPDAYVVNLGDMMQRWTRGAYRSNLHRVKNMGGGDRYSVPFFFDG
NMKTKLAPFDGGETDEEVITAEDHMMERFRTTAGF

Codon-optimized NHI 28 sequence:

ATGCCTAGCTTCACGGAAAATTCGGGCTCGACCAGCTTTCAGATTCCGACAATTGAT
ATTTCCCCGTTTATTACCAGCCCAGCATCTCTTAAAGCGGGCAAAGTAATTCAAGCC
ATCCAACCTGGCATGTCGCTCCACGGGCTTCTTTCAGATTACAGGTCACGGCGTGCC
GGAATCGGTGCAGGAACAAGCTCTGGATGCCTCGAAACGCTTCTTCAGCATGCCG
CTTGAGGACAAAGAACTTTGAACATTAACTTAATCCGGGCTTTTCGCGGCTACGAT
CCGTTGTGCACGCAGACTTATGGGCCGGATGCCCTGCCCGATCTGAAAGAAAGCT
TCTTCGTTGGCCACGACATCCCGCTCGACGATGTGCGGGTGAAAGCCCGCAGGCT
CCTGACCGGCCCAATGTCTGGCCGAGCGGGCTGCCAGCGTCCAGCTTCAAAGAT
TCCATCGAAACGTATTATAATGCGCTTCATGATTTGACGCTACGTATTTTAGAAATGAT
TGCCCTTGGTCTGCCGTATGGGGCCGACGTATTTAGTAATTTTCATCTCAAACAGCCC
GGCGGCTCCTATGCGTCTGTTGCACTACCCACCGCAGGAAATCGACGATGACAAAC
GTCAGCTGGGTGCATCCGCTCACACTGACTTCGGCGCAATTACTCTCTTGCTCCAG
GATAAGAACGAAGGACTCCAAGTGCATGATGTTCAAACGGATTCTTGATGACTGTA
CCGCCGCGCAAGGATGCGTTTGTCTTAATGTAGGGGACATGTTAGAAAAGTGGAC
GGCGGGCGATCTTAAAAGTTCTCTTCATCGCGTCCTCAACCAGGGGGGGCGCGAC
CGCTACTCGATCGCCTTCTTCTTCGATGGGAACCTGGACTGCACACTCAGCCCGCT
GGATGGGACGCGCGTCGAAGGCCGAACCGCTGACTGTTGAGAAACACATGATTCAA
TGTCTATCTGCTAGCTATAAGGAAACGGTCGAGTAA

NHI 28 protein sequence:

MPSFTENSGSTSFQIPTIDISPFITSPASLKAGKVIQAIQLACRSTGFFQITGHGVPESVQ
EQALDASKRFFSMPLEDKETLNIKLNPGRGYDPLCTQTYGPDALPDLKESFFVGHDP

LDDVRVKARRLLTGPNVWPSGLPASSFKDSIETYYNALHDLTLRILEMIALGLPYGADV
SNFISNSPAAPMRLHYPPQEIDDDKRQLGASAHTDFGAILLLLQDKNEGLQVHDVQTD
SWMTVPPRKDAFVVNVGDMLEKWTAGDLKSSLHRVLNQGGDRYSIAFFFDGNLDCT
LSPLDGTRVEGEPLTVEKHMIQCLSASYKETVE

Codon-optimized NHI 29 sequence:

ATGGCAAGCAGCGGCTTTTACATACCATCTGTTGATATTGCGCCTTTCATTACAGATC
CTTCCAGTGACGCCGCGAAAGACGTGATCGAAAAGGTCAAGAAAGCGTGCGTCAG
CACCGGCTTCTTTTCAGATTTTCGGTCATGGAGTGCCGAAGCAGGTACAGAAATCCG
CATTTGATGCATCCGCGGCGTTCTTTAGCCTGCCGGTCGATACGAAGATGGCGCTG
AAGGCAGGCCCTAGCACTGGCTTTAAGGGTTATGAACAGCTGGAGAGCCAAATCTTA
CAGCTATGATGTTAAAGGAGATATTAAAGAAGGGTTCACCATTTCACCGATTATCCT
GACGGACATCCACTCCGCACCCCTCTGAATGGCCGCCCTCGTTTTCTGACAGCTC
CGAATATTTGGCCCACCACCGAAATGATGGGCGATGCAGAGTGTTCCATGCTTCT
TTGGAAGCCTACCAAGAAGCAATGGTTACCCTGCGAGCGCATGTTTTATCTCTTATT
GCCGCGACCTTACCGTACGGCCCGCATGTCTTTGATGAAATGAATCAGGAGCCGGC
ATGCCCGCTGCGTTATCTCCACTACCCGCCGATGTCTGAAGGGACCGATATTGTGA
AACAGCAGGGAGCTTCCGCACACACTGATTTTAGCGCGGTCACTCTGCTGCTCCA
GGGTCAACAGGACGGCCTTGAGGTGCTGGATTACGACACGGGTGAATGGCATTG
GTACCGCCGAACCCGGATGCCTACGTAATTAATCTTGGCGATATGATGGCGAAATTA
CTGCGCGGACAGTATAAAAGTAGCAAACACCGCGTCATTAATCGCGCGAAAGTGGA
TCGTTACTCAATTGTTTATTTCTGGGACGGCAACCGCGATTTTGAAGTGAACCGCT
GGATGGCGCAGAAATGGATCAAAAGCACATGACGTGTGAAGAATACATGTTTCGATA
GATTGAAATTCAGCTTTGGGCGTCACGAAACCAAACAGAGTGCAGGCGCAGTAT
TAA

NHI 29 protein sequence:

MASSGFYIPSVDIAPFIHDPSSDAAKDVIEKVKKACVSTGFFQIFGHGVPKQVQKSAFD
ASAAFFSLPVDTKMALKAGPSTGFKGYEQLSQSYSYDVKGDIKEGFTISTDYPDGHP
LRTPLNGRPRFLTAPNIWPTTEMMGDAEWFHASLEAYQEAMVTLRAHVLSLIAATLPYG
PHVFDEMNPQEPACPLRYLHYPPMSEGTDIVKQQGASAHTDFSAVTLLLQGGQDGLV
LDYDTGEWHLVPPNPDAYVINLGDMMAKLLRGQYKSSKHRVINRAKVDRYSIVYFWDG
NRDFELKPLDGAEMDQKHMTCEEYMFDRCLKFSFGRHETKPEVQAQY

Codon-optimized NHI 30 sequence:

ATGGCGAATTTTCATATCCCGACCATAGACATTAGCCCATATCTCCAGGATGCCACGT
CAGATGCGGCAGACCAGGTGGTGCGTGATGTGGGTAAAGCGTGTGCCACCGTGG
GGTTCTTTAGCTTGGTTGGTCATGGCATCCCCGAAACATCCAAGACAACGTGTTC
GAAGCGGCGAAGAAGTTGTTTAACCTACCCGCCGAGGAGAAGCGTGCGCTGAGAC
ATCCACTGATGAAGAATCGTGGATATGAGCTAATTGGGAGCCAGGCACTCCAGGATA
ACACGCTGCCCAGATCTGAAAGAAGGGTTTTACATTGGCAAGCATATCCCAGCGGAA
TCGGAACGGTCGCGCGCCCATCCCGAACTGCTGGGAGAGAATATCTTTCCGGCCT
CACTTCACGATACCGTGTTTAAGAAGCCCACCGAAGCCTATTATACCGCGGTCTTGG

AGCTGTCCTTAAAAGTCTTAGAAATTCTGGCAAAGGGCCTTCCTTATGGTGATGACA
TTTTCAAGGAATTTGTCTCTAACGATCCAGTTTGTATCATGCGCCTCTTGCATTACCC
GCCACAGGCATCACAGGACGAACGTCAGCTCGGCGCGGGCGCACATAACCGATTTT
GGTGCAATCACTCTGCTGTTACAGGACACTTCCGGCGGTCTGGAAGTCCTGGATCA
TAATACAGGAACGTGGCACGCGGTAGCTCCTAACCCAGACGCGTACGTGGTGAATA
TAGGTGATATGCTGGCCCTGTGGACTAAAGGAATTTACAAAAGCGCCGTTCATAGAG
TTATTAATAAGTCCGGTGGAGATAGATATTCCGTGCCGTTCTTCTTCGATGGCAACG
CGGAAGTAAAGCTTAAGCCCTTTGATGGCAGCGAACCAGTCGGCGGGCAAATTATC
ACCGTTGAAGATCACATGTTAGAACGTCTTGGCACTTCGTATGCACGTGTGAACAA
GGTGCTGGCCAAGCGGTCAGCGCCAGTGCGTAA

NHI 30 protein sequence:

MANFHIPTIDISPYLQDATSDAADQVVRDVGKACATVGFFSLVGHGIPRNIQDNVFEAAK
KLFNLPAAEEKRALRHPLMKNRGYELIGSQALQDNTLPDLKEGFYIGKHIPAESERSRAH
PELLGENIFPASLHDTVFKKPTEAYYTAVLELSLVLEILAKGLPYGDDIFKEFVSNDPVC
MRLHYPPQASQDERQLGAGAHTDFGAILLLLQDTSGGLEVLHDHNTGTWHAVAPNPD
AYVVGDMGLALWTKGIYKSAVHRVINKSGGDRYSVPFFFDGNAEVKLKPFDGSEPVG
GKIITVEDHMLERLGTSYARVEQGAGQAVSASA

Codon-optimized NHI 31 sequence:

ATGGCCGTCCCGCTGATGCATCAGACAATGCACAGCCAGGTGCATGAAGCCAGCG
ATGCATCACACGAAATCCCGACCATTGACATCTCCGGCTTTTCGTTCCAGCCCGGAA
ACTGCGCAGGAACGTATCGAGACTCAACATGTGGTTGACCGGGTGCGTAGTGCGT
GCCTGACGACCGGGTTCTTCTGTCTCACGGGCCATGGCATTCCGCGCGAATTGCA
GGAGCGTGTTCTTGATGGTGCAGAGAAATTCTTCGACATGCCGCTTGAAGAAAAGC
TGAAACTTGTTACCCGACTCAGAAGAACCGCGGTTATGAGGTGCTTGGTGGCCAA
ACTCTGCAACGTGGTGCAAAACCGGATCTGAAAGAAGGCTTTTACATCGGCAAACA
CCTGGATGGTAACGACCCCCGAGTGCGGCGCCACCCGCGCCTGATGGGCCCAAA
TATTTTCCCGAGCGGCAGTCTTGAGGATCATGAATTTAAGAACCCACTGGAAACCTA
TTATGAAGCTATCTTAGCGTTGTCTCTGACGGTTTTTCAAATACTGGCGAAAGGCCT
GCCATACGGGGATGATGTGTTTGACCGATTCTGTGAGCGATGATCCGGTCTGCATTC
TCAGACTGCTGCATTATCCGCCGCAGACCAGTAAAGATCCTCGCCAGCTGGGCGC
CGGCGCGCATACAGATTTTGCGCGGTTACCTTGCTATACCAAACGGCACCAGGTT
TACAGGTGAAGTCTCCGAGTGGCTCAGGCTCAAGCGGTGCCGATCGTTGGATTCC
GGTGCCGCCAAACCGTGACGCGTACGTTGTCAATATAGGCGACATGCTGCACGCG
TGGACAGAAGGACTCTATAAATCCTCGATGCACCGCGTGCTAAATCTGGGTGGCAG
TCACCGTTATAGCATTCTTTCTTCTTCGATGGTAACGCAGACTGTCCGCTGACTCC
GTTCCGAAAATCAGACGACGTTGGCGCTGAGACTGCCAGCGAGCATTACTGCGT
CGCCTCGACGAAACGTATGCAATGGCAGCCAAAATCTAA

NHI 31 protein sequence:

MAVPLMHQTMHSQVHEASDASHEIPTIDISGFRSSPETAQERIETQHVVDVRVSACLTT
GFFCLTGHGIPRELQERVLDGAEKFFDMPLEEKLVHPTQKNRGYEVLLGGQTLQRGA

KPDLKEGFYIGKHLDGNDPRVRRHPRLMGPNIFFPSGSLEDHEFKNPLETYYEAILALSL
TVFKILAKGLPYGDDVFDRFVSDDPVCILRLLHYPPQTSKDPRQLGAGAHTDFGAVTLL
YQTAPGLQVKSPSGSGSSGADRWIPVPPNRDAYVVNIGDMLHAWTEGLYKSSMHRVL
NLGGSHRYSIPFFFDGNADCPLTPFGKSDDVGAETASEHLLRRLDETYAMAAKI

Codon-optimized NHI 32 (XenC homolog 1) sequence:

ATGATTCCCACGGTGGATATTTCCGCCTTCAACGACGATAGTGCTACCGCGGATAGC
AAGAACGAAGTGGTGCAGGCTGTGGCGCACACCTGCTCTACGTATGGGTTCTTCTA
TCTGGTGGGCCATGGTATTCCGGAAGCGGATCGTCAGCAGATTTTGGATTGTGCCC
GCTTATTTTCATCCCTGCCCATGGATGAGAAAATGATCATCTCGGTGAGCAAGCGCA
TGGGGCAGTCGTTTCGTGGTTATGAACCCCCAGCCCTGCAATTGCACCAGGAAGG
TCTACTTCCTGATACGAAAGAGGCTTTTCATTTTCGGTCGCGAGGTTAGCGCGGACC
ATCCGGATGCCGGCCGTTTCAGCACCGGCCCGAACCAGTGGCCGCCGTCCTACTGC
CGGATTTCGACGTTTTGTATCCCACTGCTGGAATACCAGCGTAAAATGGTGCAGCTG
GTGAAAACCATTCTGAAGATTCTGGCCTCGGGCCTGCCGGAAGAATGGAATTGTCC
ATGCGATGTGTTTGATGCCGCTACCGTTCGAACCAAGCATCCCTATGCGCTTACTCCA
TTATGCGCCTCAGTCAGTTAAGAGCGAAAACCAATTTGGCGTTGGCGATCACACCG
ATTTTGGCAACGTGTCCATTCTTCTCCAGGAAGAAGGCACGGAAGGCTTAGAAGTG
TGGTATCCGTTGACAGAGGATTGGATCCCAGTGCCCGTTATTCCGGGTTTCTACGT
AATCAATATCGCGGACATGATGCAAAAGTGGACGGCGGGCTTCTACCGTAGCGCTC
GTCACCGAGTTGTAAACCATAATAAACGTCCGAGATATAGTGCAGCGTTCTTTCCCA
ACGGCAATATTGACCTGAAGTACCGTGCGCTGGATGGAAGTGGTGTGAGAAAAC
GATTGGCAAACACATACGCCAGCGCCTGTTTGAGACAATAGGTGGCGAGGCGGGT
AAGAAGCTGGAGAACGACGCCGTCAAACGCTAA

NHI 32 (XenC homolog 1) protein sequence:

MIPTVDISAFNDDSATADSKNEVVQAVAHTCSTYGFFYLVGHGIPEADRQQILDCARLFS
SLPMDEKMIISVSKRMGQSFRGYEPPALQLHQEGLLPDTKEAFIFGREVSADHPDAGR
FSTGPNQWPPSLPDSTFCIPLLEYQRKMVQLVKILKILASGLPEEWNCPCDVFDAATV
EPSIPMRLLYAPQSVKSENQFGVGDHTDFGNVSILLQEEGTEGLEVWYPLTEDWIPV
PVIPGSYVINIADMMQKW TAGFYRSARHRVNVNHNKRPRYSAAFFPNGNIDLKYRALDG
SGVEKTIGKHIRQRLFETIGGEAGKKLENDVVKR

Codon-optimized NHI 33 (XenC homolog 2) sequence:

ATGAAAGAACAGGTGACCCCGGGCATTACCAATCCTTGGCCAGTCTGTACTTCTT
CAGCATTACGAGCATCCCCATTGTTGACATATCGCCGTTTCTCGATGAAGAAGTGTC
GCAGGAAATGATGAACCAGGTGGTGGAAAGCCATGCGTGAGGCTTGTAACACGTGT
GGGTTCTTCTATTTACTGGGCCATGGCATATTCGAGGAAGATTGTAAGAGTGCCATG
GGCAAATCCTTTTCGCGGCTACGAACCGCCTGCGCTGCAAATGCATCAGGAAGGGC
TCCTGCCTGATACCAAAGAAGGTTTTATCCTGGGTTGCGAAATTCGGGCAGATGCG
CCCGAAGCGGGCACGTTCCATACTGGCCCGAATCAATGGCCGGAGAGCATTCTAT
GATGGAATACCATAGCAAATGATAGAAATGGTGAAGGTTGTGGTCAAATCTGGC
GCTGGGTCTGCCTGATCAGTGGAAGTCTCGCCGGATGTATTCGATGAGATGACGC

ATAATCCGTCCGCGCCTCTGCGTTTGCTGCACTATGCCAGCCAACCGCTAAAATGC
GAAAATCAGTATGGTGTAGGCGATCATAACGATTTTGGGAATGTGAGTGTTCTGCTC
CAGGAAGAAGGACGTGATGGTCTGGAAGCGTTTTACCCGCCGACGCATGTGTGGG
TACAGTCCCTGCGTTGGAACACTCCTATGTTATTAACATCGGCAACATGATGCAGA
AATGGACCGGTGGATATTATAAAAGTGCCCGTCACCGAGTCATCAATCACAAGACTA
AAAGCCAATATATCACTCCGTTTTCTATGAACGGTAACTTAGGCTTAAAATGTCGCGC
ACTGGACGGATCCGGCACGGAGACTGTGATTGGTGATCATATTCGGCAGCGCTTG
GTTGAAACCATCGGTGGTGAAGCTGGGAAGAACTGGGTTACTAA

NHI 33 (XenC homolog 2) protein sequence:

MKEQVTPGIYQSLASLYFFSITSIPIVDISPFLDEEVSQEMMNQVVEAMREACTTCGFFY
LLGHGIFEEDCKSAMGKSFRGYEPPALQMHQEGLLPDTKEGFILGCEIPADAPEAGTFH
TGPNQWPESIPMMEYHSKMIEMVKVVVKILALGLPDQWNCSPDVFDEMTHNPSAPLR
LLHYASQPLKCENQYGVGDHTDFGNVSVLLQEEGRDGLAIFYPPTHVWVPVPALEHS
YVINIGNMMQKWTTGGYYKSARHRVINHKTKSQYITPFMNGNLGLKCRALDGSGETV
IGDHIRQRLVETIGGEAGKKLGY

Codon-optimized NHI 34 (XenC homolog 3) sequence:

ATGGGCTCTATCGAAGCCATTCCTACGGTGGACATCTCTGCGTGGATCGATACATCC
TCTACTGAAGAAGAACGCCAGCGTGCCGTGGATGAAATGCGAAATGCTTGCACCC
GCTACGGCTTCTTTAATTTAATTGCCACGGCGTCAGTCCCCAGGCGCAATCACAA
GCGTTAAATTGTGCTAAGTTATTCTTTACCTTGTCCGAACAGGAAAAGATGAAGGTC
TGATTAGTAAATCGGTTGGCCGTAGCTTTCGTGGGTACGAGCCCCCTGGCATTCAA
GTACATCAAGAAGGCTTACTGCCGGATACTAAAGAGTGTTTCATTATCGGACACGAG
GTGCCAGAGCATGATCCGGAATGCGGCACATTTTCGACTGGGCCGAATTTATGGCC
CAAATCTCTGCCCCGAAGATGAATTTAGGATCCCTATCATGGAATATCAGGCCCCGCAT
GGTAGCCCTGGTGAAAGTGCTTTTGAAGATCCTGGCCCCGCGGGCTGCCGAAGGAA
TGGAAGTGTCTCCAGATGTACTGGATCCGCTCGGCGAGAATAACCCGAGCATTCC
GATGCGTTTGTTCATTACGGCCCCGCAACCAATTCGTGATGACCGCCAGTTTGGCG
TTGGCGATCACACAGATTTTGGATGTGTGACCATCTTGTTACAAGACATGGGTACTG
AAGGCCTGGAAGTGTGGTATCCCCAACGGAAACCTGGATCCCGGTTCCCCCACA
AGAAAATGCGTATGTAATTAATGTGCGCGATATGTTGCAGAAGTTTACGGGGGGTTA
CTACCGTAGCGCCCGGCATCGTGTGGTGACATTCACGGCCAATCATCGCTACAGTG
TCCCCTTCTTCTGAACGGACAACCTCAAATCAAATGTAAACGCTGGATGGAAGC
GGCACCGAAACCATTTAGGCGAACACATCCGTGGGCGCTTGTTGGAGACAATGG
GAAAGGCGGGGGGAAAAGTTAAAGTAA

NHI 34 (XenC homolog 3) protein sequence:

MGSIEAIPTVDISAWIDTSSTEEERQRAVDEMNRNACTRYGFFNLIAHGVSPQAQSQALN
CAKLFFTLSEQEKMKVCISKSVGRSFRGYEPPGIQVHQEGLLPDTKECFIIGHEVPEHD
PECGTFSTGPNLWPKSLPEDEFRIPIMEYQARMVALVKVLLKILARGLPKEWNCSPDL
DPLGENNPSIPMRLHYGPQPIRDDRQFGVGDHTDFGCVTILLQDMGTEGLEVWYPP

TETWIPVPPQENAYVINVRDMLQKFTGGYYRSARHRVVTFTANHRYSVPFFLNGQLKL
KCKTLDGSGTETIVGEHIRGRLLLETMGKAGEKLK

Codon-optimized NHI 35 (XenC homolog 4) sequence:

ATGGCTGAGGTGCCTCAGCAGATCCCGACGATTGACATCAGCGCGTGGGTGAATC
CGGCGTCGTCCGCCGCAGATAAACAGCGGGTCGTTGATGAAGTCCACCATGCCGC
GACGACTTACGGGTCTTCCAGTGTGCCGGGCACGGCATTACACCCGAAACCCGT
GAACAGATTTTCAACCTGAGCAAGCGGTTCTTCTCACTTCCAATGGAAGAGCGGAT
GGAAATCTCCGTGAAGAAGAGCTTGGGCCAGTCTTTCCGCGGCTACGAGCCGCCC
TTTCTCCAGAACCACCATAAACATTTGCTCCCGGATACCAAGGAAACCTTTGTGGTG
GGCGCGGAAATCCCGGCGGACCACCCGGACGCTGGCACCTTTCTGAATGGTCCA
AATTTGTGGCCTACGAGCATTCTGAGGACGAGTTTCGTAAACCGATTATTTTCATATC
AGAGCCAAATGGTTGCGTTAGGAAATACGTTGTTAAAGATTCTGGGGCACGGCTTG
CCGAAGAGTTGGAATTGCAGCCCTGATGTTTTTCGATTCATTAGCCGAAAACCCGGC
CATTCCGATGCGCATGCTGCATTATGCGGCCACGCCGTACATGACGAGAAACAAT
TCGGTGTCCGTGCTCATAAGATTTTGGCAGCGTTACCATACTACTGCAACAGCTG
GGCACCAAAGGTCTGGAAGTGTGGTATCCACCAACGGAACGTGGATTCCGGTTC
CGCCAAAAGAAAATACCTTCGTGATTAATATCGGTGATTTTGTA

GACCGGTATACGAACGGTTATTACCGGTCTGCGCGCCATCGCGTAATTACCACCGAT
AAGGAACGGTATTCGGTGGCGTTCTTCTTTAATGGGAACATGAAGCTTAAAGCCCC
AGTTTTGGACGGCTCAGGAGATGAAGTGAAGTTAGGTGATCATATACGCCAGCGTG
CGATTGCCACTTATGGTGCAGCTCCTACTCTGATCGAGAGCAAAGTGTA

NHI 35 (XenC homolog 4) protein sequence:

MAEVPQQIPTIDISAWVNPASSAADKQRVVDEVHHAATTYGFFQCAGHGITPETREQIF
NLSKRFFSLPMEERMEISVKKSLGQSFRGYEPPFLQNHKKHLLPDTKETFVVGAEIPAD
HPDAGTFLNGPNLWPTSIPEDFRKPIISYQSQMVALGNTLLKILGHGLPKSWNCSPDV
FDSLAEENPAIPMRMLHYAAQPVHDEKQFGVGAHTDFGSVTILLQQLGKGLVWYPPT
ETWIPVPPKENTFVINIGDFVDYRNGYYRSARHRVITTDKERYSVAFFFNNGNMKLKAP
VLDGSGDEVNLGDHIRQRAIATYGAAPTILIESKV

Codon-optimized NHI 36 (XenC homolog 5) sequence:

ATGGGATCGCTCACGGAAACCGCCACAATCCCCACGGTGGACATCTCAGCATTTCT
TGACCCGAATGCTTCTGACCAGGCGCGTCAGGATATTGTTAAAGCTATGTCACATGC
TTGTCATGTCTATGGCTTCTTTAATCTGGCAGGTCATGGCATTCCGCAAGAGACACT
GAGAACAGCGCTGGAGCTTAACAAAATGTTCTTCACGTTACCGGAAGAAAGCAAGA
AGGAAGTGCTTATCTCTAAGTCAGTGGGTCAAAGTTTCCGTGGATATGAACCCCCG
GGGATCCAAACCCATCACCGGGGCCTTCTGCCGGATATTAAGGAAACCTTTATGGT
GGGCCGCGAAGTACCGCTCGATGACCCTGATTGCGGCACATTCTCACGTGATCCG
AATCTCTGGCCGACCAGCTTACCGAAAGAAAAGTTCCAGGATCGTATCATGGCCTA
CCAGGGCAGCATGCTTCAACTGGTGAAATCCATCCTGGCAATTCTGGCCCAGGGTC
TGCCGAAAGAATGGGAATGTTCTCCGACCGTGTTTCGATTCACTGCTGGAAAAGCCC

TCTATCCCGATGCGTTTTCTGCATTATGCACCAGTGCCGTCACAGCTGGAAGATGCA
CGTCAGTTTGAGGTGGCGGACCACACCGACTTTGGCTGTGTTTCTATTCTGCTTCA
GGAACCCGGAACCAGTGGGCTGGAAGTGTATTATCCCCCGTCTGACAGTTGGATTC
CGGTACCCGTGGTTAATGATGGCTTTGTTATCAACATGGGCGATATGATGCAGCGTT
ATACAGGCGGTTACTATCGCTCGGCGCGTCATCGCGTTCTGACCAATAGGGAGAAA
CATCGCCACAGCGTAACTTTCTTCCTTAATGGGAACCTGAGTCTTAAAGCGAAAGCT
CTGGATGGGTCCGATACCGAAACTGTCGTTGGTGACTGGATCCGCGGGGCGCCTGT
TGGAACAATGGGCCAGACCGGTAAATTGTTGCAGCGTGAATCCAGCAAAACCCAA
TAA

NHI 36 (XenC homolog 5) protein sequence:

MGSLTETATIPTVDISAFLDPNASDQARQDIVKAMSHACHVYGFFNLAGHGIPQETLRTA
LELNKMFFTLPEESKKEVLISKSVGQSFRGYEPPGIQTHHRGLLPDIKETFMVGREVPL
DDPDCGTFSRDPNLWPTSLPKEKFQDRIMAYQGSMLQLVKSILAILAQGLPKEWECSP
TVFDSLLEKPSIPMRFLHYAPVPSQLEDARQFEVADHTDFGCVSILLQEPGTSGLEVYY
PPSDSWIPVPVNDGFVINMGDMMQRYTGGYYRSARHRVLTNREKHRHSVTFFLNGN
LSLKAKALDGSDETETVVGDWIRGRLLTGMGTGKLLQRESSKTQ

Codon-optimized NHI 37 (XenC homolog 6) sequence:

ATGACCGTGCAGCACCAGATTCCGACCATCGATATTAGCGCGTTCCTCGATCAGAC
CGCCAGTGAAAAGGCGCGGGGAAGCAGTGGTGGCAGAGGTTTCGCGATGCGTGTAT
GACGTATGGCTTCTTCCAGCTGATTGGCCATGGCATTCTCTGGCTATCCAGGAAC
ATGTCCTTCGCTGTGCCAAGGTATTCTTCGACCAGCCCCTAGAGAAGAAACAGGCA
CTGGGCATGGATAAAGCCATGGGTCAGAGTAATCGTGGCTACGAACGGATCGGGG
GCCAAGCGCTCCAGGCAGATATGCTGCCTGACCTAAAAGAAGGTTTCTATATTGGC
CGCGAAGTGCCGGCCGACGATCCTAGAGCTGGCCAGTTTCACCAAGGTCCAAATC
TATGGCCCGAAATTCCGGAAGAAGAGTTCCGCGTACCTCTGTGCGGAATACTATCGTA
TGATGATTCAATTGGATTATCAGCTGCTGAATATCATCGCATTAGGCCTGCCGTTTTTC
CAACGACATCTTTGAAGAATTTAAGAAGGAACCGGCCGCGAGTCTGCGCTTGTTGC
ATTACCCCCCTCAGATCAGTATCGATGAGCGCCAGTTTGGTGCGGGGGGCCACAC
GGATTTTGGTTGTATGACCCTGCTGTTACAGCAGCCGGGTCAGGAAGGCCTGCAA
GTTTTCTATCCGCCGACCAAAGAGTGGATTCTGTTCCAGCTGCGGCTAACACCTT
TGTCGTCAACATCGGTGACTTGTTACAGCTGTGGACGAAAGGTCAGTATCGTAGCG
CGCTCCACCGAGTGTTAAACGCTACTAACGAGGATCGCTACTCTGTGCCGTTCTTC
TACAACGGTAACCTCGCATGTAACTGGCCCCGCTGGACGGGTCCAGTGATGTGAA
TACGCTAACAGTGGAAGAACACGTCAAACGGAAATTTCAAACATCTTATGCGGTGTA
A

NHI 37 (XenC homolog 6) protein sequence:

MTVQHQIPTIDISAFLDQTASEKAREAVVAEVRDACMTYGFFQLIGHGIPLAIQEHVLR
AKVFFDQPLEKKQALGMDKAMGQSNRGYERIGGQALQADMLPDLKEGFYIGREVPAD
DPRAGQFHQGNLWPEIPEEEFRVPLSEYYRMMIQLDYQLLNIALGLPFSNDIFEEFKK
EPAASLRLLHYPPQISIDERQFGAGAHTDFGCMTLLLQQPGQEGLQVFYPPTKEWIPV

PAAANTFVVNIGDLLQLWTKGQYRSALHRVLNATNEDRYSVPFFYNGNLACKLAPLDG
SSDVNTLTVEEHVKRFQTSYAV

Codon-optimized NHI 38 (XenC homolog 7) sequence:

ATGAGCAGCGAGAAATCGGAAGGTGGGATTCCAACTGTGGATATTGGCGCCATCCT
TAGCCCGGCCGCGAATGAAGCCGATCGTCGCCAGGTTGTGGAAGCCGTACGTTAT
GCCAGCACTACCTTCGGGTTCTTCAACTTGGTTGGCCACGGCATTCTCTCGCGCTCA
GTTGCGCCAGATTTTCGAATCTTCTAACTGTTCTTCGCACTGCCGGAAGAGAAAC
GTCGTGATGTGCATGTTGGTCAGGCATTAGGTCGCAGTTTCCGTGGTTGGGAACCG
CCGCTGATCCAACAGCATCAAGAAGGAGTGCTGCCGGATACTAAAGAAACCTTTAT
CATCGGGCGAGAGGTCCCGGCTGACGATCCGGAAGCGGGCACCTTTCTAACGGG
TCCGAATCTGTGGCCAGACCTGCCTACTGAAGATTTCCAAGAAGTCATTCTGGCCT
ACCAAGCGCGCATGATTGAGCTGGCTCGTGTGCTAGTGCGTATTTTAGTACAGGGC
CTGCCTGAAGCGTGGGGATGTCCACCGGATGTTCTGGATGGTCTGACCGTCGATC
CGGCCATCCCGATGCGCATGCTGCACTATGGCCCCGTCAAAGAAAAGGATCCACG
TCAATTTGGTGTGCTGCCCATACCGACTTTAGTGCTATTACCATTCTTTTGCAACAG
CCGGGAAGTGAAGGTCTGCAAGTTTGGTATCCGCCGACAGACGAGTGGATTCCGG
TCCCTGTATCTGAAGATGGTTTTGTGATTAACATTGGAGACTTAATGCAGCAATATAC
CGCGGGCTATTATCGATCTGCACGCCACCGCGTGATTACCTTCTCAGAAGAAGATAA
ACACCGGTATAGCGTAGCTTTCTTCCTGGATGGTAATCTGCAATTCAAGACCAAAGC
GCTGGATGGCTCTGGTACCGAGATCGTCGTTGGAGAGTACGTATATTCATGCCTCAA
TCGTACCCTGGGAGCGCGTGGCCCCGTCCCTGTAA

NHI 38 (XenC homolog 7) protein sequence:

MSSEKSEGGIPTVDIGAILSPAANEADRRQVVEAVRYASTTFGFFNLVGHGIPRAQLRQI
FESSKLFFALPEEKRRDVHVGQALGRSFRGWEPPLIQHQEGVLPDTKETFIIGREVP
DDPEAGTFLTGPNLWPDLPTEDFQEVILAYQARMIELARVVVRILVQGLPEAWGCPPDV
LDGLTVDPAPMRMLHYGPVKEKDPRQFGVAAHTDFSAITILLQQPGTEGLQVWYPPT
DEWIPVPVSEDGFVINIGDLMQQYTAGYYRSARHRVITFSEEDKHRYSVAFFLDGNLQF
KTKALDGSGETIVVGEYVYSCLNRTLARGPSL

Codon-optimized NHI 39 (XenC homolog 8) sequence:

ATGACAAGTCCGGTGGGAAGCCATTCCGACTGTGGATATTTGCGCCGTTCTGTAGCCAA
CGCACCTGCCAAAGCGCTACTGGACGTGGTAAAAGCGATGCGTCACGCCTGTACG
ACCTATGGCTTCTTTTATTTGGTCGGTCATGGGATTTGCTTGAAGAACAAAACGGC
GCGTTAGATTGCGCCAACTATTCTTCACGCTGAGTCAAGAACAGAAAATGGATGTC
TGGCTGAAGAAATCTTTAGGTAAATCGCACCGAGGGTATGAACCACCTGGCATCCA
AACCACCAAGAGGGTCTGTTACCGGATACAAAAGAAGCATTTATCTTTGGTGCTGA
AATCCCCGAGGATGATCCCGAAGCAGGTACCTTTAGCACCGGCCCGAACCTGTGG
CCAAAAGCGCTGAAGCCGCATGAGTTTCGCAACCCGATCATGGAATACCAGGGCAA
AATGCTGGAAGTGGTGAAAATCCTACTTAAATCTTGGCACTAGGCCTGCCGGAAG
AATGGGAATGCGCCCCAGACGTGTTTGATGAGCTTGCGATAAATCCATCTATGCCGA
TGCGCCTGCTACACTATGCCCCAAACCCGACGCATAACAATACTCAGTTCGGGGTG

GGGGATCATACCGACTTTGGGTGTATCACTGTGCTGTTACAAGAGCCAAACCGTGA
GGGGCTCCAGGTCTGGTATCCGCCGACTGAGACTTGGATTCCGGTACCGGTAAAG
GAAGGATCATACGTTATAAACATCGGTGACATGATCCAGAAATACACCGGTGGCTAC
TACCGTTCGGCGCGCCACCGTGTGGTCAATGGCGGTAAAGAGCACCGGTACTCTG
CTCCATTCTTCTTAAATGGTCAGTTAAACTTAAAGTTACCGCCCTTGACGGCAGCG
GCGTGGAGACAGTTGTGCGGCAACATATCCGTCAACGTTTGATCGATACGATGGGA
GAATCCGGTAACGTGTTAAATAA

NHI 39 (XenC homolog 8) protein sequence:

MTSPVEAIPTVDISPFVANAPAKALLDVVKAMRHACTTYGFFYLVGHGISLEEQNGALD
CAKLFFTLSQEQKMDVWLKKSGLKSHRGYEPPIQTHQKGLLPDTKEAFIFGAEIPED
DPEAGTFSTGPNLWPKALKPHEFRNPIMEYQGKMLELVKILLKILALGLPEEWECAPDV
FDELAINPSMPMRLLHYAPNPHTNNTQFGVGDHTDFGCITVLLQEPNREGLQVWYPPT
ETWIPVPVKEGSYVINIGDMIQKYTGYYRSARHRVNGGKEHRYSAFFFLNGQLKLK
VTALDGSVETTVVGEHIRQRLIDTMGESGNVLK

Codon-optimized NHI 40 (XenC homolog 9) sequence:

ATGGGTAGCATAAATGAAACCCAGACCACCTCAATTCCCACCGTTGACATCGGCCC
ATTTCTGGATGGACAGGCCGCTCAGGAAGCTAAGGACGCAGTCGTGGCCGCGATG
CGCCATGCCTGCTCTACCTTCGGTTTCTTCTATCTGGTGGGACACGGCATCCCAGA
AGAAGATCGCTTGCAGATTCTGAAATGCACGAAACGCTACTTTGATCTGCCGATGG
AAGAAAAGATGGAAACATGGATTGGCAAAGCGATAGGCCGATCATTCCGCGGCTAT
GAGCCGCCAGCGCTGCAAGTGCACCAGGAAGGCCTGCTGCCTGATACAAAGGAA
GGTTTCATCCTGGGTGCGGAGACTCCCGCGGACGCCCCCGAGGCAGGTACATTCC
ATACGGGGCCCTAATCAGTGGCCCAAACCTGCCGGACGAGGAATTCGCGTGCC
AATCATGAGATACCACGCGAAGATGATTGAGATGGTGAAAGTGATTCTCAAATCTTA
GCTCTTGGACTGCCGGCAGAATGGAATTGTCCATCTGATGTCTTTGACGAATTGAC
GATAAACCCATCGGCTCCCTTACGTTTACTCTACTATGCCCCCAGCCGGTCAAGCA
CGAAAATCAGTTTGGCGTCGGTGCTCATACCGACTTTGGAAATGTTTCTGTGCTGCT
ACAAGAAGAAGATTCCGAAGGCCTAGAAGTGTTTTATCCCCCCTCAGACGTGGG
TGCCGGTCCCAGCCAAACCTCATAGTTATGTTATTAATATGGGCGATATGATGCAGAA
ATGGACCGCGGGCTATTACCGTTCGCTCTGCATCGTGTAGTTAATTCAAATACGAA
GCCTCGTTATTCGGCCCCGTTCTTCATGAATGGGAACCTGGATGTGAAGTGTTACG
CGTTGGACGGTCTGGGGAAGTGACTACGATTGGCGAGCATATTCGCAAACGTTTA
ATGGAAACGATCGGGGGCGAAGAAGGTAAGAAGCTGGGCCTGTAA

NHI 40 (XenC homolog 9) protein sequence:

MGSINETQTTSIPTVDIGPFLDGQAAQEAKDAVVAAMRHACSTFGFFYLVGHGIPEEDR
LQILKCTKRYFDLPMEETWIGKAIGRSFRGYEPPALQVHQEGLLPDTKEGFILGRE
TPADAPEAGTFHTGPNQWPKTLPDEEFRVPIMRYHAKMIEMVKVILKILALGLPAEWNC
PSDVFDELTINPSAPLRLLYAPQPVKHENQFGVGAHTDFGNVSVLLQEEDSEGLEVFY
PPTQTWVPVPAKPHSYVINMGDMMQKWTAGYYRSALHRVNSNTKPRYSAPFFMNG
NLDVKCYALDGSGEVTTIGEHIRKRLMETIGGEEGKKLGL

Codon-optimized NHI 41 (XenC homolog 10) sequence:

ATGGGGAGCGTTGAAACCAATTCCAACCGTTAATATCTCCGCCTGGCTCGATCCGTC
AAGTACCGAGGAAGAAAGGCAGAACGTGGTCAAGAGATGCGTCATGCTTGTACC
ACTTACGGTTTCTTATCCCTGGTTGGTCATGGCGTTTCGCCTGAGGAGCAGAAACA
AGCAATGGACTGCGCCAAACTCTTCTTCACATTGTCTGAGGAAGAAAAGATGGATG
TGTGGATCGGTAAATGTAAAGGTCGTTCTTTTCGTGGTTACGAACCGCCTGGTATTC
AAGTACATCAGGAAGGTCTGCTTCCTGATACCAAGAATGTTTTATTATTGGCCATGA
AGTTCCAGAGGATCACCTGACTCTGGCACCTTTAGTACTGGTCCGAACCTTGTGGC
CGAAATCGCTTCCAGAAAATCAGTTCCGCATTCCGATTATGCAATACCAGGCGCGTA
TGGTGCAACTTGTTAAAGTTCTTCTCAAATCTTGGCTCGTGGCCTGCCGTGCGAAT
GGAATTGCCCCCGGATGCGCTGGACGCCTTAGCTGAAAATGAACCTCTATCCCA
ATGCGCTTGCTCCATTATGCGCCGCAGCCGGTCCGCGATGAGCGGCAGTTCGGCG
TTGGCGATCATACCGACTTCGGTTGTATTACCATTCTGCTCCAGGAAATGGGTACTG
AGGGCTTGGAAGTGTGGTATCCACCCACGGAAACATGGATTCCGGTGCCGCCTCA
GGAGAACGCGTATGTAATTAACATAGGAGATATGATTGAGAAATTTACCGGGGGGTAT
TATCGCTCTGCGCGTCATCGTGTGATTACTTTTACGGAGAAACATCGGTATTCCGTT
CCATTCTTTCTCAATGGCCAACTGAAATTGAAATGTACGGCCCTGGACGGTAGTGG
CGTCGAAACCATAGTCGGCGATCATATCCGAGGTGCGCTCATCGAAACCATGGGCA
AAACCGGCGAGAAATTGAAGTAA

NHI 41 (XenC homolog 10) protein sequence:

MGSVETIPTVNISAWLDPSSSTEEERQNVVEEMRHACTTYGFLSLVGHGVSPEEQKQA
MDCAKLFFTLSEEEKMDVWIGKCKGRSFRGYEPPGIQVHQEGLLPDTKECFIIGHEVP
EDHPDSGTFSTGPNLWPKSLPENQFRIPIMQYQARMVQLVKVLLKILARGLPSEWNCP
PDALDALAENEPSIPMRLHYAPQVRDERQFGVGDHTDFGCITILLQEMGTEGLEVW
YPPTETWIPVPPQENAYVINIGDMIQKFTGGYYRSARHRVITFTEKHRYSVPFFLNGQL
KLKCTALDGSVETIVGDHIRGRLIETMGKTGEKLG

Codon-optimized NHI 42 (XenC homolog 11) sequence:

ATGGGGCTCGATTGGGACTGCACCTGAAGCGATTCCAACCTGTTGATATTTGCGCCGTT
CATTAGCAATGCATCAGAGAAAACCTTACGTGAAGTGGTGGAAGCAGTGCGGCACG
CGTGCACCACTTACGGCTTCTTCTATCTGGCGGGCCATGGTGTAAGCCTTGAGGAC
CAGAACAAAGCACTTGAGTGCGCAAAATTATTCTTTACTCTGCCGATGGAAGATCGT
ATGGAAGCGTTTATTATTGGTGCCGAAGTGCCTGCCGACGATCCAGAAGCGGGTAC
TTTTAGCACGGGTCCGAACCTGTGGCCGAAATCTCTGCCAAAAGAGGATTTCCAAA
TCCCCATCATGGAATATCAGCGCAAGATGGTTCAGTTGGTGAAAGTCTTGTTACAGA
TTTTAGCTCTGGGGCTGCCTAAAGAATGGGGATGCCTGCCGTGAGTGTGTTGACGAA
CTGGCCGTTAATCCGAGCATCCCCATGCGTCTGCTGCACAGCGCGCCACAAACGG
TCCGTGATGAACGTCAGTTCGGCGTCGGCGACCATAACCGACTTTGGTTGTATTGCA
ATCCTGCTGCAAGAACCGGCGCGCGAAGGATTACAGGTGTGGTACCCGCCGACGG
AAACCTGGATCCCGGTACCGGTAAAAGAAGGTTCCCTATGTGGTGAACATTGGCGAT
ATGATGCAGAAATACACAGCGGGCTACTACCGCAGTGCCAGGCATCGGGTGGTTAA

CTCTGCGAAGGAACACAGATACAGTGTGCCGTTCTTTCTCAATGGCCAGTTAAAGC
TGAATGTTAAAGCACTGGATGGATCGGGGGTGGAGACAGTTGTCGGCGAACACAT
CCGTCAGCGTTTAGTTGAACCCATGGGAGAAAGCGGTAAAGTTCTGGGTAA

NHI 42 (XenC homolog 11) protein sequence:

MGSIGTAPEAIPTVDISPFISNASEKTLREVVEAVRHACTTYGFFYLAGHGVSLDQNK
LECAKLFFTLPMEDRMEAFIIGAEVPADDPEAGTFSTGPNLWPKSLPKEDFQIPIMEYQ
RKMVQLVKVLLQILALGLPKEWGCLPSVFDELAVNPSIPMRLHLSAPQTVRDERQFGV
GDHTDFGCIALLQEPAREGLQVWYPPTETWIPVPVKEGSYVNVNIGDMMQKYTAGYYR
SARHRVVNSAKEHRYSVFFLNGQLKLNKALDGSQVETVVEHIRQRLVEPMGESG
KVLG

Codon-optimized NHI 43 (XenC homolog 12) sequence:

ATGGGATCGAACTATGGTACCGGCGCGACCGCGATTCCGACCGTTGACATCTCGG
CTTTCCTGAGTCCTACGGCCAGCGACGCGACACGTCAGGCCGTGATCGAAGGAGT
ATCAAAAGCCTGCCACACCTATGGATTCTTTAACCTGGTTGGCCACGGCATCCCCC
AGGATGTGATGCGACGAGCGTTTGAATGTAACAAAATGTTCTTCGAACTGCCCAT
GACAAGAAGATGGAAGTCCGTATAAGTAAAAGCGTGGGTCGCAGCTTTCGCGGCTA
TGAGCCCGCCGGGCATCCAGACGCACCATGAAGGGCTTCTGCCGGACACCAAAGA
GACTTTCATGGTTGGTCGTGAAGTGCCCGAAGATGATCCTGATTGTGGGAGCTTCT
CTACAGGACCCAACCTGTGGCCTTCGTCCCTTTCGAAAGAGAAATTCAGGACCTT
ATCATGGATTATCAGGGGCGAATGCTTGCGCTGGTGAAGAACATCTTAGACATCCTG
GTACTGGGTCTGCCGAAAGCGTGGAAGTGCCACCAAATGTGTTTCGATAGCCTGCT
GGATAAACCGTCAATCCCGATGCGTTTTCTGCACTATGCTCCGGTGCAAGCTATGGA
TCCACGTCAGTTCGGGGTTGCCGACCATAACGATTTTGGCTGCGTGTCAATCCTGC
TTCAAGAGCTGGGCACGGAAGGTCTGGAAGTTTTCTACCCGCCGAGTGAGACTTG
GGTACCGGTCCCAGTGGTGAGCGATGGCTTCGTAGTAAACATGGGGGACATGATG
CAGAAATACACCGGGGGCTACTACCGGAGCGCGCGCCATCGCGTGCTGACGAACA
AGGACAAACATCGTCATAGTATTGCCTTCTTCTGAACGGCAACTTGAACTTAACG
CCAAAGCCCTGGATGGTTCGGGCGCCGAAACCGTCGTTGGCGAGCACATCCGTCA
ACGCCTTATCGAAACGATGGGCGACATGGGGAAATTATTGAAACGAGAGGTGGCGG
CAGCTCAAACCTAA

NHI 43 (XenC homolog 12) protein sequence:

MGSNYGTGATAIPTVDISAFLSPTASDATRQAVIEGVSKACHTYGFFNLVGHGIPQDVM
RRAFEKNKMFFELPHDKKMEVRISKSVGRSFRGYEPPGIQTHHEGLLPDTKETFMVG
REVPEDDPDCGSFSTGPNLWPSSLSKEKFQDLIMDYQGRMLALVKNILDILVLGLPKAW
KCPPNVFDSLDDKPSIPMRFLHYAPVQAMDPRQFGVADHTDFGCVSILLQELGTEGLE
VFYPPSETWVPVPVSDGFVVMGDMMQKYTGGYYRSARHRVLTNKDKHRHSIAFFL
NGNLKLNKALDGSQAETVVEHIRQRLIETMGDMGKLLKREVAAAQN

Codon-optimized NHI 44 sequence:

ATGACGACGATCAACATGCTCAAAACGGCCACCAACACTCTGGTTCCTCACTCTGGA
TCTGGCAAATGCAGAAGCGACCGGCGAGGGCCTGGCGGCGCTGGACGAAGCATG
TCGTAATCATGGATTCTTTCTGTTACAGAACCATGGCCTGGACCGCGAAATCGAGGC
CATGTGGCAAGCGTCCGAAGATTTCTTCAAACAACCCCAGGCCTCAAAGCGCGCG
ATTAACGAACCGAAAGCTCACCCTAGGATATTTTCGATCAGGAAATGACTAAACGT
CAGCGAGACTTGAAGGAAGTTTTTCGATTTTCATGCGCCCCACGGCTACCGGGCGTAA
TCTGAACCAATGGCCCGACGATGAAGCCTTTAAACAGACCCTGAGCGACTTCTTCG
AAGCAGCCTCTTCCGTCGCCCCACCGTACACTGACCCTGGTGTACCAAGCGCTGGC
CGCGGCTCCGAGCGACGCCCTGCCTATGGGCGACCCAGCGACGTCCACCGTGCG
CCTCAACTTTTACCCTACAGAAGATCCACTGCCTGCGAACGAACGGGGCACAAGCAG
GCGTCCTGGGTAAATATGGCATTACATCATACCGATCCTGGAATCCTGACCCTGC
TGTTGCAGGACCTGACGGGCGGGCTCCAGACCCTGACCCAAGACCAACATTGGAT
CGACGTGCCGCCGCGAGGCGGGGACCATTGTTGTGAATCTGGGCGATGCTATGCAA
GTCTGGTCTAATGACGCGTATCGCGCTGCGATCCATCGCGTCACCCCTCGCGACC
GCAGTGACGATATAGTACGCCCTACTTCTATAACCCGAAGTCCGATGCGGTGTTGG
CGCCCATCACACACCTTACTCAAGACCCGCCACACTTCCGCCCCGTTTACCTGGCGT
GAGTATATCAAAGGTCGCGTAGACGACAACCTTTACAGATTTAGGCGTGGATGACATT
CAGATCGATCGATTCCGTATTGCGGTATAA

NHI 44 protein sequence:

MTTINMLKTATNTLVPTLDLANAEATGEGLAALDEACRNHGFFLLQNHGLDREIEAMWQ
ASEDFFKQPQASKRAIKRTESSPLGYFDQEMTKRQRDLKEVFDFMRPTATGRNLNQW
PDDEAFKQTLSDFFEAASSVAHRTLTLVYQALAAAPSDALPMGDPATSTVRLNFYPTED
PLPANERAQAGVLGNMALHHHTDPGILTLLLQDLTGGLQTLTQDQHWIDVPPQAGTIVV
NLGDAMQVWSNDAYRAAIHRVTPRDRSARYSTPYFYNPKSDAVLAPITHLTQDPPHFR
PFTWREYIKGRVDDNFTDLGVDDIQIDRFRIAV

Codon-optimized NHI 45 sequence:

ATGACACTTCCGCAGTCGCGTCGCTTAGACTTTTCCGAAATCCCGATTGTTGATGTG
GGCCCGCTTATCGCTGGCGACCAACGACCCAACCTTTGATTAACGGCCTGCATCAGG
CTTGTACGGATGTCGGCTTCCTATATGTGCGCAACCACGGTGTTAGCATGGCACTG
GTTGAAGCGCTGAAAGAAAGTGCGGTACATTCTTCGCCGACCGATGGCAGAGA
AAATGAAAATAGTGATTGATCCGCGTATCCAAGGCTATTTACCGCTCGGGTATCGCA
GCTATGAAGGCGAGGCTCGCGCTGGCACTTCGCAACAGGAAGGCTTTTGGATCGG
TCATGATCGTCCGTTGAGCAACGATGCGTTGCTGGATGGTCCGAACCAATGGCCG
GATGGCCAGCCGGACTTAAACAAGCGATGCTTCGCTACTTCGACGCACTGGAAG
GTCTAGCAGGCGTTCTGCTGCGCGGTTTTGCCTTAGCTTTGGGCTTGGACGAGCAT
CGATTGCAACCTTCTTCAAGAAACCTCTGTCCCGCCTAAAAGTCAATCATTACCCG
CCGCAGGATACGCCGACTGCCGACAACGATATTGGCGTTGTGCCCCACAGCGACT
CTGGGGGGCTTTACCATCCTGTGGCAGGACAACCTGCGGCGGGTGTAGAGATACAGAA
CAAGAACGGCGAATGGGTAGGCGCTCCGCCACTGGAAGAGACTTTTGTCAATCAAC
CTTGGCAACATCATGCAAATTTGGAGCAACGGCCGCTTTTCGAGTACACCGCATCG
CGTGATTAACCGCTCCAACCATGATCGTTACAGCATTCCACTGTTTCGTGAATCCGTC

TGCCGATGTTTCCATTAAACCTTTAATCGACTTCGACTGCAACCATAGCACCACCTTT
AACTACGGTACCTACCAGCGAGATCTGTGGCGTCAGACGTTTCCTGTGCGCGAAAT
CCCAGTCTAA

NHI 45 protein sequence:

MTLPQSRRLDFSEIPIVDVGPLIAGDHDPTLINGLHQACTDVGFLYVRNHGVSMALVEA
LKESAVTFFAAPMAEKMKIVIDPRIQGYLPLGYRSYEGEARAGTSQQEGFWIGHDRPLS
NDALLDGNQWPDGQPDLDKQAMLRYPDAVEGLAGVLLRGFALALGLDEHRFATFFKK
PLSRLKVNHYPPQDTPTADNDIGVPHSDSGGFTILWQDNCGGLEIQNKNGEWWGAP
PLEETFVINLGNIMQIWSNGRFSSTPHRVINRSNHDRYSIPLFVNPSADVSIKPLIDFCN
HSTTFNYGTYQRDLWRQTFPVAEIPV

Codon-optimized NHI 46 sequence:

ATGGATGTTCTGAATAGCATTAGTCTGGAGCGCCTGGAAAGCACTCAAGCCGACTAT
GAAATTAGCTGCCTACGGAAATGTTGCGTTAACGATGGGTTCTTTTACCTGGCGGAC
CATGGGATTCAACCGAGCCTGATTGCAGACGCCCTGGATACCGCCCCGCCAGTTCTT
CGCCCTGCCTCTCAGTACCAAGAACCAGTTTTTGCCAGGACCGCCAGGTTGTCATC
CCGAAAACGTGCCGTGGCTATGTGGATACCTTCGGCGAGACATTGCACCCGCCGA
CCGGGCCCGGATCGTAAACAGCATTTTGACATGGGGCGCGAGGCCCCAGCGTGCG
ATTTGCCGTTCACTGGTCCGAATCTGCTCCCTTCCGAGGGCCAAGCGCCTGGATTC
GCGAAATCAATGCTCGCCTTGACGCAAGAGGTGTTAGATCGTGTGGTGCCCGCACT
GCTGCGAGGCCTGGCACTTAGTTTTGGGTTTGGATCGCGATTTCTTCCAGGCTTTCT
TTAGCAACCCGGTTCTCATACAGCGTGCCTTATATTACCCCCCTCGGGCAGCGGT
GCGGGCAAGCACACAGATAATGGTATGTTTACATTACTGTTTCAGGATCCGCGGGAT
GCCTGCGCACTGAGCGCATTACGCCAAGGTCGTTGGATTGATGTGCCGTGCCGAG
CTAATGAAGTTGTCGTGAACCTGGGGGATATGCTGATGAAATGGTCAAACGGTCTAT
TTACGTCCACTCCTCATCAAGTAATTCACCGTGCCTCTTCGGCGCGAGTCAGTCTG
CCATTCTTCGTGTATCCGGATATTGATGCCGTGTTACGCCGCTCGGGAGCGATCA
GACTGTTGCGTGCCGCCAAGTCATGTTGGATAACTTTAACAGCATTTGGGTGTCTG
GCCAGGGTGCAGGGCGCGCGCGCGAATTTGCCTAA

NHI 46 protein sequence:

MDVLNSISLERLESTQADYEISCLRKCCVNDGFFYLADHGIQPSLIADALDTARQFFALP
LSTKNQFCQDRQVVIPKTCRGYVDTFGETLHPPTGPDRKQHFDMGREAPACDLPFTG
PNLLPSEGQAPGFAKSMALALQQEVLDREVVPALLRGLALSLGLDRDFFQAFFSNPVLIQR
ALYYPSPSGSAGKHTDNGMFTLLFQDPRDACALSAFSQGRWIDVPCRANEVVVNLGD
MLMKWSNGLFTSTPHQVIHRASSARVSLPFFVYPDIDAVFTPLGSDQTVACRQVMLDN
FNSIWVSGQGAGRAREFA

Codon-optimized NHI 47 sequence:

ATGAGTAATATTTTGCGCCGCCGAGAAGTGCATAGCGTGGTGGTGGGCGGGCAAC
GCTCACAGGCCATTACTAATGAAGTACCTATTATCGATGTGGGACGGTTCTTCGCCG
GCGTCCCGGGTGCCCTGGATGAGGTTGCGCAACAGATTCATAAAGCGTGCACCGA

AATCGGTTTCTTCTATATTGTAAACCATGGCGTATCACAGCCGGTAATCCGCCGTGTT
TTCGAAGCAATGGAAAATTTCTTTACTCTTCCGGCGAGCGAAAAGTTTAAAGTGCGT
ATGAACGAACATCAGGCCGGTTGGCAAGCGCCGAACGTCGCGATTACGGCGATA
GCTTTGACAAAGTGGTCAAACCCCAAGGCCACTGAAGCGTTCAAATTCGGCCCCGGA
ACTGGACCCGGCGGATCCCGACTTCCAGGGATCTAAAAGATTTTCGCGGACATAACC
AGTGGCCCGGAAGGTCTGAGCGTAGAGGATAAACAGGCATTCCTGACCTATCACGC
CCAATTTAATGAGCTGGCTAAGCGTCTGCTGGCAGCGGTGGCTGTTGCCCTGGGC
AGCGTGCCCAATTTCTTCGATGACGCATTCGCCCGAACGGATTCTGTATGCGCTG
CGCGTACTATCCGGTCGTTAAACCCCAAGAAAATCAGTTCGGATTACCTGGACACAC
GGATCTGTCAATTCCTGACTCTTATTCCGCCGGCGACAGCGCCGGGCTTACAGATCC
TGACCGCCGCAGGCACCTGGATCGACCAGCCGATAGTAGAAGGTGGCATTCTGGT
GAACACCGGTGACGCTCTGCGCACCTGGTCCAACGATCAATTTATTGCGACCCCTC
ATCGTGTACTGGCTAGCACCAAGGAAGAACGCTACTCCGCAATCTTCTTTCTGTATC
CCGATCTGGATGTGACTATTTTCATGCGTACCGACGTGCTGGAGCGCCGATTATCCC
AAGAAATATACTGATCAGAAGTTCTCTGATTTCTATGCCGGGTATGCCGCGCGAAAC
TTCGGGTACGCGGAAAAGAAGGCTTAA

NHI 47 protein sequence:

MSNILRRREVHSHVVVGGQRSQAITNEVPIIDVGRFFAGVPGALDEVAQQIHKACTEIGFF
YIVNHGVSQPVIRRVFEAMENFFTLPASEKFKVIRMNEHQAGWQAPNVAIHGDSFDKVV
KPQATEAFKFGPELDPADPDFQGSKRFRGHNQWPEGLSVEDKQAFITYHAQFNELAK
RLLAAVAVALGSPNFFDDAFRRTDSVMRCAYYPVVKPQENQFGLPGHTDLSFLTIPP
ATAPGLQILTAAGTWIDQPIVEGGILVNTGDALRTWSNDQFIATPHRVLASTKEERYSAIF
FLYPDLDVTISCVPTCWSADYPKKYTDQKFSDFYAGYAARNFGYAEKKA

Codon-optimized NHI 48 sequence:

ATGCCACATACCCTTGACATCACTGCGCTGCCGACCTTGGATCTGTGCGCAGTTCGA
AGGGACAGCACAAACAGCGTTATGAGTTCCTGGAAGATTTGCGTCACGCCGCACGC
GACGTGGGATTCTTCTACCTGACGGGCCATGGCATTGATAGTGATCTGCTGAAGCA
GGTGCAAACGCATGCCCGCCAGTTCTTTGCGCTGCCGGATTCCGAGAAAGCGGCG
GTAGGTATGATAAACAGTCCTCATTTCGTTGGCTACAACCGTGACAGCCTCGGAAATT
ACTCGTGGTCAGCCGGAAGTTGCGCGAACAGTTCGACTTGGGTGCGGAACGTGAA
GCCCTGCCACTCACAGCAGACTCACCGTTTTGGGCACGCTTGACAGGGCCCTAACC
AGTGGCCCGGCGGCCCTGCCGGAACCTCAAACCGCTGCTGCTGGAATGGCAAGAGG
CAATGACACGCATGAGCTTACGCTTGCTTCGGGCTTTTGACAGGCACTGAGCTTG
CGTGACAGATGCATTTGACCAACTATATGGAGATAAGCCAAATGAACATATTAAATTAAT
GCGTTATCCGGGTCAATCGTCGGAATCCTCGCACCAAGGCGTGGGAGCGCACAAAG
GATAGCGGCTTTCTGAGCTTCTTGCTTCAGGATCAGCAGGCTGGTCTGCAAGTGGA
GATAGAAGAAGGACGCTGGATAGACGCTCTGCCGCGCGAGAATACGCTGGTTGTG
AACATAGGTGAATTATTGGAGCTAGCGTCGAATGGGTATCTGCGTGCCACAGTGCAT
CGCGTAGTTAGCCCGCCAGAGGGCACCGAACGCCTGTCCGTAGCTTTCTTCCTGG
GGGCGCAGTTAGATGCAGCCGTACCACTGTATCCACTGCCTACAGCCCTGCTGCGT
GAAGCTCGCGGTCCCGCTAGTGACCCCTTGAACCCTCTTTTCCGTGATGTGGGTT

GGAAGTACTTGAAAGGGCGGTTACGTAGCCATCCGGATGTAGCCCAACGCTTTTAT
GCAGACGCGCTGCTGCCGCCGCCACCGGTACGTAAAAGCGCATAA

NHI 48 protein sequence:

MPHTLDITALPTLDLSQFEGTAQQRYEFLEDLRHAARDVGFFYLTGHGIDSDLLKQVQT
HARQFFALPDSEKAAVGMINSPPHFRGYNRAASEITRGQPDRLREQFDLGAEREALPLTA
DSPFWARLQGPNQWPAALPELKPLLEWQEAMTRMSLRLLRAFAQALSLRADAFDQL
YGDKNPNEHIKLMRYPGQSSESSHQGVGAHKDSGFLSFLQDQQAGLQVEIEEGRWID
ALPRENTLVVNIGELLELASNGYL RATVHRVSPPEGTERLSVAFFLGAQLDAAVPLYPL
PTALLREARGPASDPLNPLFRDVGWNYLKGRLRSHPDVAQRFYADALLPPPPVRKSA

Codon-optimized NHI 49 (AndA) sequence:

ATGACTATCGAATCAAAAACTACCCCCCATTCGACGCGTCAATGCCTCGCAGGG
TTCCGATGCCGCCTATCAGATTCTTCAGGAAGACGGCTGTGTAATTGTAGAGCAGG
TGATTTGTCCTAATATTATTGCGAAAATTCAGACGACGTTAACCGCGTGATGGACAA
AGCCACCATTGGCGCTAAAAAGGGTGAACAGACCCATATCATCAATATGCATAATCG
TACCATTACATGGGCGACTTAGTACTGACTAGTAAACATACCGTGATGAGCTTTTG
AATCTGCCGTTTGCCCATGAAGTGTTAGAAAAAGTGTTTAAAAAGGACAGTGGCGAT
TACTGGCTGAACATGGGCAACATTCTCAACATGCTCCCGGGAGCGGAGGCGCAAC
GTCCTCACAGAGATGACTACCTGTACCCGGTAAGTCAACATATGGACCCCGCCACT
TCACCTGATCTGATGATTAACATTACCTTTCCATTAAACGAGTTTCGCCACGATAATG
GCGGAACTCTTCTGTTGCCTAAATCGCATAACCGGACCGAATGCCGATTTTTACGCCA
ATGCAGAAGATCTGCCGGCCGCGGAGATGCAGGTCGGCGATGCTCTGATTTTCAC
CGGGAAATGCGTCGACTTTTCATGAGGCCCTGGGGCTGAAATCTAAACAGCGCATG
GTGGGGGGGGCCAACCGTTCGGATAAGCCTCGCATAGGGCTGGCTCTGGCAGCGC
AGCCGGGTTATCTCACTCCGCGTGAGTCAAACGTAAATGTGCCACGGGACATCGTT
GAAACCATGACACCGCTGGCGCAACGTATGATTGGCTGGGGGACCGTTCGGACAA
AAGATACTTATGGCCTTAATATGTTACAGGACAAGTAA

NHI 49 (AndA) protein sequence:

MTIESKNYPPIRRVNASQGSDAAYQILQEDGCVIVEQVICPNIIAKISDDVNRVMDKATIG
AKKGEQTHIINMHNRTIHMGLDLVLTSTKYRDELLNLPFAHEVLEKVFKKDSGDYWLNMG
NILNMLPGAEAQRP HRDDYLPVVSQHMDPATSPDLMINITFPLNEFRHDNGGTL LLPKS
HTGPNADFYANAEDLPAAEMQVGDALIFTGKCVDFHEALGLKSKTAHGGGANRSDKP
RIGLALAAQPGYLTPRESNVNVPDIVETMTPLAQRMIGWGTVRTKDTYGLNMLQDK

Codon-optimized NHI 50 (AndF) sequence:

ATGACTGTGCCTCAACTGCACTATGTGCCTTATGATACGCCGGTCAAGACGTGAT
GAGAATTCTGGAAGAGTGTGGTACCCTAGTAATTCGTAACCTTTCTAGATCAAAAAAC
CGTTCAAAACGTGCAGAACGAGGTGGATGATTACGTTAGAACTGGAATCCAGGAC
CAAAGTATAATCATGATATCAAAACCGTAGGCAGCAAAACCAAACAGCCGTGCAACC
TTAGTCTCATGAGTAAACGTACAGATGCGAAGTTTTAAACCATCCGTGGATGCACG
CAATTTGCGAACGTATGTTTGGGCCCCACGTATGGTGATTATTGGTTTAAATGGCGGCT

CTATTTTACACTTGGAGCCTGGCGAACATACCCAGCCAATTCATCAGGACCATGTGT
TTTATCACATTAGCAAATGGCGTCGACCTACTGACCCAGACCTAACAATTAATTTTGC
CATGGCGTTGACCGAATTTACGGTTGAAAACGGTGGTACCCGTGTGTGTCCGGGTT
CGCATCTGTGGGAAAACGGCCATGCTCCGCCGGCAGAAGAAGACATGGTCCCGG
CGCTGATGCAGCCAGGTGATGCTCTGATTTTACCGGGATCAATGTGGCACTCAGCC
GGGGCCAATCGCAGTAGTGAGTATCGGCGCGGATTGCGGCCTCGTTTCACCCCT
GTCATTTTACCCCCATTGAGTCACATCATCATCTGCCCCGTGAGATGGTTGAGGAAA
TGTCGCCGTTAGTTCAGAAGATGTTGGGTTTCCGTACGCTTAATCTTCATAATAATGT
AAAGGTCTGGAAAGCGGGAGAGGGGAAACCTCGAGGATGCTACTGGTCTTAAGTCA
ATTGCAGTATGA

NHI 50 (AndF) protein sequence:

MTVPQLHYVPYDTPVEDVMRILEECGLVIRNFLDQKTVQNVQNEVDDYVRNWNPGP
KYNHDIKTVGSKTKQPSNLSLSKTYRCEVLNHPWMHAICERMFGPTYGDYWFNGG
SILHLEPGEHTQPIHQDHVYHISKWRRPTDPLTINFAMALTEFTVENGGTRVCPGSH
LWENGHAPPAEEDMVPALMQPGDALILPGSMWHSAGANRSSEYRRGFAASFHPCHF
TPIESHHLPREMVEEMSPLVQKMLGFRTLNLHNNVKVWKAGEGNLEDATGLKSIIV

Codon-optimized NHI 51 (CnsJ) sequence:

ATGACGGTTGATACTGCACCACAGAGCCATTATCAGGAAACAAAGGTATCCGAAATA
CCGATAGTAATTCTGAAAAGCTCTGCAACCGACGATGTTGCGGCGCACGAGGCCAT
TGAAGCTTTAAAGGTTGCAGGTGTTTGCATTGTGAGAAATCTGCTTGATCGAAGCAC
TGTTGATAAAGTTAGACAGGAATTACAACCATATGATAAACAGGCGGACTCATTGAA
GGGTTTCCTAAAAATTATTGCCAAGTTGCTGGTCTGCTGAGTAAAAGTCCGACTTAC
GCCCATCTATAGTTGGCAATAAACTTTTCACTGCGGTCCGGAATTATTTTCTGACTA
GTACGTACGAGTGTTGGGCCGAAAAAGGTACGTGGATGTCAGTTAAATCACCACCA
CAGTTGGATTCAACACTGGCACTGTATGTTAACCCTGGCTCAGGCGATCAGGGACT
ACATCGGGACGACGCGACTCAGCAAAATTGGAACCTCTGGCGCAAGTGAGTACAGC
CTGGGACGTGATTCAGGATGTGCGATGATGGTTGCTCTGACTGAATGTGCCCGAGA
AGATGGTACAACCCGCTTCATCCCAGGTTCTCACCTTTGGGACTACCAGTACGATCA
TCCATCAAACGATGATCCTCGTATTAGGTATGCGGAAATGCGGCCCGGCGATGCATA
CCTTATGCTGTCATCCGTGATCCATGCGGGCAGCGTGAATTACTCAACGGACCGCC
GGCGCGTTCTAGCAGCCGTGTTTCGCTGCTCGATCCCATTAAAGACAGGTAGAAAAT
CAATATCTGACCTATGATATTGAAACCGTACGTACGTTTCCGACGTGGTTACAAAGAT
TCATGGGGTACAGTTTAAGCAAGTTATTTCAAGGTTGGGTAGATAAGAAAGATCCCC
TTCTGGTCGTAGACCCGAACGCCCAACCAGAAGGTGAGGATGATGGAGGCATGAA
GCCAAATGAGGGCGAGCATGTAGTCGAAGCACAAATTTAA

NHI 51 (CnsJ) protein sequence:

MTVDTAPQSHYQETKVSEIPIVILKSSATDDVAAHEAIEALKVAGVCIVRNLLDRSTVDKV
RQELQPYDKQADSFEGFPKNYCQVAGLLSKSPTYAHSIVGNKLFTAVRNYFLTSTYEC
WAEKGTWMSVKSPQLDSTLALYVNP GSGDQGLHRDDATQQNWNSGASEYSLGRD
SGCMMVALTECAREDGTTTRFIPGSHLWDYQYDHPSNDDPRIRYAEMRPGDAYLMLS

SVIHAGSVNYSTDRRRVLAAVFAARSHLRQVENQYLTYDIETVRTFPTWLQRFMGYSL
SKLFQGWVDKKDPLLVDPAQPEGEDDGGMKPNEGEHVVEAQI

Codon-optimized NHI 52 (SptA) sequence:

ATGACGATAGAGTCCAAAACTATCCGCCAATTCGTAGGGTTAACGCATCACAAGGA
TCAGATGCGGCTTATCAGATTCTTCAGGAAGATGGGTGCGTCATCGTCGAACAGGT
TATCTGTCCTAACATCATTGCGAAAATTAGTGATGACGTGAATAGAGTTATGGATAAA
GCTACAATAGGCGCAAAAAAAGGTGAACAGACACATATTATTAACATGCATAATCGTA
CCATTCATATGGGCGATTTAGTACTGACATCTAAGTCTTATAGGGACGAACTTCTCAA
CCTTCCATTTGCGCATGAAGTCTTAGAGAAGGTGTTCAAAAAAGATTCTGGGGATTA
CTGGCTAAACATGGGCAACATATTGAATATGCTTCCGGGGGCGAGGGCACAACGCC
CTCATCGGGACGACTATCTGTACCCGGTAAGTCAACATATGGATCCGGCGACTAGTC
CGGATTTAATGATCAACATTACATTTCCGCTTAATGAATTCCGCCATGATAATGGAGG
TACTCTGCTTTTGCCCAAAGCCACACTGGTCCAAATGCTGATTTCTATGCAAATGC
CGAAGACCTGCCGGCTGCTGAAATGCAGGTAGGCGATGCACTCATTTTTACCGGTA
AATGTGTGCATGGGGGAGGTGCCAATCGTTCGGATAAACCTCGAATAGGATTGGCT
CTGGCAGCACAGCCCGGCTACCTTACTCCAAGAGAATCTAACGTCAACGTTCCCCG
CGATATAGTGGAACAATGACCCCGCTGGCTCAAAGAATGATCGGCTGGGGGACC
GTCCGTACAAAGGATACGTATGGTTTAAACATGCTGCAGGATAAGGACTTTCATGAA
GCGCTAGGTCTTAAATCAAAAACGGCGTGA

NHI 52 (SptA) protein sequence:

MTIESKNYPPIRRVNASQGSDAAYQILQEDGCVIVEQVICPNIIAKISDDVNRVMDKATIG
AKKGEQTHIINMHNRTIHMGLVLTSKSYRDELLNLPFAHEVLEKVFKKDSGDYWLNM
GNILNMLPGAEAQRPHRDDYLYPVSQHMDPATSPDLMINITFPLNEFRHDNGGTL LLPK
SHTGPNADFYANAEDLPAAEMQVGDALIFTGKCVHGGGANRSDKPRIGLALAAQPGYL
TPRESNVNVPRDIVETMTPLAQRMIGWGTVRTKDTYGLNMLQDKDFHEALGLKSKTA

Codon-optimized NHI 53 (SptF) sequence:

ATGACAGTACCGCAGCTGCATTATGTCCCATATGACACCCCTGTCTGAAGATGTAATG
CGGATTCTTAAGGAAAGCGGCACCCTGGTCATTTCGTAATTTCTTAGATCAGAATACG
GTCCAGAAAGTACAAGACGAGGTTGACGATTACGTACGAAATTGGAACCCGGGACC
GAAATACAATCATGACATTAAGAATGTTGGTTCGAAGACCAAACAGCCCAGTAATCTA
AGCTTAATGAGCAAAACGTATCGCTGTGAGGTTCTTAATCACCCGTGGATGCATGCG
ATTTGTGAGCGGATGTTTGGTCCTACCTATGGCGACTATTGGTTTAACGGGGGTTCA
ATACTGCATCTTGAGCCTGGTGAAAATACTCAACCTATCCATCAGGATCATGTGTTCT
ATCAGATTTCAAATGGCGCCGGCCGACGGATCCGGACCTTACGATTAATTTTACGA
TGGCCTTAAGTGAATTCACAGTTGAAAATGGCGGTACTAGAGTTTGCCCCGGCTCG
CATTTATGGGAAAACGGACATGCAAGCCCGGCGGAGGAAGATATGGTGCCAGTGTT
GATGCAGCCTGGAGACGCGCTTATTCTCCCCGGCTCAATGTGGCATAGCGCCGGC
GCCAATCGCACCTCGGAGTATCGCAGAGGATTTGCGACTAGTTTTTCATCCCTGCCA
TTTTACACCTATTGAATCCCACCATCATCTGCCCCGGGAGATGGTCTGAAGAAATGAC
CCCGTTGGTACAGAAAATGTTAGGTTTTTCGGACGCTGAACTTACATAATAACGTGAA

AGTATGGAAGGCGGGCGAGGGTAATCTGGAGGATGCGACCGGTTTAAAGTCAGTA
GCAGCTTAA

NHI 53 (SptF) protein sequence:

MTVPQLHYVPYDTPVEDVMRILKESGTLVIRNFLDQNTVQKVQDEVDDYVRNWNPGP
KYNHDIKNVGSKTQPSNLSLSKTYRCEVLNHPWMHAICERMFGPTYGDYWFNGG
SILHLEPGENTQPIHQDHVIFYQISKWRRPTDPDLTINFTMALTEFTVENGGTRVCPGSH
LWENGHASPAEEDMVPVLMQPGDALILPGSMWHSAGANRTSEYRRGFATSFHPCHFT
PIESHHHLPREMVEEMTPLVQKMLGFRTLNLHNNVKVWKAGEGNLEDATGLKSVA

Codon-optimized NHI 54 (2-ODD) sequence:

ATGGGAAGTACAGCGCCTCTTCGCCTACCGGTGATCGACCTCTCTATGAAAAATTTA
AAGCCAGGTACAACGAGTTGGAATAGCGTCCGGACTCAGGTTCTGTGAGGCGTTAG
AAGAATACGGTTGCTTTGAAGCGGTTATAGACGCCGTCTCCCCCGAGCTGCAGAAA
GCTGTTTGCAACAAAGGACACGAGCTTCTGAATCTTCGGTTAGAGACGAAAATGCT
TAATGGAAATAAGCCTGAGTACGACGGTTTTACCTCAATCCGAATCTGAATGAAGG
GATGGGAGTTGGTCGTATAACTGACTTAGAAAAAGTTGAACGCTTTACCAACCTAAT
GTGGCCCGAGGGCAACAAAGATTTTTGTGAAACAGTTTATAGCTACGGTAAAAGAAT
GGCAGAAGTGGATCATATCCTGAAAATGATGGTTTTTGAAGTCGTTTGGTATGGAAAA
ACATTTTGTAGCTTCTGCGAAAGTACCAATTATCTTCTTCACTTTATGCGTTATCAGC
AGCCCGGTAAAGATGGGAGGTCGCCCGCTCTGTCATTACATAAAGATAAGTCCATAT
TGACGATTGTAAACCAAAATGATGTGAAGGGTCTCGAATTCGAGACTAAAGACGGC
GAATGGATCTTACCGACGGCGGATAATCACATAGTGCTGTTGGGCGACTGTTTTATG
GCCTGGAGTAATGGCCGTTTACATTCCCCCTGCATAGGGTTACGCTGGTTGCGAA
CCAGGCACGTCTTTCCACATCCTCGTTTCAGTTTTCTTAAAGATATCATAGAAACCCC
TGCTGAGTTGGTAGACGAAGAACATCCACTGTTGTTTAAACCCTTTGAAATCACAGA
GCTGTTGGCATATTGCTTTACGAAAGAGGGCGCAAAAGCTGTTTGCATCTGAAAC
AATACAAAGCATATACTGGTGCCTAA

NHI 54 (2-ODD) protein sequence:

MGSTAPLRLPVIDLSMKNLKP GTTSWNSVRTQVREALEEYGCFEAVIDAVSPELQKAVC
NKGHELLNLPLETKMLNGNKPEYDGFTSIPNLNEGMGVGRITDLEKVERFTNLMWPEG
NKDFCETVYSYGKRMAEVDHILKMMVFESFGMEKHFDSESTNYLLHFMRYQQPGK
DGRSPALSLHKDKSILTIVNQNDVKGLEFETKDGEWILPTADNHIVLLGDCFMAWSNGR
LHSPLHRVTLVANQARLSTSSFSFPKDIIETPAELVDEEHPLLFPFEITELLAYCFTKEG
AKAVCDLKQYKAYTGA

Codon-optimized NHI 55 (Ppzd) sequence:

ATGGCGCGAACCCTGAACGCCGGGTTCTGACATTAGATTTGCTCAGTTTTGTCAT
GGGGAGCCTTCTTCTAGCCATGGCTTTTGCAGGGAAGTGGTCTGACTGTCTTCGTAG
TTTGGGGTTTGTAAAAATTCGTAACCACGGAATTAGTGGCGAAGAGATAGAAAAAGT
ATTTGTCATGAACAAATTGTTCTTTAGTCTGCCTCAGGCGGCAAAAGCCAAGGCCG
CACACCCACCCGAAGCTAATCCACATCGTGGTTATTCCTACGTGGGTCAGGAAAAG

CTGAGCCGTGTCAAAGACTACGAAAAAGGTAAACGCAGCATTGTTGATGTTTATGAT
ATTAAGGAGAGCTATGATCAGGGCCCCGGCGTTGATAAACTATATCCGAACCGTTGG
CCAGATAAACAGGACATTCCGGGTTTTTCGCGTAGTTATGGAAAAATTTTATGAGCGC
TGTCACCAAGTTCATCAGGACGTTCTGCGTGCAATAGCAACTGGATTTGACCTGTCT
CCGTCTTTTTTGACCGACCTCTGTTGCGAAAACACGTCTGAACTGAGATTAAATCAC
TATCCAGGCGTGCATCCTTCGTCCTTACGTAAAGGCGCAAAACGCATCTCAGAACAT
ACTGACTTTGGGACGGTTACCCTGCTGTTTCAAGATAGTGTAGGGGGTCTGGAAAT
CGAAGATCAAAATTCTCCGGGGACTTATTTCCCTGTAAGTTCTGAGCGCAAAAGTGA
TATGATTGTGAATGTGGGCGATTGCATCCAACGATGGACCAACGACAAAATTCTGTC
AACGAGTCATCGGGTAGTGTTGCCGGAAGACAGGGATGCTCTCATAAAAGATCGTT
ACTCAGTCGCATACTTCGGCAAACCCAGTCGTTCTCAGTTGGTGAGCCCCCTAAGA
GAATTCGTGAAAGAGGGCGAGAAGCCCAAATATAGTGCCATCTCCGCTTGGCAGTA
TAATCAGGAAAAACTCGTTCTTACTTACGGTGGAGATGAGGAGATTTTGGTTCCCAA
GCCTAGCTCAAGTGTATGCACCGGAGTGGGGCAGTTGCAATAA

NHI 55 (PpzD) protein sequence:

MARTPERRVRTLDFAQFCHGEPSSSHGFCRELVDCLRSLGFVKIRNHGISGEEIEKV
FV MNKLFFSLPQAAKAKAAHPPEANPHRGYSYVGQEKLSRVKDYEKGKRSIVDVYDI
KES YDQGPVAVDKLYPNRWPDKQDIPGFRVVMKIFYERCHQVHQDVLRAIATGFDLSP
SFLT DLCCENTSELRLNHYPGVHPSSLRKGAKRRISEHTDFGTVTLLFQDSVGGLEI
EDQNSP GTYFPVSSERKSDMIVNVGDCIQRWTNDKILSTSHRVVLPEDRDALIKDRY
SVAYFGKP SRSQLVSPLREFVKEGEKPKYSAISAWQYNQEKLVLTYGGDEEILVPK
PSSSVCTGVG QLQ

Codon-optimized NHI 56 (PpzC) sequence:

ATGGCAACTGATACCGCCGTCACCGGCGCGCCTGAGACCACGGAGTTACGTCTGG
CGTCTGGGAATGGGCCAGTTACCCGTACGGTATTGCGTACACCTGTTTCGAGATGCA
TTACCATCCGAAGTGCCATTGATCGATATTAGTCTATTTTCTCAACCTGTGTGGCAG
ATCGCAAGGCCGTAGCACGTAAGATACATGATGCCGCGACCAATATCGGCTTCTTTT
ATATTCAGAATCATCGCGTTCCGAGCCAGGATATTGATCTGGCGTATTCAGCTAGCC
AGGATTTTTTTTCGCCAGGAGATGGAAGTGAAAGTGGAAGCGGACGCTAAAGATGGT
CCGTTTGATTACAGGCTACCGGGGTCCAGGCACGCAAAGAGTTAACCCTACGGAAG
GTGCAGATCTGAGAGAGACTTATTCGATACTGTATGACCCAATGCTTGACCCAACCG
TGCCGGATCCGGCAAATATACCAGAACCGGCCCTCTCGCTTCCTTCATCTCGGGCGG
GCCCCATTTGAATCCACAGCAACTTTGCCTCACTTTAAGGATGCATTTGTCAGGTAC
TTTCAGGCGTGCCCTCGTTCTGGCCCGTGCCCTGACTCGGGCGTTTCGCCCTGAGCT
TGGATCTGCCGGAATCTGCCTTTGATGGAAAAGTCCAGTACCCGGATGCAAGCCTT
GAAATTAATTTCTACCCACCCATCTCGACCGGACATGCCGTCTCGGCCCCCGGGGA
TGCAGATACTCGGGTTTCAATTGGCAGCCATACGGACTTTTTACTGTTACGATACT
CTGGCAAGATTCAAATGGCGGTCTGCAGGTGTTAAATCGTGAGGGCCAGTGGATAC
GCGCTATACCTGTTGAGGGCACTTTTGTGGTTAACATTGGGGATTACCTGCAAAGAG
TGACAAACGATAAATACGTTTCCACTGTCCACAGAGCACAGAATTTAGTGGCCGC
GAACGTGTCTCGATGCCCTTTTTTTGGGGCTTTGGAATGCATGAGTCGTGTCAGGT

ACTGCGCAATTGCTGTGGGGAAGATGAGAAATCAAATATGATGAGGTTAAATGCGT
GGATTGGGTAAGCCGTCGTCTCGGCAATTTATTTGACCTGAGTGACAAAGGTTGA

NHI 56 (PpzC) protein sequence:

MATDTAVTGAPETTELRRLASGNPVRTVLRTVPRDALPSEVPLIDISPIFSTCVADRKA
VARKIHDAATNIGFFYIQNHRVPSQDIDLAYSASQDFFRQEMEVKVEADAKDGPFD SGY
RPGGTQRVNPTEGADLRETY SILYDPMLDPTVPDPANIPASRFLHLGRAPFESTATL
PHFKDAFVRYFQACLV LARALTRAFALSLDLPESA FDGKVQYPDASLEINFYPPISTGHA
VSAPGDADTRVSIGSHTDFLLFTILWQDSNGGLQVLNREGQWIRAI PVEGTFV VNI GDY
LQ RVTNDKYVSTVHRAQNFSGRERVSM PFFWGF GMHESCQVLRNCCGEDEKSKYD
EVKCVDWVSRRLGNLFDLSDKG

Codon-optimized NHI 57 (PrhA) sequence:

ATGGCGCCGATGATTCCCCCCCCGCCTGCAGAGGTTTCCGGCCACTGCCAGCGCG
GATGAAATATTTGCCGCCTTTCAGGAAGATGGTTGCGTGGTAATCGAAGGATTATC
AGCCCAGAACAGGTGGCACGCTTTTCGCAAGAAGTGGATCCCGCGATGGAGAAAA
TTCCGGTTGAAGTTACAAACAATGGAAATAGCAATGATCGTACTAAACGCTTTAGTAA
ATGCGTTATTGCCTCTCCAACATTCAGGAATGAAATCATAGAATCTGATCTTATGCAC
GAACTGTGTGATCGCGTGTTTTCGAAGCCGGGAGAAGGCATGGGATACCATTTTAA
TGATAATATGGTTATTGAGGTACAGCCGGGCGCACCCGCTCAGCGTTTGCACCGAG
ACCAGGAATTGTACCCCTGGTGGAATTCGATGGGTCCGGCAGGCCCGGAATGTGT
TATTAATTTTTTTTTGCGCGGTAACGCCGTTTACCGAAGAGAATGGTGCTACACGTCT
GGTACCGGGTAGTCATCTGTGGCCAGAATTTACTCAAATCAACGAACGTGATTGCC
CTCAATTTGGTAAGATTGAAACTGTTCCAGCAATTATGCAGCCTGGAGATTGTTATCT
GATGAGCGGTAAAGTGATCCATGGCGCCGGACATAATGCGACCACTACTGACCGGC
GTCGGGCGCTAGCTCTGGCGATCATACTCGCGAGTTAAGGCCGATGCAGGCTTTT
TCTCTGAGTGTGCCTATGAACTGGCACGCGAAATGTCCGAACGGTCCCAGACCAT
GTTTGGTTTTTCGTTTCATCCGTTTCAGCATTGCGACGTGGACATGGTTTATTTTGGGG
CAACGACGGTAAAGATATTGCACATCATCTCGGTCTCGAAGCTCCGTCAGTGCACG
TCTAG

NHI 57 (PrhA) protein sequence:

MAPMIPRLQRF PATASADEIFAAFQEDGCVVIEGFISPEQVARFSQE VDPAMEKIPVEV
TNNGNSNDR TKRFSKCVIASPTFRNEIIESDLMHELCDRVFSKPGEGMGYHFNDNMVI
EVQPGAPAQRLHRDQELYPWWNSMGPAGPECVINFFCAVTPFTEENGATRLVPGSHL
WPEFTQINERDCPQFGKIETVPAIMQPGDCYLMSGKVIHGAGHNATTTDRRRALALAI
RREL RPMQAFSLSVPMKLAREMSERSQTMFGFRSSVQHCDVDMVHFWGNDGKDIAH
HLGLEAPSVHV

Codon-optimized NHI 58 (AneA) sequence:

ATGACCCGGTCAAGTCCGAAATCAATACTGCGTCTGTAGCAGCACATGAAGGTGC
ACCTAAAATTCTGCAGGTGTTGAAAGAGGATGGCGCGGTAATTATCCGTGGCTTTCT
TAATCAAACCTCTTTCAGAAATTGAATAAAGAGATGGAGGATGAACTGGCAAAGCG

CCCGCTGATTTGTGAACATAATGGCGAATGGGTAAAGTGGTCTGGCCGGCTCTCAGA
CTCGCCGTCTTAATCAACTGCCGGCATTTCACCAACTTTTAGACAGGAGATTTTGA
ACCATCCCTTATTCCATGAAATCTGTGAACCGCTGTTTGGTACTCATGGATGCGACT
ACTGGATGAACACGGCGACGGTTCGTCCAGATTGGTCCGGGCAACGCAGCACAGTC
ATTACATCGGGATCAGGAACTCTATCCCGTATTCAATCCGATCGGCAAAGATGCGCC
AGAGGCATTAGTTAACTTCTTTTGTGCGCTTAGTACCTTCACAGATGAAAACGGAGC
GACCCGGGTGATTCTGGCTCTCACCAATGGGATGATTTACAACGGATGTGACAC
GTTGTGTCCCTCAGACAATTCCGGCCGAAATGGAGGCGGGGGGACTGCTTACTGTT
CTCTGGTAAAACCGTCCATGGAGGCGGCGCCAACAGATCAAAAGATAATGTTTCGGA
GAGGTATGGCCCTTGGAGTGCAGGCTACTTACCTTACCGCCGAGGAGAATTATCAG
CATATTCCTCGGGAGATTGTTGAAAGTATGACGCCTTTGGCACAGAAAATGATCCAT
TGGCGATCTTCATATTGCCCAGGTGGAGCCGGGCTCTGGCAGATTAACGGTACAGA
GCTGGCTAATGAAATCGGCCTGAAAGCAAATCAACCGAAAAAGACGATCTAA

NHI 58 (AneA) protein sequence:

MTRSSPKSILRRVAAHEGAPKILQVLKEDGAVIIRGFLNQNLFQKLNKEMEDELAKRPLI
CEHNGEWVSLAGSQTRRLNQLPAFSPTFRQEILNHPLFHEICEPLFGTHGCDYWMN
TATVVQIGPGNAAQSLHRDQELYPVFNPIGKDAPEALVNFFCALSTFTDENGATRVIPG
SHQWDDFTTDVTRCVPQTIPAEMEAGDCLLFSGKTVHGGGANRSKDNVRRGMALGV
QATYLTAENYQHIPREIVESMTPLAQKMIHWRSSYCPGGAGLWQINGTELANEIGLKA
NQPKKTI

Codon-optimized NHI 59 (HimG) sequence:

ATGGCACCAAGCACTGACGCAGAGCCTGAATGGTGCAACTCCTCAGGTTAAAGCGG
CGCTGCATCATCTTGATGCTAACACGGTATCTATAGACGACATCATTATTATTTGAA
GCATCACGGCGGAGTAGTGGTTCAAATCTGATCAGTGAAGAGATCCTTCAAGAGG
TTGGCGCTGACATCAAACCGTATTTTGTGCGCTTGTGGAACCTGGATTTTTCTCTT
CCAAAACCCGAATAGTCACACGACTGCCCAATAAATCAATTGCCTTTGTGCAAAAGA
TCTTTGGGAATCAAGTCTTTCAGGACATATGTGATCATTTTTTGACATCACATCACAG
GGGTTGGTATGGCAACGAACAGTATACATACTCCCCACATCCGGTGTTTAATAGCGC
CCTGGCATTCTCTACACTTCTGGTAATGAGACACAGTCATTACACAGAGAATGCAT
GGGCCAACACAATAAATTACCAGCTATTGCCCCTGAGGATTATCCGATCGGACGCGA
CACTGTGCTGGGCATGTTTGTAGCCGACACACGGACCACAAGAGAAAACGGAGCC
ACGAGGTTTATACCTGGCTCCCATCTGCAGAGCACATTAGATCCACCTGATGAAAGC
CAGACCGTGCCTGTAGAAATGAACAGAGGCGATGTTTTCTAATGCTGGGTTCTTG
CTATCACGGCGCTTCTGCCAACGTCTCCAGGCCGAAGAACGTATCCTTTACTCCA
CCTTTATGACCCAGTCAACAAGAAGGCAGCTGACGACTTGCCTCTTTCTGGTGCAG
GAGGAAAATATTTATCTGTGAGTCCCGTAGATCGCGTTAGAGTTTTAGCCCGCGT
ATGCAGAAACGTCTGGGTTTTTCAGCCTCAGACCCGCTTTTCGGCTGGGTTGACGT
CAAGGATCCGCGTAAAGTATTTAATCTCCCTGGTTTGTCTGAAGGTCAGCACATCGA
CGTTTGA

NHI 59 (HimG) protein sequence:

MAPALTQSLNGATPQVKAALHHL DANTVSI DDIIHYLKH HGGVVVQNLISEEILQE VGADI
KPYFDALVEPGFFSSKTRIVTRL PNKSIAFVEKIFGNQVFQD ICDHFLTSHHRGWY GNE
QYTYSHPHPVFN SALAFSTLPGNETQSLHRECMGQH NKLP AIAPEDYPIGRD TVLGMFV
ADTRTTRENGATRFIPGSHLQSTLDPPDESQTVPVEMNRGDVFLMLGSCYHGASANV
SQAERILYSTFMTQSTRRQLTTCLFLVQEENIYLSVPVDRVRVFS PRMQKRLGFSASD
PLFGWVDVKDPRKVFNLPLGLSEGQHIDV

Codon-optimized NHI 60 (CdmA) sequence:

ATGACTAAACTGCCCAGCGATGCGTCTGCGATCAAAAGAGTAGAAGCTAATACCTCC
CCGGAGCAGGTTATCGAATTATTGAAACAGGACGGCGTTGTAGTGCTCAAAGACTT
CTTAGATCAGGCCACCGTACAGTCCTTTTCGCGAGGAAATTAACCGGCAATTGACG
ATTTTGAAGGGGGGCCGAATTTCAATCCTGATGGAGTTAAAGTCGATATAGGCAGAG
GCACAAAGCATGTTGCTAATTTAACGGCCATTTCTAAAAC TTATAGGCACGACATCCT
TAATAACAAGTGGATGCACTCAATCCTGGAGCCTTTATTCCGTCCGCATTTTCGGCGA
TTACTGGATGAATCGCGGTAGCGTCTTACACATCGAACCGGGCGAACCTGCGCAGA
ATCTGCATCGTGACGATATCCTGTACAGGGTGACGAAACTCCGGCAGCCAGGCGA
CCCGGATCTGATGATTAACATCCTGATTGCTGTACAGAGTTTCGTGATGATAATGG
CGCAACACGTTTTCGTGCCGGGATCGCATGTGTGGGATGACACGCGGGGCGTTCCA
ACACCTGATCAGGCCTCTAGTGCTGCCTTAAGACCGGGCGATGCCCTATTGTTTGT
CGGAAGTCTTTGGCATGGCGCTGGCTCTAATCAGAGTGATGCGTTTCGCCAGGGT
CTGCTTCTTTGTATACATCCCTGCCACTTCACTCCGATGGAATCGCATCTCCATGTTT
CGCGCACCATCGTTGAAAGTATGACCCCTCAAGCGCAGAAAATGATTGGCTGGCGT
TCAGGTATAACGCAGCACGATGTTCCAATTTGGCTGGCGGGTGATCATAAAATGGAA
GAAACTATGGGCCTTAGATCCCAGGAGGTTTCAGTAA

NHI 60 (CdmA) protein sequence:

MTKLPSDASAIKRVEANTSPEQVIELLKQDGVVVLKDFLDQATVQSFREEIKPAIDDFEG
GPNFNPDGVKVDIGRGTKHVANLTAISKTYRHDILNNKWMHSILEPLFRPHFGDYWMN
RGSVLHIEPGEPANLHRDDILYRVTKLRQPGDPDLMINILIAVTEFRDDNGATRFVPGS
HVWDDTRGVPTPDQASSAALRPGDALLFVGSLWHGAGSNQSDAFRQGLLLCIHPCHF
TPMESH LHVPRTIVESMTPQAQKMIGWRSGITQHDVPIWLAGDHKMEETMGLRSQEV
Q

Codon-optimized NHI 61 (CdmD) sequence:

ATGACCGTCACAATTAATAAGTCCTCTGTGCCACACAAGGTAGATTTTCGCGACCCCG
TTGTCAGATGTTATCTGTCATATTAAAGAAGACGGCGCGGTTATTGTACGAGGTTTCA
TGGATGTGGAAACAATTCAGAACTGCAAGAGGAAGTTGATACTGCAGTCGAAAAA
GGATCCTTTGGACCGCGGTATCAAGAATATAATGAGGAGGCTGGGGAGATTCCGAA
ACATGAAATTTATAACGCGGAGAAAGGTAAGAAAACCAAACACATGAAGAATCTGGC
ACTTACTTCCGAAACGTTTCGTAATGATGTGCTGAATCACAATGGATGCATGCCGT
CTGTGAGCAGATTTACGGTGAAGAGTTTGGAGACTACTGGATGAATTGTGCCCATAT
TCTGCACTTAGAACCGGGTGAAAAAGCCCAATTTTTTCATCGCGACACCGGGGTCT
ATCGCGTTTCTGATTTTCGCCGTGACTCAACGATCCGGAGTTTATGATAAATTTCTT

GGTGAGTTTAAACAGAGTTTAGAGAGGATAATGGAGCAACGCAACTCATTCTGGTA
GCCATAAATGGGATGCGGCGCATCCGCCTACATTTTATGGTAGCGATGAAGCAGTTC
CTGCGATTCTCGAGCCCGGAGATGCGGTGGTGTATCTGGGGTCTCTTTTCATGGA
GCGGGTGAGAACCGCAGCCTGGATTATAGACGCGGGATGATTGTAAGTATGCATCC
GGCTCATTTTACACCAATGGAATCCCATTTCCTTACCCAAAGAGATTGTTGAATC
AATGACGCCGCTGGCACAGCAAATGGTTGGATGGCGTACCATGAATAATCAGAATAA
AATTCCGATATGGCAGGCGGGAGACGATAAGATCGAGGATGTGCTCCGTCTGCAGC
ATAAAGAAGTTTATTAA

NHI 61 (CdmD) protein sequence:

MTVTINKSSVPHKVDFATPLSDVICHIEDGAVIVRGFMDVETIQKLQEEVDTAVEKGSF
GPRYQEYNEEAGEIPKHEIYKRGEKKTCKHMKNLALTSETFRNDVLNHHKWMHAVCEQI
YGEEFGDYWMNCAHILHLEPGEKAQFFHRDVGVRVSDFRRLNDPEFMINFLVSLTE
FREDNGATQLIPGSHKWDAHPPTFYGSDEAVPAILEPGDAVVYLGSLFHGAGENRSL
DYRRGMIVSMHPAHFTPMESHFHLPEIVESMTPLAQQMVGWRTMNNQNKIPIWQAG
DDKIEDVLRLQHKEYV

Codon-optimized NHI 62 (BuaE) sequence:

ATGGAGGTGGAACGTCTGGATTATACCGACTTCTTGAATGAGGCACCCGGGCGTCA
AGAGGCCTTCGTCCAACATCTTTATGCGGCATTATCCAGAGTTGGATTGCAAAGAT
TACGAATCACCTATTCCAGAGTCTGTTATACTTCAGTTGTTTCGCATGGACGAAGGG
GTTCTTCACTCTACCCTTAGAACATAAGCGTAAGGCCGCACATCCCCGCAACCGA
ACCCGCATCGCGGATGGTCTTGCAATTGGTCAGGAAAAATTAAGCGTTATAGCGCAG
GGTAAAGCCGTCCTGGATCTGAAAGAATCATTCGACATGGGCCCAGGCGATGATGA
ATTGTATCCGAATATCTGGACCGACGAGGGGGACCTGCCAGGCTTTCGTGCTTTTAT
GGAAGATTTTTACGGCCGTTGTCAGTCTTTGCACCTACAACCTTCTTTCTGCGATCGC
TCGTTCAATGCAGTTCCAGACTCGTATTTTGCGCCGCTGTGCAGCCAAAACCTCGT
CCGAACCTTAGATTAAATCACTATCCAGCGGTGAGTCGGCACGACCTGACGACCGGA
ACTATGCGCATTAGTAGTCACACCGATTTTGGCACTATTACCCTTCTGTTCCAAGATA
GTGTGGGCGGCTTAGAAGTTGAAGATCAAACCTCGTCCGGGGCCATTACATGCCGGTC
GCCGCAGATGATTGTACAGATATTATAGTCAATGTAGGCGATTGTCTGCAACGTTGG
ACGAACGATAGGTTACGCTCCGCGAATCATCGAGTTAACTCCCGAGAGGACTGAC
CCACGGGATGGTTGACGACCGTTATTCTATCGCCTATTTTGGAAAGCCGTGCGGTG
ATGTTAGTGTGCGTACCCTGTCCGCGCTGCTGCGTGCGAATGAGGAAGCTAAATAC
CGTGAAGAAATGACCGCGTGGCAGTACAATCAGTCGCGCCTATTACAAACCTACTA
G

NHI 62 (BuaE) protein sequence:

MEVERLDYTDLNEAPGRQEAFVQHLYAALSRVGFAKITNHPIPESVILQLFAWTKGFFT
LPLEHKRKAHPPQPNPHRGWSCIGQEKLVSIAQKGAVLDLKESEFDMGPGDDELYPNI
WTDEGDLPGFRAFMEFDYGRCSLHLQLLSAIARSMQLPDSYFAPLCSQNSSELRLN
HYPAVSRHDLTTGTMRISHTDFGTITLLFQDSVGGLEVEDQTRPGHYMPVAADDCTDI

IVNVGDCLQRWTNDRRLRSANHRVTLPRGLTHGMVDDRYSIAYFGKPSRDVSVRTLAL
LRANEEAKYREEMTAWQYNQSRLLQTY

Codon-optimized NHI 63 (EasH) sequence:

ATGACATCTCAGCACCAGGAGCACACCGGAACAAAACGGTTTTCCATTTCAGAGTGA
CCCCGTGGAGATTCACCGGGCAATCGTGGAGGACGGAGTAGCAATTATTGAAGGG
TTTCTGACCCCGGAACAGGTTCAAAAGTTGAACAAGGATGTTGATGCTCCCCTCAA
GGCGGACCGTGAGCAACTGAAATTCAAAGCGGATAAGAAAGATGATCCACACTTCT
GGCTCGCTGATTTTCATCCCTGACCATGTGCGCAAGGGTTCATAATCTGGTTGACTTCT
CCCACTGCTTCCGTCATGAAATCTTGAACCACGAACTACTTCATAAAATATGCCGTCT
GACCTTCGAGGAGTCTGGAGACTACTGGCTTGGGTACGGTGCAAGTATTGAAAATG
GCCCCGGGGACACGGAACAGAAAGTGGCATAGAGACCAGCCACGTTATCCGCTCGT
GAAAGAAGGACCGGACGCACCGGAAGGTATGTTGAATTTTTTACCGCTCTGACAG
ATTTTGATGCGGAGACAGGAAAAACACAGTATATACTGGGTAGCAATAAACGTGTGG
AGCTTGGCGAACC GGATGCCGACCATCCGATTGAATATGTAGGTCTTAAACCGGGC
GATACTACCATAGTTAGTGGTAAATAACACATCGCGGTAGTGACAACCGATCTGAC
AAGATGCGGCGTGCCATGCCTATTATGATAATCCCTTCTATTCTGACTCCATTTGACG
CCACCTGCCACCTTTCCCGAGAGCTGGTGGAACGATGACTCCGCTAGCACAAAA
GATGATATGTAGACGGTCAGTTATGATTCCGGCGCCTGGGACGGTAGAAGTGAAAA
CAGGAATATGGTGCGTGAATATGAGAGAGGCTGGCGAACAGATCGGGCTGAAATCC
AACCAGCGTGCAAAAGAAGATGCCGAAGCTACGGATGCAGTTTAA

NHI 63 (EasH) protein sequence:

MTSQHQEHTGTKRFSIQSDPVEIHRAIVEDGVAIIIEGFLTPEQVQKLNKDVDAPLKADRE
QLKFKADKKDDPHFWLADFIPDHVARVHNLVDFSHCFRHEILNHELLHKICRLTFEESG
DYWLGYGAVIENGPGTTEQKWHRDQPRYPLVKEGPDAPEGMLNFFTALTDFDAETGK
TQYILGSNKRVELGEPDADHPYVGLKPGDTTIVSGKITHRGSNDRSDKMRRAMPIMII
PSILTPFDATCHLSRELVTMTPLAQKMICRRSVMIPAPGTVEVKTIWCVNMREAGEQI
GLKSNQRAKEDAEATDAV

Codon-optimized NHI 64 (FLS) sequence:

ATGGAAGTGGAGCGCGTTCAAGCGATTGCTAGCTTAAGCCATTCCAATGGTACCATT
CCGGCAGAATTTATTCGGCCGGAGAAAGAACAGCCGGCTTCCACCACGTATCACG
GTCCAGCCCCAGAGATCCCTACCATTGATCTTGATGATCCGGTACAAGATCGACTG
GTCCGCAGCATTGCGGAAGCGTCTCGGGAGTGGGGCATCTTTCAGGTTACCAATC
ATGGAATTCCGTCAGACCTGATATGCAAACTACAGGCAGTGGGTAAAGAATTTTTTG
AACTGCCACAGGAGGAAAAAGAGGTGTATCCCGTCCTGCTGACGCAAAAGACGT
GCAGGGGTATGGCACCAAGCTTCAAAAAGAAGTAGAAGGTAAAAAATCATGGGTGCG
ATCATCTTTTCCATCGAGTGTGGCCGCGTCGAGTATTAACCTACCGTTTCTGGCCTA
AAAATCCTCCTTCTATCGTGCTGTGAATGAGGAATATGCGAAATATATGCGCGAAGT
TGTCGATAAGTTATTTACTTACTTATCATTAGGCCTAGGAGTAGAGGGCGGCGTACTT
AAAGAAGCCGCTGGGGGTGATGATATCGAGTATATGTTAAAAATCAACTATTATCCTC
CGTGCCCGAGACCAGATCTTGCGCTGGGCGTTGTGGCTCACACCGATCTCTCAGC

GTTGACCGTCTTGGTGCCCAACGAGGTCCCGGGCTTGCAGGTCTTCAAGGATGAT
AGGTGGATAGACGCTAAATATATTCCCAACGCACTTGTGATCCACATTGGGGATCAA
ATTGAAATCCTAAGTAATGGAAAATATAAAGCTGTTTTACATCGCACAACTGTAAACAA
GGATAAACGAGAATGAGTTGGCCGGTTTTTCTTGAACCGCCGGCAGACACAGTAG
TTGGACCTCTCCCGCAACTGGTTGATGACGAGAACCCGCCGAAATATAAAGCGAAG
AAATTCAAAGATTATTCGTATTGTAAATTAAATAAACTGCCCCAATGA

NHI 64 (FLS) protein sequence:

MEVERVQAIASLSHNGTIPAEFIRPEKEQPASTTYHGPAPEIPTIDLDDPVQDRLVRSIA
EASREWGIFQVTNHGIPSDLICKLQAVGKEFFELPQEEKEVYSRPADAKDVQGYGTKL
QKEVEGKKSVDHLFHRVWPPSSINYRFWPKNPPSYRAVNEEYAKYMREVVDKLFY
LSLGLGVEGGVLKEAAGGDDIEYMLKINYPPCPRPDALGVVAHTDLSALTVLVPNEV
PGLQVFKDDRWDIAKYIPNALVIHIGDQIEILSNGKYKAVLHRTTVNKKTRMSWPVLE
PPADTVVGPLPQLVDDENPPKYKAKKFKDYSYCKLNKLPQ

Codon-optimized NHI 65 (F3H) sequence:

ATGGCCCCCGGGGCGACAGCTACACCATTCCTTCCGACCGTCTCTAACGAGACAA
CTTTGCGTGAAAGTTTCGTGCGGGACGAAGATGAACGCCCGAAAGTGGCGTATAAC
GTATTTTCCTCTGAAATACCCGTAATATCTCTGGAGGGTATAGACGAAATTGAAGGAC
GCCGTGTGGAGATTTGTAAAAAAATTGTGGCAGCTTGTGAAGATTGGGGAGTGTTT
CAGGTCGTTGATCACGGAGTTGACGCCGGTGTTATCGCAGATATGACGAGACTGGC
TCGTGAATTCTTTAGCCTTCCGCCGGAAGATAAACTGCGCTTTGATATGTCTGGGGG
TAAAAAAGGTGGGTTCATAGTTTCATCACATCTTCAGGGCGAGGCCGTGCAGGATT
GGCGAGAAATAGTTACCTATTTTAGTTATCCTATTAGAGCCCGAGACTATTCTCGTTG
GCCTGATAAACCGGAGGATTGGCGGTCCGTCAACGAAAAGTATTCCGAAACTCTGA
TGGAGTTGGCCTGCAAACTTCTCGGGGTTTTGTCCGAAGCAATGGGCCTGGATACC
GAGGCCATAACAAAGGCGTGTATTGATATGGACCAAAAAGTGGTTGTCAATTTCTAT
CCAAAATGCCCTCAGCCGGATCTTACCTTGGGCCTAAAACGACACACGGATCCCGG
AACTATAACTCTGTTATTGCAAGACCAGGTCTGGAGGTCTGCAAGCCACTAAGGATG
GAGGGAAGACATGGATAACTGTTCAACCGGTGGAGGGTGCCTTCGTAGTTAATCTG
GGTGACCACGGCCACTTTTTGTCAAACGGCCGGTTTAAAAACGCAGATCACCAGG
CAGTGGTGAATTCTAATACTAGTCGTCTCTCAATAGCGACTTTCCAGAACCCTGCAC
CTGATGCAATCGTATATCCGTTGGCAATTCGCGAAGGGGAAAAACCAGTGCTGGAC
GAACCGATTACCTTTGCGGAGATGTACCGGCGGAAAATGTCACGTGATCTGGAAC
AGCCAAGATCAAAAAACTTGCCAAAGTAGAAAATCAGGAAGTTCTTGAGAAGACTAA
AGTTGAAGCAGTCGCGCAGCCCAAGGGTCTGGGAGAGATCCTGGCGTAA

NHI 65 (F3H) protein sequence:

MAPGATATPFLPTVSNETTLRESFVRDEDERPKVAYNVFSSEIPVISLEGIDEIEGRRVEI
CKKIVAACEDWGVFQVVDHGVDAAGVIADMTRLAREFFSLPPEDKLRFDMSGGKKGGFI
VSSHLQGEAVQDWREIVTYFSYPIRARDYSRWPDKPEDWRSVTEKYSETLMELACKLL
GVLSEAMGLDTEAITKACIDMDQKVVVNFYPKCPQPDLTGLKRHTDPGTITLLLQDQV
GGLQATKDGGKWTWTVQPVGAFFVNLGDHGHFLSNGRFRKNADHQAVVNSNTSRLSI

ATFQNPAPDAIVYPLAIREGEKPVLDPEITFAEMYRRKMSRDLELAKIKKLAKVENQEVL
EKTKEVAQAQPKGLGEILA

Codon-optimized NHI 66 (NCS1) sequence:

ATGAGTAAGAACCTGACCGGCGTTGGCGGTTCTCTCCCAGTTGAAAACGTCCAGG
TTCTGGCGGGCAAAGAACTGAAGAATCTGCCGAACCGCTATGTCCGCCCCGAGCT
TGAGCACGATGACGTGGTCCCGATTGACAACTCTCTGGAAATTCCGGTTATTGACC
TGAGTAGACTCTTGGATCAACAGTATGCCTGTGATGAACTTGCAAATTCATAGCG
CATGCCTGGATTGGGGGTTTTTCCAACCTGATTAACCATGGAGTCCGAGAAGAAGTA
ATTGAAAAAATGAAAGTGGATACTGAAGATTTCTTCCGACTTCCGTTTAAAGAAAAA
ATGCATATCGTCAGTTACCCAACGGTATGGAGGGGTATGGCCAAGCCTTTGTGACAT
CGGAGGAACAGAACTGGATTGGGCGGACATGCATTTTCTCATTACGAAGCCGGTT
CAGGAGCGAAACATGCGATTTTGGCCGACATCCCCGACTAGTTTTCTGTGAACTAT
GGAAAAATATTCTATGGAACCTGCAAAGGTGCGCAATGTGTTTGACCGGGATGATGGC
GAAAAATCTCGGGTTGGAGTCAGAAATCTTAACAAAACCTTTGCGCACTGTTTTTAA
TCGCGAGGATGAGCTACTCCCCAGCATGAGTTCATGCGGTGAAGGCTTAGGCCTG
TCGCCACACTCAGACGCCACCGGGTTAACGTTACTGATACAGGTAAATGAAGTGAA
TGGGCTCCACATCAAAAAAGATGAAAAGTGGGTTCGATTAAGCCTATCCTAGGTGC
TTTCGTTGTTAATATTGGAGACGTTATTGAAATTATGTCTAATGGTATCTACAAATCAAT
AGAACATAGAGCTGTCATCAATACAGACAAAGAACGCCTGTCTATTGCGGCCTTTCA
TGACCCAGAATATGGCACGAAGATCGGACCGCTGCCCGACTTGGTAAAGAAAATG
GTGTCAAATATAAAACGATTGATTACGAGGATTATTTAATTCTGTTTCGAGTAACATTAAG
TTGGATGGTAAATCACTGCTGGACCAGATGAAACTGTAA

NHI 66 (NCS1) protein sequence:

MSKNLTGVGGSPLPVENVQVLAKELKNLPNRYVRPELEHDDVVPIDNSLEIPVIDLSRL
LDQQYACDELAKFHSACLDWGFFQLINHGVRREEVIEKMKVDTEFFRLPFEKKNAYRQ
LPNGMEGYGQAFVTSEEQKLDWADMHFLITKPVQERNMRFWPTSPTSFRETMEKYS
MELQKVAMCLTGMMAKNLGLESEILTKPLRTVFNREDELLPSMSSCGEGLGLSPHSDA
TGLTLLIQVNEVNGLHIKKDEKWWPIKPILGAFVVNIGDVIEIMSNGIYKSIEHRAVINTDKE
RLSIAAFHDPEYGTKIGPLPDLVKENGVKYKTIDYEDYLIRSSNIKLDGKSLLDQMKL

Codon-optimized NHI 67 (LDO1) sequence:

ATGACGGATGTGGAACCTGCGCGTGGAAGCTTTATCTCTTTCCGGGGTAAGTGCTAT
CCCGCCTGAATATGTGAGGCCCGAAGAAGAGAGAGCGGACCTTGGGGATGCGCTG
GAGTTAGCGCGAGCAGCCAGCGACGACGACGCAACAGCGCGCATACTGTTGTG
GATATATCGGCCTTTGATAACGACGGTGATGGTCGCCACGCCTGTGTGCAAGCGGT
CCGTGCAGCAGCTGAAGAATGGGGAGTTATCCATATAGCTGGACATGGCTTACCGG
GAGATGTTCTTGGTAGGCTGCGTGCTGCGGGTGAAGCTTTCTTCGCTCTGCCTATT
GCTGAAAAAGAGGCATACGCGAATGACCCGGCTGCGGGCCGCTTGCAAGGTTATG
GCTCTAAGCTGGCCGCCAATGCATCAGGTAAACGGGAATGGGAGGATTATCTTTTT
CATTTGGTACATCCAGATCATTTGGCCGACCATTTCGCTTTGGCCGGCTAATCCGCCG
GAATATGTTCCAGTTTCCCGCGATTTTGGAGGACGAGTTCGAACCTTGGCGTCTAA

ATTACTGGCAATTCTCAGTTTAGGTCTGGGACTGCCGGAGGAGACTCTCGAGCGCA
GACTTCGTGGCCACGAATTGGCCGGCGTGGATGACGATCTTCTGCTTCAACTGAAA
ATTAATTATTATCCTCGGTGTCCACGACCTGATCTTGACGTGGGCGTTGAAGCTCAC
ACTGATGTGTCTGCTCTTTCTTTTATTTTACATAATGGGGTTCCAGGACTGCAGGTG
CATCATGCAGGAAGCTGGGTGACTGCCCCGTCCAGAACCGGGCACCATCGTCGTAC
ATGTGGGCGACGCACTGGAAATTCTTACCAACGGACGTTATACAAGTGTACTGCAC
CGCGGCCTGGTGAGTAGAGATGCAGTACGATTAAGCTGGGTGGTATTCTGTGAACC
GCCGCCCGAATCAGTACTCCTGCAACCCGTCCAAGAGCTGCTGGCCGATGGAGCA
GGGAAACCGCTGTTTGCACCAAGAACGTTCAAACAACACGTCCAGCGTAAACTATT
TAAAAAGCTCAAGGATCAGCAGGATAATAATGCAGCAGCCGCTTCGAATGGCATGAT
TACCAAATAA

NHI 67 (LDO1) protein sequence:

MTDVELRVEALSLSGVSAIPPEYVRPEEERADLGDALELARAASDDDATARIPVVDISAF
DNDGDGRHACVEAVRAAAEEWGVIIAGHGLPGDVLGRLRAAGEAFFALPIAEKEAYA
NDPAAGRLQGYGSKLAANASGKREWEDYLFHLVHPDHLADHSLWPANPPEYVPVSRD
FGGRVRTLASKLLAILSLGLGLPEETLERRLRGHELAVDDDLLLQKINYYPKPRPD
AVGVEAHTDVSALSFILHNGVPLQVHHAGSWVTARPEPGTIVVHVGDALEILTNGRYT
SVLHRGLVSRDAVRLSWVVFCEPPPESVLLQPVQELLADGAGKPLFAPRTFKQHVQRK
LFFKLKDDQDNNAASNGMITK

Codon-optimized NHI 68 (LDO2) sequence:

ATGACAGACGTGGAGCTGAGAGTCGAAGCACTGTCATTGTCCGATGTGAGCGCAAT
CCCGCCCGAATACGTGCGTCTTGAGGAAGAACGCACCGACCTGGGCGATGCGTTG
GAGGTGGCTAGAGCAGCATCTGATGACGCAGATGCAGCTCGGATCCCGGTAGTGG
ACATTAGCGCATTTCGATGGTGATGGTCTGAAGGGCGTGCGTGGAGGCCGTGCGTGC
GGCCGCGGAAGAGTGGGGAGTAATGCATATAGCCGGACACGGCTTACCGGGTGAC
GTCCTAGACAGACTGCGCGCGGCGGGTGAAGCCTTTTTTCGCGTTGCCGATAGCTG
AGAAGGAAGCCTACGCCAATGACCCGGCCGCTGGGCGGCTTCAAGGGTATGGGTC
CAAGCTGGCGGCTAATGCATCAGGCAAACGGGAATGGGAAGATTATCTGTTCCATT
TAGTTCATCCGGATCACCTGGCCGATCATCTACTGTGGCCGGCAAATCCCCGGAA
TATGTTCTGTGTCTAGAGATTTTCGGTGGGCGGGTCCGAACACTGGCTAGCAAGCT
ACTAGCCATTCTCTCCCTAGGTCTGGGTTTACCGGAAGAAACCCTAGAAAGGCGCT
TACGCCGGCATGATCAACATGGTGTTGATGATGATTTACTTCTTCAGCTGAAAATAAA
CTATTATCCTCGTTGTCCAAGACCTGATCTGGCCGTGGGCGTAGAGGCCCATCTG
ATGTGTCTGCTTTAAGTTTCATCCTTCACAACGGTGTACCTGGACTGCAGGCACACC
ATGCTGGGACCTGGGTAACCTGCCCGATCAGAGCAGGGCACGATAGTAGTTCATGTG
GGAGATGCCCTTGAGATTCTGACCAACGGAAGGTATACATCAGTGCTTCATAGGAG
CCTCGTTTCCCGGGATGCTGTGAGGGTTTCTTGGGTGGTGTCTGTGAGCCTCCA
CCAGAAAGCGTACTGCTGCAGCCTCTTCCGGAACCTGTTAGCAAATGGTGCTGGGA
AGCCTCTGTTTTCGCCGAGAACCTTCAAACAACATGTTTCAGCGTAAGCTATTTAAAA
AACTGAAGGATCAGCAGGATAATAACGCGGCAGCTGCGTCGAACGGCATTATCCCC
AAATGA

NHI 68 (LDO2) protein sequence:

MTDVELRVEALSLSDVSAIPPEYVRLEEERTDLGDALEVARAASDDADAARIPVVDISAF
DGDGRRACVEAVRAAAEEWGMHIAGHGLPGDVLDRRAAGEAFFALPIAEKEAYAN
DPAAGRLQGYGSKLAANASGKREWEDYLFHLVHPDHLADHSLWPANPPEYVPVSRDF
GGRVRTLASKLLAILSLGLGLPEETLERRLRRHDQHGVDLDDLLQLKINYYPKPRPDL
AVGVEAHTDVSALSFILHNGVPGQLAHHAGTWVTARSEQGTIVVHVGDALEILTNGRYT
SVLHRSLSVRDAVRVSWVVFCEPPPESVLLQPLPELLANGAGKPLFAPRTFKQHVQRK
LFFKKLDQDNNAAAASNGIIPK

Codon-optimized NHI 69 (HxnY) sequence:

ATGGCACCGACGGCCCCGCGGATCCTTGATTTTTCTCCGTTCTATGGAACGGACGG
TGCTGCTAAAGCTAAATTGGTGCAACAAGTCCGTGAGTCATGTGAATATAATGGCTT
TTTCCAGATTACAGGTCATAGAATTCCTCGCGAATTACAAGTGAGAGTAATGGATGC
GGCGAAGCGTTTTTTTTGCTCTGCCTTTGGAAGAAAAAATGGCCATTGACAAAAATCT
CAATAGTTTCAATCGAGGTTATGAACTGCTCCGTTACAGATGCTTGAGGTAGGAAC
CGCGCCTGAGTTGAAAGAAGGACTGTACATTGGAGAAGAGATAGGTGCCGATCATC
CGTATTATATCAACGGTCGCTTGAATAGCGGACCTAATCAGTGGCCAGCTACCGTGC
CTGATGCACAAGAATTCGCGCAAACAAGTATGGAGTATTATCATGCAGTGTACGAAC
TCGCTAAGGATGTCCTTGCGGTTTTAGCCCTCACCTTAGACGTCGAAGAATCCTTTT
TTGATCCGTTGACAGAGGGCGGAGTAGCCACCATGCGGATGTTACATTATCCGAGC
CAACCGAAGGATGAAGATGAAAACTAAATCGCGGGATTGGCGCGCATACGGATTT
TGGATGTATCACGCTGCTGCTCCAAGATGAAGTTGATGGCTTACAGGTGCTTGATG
CGCCGTCCGGACAATGGCTGGATGTACAACCGGTACTTGGTGCTTATGTAGTGAAC
TTAGGTGATCTTATGATGCGGATGGCTAATGATCGGTACAAGTCTAACATCCATCGTG
TAATCAATAAATCGGGCAGAGAACGTTACTCTATTCCTTTTTTTTTCAGTGGCAACCC
GGATCATGTTTGCAAGTGTCTTCCCAATTGCTGTAAGGCAGGGGAACAGCCCAAAT
ATCCGCCGATTACAGTGGAAGATATGGTGCGTGGGGCGTATAAACAGAGCTATGGC
CGGGCCGAAGCGTACAAAAAAGAACTGGCGGAAAAGGCAAAGGCGCACAAAATCG
AAGCCGCGTCCGCCACGGCCATGGTCTCGTGA

NHI 69 (HxnY) protein sequence:

MAPTAPPILDFSPFYGTDGAAKAKLVQQVRESCEYNGFFQITGHRIPRELQVRVMDAA
KRFFALPLEEKMAIDKNLNSFNRYELLRSQMLEVGTAPELKEGLYIGEEIGADHPYYIN
GRLNSGPNQWPATVPDAQEFRETSMEYYHAVYELAKDVLAVLALTL DVEESFFDPLTE
GGVATMRMLHYP SQPKDEDEKLNRGIGAHTDFGCITLLLQDEV DGLQVLDAPSGQWL
DVQPV LGAYV VNLGDLMMRMANDRYKSNIHRVINKSGRERY SIPFFFSGNPDHVCKCL
PNCKKAGEQPKYP PITVEDMVRGAYKQSYGRAEAYKKELAEKAKAHKIEAASATAMVS

Codon-optimized NHI 70 (T6ODM) sequence:

ATGGAAAAAGCTAAACTGATGAACTCGGTAATGGAATGGAAATCCCGTCTGTTCAG
GAGCTGGCCAAATTAACACTCGCAGAAATTCCAAGCCGCTATGTTTGTGCTAACGAA
AATTTATTATTACCGATGGGTGCGAGCGTAATCAACGATCACGAGACCATTCCAGTG

ATTGATATTGAAAACCTTGTTAAGCCCGGAGCCGATCATTGGTAAACTAGAATTAGACC
GTTTACATTTTCGCTTGCAAAGAATGGGGTTTCTTTCAGGTAGTGAACCATGGGGTTG
ATGCCTCACTCGTGGATAGTGTGAAATCTGAGATCCAAGGTTTCTTTAACCTTTCTAT
GGATGAAAAAACGAAATATGAACAGGAAGATGGTGTATGTGGAAGGTTTTGGCCAGG
GCTTCATAGAGAGCGAAGATCAGACCCTTGACTGGGCGGATATTTTTATGATGTTTA
CCTTACCGTTACACTTACGGAAGCCTCATTTATTAGTAAACTGCCTGTTCCATTGCG
CGAAACTATTGAATCTTACAGCTCGGAGATGAAGAAGCTGTCTATGGTACTGTTCAA
CAAGATGGAAAAAGCGCTGCAAGTACAGGCTGCAGAGATCAAAGGCATGTCGGAA
GTTTTTATAGACGGCACGCAGGCGATGCGCATGAATTATTATCCACCTTGCCACAG
CCGAATCTGGCGATAGGGCTTACTAGCCACTCGGACTTTGGCGGCTTGACGATATT
GTTACAGATTAATGAAGTAGAAGGTTTACAGATTAAGCGCGAGGGTACATGGATTTC
AGTCAAACCCCTTCCGAACGCATTGTTGTTAATGTGGGCGATATACTGGAAATAAT
GACCAACGGCATCTACCATAGTGTGGATCATCGTGCTGTTGTAAACAGTACAAACGA
GCGTCTTAGCATCGCCACATTTTCATGACCCGTCTCTTGAAAGTGTTATCGGACCAAT
CAGTAGCTTAATCACGCCCCGAAACACCGGCGCTCTTTAAATCAGGTTCAACATATGG
GGATCTGGTGGAAAGAGTGTA AACACGAAAGCTAGACGGTAAATCATTTCTCGACA
GTATGCGCATCTGA

NHI 70 (T6ODM) protein sequence:

MEKAKLMKLGNGMEIPSVQELAKLTLAEIPSRVVCANENLLLPMGASVINDHETIPVIDIE
NLLSPEPIIGKLELDRLHFACKEWGFFQVNVHGVDA SLVDSVKSEIQGFFNL SMDEKTK
YEQEDGDVEGFGQGFI ESEDQTL DWADIFMMFTLPLHLRKPHLFSKLPVPLRETIESYS
SEMKKLSMVL FNKMEKALQVQAAEIKGMSEVFIDGTQAMRMNYYPPCPQPNLAIGLTS
HSDFGGLTILLQINEVEGLQIKREGTWISVKPLPNAFVNVGDILEIMTNGIYHSVDHRAV
VNSTNERLSIATFHDP SLESVIGPISSLITPETPALFKSGSTYGDLVEECKTRKLDGKSFL
DSMRI

Codon-optimized NHI 71 (CODM) sequence:

ATGGAGACCCCGATTTTAATAAAACTAGGTAACGGCCTGTCCATCCCTAGCGTGCAA
GAGCTAGCAAAGCTAACTCTGGCTGAAATTCCGAGTCGTTATACCTGTACAGGTGAA
TCCCCGCTGAATAATATCGGTGCCTCAGTGACTGATGATGAAACTGTGCCGGTCATA
GACTTGCAAACTTGTTAAGCCCAGAACCAAGTGGTTGGCAAATTAGAATTAGATAAA
CTACATTCTGCATGCAAAGAGTGGGGGTTCTTTCAGCTGGTAAATCATGGAGTTGAC
GCCCTGCTAATGGATAACATTAAGTCCGAAATCAAAGGTTTCTTTAACCTTCCCATGA
ATGAAAAAACAAAATATGGTCAGCAGGATGGTGATTTTCGAGGGTTTTGGACAGCCTT
ATATCGAATCGGAAGACCAGCGACTCGATTGGACGGAGGTGTTCTCCATGCTTTCA
CTGCCTCTGCATCTGAGAAAACCTCACTTATTTCCGGA ACTTCCGTTACCATTTTCGC
GAAACACTGGAGTCGTATTTGTCTAAGATGAAAAAACTGTCTACCGTCGTGTTTCGAG
ATGCTTGAAAAATCCCTGCAGTTAGTGGAGATCAAAGGTATGACTGATTTGTTTGAA
GATGGTCTGCAGACCATGCGTATGAATTATTATCCGCCGTGTCCTCGGCCGGAGTT
GGTACTTGGACTCACGTCGCACAGCGACTTTAGCGGTTTAACAATTCTTCTGCAGC
TCAACGAGGTAGAAGGTCTGCAGATTCGCAAAGAGGAGCGATGGATCTCCATCAAG
CCTCTGCCTGACGCTTTTATCGTTAACGTCTGGGGACATCCTGGAAATCATGACCAAT

GGGATTTATCGCAGTGTGGAGCATCGTGCCGTTGTCAACAGCACGAAGGAACGTCT
GTCCATTGCCACTTTTACGACTCCAAACTGGAATCAGAAATTGGTCCGATTTCTAG
CCTGGTCACACCGGAAACTCCGGCGTTGTTTAAACGTGGTCGTTATGAGGACATTT
TGAAAGAAAATTTAAGTCGCAAGCTGGATGGGAAATCGTTTTTAGACTATATGCGCAT
GTGA

NHI 71 (CODM) protein sequence:

METPILIKLGNLSIPSVQELAKLTLAEIPSRYTCTGESPLNNIGASVTDDETVPVIDLQNL
LSPEPVVGKLELDKLHSACKEWGFFQLVNHGVDALLMDNIKSEIKGFFNLPMNEKTKY
GQQDGD FEGFGQPYIESEDQRLDWTEVFSMLSLPLHLRKPHLPPELPLPFRETLESYL
SKMKKLSTVVFEMLEKSLQLVEIKGMTDLFEDGLQTMRMNYYPPCPRPELVGLTSHS
DFSGLTILLQLNEVEGLQIRKEERWISIKPLPDAFIVNVGDILEIMTNGIYRSVEHRAVVNS
TKERLSIATFHDSKLESEIGPISSLVTPETPALFKRGRYEDILKENLSRKLDGKSFLDYMR
M

Codon-optimized NHI 72 (PntH) sequence:

ATGACAGACACTTTTTCTGACTACGCGGACTGTACACCGCTGCTGGATGACCGGGA
AGCGCTCGACCGCTTTTACGACGAGCACGGATACGTTTACCTCCGCGGTGTCCTC
GATAGGGAATTGGTTCGGACGACTGCTGAGCAAATGCTCCAGGGACTGATCGCACT
GGGACATGCCGCACCCGGAACCACTTTAGATACATTGACCATAGAGTCCTATGAGG
CGGTTGACGAGGTGGCAATGCATGATCACGTCCGATACGATGATCTTTGGAATCATC
CGTCCACATTGAAAGTTTGGGAAAAAGTCTTTGGTGAACCGGTTTTTGTTCCTCAAAA
GCACGACGATTGCTATTACCCATCTGCACCCGATAGTGCTGAACCGAGCTTTTTGA
CGCCCCTGCATCAAGACGGGTTCTATATTGGACCGAACAAGATTTCCGTACCGCG
TGGATCCCACTCCTCCCGACAACCTCGGGGCACGGGCGGCGTTGCGATCGCGGAC
GGGAGTCATAAGAAAGGCCCGCGTGAACACGTTCTAACCGAAAACCTTCGTGCTT
TGGTCATGCCGTTTCGCGGGATCCCGCCTGCAGAATTCGGAGCCGATGAAGAATTG
CTTTTCTCTCCTATGGAACAGGGGATGTCCTGTTATTTTCATGCGTTCATGTGTCAC
AAAAGCCTGCCAAATGTAAGTGTTGATCCGGCAGGTATGCGTATGTCTATGGATACC
CGAATTCAGCCAGCCAGCTCGCATCGTGGCTTTAATGCCCTCACACCGTGGCCGG
AGTCTGCCAAGGACGCCTCAAAGGCATTATGTCTAAATAACGGGTACCCCAACC
ACAGCGGAGTAA

NHI 72 (PntH) protein sequence:

MTDTFSDYADCTPLDDREALDRFYDEHGYVYLRGVLDRELVRTTAEQMLQGLIALGH
AAPGTTLDLTIESYEAVDEVAMHDHVRYYDDLWNHPSTLKVWEKVFGEPVFVFKSTTIR
YYPSAPDSAEPSTPLHQDGFYIGPNKDFRTAWIPLPTTRGTGGVAIADGSHKKGPR
EHLVTENFRRFGHAVRGIPPAEFGADEELLFSPMEPGDVLLFHAFMCHKSLPNVSVD
AGMRMSMDTRIQPASSHRGFNALTPWPESAKDASKGIMSKITGTPTTAE

Codon-optimized NHI 73 (prolyl-4-hydroxylase 13) sequence:

ATGCGGAGCTATGGTAAAGAAAAAAACTTGTATTTCCGTATGTTTTTATCGCATGCT
GTTTCTTTCTCGCTATTTTTGGTTTTTGCTTCTTTAACCTTTTTTACAGGGGATCTCT

TTCTCGGAGATCCCCACTACACGTCGTTTCGGTCAACGACGAAACAGATAGTCTGGA
TCATGGGAGTTTCGGTGTCAAACATCCCGTTTCACGGTCTTAGTTGGAACCCTCGGG
TCTTTTATTTACCCAACCTTCGCCACAAAACAACAGTGTGAAGCTGTTATTGATATGGC
TAAACCGAAACTGAAACCATCAACATTAGCACTGAGAAAGGGAGAAACCGCGGAAA
CTACCCAGAACTATAGGTCCCTTCACCAACATACGGACGAAGATGAATCTGGGGTTC
TGGCGGCCATTGAAGAAAAAATCGCGTTGGCGACAAGATTTCTAAAGATTACTATG
AGTCCTTTAACATTCTTCGGTATCAATTGGGACAAAAATACGATAGCCATTATGACGC
CTTTCATTGAGCTGAATATGGACCGCTGATTTCCCAGAGAGTTGTAACTTTTTTGTTA
TTCCTGTCTTCCGTGGAAGAAGGAGGTGAAACTATGTTCCCCTTTGAGAATGGGCG
TAATATGAATGGTCGTTACGATTATGAAAAATGTGTTGGACTGAAAGTCAAACCGCG
CCAGGGTGACGCCATATTTTTCTACAATCTGTTTCCAAACGGGACAATCGATCAGAC
GAGCCTTCATGGTAGTTGTCCCGTAATTAAAGGGGAGAAATGGGTGCTACTAAATG
GATAAGAGATCAGACCTACGATTAA

NHI 73 (prolyl-4-hydroxylase 13) protein sequence:

MRSYGKEKKLVFPYVFIACCFFLAIFGFCFFNLFSQGISFSEIPTTRRSVNDETDSLHDG
SSVSNIPFHLGSWNPRVFYLPNFATKQQCEAVIDMAKPKLKPSTLALRKGETAETTQNY
RSLHQHTDEDESGVLAAIEEKIALATRFPKDYYESFNILRYQLGQKYDSHYDAFHSAEY
GPLISQRVVTFLLFLSSVEEGGETMFPFENGRNMNGRYDYEKCVGLKVKPRQGDAIFF
YNLFPNGTIDQTSLHGSCPVIKGEKWVATKWIRDQTYD

Codon-optimized NHI 74 (FLS4) sequence:

ATGGAAGTAGAACGCGACCAGCACAAACCACCCCTATCATTGCAGAACAAATAAAATT
CCCAGTTCCCAAAACTTCCCAGTGGTGGATCTTAGCAATACTAATGGAGAACTGGTT
GCGCGGAAAGTTGCGAAGGCCTCTGAAGAGTGGGGCATTTCAGGTCGTAAACC
ATGGCATAACCAACCGAATTAATTAGACGTCTGCATAAGGTGGATACTCAGTTTTTTGA
ACTCCCTGAGTCTAAAAAAGAAGCAGTTGCAAAACCGGCGAACTCCAAGGAAATCC
AGGGGTATGAAATGGATGATGTGCAAGGCCGCCGTAGCCATATCTTTCATAACTTATA
TCCTAGTAGTTTCACTTAAGTACGCCTTTTGGCCGAAGAACCCTCCAGAGTATCGCGA
AGTTACGGAAGAATTTGCTAAGCACGCTAAGCAACTGGCCGAGGAAATACTGGGCC
TGTTATCTGAAGGAGCAGGCTACCTCATGAAAATAAATTATTACCGCCCGTGTCCTG
AACCAGACTGGGTGATGGGTATTAAGGCCACACCGATTTTAATGGGCTCACTTTAC
TGATCCCGAACGAAATTTTTGGGCTGCAAGTTTTCAAGGAAGACCGATGGTTGGAT
GTGGATTATATTTATCCGGCAGTTATCATAATTATCGGAGACCAAATAATGAAAATGTC
CAATGGTCGCTATAACAACGTGCTGCATCGAGCACTAATGGATAAAAAGAAGACTAG
GATGTCATCAGTAGTACATATTAAACCCCGTATGACATGGTGGTTTCTCATTTTCCA
AATAGTCCGGCCGCTATCATATTACCAAGCTCGAGCCTGTAA

NHI 74 (FLS4) protein sequence:

MEVERDQHKPPLSLQNNKIPSSQNFPVVDLSNTNGELVARKVAKASEEWGIFQVVNHG
IPTELIRRLHKVDTQFFELPESKKEAVAKPANSKEIQGYEMDDVQGRRSHFHNLYPSSS
VNYAFWPKNPPEYREVTEEFAKHAKQLAEEILGLLSEGAGYLMKINYRCPPEPDWVM

GIKAHTDFNGLTLLIPNEIFGLQVFKEDRWLDVDYIYPAVIIIIGDQIMKMSNGRYNNVLHR
ALMDKKKTRMSSVVHIKPPYDMVVSHPNSPAAIILPSSSL

Codon-optimized NHI 75 (FLS6) sequence:

ATGAATGTGGAGCGCGATCAACATATATCTCCCCCATGCCTGTTGACTAAAAAGATC
CCTATCGTGGATTTGTCCGATCCATCGGATGAACTCGTAGCGCATGCGGTTGTGAA
AGCCTCCGAGGAATGGGGAATTTTTCAGTTAGTAAATCATGGGATTCCCGCAGAATT
GATGCGACGTCTGCAGGAAGTCGGTCGCCAGTTTTTTGAGCTGCCGGCTAGCGAA
AAAGAGTCCGTTACAAGGCCGGCTGACAGCCAGGACATAGAAGGTTTCTTTTCGAA
GGACCCGAAAAAGCTGAAAGCATGGGACGACCACCTGATTCACAATATTTGGCCCC
CGTCGTCGATTAACATCGCTATTGGCCAAATAATCCTAGCGACTATTCAGGTGACG
GATTCCGTGAAGTCACAAAAGAATACACCCGTAACGTCACCAACCTTACAGAGAAG
ATTGTAGGCGGGGATAAAGCCCAGTACGTGATGCGGATTAACATATCCACCTAGC
GATAGCGCGATCGGTGCGCCAGCCCATAACCGATTTTTGCGGATTGGCGTTATTAGT
CAGCAACGAAGTGCCAGGTTTGCAGGTCTTTAAAGATGACCATTGGTTCGATGTTG
AATATATTAATTCTGCCGTGATCGTACTGATTGGCGACCAGATCATGCGGATGTCGAA
TGGCAAGTATAAGAACGTGTTGCACAGAAGTATCATGGATGCCAAGAAGACGCGTAT
GAGCTGGCCGATCCTAGTCGAACCAAAGCGCGGTCTGGTGGTGGGGCCCCCTGCC
GGAACCTCACAGGTGACGAAAATCCGCCGAAATTCGAATCATTAAACGTTTCGAAGATTA
CGTTTATCGGAAGATCATTAAAGGTTTTACGGGATTAG

NHI 75 (FLS6) protein sequence:

MNVERDQHISPPCLLTKKIPIVDLSDPSDELVAHAVVKASEEWGIFQLVNHGIPAELMRR
LQEVGRQFFELPASEKESVTRPADSQDIEGFFSKDPKCLKAWDDHLIHNWPPSSINYR
YWPNNPSDYSGDGFREVTKEYTRNVNLTAKIVGGDKAQYVMRINYPPSDSAIGAPA
HTDFCGLALLVSNEVPGLQVFKDDHWFDVEYINSAVIVLIGDQIMRMSNGKYKNVLHRS
IMDAKKTRMSWPILVEPKRGLVVGPLPELTGDENPPKFESLTFEDYVYRKIIKVLRD

Codon-optimized NHI 76 (F6'H1) sequence:

ATGCCTGCAGTGTTATCAAGTGTTCTTTCTAATATTACGGATTTTGTGGTTCACGAGG
GAAATGGAGTCAAGGGATTAGCAGATATGGGTCTGGAAGCCCTGCCAAAACAGTAC
GTTACGCCCCGAGGAAGAACGGATCACAAACATCTACGGTTATAGTGGATGATACGATA
CCGGTGATAGATCTGTCCGAGTGGGGGTCAGATCCAAAAGTGGGAGACATGATCT
GTGAAGCCGCGGAGAAGTGGGGTTTTTTCCAAATAGTGAACCATGGTGTTCCTCTT
GAAGTTCTGGAAGAAGTCAAGGCTGCAACCTACCGCTTCTTTCGTTTACCTGCTGA
GGAAAAAATAAACACTGCAAGGATAATAGTCCTTCAAATAATGTGAGATACGGCAC
GTCTTTTACGCCTCATGCCGAAAAAGCACTAGAATGGAAAGATTTTCTGTCACTGTT
TTATGTATCTGACGAGGAAGCTGCAGCGCTGTGGCCTAGTGCATGTCGCGACGAG
GCCCTGACATTCATGCGGAATTGCGACGCGGTTATAAAACGGCTGTAAATCCCTT
ATGAAGGGGCTGAATGTTACTGAAATTGATGGGACGAAAGAGTCGTTACTGATGGG
CTCCAAACGTATCAATATGAATTATTATCCGAAATGTCCGAATCCGGAATTAACCGTG
GGTGTGTTGGTCGTCACAGTGATGTTTCGACACTGACGATCTTATTGCAAGATCAGATT
GGGGGTTTATATGTCCGTAAGTTGGACTCAGATACCTGGGTCCATGTTCTCCAATT

AATGGGGCTATCGTGATTAACGTCGGGGACGCCCTGCAGATACTGAGTAACGGTCTG
GTATAAATCTATTGAACATCGTGTTATTGCCAATGGTAGCAATAATCGAATTAGCGTAC
CCATATTTGTGAACCCGCGCCCAAATGATATTATTGGCCCTCTGCCTGAATTGCTCG
AATCTGGAGAAAAAGCGGTATATAAGAACGTTCTGTACAGTGATTATGTCAAACATTT
TTTTCGTAAAGCGCATGACGGTAAGGAAACCGTTGACTTTGCGAAAATCAATTAA

NHI 76 (F6'H1) protein sequence:

MPAVLSSVLSNITDFVVHEGNGVKGLADMGLEALPKQYVQPEERITTSTVIVDDTIPVI
DLSEWGSDPKVGDMICEAAEKWGFFQIVNHGVPLEEVKAATYRFFRLPAEEKKNKH
CKDNSPSNNVRYGTSFTPHA EKALEWKDFLSLFYVSDEEAAALWPSACRDEALTFMR
NCDAVIKRLKLSLMKGLNVTEIDGTKE SLLMGSKRINMNYYPKCPNPELTVGVGRHSDV
STLTILLQDQIGGLYVRKLDSDTWVHVPPINGAIVINVG DALQILSNGRYKSIEHRVIANG
SNNRISVPIFVNPRPNDIIGPLPELLESGEKAVYKNVLYSDYVKHFFRKAHDGKETVDFA
KIN

Codon-optimized NHI 77 (F6'H2) sequence:

ATGCCATCTACAACTTTAAGTACCGTTTTATCTGACATTAACGAATTTGTAGTTAAACA
GGGACACGGCGTAAAGGGGTTGTCTGGAGTTAGGGTTGCAGACGCTGCCAAACCAA
TATGTGCATCCGCCAGAAGAGAGACTGTCCTCGATGGATGTCGTTTCGGATGATAG
CATACTGTAAATTGATGTATCCAATTGGGAAGACCCAAAAGTAGCCAAGTTGATCTG
CGATGCCGCTGAAAAACGTGGCTTTTTTCAGATTGTAAATCACGGGATACCGTTGGA
AATGTTGGAGAAGGCTAAAGCCGCTACTTACCGCTTCTTTTCGCGAACCTGCAGAAG
AGAAGAAAAAATACTCCAAAGAGAATTGTCCTACCTCTCACGTTAGATACTCCACGA
GTTTCCTGCCTCAGATTGAAAAAGCGCTAGAATGGAAAGACCATCTGTCAATGTTCT
ATGTCAGTGACGAAGAAGCTGCACAATATTGGCCTCCGTCATGCCGCGATGATGCA
CTTGAATATCTGAAAAGTTGCGAAATGGTTTCGCGAAAGCTGCTGGAGGCATTAATG
CAGGGTCTGAATGTTAACGAGATCGACGACGCTAAAGAAAGCTTACTGATGGGTAG
TCGGCGCATCAATATCAATTATTATCCAAAATGCCCGAATCCGGACCTGACAGTGGG
CGTAGGTCGCCATAGTGATATATCCACATTAACACTGTTGCTTCAGGATGACATCGGT
GGGTTGTACGTTAGAAAACCTGGAGCACGAAGCGTGGAGCCATGTACCCCCAGTGA
AAGGTGCACTTGTGATTAACATTGGAGATGCTTTACAGATTATGAGTAATGGCAGATA
TAAGAGCATCGAGCATAGAGTTCTGGCCAACGAGACAAATGATCGGATTTCAGTTCC
GGTGTGTTGTTAACCCGAAACCTAACGACATAGTGGGACCGCTTCCGGAGGTGTTGG
CCTCGGGCGGAGAAGCCGGTATATAAACCGGTCCTCTATTTCGGATTATGCAAAACACT
TTTACCGCAAAGCTCATAATGGAAAGGATACCATTGCGTTTGCCAGAATTGAGTGA

NHI 77 (F6'H2) protein sequence:

MPSTTLSTVLSDINEFVVKQGHGVKGLSELGLQTLPNQYVHPPEERLSSMDVVSDDSI
PVIDVSNWEDPKVAKLICDAAEKRGFFQIVNHGIPLEMLEKAKAATYRFFREPAEEKKKY
SKENCPTSHVRYSTSFLPQIEKALEWKDHLSMFYVSDEEAAQYWPPSCRDDALEYK
SCEMVSRKLLLEALMQGLNVNEIDDAKESLLMGSRININYYPKCPNPD LTVGVGRHSDI
STLTLLLQDDIGGLYVRKLEHEAWSHVPPVKGALVINIGDALQIMSNGRYKSIEHRVLAN

ETNDRISVPVFNPKPNDIVGPLPEVLASGEKPVYKPVLYSDYAKHFYRKAHNGKDTIA
FARIE

Codon-optimized NHI 78 (HtyE) sequence:

ATGGCCATCACAACACTCGATTTTAATCAGTTCCGTAGTACATCGGCCGATGAGCGT
CAAATCTTTTTCGCTGATCTCTGCGAAACCCTTTCTGTTTATGGCTTCGCAAAAATAC
GTAATACGACGTTAAGTAATGAGTTGATCGATGAAATCTTTAAATACACCCGCTCATTT
TTTGCTCTTCCAAATGATATTAAAGCTAAAGCGAAACATCCAAATGCACCTAATCCGC
ACCGAGGCTGGTCAGCGATTGGCCAAGAACGAGTTTGGAAAATCAGTGGTTTCGA
GCAGAATAAGGAACGGACCGATTCTGATAATGAATTTTCGCGAATCCTTTGACCAGGG
TGCAGCGGACGATCAATTGTTCCCTAACAGATGGGTTGATGAAGATGATTTACCTGG
TTTTCAAGCTTTTCATGGAAGGATTCTATAAATCATGCGATGAACTGCATGCCCATTTAT
TACGTGCGATCTCGACGGGACTGAAATTACCAGATACACTGTTGCCCTCTAAACATC
GGCATAACACCTCCGAGCTGAGATTACTGCACTACCCGCCGATCCCTTGAGCGCA
CTGCGGAGCAATATGCGGATTGGCGAACATTCTGATTTTCGGCACTCTTACACTGCTT
CTGCAAGATAGTGTGGGTGGTTTACAGGTTGAAGATCAGAGAAATCCTCGCTCATTC
ATCCCTGTTGAGCCGGAAGACGGATATGAGGTTGTTATTAATATCGGCGACTGCCTG
CAGCGCTGGACCAACCGCCGCCTTTGTTCCGCCAATCACAGGGTGATGCTGCCTG
AAGGGAAAGATGTGGACAGCGAGGAGGTGTTGGATGATAGGTATAGCGTGGCATAT
TTCGGCAAACCGGACAGAGATGTGCTTGTGGATACCCTGCCAGAGTGTGTGGAAG
TGGGGGAACGCGTCGAGTATGGTGACCACCTCACTGCTCTCCAGTATAATCAAATTA
AGTTGACGCGCACTTATGGTTGA

NHI 78 (HtyE) protein sequence:

MAITTLDFNQFRSTSADERQIFCADLCETLSVYGFAKIRNTTLSNELIDEIFKYTRSFFAL
PNDIKAKAKHPNAPNPHRGWSAIGQERVWKISGFEQNKERTDSYNEFRESFDQGAAD
DQLFPNRWVDEDDLPGFQAFMEGFYKSCDELHAHLLRAISTGLKLPDILLPSKHRHNT
SELRLHYPPIPCSALRSNMRIGEHSDFGLTLLLQDSVGGGLQVEDQRNPRSFIPEPE
DGYEVVINIGDCLQRWTNRRLCSANHRVMLPEGKDVDSEEVLD DRYSVAYFGKPDRD
VLVDTLPECVEVGERVEYGDHLTALQYNQIKLTRTYG

Codon-optimized NHI 79 (AniF) sequence:

ATGGCCATTACTACGTTGGACTTCAACCAATTCCGCTCTACTTCCGCAGACGAGCGT
CAAAAATTTTTCGCGCGGATTTACGTGAGACGCTCTCAGTATACGGTTTCGCCAAAATT
CGGAATACTACCCTGTCTAACGAACTTATCGACGAGGTATTTAAATACACAAGATCCT
TTTTTGCACTCCCGGACGATATTAAAGCAAAGGCGAAACATCCTAAAGCACCGAACC
CCCACAGAGGTTGGAGTGCAGATTGGTCAGGAGAGAGTCTGGCGTATCTCGGGTTT
TGAGCAGAATAAGCAACGTACCGATTCTACAACGAATTTTCGCGAATCCTTTGATCA
GGGCGCAGCGGATGATCAGCTGTTCCCAAATAAATGGGTCGATGAAGATGATTTAC
CGGGGTTTTCGCGCTTTTCATGGAAGAGTTTATAAATCTTGCGATGAATTACACGCGC
ATCTGCTGCGCGCAATATCTACAGGACTGAAACTGCCGGATACACTGCTGCCAAGC
AAGCATCGTCATAATACCTCGGAAGTGCCTCTGGTTCATTATCCGCCGATTCCGTGT
AGCGCTCTTAGGAGCAATATGCGTATCGGTGAACATTCAGATTTTGGCACCCCTTACT

TTGCTTCTGCAAGACAGCGTTGGCGGCCTGCAGGTTGAGGATCAGCGGAATCCGG
GAAGCTTTATTCCCGTCGAGCCCGAGGATGGTTATGAAGTTGTTATCAATATCGGAG
ACTGCCTGCAGCGTTGGACCAATAGACGTCTTTGCTCCGCTAATCATCGCGTCAGG
TTACCTGAAGGAAAAGATGTGGAGTCCGATGAAATGTTGGACGATCGTTATAGCGTC
GCTTATTTCCGGGAAACCGGATCGTGATGTTTTGGTCGATACGCTGCCGGAGTGCCT
GGGGGTAGGGGAAAAGGTTGAGTATGGCGACCATTGACCGCCTTGCAATACAATC
AGCTTAAGCTGACGCGTACATACGGTTGA

NHI 79 (AniF) protein sequence:

MAITTLDFNQFRSTSADERQKFCADLRETLVYGF AKIRNTTLSNELIDEVFKYTRSFFA
LPDDIKAKAKHPKAPNPHRGWSAIGQERVWRISGFEQNKQRRTDSYNEFRESFDQGAA
DDQLFPNKWVDEDDLPGFRAFMEEFYKSCDELHAHLLRAISTGLKLPTLLPSKHRHN
TSELRLVHYPPIPCSALRSNMRIGEHSDFGTLTLLQLQDSVGGGLQVEDQRNPGSFIPVEP
EDGYEVVINIGDCLQRWTNRRLCSANHRVRLPEGKDVESEMLDDRYSVAYFGKPDR
DVLVDTLPECVGVGEKVEYGDHLTALQYNQLKLTRTYG

Codon-optimized NHI 80 (GA20ox1) sequence:

ATGAGTATGGTTGTTTCAGCAGGAGCAAGAAGTGGTTTTTGACGCCGCTGTGCTTTC
TGGCCAGACCGAAATACCTAGCCAGTTCATTTGGCCGGCGGAAGAAAGTCCAGGAT
CTGTTGCTGTGGAGGAACCTTGAGGTGGCATTGATTGATGTTGGTGCCGGCGCCGA
GCGGAGTAGCGTTGTGCGGCAAGTTGGTGAAGCATGTGAACGTCATGGTTTCTTTC
TTGTAGTGAACCATGGCATTGAAGCCGCGTTGCTTGAAGAGGCTCATCGTTGCATG
GACGCGTTCTTCACCCTCCCATAGGGGAAAAACAGCGGGCGCAGCGTAGAGCCG
GTGAATCTTGTGGATATGCGAGCAGTTTCACAGGACGCTTTGCGTCAAAATTGCCG
TGGAAGAAACTCTTAGCTTTGCTATTCTCTGCGGGGGATGAAGAAGGTGAAGA
GGGTGTGGGTGAGTACTTGGTTCGAAACTGGGCGCAGAACATGGGCGCCGGTTA
GGTGAAGTCTATTCGGTTATTGTCATGAGATGAGTAGACTGTCACTCGAACTGATG
GAAGTGCTGGGTGAAAGTCTCGGTATCGTGGGGGACCGACGGCACTATTTCCGTC
GTTTTTTTCAACGAAATGACTCTATTATGCGGCTTAATACTATCCAGCGTGCCAACG
CCCGCTGGATACACTGGGAACTGGCCCGCACTGTGATCCGACATCCCTCACGATTC
TGCATCAGGATCATGTAGGCGGGCTGGAGGTTTGGGCAGAAGGTAGATGGCGCGC
AATTCGTCCGCGCCCTGGCGCACTTGTCTGTAATGTTGGAGATACGTTTATGGCTCT
TAGTAATGCGAGGTATCGCAGTTGTCTGCATCGTGCCGTAGTTAATAGCACAGCACC
GCGTCGTAGCCTGGCATTTTTTCTTTGCCCGGAGATGGATACCGTGGTGGTCCCC
CAGAAGAACTCGTAGATGATCACCATCCCCGCGTGTATCCTGATTTACATGGAGAG
CACTCCTTGATTTCACTCAACGTCATTATCGTGCCGACATGCGAACCCTGCAGGCAT
TTAGCGATTGGCTTAATCATCACCAGACATTTGCAGCCAACAATCTATAGTTAA

NHI 80 (GA20ox1) protein sequence:

MSMVVQQEQEVVFDAAVLSGQTEIPSQFIWPAEESPGSVAVEEELEVALIDVGAGAERS
SVVRQVGEACERHGFLLVNVHIGIEAALLEEAHRCMDAFFTLPLGEKQRAQRRAGESC
GYASSFTGRFASKLPWKETLSFRYSSAGDEEGEEGVGEYLVRKLGAEHGRRLGEVYS
RYCHEMSRSLSELMEVLGESLGIVGDRRH YFRFFQRNDSIMRLNYYPACQRPLDTLG

TGPHCDPTSLTILHQDHVGGLEVWAEGRWRAIRPRPGALVVNVGDTFMALSNARYRS
CLHRAVVNSTAPRRSLAFFLCPEMDTVVRPPEELVDDHHPRVYPDFTWRALLDFTQRH
YRADMRTLQAFSDWLNHHRHLQPTIYS

Codon-optimized NHI 81 (D4H) sequence:

ATGCCGAAGTCATGGCCAATCGTGATATCTTCTCACTCATTTTGCTTTCTTCCTAACT
CTGAACAGGAACGTAAAATGAAGGATCTGAATTTCCATGCTGCCACCCTTTCAGAAG
AGGAGAGCTTAAGGGAACGTGAAAGCGTTTCGATGAAACCAAAGCCGGTGTGAAGGG
GATTGTTGATACTGGCATCACAAAAATTCCACGTATATTTATCGACCAGCCAAAAAAC
CTTGATAGAATTAGTGTGTGCCGTGGCAAATCTGACATCAAGATAACAGTGATTAACCT
TAAACGGACTTTTCATCGAACTCTGAGATTAGACGCGAGATCGTAGAAAAGATCGGTG
AGGCTTCTGAAAAATATGGTTTTTTCCAGATTGTTAACCATGGTATACCTCAGGACGT
GATGGACAAGATGGTTGACGGCGTTTCGTAAATTCCACGAACAAGATGATCAGATTAA
GCGTCAGTATTATTCCCGTGATCGCTTCAATAAAAATTTTCTCTACTCATCAAATTATG
TCCTCATCCCTGGGATTGCGTGTAAGTGGAGAGACACTATGGAATGCATTATGAATA
GCAATCAACCTGATCCGCAGGAGTTTCCAGATGTCTGTAGGGATATTCTTATGAAATA
TAGTAACCTACGTACGTAATCTGGGTCTGATCCTTTTCGAATTGTTGTCCGAAGCGCT
GGGCCTCAAGCCAAATCATCTCGAAGAAATGGATTGTGCCGAAGGTTTAATTCTTTT
GGGGCATTACTATCCGGCTTGTCTCAGCCTGAACTTACATTTGGTACCTCCAAGCA
CTCTGATAGCGGCTTTCTGACTATTCTAATGCAGGATCAGATTGGAGGCCTTCAGAT
CCTGCTTGAAAACCAATGGATAGACGTGCCGTTTATACCTGGCGCATTAGTCATCAA
TATTGCAGATCTGTTGCAGCTGATACCAATGATAAGTTTAAGTCAGTGGAGCATCG
GGTGTGGCTAACAAGGTCCGTCCTCGTATTTTCAGTAGCAGTCGCCTTCGGCATAA
AAACCCAGACACAGGAAGGTGTATCTCCACGCCTCTATGGTCCTATTAAAGAGTTAA
TTAGCGAAGAAAATCCACCTATCTACAAGGAGGTAACCGTCAAAGACTTCATTACTAT
ACGCTTTGCAAACGCTTCGATGATTCAAGCTCTCTTAGTCCGTTCCGTTTAAACAA
CTGA

NHI 81 (D4H) protein sequence:

MPKSWPIVISSHSFCFLPNSEQERKMKDLNFHAATLSEESLRELKAFDETKAGVKGIV
DTGITKIPRIFIDQPKNLDRISVCRGKSDIKIPVINLNLSSNSEIRREIVEKIGEASEKEYGF
FQIVNHGIPQDVMMDKMVDGVRKFHEQDDQIKRQYYSRDRFNKNFLYSSNYVLIPGIAC
NWRDTMECIMNSNQPDQPQEPDVCRDILMKYSNYVRNLGLILFELLSEALGLKPNHLE
EMDCAEGLILLGHYYPACPQPELTFGTSKHSDSGFLTILMQDQIGGLQILLENQWIDVPF
IPGALVINIADLLQLITNDKFKSVEHRVLANKVGPRISVAVAFGIKTQTQEGVSPRLYGPIK
ELISEENPPIYKEVTVKDFITIRFAKRFDSSSLSPFRLNN

Codon-optimized NHI 82 (PcbC *P. chrys.*) sequence:

ATGGCAAGCACCCCGAAGGCGAACGTCCCGAAAATTGATGTGTCACCGCTTTTTTGG
TGACAATATGGAAGAAAAAATGAAGGTGCGCGCTGCAATAGATGCAGCGAGTAGGG
ACACTGGATTCTTTTATGCCGTTAACCATGGGGTAGACGTAAAACGATTGTCTGAATAA
GACCAGAGAATTTCAATTTCTCCATTACTGATGAAGAAAAATGGGACCTGGCAATCCG
GGCCTATAATAAAGAGCATCAGGACCAGATACGTGCCGGGTATTATCTGTCTGATTCC

AGAGAAGAAGGCGGTGGAATCATTCTGTTATTTAAATCCTAATTTTAAGCCTGATCAT
CCTTTGATACAGAGCAAGACCCCAACCCACGAAGTAAATGTGTGGCCTGACGAAAA
GAAACATCCGGGGTTTCGTGAATTCGCGGAACAATACTACTGGGACGTTTTTCGGTC
TGAGCTCAGCTTTGTTGAGAGGATATGCGTTAGCACTTGGGAAGGAAGAGGATTTTC
TTTAGTCGTCATTTCAAGAAAGAGGACGCGTTATCTTCTGTCGTCCTGATTAGATATC
CATACCTCAATCCGATTCCCCCTGCCGCTATAAAAAACAGCTGAGGACGGTACCAAGT
TGTCCTTTGAATGGCATGAAGATGTGTCGTTGATCACTGTCCTATATCAGAGTGACG
TTGCAAATCTTCAGGTAGAGATGCCGCAGGGCTATTTAGATATCGAAGCAGACGATA
ATGCATATTTAGTTAACTGCGGCTCTTACATGGCGCACATTACCAATAATTATTATCCT
GCCCCAATTCATCGAGTGAAGTGGGTAAACGAAGAAAGACAGTCCCTTCCTTTCTTT
GTCAATCTAGGTTTTAATGATACGGTTCAGCCGTGGGACCCCTCCAAAGAAGATGG
GAAGACTGATCAGCGGCCGATCTCATATGGAGATTATTTACAGAATGGTCTGGTTAG
CCTCATAAATAAAAATGGCCAGACTTAA

NHI 82 (PcbC *P. chrys.*) protein sequence:

MASTPKANVPKIDVSPLFGDNMEEKMKVARAIDAASRDTGFFYAVNHGVDVKRLSNKT
REFHFSITDEEKWDLAIRAYNKEHQDQIRAGYYLSIPEKKAVESFCYLNPNFKPDHPLIQ
SKTPTHEVNVWPDEKKHPGFREFAEQYYWDVFLSSALLRGYALALGKEEDFFSRHF
KKEDALSSVVLIRYPYLNPIPPAAIKTAEDGTKLSFEWHEDVSLITVLYQSDVANLQVEM
PQGYLDIEADDNAYLVNCGSYMAHITNNYYPAPIHRVKWVNEERQSLPFFVNLGFNDT
VQPWDPSKEDGKTDQRPISYGDYLNGLVSLINKNGQT

Codon-optimized NHI 83 (PcbC *A. lactam.*) sequence:

ATGAAGATGCCTTCAGCGGAAGTTCCGACAATAGACGTTAGCCCTCTGTTCCGAGA
TGATGCTCAAGAAAAAGTTTCGTGTTGGACAGGAGATTAATAAAGCCTGCCGCGGGT
CAGGTTTTTTTTTATGCAGCAAACCATGGTGTGATGTGCAACGACTGCAAGATGTAG
TCAACGAATTCATCGAACTATGTCACCGCAAGAAAAATATGATTTGGCTATACATGC
CTACAACAAAAATAACAGCCATGTACGCAACGGTTACTACATGGCCATTGAAGGTAA
AAAAGCGGTGCAATCCTTCTGCTACTTAAATCCATCCTTCAGTGAAGATCACCCCGA
GATTAAGGCCGGTACCCCGATGCATGAGGTAAACAGTTGGCCGGACGAAGAAAAG
CATCCTTCTTTTACAGGCCGTTTTGCGAGGAATACTACTGGACTATGCATCGCCTATCTA
AGGTATTGATGCGTGGCTTCGCACTTGCACCTCGGTAAAGACGAAAGGTTCTTTGAG
CCCGAACTGAAAGAAGCTGATACATTGTCTTCAGTGTGCTCATTCCGGTATCCGTAC
CTGGAAGATTACCCTCCTGTAAAGACCGGCCCGGACGGTGAAAACTGTCATTTGA
GGATCATTTTGACGTATCTATGATTACGGTATTATACCAGACACAGGTTTCAGAATTTAC
AAGTGGAACGGTTGATGGATGGCGGGATCTACCTACAAGCGATACCGATTTTTTAG
TGAATGCTGGCACTTATCTGGGTCATCTTACTAATGACTACTTTCCGAGTCCATTACA
TAGAGTCAAGTTTGTTAACGCAGAACGTCTCTCCCTGCCTTTTTTCTTTTCATGCCGG
TCAGCATACTCTCATTGAACCATTTCTCCCCGATGGCGCGCCGGAAGGCAAGCAGG
GAAACGAAGCTGTTTCGTATGGTGATTATCTTAATCACGGATTGCATAGCCTGATCGT
GAAGAACGGTCAAACCTTAA

NHI 83 (PcbC *A. lactam.*) protein sequence:

MKMPSAEVPTIDVSPLFGDDAQEKVRVGQEINKACRGS GFFYAANHGV DVQRLQDVV
NEFHRTMSPQEKYDLAIHAYNKNNSHVRNGYYMAIEGKKAVESFCYLNPSFSEDHPEI
KAGTPMHEVNSWPDEEKHPSFRPFCEEYYWTMHRLSKVLMRGFALALGKDERFFEP
ELKEADTLSSVSLIRYPYLEDYPPVKTPDGEKLSFEDHFDVSMITVLYQTQVQNLQVE
TVDGWRDLPTSDTDFLVNAGTYLGHLTNDYFPSPLHRVKFVNAERLSLPFFFHAGQHT
LIEPFFPDGAPEGKQGNEAVRYGDYLNHGLHSLIVKNGQT

Codon-optimized NHI 84 (PcbC *L. lactam.*) sequence:

ATGAACCGTCATGCGGATGTCCCAGTGATTGACATCAGCGGTCTGTCGGGTAACGA
CATGGATGTAAAAAAGATATTGCAGCACGTATCGATCGCGCCTGCCGTGGTTCCG
GCTTTTTTTATGCCGCTAATCATGGTGTTCGATCTGGCGGCACTCCAAAAATTTACGA
CTGATTGGCATATGGCAATGTCCCCAGAGGAAAAATGGGAGTTGGCAATTCGCGCA
TATAATCCGGCCAATCCACGGAACCGGAATGGTTACTACATGGCTGTTGAAGGCAA
GAAAGCAAATGAGTCATTTTGTACTTAAATCCATCCTTTGATGCCGATCACGCCACT
ATCAAGGCTGGCCTCCCCTCCCATGAGGTGAACATCTGGCCAGATGAGGCACGCC
ATCCAGGGATGAGGCGATTTTATGAGGCGTACTTTTCAGACGTATTTGACGTAGCAG
CTGTCATCCTACGGGGATTTGCGATCGCGCTGGGTCGCGAAGAATCATTCTTTGAA
AGACATTTTTCCATGGATGATACCCTGAGCGCGGTTTCACTTATTAGGTACCCGTTTC
TGGAAGAACTACCCGCCGCTGAAACTGGGTCCAGATGGAGAAAAATTAAGCTTTGAA
CATCATCAAGACGTATCGCTGATAACCGTATTATATCAAACCGCAATTCCAAATCTGC
AGGTGGAAACCGCTGAAGGTTATCTTGACATTCCCGTATCCGATGAACACTTTCTAG
TTAATTGCGGCACTTACATGGCACATATTACAAACGGGTATTATCCAGCTCCAGTCCA
TCGCGTGAAATACATTAAACGCAGAGCGGCTGTCCATTCCCTTTTTTGCAAACCTAAG
TCATGCTAGCGCGATAGACCCATTTGCTCCACCGCCTTATGCTCCAGCTAGGGGAA
ATCCAACGGTGTCTTACGGTGATTACTTACAGCATGGTCTCCTTGACTTAATTGCGG
CTAACGGTCAGACATGA

NHI 84 (PcbC *L. lactam.*) protein sequence:

MNRHADVPVIDISGLSGNDMDVKKDIAARIDRACRGS GFFYAANHGV DLAALQKFTTD
WHMAMSPEEKWELAIRAYNPANPRNRNGYYMAVEGKKANESFCYLNPSFDADHATIK
AGLPSHEVNIWPDEARHPGMRRFYEAYFSDVFDVAAVILRGFAIALGREESFFERHFMS
DDTLSAVSLIRYPFLENYPPLKLGPDGEKLSFEHHQDVSLITVLYQTAIPNLQVETAEGYL
DIPVSDEHFLVNCGTYMAHITNGYYPAVHRVKYINAERLSIPFFANLSHASAIDPFAPPP
YAPARGNPTVSYGDY LQHGLLDLIRANGQT

Codon-optimized NHI 85 (IpnA) sequence:

ATGGGGAGTGTTTCGAAGGCGAACGTGCCTAAAATTGACGTAAGTCCGCTGTTCCG
CGATGATCAAGCAGCAAAAATGCGAGTGGCTCAGCAGATCGATGCAGCGAGCCGT
GATACGGGCTTCTTTTACGCAGTGAATCATGGCATTAAACGTTTACGCTCTGAGTCAG
AAAACGAAAGAGTTTCATATGTCCATAACGCCGGAGGAAAAGTGGGATCTGGCTATC
CGTGATATAATAAGGAACATCAGGATCAGGTTTCGAGCCGGTTATTATTTGTCTATAC
CAGGTAAGAAAGCAGTGGAAGTTTTTGTACCTGAACCCGAATTTTACACCGGAC
CATCCGAGAATTCAAGCTAAGACGCCGACTCATGAAGTCAATGTGTGGCCGGATGA

GACCAAACATCCTGGTTTTTCAGGACTTTGCCGAACAGTATTATTGGGACGTTTTTGG
CCTCTCGTCTGCACTGCTGAAAGGCTATGCACTCGCACTGGGCAAAGAGGAAAAC
TTCTTTGCGCGCCATTTCAAACCAGATGACACGCTGGCAAGCGTGGTTTTAATCCG
ATACCCATATCTGGACCCTTATCCTGAAGCAGCGATTAAAACCGCAGCGGACGGTAC
CAAACCTTAGCTTTGAATGGCATGAAGATGTCTCACTGATAACTGTGCTTTATCAGAG
CAACGTCCAGAATTTACAGGTGGAAACAGCTGCCGGATATCAAGATATTGAGGCGG
ATGATACCGGGTACCTGATTAATTGTGGAAGTTACATGGCCACCTGACCAATAATTA
TTACAAAGCTCCCATACATAGAGTTAAATGGGTAAATGCAGAACGTCAGTCGTTACCA
TTTTTCGTTAATTTAGGCTATGACTCTGTCATTGATCCTTTTGATCCCCGTGAACCAA
ATGGGAAAAGCGATCGTGAACCTTTATCATATGGTGACTATCTCCAAAATGGGCTGG
TTAGCCTGATTAATAAGAACGGTCAGACATAA

NHI 85 (IpnA) protein sequence:

MGSVSKANVPKIDVSPLFGDDQAAKMRVAQQIDAASRDTGFFYAVNHGINVQRLSQKT
KEFHMSITPEEKWDLAIRAYNKEHQDQVRAGYYLSIPGKKAVESFCYLNPNFTPDHPRI
QAKTPTHEVNVWPDETKHPGFQDFAEQYYWDVFLSSALLKGYALALGKEENFFARH
FKPDDTLASVVLIRYPYLDPYPEAAIKTAADGTKLSFEWHEDVSLITVLYQSNVQNLQVE
TAAGYQDIEADDTGYLINCOSYMAHLTNYYKAPIHRVKWVNAERQSLPFFVNLYGYS
VIDPFDPREPNGKSDREPLSYGDYLNGLVSLINKNGQT

Codon-optimized NHI 86 (CefEF A. chrys.) sequence:

ATGACAAGCAAAGTGCCGGTGTTTAGACTGGATGATTTAAAAAGTGGTAAAGTCCTC
ACTGAACTTGCCGAAGCAGTCACCACCAAAGGCATCTTTTATCTTACCGAAAGCGG
ATTAGTCGACGACGACCATACTCCGCTCGTGAAACATGTGTGGATTTTTTCAAAA
TGGTTCCGAGGAAGAAAAACGCGCAGTTACCCTGGCGGATCGAAATGCCAGAAGA
GGATTTTCAGCTTTAGAATGGGAGAGTACTGCCGTGGTTACAGAGACTGGTAAATAT
TCAGACTACTCGACCTGCTACTCAATGGGAATTGGTGGTAATTTGTTCCCGAATCGT
GGCTTTGAGGATGTGTGGCAGGACTATTTTGATCGTATGTACGGCGCAGCGAAAGA
TGTTGCCAGGGCAGTATTAATAGTGTTGGCGCACCGCTTGCCGGTGAAGATATTG
ATGATTTTGTGCAATGTGACCCACTACTGCGCCTGCGGTACTTTCCGGAAGTTCCC
GAGGATCGTGTAGCGGAAGAAGAGCCGCTGCGTATGGGACCTCATTATGACCTTTC
TACGATAACCTTGGTCCATCAGACCGCATGTGCAAACGGTTTTGTAAAGCCTTCAGTG
CGAAGTCGATGGTGAGTTTGTGGATTTGCCAACGCTGCCAGGTGCTATGGTCGTAT
TTTGTGGCGCCGTTGGCACCCCTCGCAACCGGTGGGAAAGTTAAAGCTCCTAAACAT
CGCGTTAAATCACCGGGCCGTGATCAGAGAGTTGGGTCCAGTAGAACGTGCGAGTG
TATTTTTTCTTCGTCTAAACCAGACTTCAGTTTTAATGTCCAACAGTCACGCGAGTG
GGGGTTTAACGTCCGCATTCCGTCCGAACGTACGACCTTTAGAGAATGGCTTGGCG
GCAATTATGTTAACATGCGGAGGGATAAGCCGGCGGCGGCGGAAGCTGCTGTCCC
CGCCGCGGCCCCAGTGAGTACCGCTGCTCCCATAGCCACTTAA

NHI 86 (CefEF A. chrys.) protein sequence:

MTSKVPVFRLLDLKSGKVLTEAEAVTTKGIFYLTESGLVDDDHSTARETCVDFFKNGS
EEEKRAVTLADRNRARGFSALEWESTAVVTETGKYSDYSTCYSMGIGGNLFPNRGFE

DVWQDYFDRMYGAAKDVARAVLNSVGAPLAGEDIDDFVECDPLLRLRYFPEVPEDRV
AEEEEPLRMGPHYDLSTITLVHQTACANGFVSLQCEVDGEFVDLPTLPGAMVVFCGAVG
TLATGGKVKAPKHRVKSPGRDQRVGSSRTSSVFFLRPKPDFSFNVQQSREWGFNVRI
PSERTTFREWLGGNYVNMRRDKPAAAEAAVPAAAPVSTAAPIAT

Codon-optimized NHI 87 (CefE *S. clav.*) sequence:

ATGGACACGACTGTTCCAACTTTTTCTTGGCCGAACTTCAACAGGGCCTGCATCA
GGACGAATTCCGTCGATGCCTTCGTGACAAAGGGCTGTTCTATTTGACCGATTGTG
GTCTGACGGACACCGAGCTGAAAAGTGCAAAAGATCTCGTAATTGACTTTTTTGAA
CATGGAAGCGAGGCAGAAAAGCGCGCAGTGACGAGCCCGGTACCTACAATGCGTC
GGGGTTTCACAGGTCTTGAAAGCGAGTCTACTGCTCAGATCACCAACACAGGCTCA
TATTCAGATTACTCGATGTGCTACTCCATGGGTACTGCCGATAATCTTTTTCCATCTG
GAGATTTTGAGCGCATTGGAAGTCTGATCGTCAGTACACAGCGTCTCGCG
CCGTGGCAGGGAAGTTCTACGTGCGACAGGCACGGAGCCTGACGGCGGTGTAG
AAGCATTTTTGGATTGTGAACCACTGCTCAGATTTGCTATTTTCTCAGGTCCCGG
AGCATCGGAGTGCCGAGGAGCAGCCGTTGCGCATGGCCCCCATTATGATCTGTC
AATGGTTACGCTTATTCAGCAGACACCGTGTGCTAACGGATTCTGATCTTTACAGGC
CGAGGTTGGTGGTGCCTTACCGACCTTCCCTACCGCCCCGATGCTGTCTTAGTCT
TCTGTGGGGCAATTGCAACACTTGTAAACCGGTGGCCAGGTTAAAGCGCCGAGACAT
CACGTGCTGCGCCCCGTGCGGATCAAATCGCTGGAAGTAGTAGGACATCTAGCGT
CTTTTTTCTGCGTCCGAATGCCGATTTTACATTCTCAGTGCCACTCGCAAGGGAATG
TGGCTTTGATGTATCTCTGGACGGTGAAACCGCAACATTTCAAGATTGGATTGGTGG
CAACTACGTGAACATCCGCCGAACGTCGAAAGCCTGA

NHI 87 (CefE *S. clav.*) protein sequence:

MDTTVPTFSLAELQQGLHQDEFRRCLRDKGLFYLTDCGLDTELKSAKDLVIDFFEHGS
EAEKRAVTSPVPTMRRGFTGLESESTAQITNTGSYSDYSMCYSMGADNLFPSPGDFE
RIWTQYFDRQYTASRAVAREVLRATGTEPDGGVEAFDCEPLLRFYFPQVPEHRSAE
EQPLRMAPHYDLSTMVTLIQQTPCANGFVSLQAEVGGAFDLPYRPDAVLVFCGAIATLV
TGGQVKAPRHHVAAPRRDQIAGSSRTSSVFFLRPNADFTFSVPLARECGFDVSLDGET
ATFQDWIGGNYVNIRRTSKA

Codon-optimized NHI 88 (CefE *A. lactam.*) sequence:

ATGACGGATGCAACAGTTCCGACCTTCGATCTGGCAGAACTGCGTGAAGGTCTGC
ATCAGGAGGAGTTTCGGCATTGTCTGCGCGAGAAAGGAGTGTTTTACCTGAAAGGC
ACAGGTCTGCCGGCTGAAGCAGATCATGCCTCTGGTCGCGAAATTGCGGTAGATTT
TTTCGATCATGGCACGGAAGCCGAAAAAAGCTGTTATGACTCCAATCCCGACGA
TTCGTCTGGATATGCTGGTCTTGAAAGTGAATCGACAGCGCAGATCACAAACACC
GGCAAATATACGGATTACAGTATGAGTTATTCCATGGGGACGGCCGATAACCTGTTT
CCATCTGCCGAGTTTGAGAAAGCGTGGGAAGACTATTTTGC GCGCATGTATCGCGC
GTCTCAAGATGTAGCGCGTCAAGTTCTGACATCAGTTGGCGCCGAACCCGAAGTC
GGCATGGACGCGTTCCTGGATTGTGAACCGCTGCTCAGGTTACGTTATTTCCCTGA
AGTGCCGGAAGATCGCGTAGCGGAGGAACAGCCTCTTCGGATGGCCCCACATTAT

GACTTGTCCATCGTGACATTGATCCACCAGACACCCTGTGCTAATGGCTTCGTGAG
TCTTCAAGTTGAGGTGGATGGGAGCTACGTTGACATTCCGGCGCAACCAGGAGCC
GTCCTTGTATTTTTCGGCGCTGTAGCGACCCTTGTGGCCGATGGTGCATCAAGGC
GCCGAAACATCATGTCGCGGCACCGGGCGCTGACAAACGAGTGGGCAGCAGTCG
TACCAGTTCAGTTTTTTTCTGCGTCCGAATGGTGATTTTCGGTTCAGTGTGCCGCG
AGCACGAGAATGTGGCTTTGACGTTTCTATCCCGGCAGAGACGGCCACCTTTGATG
ATTGGATAGGTGGGAACATATAAATATCCGTAAAACCGCTGCCGCCCGTTAA

NHI 88 (CefE A. lactam.) protein sequence:

MTDATVPTFDLAELREGLHQEEFRHCLREKGVFYLKGTGLPAEADHASGREIAVDFFD
HGTEAEKKAVMTPIPTIRRGYAGLESESTAQITNTGKYTDYSMSYSMTADNLFPSAEF
EKAWEDYFARMYRASQDVARQVLTSGAEPEVGMDFLDCEPLLRLRYFPEVPEDRV
AEEQPLRMAPHYDLSIVTLIHQTPCANGFVSLQVEVDGSYVDIPAQPGAVLVFCGAVAT
LVADGAIKAPKHHVAAPGADKRVGSSRTSSVFFLRPNGDFRFSVPRARECGFDVSIPA
ETATFDDWIGGNYINIRKTAAR

Codon-optimized NHI 89 (H6H) sequence:

ATGGCCACTTTTGTGAGCAATTGGAGCACGAAAAGTGTCTCTGAATCATTATTGCT
CCCTTGCAGAAACGCGCGGAGAAAGGATGTGCCGGTGGGAAATGATGTGCCTATAAT
AGATCTGCAGCAACATCATCACCTTCTCGTCCAACAGATAACCAAGGCCTGTGAGG
ATTTTGGGCTTTTTCAGGTTATCAATCACGGCTTCCCGGAGGAACTGATGCTGGAAA
CAATGGAGGTTTGCAGGAGTTTTTTCGCATTACCGGCAGAAGAAAAAGAAAAGTTC
AAACCAAAAGGTGAAGCAGCTAAATTTGAGTTGCCGCTTGAGCAGAAAGCTAAACT
TTATGTAGAAGGTGAGCAGCTTAGTAACGAGGAGTTTTTGTACTGGAAAGATACCCT
GGCCCATGGTTGCCACCCCTTAGATCAAGATTTGGTTAATTCCTGGCCGGAGAAAC
CGGCAAAATACCGTGAGGTGGTCGCCAAATATAGCGTTGAAGTCCGTAAGCTGACG
ATGAGAATGCTAGACTATATTTGTGAAGGCTTAGGCTTAAATAGGGTATTTTGATAA
TGAAGTGTCTCAGATCCAGATGATGCTGACTAATTATTATCCGCCCTGTCCGGATCC
TTCGTCAACGCTGGGCAGCGGGGGACATTACGATGGGAACTTGATAACACTGCTG
CAACAAGACCTTCCAGGGCTGCAGCAACTGATTGTCAAAGACGCAACGTGGATCG
CTGTGCAACCTATTCCGACTGCATTTGTGGTAAACCTTGGACTGACATTAAGTTAT
AACTAACGAAAAATTTGAAGGCTCTATCCATAGAGTTGTGACTGACCCGACTCGTGA
TCGGGTGAGCATTGCAACATTAATTGGGCCCGATTATAGTTGTACGATCGAGCCGGC
AAAAGAGCTGTTGAATCAGGACAATCCACCGTTATATAAGCCGTATTCCTATTCAGAA
TTTGCAGATATTTATTTAAGCGATAAAAGTGATTACGATTCAGGTGTCAAACCTTATAA
AATTAATGTATGA

NHI 89 (H6H) protein sequence:

MATFVSNWSTKSVSESFIAPLQKRAEKDVPVGNDVPIIDLQQHHHLLVQQITKACQDFG
LFQVINHGFPPEELMLETMEVCKEFFALPAEEKEKFKPKGEAAKFELPLEQKAKLYVEGE
QLSNEEFLYWKDTLAHGCHPLDQDLVNSWPEKPAKYREVVAKYSEVRKLTMRMLDYI
CEGLGLKLGYFDNELSQIQMMLTNYPPCPDPSSSTLGSGGHYDGNLITLLQQDLPLQL

QLIVKDATWIAVQPIPTAFVVNLGLTLKVITNEKFEGSIHRVVDPTRDRVSIATLIGPDYS
CTIEPAKELLNQDNPPPLYKPYSYSEFADIYLSDKSDYDSGVKPYKINV

Codon-optimized NHI 90 (CefF *S. clav.*) sequence:

ATGGCAGACACACCGGTTCCGATTTTTAATCTCGCTGCACTCCGTGAGGGCGCTGA
TCAAGAAAAATTTCCGGAATGTGTGACAGGTATGGGCGTATTTTATCTGACAGGTTAT
GGTGCTGGCGATAAAGATCATAGACTGGCTACCGATACGGCGATGGATTTTTTTGCG
AATGGGACTGAGGCAGAAAAGGCGGCAGTGACGACGGATGTTCCAACCATGCGTA
GAGGTTATTCAGCTCTTGAAGCAGAATCCACCGCCCAGGTTACTCGCACAGGAAGC
TATACAGATTATTCTATGTCGTTTTCTATGGGAATTTCTGGGAACGTGTTTCCATCGC
CTGAATTCGAGAGAGTGTGGACTGAGTATTTGATAAGTTATACGCAGCGGCCCAA
GAAACCGCCCGTCTGGTATTAAGTGCCTCCGGCGGGTACGACGCTGAAATTGTGG
GATCTCTGGATGAATTGCTGGATGCCGATCCGGTCCTCCGGCTGCGCTATTTTCCG
GAAGTGCCCGAACATCGATCCGCGGAACATGAACCTAGACGCATGGCCCCACACT
ACGACCTGTCTATAATAACTTTTATTCATCAAACCCCGTGTGCGAATGGTTTTGTGAG
CCTGCAGGCAGAGATTGGAGGTGAAGTAGTGTCTTTACCTGTGGTTGAGGATGCTG
TTGTGGTTATGTGCGGAGCTATGGCACCGCTGGCCACCCAGGGTGCCCTGCCCGC
GCCCCGCCATCACGTCCGCTCTCCTGGTGCCGGTATGCGTGAAGGATCGGATCGT
ACAAGTTCGCTGTTTTTCTTCGACCAACGACAGATTTTTCGTTCTCAGTAGCAAAA
GCCCGTTCTACGGTTTGGCTGTGGATCTTGATATGGAGACGGCGACTTTTCGGGGA
CTGGATCGGAACGAAGTATGTCACAATGCACGCGAAGAATGAGCCACAGGCGGGT
TAA

NHI 90 (CefF *S. clav.*) protein sequence:

MADTPVPIFNLAALREGADQEKFRECVTGMGVFYLTGYGAGDKDHLRLATDTAMDFFA
NGTEAEKAAVTDDVPTMRRGYSALEAESTAQVTRTGSYTDYSMSFSMGISGNVFPSP
EFERVWTEYFDKLYAAAQETARLVLTASGGYDAEIVGSLDELLDADPVLRLRYFPEVPE
HRSAEHEPRRMAPHYDLSIITFIHQTPCANGFVSLQAEIGGELVSLPVEDAVVVMCGA
MAPLATQGALPAPRRHHVRSPGAGMREGSDRTSSVFFLRPTTDFSFSAKARSYGLAV
DLDMETATFGDWIGTNYVTMHAKNEPQAG

Codon-optimized NHI 91 (Trt7) sequence:

ATGACTGGTCAAGCGGAAGCAATACGGCGCGTTACCCGACCGTAAGCCCCAAAAC
AGGCGGCCCCAGATGCTTCAGGAAGACGGTGTAATCATCCTTAAATCATTTCTGGCG
CCAGATGTTATGCAGCGCTTCCAGGCGGAGGTGGATGAAGATGTCGAAAAAACTTC
AACAGGTGCGCGTATGAAAGCGTATAAATTAGTTAATGATAAGACCAACATATGGCA
GATTTAATTGTTTCGTTCCGAAGTTTTTAGATCTGATATTCTGACACATCCCTTGACCA
TGCAATAGCGGATGAACTCTTCCGAGCAGACTACGGGGATCATTGGCTTAACGCTT
CGGCAGTGCTTCAGTTGATGCCAGGCGCGCCCGCGCAGCAGCTCCATAGAGACG
AAGAAATATTTGCCGCCTCCAAATTCGTTCTCCAACCGACCCTCAGTTAAGCCTGA
GTTGTCTGGTAGCACTTACTGAATTTACTGAAGAAAATGGTGCAACACGCCTGATTC
CCGGTTCCCATCTTTGGGATTCGGCCCATCCCGCACCTTCCCCGGACCAGACCGT
TCCGGCCATAATGCAGCCTGGCGAAGCAATTCTATTTCTCGGTTCTCTGTTTCACGG

AGGAGGCGAGAACCGGACTGAAAATGTGAGACGGGGACTTGGGATGTCATTAATT
CCTTGCCAGTTTACCCCCTATTCCTCTCATATGCATGTGCCCCGCACCATTATCGAAA
CCATGACACCGCTGGCTCAGAAATTGGTTGGCTGGCGGACCGTTGAATCACACCG
ACAGTATCCTTTTTTGGCAGGGAGGTGATCGCCGACTGGAGGATGTGCTTGGGTTAG
CCAGCCGCGAGGCCTAA

NHI 91 (Trt7) protein sequence:

MTGQAEAIRRVHPTVSPKQAAQMLQEDGVILKSFLAPDVMQRFQAEVDEDVEKTSTG
ARMKAYKLVNDKTKHMDLIVRSEVFRSDILTHPLYHAIADLFRADYGDHWLNASAVL
QLMPGAPAAQLHRDEEIFAASKFRSPTDQLSLSCLVALTEFTEENGATRLIPGSHLWD
SAHPAPSPDQTVPAIMQPGAILFLGSLFHGGGENRTENVRRGLGMSLIPCQFTPYSS
HMHVPRTIETMTPLAQKLVGWRTVESHRQYPFWQGGDRRLDVLGLASREA

Codon-optimized NHI 92 (VldW) sequence:

ATGACCGGTTTCAGTCCCCATAGTGGATCTTGAGGCATGGCGCGCGGCTGACGAAG
AAAACCGTGTCATCCCTGGCAGAGATCATTGACGGAGCGTTGCACACAGTTGGAAC
ATTTCTGCTGGCAGGGCATGGTGTTCCTGCAGAACTCACCGCGCGCATGCGTACC
GCGGGTAGAAGTTTTTTCGATCTGCCTTGGGAAAAAAAAGAACCACACGCGGTTCA
ACGACCACACGACAACGGATGGCGTGGCCTGGTTAAACACCGTACCGATACCATAG
AAGGTACCGGAGGAGCGCCCGACCTGCATGAAGCTTTTCACATGGGGCCTACACA
CCGCACGGGGCGACGATGCATTTGATGCGCTGTACTATCCGGCCAATAAGTGGCCG
GCCGAGCTGCCGGAAGTGAAGAGAACTGCCCTTGCATACACCGCACATATGACGC
GGGTAGCAGGTGCTGTAATGGAGATGCTAGCGGGCGTTCTTGGTCTGGAACCGGC
TTTTTTTACATCTCGCTGCGAGCATGCCACTTGGACCCAGTCAGTGAAGTGGTATCC
GTCCTTGGATACGGTGGGGCAGACCGCAGAAGGTCAGATGCGAGTAGGTCCCCAC
ACTGATTTTGAACAATTACGCTGTTGGATCGACAACAAGGAGTATCTGGCTTAGAA
GTGTGGTCAGAAGAGGACGGATGGTTTGCACCGCCATTTGTGGAAGGTACCTTACT
GGTTAACCTCGGCGATCTGATGCATCAGTGGACAGATGGCCGCTGGCGCAGTCTA
CGTCATCGGGTACTTGCACCGAGTGCCTCGGCGCCACAGGAAGAGCTGGTTAGCC
TGGTATACTTCTTCGATGCGGATCCGGAGGCTGAACTTGTTCCGCTTGCGGCTCCT
GTAGGGGGAGGGGCGGCATGCCTACCGTGAATGTTGGCGAAACCATTCTGAAAA
AGAATATACAAATGTTGACTGATTTGAAAGGTCACGGGCTTTTCCAGGGTGAGCTTT
CGTTATCGCGTCCTGGAAGCGCTGATTCACCCGGTTCTTCACCTGCCGATGATCAT
CCTTCACGTCCTGGTCGTCACCCGGCTCAAGGGCCACAATAA

NHI 92 (VldW) protein sequence:

MTGSVPIVDLEAWRAADEENRASLAEIIDGALHTVGTFLLAGHGVPAELTARMRTAGRS
FFDLPWEKKEPHAVQRPHDNGWRGLVKHRTDTIEGTGGAPDLHEAFHMGPTHRTGD
DAFDALYYPANKWPAELPELRETALAYTAHMTRVAGAVMEMLAGVLGLEPAFFTSRCE
HATWTQSVNWYPSLDTVGGQTAEGQMRVGPHTDFGTITLLDRQQGVSGLEVWSEEDG
WFAPPFVEGTLLVNLGDLMHQWTDGRWRSRLRHRVLAPSASAPQEELVSLVYFFDADP
EAEVPLAAPVGGGAGMPTVNVGETILKKNIQMLTDLKGHGLFQGELSLSRPGSADSP
GSSPADDHPSRPGRHPAQGPQ

Codon-optimized NHI 93 (EctD) sequence:

ATGGAGGATCTGTATCCGAGTAGACAAAACAACCAGCCAAAAATACTTAAACGTAAA
GATCCAGTTATCTATACGGATCGGTCAAAAGATAACCAAGCGCCGATAACTAAAGAA
CAGTTGGACAGCTATGAGAAAAATGGTTTTCTCCAAATTAAAAATTTTTTTCTGAAG
ATGAAGTCATTGACATGCAAAAAGCAATATTCGAACTGCAGGATAGCATTAAAGACG
TTGCCTCGGATAAGGTCATTTCGTGAGCCCGAATCGAATGATATACGTAGTATTTTTCA
TGTGCATCAGGACGATAACTATTTTCAGGACGTAGCGAACGATAAGCGAATCTTAGA
TATAGTACGCCATCTGCTTGGGTGAGATGTATACGTGCATCAGTCTCGGATCAATTAT
AAACCAGGATTTAAAGGAAAAGAATTTGACTGGCATTCCGATTTTGAACTTGGCAT
GTAGAGGACGGTATGCCCAGGATGAGAGCGATTTCCGTCTCAATTGCTCTGAGTGA
TAACTACTCCTTTAACGGACCCTTAATGCTTATTCCTGGCAGTCATAACTATTTTGT
CATGCGTGGGAGAGACTCCAGACAACAACATAAAGAATCCTTAAAAAAGCAGAAA
CTTGAGGTGCCAGATGAAGAGTCTTTGCGCGAGCTTACGCGAATCGGTGGAGGTAT
ATCAGTCCCGACGGGGCAAAGCTGGCAGTGTTACATTGTTTCGAGTCGAACACCATGC
ATGGCAGTACGTCTAACATCACGCCGTATCCGCGCAACAATTTATTCATGGTATACAA
TTCCGTTAAAAACCGGCTGGTGGAACCGTTCTCTGGCGGCGAGAAACGTCCGGAA
TATATCGCTGTACGCGAAAAACAACCAGTTTATTCCGCAGTGAATTAA

NHI 93 (EctD) protein sequence:

MEDLYPSRQNNQPKILKRKDPVIYTDRSKDNQAPITKEQLDSYEKNGFLQIKNFFSEDE
VIDMQKAIFELQDSIKDVASDKVIREPESNDIRSIFHVHQDDNYFQDVANDKRILDIVRHL
LGSDVYVHQSRINYKPGFKGKEFDWHSDFETWHVEDGMPRMRAISVSIALSDNYSFN
GPLMLIPGSHNYFVSCVGETPDNNYKESLKKQKLGVPDEESLRELTRIGGGISVPTGKA
GSVTLFESNTMHGSTSNITPYPRNNLFMVYNSVKNRLVEPFSGGEKRPEYIAVREKQP
VYSAVN

Codon-optimized NHI 94 (EctD homolog) sequence:

ATGGCAGAAAATAAGTTTTATTTGGATTCCTATCCTACGAGAACGTCATCTGAAAGCA
AGATCATACAACGCACGAATTCAGTGGTCTACTGTGACTCGTACGATGAACTCAACA
AGGAGCAGGTGGACTTTTTTCGAGAAAAACGGTTATCTGATTTTTGAGAATTTGTTCA
GCTCTGATGAAATTACTAACTTTTCGATGAGCTCACCTCACTTTCTCAGGACGAAAT
TAAAAAAGATCTGCCTCAATTCATATTAGAGGAAACCCAGCGGGATGTGAGGTCAAT
TTTTGAGATACATAAAATTAGCAAACCTCTATGGCAAACCTTTGCCGTGATAAAAGAATTT
TGAAAATTGCACAGCAACTGCTGGGGTCTAAGGTCTATATTCATCAAAGCCGTGTTA
ATTTAAAACCGGGTTTTGATGGTAAAGAATTTTACTGGCATTCTGACTTTGAAACGTG
GCATAGCGAAGACGGGATGCCCAACATGCGCGCAGTCTCATGCTCTATTTCACTGA
CGAAAAATTACGAATTTAATGGACCGCTGATGGTCATTCCCTGGCAGTCATAAGGAAT
TTGTCTCCTGCAGTGGCAAACTCCGGATGAACACTATAAGCAAAGTTTAAAACGCC
AGGAGATCGGTACTCCGCAGAAAGAATTCCTGGAGCAGATGGTTGAGAAAAGCAAA
ATCGTTTCAGCAAAAGGAGATGCCGGTAGCGTGATATTTTTCGACTGTAATATCATGC
ACGGTTCGAACGGTAACATTACCCCTATCCTCGCTCGAACGCCTTCTTCGTGTACA
ATTCTATTCATAATACTTTGGTCGATCCTTTTTGCGGATTAAATCCGCGCCCTACGTAT

ATTGCCGAACGTGAATTCTTGCCGTTAGAACCGGTCGATAATTTCTTAGACTCAAAAT
AA

NHI 94 (EctD homolog) protein sequence:

MAENKFYLDSPTRTSSESIIQRTNSVVYCDSDYDELNKEQVDFFEKNGYLIFENLFSS
DEITKLFDELTSLSQDEIKKDLPPQFILEETQRDVRSIFEIHKISKLYGKLCRDKRILKIAQQL
LGSKVYIHQSRVNLKPGFDGKEFYWHSDFETWHSEDGMPNMRAVSCSISLTKNYEFN
GPLMVI PGSHKEFVSCSGKTPDEHYKQSLKRQEIGTPQKEFLEQMVEKSKIVSAKGDA
GSVIFDCNIMHGSSNGNITPYPRSN AFFVYNSIHNTLVDPFCGLNPRPTYIAEREFLPLE
PVDNFLDSK

Codon-optimized NHI 95 (FtmOx1) sequence:

ATGACCGTTGACTCAAAACCGCAGTTGCAGCGTCTCGCTGCAGATGCCGACGTGG
ATCGTATGTGTCGTCTCCTCGAGGAAGATGGAGCGTTTATATTAAAAGGGCTGTTGC
CATTTGATGTTGTGGAGTCATTTAATCGTGAATTAGATGTGCAGATGGCCATCCAC
CCCCAAGGGTGAGCGGCTGCTAGCCGACAAGTACCCACCTCACTTCAAGTACGT
CCCCAATGTTGCAACTACTTGCCCGACATTTAGAAATACCGTTCTGATAAACCCCGT
GATTCACGCCATTTGTGAGGCGTATTTTCAGCGTACCGGTGACTATTGGTTAAGCGC
CGCTTTCCTCCGTGAAATAGAAAGTGGGATGCCCGCGCAGCCATTTTCATCGCGATG
ATGCAACCCACCCGCTGATGCACTACCAGCCTTTGGAGGCCCTCCTGTCAGCCTA
AGCGTCATTTTCCCGCTGACAGAGTTTACTGAGGAGAATGGTGCCACGGAAGTAAT
CTTAGGTTCTCACCGATGGACCGAAGTCGGCACCCCCGAACGCGATCAGGCTGTG
TTAGCGACCATGGATCCGGGTGATGTTTTGATCGTGCGCCAGCGGGTGGTGCATG
CGGGAGGAGGTAATCGAACGACAGCAGGTAAGCCACGGCGCGTCTGTTCTGGCCTA
CTTTAATTCTGTTCAACTTACTCCATTTCGAGACGTACCGGACCATGCCGCGTGAGAT
GGTTGAGTCCATGACCGTCCTGGGTGAGCGGATGTTAGGATGGCGAACAATGAAA
CCGAGTGACCCAAACATCGTTGGTATTAATCTTATTGATGATAAACGCCTGGAAAAC
GTTCTGCAGCTGAAAGCAGCGGATTCTCCAGCATAA

NHI 95 (FtmOx1) protein sequence:

MTVDSKPQLQRLAADADVDRMCRLLEEDGAFILKGLLPFDVVESFNRELDVQMAIPPP
KGERLLADKYPPHFKYVPNVATTCPTRNTVLINPVIHAICEAYFQRTGDYWL SAAFLRE
IESGMPAQPFHRDDATHPLMHYQPLEAPPVSLSVIFPLTEFTEENGATEVILGSHRWTE
VGTPERDQAVLATMDPGDVLIVRQRV VHAGGGNRTTAGKPRRVVLAYFNSVQLTPFET
YRTMPREMVESMTVLGQRMLGWRTMKPSDPNIVGINLIDDKRLENVLQLKAADSPA

Codon-optimized NHI 96 (AusE) sequence:

ATGGGCAGTGCGACCCCTAGCAGACTGCAAAAGTTTCCCGCGACCGCCCCCGCAG
ATGAAATTTATGCAGCCTTTAAAGAAGATGGGTGTGTAATTATTGAGGGTTTTGTCCC
GCCTGATCAGATGGCGAGATTCTCGCAGGAAATACAGCCTGCAATGGAGAAAATTC
AGGTACAGGTTACTAACGATGGGAACAGCAATGATCGCGTAAAACGTTTTTTCGAAAC
TAGTTACCACTAGTCCAACCTTTTCGCCATGAAATTCTGGAGAACGATCTTATGCACG
AACTGTTACAGCGTGTCTTTTCAAACCGGGTGAGGGAATGGGTTACCATTTTAATG

ACACCATGGTGATCGAGGTACAGCCGGGGGCACCCGCTCAGAGGCTGCACCGCG
ATCAAGAACTTTATCCGTGGTGAATTCCATGGGTCCAGATGCTCCTGAATGTTTAG
TGAATTTCTTTTGC GCGGTGACTCCGTTACGGTTGAAAATGGTGCGACGCGTTTG
GTTCCAGGATCTAACCGTTGGCCAGAACTTACCTTAATTAATGCTACAGATTGCCCC
CAGTATGGTAAAATCGACTCCGTGCCTGCCATCATGCAGCCTGGGGATTGCTATATG
ATGTCAGGCAAAGTTATTCATGGCGCTGGCCACAATGCAACACTTTCCGACCAACG
TAGGGCTCTAGCATTTAGTACGATCCGAAGAGAACTCAGACCCGTGCAAGCCTTTC
CATTGTGGATTCCGATGCAAATCGCCACCGAACTCTCACCGAGAACGCAGGCAATG
TTCGTTTTTCGTTCTTCAACACAGCACTGTGACGTTGACACGGTACACTTTTGGGG
TAACGATGGCAAAGACATAGGTGAACACCTCGGATTAATATCAAGCGCCTAA

NHI 96 (AusE) protein sequence:

MGSATPSRLQKFPATAPADEIYAAFKEGDCVIEGFVPPDQMARFSQEIQPAMEKIQVQV
TNDGNSNDRVKRFSKLVTTSPTRHEILENDLMHELLQRVFSKPGEGMGYHFNDTMVI
EVQPGAPAQRLHRDQELYPWWNSMGPDAPECLVNFFCAVTPFTVENGATRLVPGSN
RWPELT LINATDCPQY GKIDS VPAIMQPGDCYMMMSGKVIHGAGHNATLSDQRRALAFS
TIRREL RPVQAFPLWIPMQIATELSPRTQAMFGFRSSTQHCDVDTVHFWGNDGKDIGE
HLGLISSA

Codon-optimized NHI 97 (AsqJ) sequence:

ATGGGATATCCTAAAGCCTTTACGAGTTCGGATTCTGAGCCGGAGCCTGACTTGAG
CAGAGATCTTGGAACCCGGTTATGGGAAATCCGGGTGTTGTGAGCCGCTCCTCAT
CCACCGTGGCGCAGCACAGCGTACGTAACAATCCAACGGGCCCCGATGGTCGCTT
GGCGGGGGCTTTGGAATGCCCGTGCTCTGCTACGTTTCCCCGCGGAGGTGAATGGC
GTACGCTTAGATTTCAAAGTGTAAGTTCCCGCCGCCCTACATCACGGCTCAGCTTG
TTACCTTACTCTCTGTGTTCTATCTGCCCTTCATCTCAGGCAACAATGACTAGTAAGG
ATCATGTGAAGTCACAAATTCCGCGGCTCTCCGCAATCAATGACTTACATAAAATTTG
GCCTACTGTAGAAGAGCACGGTGCTGCCATTATAGAAAGTTTCCTGTCTTAGATAT
CGTACGTCGGCTGAATGAAGAGGTGGATCCGTTTGTGAAAATTGAGCCGATTCCGG
CAGCCAAAACGAAGGATCATCAAACCATGTACTGTCTACAACGACCAGGCTGGTG
AATGTGCTGGCACCCATTAGTAAAGCCTACCGCGAAGATGTTCTGAATTCAAAGTG
CTGCATCGCATTTGCTCAGATGCTTTCCATGTTTACGGGGATTATTGGGTTCTTATGG
GGGCTGTAATGGAATTAGCCCCATCAAACCCCGCTCAACCGTTACACAGGGATATG
CGGTTTAGTCACCCTATTGTAGAGTATCTTAAACCGGACGCGCCAGCAACCAGCATC
AATTTTTTAGTGGCTCTGAGCCCGTTTACCGCGGAAAACGGAGCGACCCATGTAAT
CCTGGGCTCTCATAAGTGGCAGAATCTTCAAATGTGAGCATGGACGCAACGGTCC
GGGCGCTGATGAATCCGGGGGACGCGCTGTTGATAACCGACTCGACCATTCAATTGT
GGCGGTGCCGAACTACTGGAACGGAGACACGGCGGCTGTTAACGATTACGATGG
GTATCTCGCAGCTGACCCCACTGGAATCGAACCTTGCCGTCCCCCGTCCGGTTATT
GAGTCCCTACCCCGCTAGCTCAACGCTTGCTGGGATGGGCCTCGCAGCGTAGTG
CTGCCCCCGTGATATAGGCCTGTTGACTATCCGCGGTAATAGTATTGAAAAAACGA
TGAATCTCAAAGCAGAGCAGCCTTTGCATGATGATGAAGCCGAACCTCTGTGTCGT
GAAACGATTTGA

NHI 97 (AsqJ) protein sequence:

MGYPKAFSTSSDSEPEPDLSRDLGNPVMGNPGVVSRSSSTVAQHSVRNNPTGPDGRL
AGLWNARALLRFPAEVNGVRLDFKSVSSRRPTSRLSLLPYSLCSICPSSQATMTSKDH
VKSQIPRLSAINDLHKIWPTVEEHGAAIIESFLSLDIVRRLNEEVDPFVKIEPIAAKTKDH
PNHVLSTTTTRLVNVLAPISKAYREDVLNSKVLHRICSDAFHVG DYWVLMGAVMELAPS
NPAQPLHRDMRFSHPIVEYLKPDAPATSINFLVALSPFTAENGATHVILGSHKWQNLSNV
SMDATVRALMNP GDALLITDSTIHCGGAETTGTETRRLLTITMGISQLTPLESNLAVPRP
VIESLTPLAQ RLLGWASQRSAAPRDIGLLTIRGNSIEKTMNLKAEQPLHDDEAEPLCRET
I

Codon-optimized NHI 98 (LoIO) sequence:

ATGACAGTCACAAACAAACCAGTTGAACCTGCCAATGTCCCGGTTATGGATTTTGAA
GCAATTCATGCATCGGTGGGCAATGAGCGTAAGGAATATTTGCGTCAGCTTGACGA
AGCGTGGTCTCATCACGGCGCAGTCTATGTCATTAATCATTCTATAGGGACAGAGAC
CTTAGAAGAGGCCTTCGTGTGGTGCAAAAAATTTTTTACTTGCCTCTTGCCGTGAA
AAATAGCGTCCATATCCCTCCTGATGTATCCAAACATTTTCAGGGCTGGACTGGCAC
CGGTGAAGCGATTTCAAGTCAAGGCGTCTGGGATCCTGACGAAATTGAAAGACTGC
GTAAGGAAATGCCAACGGAAGTGAAGAAGCCATGGAATTGCAGGATCCATGCGGA
ACTTACCCCCCTGGCAATCCCGATCTGAATCTCGTGGAACAACATCTACCCGGCTA
CCTCGATTTTCTGAAAAAATGGTTCGCAGCTTGCTATAAACAGTCCCTGCAGAATAT
GCGTCTGGTTTGTGAAATCTTGGGGATGGAAGACCTTGATTACATTGGAaaaaaatt
TGAACCACGGCATATGAGTACACACTCAACTTGGAATTATTTTTTAGGCCAGCCGGT
ATCTCAGCTTGCCTCAGGATCTTCAATCGCTTAAATGCGCATAACAGATTACTGCCA
GTTACAGATGCTATTTTCAAGACATGGTTGGCGGTTTAGAGCTTCACGACTATGAAGA
AGACATTTATCGTCCAGTGCCGCCGATCAAAGGCGCCATGATAGTGCAGGTCGGTG
ATCTGCTGGAGAAACAGACGAATGGACGGTGGAGAAAGTGCTTTGCACAGGGTCAC
CGCACCCCTCTCGTTACATGTATGGGGGAAGTCCAGGTGGAGATGATGAACTGGTGC
AGCGCTATAGTCTGGTGTTTTTTGGACACCTTAACCTGGATGAAATGATTAAACCACT
GCCGGGATGTGAGAAACCGGGGCAAATGGTCCACCCTGGAATGGAAAGATCTCATG
ACAGCCGGCCAGTGGTTAGCAAGGCGTGTAGCCCTGGAGTATGAGCGTAAAACCG
CAGCCACGGTAATGTGA

NHI 98 (LoIO) protein sequence:

MTVTNKPVEPANVPVMDFEAIHASVGNERKEYLRQLDEAWSHHGAVYVINHSIGTETL
EEAFVWCKKFFDLPLAVKNSVHIPPDVSKHFQGWGTGEAISSQGVWDPDEIERLRKE
MPTELKEAMELQDPCGYPPGNPDLNLVEQHLPGYLDFLKKWFAACYKQSLQNMRLV
CEILGMEDLDYIGKKFEPRHMSHSTWNYFLGQPVSQLASGSSNRLNAHTDYCQFTM
LFQDMVGGLLELHDYEEDIYRPVPPIKGAMIVQVGDILLEKQTNGRWRWSALHRVTAPSR
MYGGSPGGDELVQRYSLVFFGHLNLDDEMIKPLPGCEKPGKWSTLEWKDLMTAGQW
LARRVALEYERKTAATVM

Codon-optimized NHI 99 (LoIE) sequence:

ATGACAGCCGCAAGTTCACCGCATCCGGGGGTATCCGCGGAAGACATTGAGTTTTTA
CCAGGCCAAACGGGTATTTGCGTCTCCCGCAAGAAGCTCATGGACTGTTTGACGAC
CTGGCCAAGTTACAGGCATGGGTGCGCCGAGATTTTCGCAGTGGGGCCTTGAAACAG
GTAAGTGGCGTCATTATTATGAAACAACGAACGGGCAAACATCTCCTGTGGGGGTACG
GAGAAGCTGATGGAATATCATGCTCCGATGCGTGACCTCATTGCTGGAGAAGCGCC
CCTGACCTTGTTAAAAAGCCTAACGGGGCAAAGATATGGTAGTCTTTAAAGACGAAAT
AGGTTGGAAACTGCCGGGTGGTAAAGGTGCCGTCCCCCATTGATCGGCCTGCG
TACAGCATGTTTGCTCCTGAGTTTATCGAAATTATGATCGCTGTTGATGCGCACACC
GTCGAAAACGGGTGTCTCCAGTTTGTGCCAGGTAGTCACAAAGAGGCAGTTCCGA
TCAGCGCCGATGGGCGTATTGCCTCTGCATGGCTGGAAGGGAAGGAATTCATTCCG
ATGGTCCTGGATCCGGGAGATGTTCTGATCTTCAATGAATCTATGGCGCATCGTCTC
GATCCAAATAAACTGATCAGCGCCGGGCAGCTGTATTTGGTACGTACCATTTTGAT
CGATCACAGCCCGACTTACGAGATAAATTCTACGCTCATAGATTAATTCATTCTCCAC
CAGAGAACGCCTGGGTAGAAACAGTGGAAGCTCAGACTTGA

NHI 99 (LolE) protein sequence:

MTAASSPHPGVSAEDIEFYQANGYLRLPQEAHGLFDDLAKLQAWVAEISQWGLETGK
WRHYYETTINGKHLWGTEKLMEYHAPMRDLIAGEAPLTLKSLTGKDMVVKDEIGWK
LPGGKGAVPHLDRPAYSMFAPEFIEIMIAVDAHTVENGLQFVPGSHKEAVPISADGRIA
SAWLEGKEFIPMVLDPGDVLIFNESMAHRLDPNKTQRRRAAVFGTYHFDRSQPDLRDK
FYAHRLIHSPPENAWVETVEAQT

Codon-optimized NHI 100 (KanJ) sequence:

ATGGCTCTCGCCGCGCCCCCAGGCGAACTAACCCTGGCTCTGACACCGGATGATA
AGACTTTGGATCCGGCAAGCCTTGATAGAGCACTGGCCATTCTCGCCGAACACGG
GATTCTGGTCCTGACAGGTATGCTTCGTACGCGTCTTACGGACCAATTAAGAACCG
CTATGCTGGATGATTTGCCTGAAGTCCTGAGGCAACAAGATGTGCCTACCAATTTTG
TACCTGGGCATGTGCAGCAAGATCCGCCGGTACGCGAATCCTTGCTGTTTCCGGAT
GTACTCCTAAATCCGGTGGTATACCAGATCACGCATGCCGTCTGGGCGCAGATGC
GCGGAATGCGGTATATTCCGGAAATATGAATCTGCCTGGCTCTCACGAACAGCCTGT
TCACTTAGACGAACCGCATCTGTGGCCGGGCATCTCACATCCCCCGTACTGTCTTT
GTGTAGATGTTCCGCTTATTGACTTTACCCTGGAGAATGGCTCTACAGAATATTGGC
CGGGGTCGCATGTCCTGAACCCTGATGAATGTTATGACGAGCGTGGGTGCGTTCT
GCCGGCGGAGCTGGAACGCAGGCGCGCAGTTGCTCCACCGGTACGGTTTCCGAT
TCCGGTGGGCTCCGTTGTCATCCGCGATGGCCGCCTCTGGCATAGAGGGGTTCT
AATCTGAGCGCGGCCCGCGGCCCTGCTCGCCATGACCCATTATACCGAGTGGT
TCGATATGCCGCCTATCCAACCTGCCGATACGGTGAAATCCTGGGTAGACGGGTCC
GACAGACATACCCATGCCCATTTCTGTGGCGGGTGATGTGGATCATCTTACGGGTGA
CCATCCGTTTGCAGTTAGATAA

NHI 100 (KanJ) protein sequence:

MALAAPPGELTLALTPDDKTLDPASLDRALAILAEHGILVLTGMLRTRLTDQLRTAMLDDL
PEVLRQQDVPTNFPVGHVQQDPPVRESLLFPDVLLNPVVYQITHAVLGADARNAVYSG

NMNLPGSHEQPVHLDEPHLWPGISHPPYCLCVDVPLIDFTLENGSTEYWPGSHVLNPD
ECYDERGCVLPAELERRRAVAPPVRFPIPVGSVIRDGRLWHRGVPNLSAAPRPLLAM
THYTEWFDMPPIQLPDTVKSVDGSDRHTAHFVAGDVDHLTGDHPFAVR

Codon-optimized NHI 101 (Downy Mildew Resistance) sequence:

ATGGCGGCCAAATTAATTTCAACCGGCTTCAGACATACAACGTTACCAGAGAATTAT
GTGCGTCCGATATCAGATAGACCACGCTTAAGCGAAGTCTCCCAACTGGAGGATTT
CCCCCTTATTGATCTCTCATCTACTGATCGCTCGTTTCTCATTACAGCAGATACATCAA
GCTTGTGCTAGGTTTGGATTCTTCCAGGTGATTAATCACGGAGTAAATAAACAGATC
ATCGATGAAATGGTATCTGTGGCAAGAGAATTTTTTCCATGAGCATGGAAGAAAAA
ATGAAACTGTATTCGGATGATCCCACTAAACTACACGGTTAAGTACGTCCTTCAATG
TTAAAAAGAAGAAGTTAATAACTGGCGCGATTACCTGCGATTGCATTGCTATCCAAT
ACACAAATATGTTAATGAGTGGCCATCCAATCCTCCGTCATTTAAAGAGATCGTTAGT
AAATATAGTCGGGAGGTCCGTGAGGTGGGTTTTAAATTGAAGAACTGATCTCCGAA
TCTTTAGGTTTGGAGAAAGACTATATGAAAAGGTTCTTGGGGAACAGGGGCGAGCA
TATGGCGGTCAATTATTACCCACCTTGCCCCGAACCGGAATTGACTTATGGCCTGCC
GGCGCACACCGATCCCAACGCATTGACCATTCTGTTGCAGGATACTACCGTTTGC
GGTTACAGATCTTAATTGACGGTCAGTGGTTCGCAGTTAATCCCATCCCGACGCTT
TTGTTATTAATATTGGAGATCAGCTCCAAGCACTTAGCAACGGTGTATACAAGAGCGT
GTGGCATCGGGCAGTTACGAATACGGAGAATCCGAGACTTTTCAGTCGCTTCTTTTTT
ATGTCCAGCGGATTGTGCGGTGATGTCCCCGGCGAAGCCATTATGGGAAGCCGAG
GATGATGAGACGAAGCCGGTGTACAAAGATTTTACTTATGCGGAATATTATAAGAAGT
TCTGGTCCCGAAATCTGGACCAGGAGCATTGTCTTGAAAACTTTTTGAATAATTGA

NHI 101 (Downy Mildew Resistance) protein sequence:

MAAKLISTGFRHTTLPENYVRPISDRPRLSEVSQLEDFPLIDLSSTDRSFLIQQIHQACA
RFGFFQVINHGVDNKQIIDEMVSVAREFFSMSMEEKMKLYSDDPTKTTRLSTSFNVKKEE
VNNWRDYLRLHCYPIHKYVNEWPSNPPSFKEIVSKYSREVREVGFKIEELISESLGLEK
DYMKKVLGEQGQHMAVNYYPPCPEPELTYGLPAHTDPNALTILLQDTTVCGLQILIDGQ
WFAVNPFPDAFVINIGDQLQALSNGVYKSVWHRAVTNTENPRLSVASFLCPADCAVMS
PAKPLWEAEDDETKPVYKDFTYAEYYKKFWSRNLDQEHCLNFLNN

Codon-optimized NHI 102 (scopoletin-8-hydroxylase) sequence:

ATGGGTATTAATTTTGAAGACCAGACTACATTATTTAACTTCGTGGTGCGGGAAGGC
AATGGTGTCAAGGGTATGATTGACAGCGGACTGTCTTCAGTCCCGCGGCCTTTTGT
TCAACCGTTATCGGAACGGATCCCAACCCAGAAAGCCTTGACATGTGAAGCCACCC
AGCCTATTGATTTGTCAAATCTTGACGGGCCTCAACACAAAGAAGTGGCAAAGCAG
ATCGTTGAAGCAGCTGAGACGCTCGGCTTCTTTCAGGTTGTTAACCACGGCGTGTC
AGTTGAGTTATTGGAATTCTGAAGTCATCTGCACATGAGTTCTTTGCACAAGCTCC
GGAGGAAAAATCTATGTACCTGAAAGAAGTTTCACCATCGAAGCTGGTTAAGTATGG
CACTAGCTTTGTCCCAGATAAAGAAAAAGCGATTGAATGGAAGGACTATGTTAGTAT
GTTGTATACCAATGATAGCGAAGCTTTACAGCATTGGCCGCAGCCATGTCGTGAAGT
CGCATTAGAGTTCCTGAACAGCTCGATGGAAATGGTCAAAAATGTGGTCAACATTCT

GATGGAAAATGTGGGTGTGACACTGGAGGAAGAAAAAATGAATGGTTTGATGGGAA
CGAAGATGGTTAACATGAATTACTATCCGACTTGCCCTTCTCCGGAACATACTGTGG
GAGTAGGAAGACACTCAGACATGGGGATGTTGACTGTCCTGTTGCAAGATGGGATC
GGGGGACTGTACGTGAAATTAGACAATGGGGAATGGGCAGAAATACCACCGGTACA
TGGCGCTCTCGTTATTAATATTGGTGATACGCTACAGATTCTGTCAAATGGAAAGTAT
AAATCTGCAGAGCATCGTGTCAGAACCACAAACATCGGTTCCCGCGTTTCCGTTCC
GATTTTCACCGCACCTAATCCGAGCCAAAAGGTAGGTCCGTTGCCAGAAGTTGTTA
AACGTGATGGTGTTGCACGCTATAAAGAGTTTCTCTTCCAGGATTATATGAATAATTT
CTTCGGTCAGCCGCACGATGGTAAAAAGAGTCTGGACTTTGCACGTGCCGAATAA

NHI 102 (scopoletin-8-hydroxylase) protein sequence:

MGINFEDQTTLFNFVVREGNGVKGMIDSLSSVPRPFVQPLSERIPTQKALTCEATQPI
DLSNLDGPQHKEVAKQIVEAAETLGFFQVVNHGVSVELLELLKSSAHEFFAQAPEEKS
MYLKEVSPSKLVKYGTSFVPDKEKAIEWKDYVSMLYTNDSEALQHWPQPCREVALEFL
NSSMEMVKNVVNILMENVGVTL EEKMNGLMGTKMVNMNYYPTCPSPELTVGVGRH
SDMGMLTVLLQDGIGGLYVKLDNGEWAEIPPVHGALVINIGDTLQILSNGKYKSAEHRV
RTTNIGSRVSVPIFTAPNPSQKVGPLPEVVKRDGVARYKEFLFQDYMNNFFGQPHDGK
KSLDFARAE

Codon-optimized NHI 103 (GloF) sequence:

ATGGCTATTCAGACCCTTGATTATCGTGACTTCCAGTACGGGGGCCAGGAACAACAT
CGTACCTTTTGTCAACCTGTGTGAAACGCTCAGCACGTGGGGATTATCAAAATC
CAAAACACCAGTATTCCTGACGCCGTTATAGACGAGTTATTTTCGTACAACAAAAAAT
TTTTTGCACTTCCCGAACATATTAAACAGAAGGCAAGGCATCCGGCAGCTCCAAATC
CGCACCGTGGTTGGAGTGCAAGTCGGTCAAGAACAGTTGTGCGGTATCGCAGGTTT
TGAAAAAGACGAAGAAACAGATGGCTTCGTTCCGGAATATCGTGAATCTTTCGATCA
GGGTGCCGCGGACGATGAACTCTTCCAAATCGGTGGATTGATGAAGACGATTTAC
CGGGTTTTCGGAAATTTATGGAAACTATTACGAGATGTGTTATAACTTTCACTCA
GCTGTTACGTGCAATCTCCACCGGTCTCTCGCTCCCTGAAGATCTCTTACTGAGTC
GCCATCAGACGGATACCTCAGAACTGCGCATGAATCACTATCCAGCCATCGCCTGT
GAAAACCTGAAATTCGGCATGCGTATCGGCGAACACTCTGATTTTGAACTCTTACG
TTATTACTCCAGGATTCCGACCGGGGGTTTACAGGTGGAAGACCAGAAGAAGCTTGG
AACTTTTATTCCCGTCGAGTCGGACAGTCGTTACGAGGTCATTGTAAACGTTGGCG
ACTGTCTTCAACGCTGGACGAATCGTCGTCTGCGTTCAGCCAATCATCGGGTACAT
CTTCCTGAAGGCAAAAACCTTCAAATCCGACGAAGTCCTTGCGGATCGCTATAGCGT
TGCTACTTTTGGTAAACCGGACCGTAATGTACTGGTCTGACTCATTCCCCGAGTTTTC
CCGCGGTGGTGAAAGCAAATATAACGATCATATGAATGCACTTGAATATAATCAGACA
AAGCTGCTTCGTACTTACGCCTGA

NHI 103 (GloF) protein sequence:

MAIQTLDYRDFQYGGQEQHRTFCHNLCETLSTWGFIKIQNTSIPDAVIDELFSYNKKFFA
LPEHIKQKARHPAAPNPHRGWSAVGQEQLSRIAGFEKDEETDGFVPEYRESFDQGAA
DDELFPNRWIDEDDLPGFRKFMENYYEMCYNFHTQLLRAISTGLSLPEDLLLSRHQTD

TSELRMNHYPAIACENLKFGMRIGEHSDFGTLTLLQDSTGGLQVEDQKKLGTFIPVES
DSRYEVIVNVGDCLQRWTNRRLRSANHRVHLPEGKNFKSDEV LADRYSVAYFGKPDR
NVLVDSFPEFCRGGESKYNDHMNAL EYNQTKLLRTYA

Codon-optimized NHI 104 (GloC) sequence:

ATGAGAATCGAGACACTGGATTTCTCTAAGTTTCATGTGGGGACTGATGCTGAACG
GTATGAATTTTCTACCGCTCTTCTTACAGGATTTGCGAATGCCGGCTTCATCAAACCTT
ATTAATCATGGCTTCAGTCAGGAAGAAATTAGTCAACTGTTTGAAGAAAACCGACGC
TTTTTTGATCTCACCAGACAGTATTAAGGCACAAATAGCTAACGAAGATGGACCAAAG
CCGCAGCGGGGTTGGTCAGCCGTGGGCGCCGAAAAAACCGGACTCCTAAATACTG
GTGGCAAAACGAATTTAACCAAACCCGACATTGAGGCCCCGCGAGGACGCGAAGGA
ACACTTTGATATTGGACCATCTGAGGATACTGAATCCCAAATAAATGGCCAGATAAT
GAACTGATACCAGGATTTGAGCCATGGTTAGAATCCTTTTACGAACGCTCACGCTAT
ATAAGCTTTGCCCTTATGGAGGCGCTCGAGATAGCAATGCATCTGCCGAAGGGTGC
TTTTATCCAGAAATGCCAGGGCCATGCCAGTGAAC TACGGCTGAATCATTACCCTAA
GATTACGGTAAAGACCTTGCAAGGCGGTAAAACCAAGTCGTATTTGGCCCCATACCG
ATTTTGGCGTAATTACGTTACTTGCACAGGACGATGTAGGCGGACTTGAGATCCAGG
ACAAAGACAATCCGACCAACTTCTTACCGGTCAGCCGGGAGGACGTCTCAGAGTTA
GTCGTTAACATAGGGGACACACTGGAACGGTGGACGAATGGCTTTCTGAAAGCCG
GCTTACATCAAGTGACTACCCCTAGGGAAATTGTCAACCAGAGCACAGAAACGGTT
CGGCCGCGGCGTAGCATTGCCTTCTTTCTGAAAGCGCATCGCCAAATGTCTGTGG
GACCGCTACCGCAGTTCGTTGCCGAAAAAACCCCGGCAAAATATGAGGATATCACC
GCCTTGGCATACCAACAACGCCGAAC TGCATTGTTTATTAA

NHI 104 (GloC) protein sequence:

MRIETLDFSKFHVGTDAERYEFSTALLTG FANAGFIKLINHGFSQEEISQLFEENRRFFD
LTD SIKAQIANEDGPKPQRGWSAVGA EKTGLLNTGGKTNLT KPDI EAPQDAKEHFDIGP
SEDTEFPNKWPDNELIPGFQWPWLESFYERSRYISFALMEALEIAMHLPKGAFIQKCQGH
ASELRLNHYPKITVKT LQGKTSRIWPHTDFGVITLLAQDDVGGLEIQDKDNPTNFLPV
SREDVSELV VNI GDTLERWTNGFLKAGLHQV TTPREIVNQSTETVRPRRSIAFFLKAHR
QMSVGPLPQFVAEKTPAKYEDITALAYQQRRTAIVY

Codon-optimized NHI 105 (GloM) sequence:

ATGTATAAAATTTCTTCGCCGGGTACTACGTGGAAATGATAAAGACGTTGAGGCG
CTGGCCGAGGCTGCTACTACGCAGGGGTTTCTTGAATTGGAAC TGGATTGTATGGA
AGCGAAAGCGCTGCAGAAAGATGT TAAATTTCTTGAATCATTTGCGGAATCAATACTT
GATTCCCCCGAAGATGTAAAAGCAGCGTATCATTTTCAAAAACCGGTGCTTTTCGA
ACTACTGGTTTTAAACCGCTGGGGAGTGAGGAAGGTGCGAAAGCAGGAAATCCAG
ACGGCTTTGAATCTTTTTTCTGCCCAA AAGGAGATATTACTTTCTGAATTTGAGA
ACAGCTGAATTGTCCGCCACTGGCGATGTCGAATATTGATACGTTAACAGACTGTTT
CAAACATTATGAAAGAACCGCC CAGATGCTTTTACAAAGGCTGACAGAAGGTTTAA
ACTGGGAAACGAGCTCCTGAATGCTCACAACACCAGCCTGCCCTGCGTAACCAATA
TGGGTCTGATCCGGTATCCACCC CAGCCGAAAGAATCGGGGAATTTCCGGTCATATC

GCACACACCGATGTCGGGAGTCTGACCATTCTTGCAGCTACCCAGCGTGGTCTTCA
GACACTGGATAATATTACACAAAAATGGATCTGGGTCGATCCCAGTGATGAATGCCT
GTTCTGTCAGTTGGGAGATTCAATTGAAGTTTCTAAGCCGCGGTAAAATCCTGCCCT
CCTTGCACAGAGTTTTGCCGTCTGATGTTGCCCCCGAAGCAACCAAATATACAATCG
CCTATTTTCTTCGTCCGAACGAAGAAGCCGAGATCACAAGCGATGATGGGAAAGTG
TGGCTGTATAAAGATTACCAATTGCCGAAAATTTGATGCCTTCGCACGTCCGCTGGGT
TACCGTCCGGATGGTGAAGAATCACTTATCTCATTGCGCGACTATACGGGTGTCGAG
TAA

NHI 105 (GloM) protein sequence:

MYKISFAGLLRGNDKDVEALAEAATTQGFLELELDCMEAKALQKDVKFLESFAESILDS
PEDVKAAYHFHKTGRFRTTGFKPLGSEEGAKAGNPDGFEFFFLPQKEILLSEFREQLN
CPPLAMSNIDTLTDCFKHYERTAQMQLLQRLTEGLKLGNELLNAHNTSLPCVTNMGLIRY
PPQPKESGNFGHIAHTDVGSLTILAATQRGLQTLDNITQKWIWVDPSECLFVQLGDSL
KFLSRGKILPSLHRVLPDVAPEATKYTIAYFLRPNEEAEITSDDGKVWLYKDYHCRKFD
AFARPLGYRPDGEESLISLRDYGVE

Codon-optimized NHI 106 (GloE) sequence:

ATGGATTATATTAACAGGCGGAAAGCACCCAATTGTCGTCTCTGTCACTGTCTAGG
CTGGAAGGTAATAACGCAGAGGAGTCCAAACGTCTGTTAGAAGCATGTGCGCAGGA
TGGCTTCTTCTACTTAGATCTGAGAGACCACAAACAGTTATTGGTGGATTATGAAGC
ACTCTTAGAGATAATCAAACAGTATTTTAATGAACCACTGGATCAAAAAATGAAAGAT
GATCGTAAAAGCGACACCATAGGTTACGAACCGGTCGCGACCAGCGCGGGCGTAC
TGGACGGCCTGCCGGACTATTACGAATCTTTTAAGGTGTCCTGGAACCAATTACGC
GATCATGTGCAGGAGCTTCCAACGGTAGTTGAGACAAACATTGAAGTGTTTCGATCG
GTTTGCCAAGTATGTCCATAGCATCTTACTGATGATCCTGAGTCGTCTATCACAGACA
ATGGGTAGAAATAACGACAACCGGTTTCGAAAGTTACCATCGTGATTCTATTGCTACA
CGCACCAACCTTACATTTCTGAAGTATCCCAAACAAGACACGACAGAGCACGGCGT
CGGGCATAACAAACACACCGATGTGGGAACCTCACATTCCTGCTTTCCGGTCAGC
GCGGGCTTCAGAGGCTGACGCCTGAGGGGTGGTGTACGTGGAGCCAAGATCTG
GCTTTGCTGTTGTAAATGTTGGAGATAGTTTACGGTTCCTGTCTGACTGCGTACTTA
GCTCAGTTATTCACCGCGTTTTGCCGGTGGGCGCGCATCAGACCGAGGATCGTTAT
ACCCTGGCCTATTTTTTACGTCCTGAAGATGACGCCGTATTCAAGGACATTAACGGG
AATCTGGTTAGCGCGCGCAGCTGGCATGACCGGAAGTTCGATCATTTCCGCGCGA
GCCATAATCAGCAGAAAAATGATACAATCTTGATGGGCGGTATGGAAGAAAATCAA
AGTTTTTACAGTATAAATTTACAGGCTTGA

NHI 106 (GloE) protein sequence:

MDYIKQAESTQLSSLSLSRLEGNNAEESKRILLEACAQDGGFFYDLDRDHKQLLVGYEALL
EIKQYFNEPLDQKMKDDRKSDTIGYEPVATSAGVLDGLPDYYESFKVSWNQLRDHVQ
ELPTVVETNIEVFDRFAKYVHSILLMILSRLSQTMRNNDNRFESYHRDSIATRTNLTFL
KYPKQDTTEHGVGHNKHTDVGTLTFLLSGQRGLQRLTPEGWCHVEPRSGFAVVNVGD

SLRFLSDCVLSSVIHRVLPVGAHQTEDRYTLAYFLRPEDDAVFKDINGNLVSARSWHDR
KFDHFRASHNQQKNDTILMGGMEENQKFLQYKFQA

Codon-optimized NHI 107 (Diox4) sequence:

ATGGCCCCTACAAAAGACTCTGTTATCCATATGGGTGCCGAGTCCTGGGACGAAATA
AGCGAATTTGTCACCAAAAAAGGTCATGGGGTGAAAGGATTGTCTGAACTGGGCAT
TAAGACATTGCCTAAACAGTTTCATCAGCCGCTGGAAGAACGCTTTAGTGAGAAGAA
AATCCTGGAACGTGCAAGCATACTCTTATTGATATGTCTAAATGGGATTCACCTGAA
GTCGTGAAATCCATATGTGATGCGGCGGAGCATTGGGGATTTTTTTCAGATTGTGAAC
CATGGCGTGCCATTAGAAACACTGCAGCGTGTGAAAGAAGCGACCCATAGATTCTT
CGCGCTACCAGCAGAAGAGAAAAATAACAGCAAAGAGAACAGCCCTATTAACAA
TGTCCGCTTTGGTTCAAGCTTTGTCCCGCATGTGGAAAAAGCCCTGGAGTGGAAAG
ATTTTTTAAGTATGTTTTACGTTAGCGAGGAAGAGACGAATACATATTGGCCTCCGAT
TTGCCGCGATGAGATGCTGGAGTACATGCGCTCATCTGAAGTGTTAATAAAACGGCT
CATGGAAGTTCTGGTCGTTAAAGGCCTCAAAGTAAACAAATTGATGAGATACGTGA
GCCGATGCTTGTGCGATCGCGTCGTATCAACTTAAATTATTATCCGAAATGTCCTAAC
CCGGAATTAACCTTGGGGGTGGGTGCGCATAGCGATATTTCAACATTTACGATTCTC
CTGCAGGATGAAATTGGCGGCCTTCATGTTCTGTAAATTGGATGATACAGGTAATACT
TGGGTTACGTTACGCCGATCTCAGGTAGCTTAATTATTAATATCGGAGACGCACTG
CAAATTATGTGCAACGGTCGTTATAAGAGCATAGAACATATGGTGGTTCGCAAATGGC
ACCCAAGATCGGATTTCTGGTCCCATTATTTGTGAATCCAAAACCCCAGGCAATTCTC
TGCCCGTTCCCGGAAGTTTTAGCAAATGGGGAAAAGCCGGTATACAAGCCAGTGTT
GTGTAGCGATTATTCCAGGCACTTTTATACTAAACCACATGACGGTAAAAAAACCGTC
GACTTTGCCCTAATGAATTAA

NHI 107 (Diox4) protein sequence:

MAPTKDSVIHMGAESWDEISEFVTKKGHGVKGLSELGIKTLPKQFHQPLEERFSEKKIL
ERASIPLIDMSKWDSPEVVKSICDAAEHWGFFQIVNHGVPLETLQRVKEATHRFFALPA
EEKNKYSKENSPINNVRFGSSFVPHVEKALEWKDFLSMFYVSEEETNTYWPPICRDEM
LEYMRSSEVLIKRLMEVLVVKGLKVQIDEIREPMLVGSRRINLNYPKCPNPELTGVLG
RHSDISTFTILLQDEIGGLHVRKLDDTGNTWVHVTPISGLIINIGDALQIMSNGRYKSIEH
MVGANGTQDRISVPLFVNPKPQAILCPFPEVLANGEKPVYKPVLCSDYSRHFYTKPHD
GKKTVDFAALMN

Codon-optimized NHI 108 (Fum3p) sequence:

ATGAACAAGGAAAAAGTACCGCAGAACGCCGTACCGAATGGTAGGACGAAACTCC
GTCAGGTGACGTCTGCAACTCCTCTCGATGAAGTTTTTCAATACTGGGAGGAAGAC
GGTGCCATTGTAATAAAAGGCCTGTTGACTTCGGCCCAGGTTGAACAGCTGAACCA
GGAGATGGGCCCAATCCTGCAGAAAGTTGCCATTGGAGGCCATGCATCAGATGTTT
GGTTACAGAATTTTCATGGGATGAAGACGAAACGTGCAGGCGATCTGACCAATAAC
AGTGCTGTTTTTCGGGATCATTTACTAGATAATGACTTTATTCATGCCGTCTCTCAAC
GTTGCTTCGCCTATAGGGGAAAAATGGGCCCGGATGCGTATTGGTTAGGTTCCGCG
AGCACTATCCATGTTGGCCCGGGTCAAAAACCACAAACCCTCCATCGTGATCTTGG

CAGCTATCCTATCTTTTGGATGCTAGGACCTCAGGGCCCGGAATCCCAGATTAACCTT
TTTAGTGGCAACAACCTGATTTACAGAGGCTAATGGCGCTACGAGGATTATCCCCG
GCTCACATAAGTGGGAATTTAATCAGCATGGCGATCGCGATATGACAATCCCGGCAG
AAATGAAAGCAGGCGATTGTCTGCTGATTTACGGTAAGGTTATTCATGGAACCGGAG
GAAACAAAACGGATCAGGAGCGCGGCTGCCTCGCCGTGACGATGTGCGCGAATTT
CCTAGCACCTGAGGAGGCACATCCGTTTATTGTATCTATGGGTACCGCTAAAAAACT
TCCTGTCCGTTACAGCGATGCTTGGGCTTTCCGGTCACAGTGGCCACAGAGTTCTC
CGGGCTTATGGACGAAAGATTATTCTGAGCTGGCTTTGCACTTAGGACTGGATGATT
AA

NHI 108 (Fum3p) protein sequence:

MNKEKVPQNAVPNGR TKLRQVTSATPLDEVFQYWEEDGAIVIKLLTSAQVEQLNQEM
GPILQKVAIGGHASDVRLQNFHGMKTKRAGDLTNNSAVFRDHLLDNDFIHAVSQRCFAY
RGKMGPDAYWLGSASTIHVGPQGKPQTLHRDLGSYPIFWMLGPQGPESQINFLVATTD
FTEANGATRIIPGSHKWEFNQHGDRDMTIPAEMKAGDCLLISGKVIHGTGGNKTDQER
GCLAVTMCANFLAPEEAHPFIVSMGTAKKLPVRSQRCLGFRSQWPQSSPGLWTKDYS
ELALHLGLDD

Codon-optimized NHI 109 (PODA) sequence:

ATGGAAACGGCGAAACTTATGAAACTGGGCAACGGTATGTCAATTCCGTCCGGTTCA
GGAACCTGGCCAAATTGACGCTGGCTGAAATACCGTCGCGCTATATTTGCACCGTTG
AGAATTTACAGCTGCCGGTAGGTGCATCTGTTATTGATGATCATGAGACCGTTCCGG
TAATCGATATAGAAAACCTTATTAGTAGTGAACCGGTACAGAAAAGCTTGAACCTGGA
CCGCTTACATTCAGCGTGTAAGAATGGGGCTTCTTCCAAGTGGTGAATCACGGCG
TTGATACTTCATTAGTGGATAATGTTAAAAGCGATATTCAAGGTTTTTTTAACTATCGA
TGAATGAGAAAATCAAATACGGTCAGAAGGATGGGGATGTTGAGGGGTTCCGGACAG
GCATTCGTAGCAAGTGAAGACCAGACTTTGGATTGGGCGGATATTTTCATGATCCTG
ACTTTGCCGCTGCATCTGCGGAAGCCCCACCTGTTTAGTAAACTGCCGTTACCCTT
GCGCGAGACCATCGAATCCTATTCGAGTGAAATGAAAAAATTATCGATGGTCCTGTT
CGAAAAAATGGAGAAAGCCTTACAGGTTTACGGCGGTTGAAATCAAAGAAATTTCTG
AAGTCTTTAAGGACATGACCCAGGTAATGCGCATGAACTATTATCCACCTTGCCCTC
AACCGGAATTAGCAATAGGCCTTACACCGCATTCTGATTTTGGCGGTCTCACTATTC
TTCTCCAGTTAAATGAAGTCGAAGGCCTGCAAATCAAGAATGAAGGCCGGTGGATC
AGCGTTAAACCCCTTCCTAATGCATTTGTAGTAAATGTTGGAGACGTTCTGGAAATC
ATGACGAATGGTATGTATCGTTCCGGTAGATCATCGCGCGGTGGTGAACCTTACCAAA
GAACGCTTGAGTATTGCTACATTTTCATGATCCCAACCTTGAATCAGAAATTGGCCCA
ATCTCTAGCCTTATTACACCAAATACGCCAGCACTCTTTCGTAGTGGCAGCACATATG
GCGAACTGGTTGAAGAGTTCCATAGCCGCAAGTTGGACGGTAAATCCTTTCTAGATA
GCATGAGAATGTAA

NHI 109 (PODA) protein sequence:

METAKLMKLGNGMSIPSVQELAKLTAEIPSRYICTVENLQLPVGASVIDDHETVPVIDIE
NLISSEPVTEKLELDRLHSACKEWGFFQVVNHGVDTSLVDNVKSDIQGFFNL SMNEKIK

YGQKDG DVEGF GQAFVASEDQTL DWADIFMILT LPLHLR KPHLFSKLPLPLRETIESYSS
EMKKLS MVLF EKMEKALQVQAVEIKEISEVFKDMTQVMRMNYYPPCPQPELAIGLTPH
SDFGGLTILLQLNEVEGLQIKNEGRWISVKPLPNAFV VNVGDVLEIMTNGMYRSVDHRA
VVNSTKERLSIATFHDPNLESEIGPISSLITPNTPALFRSGSTYGELVEEFHSRKLDGKSF
LDSMRM

Codon-optimized NHI 110 (BX6) sequence:

ATGGCCCCGACCACAGCGACGAAAGATGATTCTGGATATGGCGATGAACGGCGCA
GAGAACTCCAGGCATTCGACGACACAAAATTGGGTGTCAAGGGATTAGTGGATAGC
GGAGTGAAAAGCATCCCTAGCATCTTTCATCACCCACCGGAAGCTCTGTCCGATATT
ATCTCTCCCGCTCCACTTCCGTCATCGCCTCCAAGCGGGGCCGCGCCATTCCGGTTGT
GGATCTTAGCGTCACGCGCAGGGAAGATTTGGTCGAGCAGGTGCGGCATGCGGCT
GGCACCGTTGGCTTCTTTTGGTTAGTAAACCATGGCGTCGCGGAGGAACTGATGG
GGGGCATGCTCCGCGGGGTACGCCAGTTCAACGAGGGACCGGTAGAAGCAAAGC
AAGCACTGTACTCAAGAGATTTAGCTCGTAACCTGCGATTTGCGTCGAACTTCGACC
TGTTCAAGGCGGCCGCCGCGCAGATTGGCGTGACACACTGTTCTGCGAAGTGGCACC
GAATCCACCACCCCGTGAGGAATTGCCTGAACCCCTTAGGAACGTGATGCTGGAAT
ATGGTGCTGCGGTTACGAAACTGGCCAGGTTTGTATTTGAACTACTTTCCGAGAGT
CTGGGAATGCCATCGGATCACTTGTACGAGATGGAATGTATGCAGAATCTGAATGTC
GTCTGCCAGTATTATCCCCCCTGTCCGGAACCACACCGCACAGTGGGCGTAAACG
CCATACGGATCCGGGTTTTTTTCACAATTTTACTTCAAGATGGCATGGGTGGTCTTCA
AGTGCGTCTTGTAATAACGGCCAGTCAGGTGGCTGCTGGGTGATATAGCCCCTA
GACCAGGTGCTCTCATGGTTAATATAGGCGATCTGCTTCAACTGGTCACAAATGATC
GCTTCCGCGAGTGTGGAACATAGGGTGCCCTGCAAACAAATCAAGTGATACCGCCCGC
GTGTCGGTGGCTTCCTTCTTTAACACCGACGTGCGGCGCAGTGAAAGAATGTATGG
TCCTATTCCGGATCCGAGCAAACCGCCTCTCTATCGCAGTGTACGAGCTCGTGATT
CATTGCAAAATTCAACACTATAGGGTTGGACGGCAGAGCCTTGGATCATTTTCGGCT
TTAG

NHI 110 (BX6) protein sequence:

MAPTTATKDDSGYGDERRELQAFDDTKLGVKGLVDSGVKSIPSIFHHPPEALSDIISPA
PLPSSPPSGAAIPVVDLSVTRREDLVEQVRHAAGTVGFFWLNVHGVAEELMGGMLRG
VRQFNEGPVEAKQALYSRDLARNLRFASNFDLFKAAAADWRDTLFCVAPNPPPREEL
PEPLRNVMLEYGAAVTKLARFVFELLSGLMPSDHL YEMECMQNLNVVCQYPPCP
EPHRTVGVKRHTDPGFFTILLQDGMGGLQVRLGNNGQSGGCWVDIAPRPGALMVNIG
DLLQLVTNDRFRSVEHRVPANKSSDTARVSVASFFNTDVRRSERMYGPIPDPSKPLY
RSVRARDFIAKFNTIGLDGRALDHFRL

Codon-optimized NHI 111 (BX13) sequence:

ATGGCTCCCACGGCAGCAAAGGACAGCGGTTCCGGCCGACAGGTGGCGCGAGGTT
CAGGCTTTCGAAGATTCCAAGTTGGGCGTGAAGGGGTTAGTTGATTCAGGCGTCAA
ATCAATTCCGGCGATGTTTCATCACCCCTCCTGAATCCCTGAAAGATGTCATCTCGCC
TCCGGCCTTGCCGTCATCTGACGATGCGCCGGCGATTCTGTTGTTGATCTGTCGG

TAGCTCGCCGTGAGGATCTGGTCGCACAGGTAAAGCATGCTGCGGGAACAGTTGG
CTTCTTTTGGGTTGTTAATCATGGAGTCCCGGAGGAATTGATGGCGTCTATGCTTAG
TGGCGTACGCCAATTCAACGAAGGGTCTCTGGAAGCTAAACAAGCTTTATATAGTAG
AGATCCGGCCCCGCAACGTGCGTTTCGCTTCGAACTTCGATCTTTTTGAAAGTGCGG
CGGCCGACTGGCGAGATACTCTATTGCAAAATTGCTCCTGATCCGGCTCCTCGC
GAATTAGTACCCGAGCCACTACGTAACGTCATGATGGAATATGGAGAAGAATTAATA
AACTGGCACGTAGCATGTTTCGAGCTTCTGTGCGAGTCTTTAGGCATGCCTAGTGAT
CATCTACATAAAATGGAATGTATGCAACAACCTCATATTGTCTGCCAATATTACCCACC
GTGCCCTGAACCGCATTAACTATCGGGGTTTCGAAACACAGTGACACCGGCTTTT
TCACTATCTTGCTCCAGGACGGTATGGGAGGTCTACAAGTACTGGTCGATAGAGGA
GGTGGCCGTGACACCTGGGTTGACGTAACACCGCGCCCTGGTGCCCTTATGGTTA
ATATGGGTAGTTTCTCCTCAGTTGGTCACTAATGACCGCTACAAAAGCGTAGACCATA
GGGTGCCAGCTAACAAATCAAGTGACACAGCTCGGGTGTGAGTTGCTGCCTTCTTC
AACCCGGATGAGAAGAGAACGGAAAGGCTCTATGGCCCGATCCAGATCCAGCA
AACCTCCTTTATATCGTAGCGTTACCTTTCCGGATTTTATCGCCAAATTCAATAGTATT
GGTCTGGATGGTCGTGTTCTGGACCATTTTCGACTTGAAGACGATGGTCCGACCCG
ACTGGCCGCACCTGCCCATCACGTGTAA

NHI 111 (BX13) protein sequence:

MAPTAAKDSGSADRWREVQAFEDSKLGVKGLVDSGVKSIPAMFHHPPESLKDVISPPA
LPSSDDAPAIPVVDLSVARREDLVAQVKHAAGTVGFFWVNHGVPEELMASMLSGVR
QFNEGSLEAKQALYSRDPARNVRFASNFDFESAAADWRDTLYCKIAPDPAPRELVPE
PLRNVMMMEYGEELTKLARSMFELLSESLGMPSDHLHKMECMQQLHIVCQYYPPCPEP
HLTIGVRKHS DTGFFTILLQDGMGGLQVLVDRGGGRQTWVDVTPRPGALMVNMGSFL
QLVTNDRYKSV DHRVPANKSSDTARVSVAFFNPDEKRTERLYGPIPDPSKPPLYRSVT
FPDFIAKFNSIGLDGRVLDHFRLEDDGPTRLAAPAHV

Codon-optimized NHI 112 (HPPD) sequence:

ATGTGTTTATCTCTGGCCAGCACCGCCCAGCGCAATACCCAGTTTAGAAGCCGCGT
CCTGGTACTTGCTGAGCTTGTTAAAAGTATGGGTCATCAGAACGCCGCGGTGAGT
AGAATCAGAATCATGATGATGGCGCAGCAAGCTCGCCCGGCTTTAAGTTAGTGGGG
TTTAGCAAATTTGTCCGTAAAAATCCCAAATCAGATAAATTTAAGGTTAAACGCTTTC
TCACATTGAATTTTGGTGTGGAGATGCGACAAATGTTGCCCGGAGATTTTCCTGGG
GTTTGGGTATGCGCTTTTCGGCTAAAAGTGACCTGAGTACGGGAAATATGGTTCAC
GCAAGCTATCTTCTGACCAGCGGAGACCTGCGATTTTATTTACTGCTCCATATTCTC
CATCTCTTAGCGCCGGCGAAATAAAACCGACAACGACCGCCTCAATTCCGTCCTTC
GACCATGGGTCTTG TAGATCGTTCTTCTCCTCTCATGGCCTGGGAGTGCGGGCCGT
CGCGATTGAGGTTGAAGACGCTGAATCCGCTTTCTCGATATCAGTTGCAAATGGGG
CTATTCCGAGCTCACCGCCTATCGTGCTGAATGAAGCAGTGACCATCGCTGAAGTC
AACTATACGGTGATGTTGTCCTGCGTTATGTAAGTTATAAAGCAGAGGATACGGAAA
AGAGCGAATTCCTGCCGGGCTTTGAGCGTGTAGAAGATGCCAGTTCTTTTCCACTC
GATTATGGTATTCGCCGACTGGATCACGCGGTGGGCAATGTTCCCGAACTGGGTCC
AGCCCTTACCTATGTGGCGGGCTTTACGGGTTTTTCATCAGTTTGCTGAGTTTACCGC

CGACGATGTGGGGACCGCGGAAAGTGGTCTGAACTCGGCGGTTCTTGCAAGCAAT
GATGAAATGGTCCTGCTTCCGATAAATGAGCCTGTGCATGGGACCAAGCGTAAGAG
TCAGATCCAAACATATCTAGAGCACAATGAGGGAGCTGGGCTGCAACACCTGGCCC
TTATGTCCGAAGATATTTTTAGAACTTTACGGGAAATGCGCAAGAGGAGTAGTATAGG
TGGTTTTGATTTTATGCCTTCGCCTCCCCCTACTTACTATCAAAACCTAAAAAACGT
GTGGGGGATGTGCTGTCTGATGATCAGATAAAAGAATGCGAAGAGCTGGGTATACT
GGTTGACCGGGATGATCAGGGTACTTTACTACAGATCTTCACTAAGCCGCTTGGTC
GTCGCTACTTATAA

NHI 112 (HPPD) protein sequence:

MCLSLASTAQRNTQFRSRVLVLAELVKSMGHQNAAVSENQNHDDGAASSPGFKLVGF
SKFVRKNPKSDKFKVKRFHHIEFWCGDATNVARRFSWGLGMRFSKSDLSTGNMVHA
SYLLTSGDLRFLFTAPYSPSLSAGEIKPTTTASIPSFHDHGSCRSFFSSHGLGVRAVIEVE
DAESAFSISVANGAIPSSPPIVLNEAVTIAEVKLYGDVVLRYVSYKAEDTEKSEFLPGFER
VEDASSFPLDYGIRRLDHAVGNVPELGPALTYVAGFTGFHQFAEFTADDVGTAESGLNS
AVLASNDEMVLPPINEPVHGTKRKSQIQTYLEHNEGAGLQHLALMSEDIFRTLREMRKR
SSIGGFDFMPSPPTYYQNLKKRVGDVLSDDQIKECEELGILVDRDDQGTLQIFTKPL
GRRYL

Codon-optimized NHI 113 (KthP) sequence:

ATGACAACGGCCGATGAATTCGCCTTAAGCGACGAACAGCTGGACTCCTTTTCGACG
CGACGGTTTTTTTAGGCCCTTTGACTATCTATCCTCCGGATGAAATGGCGGAACTCTG
GAAATCAGTCCAGCGTCAAGCATTAGATCGAACACATGCAGCTTATCCGGAAGTGG
ATGCGGGTATGGGCGCCCCCTAATATATTCAATTATGATAGACATCTGGATATAGATCTG
TTGGCAGAGCACGCTTGTCATCCTAAAATTGTTGGCGCGGTTGGCTCAATTCTGGG
CCGCGACCTACTGTGTTGGAGGTCAGAATTCTTCCCAAATATCCTGGTGATGAGG
GCACAGATTGGCATCAGTCAGATACCTTCGCAAATGCCTCCGGTAAGCCACAATTG
CTATGGCCGGGTGAATCCGAGGAATCTTTCGGCGCCGGCACGATCACGGTGTGGA
CCGCCTTTACAGACTCAACCCGTGAAAACGGTTGTCTTCAGCTCATTCCGGGAACA
CATAATAAAATGAATTATGATGAAACGAAAAATATGGATTATGACCAAGAACGTATCAA
TGCCTCCGTAAAAGATGGTGTAACCGTGTTTCTTTGGGTATGACTATCGGCAGCT
CCAAATTGATCCAGATTGGACTCCGGATGAAGATGATGCCTTTTCGTTAGTCATGGA
AAAAGGACAGTGTGTTATATTCTGGTCTACACTAATGCATGCAAGTCTGCCGAACAC
TAGTACGGGCCGGGATTACCGTATGGGTTTTGCTACGCGTTACGTTCTACGGCGG
TTCGCGTGTATCCGGATACAGATCATATTGAAGAGTACGGTGGGAATATCTCTCTAGA
TCGTTGGGGAGCTATGTTGGTGAGCGGCCGTGACGAACACGGCCATAATAAAGTG
GTGGATCGGACGCTCCGTGGGACTCCAGTGGGGCCTGCAGCGGCTAGACGTCCG
TAA

NHI 113 (KthP) protein sequence:

MTTADEFALSDEQLDSFRRDGFLGPLTIYPPDEMAELWKSQVRQALDRTHAAYPEVDA
GMGAPNIFNYDRHLIDLLAEHACHPKIVGAVGSILGRDLLCWRSEFFPKYPGDEGTD
WHQSDTFANASGKPQLLWPGEESEESFGAGTITVWTAFTDSTRENGCLQLIPGTHNKM

NYDETKNMDYDQERINASVKDGVKRGFFGYDYRQLQIDPDWTPDEDDAFSLVMEKG
QCVIFWSTLMHASLPNTSTGRDYRMGFATRYVPTAVRVYPDTHIEEYGGNISLDRWG
AMLVSGRDEHGHNKVVDRTLGRTPVGPAAARRP

Codon-optimized NHI 114 (WelO5) sequence:

ATGAGCAATAACACAGTTTCTACAAAACCGGCGCTACATTTTTTAGACATCAACGCTA
CTGAAGTAAAAAATATCCGACTGCCATACAAGATATTATTATTAACCGTAGCTTCGAT
GGGATGATAATTCGCGGAGTCTTCCCCCGGGATACCATGGAACAGGTGGCACGCT
GCTTGGAGGAGGGTAACGATGGAGGGATGAAATCAATTCTTAATAAGAACGAAGAAT
TCGGCACCAAAGTTGCCCAGATTTATGGCCACGCCATTGTGGGACAGAGCCCGGA
TTTAAAGATTATTTTGCATCGTCGGCAATCTTTCGCCAGGCTTGCCGAACGATGTTT
CAGGGAAGCCCTGACTTTGAGGAGCAGGTGGAATCAATATTTTATTCTCTGTCTGG
GTTGCCGGTGGAAATTCCAACCTGGCCCAGAAGGCCAAACATATACGCCTGCAACTA
TCCGTCTTTTACTGGAGGGGCGCGAGATCGCCGTGCACGTCGGTAACGATTTCTTA
CTCATGCCGGCTGCCAATCATCTGAAAACACTGCTGGATTTAAGTGATCAACTGTCTG
TACTTTATACCATTGACCGTACCTGAGGCAGGTGGGGAATTAGTCGTTTATTCTCTG
GAATGGAACCCACAGGAAGCAAGCAAATACGCACAAATGCAGGAGTATATGGATGA
TGTTGAATTCAAGATTAAGAGTAACCAGAGCCAGAGCGTAGCCTATGCACCGGGGCC
CTGGTGACATGTTACTCTTTAATGGGGGCCGCTATTACCATAGAGTTTCAGAAAGTTAT
TGGAATTCTCCTCGGCGTACAATAGGCGGTTTCCTGGCATTCTCGAAACAACGAG
ATAAGATATACTACTGGTCTTGA

NHI 114 (WelO5) protein sequence:

MSNNTVSTKPALHFLDINATEVKKYPTAIQDIIINRSFDGMIIRGVFPRDTMEQVARCLEE
GNDGGMKSLNKNEEFGTKVAQIYGHAIVGQSPDLKDYFASSAIFRQACRTMFQGS
PD FEEQVESIFHSLSGLPVEIPTGPEGQTYTPATIRLLLEGREIAVHVGNDFLLMPAANHLKT
LLDLSDQLSYFIPLTVPEAGGELVVYSLEWNPQEASKYAQMQEYMDDVEFKIKSNQSQ
SVAYAPGPGDMLLFNGGRYYHRVSEVIGNSPRRTIGGFLAFSKQRDKIYYWS

Codon-optimized NHI 115 (AmbO5) sequence:

ATGAAAAACAACACTCGTAATTTCAATATTGGCAACGTGATCCACGGTAAACGGCTG
ATTATCACGGAAGAAAAAATGATGAGTAATAATGCCGTGTCGACTAAGTCAGCACTG
AATTTTCTGGACATTAATGTAACGGAAGTTAATAATTATCCCACTGCTATTCAGGATAT
TATTATTGATCGTCGTTTTGATGGAATGATCATAACGGGGGGTTTTGCCCTTAGATAGC
ATCGAACGTGTGATCCGTGCCTGGAAGACGAAGATGAAGGGGGTATGAAACTCAT
TTTTAATAAAAAACGAAGAGTTTGGCACGAAGGTTGCACAGATTTATGGTCATGTTATC
GTTGGTCAGAGTCCAGACCTAAAAGACTACTTTGCGAGTTCCGCAATTTTTAGACAA
GCTTGTAGAGCCCTGTTTCAGGGTAACCCAGATTTTGAAGAACGCGTCGAATCCGT
GTTTCATAGCCTGTGTGGGTACCTGTAGAAATCCCCACGGGGCCAGAGGGTCAAT
CATATACCCAGCCACTTTACGCCTCTTGACCGAAGGACGCGAAATTACTGTCCATG
TGGGTAACGATTTCTGCTTATGCCTGCTTCAGAACATTTAAAGACACTGTTAGATTT
TTCCGACCAGCTGAGCTATTTTATCCCTTTATCCGTCCCACAAGCTGGGGGCGAATT
AGTCGTCTATAATTTGGAATGGAATCCTGAGCAGGCTGAACAGTCTGGTGACCTGC

ACAAGTACATGAACGACGCAGACAGCCGCTTTCAAAGCCAACAAAGCCAGAGTGTT
GCTTTTGC GCCCGGACCAGGTGACATGATCCTGTTCAACGGTGGCCGCTATTATCA
TCGTGTTAACCAGGTGATCGGAAATTCGCCGAGACGCACCATTGGTGGTTTCCTGG
CTTTTAGCAAAGAGAGAAACAAAATTTACTATTGGTCTTGA

NHI 115 (AmbO5) protein sequence:

MKNNTRNFNIGNVH GKRLIITEEKLMSNNAVSTKSALNFLDINVTEVN NYPTAIQDIIIDR
RFDGMIIRGVLP LDSIERVIRLEDEDEEGM KLIFNKNEEFGTKVAQIYGHVIVGQSPDL
KDYFASSAIFRQACRALFQGNPD FEERVESVFHSLCGLPVEIPTGPEGQSYTPATLRLL
TEGREITVHVGNDFLLMPASEHLKTL LDFSDQLSYFIPLSVPQAGGELVVYNLEWNPEQ
AEQSGDLHKYMN DADSRFQSQQSQS VAFAPGPGDMILFNNGGRYYHRVNQVIGNSPR
RTIGGFLAFSKERNKIYYWS

Codon-optimized NHI 116 (DAO) sequence:

ATGGTCGAAATTCCAGCTATCGATCTGCGCCTTGCGGGTGGCGGCGGTGGCGCGG
AAGAGACCGCACGCCTTCGCGATGCCTGTGCCCGTTTAGGATGTTTTCTGTGTCTCC
GGGCATGGCGTACCGCCGGGCCTGCAGGCTGAGATGAAAGCGGCCGTCAGGGCA
TTGTTTCGATCTGCCAGACGACGCAAAAAGGCGTAATGCGGATATAATTCCTGGTAGT
GGGTATGTCCCTCCTGGTACCGCCAACCCTCTATACGAAGCCTTTGGGCTTTGTGA
CGCTGCCGCTCCTGCTGATGTAGATGCGTTCTGCGCAAGACTCGATGCACCGCCG
CACGTCCGGGAAACCGTGAAAGCATACGCGGAACGTATGCATTCCTCATTGTTGA
TGTCGCTGGAAAAGTAGCTGCGAGTTTAGGTCTCCACGGTGCCTCTTTTCAGGATT
GGCCTTGTCAGTTTAGAATGAATCGCTATAACTATACACAGGATTCAGTTGGCTCAC
CGGGCGTTCAAGTTCACACTGATTCCGGTTTTCTGACAGTTCTGCAGGAGGATGAA
TGTGTGCGGAGGGCTTGAAGTCCTAGATCCGGCGGCAGGCGAGTTTGTACCGGTAG
ATCCGCTGCCGGGATCTTTTGTGGTGAACGTTGGGGATGTTGGTCAAGCATGGTCC
AACGGTCGGCTGCACAATGTTAAGCATCGTGTGCAGTGTGTGCGCCGCTGTCCCAC
GGGTCAGTATTGCTATGTTTTTGTGGCGCCAAAAGATGACACGGTTAGTGCACCG
GGGGAAC TTGTAGATGGTGAGCACCCGCGAAGGTATCGGGAATTTAAATATGACGA
CTACCGCCGTTTACGCCTCTCCACCGGGGAGCGTGCAGGTGAAGCACTAGCGCGT
TTGGCCGCATAA

NHI 116 (DAO) protein sequence:

MVEIPAILRLAGGGGGAEETARLRDACARLGCFRVSGHGVPPGLQAEMKAAVRALFD
LPDDAKRRNADIIPGSGYVPPGTANPLYEAFGLCDAAAPADVDAFCARLDAPPHVRET
VKAYAERMHSLIVDVAGKVAASLGLHGASFQDWPCQFRMNRYNYTQDSVGS PGVQV
HTDSGFLTVLQEDEC VGGLEVLDPAAGEFVPVDPLPGSFVVNVGDVGQAWSNGRLHN
VKHRVQCVAAPRVSIAMFLLAPKDDTVSAPGELVDGEHPRRYREFKYDDYRRLRLST
GERAGEALARLAA

Codon-optimized NHI 117 (M2H) sequence:

ATGGTAGCACCTAGCCGCCTACCGAGTCATGAAGAACAGAGCGCAGCGGCAGCAG
CCGATGGCAGCGCGACACCCTACAAGGAATTCGGTGGTTGATCTGGGGGTTCT

GATTAACGGTGCGGCTGATGAACGGAGCCGTGCTATTTCGTGATCTAGGTCGCGCCT
GTGAAGACTGGGGCTTTTTTCATGGTTACCAACCATGGTGTACCGGAGGCCTTACGA
GAAGCGATCATGGATGCATGCAAAGAACTGTTCCGTTTGCCTTTGGAGGAAAAGAA
AGAATACATGCGTGCAAACCTATGGATCCTATAAGGATTGGGACCGGGTTTTATTCT
GTCGTTGACGCAGTGCCCTGTCGCAGAGACTATCTCAAATGTTCTCACATCCCGA
GTTTCACTGTCCGGAGAAACCGGGCGAAGTTACGGGAAATTGCTACTGAATATGCGA
CCTGCACCCGCGCCTTGCTGCTGGAATTAACCAAAGCTATATCTGAATCACTTGGCT
TGGCAGGGGGCCGCCTGTCGGAGGGCGCTAAACCTCGAGTCGTGCTTTCAGATCCT
TGTTGGCAACCATTATCCGGCTTGTTCAAGACCCGATGAACAAGCAATGGGGTTGA
GCGCGCACTCTGATCATGGGTACTGACCCTGCTTTTCCAGAATGGAGTGGATGGG
CTGCAGGTAAACATGATGGCGAATGGTTGCTCGCGAAACCACTGCCTGGCTCATT
TTTCGTAATCGCTGGCGATCAGCTTGAGATCGTCACCAACGGTAGATATAAAGGTGT
GCTGCATCGTGCAGTCGTGCGCGGTGAACAGTCTCGCATGAGCTTCGTGTCACTG
ATAGGTCCATGCATGGATACGGTAGTAGAGCCGCTGCCGGAATGGCAGCCGATGG
TCGTGGCCTGGAATTTCTGTTGGGATCAGATACAGAGATTATATGGAAATGCAACAGTC
TAATAGCATAAACGAAAAACGGCTTTAGATATCGTTCGGGTATGCATCAAGCTGGC
TAA

NHI 117 (M2H) protein sequence:

MVAPSRLPSHEEQSAAAAADGSATPSQGIPVVDLGVLINGAADERSRAIRDLGRACED
WGFFMVTNHGVPEALREAIMDACKELFRLPLEEKKEYMRAKPMPIRIGTGIFYSVVDA
VPCRRDYLKMFHPEFHCPEKPAKLREIATEYATCTRALLLELTKAISESLGLAGGRLSE
ALNLESCFQILVGNHYPACSRPDEQAMGLSAHSDHGLLTLFQNGVDGLQVKHIDGEW
LLAKPLPGSFFVIAGDQLEIVTNGRYKGVLRHAVVGGEQSRMSFVSLIGPCMDTVVEPL
PEMAADGRGLEFRGIRYRDYMEMQQSNSINEKTALDIVRVMHQAG

Codon-optimized NHI 118 (EcdG) sequence:

ATGGCTTCAATCTCATACGATGGCCTGTTACGAGGCGCGGCAAAGGACATCCAGAC
CCTGACTGAAGCCTCGCTGGTGCATGGGTACTTTAATCTCGAGCTAGACTGTCCTG
AAGGTAAACAGTTAAGAGAAGATGTGCTGTTTCTGGAGTCCTTCACCAAAGAAATTT
TTGATACCCCGTCTCCCCAGAAAACCATTTATGATTTTAAAAAACTCGGCAGGTTCC
GTACTACCGGTTTCAAGCCATTGGGTATCGAAGAAGGCGCGAAAGGTAAATCGGAT
GGTTTTGAAATGTTTATGCTACCGCAGAACGAGCTGCTTTTACCTTCCCACCAGGAT
AACTTCAATCCCCCTCGGCCGTTTTTAGTCACCGGGCAACCCTGACAAGATGTAT
GTCCAATTATGATGCGGCAAGCCAGCTGATTCTTAGACGCATTACGGAATCACTGAA
TCTGGATAATGCATTGTTGGCTGCTCACAATCCTAGTGAACCCTCAGTAACTAATT
GGGTTTTTTACGTTATCCGCCACAGCCTGCGACTTCAGAGAATTGCGGGCATATCG
CACATACTGACGTTGGCACCCTGACGATCCTTGCAGCGACACAGCGGGGTCTGCA
GGTGATAAATTCTGAGACCCAGGATTGGACCTTTGTTGATCCGCATCCGCAAACG
AATTTCTGCTAGTCCAATTTGGAGATTGTCTGAAATTTTAAGCGGCGGTTCGTGTGG
TGCCGTCGATTCATCGCGTTATTCCGTCCGATAATCCCGAAGAAAGGGAAGGTACG
AAATATACCCTTGCAATTTTGTAAAGACCTAACGAGGAAGCGGTTATCAAAGATGATC
GTGGCGAGGAATGGGTCTATGGTGACTATCACTGTCGTAAATTTGGTGCATTTGCG

CGTCCACTCAAAGAGGGCGATCGTGACCGGATGCCAGATAGGCCTGTGGGCGAAT
ATGATATGATTAAACATTCGCAAATTTAAGGGTCTGGCTGACGGAGTCAGTGCGGCTT
AA

NHI 118 (EcdG) protein sequence:

MASISYDGLLRGAAKDIQTLTEASLVHGYFNLELDCPEGKQLREDVLFLESFTKEIFDTP
SPQKTIYDFKKLGRFRTTGFKPLGIEEGAKGKSDGFEMFMLPQNELLLPSHQDKLQSP
SAVFSHRATLTRCMSNYDAASQLILRRITESLNLDNALLAAHNPSEPSVTNLGFLRYPQ
PATSENCGHIAHTDVGTLTILAAATQRGLQVINSETQDWTFVDPHPQNEFLLVQFGDCLK
FLSGGRVVPSIHRVIPSDNPEEREGTKYTLAYFVRPNEEAVIKDDRGEWVYGDYHCR
KFGAFARPLKEGDRDRMPDRPVGEYDMINIRKFKGLADGVSA

Codon-optimized NHI 119 (AniG) sequence:

ATGGCGTCTATTAGTTATGAGGGTCTTCTGAGAGGAGCAGCTAAGGACATTCACGC
CCTTACCGAAGCCGCATTAGTGCATGGATACTTTAACCTCGAGCTAGATTGTCCAGA
AGGACAACAGCTCAGACAGGATGTGTTATTTCTAGAATCTTTTACCAAAGAGATCTT
CGATACACCGTCCCCCAGAACTGTGTATGATTTTAAAAAACTTGGTAGATTTTCG
CACCCTGGTTTCAAACCACTAGGTATCGAGGAAGGGGCTAAAGGTAAGAGCGATG
GCTTCGAGATGTTTCATGTTGCCTCATAATGAGCTGCTGCTGCCTACGCATCAGGATA
AGCTGCAGTCCCCATCTGCCGTCTTTGCCACCGGGCAACATTAAGTAGATGCATG
AGTAATTACGAGGCAGCATCACAGCTAATTTTACGTAGAATAACGGAATCGTTAAACC
TGGACAACACTCTGTTGGCGGCGCACAATCCAAGTGAGCCATCCGTCACCAATTTG
GGTTTTTTACGTTATCCTCCGCAGCCTGCGACCAGCGATAATTGTGGACATATTGCG
CATACCGATGTAGGGACCTTAACGATTCTCGCTGCAACCCAACGCGGTTTACAGGT
TATTAATTCGCAGACACAGGACTGGACGTTTGTGATCCGCACCCTCAAAACGAAA
GTTTACTCGTGCAGTTTGGTGATTGTCTGAAATTTCTGTCAGGAGGACGTGTTGTAC
CTTCTATCCACCGGGTTATCCCGAGCGATAAACCGGAGGAACGGGAAGGCACTAAG
TACACACTCGCTTATTTCTGTCGCGCCCGAACGAAGAAGCTGTCATTAAAGACGGGCG
AGGCGAAGAATGGGTATATGGCGACTATCATTGCCGTAAATTTGGGGCTTTTGCACG
TCCTTTGAAAAAGAAGGTGATAGAGATAGGATGCCGGATAGACCTGCTGGCGAGT
ACGATATGATTAAATTCGTAAGATAAAGGGCTTATCTGAGGGCGTGTCCGCCGCTTA
A

NHI 119 (AniG) protein sequence:

MASISYEGLLRGAAKDIHALTEAALVHGYFNLELDCPEGQQLRQDVLFLLESFTKEIFDTP
SPQKTVYDFKKLGRFRTTGFKPLGIEEGAKGKSDGFEMFMLPHNELLLPTHQDKLQSP
SAVFAHRATLTRCMSNYEAASQLILRRITESLNLDNTLLAAHNPSEPSVTNLGFLRYPQ
PATSDNCGHIAHTDVGTLTILAAATQRGLQVINSQTQDWTFVDPHPQNESLLVQFGDCLK
FLSGGRVVPSIHRVIPSDKPEEREGTKYTLAYFVRPNEEAVIKDGRGEWVYGDYHCR
KFGAFARPLEKEGDRDRMPDRPAGEYDMINIRKIKGLSEGVSA

Codon-optimized NHI 120 (PhqC) sequence:

ATGTCTAAAACCAACCGAAGTGAAAGTGGTTGAAGTCCAGCTGGAAGACCCGCTGC
CGGGGATTGTAGACGCGTCACGTTTTATCTCCGGTTACCAACTGAACAACGCGCT
TTCGCGGTTGAGCTGGTGGATTCCGTTTCGACGTTGCGGATTTGTAAAGGTGATCAA
TCATGGGCTTTTCGGACGAACCTCATAGATGAATTATTCGCCTGGTGTCACTCCTCTCA
CGTAGTACAAAACGAAAGGTTTTTTGCAATTGACCCAGAGCAAAAGTTGGCGGTAG
TCAACCCGCCAGGACCGTCTCCACAGCGAGGCTGGTCTTGTGTGGGAGCGGAAA
AGGCCAGCCGTTTATTTTCCCGTGGGCAGACAAGTTTGGATTTAACAGACGCCCGC
CGAAACCAGGAACACTTCGACGCCGGATCTCCGAGCGATACAAAATGGCCTAGTAG
ATGGCCGGATGAGGCGGTCATACCCGGTTTCAAAGCATTTCTAGAAGACTTTTATGT
GCGGTCTCATCAGGCAGCTCTGCTTATACTGGAAGCTCTGGAAATGGGTTTAAACC
TGCCGGCTGGAGTTCTTAAGAGCCGGTGTGGGGGCTGCGCGTCGGAAGTGAAGATT
AAATAACTACCCCGAAATTGATATTGAAGAACTGCGGCGTGGAAAGATTAGCCGCAT
TCATCCACATGCAGATCTCGGTGTTATTACGTGCCTTTTTCAAGACGTTTGGGGGG
ACTGGAATTAGAACATCGTAGCCATGCTGGTCTTTCTGCCTGTTCCACCAGGGG
CGAGATCAGAGATGGTAGTAAATATTAGTGAGACATTCCAAGTGTGGACCAACAATG
TTATTACCGCTGGCATAACATCAGGTGACAGTCCCTCCGGAAATGAAAACACGCACC
GAAGGACGTATTAGCGCTCGCCGCTCATGCGCCTTTTTTTTAAAGCTAATGGAGAT
GCGTCTGTTGCCCCCTGCCACAGTTTGTTACACAAGAACGTCCGGCTGCATATTC
AGAAATGACGGCTCTTGATTACCACCAGAAACGGTTAGCAACAGCTTACTGA

NHI 120 (PhqC) protein sequence:

MSKTTEVKVVEVQLEDPLPGIVDASRFISGSPTEQRAFAVELVDSVRRCGFVKVINHGL
SDELIDELFAWCQSSHVVQNERFFAIDPEQKLAVVNPPGPSPQRGWSCVGAEKASRLF
SRGQTSLDLTDARRNQEHFDAGSPSDTKWPSRWPDEAVIPGFKAFLDFYVRSHQAA
LLILEALEMGLNLPAGVLKSRCGGCASELRNNYPEIDIEELRRGKISRHPHADLGVITC
LFQDGLGGLELEHRSHAGSFLPVPPGARSEMNVNISETFQLWTNNVITAGIHQVTPPE
MKTRTEGRISARRSCAFFLKANGDASVAPLPQFVTQERPAAYSEMTALDYHQRLATA
Y

Codon-optimized NHI 121 (ImqE) sequence:

ATGACGATGTTGTCTGACCTGCCAAAGATACCTACCCTGGATTGGGCTGATTTTGCC
GAGGGTGATACGGATCAGCGTTTAAACTGGCACAGGGTCTGGTACAGGGTTTTAA
ACGCTTTGGCTTTGTAAATTAGTGAACCATGGTCTGTCTGATGAACTGATTCAGCA
GTTGTTGCGCGAGGTTAAACGCTTCTATCGTCTGCCTGACGAACTGAAACAGAAAG
CTGCGCACCCGCCCGGGCCTAATCCTCAGCGCGGTTGGTCTGGCATTGGAGTTGA
ATCCACGAGTAAGTTGTATGGTGAACAAACCGAGCGGCCTTCTGGTAAGTTAAAG
ACGCCAAGGTCGGGACAGATTTCTCAAGCAGTAACTCTCCCGAGAGCTGACCCA
CATGAAAGAACATTATGACATTGGACCACCGACAGACACGCAGTTTCCGACCCGCT
GGCCAGATGAGCAGGACATCCCAGGATGGCGGGCGTTTATGGAGAGCTATTACGC
CCGTGGGCAGTCTTTCTGTTTAGACTTAATGGAAGCGTTAGAAATCGGCTTAGAACT
CCCTAAGAACACACTGCGTTCCATGTGCATACCCGATGGATCCGAGCTTCGGCTGT
TACACTATCCTGAGATTCCTGCAGCCGAATTGAGAAGTGGTGACACGGCGAGAATT
TGCCACACACCGATTTTGGACTGATCACACTGCTTTTCCAGGATGGAGTGGGTGG

TCTGGAAGTTGAAGATCCCCTCCAGCAGGGGGCATTACATTGAAGTTGCTCGGGAGC
AGCCTTATGAAATGATTGTCAATGTTAGCGCTACCTTTGAGCGTTGGATGAATGGTG
TAATTAAAGCGGCGGTACACAGGGTAAATATAACGCCAGAAGGCAAACACGTCGAG
GATGCAGTTGTGCCTGAACGCTGGTCAGCAGCTTATTTTTTTAAAGCTCACAAAATG
GCGCATGCTGGTCCTCTTCCAGCGTTCGTTACTCCCGAACGCCCCGGCCTTATATGA
TAACATAACAGCACTTGAGTTTCAGAAACGGCGTACTGATCTGGTATATACAGGCCA
GCAGCTAAAAGTTGAGGAAGCAGCCTGA

NHI 121 (ImqE) protein sequence:

MTMLSDLPKIPTLDWADFAEGDTDQRLKLAQGLVQGFKRFGFVKLVNHGLSDELIQQL
FAEVKRFYRLPDELKQKAAHPPGPNPQRGWSGIGVESTSKLYGEQTERPSGKLKDAK
VGTDFFSSSKLSRELTHMKEHYDIGPPTDTQFPTRWPDEQDIPGWRAFMESEYYARGQS
FCLDLMEALEIGLELPKNTLRSMCIPDGSELRLHYPEIPAAELRTGDTARIWPHTDFGLI
TLLFQDGVGGLEVEDPLQQGHYIEVAREQPYEMIVNVSATFERWMNGVIKAAVHRVNIT
PEGKHVEDAVVPERWSAAYFFKAHKMAHAGPLPAFVTPERPALYDNITALEFQKRRTD
LVYTGGQQLKVEEAA

Codon-optimized NHI 122 (UcsF) sequence:

ATGGGCGTCCCCTCCTGGACGTGTCACAGCTGCAGGCGGGCCCAGAAGCCCGT
AAACAATACTTACAAGCCCTGGTCCAGAGTTTTTCGGGACTATGGCTTTGTCCGTCTT
ACCAAACATGATGTTCCGGCTAAGCGGGTACAACGCATATTCGACTTAAGCACCCA
GATGTTTAATCTTGATATCGATAGCAAATTAGAGTTCGCGAATATTGCGGATGGCAGT
CCACAGCGTGGGTATTCAGCTGTGGGCGTAGAAAAAACCGCATCACTGCATGGCAA
TCTGATAGGCAGACGTGTGGATGAGAACTGACAGACGCTCGGGAACACTTTGATT
GCGGATCCCCACTGGACAAGTCGTTGCAAAATCGTTGGCCTGAGAACTGCAGGG
TTTTCAGCAGGAATTGGAGTCTTTTTATTTTCTGAACTGGAACAGGTAACAGCTGGGAT
ATTGGGCAGCTTGGAAGAAGCTTTAACTGCCCCCCTGGTACGTTGAACAACATGA
TTACGAAAGAGAACAACGCTTCTGAGCTGCGACTCAACCACTACCCGCCTGTCCCT
GCAGGCACGCTGAGAAATGGTAACGTAGCACGGATCTGGCCGCATTTTGACCTGG
GCGTCATTACCTTGTTATTTACTAGTGCCGTTGGGGGATTGGAAGTTGAGGATAGAA
ATGCACCCGGGCCACAGACTTTTATACCCGTGGAACCAGAGACCGAAGCGGAGTT
GATTGTCAATATCTCAGAAACCTGCAGCGGTGGACAGATGACCATTACCTGCCG
GGCTGCACAGAGTTACAATACCGAAAGATCTGGACACCGAAATACAGAACGATGCA
AATGTAGAAATTCCAGGTCGTTATAGCATTGCGTATTTGTGCAAGGCCGACCGCGAG
GCAGATGTTGGAACCTCTGCCCGTTTTCCAGACGGGTGAAGCTCCGAGGTATAAAGC
CATGACGGCTTCTGAATATCATCGATCTCGGTTGCTGACCGCGTACTAA

NHI 122 (UcsF) protein sequence:

MGVPLLDVSQLQAGPEARKQYLQALVQSFRDYGFRVRLTKHDVPAKRVQRIFDLSTQMF
NLDIDSKLEFANIADGSPQRGYSAVGVEKTASLHGNLIGRRVDEKLTDAREHFDGCSPL
DKSFANRWPEKLQGFQQELESFYFELEQVTAGILGSLEEALNCPPGTLNNMITKENNA
SELRLNHYPVPAGTLRNGNVARIWPHFDLGVITLLFTSAVGGLEVEDRNAPGPQTFIP

VEPETEAEELIVNISETLQRWTDHLPAGLHRVTIPKDLDTIQNDANVEIPGRYSIAYLCK
ADREADVGTLPVFQTGEAPRYKAMTASEYHRSRLLTAY

Codon-optimized NHI 123 sequence:

ATGGGCTCGTTGGAAGTTCCATGTATTGACCTGAGCGAGAATGATACTAGTATCGTT
GTTAAAGAATTACTGGATGCGTGTA AAAATTGGGGTTTTGTATCCTTGAAAAATCATG
GCATCCCGCTGGACGAAATTGATCGGACTTTTAAACTTGCAGATAAATTCTTTGATAT
TCCAGTCGAAGAAAAACAGAAATACCTGTTTAAAGGTGGTCGTCTGCACTCAGGCT
ATACTGGTCACTTTGGTGAAAAGCTCGATATGGAACATCAGTCAAGGGGTGATCTGA
AAGAAAGTTACGACTTGGCAGGATTCCCAGATCCTAAACTTGAGAACCTCTGTCCCT
TTATCGCTGAACACATGGATGAGTTTCTGCAATTTACGCGCCATTGCTATAAGCTTAC
CTTAAGATTGTTAGATTTTTTTTGTCTATTGGATTCCGGTATCCCCCAGATTTTTTTTCAA
AATCACACTCTTCAGAAGAAGACGTGCTGCGACTGTTGAAATACTCAATACCAGAAG
GTGTTGAACGGCGTGAAGATGACGAAGATGCCGGTGCCCATTCAGATTATGGCTCA
ATCACTTTACTGTTTCAGCGCGATGCAGCTGGGTTGGAAATACGCCCCCCTAATTTT
GTGAAAGACATGGATTGGATAAAAGTGAATGTGCAACCTGATGTTGTCCTGGTGAAT
ATTGCGGATATGCTCCAATTTTGGACAAGTGGGAACTAAGGTCAACAGTTCACCGT
GTCCGCATTGATCCGGGAGTAAAACTCGTCAGACTATCGCGTACTTTGTTACTCCC
GATCCTGAAACTCCACTTTCACCTCTGTTTGAAGAAAAGACAGGCAAGGATATTGAA
ACCGTAACCGCCGGTGAATGGATTGATGGCCGGATCAATTTTACATACGGGTATAGT
GCCCCGCCGAAAGGGTATCTCGGAAGCCAGAACGATGGCATTGTGGCTTGA

NHI 123 protein sequence:

MGSLEVPCIDLSENDTSIVVKELLDACKNWGFVSLKNHGIPLDEIDRTFKLADKFFDIPV
EEKQKYLFKGGRLHSGYTGHFGEKLDMEHQSRGDLKESYDLAGFPDPKLENLCPFIA
EHMDEFLQFQRHCYKLTLRLLDFFAIGFGIPPDFFSKSHSSEEDVLRLLKYSIPEGVERR
EDDEDAGAHSDYGSITLLFQRDAAGLEIRPPNFVKMDWIKVNVQPDVVLVNIADMLQ
FWTSGKLRSTVHRVRIDPGVKTRQTIAYFVTPDPETPLSPLFEEKTGKDIETVTAGEWID
GRINFTYGYSAPPKGYLGSQNDGIVA

Codon-optimized NHI 124 (ANS) sequence:

ATGGCCACATGCTGGCCCGAGCCGATTGTAAGTGTACAGTCGCTCAGTCAAACCG
GAGTACCAACGTTCCGAATCGGTACGTTAAACCGGCCCATCAACGCCCCGTATTT
AATACGACACAGAGCGATGCGGGTATAGAGATACCTGTGTTAGATATGAATGATGTG
TGGGGAAAACCGGAAGGGTTGAGGCTGGTTCGCTCAGCTTGCGAGGAGTGGGGG
TTCTTTCAAATGGTCAACCACGGTGTACCCATTCGTTAATGGAACGTGTGAGAGGA
GCCTGGCGGGAGTTTTTTGAACTCCCCTTGAGAGGAAAAGCGTAAATACGCAAATTC
GCCTGATACGTACGAAGGATATGGAAGCCGACTGGGTGTGGTGAAAGACGCCAAA
CTGGATTGGTCGGACTATTTCTTCCTGAATTATCTTCCAAGCAGCATCCGAAATCCAT
CGAAATGGCCAAGTCAACCTCCTAAAATTCTGTAACCTATCGAAAAATACGGCGAGG
AAGTGAGAAAACCTCTGCGAGCGGCTCACCGAACTCTGAGCGAATCGTTGGGACT
CAAACCGAATAAACTTATGCAGGCTCTGGGCGGCGGAGATAAAGTGGGTGCCTCG
CTGAGAACCAATTTCTATCCGAAATGCCCCCAGCCGCAGCTGACACTTGGTCTGTC

CTCACATAGTGATCCGGGTGGCATTACTATCCTGCTGCCGGATGAAAAAGTTGCGG
GCCTGCAGGTACGCAGAGGCGACGGTTGGGTCACGATCAAATCTGTCCCGAATGC
GCTGATCGTGAATATTGGTGATCAGCTCCAGATTCTGAGCAATGGTATATACAAATCA
GTTGAACATCAGGTTATCGTCAATTCAGGCATGGAAAGAGTCTCTCTGGCATTTTTC
TACAATCCACGCTCTGATATCCCGGTTGGTCCCATTGAAGAGTTAGTAACCGCAAAT
CGTCCGGCTCTGTACAAACCTATCAGATTGACGAATACCGGTCACTGATTGCGCA
AAAAGGACCTTGCGGCAAGAATCAGGTGGACAGCTTGCTTTTGACTAGATAA

NHI 124 (ANS) protein sequence:

MATCWPEPIVSVQSLSQTVPTVPNRYVKPAHQRPVFNTTQSDAGIEIPVLDMMNDVW
GKPEGLRLVRSACEEWGFFQMVNHGVTHSLMERVRGAWREFFELPLEEKRYANSP
DYEYGYGSRLGVVKDAKLDWSDYFFLNYPSSIRNPSKWPSQPPKIRELIEKYGEEVR
KLCERLTETLSESLGLKPNKLMQALGGGDKVGASLRNFPYKCPQPQLTLGLSSHSDP
GGITILLPDEKVAGLQVRRGDGWVTIKSVPNALIVNIGDQLQILSNGIYKSVEHQVIVNSG
MERVSLAFFYNPRSDIPVGPIEELVTANRPALYKPIRFDEYRSLIRQKGPCGKNQVDSLL
LTR

Codon-optimized NHI 125 (VrtI) sequence:

ATGGGTGCACGCTCTAAAGAGGCTAGACACCGCTGGTTCAGCCGCCTACCTCCTG
ATGCATCGAGCTCCTTTCTGCCATGCAAAAGACCGAAATCTGTTGCATTTGTGCTAT
GGATCCGACCGAGTTGCTTTTTGAGGGAGTCTTTACCGGCCGAACCTCAGTGGGCT
CCCAGGTCTGCCCCTCCCGCCCCAGGCTCCGACTAACACCATGAACTGTGGGTTG
CCCGAGGCAGCCCTTAGTGTCATTGATTGGGGTCGCCTGAAAAGTGGTGATGTGAA
CGAAGGTGCGCGCTTGCTGTCTGCCTGTGATGATCAAGGTTTTTTTTTACTTAGACCT
GTCTTCTGAGCCGTCTTTCCTGCATGACCACAAAAGCGTCCTGCATTTTATGGATCA
GTATTTCCATCAAGGGCTGGCGGATAAGATGAAAGATGATCGCCAGAGTGACACCC
ACGGCTACGAACCAGTCGCCACCAGTACAGGCGCGCTGAACACCCTGCCGGATTA
TTATGAATCATTGAAAGCTAGTCGCGATGAGCTGCATGGAGATGGACGTAATTTAGC
GCCGGCAGTGTGTGACCGCCAAGATTTATTTCTTAGATTTTCCGACATGATGCATCA
AATGGTTATCGCCATACTTCAAGAGTTAGACTGCCAGCTGGGCTTTGGCGGTGATC
GGGCTTCTTTTCAAGATTTCCATCGGAAAGACGCTGAGTCCCTTACGACATTAAGCA
TGTTTCGGTATCCTAAACAGGAAACACTGGATCTAGGAGTAGGGCATAATAAACATA
CTGATATAGGCACATTAACCTTTCTGCTGTGTGATCAGTGGGGCCTGCAGGTCCTGT
CAAAAGACCCTGCGGGATGGCGATTTGTGCGACCGAGGGAAGGTCACGCAGTGAT
TAATGTTGGCGACACACTGCGCTTTCTATCCGGTAACCGCTTTCGGAGTGCGGTTC
ATCGTGTAATTCCAACGCAGCGCTTACAGCATGAAGATAGATACTCGATTGCCTACTT
TCTGCGTGCCGCTAATGACACTCAGTTCACCGACTCAGCAGGACGCCAGGTTTCA
GCTAAGCAGTGGCACGATGAAAAATTTGACGTATTCAGAGAACTCACGAGGAGCA
GGAAAAAATGCCAATTCTAACCGGGGGAATGGAGCGGTTAGAAAACCTGCCCGGG
ATATGGTGCGGAGTCCGCTTAACGGATGGAATTTGTCATCGCCTCGGTGCGGAAGA
CTGCTGA

NHI 125 (VrtI) protein sequence:

MGARSKEARHRWFSRLPPDASSSFLPCKRPKSVAFVLWIRPSCFLRESLPAELSGLPGLPLPPQAPTNTMNCGLPEAALSVIDWGRLKSGDVNEGARLLSACDDQGFFYLDLSSEPSFLHDHKSVLHFMDQYFHQGLADKMKDDRQSDTHGYEPVATSTGALNTLPDYYESLKASRDELHGDGRNLAPAVCDRQDLFLRFSDMMHQMVIAILQELDCQLGFGGDRASFQDFHRKDAESLTTLSMFRYPKQETLDLGVGHNKHTDIGTLTFLLCDQWGLQVLSKDPAGWRVAPREGHAVINVGDTLRFLSGNRFRSAVHRVIPTQRLQHEDRYSIAYFLRAANDTQFTDSAGRQVSAKQWHDEKFDVFRETHEEQEKMPILTGGMERLENLPGIWCGVRLTDGI
CHRLGAEDC

Codon-optimized NHI 126 (GA2ox3) sequence:

ATGGTTGTGCTCGCGGGCCCTCCAGCTGTAGATCATATTCCACTGCTTCGCTCACCAGACCCGGGCGACGTGTTACAGCGGTGTACCGGTCTAGACTTGGGTTCGCCAGGCGCGGCAAGGGCAGTAGTGATGCATGCGAGAGATACGGTTTTTTTAAAGTTGTATACACGGCGTGCGACAGATACCATGGATAAAGCCGAATCTGAGGCGGTGAGATTTTCTCAGACCCAACCGGACAAAGACAGGTACAGTCCCGCGTATCCTTTTGGTTACGGCAGCAAGAGAATCGGCTTTAATGGTGACATGGGTGGCTGGAATATCTGTTGTAGCCCTAGATGATGCCTCTCTGGCGGATGCTTGCACTGTCCCAAGCTGCGCAGTATTTAGGGCGGCATTAAATGAATATATTAGTGGCGTTAGGAAAGTCGCCGTGCGCGTTATGGAGGCGATGAGTGAAGGCCTGGGTATTGCCCAGGCAGATGCGCTGAGTGCGCTGTAAGTGAAGGTTCTGATCAGGTATTCCGCGTTAATCATTACCCCCCATGCCGTGCCCTGCAAGGCTTGGGATGCAGTGTTACGGGTTTCGGGGAGCACACAGATCCAGCTGGTGAGCGTTCTGAGAAGTAACGGGACATCAGGCCTTCAGATTGCTTTAAGAGACGGTCAGTGGGTCTCGGTGCCTTCGGACCGGGATTCAATTTTTGTGAATGTCGGCGATTCTCTGCAAGTCCTTACCAACGGCCGCTTTAAGTCAGTCAAACATCGCGTAGTTGCGAACTCGCTGAAATCACGCGTGTCCTTCATCTATTTTGGCGGCCCTCCTTAGCCAGCGTATCGCTCCGTTGCCTCAACTGTTAGGCGAAGGAGAACAGTCATTATATAAGAGTTCACGTGGGACGAATACAAAAAGGCAGCCTATAAATCACGTCTCGGAGACAACCGCTTAGCACAGTTTGAGAAAAAATAA

NHI 126 (GA2ox3) protein sequence:

MVVLAGPPAVDHIPLLRSPDPGDVFSGVPVVDLGSPGAARAVVDACERYGFFKVVNHGVATDTMDKAESEAVRFFSQTPDKDRSGPAYPFGYGSKRIGFNGDMGWLEYLLALLDDASLADACTVPSCAVFRAALNEYISGVRKVAVRVMEAMSEGLGIAQADALSALVTAEGSDQVFRVNHYPPCRALQGLGCSVTGFGEHTDPQLVSVLRSNGTSGLQIALRDGQWVSVPSDRDSFFVNVGDSLQVLTNGRFKSVKHRVVANSLKSRVSFIYFGGPPLAQRIAPLPQLLGEGEQSLEYKEFTWDEYKKAAYKSRLGDNRLAQFEKK

Codon-optimized NHI 127 (GA2ox5) sequence:

ATGTGTATTTATGCCTCCCGTCAGACCGTCTGCCCTTATTTGACTCCCTTTAAGGTGAAACGTCCCAAGTCACGAGAAATGAATAGCAGTGACGTAACTTCTCGCTGCTGCAAGAGCAACCGAATGTTCCCGCGGAGTTTTTCTGGCCTGAAAAAGACGTTGCGCCTCCGAAGGTGATTTAGATTTACCAATTATTGATCTGAGCGGTTTCCTGAACGGCAACGAAGCGGAGACACAGCTGGCTGCCAAAGCAGTTAAGAAAGCGTGTATGGCCCATGG

TACTTTTCTGGTCGTAAATCACGGCTTTAAAAGCGGTCTTGCAGAAAAGGCGCTAGA
AATTAGCTCTTTATTCTTTGGACTTAGCAAGGATGAAAAATTAAGGGCGTACCGTATC
CCAGGCAATATCTCGGGCTACACCGCGGGGCATAGTCAACGGTTTTTCGAGCAACCT
GCCTTGGAACGAACTTTGACCCTGGCATTAAAAAAGGGCCCCCCCCACGTTGTGG
AAGATTTTTTGACATCCCGTTTAGGTAATCATCGCCAGGAAATAGGGCAGGTCTTCC
AAGAGTTTTGCGACGCTATGAACGGTCTGGTAATGGATCTTATGGAGTTGCTGGGA
ATTTCTATGGGACTAAAGGACCGTACATATTATCGTAGGTTTTTCGAAGACGGCTCAG
GAATTTTTAGATGTAATTACTATCCGCCTTGTAAGCAGCCGGAAAAAGCCCTGGGTG
TTGGTCCGCATAATGATCCAACCGCTATAACCGTACTGTTACAAGATGATGTAGTAGG
ACTAGAAGTTTTTCGCGGCGGGGTCTTGGCAGACGGTTCGCCCACGCCCGGGTGC
TCTGGTCGTTAACGTCGGTGATACGTTTATGGCTCTGTCCAATGGGAATTATAGAAG
CTGCTACCATCGTGCCGTCGTCAATAAAGAAAAAGTGCGTCGCTCTCTTGTATTTTT
TAGCTGCCCCGCGTGAGGACAAAATTATTGTTCCGCCTCCGGAAGTGGTCGAGGGC
GAGGAAGCATCACGTAAATACCCGGACTTTACTTGGGCTCAACTCCAGAAATTTACT
CAGTCGGGCTATCGAGTGGATAATACTACGTTACACAATTTACGCTCCTGGCTTGTT
TCAAACAGCGATAAGAAATCTACTTAA

NHI 127 (GA20ox5) protein sequence:

MCIYASRQTVCPYLTPFKVKRPKSREMNSSDVNFSLLQSQPNVPAEFFWPEKDVAPSE
GDLDLPIIDL SGFLNGNEAETQLAAKAVKKACMAHGTFLLVNHGFKSGLAEKALEISLFL
FGLSKDEKL RAYRIPGNISGYTAGHSQRFSSNLPWNETLTLAFKKGP PHVVEDFLT SRL
GNHRQEIGQVFQEFCDAMNGLVMDLMELLGISMGLKDRTYYRRFFEDGSGIFRCNYY
PPCKQPEKALGVGPHNDPTAITVLLQDDVVGLEVF AAGSWQTVRPRPGALVVNVGDT
FMALSNGNYRSCYHRAVVNKEKVRRLVFFSCPREDKIIVPPPELVEGEEASRKYPDFT
WAQLQKFTQSGYRVDNTTLHNFSSWLVSNSDKKST

Codon-optimized NHI 128 (GA2ox1) sequence:

ATGGTACTGCTCTCCAAACCCACTTCAGAGCAGTACACTTACGTTTCGCAACAATATG
CCCATTACATTTAGCTCTAGCATCCCATTGGTGGATCTCAGTAAACCAGACGCGAAG
ACTCTTATCGTAAAAGCCTGTGAGGACTTTGGTTTCTTTAAAGTGATAAACCATGGCA
TCCCTTTGGATGCCATATCACAATTGGAGAGCGAAGCCTTTAAGTTTTTTAGTCTGC
CGCAGACTGAAAAAGAAAAAGCTGGTCCAGCCAATCCATTTGGATATGGGAATAAAA
GAATTGGTTTAAATGGGGATATAGGTTGGATAGAATATTTACTTTTAACGACGAATCAG
GATCACAACCTTTTCACTGTACGGAGAGGATATAGATAAGTTCCGGGGTTTACTGAAA
GACTATAAATGTGCAATGCGTAATATGGCCTGCGAAATCCTGGATCTCATGGCCGAA
GGTCTTAAGATTCAACCTAAAAATGTGTTTTCAAATTGGTCATGGATAAACAGTCTG
ATTGCCTGTTTCGCGTCAATCACTACCCAGCCTGCCCCGAGCTGGCAATCAACGGC
GAGAACCTGATTGGGTTTGGTGAACACACGGATCCTCAAATCATCAGTATTCTCCGT
AGTAATAATACGTCGGGGTTTCAGATATCGTTACGTGATGGTTCTTGGATTTCAGTTC
CGCCTGATCATTCTTCTTTTTTTATAAATGTCGGGGATTCTCTCCAGGTCATGACCAA
TGGTCGTTTCAAGAGCGTTTCGACACCGCGTTTTAGCTAACGGCATCGATCCAAGAT
TGTCATGATCTACTTTTGCGGCCCACTGTCCGAAAAAATTGCACCGCTGCCTA
GCCTTATGAAAGGCAAAGAGAGTCTCTATAAAGAGTTTACGTGGTTTCAATATAAATC

GTCTACCTATGGCAGCCGTTTAGCGGATAACCGGCTGGGCAACTATGAACGTATTG
CGGCCACCTAA

NHI 128 (GA2ox1) protein sequence:

MVLLSKPTSEQYTYVRNNMPITFSSSIPLVDLSKPDAKTLIVKACEDFGFFKVINHGIPLD
AISQLESEAFKFFSLPQTEKEKAGPANPFGYGNKRIGLNGDIGWIEYLLLTTNQDHNFSL
YGEDIDKFRGLLKDYKCAMRNMACEILDMAEGLKIQPKNVFSKLVMDKQSDCLFRVN
HYPACPELAINGENLIGFGEHTDPQIISILRSNNTSGFQISLRDGSWISVPPDHSSFFINV
GDSLQVMTNGRFKSVRHRVLANGIDPRLSMIYFCGPPLSEKIAPLPSLMKGKESLYKEF
TWFEYKSSTYGSRLADNRLGNYERIAAT

Codon-optimized NHI 129 (GA2ox1 2) sequence:

ATGGCCGTGCTCAGTAAACCCGTAGCTATTCCCAAAGTGGTTTTTCACTAATTCCG
GTAATTGATATGAGCGATCCCGAGAGCAAACATGCCTTAGTTAAAGCATGTGAAGAT
TTCGGGTTTTTTAAAGTGATTAATCATGGTGTATCAGCCGAGCTGGTCTCAGTTTTG
GAACATGAAACAGTGGATTTTTTTAGCCTGCCTAAATCTGAAAAGACACAAGTGGCC
GGTTATCCGTTTGGCTACGGCAATTCGAAAATTGGCCGGAACGGAGACGTTGGGTG
GGTAGAGTACCTGCTGATGAATGCCAATCATGACTCAGGTTCTGGACCCCTTTTTCC
ATCACTTTTAAAGTCTCCCGGGACTTTCCGTAATGCCCTGGAGGAATACACCACAAG
CGTGCGGAAGATGACGTTTGTGTGCTGGAAAAAATCACGGACGGTCTCGGCATCA
AGCCTCGCAATACCTTGTCAAAGCTCGTTTCCGACCAGAATACGGACAGCATATTAC
GTCTGAATCATTATCCGCCTTGCCCACTGAGTAATAAAAAAACAACGGTGGTAAAA
ACGTGATTGGTTTTGGCGAACATACTGATCCACAGATAATTTCCGTGCTTAGATCCAA
TAATACGTCTGGCCTGCAGATTAATCTCAACGATGGTTCCTGGATTTCCGTTCCCCC
GGACCACACGTCTTTCTTTTTTAATGTAGGGGATTCTCTTCAGGTAATGACCAATGG
AAGATTCAAGAGCGTGCGCCATCGTGTATTGGCTAACTGTAAAAAAGTCGCGTTTC
GATGATTTATTTTGTGCTGGGCCATCGCTGACGCAGCGTATTGCTCCGTTGACCTGCTT
AATCGATAACGAAGATGAACGTCTGTATGAAGAATTCACATGGAGCGAGTATAAAAAT
AGTACCTATAATTCACGCCTGTCAGATAATCGGCTGCAGCAATTTGAGCGGAAAACC
ATCAAAAATCTGCTGAATTAA

NHI 129 (GA2ox1 2) protein sequence:

MAVLSKPVAIPKSGFSLIPVIDMSDPESKHALVKACEDFGFFKVINHGVSAELVSVLEHE
TVDFFSLPKSEKTQVAGYPFGYGNKIGRNGDVGWVEYLLMNANHDSGSGPLFPSLL
KSPGTFRNALEEYTTSVRKMTFDVLEKITDGLGIKPRNTLSKLVSDQNTDSILRLNHYP
CPLSNKKTNGGKNVIGFGEHTDPQIISVLRNNTSGLQINLNDGSWISVPPDHSTFFFN
VGDSLQVMTNGRFKSVRHRVLNCKKSRVSMIYFAGPSLTQRIAPLTCLIDNEDERLYE
EFTWSEYKNSTYNSRLSDNRLQQFERKTIKNLLN

Codon-optimized NHI 130 (GA3ox2) sequence:

ATGCCGACGCCGAGTCACCTGTCTAAGGATCCACGTTATTTTATTTCGCGCCGC
ACGTCGTGTACCTGAAACGCATGCCTGGCCCGGTCTTCACGATCATCCCGTAGTCG
ATGGTTCTGGTGCCGGTGGCGGTCCCGACGCAGTTCAGTTGTGCGACATGAGGGA

CCCGTGTGCAGCGGAGGCAGTCGCACTGGCCGCACAGGATTGGGGTGCTTTTTTG
CTGGAAGGGCATGGTGTACCCTTGAGTTGCTGGCAGGCGTTGAAGCAGCCATTG
GGGGCATGTTTCGCATTACCAGCATCCGAAAAGATGCGTGCAAGTTCGCCGTCCTGG
CGACAGCTGTGGTTACGGCTCGCCCCCGATCAGTTCTTTTTTTTCCAAATGCATGTG
GTCGGAAGGTTATACTTTTACGCCAGCGAACCTGCGGTCTGATCTGCGGAAGCTTT
GGCCGAAAGCCGGCCATGATTATCGGCACTTTTTCGCGCAGTAATGGAAGAGTTCCAT
CGCGAAATGCGAGCATTAGCAGATAAATTATTAGAGTTATTTCTGGTAGCGTTAGGAC
TGACCGGTGAGCAGGTTCGCGGCTGTTGAAAGTGAGCATAAAATCGCGGAGACTAT
GACAGCCACCATGCACTTAACTGGTACCCAAAATGCCCAGATCCAAAGCGTGCAC
TGGGCTTAATCGCACATACAGATTCTGGGTTTTTTACATTTGTTTTGCAGTCCTTAGT
GCCGGGTTTACAGTTGTTTCGCCACGGCCCCGATCGGTGGGTGACCGTACCAGCT
GTGCCGGGCGCTATGGTTGTGAATGTTGGGGACCTGTTTCAGATTTTGACCAACGG
ACGTTTTTCATTCTGTATATCATCGCGCTGTTGTCAACCGAGATTCAGACCGTATTAGT
CTTGGCTATTTTCTCGGGCCGCCCGCTCATGTTAAGGTGGCCCCGTTGCGTGAAGC
ATTGGCAGGTACCCAGCAGCATATCGGGCCGTGACCTGGCCGGAATACATGGGT
GTGAGAAAAAAGCTTTTACAACCTGGAGCCTCAGCGCTGAAAATGGTTCGCCATTTT
TACTGATAACGATGCCGCGAACGATACAGATGATCTGATAAGCTCATAA

NHI 130 (GA3ox2) protein sequence:

MPTPSHLSKDPYFDFAARRVPETHAWPGLHDHPVVDGSGAGGGPDAPVVDMDR
PCAAEAVALLAQDWGAFLLLEGHGVPLELLAGVEAAIGGMFALPASEKMRVRRPGDSC
GYGSPPISSFFSKCMWSEGYTFSPANLRSDLRKLWPKAGHDYRHFCAVMEEFHREMR
ALADKLELFLVALGLTGEQVAVESEHKIAETMTATMHLNWYPKCPDPKRALGLIAHTD
SGFFTFVLQSLVPGLQLFRHGPDRWVTVPAVPGAMVVNVGDLFQILTNGRFHFSVYHRA
VVNRDSDRISLGYFLGPPAHVKVAPLREALAGTPAAYRAVTWPEYMGVRKKAFTTGAS
ALKMVAISTDNDANDTDDLIS

Codon-optimized NHI 131 (GA3ox1) sequence:

ATGCCATCGTTAAGCGAGGCTTATCGCGCCCCACCCTGTACACGTAAATCACAAACAT
CCGGACTTTAATTCATTACAGGAATTACCAGAATCATACAATTGGACGCATTTAGATG
ACCACACGCTGATCGATTCTGAATAACATTATGAAAGAGTCGACCACAACCGTGCCG
GTGATTGACTTAAACGATCCCAATGCATCTAAACTGATCGGGTTAGCATGTAAACTT
GGGGCGTTTATCAAGTGATGAACCATGGTATTCCTTAAGCCTGTTAGAGGACATTC
AGTGGCTGGGACAAACATTATTTTCTCTGCCAAGCCACCAGAAACATAAAGCCACC
AGATCTCCAGATGGCGTATCTGGCTATGGGATTGCCCGAATTTCTTCATTTTTCCCG
AAGCTGATGTGGTATGAGGGGTTCACTATTGTGGGTAGTCCGTTAGATCACTTCCGC
GAACTGTGGCCGCAGGATTATACACGGTTCTGTGACATAGTGGTACAATATGATGAG
ACTATGAAAAAGTTAGCCGGAACGCTTATGTGTTTGATGCTGGATTCTTTAGGCATCA
CGAAAGAGGATATCAAGTGGGCGGGTTCTAAGGCTCAATTTGAGAAAGCATGCGCA
GCGCTCCAACCTGAACCTCCTACCCTAGTTGCCCAGATCCAGATCACGCCATGGGGCT
GGCACCACACACGGATAGCACTTTCCTGACGATTCTCTCTCAGAACGACATATCGG
GTTTACAGGTTAATAGGGAAGGATCGGGTTGGATCACAGTACCGCCGCTGCAAGGC
GGCCTGGTGGTGAATGTTGGCGATCTCTTTCACATACTCAGCAATGGTTTATACCCG

AGTGTGTTACATCGGGTTCTGGTAAATCGTACGCGGCAACGCTTCTCTGTGGCTTAT
CTCTATGGTCCGCCGTCCAACGTGGAAATCTGTCCCCATGCAAACTGATTGGGCC
GACGAAGCCCCCGCTGTACCGCTCAGTCACATGGAATGAATATCTCGGCACTAAAG
CCAAGCACTTTAACAAAGCGCTGAGCAGTGTGCGTCTGTGCACCCCAATTAATGGA
CTGTTTCGATGTGAATGATTCTAACAAAAATTCTGTACAGGTTGGGTAA

NHI 131 (GA3ox1) protein sequence:

MPSLSEAYRAHPVHVNHKHPDFNSLQELPESYNWTHLDDHTLIDSNNIMKESTTTVPVI
DLNDPNASKLIGLACKTWGVYQVMNHGIPLSLLEDIQWLGGTLFSLPSHQKHKATRSP
DGVSGYGIARISSFFPKLMWYEGFTIVGSPLDHFRELWPQDYTRFCDIVVQYDETMKK
LAGTLMCLMLDSLGITKEDIKWAGSKAQFEKACAALQLNSYPSCPDPDHAMGLAPHTD
STFLTILSQNDISGLQVNREGSGWITVPPLQGGLVNVGDLFHILSNGLYPSVLHRVLVN
RTRQRFSVAYLYGPPSNVEICPHAKLIGPTKPPLYRSVTWNEYLGTKAKHFNKALSSVR
LCTPINGLFDVNDSENKNSVQVG

Codon-optimized NHI 132 (OsGA3ox1) sequence:

ATGCAGATCATGACATCATCGTCTACCAGTCCAACCTCTCCAACCTCCCCTCTGGCG
GCCGCGGCTGATAATGGTGTAGCCGCTGCCTATTTCAATTTCCGTGGTGCAGAGCG
GGTCCCTGAGAGTCATGTGTGGAAAGGTATGCACGAAAAAGATACGGCACCCGTA
CTGCCGCTGATGCGGACGGGGGTGATGCAGTGCCGGTTCGTTCGATATGTCAGGGG
GGGACGATGCTGCGGTGGCCGCAGTTGCACGTGCAGCTGAGGAATGGGGAGGTT
TTCTTCTGGTTGGCCATGGTGTAAACGGCAGAGGCGCTGGCACGCGTTGAAGCACA
AGCCGCCAGGCTCTTCGCATTGCCTGCGGATGACAAGGCGAGGGGTGCCCGACG
CCCGGGGGGAGGCAATACCGGTTATGGCGTACCGCCGTATCTGTTGAGATATCCCA
AACAGATGTGGGCTGAAGGATATACCTTTCCGCCGCCCGCAATACGCGATGAGTTT
CGTCGTGTGTGGCCAGATGCAGGTGATGATTATCATCGTTTTTTGTAGTGCCATGGAA
GAATACGATTCTTCTATGCGTGCTCTGGGTGAACGGCTGTTGGCGATGTTCTTTAA
GCATTGGGGCTGGCAGGCAACGATGCTCCGGGGGGCGAGACCGAACGTAAAATTC
GTGAAACCTTAACAGTACCATACATCTTAACATGTTTCCGCGGTGTCCCGACCCCTG
ATCGAGTGGTTGGTTTAGCCGCACATACGGATAGCGGCTTTTTTACGTTTATTCTTC
AGTCCCCAGTGCCAGGACTGCAATTACTGCGCCACCGCCCTGACCGTTGGGTAA
AGTTCCTGGAACGCCGGGAGCCCTGATAGTCGTAGTAGGCGATCTATTCCATGTGC
TGACCAATGGACGTTTTTCATAGTGTTTTTACCAGGGCTGTTGTGAACCGCGAGCGC
GATCGCATTAGTATGCCCTACTTCCTGGGCCCCGCCAGCCGACATGAAAGTCACGCC
CTTGGTGGCAGCTGGAAGTCCAGAGTCAAAAGCTGTCTATCAGGCCGTGACCTGG
CCGGAATATATGGCAGTCCGCGATAAATTATTTGGGACAAACATCTCTGCCCTATCTA
TGATCCGGGTGCGGAAGGAAGAAGATAAAGAAAGTTAA

NHI 132 (OsGA3ox1) protein sequence:

MQIMTSSSTSPTSPTSPLAAAADNGVAAAYFNFRGAERVPESHVWKGMHEKDTAPVA
AADADGGDAVPVVDMSGDDAAVAAVARAAEEWGGFLLVGHGVTAELARVEAQAA
RLFALPADDKARGARRPGGGNTGYGVPPYLLRYPKQMWAEGYTFPPPAIRDEFRRVW
PDAGDDYHRFCSAMEEYDSSMRALGERLLAMFFKALGLAGNDAPGGETERKIRETLT

STIHLNMFPRCPDPDRVVGAAHTDSGFFTFILQSPVPGQLLLRHRPDRWVTVPGTPG
ALIVVVGDLFHVLTNGRFHSHVHRAVVNRERDRISMPYFLGPPADMKVTPPLVAAGSPES
KAVYQAVTWPEYMAVRDKLFGTNISALSMIRVAKEEDKES

Codon-optimized NHI 133 sequence:

ATGCTGGATCAAACACTAGCATTCTTAGTGAAGAGACATTAAACGGCCCTGAGACA
GTAGTTTTGAAAGAACCGTTCCCTATTCTGATCGCGTCTAATCAGCTCCCGGCCAAT
TTGGCTGAAGAAATTGACCGCGACTTCCCGCAATACACCGGAGCAGGGTATCTGCC
ACATGAAGAAAAAGACTGCGGACCAACAATCAATACTATCGTTAGTGAAATGACCTC
ACGTGAGTTTGCCAATTCCTTAGGCAAGCACCTCGGAGTCCCTAATCTGGGTGATTA
TCCTACAATGGTGAGCATATGTCGTAGCTTGAATAAAAAACATGGAAATATCCACACA
GACTCGAATTCTAAAATCGTCACTTGCCTGCTGTACTTAAACCCGGGTTGGATGCAA
AATAGCGAGGGTTGTCTGAGGTTCTTGAGACGTATTGATAGCTTTGAAGACATGGT
GGCGCCTGAAATTCGTCCGGTGTACGGTTCATTGGCGGCTTTTCGTGAGCTGATA
ATTCGTTCCATGGTCATCTGCCATACGAAGGCGAGCGGCGCGTTATTCAAGTTGCAT
GGTTAACGTCTGAGGAGGAAAAACTCCGTAAAACGAAGCGGGGCAAAATGAGCCG
CTTTCTGAAATGGATGACATCTAGCTTGGGGAAATAG

NHI 133 protein sequence:

MLDQTTSILSEETLNGPETVVLKEPFPILIASNQLPANLAEIDRDFPQYTGAGYLPHEE
KDCGPTINTIVSEMTSREFANSLGKHLGVPNLGDYPTMVSICRSLNKKHGNHTDSNSK
IVTCLLYLNPGWMQNSEGCLRFLRRIDSFEDMVAPEIRPVYGSAAFRADNSFHGHL
PYEGERRVIQVAWLTSEEEKLRKTKRGKMSRFLKWMTSSLGK

Codon-optimized NHI 134 sequence:

ATGGCCACCGCGGCCTATACAACCTTTACGCTGGGTGATACACTTACAGAAGAGCA
GAAGGCATTCTTTGAAAAACATGGGTATATCACTTTTCGAATGTGTTGGATGCAGC
AACCGTGCACTCTATCATAGATGCCAGCGAAAAAGTGCAGGAGAAATGGATTTAG
AAAATGTTGAAAAGATCAACGGTGTGCCGATAAAATACGGCACCGATCTCAACGGAA
ATAAAATCGTTCAACGTTTTCGCCTTTATAAATCAGCATACTACTGTTTTAAGCGAACTG
CTAAAGGATGCACGTATTCAATCACTGCTCGAGCTTATCGGCCCAGGTGCTAGAATT
GGTGAGAATGAGAAGGACGGTATGGTCTTCAATCATTATGTTAACGGTCCGGAGAG
TAAGTTCACTAAGATGGGTGGCACACGGATGGACTCCGTGATATCTTCATGGGCG
GCCGACTGAATCCGATGTTGAATATGGGACTGCATCTTAGCACCTGCAGCCGGAA
AACGGTGGCCTGCGTATCATACCGGGCACACATAAACAGGGTCTATTTTCAATGCTA
TTTAAAAAACGGTATTTTATAGATCATCGTCCGGATAAAATGAAATAGCCATTATGCC
GAAAGCGGGGGATCTAACGGTGCATGACGGCCGCCTGTGGCATCGCGTTGCCCA
GAGCAGTGTGCTGGGTGAAGCCTCACGGCGTCGCGTCATTTACGTGCCGCTGATT
GCGGGTAAGTACGAACCCAAAAATGAAAACAGCCCTACGCTGTTCTATCAACGCTTT
GCAGGCATCGTGAAATAA

NHI 134 protein sequence:

MATAAYTTFTLGDTLTEEQKAFFFEKHGYIHFSNVLDAATVQSIIDASEKVQEKWISENVE
KINGVPIKYGTDLNGNKIVQRFAFINQHHTTVLSELLKDARIQSLELIGPGARIGENEKDG
MVFNHYYVNGPESKFTKMGWHTDGLRDIFMGGRLNPMLNMGLHLSTLQPENGGLRIIP
GTHKQGLFSMLFKKRYFIDHRPDKNEIAIMPKAGDLTVHDGRLWHRVAQSSVVGESR
RRVIYVPLIAGKYEPKNENSPTLFYQRFAGIVK

Codon-optimized NHI 135 sequence:

ATGGCGACACGTCATCGCCGTCGTGGAGGGAGCGCTCCGTCTTGGGCAAAACCA
GGTAAACCTGGTGAAAGACCGGGAGGACCAAAGAAATCACGTGGTAGAACGTCTT
GGAAAAGTCTGTTAATTTGGGGTGTTTTCGGGGTAACCTCTGGGACTGATGGCGGGG
TATTATTTATGGGGCGAACTGATTACCGACGACTCGGTAACCTGAAGTATTAGCAGCG
CAGCGTGATGCGGTGGCGCAGAGATTCTTTCATGTGCCGTGCTCACAGGATTATGA
ATCCCTCAAACAATTTGAAGCGTGCAACCCCGCGCAAATGTGGGCGCGCTGTAACAG
ATTCTGTTATCACCTACAGGAAGCCGAGAAAATGCGTCGGCTAGCAGAAGCAGGG
CTGTCACTTGGGGGTTCCGATGGCGGTGCTAGCATCCTCGATCTCCACAGTGGGG
CTCTATCTATGGGTAAAAAATTTGTGAACATGTATCGCTTCTTTGGAGACAACTTAA
AGATGTCATGTCAGATGAGGATTTTAATCTTTATCGCGAAGTCCGCCTAAAGATACAA
CACGAAATTGCGCGAACATTCAATATTAGCGTGTCAAGTCTGCACCTGACGAAGCC
TACATTTTTTTCACGGATGAACAGTAGTGAAGCGAAGACAGCCCATGATGAGTATTG
GCACCCGCATATCGACAAAGTGACCTACGGGTCATTTGACTACACGAGCCTCCTTTA
CTTATCAGATTATTCACAGGATTTTGGGGGCGGCCGGTTCGTCTTTATAGACGAGGG
GGCCAACCGCACAGTCGAGCCGCGAACGGGCCGGCTGAGCTTCTTTACAAGTGG
TTCAGAAAATCTGCACCGCGTGGA AAAAGTAAGCTGGGGTACGAGGTATGCCATAA
CTATCTCGTTTACCTGTAATCCGGAACATGCAATTGGGGATCCTACGTGGACATAA

NHI 135 protein sequence:

MATRHRRRRGGSAPSWAKPGKPGERPGGPKKSRGRTSWKSLLIWGVFGVTLGLMAGY
YLWGELITDDSVTEVLAAQRDAVAQRFFHVPSCSDYESLKQFEACTPRKCGRAVTDVS
ITLQEA EKMRRLAEAGLSLGGSDGGASILDHSGALSMGKKFVNMYRFFGDKLKDVMS
DEDFNLYREVR LKIQHEIARTFNISVSSLHLTKPTFFSRMNSSEAKTAHDEYWHPHIDKV
TYGSFDYTSLLYLS DYSQDFGGGRFVFIDEGANRTVEPRTGRLSFFTSGSENLHRVEK
VSWGTRYAITISFTCNPEHAIGDPTWT

Codon-optimized NHI 136 sequence:

ATGAATCATCCAAATCGTGATAAAATTGCTGACCTAATTTTTCTGAAACTTAAAGACAA
TGAAGAAGTCCTGAAGAAACACTTCGCACACACCTCCGAAGGCATCGGCTATTTTT
ACCTGGACGATCTGCTGCCGAAACATCTAACCGACGAAATTTTTGATAAATTTCCGA
GCACAAAAGATATTAAAGTCAAACGTAGCTTACGGGAATATAAATATGTGGCAGCTCA
AATGAATCGATATCATCCACTGCTGGAGGAAACCATTTATGCCTTTCAAGACAAAAG
AATCGTGGATCTCATCAGTAAGATCTGCGGTTACTGCAGCGTGAAGCCGGATGGTG
ATCTGTACGCAGGGGGTATTTTCGATGATGGGCAAAGGAAATTATCTGAATCCGCATC
TGGATAATTCTCATGACAAAGACGTAAAGTTATGGAGAGTGCTGAATCTGCTGTATTA
CGTAACACCCGACTGGATGCCCGAAAATGGTGGCAACCTGGAGCTATGGCCTAAA

GGCTTAAAAAGTGAACCAGTGAGTATCAATTCCCGGTATAATCGTCTGGTCGTTATG
TCGACCCATAAAGATAGTTGGCATTCTGTTGGCAAGGTGAAAGTCGACAAAGTCAG
ATGTTGTATTTCAAATTATTATTTTAGTACGGAACCGGTAGATGCCAAACGAAGTTTCC
ATGTCACGACTTTCCGTGGTTGGCCTGGCCAATTTTTTCGAGATAAAATTCTTAAAT
AGACTCATCTATACGTATGGGCGTACGTAAATCTTTAAAAAAGGTATTCGCGAAAAT
CCACACGTCTACAAAAAGGACGATTAA

NHI 136 protein sequence:

MNHPNRDKIADLIFLKLKDNEEVLLKKHFAHTSEGIGYFYLDDLPLKHLTDEIFDKFPSTKD
IKVKRSLREYKYVAAQMNRYHPLLEETIYAFQDKRIVDLISKICGYCSVKPDGDLYAGGIS
MMGKGNYLNPFLDNSHDKDVKLWRVLNLLYYVTPDWMPENGGNLELWPKGLKSEPV
SINSRYNRLVVMSTHKDSWHSVGVKVKVDKVRCCISNYFSTEPVDAKRSFHVTTFRG
WPGQFFRDKILKIDSSIRMGVRKIFKKGIRENPHVYKKDD

Codon-optimized NHI 137 sequence:

ATGACTGAATCTAGTCTGCATGACCTCTTCTCGGGTTTGATAGACGATCTGGCATCA
CAGGGATGGTCACAGCGTGCTCTTTTTGCGCCTGAGGTACTCACACGTGAATTAGC
CGTAGAATGTCGCACACGTGCGCAGAATGGAGAGCTAAACGCCGCCGGGGTAGGA
CGTGGCGGCACCCAGCAGGTTCAAGGAAGGGATTTCGTGGTGATCGTATTCAATGGT
TAGAACCCGGTCAGTCACAGGCCTGCGACCAATATCTGCAGCTAATGGATGGGCTG
CGCCAGGCACTTAATCAAGCTCTATACTTAGGCTTAGAGGATTATGAGTGTCATTTTCG
CTTGCTATGCCCCAGGTGCATTTTATCAGCGACACTTGGACAGGTTCCGAGATGAT
GATCGTCGTACCGTGTCCGCGGTTTTTTATTTGAACGAAGAATGGCAAGCCGAACAT
GGTGGCGCGCTGCGTCTGTATTTGGCCGACGGTTCGCGAACATGATGTACTGCCCC
AAGCCGGTACCCTGGCGTTATTCCTGAGCTCAGAGATGCCACATGAGGTACTGCCG
GCTACACGAGAGAGATTGTCACTGACAGGTTGGTTTAGACGTCGCGGAGGGGCAG
TACTGTAA

NHI 137 protein sequence:

MTESSLHDLFSGLIDDLASQGWWSQRALFAPEVLTRELAVECRTRAQNGELNAAGVGR
GGTQQVQEGIRGDRIQWLEPGQSQACDQYLQLMDGLRQALNQALYLGLEDYECFHA
CYAPGAFYQRHLDRFRDDDRRTVSAVFYLNEEWQAEHGGALRLYLADGREHDVLPQA
GTLALFLSSEMPHEVLPATRERLSLTGWFRRRGGAVL

Codon-optimized NHI 138 sequence:

ATGGATCCATCTCATATGAATTTGCCAGTTATTGACTTAGACGTTTATCTGAATAAACC
GCGTGATTGCGCTGAGGTGCGCCAAGAATGTGCAAAGGCAGCAGATGCTCTCATT
CTTATGGCGCCCTTGTTCTTCATGATTCACGCGTAGACTCTAAAGACAATGATACGTT
CCTGGATCTCCTGGAGGATTATTTTGTCAACCGTCCGACGTAATTCGCAACGATGA
GCGCCCCGAAGTGAAGTATCAGATTGGAGTTACATTGGAAAATACTGAAAAACCGAA
ATGCGCAGTAGACGAGCCTTGCCCTCGATGTAGTTAGACGTCTTCATCCTAGTCAAC
GTCCACTCGATATTTCAAGGTCATCAACCTGATCCGAAATGTCGCTTTTTTTGGCGTAT
GGCTGACCGGCCACCGTACGAGACAGATTTCCGTCACTGAATGCACCGAACGTG

GTACCGGATGCTCCGCACATCAAACCGCGATGGACACCCACCATGGACATGTGGG
GTACGTCTATGAAAAATGCCGTTAGCAAGTTATCAGAAATGACTGCAGTGGGGTTAG
GCCTTCCCGCAGAGTATTTTTCTGACGCGGCAAGATACGGGGCCGCATCTCTTGGCG
CCTACAGCATCTGATCTCGAGAAATATGGGGCGAAAGATACAATCCTGGCCGGTTTT
CATACCGACCTGAATTTTTCTGACAATTCATGGCCGTTTCGCGGTATCCAGGCCTGAAC
ATTTGGGCCCCGAATACGGGGGACCGCATTGCCGTGAAAATTCCGCCGGGGGAATT
ATCTGCTTGTACAAGCCGGCAAACAGCTAGAATATGCAACGGGTGGTCTGGTGAAG
GCCGGATACCATGAAGTGGTGATTAACGACAAGACGATTGAAACGATAGAAAAGAG
GAAACAACAATTTCTGATCGCCCTCTGGTGCGCATTAGCTCCACGTTATTTTGGCA
CCTTAATAGCGATTTTGATCTGGCCCCCGTCCCAAAGCTTGCCGAACGTGCTAAACA
GGTGCGCGACGAACAGGCACGACTGGGAAGGGATGAAGGTGAACCATTACATAC
CCACCGACGAAAGTAGGCCACCAGGTGGAGGGAGAATTAAGCATATCGCACTGAT
GGTTTGA

NHI 138 protein sequence:

MDPSHMNLPVIDLDVYLNKPRDSPEVRQECAKAADALITYGALVLHDSRVDSKDNDTF
LDLLEDYFAQPSDVLRNDERPELSYQIGVTLENTEKPKCAVDEPCLDVVRLHPSQRPL
DISGHQPDPKCRFFWRMADRPPYETDFPSLNAPNVVPDAPHIKPRWPTMDMWGTS
MKNAVSKLSEMTAVGLGLPAEYFSDAARYGPHLLAPTASDLEKYGAKDTILAGFHTDLN
FLTIHGRSRYPLNIWARNTGDRIAVKIPPGNYLLVQAGKQLEYATGGLVKAGYHEVVIN
DKTIETIEKRKQQFPDRPLVRISSTLFWHLNSDFDLAPVPKLAERAKQVRDEQARLGRD
EGEPFTYPPTKVGHQVEGELKHIALMV

Codon-optimized NHI 139 sequence:

ATGGCTGCAGAAACCTTACTTGGAATGAAGTCAGTCCTTTTGGCTCCTTATCAT
GGGCTCCAGATCTTTACCGGCACGAAGGATTTTCGGAGCCCCATACTGGGGAGCG
AGAGATTAAATAGAAAAGGTTTGCATATCTGGCGCGTTAAGACTGCGTCGCGATTGG
CTGAGTGGCGTCGGCGTCGGCTGAAACATCTCCTTCCCGAAGCGGAACGAGCCTT
TTTTGCGGAAAATGGTTACATAGAACGACGCAATCTCTTGGCCGAAAACGAATTCG
GCAACTGGTGGCAGAAGTAGAAGCCCTGCAAGCGCCAGCTCGTGAAATGCGTGAA
GGCGGCGCTGTTACACGTCGTATTCCGTTGACACCCGACGTATTAACCCGTAGCCC
GGCGCTCGCAGCATTATTAAAAATGAAGCCTGGACCGCCCCCATTTCGCTATGTGG
GGGGCTTTAATGTAGAACCGGTACTTTTGATTCAAACAATATTTGGAGAGCCGAGCG
CCCGGCCGGGGGAAAAAGACCCACAGACGGATTTACATATGGATACTTTTCATTCAA
CTGCAAAGGCATGGTTCTTCCTTTACGACGTACCGGAGGACGAAGGCCCATTCACC
TACGTAGCTGGGTCCCATAGATTAAGCGACGTCTGGCGTGGCAAAAACGGAT
GTCCGTTCTAGCGGCTGCTAGACACGGGGGCGGGGCATTTTCGGATCCCGGCCGC
ACAACTGAAGAGATTACACATGCCTGCTCCACGAAATTCGCGGTCCCTGCAAATA
CCCTTGTGGTCGGTGATACCTTTGGGTTTCATGCCCGTGGTTTGAGCGTGCGTAAA
TCTATCCGAGTTGAATTATACGCAAGCCAACGTCCAAATCCTTTTCTGCCATTTATTC
ACCTGGACAGTGCGTGGCTCCCGTTTGTGTCCGGGCGCAAAGAAGTGATCAGTTG
GTTTTTCGAAGATTGGCTGGTAAAGTTAAACTGCAACGACTTATCTGGAGACCTGT
CGGCCGAGTTAGCGCCAGGCAAACAGCGCCGGAACAAGGGTAA

NHI 139 protein sequence:

MAAETLLGKLKSVLLAPYHGLQIFTGTKDFRSPILGSERLNRKGLHIWRVKTASRLAEW
RRRLKHLLEAERAFFAENGYIERRNLLAENEFRLVAEVEALQAPAREMREGGAVT
RRIPLTPDVLTRSPALAAFIKNEAWTAPIRYVGGFNVEPVLSIQTIFGEP SARPG EKDPQT
DLHMDTFHSTAKAWFFLYDVPEDEGPFTYVAGSHRLTKRRLAWQKRMSVLAAARHGG
GAFRIPAAQLKRLHMPAPTKFAVPANTLVVGDTFGFHARGLSVRKSIRVELYASQRPNP
FLPFIHLDSAWLPFVSGRKEVISWFFEDWLVLKLLQRLIWRPVGRVSARQTAPEQG

Codon-optimized NHI 140 sequence:

ATGCCCCCGAAAACGTCTAAGTACCCTGCCCCGCAGAAAGTTACTTATACCTCAAAA
GACATACCGATACCGGATAGTTTCCTGAGTAGTCCAAGTACCACGCCGATTACCTCT
ACTCATATTGACTTCGCGAACAGTCCGCTGCCTCAATATGATGGTCATACCGCCATT
GTGCTGGATAACGTTCTGTGCGCTGAAGAGTGCAAAGAGCTACTTTCGTTGGCGGA
GCAATCAGTTGAAGGTGATGAAGCTTGGCAGCCGGCTCTCGTGTCCGGGGGTAAAT
GGGTACGAGGCCCGTGCTCCGGGCTATAGGGAATCAGACCGTATTATCTGGGATCA
ACAGACTATCGTAGATAGGTTATGGGAACGGTGCTTGGAAGCAGACGGATTACGCG
ATTTACTCGCAGTTGCCCCGCATGAGACATGGATGAAAGGAGGTAAATGGGTCTTTT
CACGCTTGAACGATCGTATGCGTTTTTTTGAATATTCTCCCGGGCAGTTTTTTAAACC
GCATTGCGATGGCGCCTATTACTATACCGAAGGTCCGGGCAAAGAATTTGAGACTTT
CTACACCCTACACCTTTATCTGAACGATTCCGTAGAGAATGACCCGACTAGCGACCT
CCAGGGCGGGCGCCACATCGTTTCTCGATCGGAAAGGTGAAAAACGTGTGGATGTG
AACCCGAAGGCCGGCTCAGTTCTCATTTTCCAACATAAAGGCTTATTCCACGAGGG
TGCGTTGGTGAAAAAGGGGATCAAGTACACTATGCGAACTGATATCTTGACGAATG
GATCCCGGACAAAGATGCCGAGGAGAAACCTGAGACGAAAGCACCTTGGTGGGAA
TGA

NHI 140 protein sequence:

MPPKTSKYPAPQKVITYTSKDIPDSFLSSPSTTPITSTHIDFANSPLPQYDGHTAIVLDN
VLSPEECKELLSLAEQSVEGDEAWQPALVSGGNGYEAPGYRES DRIIWDQQTIVDR
LWERCLEADGLRDLLAVAPHETWMKGGKWWF SRLNDRMRFLKYSPGQFFKPHCDGA
YYYTEGPGKEFETFYTLHLYLND SVENDPTSDLQGGATSFLDRKGEKRV DVNPKAGSV
LIFQHKGLFHEGALVKKGIKYTMRTDILYEWIPDKDAEEKPETKAPWWE

Codon-optimized NHI 141 sequence:

ATGCCTAGTCGCCGTCTGAAGTCCGAGAGTGTTGCAGGCGCGGGGGGAAATCGTG
CACGTGCGACTAGATCGCATCGTTCCATCCCCTGGCGTATGCGCCGCCCGTGCTG
GACATGGTCACCGAACCCTAGTGCACGGTCGCGACGTTCCATGGTCGAAACTTCA
GCTGATCTGAGTGGGGACGTTAGTCGAAGTATACCGCTACTTGTCCGTGACCTGCC
GCGTGGATACCCCGCTCATGACACGGCCTTTTCGAGCCGTCATGGTCTGTCATGCA
AAGGCAGGACCATGGACGTAAATCGGCAACAGTTCTTAGAAGATGGTTACGTTATAC
TGCGCCAGGTAATACCACCAGATCAACTCGACAGCCTGAGGCACTCGTATGAAATT
ATGGTGGACAGGCAACGTGCTACTTGGGCCCAGGAGGATCCACCGGACGGAGTG

TGGGGGCGTGCGGCACAGCCCAGGTAAATCTGACATCCCTTGCGGATCAGATCG
ATGAGTCAACAGCCTCAACAATCGAGATATGGCTGCATGAAAATACGCAGGGAGCT
AGCAGCCAGCTTCTGGGTGTTGAAGATGCAGCCGTCACGGAAATGATGCTCATGTG
TTCGCCGGTTTTCTGATCACGGACCCGCACGATGGCATCGCGACTTCGCCCCCTGCC
CATAGTTCCCCTGTGATGGGGTATGCAGAAGATATTCTGGATAATGGACCCCGGTAT
GTGCAGTGGAAATTTATCCCTGTACGATGATGACGTAATGTGGGTCGTTCCAGGCTC
ACATGTTCCGGTGTCTGACCTCGGACGAGAACGCCCAACTAGAGCGCGATGCTCAG
GCTCCGCTGAGTAGTGGAGTACAAACGCACCTCAAAGCAGGGGACGGTGTAGCGT
ATATCCTTCCGATACTGCACTGGGCATCAAATTATAGTACTCGTATGCGGCGTACAAT
CCATGGTGGCTTTTCAATGTTCACTCATTATCCGAACCTTATCATACTTAAAGCAGCTT
TCACCACAGGCCCGAGAGACCTTTGGTCGTTGGGTGAAGCGTTCGGACGATACGA
TGACTCGGACTGAAATTGCATTGCGTGCCGCGTTAGAAGCGGATGGTGCAGCATAT
CAGCAGGCCTTACAGGATCTTCATCCTGAAAGCGGCAGTATGGCAATGCGGCAAAG
TACGATTTTGCTTAGCAAGTCCGCCAAACGTATTCAGGATCTGAAGCGTCCAGCGA
GCGGGGTCTCGGAAGAAGAAGTGTCTTATGCGACGGGCATCCACCCTATGACCAT
GCAGTGCGGCCAGCCAATGGCTGATCGGTTTTCTGTGCGCCGAGGCAGGTGCGCT
GTGGGAGCGGTTTTCTGTTGGGTTGACGATCGTCTGCGCGGCGATCAAGACGCAACT
TGTCCTGGTTTTTCAGGGAGGACCCACTTCATATTTTTTTAATGATGTACCTGCCGAA
CTCACAGTTGATAACTTCATCGCCCAATGGTCATAG

NHI 141 protein sequence:

MPSRRLKSESVAGAGGNRARRTRSHRSIPWRMRRPCWTWSPNPSARSRRSMVETS
ADLSGDVSRSIPLLRDLPRGYPAHDTAFSSRHGLSCKGRTMDVNRQQFLEDGYVILR
QVIPPDQLDSLRSYEIMVDRQRATWAQEDPPDGWVGRAAQPRNLNLSLADQIDESTA
STIEIWLHENTQGASSQLLGVEDAAVTEMMLMCSPVSDHGPARGWRDFAPAHSSPVM
GYAEDILDNGPRYVQWNLSLYDDDLWVVPGSHVRCLTSDENAQLERDAQAPLSSGV
QTHLKAGDGVAYILPILHWASNYSTRMRRTIHGGFSMFTHYPNLSYKQLSPQARETFG
RWVKRSDDTMTREIALRAALEADGAAYQQALQDLHPESGSMAMRQSTILLSKSAKRI
QDLKRPASGVSEEELSYATGIHPMTMQCGQPMADRFSAEAGRLWERFRWVDDRRLR
GDQDATCPGFQGGPTSYFFNDVPAELTVDNFIAQWS

Codon-optimized NHI 142 sequence:

ATGTCTACAACACAAACCCAGACGGTTACGACAACGGTAGAGGCCCCGAAACCTAA
ATTGAGATATCTACATATTAGTGATGGGAAAGACGCAATTCTGAAAGTCATGCAGGA
GGACGGTGGCGTGGTGATTCGTGAATTCTTAACGGCGGAGCAGACCAGGCAAATT
AATCGCGATATTGATGCAGTTATCGATAAAATAGTGGATGGGGGAAAACATGCGGAT
AGTAAAATGCGAGATTTCCACGGTCTTGTGACCAAACGTATGACAAATCTGCCCACA
CATAGTAAAACCTTTCGCGACGAAGTGCTCAATCACAACTACTACATGAGATATGC
GAGGATATTTTTAGGGCTGACTCCGGTGATTACTGGATGAACACAGCACAGGTGATA
GATATCGGTCCAGGCAATGTGGCGCAACCTCTTCATCGTGATCAGATGCAATACGCT
GTCTTTTTCGGACGTTGGTTCTAAACTAAACATGAAGCCACTATAAATTTTTTTGTTG
CGCTGACTGAATTTACAGATGAGAATGGAGCCACACGCATTATTCGGGGGAGTCATA
TATGGCCTGATCATCGTGATTACGGTACCCAAGAGCAGACTATCCCTGCCTTAATGC

ATGCCGGGGACGTGACCCTCTTTAGTGGCAAAACCTCACACGGAGGTGGCGCAAA
TCGCACAAAAGACTTTTCGACGCCGTGGAATAGCGCTCAGCATGCAGGCAAGCTACT
TGACACCGGAAGAAGCGTACCCATTTATAGTTCCTCGCGAAATCGTTGAAACTATGA
CACCGCTGGCGCAACGAATGATAGCTTGGCGTTCTCAATATCCAGTTGACTCTGGG
GGCTTATGGCAATCAGACTACTGCGAAATTGCCGACGTTTTAGGACTGAAGGCAGC
TCCACCCGTGGCGTAA

NHI 142 protein sequence:

MSTTQTQTVTTTVEAPKPKLRYLHISDGKDAILKVMQEDGGVVIREFLTAEQTRQINRDI
DAVIDKIVDGGKHADSKMRDFHGLVTKRMTNLPTHSKTFRDEVLNHKLLHEICEDIFRA
DSGDYWMNTAQVIDIGPGNVAQPLHRDQMQYAVFSDVGSSTKHHEATINFFVALTEFTD
ENGATRIIPGSHIWPDHRDYGTQEQTIPALMHAGDVTLFSGKTSHGGGANRTKDFRRR
GIALSMQASYLTPEEAYPFIVPREIVETMTPLAQRMIAWRSQYPVDSGGLWQSDYCEIA
DVLGLKAAPPVA

Codon-optimized NHI 143 sequence:

ATGAATGTGCTGCGTATCGATTTTGAAAAATCGGATGCGGTATCATTATTCGCCTCAT
CCCTTCATCAGACCGGTTTTGCGGTTCTGGAAAATCACCCGATCGACATACAGTTG
ATTGATGATATATACAAGGAGTGGATGACCTTTTTTCACTCTGAAGATAAATTTAACTA
TCAATTCATAGTAGAGACACAAGACGGATATTTTCTGCAAGTATTTCTGAAAAAGCC
AAGGGTGCCGAACAGAGAGATCTTAAAGAGTTTTATCAATACTATCCGTGGGGGCAA
TATCCGAAGAAGTTGGGTGATAATACAAAAAACTGTATCAGCAGCTGAATCAGTTG
GCTGCGACTTTACTTTTCGTGGCTTGAAGATGAGCTGCCAGCCGCGATTGCATCGAA
ATTGTCCATGCCTCTGTCAGCAATGATAGAGCACAGTAATCAGACTATGTTAAGAATC
TTACACTATCCGCCCTTATCCGGAAACGAACCGGCGGGGGCCGTGCGCGCGGCAG
CGCATGAGGATATAAATTTAATAACCCTGCTTGTAGGGGCGACATCGAGTGGTCTTC
AGGTAAAGGACAATCACGGTAATTGGCACGAAGTTCCGTGCGCAAAGGAATCGATC
GTAATTAATATCGGTGACATGCTGGCTCTGGCGACGGATACTACTACCGCTCTACA
ACGCATCGCGTCATAAATCCTGATGCAAACCAGGAAGCCCGCTTGAGCATGCCTTT
GTTTCTGCACCCTGACACGAAAGTTCGTTTATCCCCGGAAAAGACTGCCGGCGAAT
ACCTCCACGAACGGCTCGTCAATTGGGAGTCAAATGA

NHI 143 protein sequence:

MNVLRIDFEKSDAVSLFASSLHQTGFAVLENHPIDIQLIDDIYKEWMTFFHSEDKFNYQFI
VETQDGYFPASISEKAKGAEQRDLKEFYQYYPWGQYPKNLGDNTKKLYQQLNQLAAT
LLSWLEDELPAAIASKLSMPLSAMIEHSNQTMLRILHYPPLSGNEPAGAVRAAAHEDINL
ITLLVGATSSGLQVKDNHGNWHEVPCAKESIVINIGDMLALATDNYRSTTHRVINPDAN
QEARLSMPLFLHPDTKVRLSPEKTAGEYLHERLVELGVK

Codon-optimized NHI 144 sequence:

ATGACGACCCCGACACAGACCAGTGCCGATGAACTGGGTACACCGTATGGGGGGC
TCACCCCAACGGCAACCGCTGATCTGAACCGCGATGGCTTTGTCCACCTGTCCGG
CGTGCTATCTCCCGAAACGATAGCGCGTTACGAGCCGACCATAACTTCAGAAGTCA

TCCGTCTTAATACTCAACATCTGCCTCTGGCGGAACGGGATACCTACGGCAAAGCAT
TCCTTCAGGTTTCTAACCTCTGGCGCGAGAACGAACGTGTGAAGGAATTGGTGT
TCGCCTCGCTTAGCAGCGATTGCAGCCCGCTTACTTGGAGTTCATGCTGTCCGGCT
GTACCATGATCAGGCGCTCTATAAGGAGCCTAGCGGTGGCATCACTCCGTGGCATG
CTGATCAATACTATTGGCCACTTTCTTCAGATCGTACAATTACGTTATGGCTCCCGTT
ACAGGAAACTGGCTCCGACCTGGGTCCTTTAGCCTTCGCTCGCGGCTCCCACCAT
TTTTCTTCAGGACGTGATCTGCCAATTTAGACGAAAGCGAACGCGTTATCGAGCA
AGCGGTTGCAGAACAGGATTTTGAGGATGTCGTGAACCGTTTGCTTTAGGCGATG
CATCTTTTCATCATGGGTGGACATTCCATCACGCAGGCCCAACCGTGGTGCAGTA
CCGCGTCGCGTAATGACTGTAATTTATATGGATGCAGATATTCGTGTTACCGAGCCG
ACCAATGCACACCAAGTGAACGATCGTGATGGATGCCCGGCACGGGCATTGGCC
AAGTCCCTGATACTCCGCTGAACCCAGTACTGTTTGACGGCCGGAGTTCATGA

NHI 144 protein sequence:

MTTPTQTSADELGTPYGGTPTATADLNDRDGFVHLSGVLSPETIARYEPTITSEVIRLNT
QHLPLAERDYGKAFQVSNLWRENERVKELVFSRPLAAIAARLLGVHAVRLYHDQALY
KEPSGGITPWHADQYYWPLSSDRITLWLPLQETGSDLGPLAFARGSHHFSSGRDLPI
SDESERVIEQAVAEQDFEDVVEPFALGDASFHHGWTFHHAGPNRGAVPRRVMTVIYM
DADIRVTEPTNAHQVNDRAWMPGTGIGQVPDTPLNPVLFDRSS

Codon-optimized NHI 145 sequence:

ATGAACTCCCTGTTTGAACCAATAATCGACGGGATTCTAGCGGAAGGTTATGGTATT
GTCGATAATTTTCTTACAGCCAATGAGGTATCCGCACTGGCTAAACGCCTGCACGAA
CGTCGTGAGGCTGGTCAGTTTCGTGCAGCAGGGATAGGAAACCAGCAGCGCACC
GTCGAAAACACCATTCGAGGTGATGAAATCCTGTGGCTGGAAGAGGCATCAGCGA
CCCCAGAGGAGACAATGTTTCTGCAACGTATAGGTGAATTTGTTGAATATGTTAACC
GTACCTGTTATTTGGGCCTGCGGGATTATGAGTTTCATTACGCATTGTACCCCCCAG
GTACCTTTTATAAACGGCACCTTGACCAGTTTCGTTCTGATTCCCGTCGGAGATTGT
CAGTTATCTGCTATTTGAACACGGATTGGCAGGAAACAGACGGAGGACAGCTGGCC
CTCTTTCTTCCAGAGTCTGATGGTAGCACAGAACACGAACTGACTATCCAGCCTATT
GGTGGCCGTCTCGTGTGTTTTGAAAGTGGTCGCTTAGAACATGAGGTGCGGCCTG
CAGGTCGTGAACGTCTTCCGTAACAGGCTGGCTGAAAACGGCGTAA

NHI 145 protein sequence:

MNSLFEPIDGILAEGYGIVDNFLTANEVSALAKRLHERREAGQFRAAGIGNQQRTVENT
IRGDEILWLEEASATPEETMFLQRIGEFVEYVNRTCYLGLRDYEFHYALYPPGTFYKRHL
DQFRSDSRRLSVICYLNTDWQETDGGQLALFLPESDGSSTEHELTIQPIGGRLVCFES
GRLEHEVRPAGRERLSVTGWLKTA

Codon-optimized NHI 146 sequence:

ATGTCACGGGTTTTTAACCGATACCGCCCTGGCACAGTATCGTGACCAAGGCTACTA
CAGTCCGTTGCCAGTGCTGTCGGATGATGAAGTTGCGCGCCTGCGCGGAAAGTTA
GAGGATTTTGAATCGGCACAGGGTCACCCGATAGAGGGCGCTCAGCGCTCCAAA

GTCATCTCCTGTTTAAATGGCTAGATGATTTGATGCGTGAGGCCCGCATTCTGGATG
CAATAGAAGACCTGATCGGCCCGGATTCTCTCTGTTGGAATTCCATATTTTGGATAAA
GGAACCCAAGAGTCCCAGCTATGTTTCTTGGCACCAAGACTTAAATATTGGGGTCT
GGACTGTGACGACCTGGTTACTGCCTGGGTTGCTCTGTCTCCGGCATCTGTGGAAT
CGGGTTGTATGCGCGTCTTGCCAGGTTACATAGGAATGAAATGTTACCGCATGAA
GATATTTTCGATGCAGAGAACATGCTTACCCGGGGACAAGAAATTGCTGTAGAAATG
GATGAGGGTGCTGCAGTCACCATGGCACTACAGGCAGGCCAAATGTCTATCCATAA
CGTTAGACTGGCGCACGCAAGTGGCCCGAACAATCAGCGGATCGGCGAATCGGT
GTGAGCTTCCATTATATGCCGACAAGAAGTAGGCAACTGGTTGGTGAATGGGATAG
CGCTGCGCTCGTAAGGGGGGAAGACCGTTTTCCACCATTTTGTGCGGCCCTGTC
CCATCCTGTGATTTGGATCCGGAGGCGGTAGCGTTTCATGAGAAAGCCTCAACAGC
GGTACGTGACATTCTTTTAAAGGTGCTCACCGCGTTCGCCAGACTCTTTAA

NHI 146 protein sequence:

MSRVLTDTALAQYRDQGYYSPLPVLSDDEVARLRGKLEDFESAQGHPIEGAQRSKSHL
LFWLDDLMREARILDAIEDLIGPDSLWNSIFWIKPKSPSYVSWHQDLKYWGLDCD
DLVTAWVALSPASVESGCMRVLPGSHRNEMLPHEDIFDAENMLTRGQEIAVEMDEGAA
VTMALQAGQMSIHNVRLAHASGPNNSADRRIGVSFHYMPTRSRQLVGEWDSAALVRG
EDRFHHFVAAPVPSCDLDPEAVAFHEKASTAVRDILFKGAHRVRQTL

Codon-optimized NHI 147 sequence:

ATGGATGATGAAGAGGAAGAATTTATTCCACCGCCTCCGCCACCAGGGCCTCCCCC
CCTGTAAAGAGAGGATCAGCTCCTGCATTTAGCGCGGCAGGGATGGCTTTTTCTAG
AACTTCCGGATGCTCTGAGCCAGTCCATGACTGATCTGCTGCGGCATTCTTCGCAG
TATTTTGATGCGCCGGATATTGAAAAAAGAGCTCTACCCAAGTAAATTAGGGACC
GAGTTTGGCTATTATCCAATTGCTGAAGAAAAAGAATACATTACCTTGCGCTGTCGG
ATTCACACCGATCCGAACGCTAAAGCGAAACCGGCGGCAAACTCGCTGCCTCAC
TTGAAGCTTCGACGGTGGACGCCTGGCACCACGCAGGAGTGTTATTATGGAGGATT
TTATGTGATATTGCCAGAGCTAGCGACTTGGATGTTTCCGTGTGGAACGATATCCTA
GACGGAACACTGGATATGCCCGATAGCGAAGATCAACTGACATATGCTTTGATGCGC
CTTTTTCGTTATTTTCCTTCTACAGGCCACGCAGAAGAACACACTGACTTAGGTCTG
CTGACCTTATGCATAGGTGATGGGACAGGACTGCAGGTCTTAGATAGGTTCAACAG
CGATCAAAAACCCGCTTGGGTGGATGGTCCAGTCGGTACTCACACAGCAACGGTAT
TAGTCGGACAAACATTAAGGGCACTCTCTGCACGAACATGACCGCCGGGGCACAT
CGCGTTGTTGGAAATAGTCAGGGCCGCCAGTCGGTGGTTTTTTCGCTGCGTCATTC
AACCCGCCATGACGTTGATTTTCACTGTTTGGCGGGGAAGGCAAACTCCATCCCA
CGGAACGTGGAAAACCATCCAGGCAGGTAAGGTCAATATTAACACCGAGAAGGAA
AAACGCGATGCGCAACGGGCACGTTTTAGAACAAACGCGCCCAGGCGCCGAAGCC
GAAGCGCTGGAGCATGCCACCCGTGGTCAGGGATAA

NHI 147 protein sequence:

MDDEEEEFIPPPPPGPPPLLREDQLLHLARQGWLFLPLDALSQSMTDLLRHSSQYF
DAPDIEKKELYPSKLGTEFGYYPIAEEKEYITLCRIHTDPNAKAKPAKLAASLEASTVD

AWHHAGVLLWRILCDIARASDLDSVWNDILDGTLDMPDSEDQLTYALMRLFRYFPST
GHAEETHDLGLLTLCIGDGTGLQVLDRFNSDQKPAWVDGPVGTHTATVLVGQTLRALS
ARTMTAGAHRVVGNSSQGRQSVVFALRHSTRHDVDFSLFGGEGKLHPTTELWKTIQAGK
VNINTEKEKRDAQRARFRTRPGAEAEALEHATRGQG

Codon-optimized NHI 148 sequence:

ATGGATTATGGCTATGCCGCAGAAGAACCGGCTGGGGATGCACCCACTGCATTTCTG
TGAAGCCGAAAAAATACCAGCTATACCGTGAGCAGCAAATGAAGATGAAGAACG
GTAAATTACGTGCCGGCAAATTAGTGACCCGGCCACAGATATGTCAGAAGTCCTG
GATTTGCACCCCGACGCGGCTGGCAATGGCTCATTGAATGAAGCGGGGGCCACCA
GCTCTTCTGGCCAGTTCACAGCAGATGCCCTGCAGCGCGGTGAAGTTGTGAGGCA
TGTCGTTCCAGGATGTCCAGGGGCAGTATATACGTTTCGCGCAGCATGCAGGGTTAG
TATTTCTGCCGGCTGCACTTCCCGCGCAACAGCAGGCTGCCTTAGTGGCAGCCGC
ACTGGAAGATTATTGTCAGCCTCCGGCACGGACTAACCATACTCGTCTGTATGGTCC
GCTTCCGGGTCTGTGGCGGGCCGCAACAGAACGTCTGCGTCTAAATTGGACGCGT
CGCGTCGAAGAGCCTGCGGCTCCACAGACTGGCGCTGCTGCAGGCTCAAGGGGT
GGTGCGGATGAGGGTGAAGAAGAAGACGGTGCGGGTGGATGTGCTGGCCCGTCA
TATGAGCAGTGCTGGTCTCGCGACGGGCCTGGCCCTGCCGCAGAACAATTGCTGC
GCAAGCTGAGATGGGCAACACTGGGGCCACAATTTGATTGGACCGAACGGCAATA
CGATTTCACTGGAGCCTATCGCCAGCTGCCCCCTTCCCTGTCCGATCTGGCAAGGC
AGATGGCTGCAGTAGTGATGCCCTGCAGGCTGCGGGTATGCAGCTGATGCCGGC
AGCAGCGGAAGCGGAAGTCCCTCAACAGCCGGCAGCACCGCGCGTACAGGCTAC
TGAGCCTTCGCAAGCTGCAGCTGGTGCCGTTTCGGCGGGACTGGCACCGCCTACT
GGCGCGGCCCCCGGCGGCCCTCGTGGCTACGAACCTGATGCAGCAATTGTAAATT
ACTATCAGATCGGTGATGTCCTAGGTGGCCATGTGGATGACGTTGAGTCAGATCTC
GCACAGCCGATCGTCTCTGTGAGCCTCGGTTGTCCGGCGCTGTTTTTAATGGGAG
GCCGAACGAAAGCAACTCATCCGTCAGCCCTACTTTTACGTGGTGGTGTGACTC
GTGCTTGCGGGCCAGGCCCGCTCTTGCTATCATTAG

NHI 148 protein sequence:

MDYGYAAEEPAGDAPTAFREAEKKYQLYREQQMKMKNGLRAGKLVTRPTDMSEVLD
LHPDAAGNLSLNEAGATSSSGQFTADALQRGEVVRHVVPGCPGAVYTFQAHLVFL
PAALPAQQQAALVAALEDYCQPPARTNHTRLYGPLPGLWRAATERLRLNWTRRVEEP
AAPQTGAAAGSRGGADEGEEDGGGGCAGPSYEQCWSRDGPGPAAEQLLRKLRWA
TLGPQFDWTERQYDFTGAYRQLPPSLSDLARQMAAVVDALQAAGMQLMPAAAEAEV
PQQPAAPRVQATEPSQAAAGAVSAGLAPPTGAAPAAPRGYEPDAAIVNYYQIGDVLGG
HVDDVESDLAQPIVSLSLGPALFLMGGRTKATHPSALLLRGGDVLVLAGQARSCYH

Codon-optimized NHI 149 sequence:

ATGTCCACGTCTGAGGCACTGGTAGCAGAACAGGTTATAGACACAGGCCGCTATCC
GTTGTCAGAACCAGGAGGTGCAGCCTGGCGGGAAGCAGTGTCGGCAACCCATCG
CGAATTAAGGGAAAGTGTTGTAGTGTTCTGCCAGACTTCGTGCGCGCCGAGCTG
ATTGGTGCGCTGCGCGCAGAATGCGTCCAGGTGGCCCCATTGGCACACTTTGACG

TCGAGACAGTGAACGCATACAATATCGCCTTAGATACACCGCTTCCAGAAGACCATC
CCGGGCGGACAACAATGGAGCGGGGGAATGCGTTCGTTCGCACGCGACAGAATAC
CGGCACATTATTTGATTCAGCAGCTGTACGGTTCGGTCTGTTTCAAAGATTTGTAG
CCGACTGCTTTGAACTGCCTCAGGTGCATGAGCTCGCGGACCCTCTGTCCGGGTT
AGTCTTAAACGTTGTATCTCCCGGATTATCACACCCTTGGCACTTCGATACTAATGAA
TTCACGGTGAGCATGCTTACCCAAGAGTCAGAGAGTGGAGGCGTGTTTGAATACTG
CCCGAATATAAGGAGCGCGGCGGCGGAGAATTTTGATGACGTTTCGTGCGGTTCTG
GCTGGTCGCGGTGATCACTTGATACGGCGTTTATCCTTACGGCCAGGCGACCTCCA
GCTGTTTAAAGGACGATATAGTTTACATCGCGTTTCAACAGTTGATGGGGAAACGGC
ACGCCACTCCGCAATATTTGCGTACTCGGAACGCCCGGGCGTGATTGGCTCGGTT
GAACGTACCCGCCAGTTGTTTGGGAGAGTGTTGCCGGATCATTTAGCCGCCGCGG
GTAACGCAGTACGTGGGGATCAGCTCCTTGATTAA

NHI 149 protein sequence:

MSTSEALVAEQVIDTGRYPLSEPGGAAWREAVSATHRELRESGCSVLPDFVRAELIGAL
RAECVQVAPLAHFDVETVNAYNIALDTPLPEDHPGRRTTMERGNAFVARDRIPAHYLIQQ
LYGSGLFQRFVADCFELPQVHELADPLSGLVLNVVSPGLSHPWHFDNEFTVSMLTQE
SESGGVFEYCPNIRSAAAENFDDVRAVLAGRGDHLIRRLSLRPGDLQLFKGRYSLHRV
STVDGETARHSAIFAYSERPGVIGSVERTRQLFGRVLPDHLAAAGNAVRGDQLLD

Codon-optimized NHI 150 sequence:

ATGTCCTCGCTTGAGGAACAGCTGCTGGGCTTGCGCGACTGTATACAAAATAAACC
CCCGTATTGCTCTGGTACCCTAAAGTTGCCTAGTGAAGATTTCTCTTGTAATATGG
GAAAGGTGATGATGTTTCGTCTGATTTGATTTATCTACGGCTACCGAGGAGTCGGCAAA
ACACCTGGCAGAGAGTTGTGACAGTGCGACTTTTGGGTTGAACAACCAGGACGTAT
TAGACGAGAGTTATCGTAAAGCTGGAAAGCTGGATCGAAAACACTTTAGTGCGGCC
TTTGACCTGGCCTCTACCGGCCTGTTGGGCATCATTGAAACCGAGTTGATCGAACG
CAATAAGGAATCCCCTTCAATAAGAGCTGAACTGTATAAACTTAATGTTTATGGACCG
GGGTCGTTTTTCAAAGCTCACCAGGACACCCCGCGGTCAACAGATATGTTTGGCAG
CTTAGTGGTGGTGTATCCAACAGTGACGAAGGGGGGGCGCTGGTATTGAGGCAC
GGCGGCAAAGAATGGACATTTGATTCGTCTCGGTATCATAGCCGATCAAAAAGAACC
GCGCCTAGCATATATAGCATTTTTTAGTGATGTGCAACATGAGGTGCTGCAGGTTCA
ATCAGGTTATCGCATTACCATCACATATAATCTGTACTATAATAGCCCGCAGCCCAGC
GATTTACCTGTGCGGCCGTTGACCTTTGAATCTAGTCTCAAAGCTGCTCTGCAGCA
CCTACTTGATGATCCAACGTTTTTACCCGAAGGTGGTCATCTGGGCTTTTGTCTAAG
CCATCAGTACCCGGTTGAAAGCGATCGCGGGGCAGGTGCAAAGCTCTGAGCGAA
GTATCCGGCATGTAAAGGGATCTGATGCGTCCTTGGCTTCGGTTTTTACAGCGCT
GGGACTTAGAAGCAGACTGGAATTCGTCTGGAGGATAACGAGGTCTGGATTGAG
AGTGAGGAGGGCGGATCAACTCACATGATTGCGTATGTTTTATTTGATGAATATGCG
CCGGATATAGAAGAAACAACGGTAGAAAGTCAGCTGTTTCAAGTTCTGCGCGACAT
CGGTGGCAAACCTGGCGACCAGAATTCCGCCGCTCTACCCAGGCAGGAACGCGATT
GAACCCGCTGTACATGTGCATTGGGTGGTCAGTCCAAGTAGCAATACAGGTTTTAA
TTCAAACCCGATACCTCACATATGGGAATGAGGCTTTACCTGGAATGCAGTATACC

CATCTTTGCATTTTGGCTGATGTGGGGGATTCCAGCGAACGACGTCGTTTCATGGTAT
AGTCGTGTGGGCGCGCGAAGATAA

NHI 150 protein sequence:

MSSLEEQLLGLRDCIQNKPPYCSGTLKLPSEDFLLYYGKGDDVRRIDLSTATEESAKHL
AESCDSATFGLNNQDVLDESYRKAGKLDRKHFSAAFDLASTGLLGIIETELIERNKESPS
IRAELYKLNVEYGPSSFFKAHQDTPRSTDMFGSLVVVYPTVHEGGALVLRHGGKEWTFD
SSRIIADQKEPRLAYIAFFSDVEHEVLQVQSGYRITITYNLYYNSPQPSDLPVRPLTFESS
LKAALQHLLDDPTFLPEGGHLGFCLSHQYPVESDRGAGAKALSEVSGMLKGSDASLA
SVFTALGLRSRLEFVVEDNEVWIESEEGGSTHMIAYVLFDEYAPDIEETTVESQLFEVLR
DIGGKLATRIPLYPGRNAIEPAVHVHVWVSPSSNTGFKFKTRYLTYGNEALPGMQYTH
LCILADVGDSSERRRSWYSRVGARR

Codon-optimized NHI 151 sequence:

ATGTGGCTGAGTAATAACCGTAAACCAGCAACTACGGGGCCACCTGTGATTTTAGA
GACTTCATATACGAGCAAAGAGGTGCCCATACCAGATACATTCCTTACGACTCCGCC
GCCTGATGCCGAGCCGATCACCGCCAATATTATCGATTTTTCTTCATCATCGCTGCC
GGAATATAAGGATTATTACGCGGTTATATTAGATAACGTGTTGAGCCAGTCGGAATGT
GCCCAGTTATTATCGCTGGCGGAACAGTCCGTTGAAACAGGTGAAGCCTGGAAACC
GGCATTGGTTAGTGTTGGTCCGGGTCTGTGAGGCTTTGATGACAACTATCGTAACT
CGGATCGCATCATTTGGGACAATGATGTTATCACGTCACGCCTCCTGGAGCGGTGC
TTTCAGGCAGACGGAATTCGTAAACGTCTTTCTGTAATCGCCGGCGAACAGTATCG
CGGGATTCTTGGGGCGTGGGCACTGGAGAGTGCACAAAGCTGGCAGGTAGTCAA
GCTAAATGAACGCATGCGCTTCTTAAGGTATACTAACGGGCAATTCTTTAAAGCACA
CTGTGATGCGCCGTACACGACCGGCAATCAGTGTACCTTTTATACTGCAACTGTA
CCTATCTGATGGAAAAGAAGACGAATCAAAACGCTTGGTGGGCGGTGCTACCACCT
TTTGGAGTCGCGATGAGCGTCGCCGCTTGGATGTTTCATCCCAAACCCGGGCGGGT
CCTGATTTTTTCAGCATAGGGGATTATATCATAGTGGCGATCAAGTACTCCAAGGGGTT
AAATATACGATGCGTTCGGATATTCTGTACAAACCGGAAACACAGGGGAAATGA

NHI 151 protein sequence:

MWLSNNRKPATTGPPVILETSYTSKEVPIPDFTLTPPPDAEPITANIIDFSSSSLPEYKD
YYAVILDNVLSQSECAQLLSLAEQSVETGEAWKPALVSVGPGRREALMTNYRNSDRIIWD
NDVITSRLLERCFQADGIRKRLSVIAGEQYRGILGAWALESQSWQVVKLNERNMRFLR
YTNGQFFKAHCDAPYTTGNQCTFYTLQLYLSDGKEDESKRLVGGATTFWSRDERRRL
DVHPKPGRVLIFQHRGLYHSGDQVLQGVKYTMRS DILYKPETQ GK

Codon-optimized NHI 152 sequence:

ATGACCCTGTGCCGTATCTCAGCCTCGGCTATTCTGTCGTAAATCGTATGACGCTT
TGCCGTTTTATTTGCTCCGAGCTACCGTGTGCGTGTGAGCTGTTATTCGGTGGGTAA
CCGCGCCATGTCTTGCTCCCGATCATCGGAGCTGATGAGTGTTTCCCCACCATTAC
AACCGCAGAAGTCAGCCGATTCTGCAAGTAGCAGTAGTGGTCCCAAGATCGGGGG
TTGCCTGGAAAACGAAAGCTCAGATGACTGGAATAGCCATGAGAAATGGTTCTCCC

TGCGCAATCCCCAGATCCAGGACGATTATCGCGCCTATCCTATGCCAACGCCAGGA
CGTATTTTTAGTCCAAAACGCGTAGTGAATAACGATTTGCACAATGAAACCTCACCAA
GCTTCCGCCCCCTTCAATATCTGCTACCCGGAAATGCGTATCTCCGGGCGGGGCACC
CCGGTTAAATTAATGCCAGCTTGTTTACCCAGAACAAGCAAAAGCGGATGGAGAA
CGAATGCCTTAAGAACGGCATTGAATGGCATAAACTAGGCCCTGGAATGATCCTGTT
AAAAAGTTTTTTAGGTACGATAATCAGATCGAAATAGTGCGTATTTGTCAAAAACCTG
GGGGCGGGTCCTGGCGGCTTTTATCAGCCGGGCTATTCCGACGGTGCTAAGCTTC
GCCTAAAGATGATGTGTCTTGGGAAAAATTGGAATCCACAGACAAGCGCATAACGAG
GATACATGTTTCGACGGGAGTAAACCGCCGGACATTCCGAACAAATTTAAGGATCTG
GTACTACGGGCCATCACATCAAGTCATGAGTTACTGAAGCAGGAAAATCGCTTCGC
GGACGTAAAAAAAATTTTACCTAGTATGACACCAGACCTATGCATAGTAAATTTCTATA
GGAATACTGGTTGTCTTGGCCTGCACCAGGATAAAGATGAAAGTAAGGAATCCATTG
GTCGGGGTTTACCAGTAGTTAGTTTTAGCATTGGCGACTCAGGTGATTTCTTTATG
GTAAAAACGCGATGAGAAAAAAGCGTCTCGTATAAAGTTGGAATCGGGCGACGTA
CTGATTTTTTGGTGGAATCCCGGGATGTATTTTCATGGCGTACCTTGTATCCACAGG
GATACCATGCCCAACTGGTTACGTGATGAAAGCCGTCTATTTCCGGGCGGCCTGAA
TCTGACGTTTCGCAAGTATTGA

NHI 152 protein sequence:

MTLCRISASAILSLNRMTLCRLFAPSYRVRVSCYSVGNRAMSCSRSELMSVSPPLQP
QKSADSASSSSGPKIGGCLENESDDWNSHEKWFSLRNPQIQDDYRAYPMPTPGRIF
SPKRVVNNDLHNETSPSFRPFNICYPEMRISGRGTPVKLNASLFTQNKQKRMENECLK
NGIEWHKLGPGMILLKSFLGHDNQIEIVRICQKLGAGPGGFYQPGYSDGAKLRLKMMC
LGKNWNPQTSAYEDTCFDGSKPPDIPNKFKDLVLRITSSHELLKQENRFADVKKILPS
MTPDLCIVNFYRNTGCLGLHQDKDESKESIGRGLPVVSFSIGDSGDFLYGKKRDEKKA
SRIKLESGDVLIFGGKSRDVFHGVPCIHRTMPNWLRLDESRLFPGRNLNLTFRKY

Codon-optimized NHI 153 sequence:

ATGTCAGATAAAATGAACCTTCCTATCATTGATCTGTCTGGATATCTGAATCCGGAGT
CGCTGGAGGATCGTGAGCGTGTGATCGCCCAAGTGCGCGATGCATCCCGGCAGTA
TGGTTTTTTCCAGGTACGCGGTCATGGAGTGCCCCCGAGCCTCCAACACGATCTGA
TTCGATCAATGGGTAATCTGTTTCAGTTTACCTAAAGAAAAGAAGTTAGAAATGTCTTT
TCTCAAGAACCCGTGTAGGCGGGGATATGAAGCGGCGGGTATGTCTCTGCGAGAA
GGGGATCTACTTCCTGACGCCAAAGAATGTTTTTATATCGCCCGCGATGATCCGATT
ATCGAGCTTAGTGGCTTCTATGGTCCCAATCTGTGGCCTAATCTGCCAACCGAAGAT
TTTCGCGACCCGGTTTGGAATATTATGAACAGACTAGCGAACTTGGTAAGAAGATT
TGGTCTATTTTGTAGAGGGGCTCGGTCAACCGCCCAGTTTAGTGAGAAATTTGC
ACGCCGCGCGATTGTACCGATGAAGATGATTCGCTATCCCCCGCAGTCAATTACCT
GCCGGGCCAATTTGGCATTGGCGCGCATACCGACTTCGGTGGCGTCACGATTCTC
TTCCAACAACCAGGTAAAGATGGATTAGAAGTGTGGCATGAAGGCAAAGAGACGTG
GATCGAGGTTCCGGCTCTGGAAGATGTTTACGTCATTAATTGCGGTGATATGATCCA
ACGATGGTCCGGAGATCGGTACAAGAGTGCGCGTCACCGTGTAATTAATAAGTCCC
AGGGCGAACGGTTATCCTGCGCTACGTTTTGGCATGGCGACATTAACGCCACTAAT

CCTTTAAATCCCGACGATCTTAATTCAGAAACAGTTGGCCAGCTGCTGGTGAAGAG
ATTTAGAAACCAGTATAGTATGACCGAAGAATCTATTGCGCAGGTTGGTGCGGCGTA
A

NHI 153 protein sequence:

MSDKMNLPIIDLSGYLNPESEDRERVIAQVRDASRQYGFFQVRGHGVPPSLQHDLIR
SMGNLFSLPKEKKLEMSFLKNPCRRGYEAAGMSLREGDLLPDAKECFYIARDDPIIELS
GFYGNLWPNLPTEDFRDPVWKYYEQTSELGKKIWSILLEGLGQPPSLVEKFARRAIVP
MKMIRYPPQSITLPGQFGIGAHTDFGGVTILFQQPGKDGLEVWHEGKETWIEVPALED
VYVINCGRMIQRWSGDRYKSARHRVINKSQGERLSCATFWHGDINATNPLNPDDLNSE
TVGQLLVKRFRNQYSMTESIAQVGAA

Codon-optimized NHI 154 sequence:

ATGCGTGGAGTAGATTTTTACACAAAGCGGGCCAAGGATATGTTGATCTGCCTCC
CCGTCTTGTGCATTAGATGACCTGCCCGAAGTTCGTCGTCATTAGATGGCTTAGC
AAGACTGCGCCGCAGGCCTGGCCCTGATGCTAGTGCTCGCACCACTGCCGCATTA
GCAGCATTGTTATTGAAATACACGCTCAGGCGGCCAGAGCCGCAGGCGATTACC
GGCGCTGATGGATCGCAGGGTGCACGATGTAGTGGAATGGCTGACTAGGCACCCC
GCACGGAGACTAGATGTTACGCGAGCGGCGGCGTCAGCGGAGTTATCAGCAGATC
AGTTTTCAAGGCTGTTTCGCCGCGAGACCGGGCAGTCACTGAGAGCCTACCAAATA
GCCGTGCGTATTGATCGCGCGCGGGAACCTCCTCGCCGAATCGGGTATGACAGTCT
CCCAGGTAGCGCGCGTTCTCGGTTATGCAGATCATCGCCTTTTCGCCAGACAGTTT
AAAGATCACACCGGCACCGCACCGGCATCCGGCGGAGCTCCGAGAGAGCGTACA
CCCGATCGGTCATCAGCCCGCAATCCACGTCTCGCTCCGTCTCCTGCGGGATTG
CATGGGCATATAGACGACCGATGACATATATCGTTCCTGCAGATCTGTACCGTGCGG
ATCGGCTGGCTCCGCTGCTGGATAGCATTAAACGAAATCGGTGCGGCAGAGACGGA
TGCCTATCGCTCTCTTGGGTTTCTCGCAGTTAGATCCCTGCTTACCGATCGCGATGT
TCGGGGAGTGTTAGACGGACTGGCCGTCGTAGCGCGGCCGCCGAGCCCGGCCTC
AGTGGAATTTGAGGAATGGGCGAGACCGGAAGAACGTGACCGCGATGAACTTTTA
GATGGAGTACGCAAACTACAACATTTTTGTGTCCATGAACCGCGGCTAGCCGAGCT
CGCACGGCACCCGGCTATCCTGGCGACCGTTGCCAGATTTCTCGATGCTGAACCT
GTTTTGATCCAGGATATGGCGCTTTTGAAACCACCGGGACGCGGTGCGGAAAAACC
GTGGCATCAGGACAAAGCTTTTTTTGATATTCGCTCGGTGCACCAAGTGGTTGGAG
TTTGGATTGCTCTCGATGAAGCCACTATCGATAATGGATGCATGCACATTCTGCCTG
GAACACACACTGAAGGTCCGGTGGTCCACTTTGTCCGCAGAGATTTTCAGATTTGC
GACACGCAAGTCCAGACACACCGTGATACAGCCGTTCCACTTCCGCCTGGTGGCG
CCCTGTTTCTGGACGGATTACTGCACCACGGGACCCCGCCAAATCTGACGGACAC
TCGTCGGCGGGCAGTTCAGATGCACTATGTTCCGGCCGGGACCATGCGGACACCG
CAGGAAGATAGACTGGAGGTTTTTCGGCGCTGATGGTAAAGATGCCACTTGTTAA

NHI 154 protein sequence:

MRGVDFSHKAGQGYVDLPRLVHLDDLPEVRRHLDGLARLRRRPGPDASARTTAALA
ALLLEIHAQAARAAGDLPALMDRRVHDVVEWLTRHPARRLDVHAAAASAELSADQFSR

LFRRETGQSLRAYQIAVRIDRARELLAESGMTVSQVARVLGYADHRLFARQFKDHTGTA
PASGGAPRERTPDRSSARNPRLAPSPAGFAWAYRRPMTYIVPADLYRADRLAPLLDSIN
EIGAAETDAYRSLGFLAVRSLLTDRDVRGVLDGLAVVARPPSPASVDFEWARPEERD
RDELLDGVRLQLHFCVHEPRLAELARHPAILATVARFLDAEPVLIQDMALLKPPGRGRE
KPWHQDKAFFDIPLGAPVVGWVIALDEATIDNGCMHILPGTHTEGPVVHFVRRDFQICD
TQVQTHRDTAVPLPPGGALFLDGLLHHGTTPNLTDTRRRAVQMHYVPAGTMRTPQED
RLEVFGADGKDATC

Codon-optimized NHI 155 sequence:

ATGATTCAGCTCCCATTAATGTTCTGTTCAATATGATGACCAGTTTCTGAGCCAGA
AAGAATCAACAGCCCTGTACCAAAAACCTGAAGCTTTGTATTGCGACTGCACCTCAGA
AAATCGATCTCCCTAACGGGGAACACATTCTCCTGGAAGCCCCGAAGACACAGTTC
GCACTGCCTGAATTCATTGCCACCGAAAAAATGCATAAGGCCCCAGGGCGAAATACA
AGTCATGTTCCCAGAGCTGGAACGTATTAAGCAGAAGCTGGAGACACTTACTCAGA
GGCGGTTAGATGTCTGTGTTGCAATTTATTATTCAAGCGGAGCGGTTGGCGTTGACT
ATCATTCTGATTTGAGCGCCTTCGGTGATACTTCTTGCATCCCAAGCATCTCGCTCG
GTGCGACCAGGACGTTTAGCCTCCGGGAAAACATTAATCACGCCAATTCATATGACC
TGGAGCTGGTACACGGCTCGTTAGTTATTATGGGAGAAAATTGTCAAGAACATTATG
AGCACGCTATTCTTGCTGATCCAAGCATTCAAGAAGACAGAATCAATCTCACATTC
GTACATTCGGCGGTCTGAACTAA

NHI 155 protein sequence:

MIQLPLNCSVQYDDQFLSQKESTALYQKLKLCIATAPQKIDLPNGEHILLEAPKTQFALPE
FIATEKMHKAQGEIQVMFPELERIKQKLETLTQRRLDVCVAIYYSSGAVGVVDYHSDLSAF
GDTSCIPSISLGATRFTSLRENINHANSYDLELVHGSVIMGENCQEHYEHAILADPSIQE
DRINLTFRTFGGLN

Codon-optimized NHI 156 sequence:

ATGCGCCTTAAATTGAATGAGCTCCAGGATCAGTTTGAAGAAGTTGGAGCCATTCTG
CATACCGAACTTTCTTTCAAATCCGACCTAGACACGATAAAAGAACTCTATTCAGAA
CTGAAAATAAAAACTTAAAGGAATATACTCAAATGTTAAAGATAAGAAGCCGGAAC
TCAACTACAAAATAAATGAGACGTTGAAGTCGTGTTACAAAAAGGCAATAGATCAGC
ACTTTATCGACTATCAAATTGGCGGAGGCGCCTTTCTTATAAAGGGCACTGGGGAG
GGGAGCGAATCAAGCCTACACCAGGATTGGAATATTGTAGACGAATCACAATATAAA
TCGGCCGCTATATTCTGTCCAATCCAGGACGTTGATGAGAAGAATGGCTGTATTCAA
GTTGTGAAAGGCAGTCACAAATGGTTTAATAATATCAGGAGTTTCAATATCAGTAGCA
TATTTTTGAACTTTAACGACGTAGAGAAACAGCTGATTCCCTTCCCGGCAAAGGCAG
GTGATGCGATTGTGTTTAGGCATAACGTGTTTCATGGCTCTAAGCCTAACTTCACCA
ACGATATTCGCATTGCAGCTTCCATGTCTATAGCCACCAAGAACAGCCCGTTGATAC
ATTATAAAAAATCAGGAGATAATGTGATTACGTTTCGTGTTAACGACCACTTCTTTAAC
GAAGAAGTTCAGCAGCTGTACCAGAACAAGGAGATAAATATTGATATTATAGACGAAA
AACCCATTTCAGAATTTCAAATTCTGGATCAGAAAACCTTTCGCGAATCATTATAAAAA
GAATAAATCAAAAGGCATCGTGAATCGAATTAAACACATCTTGTTCCGCTAA

NHI 156 protein sequence:

MRLKLNELQDQFEEVGAIKIPNFLSKSDLDTIKELYSELKIKNLKGIYSNVKDKKPELNYKI
NETLKSCYKKAIDQHFIDYQIGGGAFLIKGTGEGSESSLHQDWNIVDESQYKSAAIFCPI
QDVDEKNGCIQVVKGSHKWFNNIRSFNISSIFLNFNDVEKQLIPFPAKAGDAIVFRHNVF
HGSKPNFTNDIRIAASMSIATKNSPLIHYKKSGDNVITFRVNDHFFNEEVQQLYQNKEINI
DIIDEKPISEFQILDQKTFANHYKKNKSKGIVNRIKHILFR

Codon-optimized NHI 157 sequence:

ATGAACTTACATAGCATCGATGATGGGTTTACACATGCCGAGTGTTCTGCGATCATT
CTGCCGTTGCAAGTGCTCCAGCGAGAGAAGCTCTGCTGGTTGGTCAAAATAGGGA
TCATAATCTGCGTCGAGCTAAACTGGTCTGGCTCGATGAAATTGATGGTATGGCTTG
GGTGATGGAACGTCTGATTGATGTAGTACGTACAAGTAATAGGAATTGGTTTGATTTC
GATCTTCGAGAGTTTCGCAGAGTCCCCTCAAGTCGCGAGCTACGATGCTACCGATGG
TGGTCATTTTACGTGGCACAGCGATATTGGTGATGGTCCCATGGCCCATAAACGCAA
GCTGACCCTGGTTGCTCAACTGTCTAAGCCGGGTTTCATATGATGGTGGCAACCTCG
AAGTAATGCCTGGCGCACACATTATTGCCGCTAATCGCACCCAAGGAAGTATCACCA
TCTTCCCTTCCCTTCACTTGTCATCAGGTCACCCAGTCACCCGTGGCACTCGGAAT
TCCCTGACTGTGTGGGCGCATGGGCCGAAATTCCGGTGA

NHI 157 protein sequence:

MNLHSIDDGFTHAECSAITAVASAPAREALLVGQNRDHNLRRAKLVWLDEIDGMAWVM
ERLIDVVRTSNRNWFDLREFAESPVASYDATDGGHFTWHSIDGDGPMAMHKKRLTL
VAQLSKPGSYDGGNLEVMPGAHHIAANRTQGSITIFPSFHLHQVTPVTRGTRNSLTVWA
HGPKFR

Codon-optimized NHI 158 sequence:

ATGGCATCTGACCTCAAAGACTTTATACGCGTTTACGATGGCGTGCTGGATCCCAAT
CTTTGCAACAACGCCATCAAAGTGTGATGAAACACCAGATGCCCAGGACCGCTG
GGATAATCAGCATAAACCACAGTTCACGCAGATGAATATAACTATGTTATCCGAACAG
CAGAGAGACCCTAAATGGGGTGTGATCCATAACAACATTATAGCATCAATTCAGACC
GTTAGCGAATACTATATGCGAGATACTGGATGTTCCCCGTTCTGGCCGCCGAAGAAC
TCTATAGAGCAAATTCGTATGAAAAAATATACCGCTGAATCAGGTGACCGGTTTGATC
TGCATATAGATGTTGGAGATTACGCCACTGCCCGCCGGTTTTAGTGATGTTTTCTA
TTTAAACGATGTTGAAAAAGGTGGTGAAACGTCGTTTCCGGCGATTGAACTTGATAT
TAAACCGAAACAAGGCTCGGTCTTGTGTTTTCTCCCGTGTGGATGTATCCACATTT
AGGCAAACAGCCGATTTTGAATGATAAATATATCGTCGGCACCTACCTTCATTATCAG
TGA

NHI 158 protein sequence:

MASDLKDFIRVYDGVLDPNLCNNAIKLFDETPDQAQDRWDNQHKPQFTQMNITMLSEQ
QRDPKWGVHNNIIASIQTVSEYYMRDTGCSPFWPPKNSIEQIRMKKYTAESGDRFDLH

IDVGDYATARRFLVMFFYLNDVEKGGETSFPALIELDIKPKQGSVLCFPPVWMYPHLGKQ
PISNDKYIVGTYLHYQ

Codon-optimized NHI 159 sequence:

ATGGCTCAAGCTCTGGATCTGGTTGCCTATGAGACTCTTGAAGAACAGGTCGAAGC
AATGTATCGTGATGGCTTTGCATATTTACCGGACGTTCTCAATGCTGACGAGGTGGC
AGCGCTGCGCGCCCAGATGGACGCACTTACACCGAGACCCGAATCATACGATAAAA
ACATGACTGCGGAGGAGGACGGGTTTATCAATCGTCACGTTAACAATGCCTTTAACC
GGCATCCCCACTTTCTGGGCTTTCTCGATCGGGAAGCGGTAGTCGATATCGCAGAA
GCGATCCATGGGGACGATTGTCATTGCTTAGGAATGACTGCATGGATGACTGGCCC
TGGTCGTCCGGACCAGGCACTGCATACAGATTGGTTACCGCTAGAACTTCCACCAG
ACATCCTTGCCGACCCCAGAGTGCGTACGCCTATTTTTATAACCACCGCGCACTATT
ATCTTGATGATCTGTATGAAGAATTAGGACCAACCAATTTTATCCCGGGCTCGCATCT
GTCCGGTTGTAAGCCTGACGGGGCTACAAGTTGGCAGGGCCAGGAAGAAAGAAG
CATTATGTGCAAAGCAGGAGATGTTGTGATTTTCCGTTCCGAGGTTTGGCATCGCG
GGACTGCAAATACCTCTGACCGCACGCGTTTCCTGCTGCAGGTACACTACTCCATG
CGGATGATCACCCAAAAGTATCCGCCATACATGGGACGCTTTAAGTTTGACCCCGC
GATTCTGGAAGCCGCAAATGAACGGCAACTTCGACTTATGGGCGACCATAAGAAGA
GCAACTATGATTAA

NHI 159 protein sequence:

MAQALDLVAYETLEEQVEAMYRDGFAYLPDVLNADEVAALRAQMDALTPRPESYDKNM
TAEEDGFINRVNNAFNRHPHFLGFLDREAVVDIAEAIHGDDCHCLGMTAWMTGPGRP
DQALHTDWLPLELPPDILADPRVRTPIFITTAHYLLDLYEELGPTNFIPGSHLSGCKPD
GATSWQQQEERSIMCKAGDVVIFRSEVWHRGTANTSDRTRFLLQVHYSMRMITQKYP
PYMGRFKFDPAILEANERQLRLMGDHHKKSND

Codon-optimized NHI 160 sequence:

ATGCTGTCTAAAGAACAGGTCACCCATTTGAAACCTTTGGATTCTGGTACTGAGA
AACGTACTIONTAGAGACGAGGTGGACATTATGATAAAGGAATCGGAGGAAATATTT
TCAGAGCTACGCCAGGGGAAACCGTTACGGCCCAAGAAAGAGAAGCTGTGCAGC
CTTTTTTCGAGAGAAAACCTTTTTTACACTCATTGCTTGCGGACGATCGTATTTATAG
TATCGGAGAATCGATTCTGGGCTCTGATTTTATGTTGGTTGTTACAGAGGGCAACTT
ACATGTAGGTGACACGCCGTGGCATGCGGGAGAGGAGGCAAGGGTTCTACGCACC
GTTAAAATAGGTTTCTATCTGGAACCGTTAACAATTAACACCGGTAGCCTACGCGTG
ATTCCTGGCACGCACAAGATTAGTAGCCAGATTATTTTAGTATTCTCAAAACAGCAA
ACAAAACAGTGACTTTATGCCCTTCGGCGTCTCACCCAAAGAAATCCCCTCGTAC
CCTATTCAAAGCGATCCGGGTGATCTGATAGTATTTACCGAATCTGTTGTCCACGCA
GCTTTTGGAGGTAGGGATGGGCGGCACCAACATGCTATCAATTTTTGTGAGTATCCT
AAATCTGAAGAGCTGATAGCGTTTACCAAAGACTGGTATTCGAGAACCGTTAACATAT
TTCGACCTCCGATTTCAAATATAAATAGCGACAATGCCAGATTACGACGTATGTCTGA
AGGCTGGCTTCAGCTAGGATTTGAACCAGTGAACATTTGA

NHI 160 protein sequence:

MLSKEQVTHFETFGFLVLRNVLSRDEV DIMIKES EEIFSELRQGKPFTAQEREAVQPFFE
RKPFLHSL LADDRIYSIGESILGSDFMLVVTEGNLHVGDTPWHAGEEARVLR TVKIGFY L
EPLTINTGSLRVIPGTHKISSPDYFSILKTANKNSDFMPFGVSPKEIPSYPIQSDPGDLIVF
TESVVHAAFGGRDGRHQHAINFCEYPKSEELIAFTKDWYSRTVNIFRPPISNINSDNAR
LRRMSEGWLQLGFEPVNI

Codon-optimized NHI 161 sequence:

ATGGTAGAAAATT CGCAGCAGCTGTCCTCGCAATACGACGAGCATGGATACTTCGT
CATTCGTA ACTATTTTAATGAGGCCCAAATTGCAAGCCTCAGAAAAGTTGTGCTTAAA
TTTCACGAAAGTTGGAAGGCTGATAATGAGGAGTTTTACCAGGAAGAGGCATTTAAT
TCCTCCTTAATTACAGGTT CAGAATATCTGCCTACCGAAGACCGCAGGGTTCTTTTAA
ACTTTATATCCAGCAAAAAGGTTATGGAAGTTGTGCGATGCCGTAATACCTAACAAACC
GGCTTTTATGAATACCCAATTATTTTTTAACCCGGTAAATCCTCAACAGAAAGATTTTT
GGCACCGCGATTGCCAGTACGATTATGATATTGACGATCAGATGAAAGTTATTATGGA
AACGCAGGTTCTGCATCTGCGGGTGCCACTTTTTGACGAGCCGGGCATGGAGCTG
ATCCCTGGTACCCATAAACGGTGGGATAATGAGGAAGAATATAACGTTCCGCCAGGAA
GAAATGGTAAAGTATCGTCTGATGATATTTCCGGTGGCAAGCAAATTCGCTGGCG
GCGGGTGATCTATTAGTCTTCAGTGCGGATATGATCCACCGCGGGCGTTACGGTCT
AGATCGTCTGGCGTTAGACATCCTCATCTTTGATAGCGCTGCTGACTACGTTGATTA
CGTTGATGACGATTGCCTTCCTACACCAGCAATGTTAAGCAATATAAGCGACCCGCG
ACTGTTTATGAACACGCTGCACTTGAAGTCCATGCAATACTCCTGA

NHI 161 protein sequence:

MVENSQQ LSSQYDEHGYFVIRNYFNEAQIASLRKVVLKFHESWKADNEEFYQEEAFN
SSLITGSEYLPTEDRRVLFNFISKKVMEVVDAVIPNKP AFMNTQLFFNPVNPQQKDFW
HRDCQYDYDIDDQMKVIMETQVLHLRVPLFDEPGMELIPGTHKRWDNEEEYNVRQEE
NGKVSSDDISGGKQIPLAAGDLLVFSADMIHRGRYGLDRLALDILIFDSAADYVDYVDDD
CLPTPAMLSNISDPRLFMNTLHLKSMQYS

Codon-optimized NHI 162 sequence:

ATGAAA ACTGACGCGCGCGCCCTGGATAGCGCGCTTGAGCAGTTTCGTGCCGAAG
GCTATACCATTATCCGTAATTTTATCC CAGCAGATGAGGTGGCGACCTTGCGGGCCG
AAACCAAAAAATTATACGAGACTGCGATCGCGCATCACGCGACCTACAAGCATGGG
AACCTTACCTATGAGATCTTACCGGAGGCCCATTTCCGGTGCTCGCTATGTCATCCAA
GCATATTGGTTCGCGTGGGCCAGTGAGTATTTTGATAAATTCGGTCACATCCGGCA
AATCTTGAGTTACTGGAACCGCTGATTGGTAGGAATATTAAACAGGTAACGCAACAG
ATCCATTGGAAGCCGCGGGAGCCTCTCTGACCGGATACCGATTTACCAGGACCT
GCGCTTTCGGGAGTCGAAGGAAGCATTTGAGGATGTGGTGCGCGATTCCGGTTACC
ATTGGTGTGCTATCGATCCGGCAACCAAAGATAATGGATGTCTGAGAATCGTCCCC
CGGTCTCATACTAACGGTTATTTGGGCCTATCAGATAACGGCGAAGGGAGCATTATG
AAAGGGCTGACCCATAACGATGAGCTGCTGGCCGTCGGTTTAAATCCTGACGACAT

ACTGGATGTTGAGCTGGAACCAGGGGATGCTGTGGTTTGGGGTCTTCTGACGGTG
CATGGTTCCCTGCCAAATTCCTCCGCAAACGATCGAGCCTTTGCTTTAAGTTCTTAC
GTGCGTGGCGAAACGTCCCAACGTGGTGAGTGGGCATTTAAAGATGGTGTGGGAA
CTCGTCTAGGTGCAACACCACAGCTGTGTAAAAATGAACGCCTGTTTGATAACCTTG
AACCACATTATGATGAAACTAAATGGTTCCTGTAA

NHI 162 protein sequence:

MKTDARALDSALEQFRAEGYTIIRNFIPADEVATLRAETKKLYETAIAHHATYKHGNLTYEI
LPEAHFGARYVIQAYWFAWASEYFDKFRSHPANLELLEPLIGNIKQVTQQIHWKPPGA
SLTGYRFHQDLRFRESKEAFEDVVRDSVTIGVAIDPATKDNGCLRIVPRSHTNGYLGSL
DNGEGSIMKGLTHNDELLAVGLNPDDILDVELEPGDAVVWGLLTVHGSPLNSSANDRA
FALSSYVRGETSQRGEWAFKDGVGTRLGATPQLCKNERLFDNLEPHYDETKWFL

Codon-optimized NHI 163 sequence:

ATGCGGGTTCTTCTGGCCGATGAGCCTGGGTTCATGAAGGGCTCGTGCAAGCAG
ATTCCACCGCGAAGCCTCAGTTACCGAGATCACCTCCTATTTGGGCGGGTTCAAGA
CAGGAAATTTGTGAGAATTTAGAATACTTCCGCTCATATCAGGGTGGTGTCTTACTTCA
AAAATGATATAGTTCGGGGCTACCTGCTGGGGGCTTTTTCGGGCAAGCCGTGATCTT
TTCTGCCATGATGGGAAACTCATTATCTCGCACGGTGGTGGAAAAAGTGAAAACTTT
GTAGGAAGCAAAGAAGGCCCGCCAGCTGCGTAGCGCGCAGGACCAGACTACGTCT
GACACATCTGTGCGTGCTCTCCTGCGTAATAAAGAAGGACGTCCGTTAGTGTTA
TTAGCAGATGATCAGTACACCCTGTTTCCTTATGATCTGTCAGCATCAGGTTATACGT
ATGTGGTCCTGGGTATCTATTGGATATCCCATGCTTGGGCCGAATGCCAGCCGACC
CTGACTAACAGTACATGCGTAATCCGTTGGAAATTCGCTTTTCAGTGGTGGGAGGAT
CAGGGCTCCCCGTGGTGGCTGACTCCGGAGGGGGTTCGTGGCTTCATTACCGGAG
CCATTAGTCGGGAAAACAGATCCGGATTTTCATTGTACCCAGTGCCTATGGGAGTCA
CCAGTTGTATACAAACAGAGTGGGATGTGTCTGATTCCGGGCTGCGCCGGTTTTTG
GAAGATAAATGGCAAAGAACCTGAAAGCCTGGAGTATAGCGTTGAATTTCTGCGTCT
GCAACCGAGAACTTTTCGAAATCTTACCGACGCTCACTCCAGCCGCTGGGACCCAC
CAGATTACTTCATCTCTGAGCTGCCCAGGCTGGCATTGCACTGATTGCGGTCGTCT
ATCTTCTCGTTTCAAGTGGGAGTGCTGGGAGTGCGCACATTGCCAGGCAAACCAA
CCTATTAAGTGGGAAATACGCCATGCCAAAGATTTTTGGCATTGAGACCAACCAAT
TATTTTACCGATCTACGCATTGCCCGAAAGTCCGGTATAACCAAAGCCTCCCGGAT
GTCATAGATATGGGTGGGGGTAGGGGATTATGACTCGCCAAACCTTTGTACTTCCA
TCCGACCGCGGAAAGATTCATCTGATAACCACAGGCCCACTGTGCAACCGAGAAGT
GGACGAAATTTTCCAGGGTTATCAGAAGCAAGCGTCTTGTGGCGAACTGAACTTC
AACGATATCCCTTACGCAGCGTGCGGCTCTTTCATCCGGCTATCCGTGGCACACTG
CTGGCCAACTACTTTTCCCAAATTCGGGCGAACCGTATAAATACGTTGGTGAAACC
GCAAACCTCTGTGCCTTTAGACGAAGCACCCAACTGTGTGCAGGCAGCCCTGGACG
TTATTAAGGCGGGTGGATCTGGCCGTAAAGCAAAAATCTCAGTTCAATGAAGTCC
TTACAGCGGCGTATTTGGCTACCAAAAAATGTCTTTTCATAGTGATGACGAAAAAG
GATTAGGTCCGATCGTTGCCTCCCTGTCCCTTGGCAGCCCTGCCGTGATGCATTTT
AGACACCACACAAGTGGAGCTGGGCAAAATCAACCTAACGAAATCGTCATGAGCTT

AATTCTACGCCACGGTAATGTGCTTGTTATGGAAGGATGCGGGGTACAGATACACTA
TGAGCATAACAGTTATACCGACCGATTTCCTGATAGCAGCCACAGCACGTTGGATCAA
CCCATCTTAA

NHI 163 protein sequence:

MRVLLADEPGFHEGLVQADSTAKPQLPRSPPIWAGSRQEICENLEYFRSYQGGVYFKN
DIVRGYLLGAFAASRDLFCHDGKLIISHGGGKSENFVGSKEGRQLRSAQDQTTS DTSV
RALLRNYKEGRPLVLLADDQYTLFPYDLASGYTYVVLGIYWISHAWAECQPTLTNSTC
VIRWKFAFQWCEDQGSPWWLTPEGVVASLPEPLVGKTDPDFHCTQCLWESPVVYKQ
SGMCLIPGCAGFWKINGKEPESLEYSVEFLRLQPRTFEILPTLTPAAGTHQITSSLSCPG
WHCTDCGRLSSRFKWECAHCQANQPIKWEIRHAKDFWHLRPTNYFTDLRIARK
SGITKSLPDVIDMGGGRGFMTRQTFVLPSPDRGKIHLITTGPLSNREVDEIFQGYQKQAS
CGELKLQRYPLRSVRLFHPAIRGTLANYFSQNSGEPYKYVGETANSVPLDEAPNCVQ
AALDVIKKRVDLAVKQKSQFNEVLTAAYLAHQKMSFHSDDEKGLGPIVASLSLGSPAVM
HFRHHTSGAGQNPNEIVMSLILRHGNVLMMEGCGVQIHYEHTVIPTDFRIAATARWIN
PS

Codon-optimized NHI 164 sequence:

ATGAGCGGTTCGCGATTTATTAGGCTTACCGCTTGCCGCCCTTGCGGTATTTACCAGC
GCGAAATCTTTTCGCGATAATCCAGTGCTGGGGAACCGCTGGCTGAATCAGCGGG
GGCTGCATGTTCTGCGCTTGAAGCTTGCTGCCGGCCTCGCAGCCCGCCGTAGACG
CCGCATTTCCCGGCGTATCCCAGCGAGCTTGCTGCATCAGTTCAACAAAACGGTT
TCATTTGTATTGAAAATTTCTGCCGCAAGCTGAATTCGGAGCCTGCAGAAAGAAG
TGTTAAACCAGCCATGGGATAGATATGATATGACTCAAGGCACAACAACCACCCGCC
GAGTTTTTTTAGATGGTGTGGAAGTAAACGGTTATCACCGGCGCTGCACAATCTG
ATATCTCGGCGTGATCTGAAAGATCTCATTGCTATGTTGCCGGTACTCTTGCTGAA
CCTGTGTTTAGCCTGCAGGCGATATTTCTGGTGGGACGGGTAAACACCCTGACCC
GCAGGCGAGCGTCCACAGTGACACATTTCACTCTACAGCCAAAGCTTGGCTGTTTC
TTGAGGACGTTTCAGGATGAGGCAGGACCTCTGTCATATGTGCCTGGCTCTCATCGT
TTAACTAGCCAGCGTCTGGCGTGGGAGAAAATTCAGTCCTTGACAGCAGCTGAATC
GGCAATTACTTATCATGCCCGTGGTAGCTTTCGCGCCGCGATGAAGGACCTAAATG
CGATGAATTTGCCGTGCCACACCGTTTTGCTGTACCTGGCAACACGCTAGTGATC
GCGGATACACACGGCTTCCATTGCCGTTACGCCACTAGAATAACAAACACGGGT
GGAAGTGTATGCGAGTCTTCGTCGAAATCCTTTCGTGCCGTTTGGCGGTCTGGATA
TACTGTCGTGGCCGTGGTGCCGCGAAAGGTCAGGCTCTCTTTACTTAAAAGCGATG
GAGTTCGGTAGAAGGTTTCGGTTCAATTCGTATGCCGTGGAAGCAAGTTGGGCATGG
TCTCCTGAAGCAGAGCTAA

NHI 164 protein sequence:

MSGRDLLGLPLAALAVFTSAKSFRDNPVLGNRWLNQRGLHVLRLKLAAGLAARRRRRI
SRRIPASLLHQFEQNGFICIENFLPQAEFRSLQKEVLNQPWDYDMTQGTTTTRRVFLD
GVELKRLSPALHNLISRRDLKDLIRYVAGTLAEPVFSLQAIFSGGTGKHPDPQASVHSDT
FHSTAKAWLFLEDVQDEAGPLSYVPGSHRLTSQRLAWEKIQLTAAESAITYHARGSFR

AAMKDLNAMNLPCHRFVPGNTLVIADTHGFHCRSATRIQTTRVEVYASLRRNPFVFP
GGLDILSWPWCRERSGSLYLKAMEFGRRFGSIRMPWKQVGHGLLKQS

Codon-optimized NHI 165 sequence:

ATGTACCAAATTTCTAATAACATCTATTGCCAAAAAGACATTTTAGATCTGGAGATCCC
CACCAAACCTTCTCACCAACCCATACTATGACTATCCGTATCTGATCATAGAGAATTC
TTTTCTCAGAAAATTCTGAAAGAGCTGTGTGAGATGTTAAAATCCAATTCCGATGTCTG
AAGATGCCAAAATTGCAAAAATTAACGATCTGAACATCTTAGACAAGAAATTGGATAA
GTCAATCCGGAAAACGAAGATACATAAATTACCTAAACGTTATTTAAATATTTATGAGA
AAAAATTCATTTTTTCATCAGAAAAAAATTGAGGAGTTTTTTAAATTAGCGCTCACCACT
AGTACCAAGGTGCAAGTTTTACAGTATACTGAGGGATGTTTTTACAAGGCTCACAGC
GACGACTCGAATATGATCGTAAAAGATGATAAATTAGTGGGGTTTTTGCCGGTGGCC
CTGCAGAGAAAAATTACCACCGTTTTTATTCCTCGACGACGAATATGAGGGGGGCGA
ACTGGTGTTTAATTATCTGTATGACAAAGATGGAAATGTGGTTCAACTGAAAGGGCG
GAGCGGCGATATGATTGTTTTTCTGTCAAACCCAGTTTATACCCACGAGGTTAAAGA
GGTTATTTCAAGGCAACAGGTTTACTTTGGTGCAATGGCACGATGCCATCATCAACTG
A

NHI 165 protein sequence:

MYQISNNIYCQKDILDLEIPTKLLTNPYYDYPYLIENFFSQKILKELCEMLKSNSDVEDAK
IRKINDLNILDKKLDKSIRKTKIHKLPKRYLNIYEKKFIFHQKKIEEFFKLALTTSTKVQVLQ
YTEGCFYKAHSDDSNMIVKDDKLVGFLPVALQRKITTVLFLDDEYEGGELVFNYLYDKD
GNVVQLKGRSGDMIVFLSNPVYTHEVKEVISGNRFTLVQWHDALIN

Codon-optimized NHI 166 sequence:

ATGCTAACGCCTGACGAAATTGGGCGTCTGCGAGATTTTTGTGATGAATGGTTCGG
GGCAGCAGACACCGACGAGATGCCGGCCTCAGTCTTCCTGCGTACCCCGGAACTG
GCTCGAATTCCTTTACGTGAAGACGTAGTGGCGCTTCTTAAGACTGTGTATGGCGAA
TCCTATGTGCTTTATCCTAATTTTATAGTAAGGAAAAACCTGTATGTCGGCTGGCACA
TTGATCCTGCATTGCGAGGTCCGGATAAGCGCTATGTTTGAAACCTGGGTTTTTAC
ATTTCCAATGTGTTATATATTTACAGGACAATGACCGCGAATTGGGCGGCGGGCTAG
ATGTTATATCAGGTTTCGCATCGTCCGGTGCTTCGCTTCATATCAGGACGCAGTCCCT
TTGGTTATTATGGCGTTCAGTTTGCGAGACGGTTACTTATGCCCCGCGCGAAACATA
CCCTGCGCATCAGAGCAGGGGACCTCGTGATATGGCATGGGCGCGCAGTGCATCG
GTCGACTCCAATGCGTCGCCAGACTCGAGAAGCAAATACGCTCTCTTTTCTCTAC
AGGACAACGTGACCGCTTCGCTACGAACCAAGTTTTTGGCTGACATGGTTGGCCATC
GATACCAGTTAAAAAAGGGGCGACACAATACAGTCCCCGATACGCTGAGCTGCCT
GACCTGCACTATCCAGACGCATTCCCTCCGGAAATGGTGGAAAGTGCAACAAAACA
TTCCGTGGAAGTTGTCACTTTAA

NHI 166 protein sequence:

MLTPDEIGRLRDFCDEWFGAADTDEMPASVFLRTPELARIPLREDVVALLKTVYGESYV
LYPNFIVRKNLYVGWHIDPAFAGPDKRYVWKPGFLHFQCVIYLQDNDRELGGGLDVISG

SHRPVLRFISGRSPFGYYGVQFARRLLMPRAKHTLRIRAGDLVIWHGRAVHRSTPMRR
QTREAKYALFFSTGQRDRFATNQFLADMVGHRYQLKKGATQYSPRYAELPDLHYPDAF
PPEMVESATKHSVEVVTL

Codon-optimized NHI 167 sequence:

ATGGCACCGGCAACAGAAGCGGCCCTACAGCAGACCCGGCGGGCGGGCGGCTTTG
GCAGCAGTCGAACGCCAGGCAGCTCGCGCAGCATATTTGCGAGACGGTCGTCTGT
TATTTCTTGAACGTTTCTTGCCAGAAGCTGCATTGCGCCCGTTACAGGAGGAAGCA
CAGCGTTTGAAACCTAGATTACATCGTAATTATATACCTCGTCATAAAAAGGGGGGCG
CCGTGGGTATCCGACATTACGTGCAGAAGCGCCCGCCATCTTAGCGTTGTACCGG
AGCGCGGGCGCTTTTGGATTTTGTGCGGGCGTGACCGGGGAGGCAGTACAGCATT
GTCCGGAAGATGACCCCCATGCATGTGCTCTTTACTATTATACTGAACCTGGGGATC
ATATCGGCTTTTATTACGATACCTCTTATTATCGGGGGCTGCGTTTTACCTTGCTGGT
TGGCTTAGCGGACCGTTTCATCGTCGCGACTGGTTTGCAGGCTTCACACGCGCGAC
CCGGCGCCGGAAGAAGAGGTGGCTCTGGCTACACCGCCGGGTTGTCTGGTTTTGT
TTGATGGAGATGCACTCTATCATGCAGTTACGCCTCTTGGGCCGGGGGAGGAGAG
GATTATGCTGACACTGCAGTACGTCACCGATCCCCGGATGGGTCTTGCCCATCGTT
TCGTTTCCAACATGAAAGATGCTATAGCCTATTTTGGCCTCCCGGCTTTACGCAGAC
GCCGACCATAA

NHI 167 protein sequence:

MAPATEAAPTADPAAAAAALAAVERQAARAAYLRDGRLLFLERFLPEAALRPLQEEAQRLK
PRLHRNYIPRHKKGGAVGYPTLRAEPAILALYRSAALLDFVAAVTGEAVQHCPEDDPH
ACALYYYTEPGDHIGFHYDTSYYRGLRFTLLVGLADRSSSRLVCR LHTRDPAPEEEVAL
ATPPGCLVLFDG DALYHAVTPLGPGEERIMLT LQYVTDPRMGLAHRFVSNMKDAIAYFG
LPALRRRRP

Codon-optimized NHI 168 sequence:

ATGGCCCCACAGCGCCGAGGTGCCCCGAAAGCTGAAGGCAGCGGTGCGGCCGA
GCGTGGTCGACCCTCTTCTACAAAGAAAGATCAAGTCCCCCGTGAGGCACAGCGT
CCATGGCTCAAGACCGTAGGTTTAGGAGCAGGACTGGTCCTTGTGGCGTTACTTCT
GTGGAACAGTCTGGGCACTGACGATGGTATCGCAGAAGTTCTGGCACTGCGAGGA
GAAGTACTTGCGGGCCGCTTCATTGAAGTTCCGTGCAGTGAGGATTATGACGGCCA
CCGTGCGTTTGAAGGATGCACCCCGCGAAAGTGTGGTCGAGGAATAACCGATATT
GTTATAACAAGGGAAGAGGCAGATCGTATCAGAAGTATCGCCGAAAAGGGCCTTTC
TTTAGGAGGCTCCGATGGCGGCGCTTCCATTTTGGACCTGCACAGTGGGGCACTAT
CGGTGGGGAAACACTTTGTGAATCTTTATAGGTACTTTGGCGATAAAATACAAAACAT
TTTTTCCGAAGAAGACTTTCAGTTGTATCGCGATGTTAGGCGTAAAGTTCAGCTCAC
GATCGCAAATGCATTTGGTATTTAGCTTCCAGTCTGTACTTAACGAAGCCGACATT
CTTAGTCGTATTAATTCTACCGAAGCGAGAACTGCGCACGATGAGTACTGGCATGC
ACATGTTGATAAAGTAACTTACGGCAGCTTTGATTATACATCTCTTCTCTATCTGTCAG
ATTATCTTGAGGACTTTGGAGGTGGGCGCTTTGTTTTTCATGGAGGAGGGTGCTAAC
AAAACAGTAGAGCCTCGCGCAGGCCGAGTTTCTTTTTTTACGTCCGGAAGTGAGAA

TTTGCATCGGGTTGAAAAAGTGAGATGGGGCACGCGGTATGCTATAACTATCGCCTT
TACTTGTAATCCAGATCACGGTATCGAAGACCCGGCCCTCACTTAA

NHI 168 protein sequence:

MAPQRRGAPKAEGSGAAERGRPSSTKKDQVPREAQRPWLKTVGLGAGLVLVALLW
NSLGTDDGIAEVLALRGEVLAGRFEVPCSEDYDGHRRFEGCTPRKCGRGITDIVITRE
EADRIRSIAEKGLSLGGSDGGASILDLHSGALSVGKHFNLYRYFGDKIQNIFSEEDFQL
YRDVRRKVQLTIANAFGISASSLYLTKPFFSRINSTEARTAHDEYWHAHVDKVTYGSF
DYTSLLYLSDYLEDFFGGGRFVFMEEGANKTVEPRAGRVSFFTSGSENLHRVEKVRWG
TRYAITIAFTCNPDHGIEDPALT

Codon-optimized NHI 169 sequence:

ATGGCGACCAGCTATCCCACATTCACGCTGAGTGAACAACCTACCGCGGAACAGCT
GGACTTCTTTAACACGCATGGTTTCATCCATTTCAAGAACTTTATTAAGCCCGAGACC
GTATCATCGATTATCGATGCTTCGAAACAGGTTGAACAGAAATGGATTGAACAGGAC
ATTAAAAAATCAATGGCGTTCCTATTAAGTACGGTAAAGACTTGGATGGGTCAGCC
ATTGTGCAGCGCTTCGCCTTTATAAATCAACAACACCAGACCCTGAGCGGTCTGTTA
CTGGATCCTCGCTTTAATGCCTTACTTCAACTGGCCGGAGATGGAGCTCGGCTTGG
AACGGAGGAAAAAGATGGGATGGTATTCAATCATTATATAAACGGCCCGGATTCTAAA
TTAGCAAGATGGGCTGGCATACTGACGGTCTGAGAGATATCTTTTATGGTGCTAAA
TTAAACCCAATGCTTAATGTAGGCATTCATTTATCGACCTTAAACCGGAAAATGGGG
GTCTGAAAATCATCCCTGGCACCCATAAGCAGTCTATCTATCAAATGCTTTTTAGAAA
GAAGTACTTCTTAGATAATAAACCGGATGCAGAAGAGGTGAGCATACTGCCGGAGG
CCGGAGACCTGACTATTCATGATGGGCGACTGTGGCATCGCGTCGCCGAGAGCGC
CATTAGAGGTGAAGAAAGCCGTAGGCGTGTGATTTATGTGCCAATCATTGCTGGCAA
ATATGCACCTAAGAATGAGAATAGTCCTACCGTATTTTATCAGCGGTTTGCCGGTATA
GTCAAATAA

NHI 169 protein sequence:

MATSYPTFTLSEQLTAEQLDFFNTHGFIHFKNFIKPETVSSIIDASKQVEQKWIEQDIKKIN
GVPIKYGKDLDGSAIVQRFAFINQQHQTL SGLLLDPRFNALLQLAGD GARLGTEEKDG
MVFNHYINGPDSKFSKMGWHTDGLRDIFYGAKLNPMLNVGIHLSTLKPENGGLKIIPGT
HKQSIYQMLFRKKYFLDNKPDAEEVSILPEAGDLTIHDGRLWHRVAESAIRGEESRRRV
YVPIIAGKYAPKNENSPTVFYQRFAGIVK

Codon-optimized NHI 170 sequence:

ATGACAACGGACTATTGCCCCCATGGTCGGGACTATTATAGAAGAATTGGATTTGGC
AAGGCTGAAATCGTCGTTACAGGTTCTAAACGCGCCGTGCTGGTCGCGATGTTAGC
AGGTCCGGATGGACAGGCAAGGCGTGAAAGCGCCGTTGCGGGCCGCATAAGGCG
CCATAATCGAATTGGTTTTGTAGTGCACGCGCCCCCACTGATATACGCTCCCTGGC
ACGTAGCGGCGTCGAAGCCCTACGTGCGGGGACGCCAGAGCTGCAAGAGAAAG
CTTCGAACAGATAGTTGCTACGGGTGAGGCAGATGCCAGTGCCTGCGTTGGGCTG
GCATATGCGTGTGCTCCTTGAAAGACAATGCCGCTTCGAAAGCCGCGTTAGACAA

AGCACTGGCCTTAGAACCGCGAAACTTGCGTGCACTTATACTGAAAGCAGACTATTT
TGCAGAAGAAGGTGATGCGCGAGCCGCAAGTGCCTTCTATCTTGCCGCGGTAAAG
ACCGCACCGCCAGCTGATCAACTGCCACCTGACCTTAGAAATGAAGTGGGTGCGC
CAAAAGCAATGTGTGATAGGTACGCCGCGCAATTTGAGGCGTTTCTCAAAGATCGT
CTCTTAGGGGAACGCCTGGCTGATGGGCGTGACGCGCGCCGCTTTACGCACTCTT
TAGATATTCTGTTCCGTAAAAAAGATTTATTTCCAGGATCCCCGATACTATTTCTTC
CCGGGTTTACCGCAAATACAGTTTTATGATAGAGAGGACTTTCCATGGTTAGATAAAA
TAGAAGCAGCCACAGCAGATATCCGCGCCGAAGTATCGAGGTTTTAAAGAAGAA
TCTACCTTTAGTCCTTATGTCCAAGGTGAGTCTCGCCGCCACATAATGAACAAGAG
GGGATGCTCAACAATCCCGCGTGGTCCGGCATTATCTGTGGAAAGACGGAGCAGT
TGTAGCTGATCATGCCGCTCGCTGCCCTGGTACAATGAAGGCCCTGGCCGAGGCG
CCGCTTACTCATGTTCAGGGCCGTAGCCCGGCCGTTCTTTTTAGCCTTCTCCGTCC
CGGTACCCACATCCCCCACACAATGGTTTTGTAAACACGAGATTAATTTGTCATTTA
CCTCTGATCGTACCCGGAAGTTGTAGTTTGCGGGTGCGAAACGATACCCGTGATCT
GGTTGAGGGAAAAGCGTGGCTTTTTGACGATACGATGGAGCATGAAGCATGGAAC
GACTCGGATCAGACTCGTGTTATCTTGCTATTTGAAATATGGAGACCGGAGCTAACG
GCAGAGGAGAGGCGACTGGTGTCCGCGATGTTTGAAGCGATAGATGATCATAGCG
GTGAAAAAGCCGCGTGGGGGATATAA

NHI 170 protein sequence:

MTTDYCPHGRDYRRIGFGKAEIVVTGSKRAVLVAMLAGPDGQARRESAVRGRIRRH
RIGFVVHAPPTDIRSLARSGVEALRRGDARAARESEFQIVATGEADASACVGLAYACRS
LKDNAASKAALDKALALEPRNLRALILKADYFAEEGDARAASAFYLA VKTAPPADQLPP
DLRNEVGRAKAMCDRYAAQFEAFKDRLLGERLADGRAARRFTHSLDILFGKKKIYFQ
DPRYYFFPGLPQIQFYDREDFPWLDKIEAATADIRAEILVKEESTFSPYVQGESSRPH
NEQEGMLNNPAWSAFYLVKDGAVVADHAARCPGTMKALAEAPLTHVQGRSPAVLFSL
LRPGTHIPPHNGFVNTRLICHLPLIVPGSCSLRVGNDTRDLVEGKAWLFDDTMEHEAW
NDSDQTRVILLFEIWRPELTAEERRLVSAMFEAIDHSGEKA AWGI

Codon-optimized NHI 171 sequence:

ATGCTGACAATTACGACTGTTGAGGAACTGACCGCCGCGCAGTTAGACGATGTCAT
TCATGCAAGGACTCTTGGTATACTGATAAAAGGTTTCGCCCCGGCAGACACCGTAG
CTCTTGCGTTAGAACGTCTGTCCCATCATGAATTACGTGGTGCATTTTCTGAACAGA
CCGAATTCGTACGCCTTGGTAAAGCCTATATCGAGATACGGGACGAAGCCTCCAGA
GCTGAATATCACGGGCAGGCGATAGCGAATATAAGAAAGATCCGCAGTGCGTTTGG
CCCACTGGCGTCGCCTATTGATCATCTGCGCTTGTTATTAGACGAAGTGTGGCCCA
AAGGAGCCAGACTCCTAGATGTGAACGGCCGAAAATGTTTCGTTGGGATTGCACGG
TTTCAGGGTAACAATGTTGACCTGACACCGCATACCGACGATTAGCCCGTAATGCA
CCGGCCGATCACCAGCCTAGACTCTTGACTCAACTGTCCACGAACATCTACCTACA
GATTTCCCGAGGATGGCGGCCGAAGTAGAGATTTGGAACCTCCATCCGGATGAAGCC
GAATATGAACGATTACGTGGGCAGCGTGCTTATGGTATCGAACGCCACCTATTACCT
CCTCCTGATCTCGTGGTTAAGCCAGAGGCGGGAGATCTGATTTTTTTGAACCCACG

TTTAATTCACGCGGTGCGTCCTTCCGCCACATCAACCCGCGTTACACTCGGTGTATT
TATAGGTTATTTTGGAGACGATCAGCCTCTCGCCTATTGGTCGTAG

NHI 171 protein sequence:

MLTITTVEELTAAQLDDVIHARTLGILIKGFAPADTVALALERLSHHELRGAFSEQTEFVRL
GKAYIEIRDEASRAEYHGQAIANIRKIRSAFGPLASPIDHLRLLLDEVWPKGARLLDVNG
RKCFVGIARFQGNVLDLTPHTDDLARNAPADHQPRLLTQLSTNIYLQIPEDGGELEIWN
LHPDEAEYERLRGQRAYGIERHLLPPDLVVKPEAGDLIFLNPRLIHAVRPSATSTRVTL
GVFIGYFGDDQPLAYWS

Codon-optimized NHI 172 sequence:

ATGATTCAAATCAGTAATGAAATTTTTTTCGACAAACGCCTCCAAGAATGGGATATTG
CGAATAAAATCTTGCCCAATCCATACTATGATAAACCGTTTCTGATTATACCTAATTTTC
TGGATAGCGCCCTGGCACATGAAATCGCTGATACTATTTACCAAGAGCAATATGCTC
AAAAAGCTCAGATTAAAAAATGCTTCTGCAATCAGTTGTTGAGCCAGATGTTAATGA
AACGATCAGAAAGACAAACCTCTATACATTGAATCCATTTACACGCGCGTTATATGAA
ACACTGTTTAACTCCACCAGCCACAGATAGAACGTTTCTTTAATTTGTCCTTGACC
CTGGCCACGCCGATTCAAGTTTTAGAAATACACGCAGGGCAGCTTTTATATCAAACAT
GCGGATGATTCTAACGAACTGATCGATAAAGAGGGTAGTACAGTGGGTTTCACCTG
CGTTGCACCCCAACGTAAATTAACGACGGTTTTATTTCGCGACTAGCTTTTCATGAAGG
GGTTGTCGCACAGGATGATAACAGCATTTTTCAGGTGGGGAAGTCTGTTTAACTA
CCTGTATGATAAGGACGGTAACGGCATAGCTTTTAAACCACGCGCGGGCGATATGAT
ACTCTTTCCAGTAATCCATATTTTAGCCATGAGGTCAAAAAGGTTACATCCGGTTAC
CGGTTAACTCTGGTCCAGTGGCATAATGCTATTGTGAGCGGCTGCTGA

NHI 172 protein sequence:

MIQISNEIFCDKRLQEWDIANKILPNPYDCKPFLIIPNFLDSALAHEIADTIYQEYQAQKAQ
IKKMLLQSVVEPDVNETIRKTNLYTLNPFHTALYETLFLKLHQPQIERFFNLSLTLATPIQVL
EYTQGSFYIKHADDSNELIDKEGSTVGFTCVAPQRKLTTVLFATSFHEGVVAQDDKQHF
SGGELLFNYLYDKDGNIGIAFKPRAGDMILFPSNPYFSHEVKKVTSGYRLTLVQWHNAIV
SGC

Codon-optimized NHI 173 sequence:

ATGAACGAACACTACGATATCAATCAGCACGGTTCCAAAGCGAAACGTCTACGCCC
CTCAGAGGAAGCAAAACAGATTGCCTCAACCGCTCATACAGATGATAATCAGGAGA
GCACTGAGTTCAAGCTGGCCGTA CTGCGCATCTCTGCATCCGGATAAGTCGCCCCGAA
GTTCACTTAGATTACCTTCTGGCCTATCAAGGAAGCGCCTATAAAGCTTCCCAGGCT
CTGTCTGGGCAAAGCAGACCACGCAGTTAAGCGTAAAAGGCCGTCAACGATAGCATA
CCAAAGTAGCCTTGACTCGTTTGCTAGTAGCCGGGGACCGGCCAACGGCCCTGCG
ACAGGAAGCGGCAAACTGTTAACAAAAAAGGGCAAAACGCTGCATCTGTACACCCC
ACGTGAGATAGAGGAAAATACTCCATGTTCTATTATACATAATTTTCTTCTACTGCAC
AGGCTGATGCGTTATTGAAAGAACTATTGGAAGAAGCACCGAGCATGAAAAGAGGG
ACATTCCAACGTGTTTGAACGTACCGTTCAGAGTCCTCACACCTGGAAATTCTATGTG

GATACGCTTGAAGATGTGAAAGTGCAAAAAACAGAATATGTGTATGACGGTATATATG
AAGCGGACGTCAGTCAAACAACCCCGGAAATGCTGAAAGTCAGTAAAATCGTGAAG
CAGACGGTAAACAAAGAGGTGGAACGCCGTATGCGTGATTTTCAGCCTGAGGGCA
AAAAATTACGCTTTTCAGTCTCCGACATCCTGGGAGCCTAATGCAAGTTTTGTAACT
GCTATGACGGAGCAAAAGAAAATGTTGGTTGGCACGCAGATCATTTGACATATCTGG
GTCCCCGTGCTATAATAGGGTCCTTATCACTGGGTGTGGCGCGCGAATTTTCGTGTA
CGTCGGATCGTCCCGCAAGTGAAGCGGCTAGCGCCGATGAACAGGGTCAGATCG
CTATCCATCTGCCACACAATCACTCCTTGTTATGCACGCAGAGATGCAGGAAGAAT
GGAAACATTCAATCGCAGCGGCTCAGACAATTGAACCTCATCCGTTGGCCGAAAAC
AAGCGCCTTAATGTTACCTACCGATGCTACCAGCCTTATTTAAATCCGAAATTCACCC
CACGTTGCCGCTGTGGCGTCGCGTGTGTACTGCGATGTGTACAGAAAAGGCAAGT
GACCCGTGGCCGGTACATGTGGATGTGTTACGCGAACTATACCCGGCCAGAAAG
GATGCAGCTATTTTGTGGGCGGAGTTTGACGAGGATGGTCGCCCCGCTTGGGC
CGAAGGTTACCTAGGTAACGCCAACAAGCCTGAACCAGGAACAGGCAGATGGGAT
GAAGATTCCACGCTACCAGCATAA

NHI 173 protein sequence:

MNEHYDINQHGSKAKRLRPSEEAKQIASTAHTDDNQESTEFKLAVLASLHPDKSPEVH
LDYLLAYQGSAYKASQALSGKADHAVKRKRKPSTIAYQSSLDSFASSRGPANGPATGSG
KLLTKKGKTLHLYTPREIEENTPCSIIHNFLPTAQADALLKELLEAPSMKRGTFQLFERT
VQSPHTWKFYVDLTEDVKVQKTEYVYDGIYEADVSTTPPEMLKVSKIVKQTVNKEVER
RMRDFQPEGKKLRFQSPTSWEPNASFVNCYDGAKENVGWHADHLTYLGPRAIIGSL
LGVAREFRVRRIVPQVEAASADEQGQIAIHLPHNSLLVMHAEMQEEWKHSIAAAQTIEP
HPLAENKRLNVTYRCYQPYLNPKFTPRCRCGVACVLRCVQKRQVTRGRYMWMCYAN
YTPGQKGCSYFVWAEFDEDGRPPWAEGYLGNANKPEPGTGRWDEDSTLPA

Codon-optimized NHI 174 sequence:

ATGGCAACCCAGACTAGTAAGAATGAATTTTCTGCTGCAGATTTGCTGAATAGTGCT
CTGACATTTATTAATCCGATTGTCACCGCACCCCTTTAGTGCGATTCAGAATCTTTTTG
ATTATCTGGGAAAAAATCCGCAAACCGCTGCTGCGCTCAATAAAACATACAGCTTAA
GAGGGATATTTAAAGCAGCTGCGACTACTAATCCAGCCTGCGACCAGAAATTAACAA
TTGATCTAAGCCCGACTAGGATTGCCCGAATTCGAAGCGCCATCAAATGAAGTGG
CTCCGCATGGTTTAGTGGAGATTTTAGCTTTCTTTGACGCTATCAGTCAACATTACGT
TGTGCAGATTTTGACTTTACTGTCCACGATCGCAGGGCGCGATCTGATTTCTTTACA
TGCGAATTTTAACGTTAATTTTCGTATCATCGACTACACTACAGCAACAGCTTCCCCG
CAGTCCCTGAATGGATGCGGAGCGCATACCGATTATGGTACGTTTTCAATTATTTTC
AGGACGGTAATGCCGGCTTAGAGATGGAGGCACCGGAAGCGCCTGGCACATGGGT
CTCTGTTCCGGGTAACGCGACGGTGCTGCTGTGCGGTTGGTGTGCGGTCGTTCTT
AGTGGCGGAAAAGTGATGGCAGTGCGTCACCGTGTCCGTCTGTGTACCGGGTGTTT
GCAGGCTTTCTGCTGTTTTATTTCGTAGCGCCTGATTTAAACGTAAACTCGCGCCAG
CCGAACGAGTAGACGCGTTTCAAGTATCCGGTTAATCAGGGACTTTTTGATGTGAGA
TGGTTCAAAGAAGAAATGGGAAAAAAGTGGCGGTATCGGGAGGGAAATGGGAAGT

TGGAGGATGACGGCTTAGAGCAGGATCATGACATCGGAGATTTGATTTGGAAGAGT
TTCTAA

NHI 174 protein sequence:

MATQTSKNEFSAADLLNSALTFINPIVTAPFSAIQNLFDYLGKNPQTAAALNKTYSLRGIF
KAAATTNPACDQKLTIDLSPTRIARIPSAIKIELAPHGLVEILAFFDAISQHYVVQILTLLSTI
AGRDLISLHANFNVNFRIIDYTTATASPQSLNGCGAHTDYGTFSIIFQDGNAGLEMEAPE
APGTWVSVPGNATVLLCGWCAVVLSSGGKVMAVRHRVRRVPGVRRLSAVLFVAPDLNV
KLAPAERVDAFSDPVNQGLFDVRWFKEEMGKKWRYREGNGKLEDDGLEQDHDIGDLI
WKSF

Codon-optimized NHI 175 sequence:

ATGATCGAATCGTTCGCCCAGGATGGTTTTTGTGTACTCCCGGGTGCTCTTTC AATA
GAAGAGGTTACGGGCCTGAACGAAGCTATAGACGACCACCAGCTGCAGTTTCCAG
ATGGATGGCCAGACCGCGGCGAGGGTGGTCGCCGTCAGTGCGTCTTCTATCCTCCT
GAATATGCCTGTCTTTGATCATAGTTTACTACATCCCGAAATTCTTCCGCTGGTTAGC
AGACTGGTGGGTAAAGAACTGGTGGTAGAAGAGCATAGTGTAAATGGTGCGGGGCGC
CGATTGCGGGTAATCCGCCAGTGTCAAATGGCATCGAGACCGACCTAATAACCCG
GATCATGAATTTGGTCTAGAAGCGTTAAGCTTGGTGTATTATTTAACGGACGTCGATG
AGACTACCCATTGTTTTTCAGTTGTTCCCGAGACCGTGGAACCAAACGTGGAGAA
AATTTGCCCGGTTGTGACGGCAGCAGTGCCCGCGATGTGCTGGGGCCTGCGGGC
ACTGCGTTACTTTTTCAATGCAGGGTCCTGTCATGCAGGACGCCTGAGAACAACAAC
CAGCGAACGCCGCACAATTCATATTTATTTCCGGTCACGCTTCACATGCTCCGTTATC
AAATCACACACCTATGCCTCGGAGGCTGGTTGAACATCCAGATGAAGACGTTCCGCC
GTCTGTTTAGCCAGCCGAATGAAGTTACTCGTTCTTCCTTATGTCCGACCGTCCCGT
GA

NHI 175 protein sequence:

MIESFAQDGFVLPGLSIEEVTGLNEAIDDHQLQFPDGPDRGEGGRRQCVSILLNM
PVFDHSLHPEILPLVSRLVGKELVVEEHSVMVRAPIAGNPPVSKWHRDRPNPDHEF
GLEALSLVYYLTDVDETHCFSVVPETVETKRGENLPGCDGSSARDVLGPAGTALLFNA
GSCHAGRLRTTTTERRTIHIYFGHASHAPLSNHTPMPRRLVEHPDEDVRRLFSQPNEV
TRSSLCP TVP

Codon-optimized NHI 176 sequence:

ATGGATATAGGCGTCCCCGTCAAAGAGCTGGGTGATATAGAAGTGACACCCCTGAT
CGACGCTATTATGTCTATAGACGACGAAGCGTGGTTAGCAAATAATCATCGTCAGAAT
GCGTATGAGGTGCATGCCCAGACGCAGTCTTTAGTGATGCTATTTGTGATACTAAA
GTTTGGCCGACAGTTAAAGTCACTAAAGAATCAGCCTGGGATGAGCTCTCCGAAGT
GGCGCTTCCGATCATGGATTGATTCTTGAGAAATATTATCCTCCGGGCGGGGACTGT
CATTCGTGCCATGGCAGCGAAGCTGGTCAGCGGCGGAGTGATTAAGCCGCATGTA
GATAGTCATCAGAGTTTTTCATGTGGGCCATCGCATCCATATACCGATTACGGCAAATC
CACGCGTTCGGTTTATGATCGATGGTAAACCCACCGTTTCGAGATTGGCAAAGTG

TACGAACTGAATAATCAAATTCAGCACTCTGTCATGAATAAAGGGAAGGACGACCGC
ATTAATTTTATTTTTGACTATATTCCGCCTGTTGAGCTCGCAAACTTAATAGCTAA

NHI 176 protein sequence:

MDIGVPVKELGDIEVTPLIDAIMSIDDEAWLANNHQRNAYEVHAQTQSLVMLFVDTKVW
PTVKVTKESAWDELSEVALPIMDSILEKYYPPGGTVIRAMAAKLVS GGVIKPHVDSHQS
FHVGHRIHIPITANPRVRFMIDGKPHRFEIGKVYELNNQIQHSVMNKGKDDRINFIFDYIP
PVELAKLNS

Codon-optimized NHI 177 sequence:

ATGAAAAATGAAATTATTCTTAGTGATTCCCTGCAGCAGAACGACCTGACTCGCCTG
TTCAATGATGAGGCACTCGCTATTCATGTAAAAAATATGTCGATTACATATTTCCGA
CAAACCTTTCTGACTTTTTCTGTCTCACGAAAACCTTAGAGTTGCACGCCCATGACTT
GAAAAAAGGCGATAAAACAATTTTTGTGGATTATGGTGTGGATCGCGTAGGCACGAG
CTATAACTACCTTTGGGAAACCGAAAGATAGCGATT CATATAAAAAATATTTGAAA
ATGCCCTGTCGGCGATTAGGAATGTTTCGTAATTTTTGCGCTCCAATGTTAGCTCCGT
TCGATAAACTGAGACTGGAGCTGGATGAAGTTTGGTCTGCTGGCGCCTCCATTGCT
AACTTTGAAGGGAAGAAGATGCATGTGGGTATTGCGCGCGTGATGAAACGCCCCG
AGTTATCGTTTATGGTCGAAAAACAGCCCCACTTTGACGGATTACAGCAGCAGTCAG
TCAATTTACTGGGTCAGTTCAGTGTTAATATTTATATT CGAATGCCGGTCAACGGGGG
CGAACTGGAGCTGTGGGATGTGCCAGCCATTCCGATTACAGAGCTGATTGAGGATA
ATGCCGAAAATGATTGGCGCGGTACATTACCCCTAGTACGTTAATCAAACCCGAAC
AGGGCGATCTTATAATTATAAATACCCGTAAACCGCACGCAATTCGTAGCTTCTCGGA
CGGTTGCCGGATTTCTCTGCAGAGCTTTATCGGACTGAGCCTTAACAAACATCTTAT
GCTGTGGAATTGA

NHI 177 protein sequence:

MKNEILSDSLQQNDLTRLFNDEALAIHVKKYVDSHISDKLSDFFLSHENLELHAHDLKK
GDKTIFVDYGVDRVGTSYNTTFGKPKDSDSYKKYFENALSAIRNVRNFCAPMLAPFDK
LRLELDEVWSAGASIANFEGKKMHVGIARVMKRPELSFMVEKQPHFDGLQQQSVNLL
GQFSVNIYIRMPVNGGELELWDVPAIPITELIEDNAENDWRGTLPPSTLIKPEQGDLIINT
RKPHAIRSFS DGCRI SLQSFIGLSLNKHLMLWN

Codon-optimized NHI 178 sequence:

ATGGAAAACGACAAAGTCAACGAAGCAGCTACGGGCCTAACTCCTGAAGAAGTGG
ATCATTTTATAAAACATGGTTACGTACTGCTGCGGAATGCGTTTAGTACAGACACCGC
AGCCGCCTGCCGTGAATTTCTCTGGGCGAGGATAGAGCAAGATGGTGTGAAACAG
GACGACTGCAGTACCTGGCCAGAAAAACATTGGATTTCTGAGAGTTACTCTGATGC
TACTGGTGAACCATGGACGGAGGTTTTGACTCCCCGCCTGAAATCAGCTATGGACC
AGCTTTGTGGAAAAGACCGCTGGAATGAGTCGGACCTAGTTTGCGGATGGTGGGT
TATCACATTTCTGGCCATGCCATGCTTCCTTGGGGTCCTGCCGGCCATTGGCACG
TTGACGGAGCTCATTACATACATAAACTCGATTCCAAGGAAAGCGGTCTGCTTCCGA
TTTTTATCTTCTCTGACATCAACCCCGGTGACGGGGGAACGGCGTTAGCTCCAGGA

AGCCACATGCATGTGGCCCATATTTTACAAGATGCGATCCCGCATGGAGGCTTAAAA
GGAGGAAAGGTCAGTGCGAGCGCGAAGTCTTGGGTTTATCAACACAAAAAGGCTT
GGGTTGAAGTCCAAGGTCGGGCAGGAGATATCATGCTGACGCATCCCTTTCTGATG
CATGCACGCAGCAAAAATTGTGGTAGTCAGGTCCGTTTCATGTGTAATCCAAATGTC
GGGCTCAAGGATCCGATGCAATTCGAACGTGACGACCCCGCCGATTATTCACCCGT
AGAAGCCGCTATTGTGGAGGCTTTAAGACAGCCGGTGTGCGAAAAGAAAAGAAAAC
GCGGTTTGC ACTATTAA

NHI 178 protein sequence:

MENDKVNEAATGLTPEEVDHFIFKHGYVLLRNAFSTDAAACREFLWARIEQDGVKQDD
CSTWPEKHWISESYSDATGEPWTEVLTPLKLSAMDQLCGKDRWNESDLVCGWWVIT
FPGHAMLPGWGPAGHWHVDGAHYIHKLDSKESGLLPFIFSDINPGDGGTALAPGSHMH
VAHILQDAIPHGGLKGGKVSASAKSWVYQHKKAWVEVQGRAGDIMLTHPFLMHARSK
NCGSQVRFCNPNVGLKDPMQFERDDPADYSPVEAAIVEALRQPVVSQKKRKRGLHY

Codon-optimized NHI 179 sequence:

ATGAATTATATCAACTTCCAATCCGTAGATGATTGTTTAGCCACTCTGTCGGAGACAG
GCTTCGTTAGAGTCCCGAACTTTCTTGACGAGCACACGGCAACCACCTTAGAGAGT
ACGCTGAAAGAATTAAATTTCAACACCGCCGCGATAGTCAACGCCCGTCCCCATGA
ATTCACAGCGAGTGAATTGCAGAACAAAACCAACCAGGAGAAGCAGCAATTGTTTT
CCCATATCTATGAACAGGCCTCGCAAGGTATTGGCTTTGTTTACGGACGGCACGCA
CTGTCTGAATACTACAAACGAAGAGCTGATTGATTTTAAGAAGCATCTGAACTCAAGC
GAACTTTTAGAGCGAATGTCTCGGGCCTCCCAGCAAAAAATTATCAAAGCGAGTGC
TCAAGCTACGTGCTATCAGAAAGGTGATTTCTCACACGTCATAACGATGTGATACAT
AGCGAGGGACGGGTATATGCCTACGTTCTGGGTCTGACAGAACAGTGGCACCCAG
ATTGGGGCGGTCTGCTGCAGTTTTACGAACCGGACGGGACGCCAACGAACTCGTA
CGCGCCGCAGTTCAATTCCTCACTATTTTTGACGTAACGAAATTCCTCGGTCAG
CTATATCACCCCCTTCGCTAAGTCAAAACGCCTGTCAATTACAGGTTGGTTCAGAAC
CAAATAA

NHI 179 protein sequence:

MNYINFQSVDDCLATLSETGFVRVPNFLDEHTATTLESTLKEINFNTAAIVNARPHEFTA
SELQNKTNQEKQQLFSHIYEQASQGIGFVYGRHALSNTTNEELIDFKKHLNSSELLERM
SRASQQKIIKASAQATCYQKGDFLTRHNDVIHSEGRVYAYVLGLTEQWHPDWGGLLQF
YEPDGTPTNSYAPQFNSLTIFDVTKIHSVSYITPFAKSKRLSITGWFR TK

Codon-optimized NHI 180 sequence:

ATGAATGAGATTCTGTCCTCGGAACAAATTGAAATTTTTTCATCGCGACGGCGTGGTC
GTAATCCCGAACTTTTATAACATGGATAAAATTACGCCCATTCAGAAGGGTATCTATGA
CATTATCGGTCAAATAATAACCCGTGAAGAGATCTCCCTCAAACAGAGCCCTTTTAA
CCCGGATACTTTTGATCAGCATTATATCGACTTAATTGGAATCAATCGCGATTGGGGC
GGCGAGATTTATGATGCGGTAAACAGATTCCGGCTTTTATTTCGGCTTGTGGCGGAC
GAGTCGCACGAAAAGTTATTTAAACAAATTCGCTGTACCCGCAAACCGGGGATCGC

CGGGGGGGGATATGGTATTCGCATTGATAACCCAAATGAAGATAAGTTCAGAGCAAT
GTGGCATCAGGAGTACCCAGCGCAGTTGCGCAGTGTGGACGGGTTAGTTTTCTGG
TCGCCATTAGTGCCTATCACGGAAGAAATTGGTCCAGTAGAATTTTGTCCAGGATCC
CATCATGAGGGTCCGCTGCCGGTATGTTACGCTGACCCGGAGCACACAGGCCGCA
CCGGAGCCTATTCACTGAAAATCCTGGACGAGGCCGTGCACCTGAACAAGTACCG
GAAGATAGCGCCGCTGACCAGCCCCGGGGGACCTCGTGATAATAGATTTTCTAGTTC
TCCACGCAAGCGGCGTCAATAGATCACATCGCGCCCGTTGGTCCATGCAGTTTCGG
TATTTCAACTTTGAGGATCCCATTGGACGCAAACATGGATGGAAAGGCTCCTTTGCG
GCGGGCACCGATTTTCGTAAGATTCATCCGGAACATTTGCAGAAAAACCGCTGCA
GTAA

NHI 180 protein sequence:

MNEILSSEQIEIFHRDGVVVIPNFYNMDKITPIQKGIYDIIGQIITREEISLKQSPFNPDTFD
QHYIDLIGINRDWGGEIYDAVKQIPAFIRLVADESHEKLFKQIRCTRKPGIAGGGYGIRIDN
PNEDKFRAMWHQEYPAQLRSVDGLVFWSPLPITEEIGPVEFCPGSHHEGPLPVCYAD
PEHTGRTGAYSLKILDEAVHLNKYRKIAPLTPSGDLVIDFLVLHASGVNRSHRARWSMQ
FRYFNFEDPIGRKHGWKGSFAAGTDFRKIHPELFAEKPLQ

Codon-optimized NHI 181 sequence:

ATGGAGGATCAGAGTGAGTCCTCCCAGAACCAGAGCACTTCGCGGGCGCAATAGGC
CACCGCGGCCTCCGGTCTTAGGTGTGGGTGACCGATTGGGGGCCGGGGACTCTT
ACATAGTCAACGATTTGTTGCCGAGTGACATCGCGGATAATGTATTTCATAAGCTCAA
GGAAGAAGTAAAATGGCAGGTGATGCATCATCGCGGTGGTGAAGTTCCCCGCCTT
GTGGCGGTGGAAGGAGAAGTTGGCCAAGACGGTAGTTTTCCGGTGTATCGCCATC
CGGCTGACGAAAGTCCGCCACTGCTGCCTTTCTCGCCGACTGTTTCATTAATACGG
GAATATGTCCAGAAAGTTGTGCAGCATCCTGTTAATCACGTACTAATCCAGTTCTACC
GTAGTGGAAGATTACATCAGTGATCACAGCGATAAACGATAGATGTTGTACGCG
GGAGCAACATAGTTAACGTTTCCATGGGAGCGCAACGCGTTATGACGTTGAAATTAA
AAAAAGATAAGCACAAAGGGAAGTGATATCGACGCACCTCATGAGGGTGAGATTCTG
CCAACACAGGCAAGCAATGTACCGGCGCCTCCGAGGCCATCCCCAAAAATTCCGTT
GCCACATAATTCAATGTTGGTTATGGGTCTCGAAACAAATGCAAAGTGGTTGCATGG
AATAAATCATGATAATCGTCCGTTTCAGACGAAGAGTCCGGAGGAACAGCGCGAAG
AGGGTGGCAGAATTTCTCTGACTTTTCGACATATTGGTACCTTTTTGTCAAAGGACG
AGGCATTTATCTGGGGCCAAGGGGCGAAGGGCAAACGCGAGAAGAAGCGGGTAA
GGTGATATTTACCAGGATGAAGTGGAAGAACTTCTGATAGCGTTCGGTGATGAGAA
CCAGCAGTCGGATTTGATTGGGACCAGGCATATGGTCCTGGGTTTGACGTATTGC
ACCTGACAAGTAAAGAAGGTAGCGGTGGGCCTCCAAGTTAA

NHI 181 protein sequence:

MEDQSESSQNQSTSRNRNPPRPPVLGVGDRLGAGDSYIVNDLLPSDIADNVFHLKE
EVKWQVMHHRGGEVPRLVAVEGEVGQDGSFPVYRHPADESPLLPFSPVSLIREYV
QKVVQHPVNHVLIQFYRSGKDYISDHSDKTIDVVRGSNIVNVSMGAQRVMTLKLKDK
HKGSDIDAPHEGEILPTQASNVPAPPRPSQKIPLPHNSMLVMGLETNAKWLHGINHDN

RPFQTKSPREEQREEGGRISLTFRHIGTFLSKDEAFIWGQGA KGK TREEAGKVIFHQDEV
EKLLIAFGDENQQSDFDWDQAYGPGFDVLHLTSKEGSGGPPS

Codon-optimized NHI 182 sequence:

ATGATTGATGTAGATAGTTTTTTGGCCCGCGGTTATGTCCGTGTTGAACAGGCTTTTA
GCAGAGAAAGTGCGGCCGCTTGTAGAGATATTCTATGGCGCGATACAGGATGTGAT
CCGGATGATCCGGGCGACATGGACCCGTCCAGTTGTACTGCTGGGAGACCACGCGG
AAGAACCCTTCCGTGCCGCGGCCAATACCGGTCCGCTGCCGGGCGGCTTTTCGATGC
GGTTGTAGGTAAGGGCGCTTGGCTCCCAAGAGGTGGCCTTGGCACCTTTCCGGTC
CGCTTCCCCAGTGATCAGCCACCGGGTGATGATGGTTGGCACATTGACGCTTCTTT
TCCTGGAGACGATCCAGCCGACTTTATGTCTTATAGAGTTAATGTGGCCTCTAAGGG
GCGAGCCTTACTGATGCTATTTCTGTTACGCGACGTCCGGGAAGATGATGGCCCGA
CTCGAATTCGTGTTGGCAGTCATCTGGACGTCCCTCCTATTCTGGAGCCTGCCGGA
GAGGCTGGACTGGCATTTCATGGATCTCGCTGCTAAAGCAGTTCCGGCAACGGAAC
ACCGTCCACTAGCCTCCGCTACCGGCCGGGCTGGCGATGTGTACTTGTGTTCATCC
GTTTCTGGTGATGCTGCACAAGCACATCGAGGTACTACCCCGCGCTTTATGGCGC
AGCCTCCACTGTTCCCCGCGGGGCCCTTACCGCCAAGCAGCCCAGTACGCGTAGC
TATCGACCGTGGCTTAGGTCGTAGTTGA

NHI 182 protein sequence:

MIDVDSFLARGYVRVEQAFSRESAAACRDILWRDTGCDPDDPATWTRPVLLGDHAE
EPFRAAANTGPLRAAFDAVVGKGAWLPRGGLGTFPVRFPDQPPGDDGWHIDASFP
GDDPADFMSYRVNVASKGRALLMLFLFSDVGEDDGPTIRIVGSHLDVPPILEPAGEAG
LAFMDLAAKAVPATEHRPLASATGRAGDVYLCHPFLVHAAQAHRGTTPRFMAQPPLFP
AGPLPPSSPVRVAIDRGLGRS

Codon-optimized NHI 183 sequence:

ATGGCCAAGCGCAAATTAGCTCAAGAGGGGCGCGTCCGCAAAACGGACCAGGAAAG
GGGGCGAAAAAGAGGCAGTGGAAGTGTACCTCCAAATGGCGCCCGGGAACCGG
TGTGCGGCGAACCACCAGCTTGGGCCGAGTGCCGTCCTGATCTTTGCGATGCACT
GCCGTGGTTCCGGTCCGTACAGGGTGGCGCTTATCATGTTGGTGAACCTCTCATTCG
GTTGCTTGATCGATGCGGATTCCGGTAGTCGTGCTCACGTAGATGATGAAACTATTA
TAACAAGGATAGGCGGGGGATGCGAAAAAGATCCTAACGGGAATTTGATCCTGCTG
AAGGACCAGGACGACAGCGATCGCGCAGTCGCATCCTATATCAACTCACAAGCGGT
TAAACTGCCGGTGGGCCTTATTATTGGCGATCGTAACCCCGTGCTAGGTCCGCAATT
GCCGCATCGGTATAATGTAATGGGTACTTCCGCATTACGCACATTTGGTACGAAAAA
GTCGGCAAGCGGACGGGTGCTAAAATCAGATTTCAAAGCTTGATTTAGAAAAAAG
GTCGTGGTGGGCGGCGAAAGGTTCTCCCGATCCTCCGCCGGTCTCAGAGCGCGA
ATTTCTGCCACCTGAAAGTCAAATATGTCCGGAATGCACTGTAAGTAGTCCCCGCGT
TTACAATGAAGGTTGGATGTGTCTCCAGCCTTCTTGCTCACAATTCTGGTCTATCAAT
GGAAGTCCGCCGCCGTCTGACCTTACCTATCATCCGCGTTTTCTGTGCGGCGCGCG
AGCCTCCAGATGAGGAAATTAGGCCCCAGTATTCCTTAGTCCCGGATCTCTTATCCA
CAATTCCTGAAGAAGACCGTGATGCGGGTTTATCGCGTATTGCCTGGAAAGGTATTG

TCTGCCCCGACTGCCACAAATGCATTTACGCAGATTTTGGCGCGGATGGAAGTGC
ACCGATCCGCTGGACAATAAAAGTAGTAGCCGCACTTCGACGAGTAATAACAACTCC
CCGACTGGATCTACCTCTCCGTGCACTTTCGAAAAAATGATGCCAATGAGGCCGGT
CTCACTCCGAAGCGTCGTGTCCGATTTTCGAAATTGGCGCCATTACGCGCGCTGTTT
TCGTTAATTCAAAACCTATTCAGCCAAAACGGGATTTCTCGGGGCCCTTATGCTAAATA
TACCTATGAACTGAAAGATATCGGCACAGTCACTCATTTATCGCCAACCGGGAGGT
ATTAAGCCGTCCTAATGGGCCAAATGACTTATTTGAACAGTTGCAACTGGTGGACCT
GGGGCTGCGGCGTTACCCCTGCGAGCAGAGTGTGCGCAGCAGTGGCCGGAACGTT
GACTAGCCACTTTGCCGTTAACTACGGTATGCCGTATAAGTATGTGGTCTCTGTCTGA
CTCAAAAGGCTTCGATGAGGCTCCTCCAGAAATCATGCGTGCTTTAGGCCGCTGA
CCTGGGCAACTGAACAGGCGGTGACGGCAACCGGTGATGAATTTTTACCGCCAAA
CGAGTTACTGGCCCTGGCCTACTTCGAAGAAATGTCAATTGGTTATCACGATGATGG
TGAGAGTTCCTTAGGCCCTACCATAGCGACGTTGTCATTAGGTGCAAAAGCAACTAT
GAAAATACGTTTAAAAGATAAGTACTTCAGAGGGTGTTCGGCATCAAAGAACTTCT
GCCGGAGGACATTGTTCTGCCTGGCTGCGAGAAATACGATGAACGCAAAGAACTAA
AAGCTAAATTTGCTAATAAAGAGATAACTGAAAAAGAATATAACGAACGCAGACTGGA
GATTATCAATCTATGAAGAGGCGCGAGGGCAGTCCCGTTATCAGCCTCGAAATTCA
TCACGGTGATTTAGTTGTAATGCATGGAGCTGATATGCAGAAATATTATGAACATAGT
GTGATCCCGGAGAATAAATTACGTTTTCGCTCTGACCTCACGTTATATTAAACCTGACC
AAGTTGAACAAAAAGACCTGCACAAAGGTAATTTACCCTTGGGCCGGATGAAGTTT
ATCATGGGCAGTAA

NHI 183 protein sequence:

MAKRKLAQEGPVAKRTRKGGEKEAVEVSPPNGAREPVCGEPPAWAECRPDLCDALP
WFRSVQGGAYHVGELSFGCLIDADSGSRAHVDDETIITRIGGGCEKDPNGNLILLKDQD
DSDRAVASYINSQAVKLPVGLIIGDRNPVLGRQLPHRYNVMGYFRITHIWYEKVGKRTG
AKIRFQKLDLEKRSWWAAKGSPDPPPVSEREFLPPESQICPECTVSSPRVYNEGWMC
LQPSCSQFWSINGSPPPSDLTYHPRFLSAREPPDEEIRPQYSLVPDLLSTIPEEDRDAG
LSRIAWKGIVCPDCHKCISRRFWRGWKCTDPLDNKSSSRTSTSNNSPTGSTSPCTFE
KMMPMRPVSLRSVSDFEIGAIQRAVFNNSKLIQPKRDFSGPYAKYTYELKDITVTHFI
ANREVLSRPNGPNDLFEQLQLVDLGLRRYPLQQSVAAGTTLTSHFAVNYGMPYKYVV
SVDSKGFDEAPPEIMRALGRLTWATEQAVTATGDEFLPPNELLALAYFEEMSIGYHDDG
ESSLGPTIATLSLGAKATMKIRLKDKEYFRGCSASKKLLPEDIVLPGCEKYDERKELKAKF
ANKEITEKEYNERRLEIIKSMKRREGSPVISLEIHHGDLVVMHGADMQKYYEHSVIPENK
LRFALTSRYIKPDQVEQKDLHKGNFTLGPDEVYHGQ

Codon-optimized NHI 184 sequence:

ATGGAATCCTTTAATGAATATCACATTGATCCGAAACAGTTTCCCTTTCTGAAAAATTT
CCAGGACAACCTGGCAGGTGATCCGTGACGAATATACCTCATTTATCAACAACGCATC
TAATGAAGAATTGAAATTTACTTATGATGTTTTAGGCCCGAAAAGTCAGACGATAAAG
ACCAAAGGCAAATCAAATATAGCGCCTTCGGCATTCTGTTTCAAGGTATGTTTCATTG
AAAAGTACATTCAGGCACATCAGATTAAATACCCGGACTACGGCCCGGACGAAGCC
TCAAAGAAAGCGCTGGTTTTGCGGGAAAAATACTTTACCAATTTGGCCAATGTAATTA

AAGATGTAAATCTCATAGAAGATGATATCATCCGTAATGTATATTATGGGACCTTTTCAT
CCCGGTCTTGATATTAAATTACATGTAAATTATAACCCGCATACTAATCGTGGATATCT
GGGTCTGATTGTACCTGAAGGCGATGTGGCGATGAAAATATGCCACGATCAACTGT
ATTGGCATGAAGGGAAATTTCTGGTACTGGATCATTCTTACCCTCATTGCCCCCATAA
CTACACATCCTATGACCGCACGGTACTAGTAGTCGATTTTTTTAAGCCCGATAAGCCA
AGAGATGAAGTTATAGAATTCGAAAAAGAACAGGTTACGCAGAGAATGCAAGATAAC
CCATACAGCTTGGGCGTATTTGGAAAAAGCGATAAAGCTAAAGAGGAGGACTTTATT
AAATATGGATTAGCTCACCAGCTTGAATGGGATAAGGCTTTGATTGA

NHI 184 protein sequence:

MESFNEYHIDPKQFPFLKNFQDNWQVIRDEYTSFINNASNEELKFTYDVLGPKSQTIKT
KGKSKYSFAFGILFQGMFIEKYIQAHQIKYPDYGPDEASKKALVLRKYFTNLANVIKDVN
LIEDDIIRNVYYGTFHPGLDIKLHVNYNPHTNRGYLGLIVPEGDVAMKICHDQLYWHEGK
FLVLDHSYPHCPHNYTSYDRTVLVVDFFKPKRDEVIEFEKEQVTQRMQDNPYSLGV
FGKSDKAKEEDFIKYGLAHQLEWDKALI

Codon-optimized NHI 185 sequence:

ATGCAGTTAGACAAACAGCAAATAGACTTTTTTGGATGCGTTCGGCTATCTTGTCGTAC
GCCAGTTACTGACCCCAGAGGAAACCGAGCGTGTCATTGATGCATTTGAGTGGTCT
ATACAAAATTGGGGTGGCGGGACCGATCACGATGGTTCCAAACGTACGATGTTTGG
CGGACCAATTGAACACTCCCCTGATTTGTGCGCTTTAATGGATCACCTGGGATTCT
GGGCTTGATTGGTGGTGTAAACAGGTGAGAGATTTAACTATTGCGGCGGTGATGGCA
ACTACTATACGGGCAATACCCGGTGGCACCTGATGGTAACTGGGGCCAGGCTTGG
GCAACAAAGACTGCATTTTATCTCGATGAAGTGGGAGCCGATTCCGGTTGTCTGCG
TGTGATTCCTGGGTCCCATCATCCGGATCACTTCATCCGTAAAGAAGGTATTCATCC
CCGCGATTCTCTCGAAATGTTTGGGGTACCACCGGAAGATTTCCCCGGTAGCGTGG
CGCTGAGTACGTGCCCAGGAGATGTGGTTATATTCAATCATGATCTGTATCACGCGA
GTTTTGGCGGCGGTAACAGGCGCCGTATGTTTACTATGAACTGCACCCGTCATGCC
GAAACACCTGACGAGATGGACTTGCTGCGCGAGTATGTTGCGGTTTATTCTCCGGG
TGGTTATGATATAGATACCGGCGCGGGCATGTACTTTCCGACGATGTTACATACCGC
AGATGATGCTCGGATGACCCATCTGCAGCAGGCTCGTGATATGCATGACGAGCTGT
ATCCGCAGTACGCTCGCGACGCAGGTGAACGCGATAGTCACAGAGGCCGCATGCG
CTAA

NHI 185 protein sequence:

MQLDKQQIDFFDAFGYLVVRQLLTPEETERVIDAFEWSIQNWGGGTDHDGSKRTMFG
GPIEHSPDLCALMDHPGILGLIGGVTGERFNYCGGDGNYYTGNTRWHPDGNWGWQAW
ATKTA FYLDEVGADSGCLR VIPGSHHPDHFIRKEGIHPRDSLEMFGVPPEDFPGSVALS
TCPGDVVIFNHDLYHASFGGGRNRRRMFTMNCTRHAETPDEMDLLREYVRVHSPGGY
DIDTGAGMYFPTMLHTADDARMTHLQQARDMHDELYPQYARDAGERDSHRGRMR

Codon-optimized NHI 186 sequence:

ATGTTAACTCCAGAAGTTACGTAAAAAAGGAAGCGCAGTACTAGTGAAACTGAAGGT
GATCAGGAGTATACTCATGGTATAGCTGGTGCCGTTATACTGAAATCATCCCCGACT
CAAGACAATAAAAAACAGAAGTTGTTAAAAATTGATATTCCGACCTCTTTGAACTCGG
GTTTAGAACAAAGAATCACCGGGCGCCGACAACTCGTATAGCAGCCTTTTTTCTGAC
ACTGACAATGACAGCCTTTTTGGAGACCTGACCGACTCATCCACAACAAGTCACTG
CGAAGATGACACGGTTGCGTCCCGCACGGCACCATCAATCATGGGAATGTTTTTCG
AACCTACCATTCTGCTTTCCCCCGAACTGGCCGACACAGTGATGACCTTCTGCATG
AATACATACTTTCTGTCGGCCCCGAAGACAACCAAGCGATGTTATTTGGGCGGCTATGT
TCGCCGGACCCCTCTGAACCTCTGGATGTTAATAAGGGTTTGCCTGAAGTTCTGCT
GGACTTACTGGAATCAGTGCGGGTGCGTCTGGAACCGGTTTTACCAACTGACAAGT
ATGAGCTCCTGTTTCCCAAATGCCCTACCATGGCGCGTCAGGCTATTATTAATCTTTA
TCAGCCAGGAGAGGGAATTACCCCCCATGTTGATCTGTTGAGTCGATATGGTGACG
GTATTATCGGCGTCTCTTTAGGTTCTGGCTGCGCAATGCGCTTCGATAAGACGGCAA
CAACAGATAATAGTTTTCCAAGTGAAGGTGAGCGGCATGACCTGTATCTGCCAGAAA
GGTCCGTAATTGTGCTGTCGGAGGACGCACGTTACCAAGTGGACGCATGGAATTGAT
AAAAAGACACGTGACTTTGTTGCGAACGATACTGATTTCGACCCAAGGCGCATGGGT
TGAGCGATCTGTTGCTTAAGCGTTACATATCGGTGGCTGTTACCAGGTGCGGACG
TTGTAGGCGAAACAGCAGATTAA

NHI 186 protein sequence:

MLTPEVTLKRKRSTSETEGDQEYTHGIAGAVILKSSPTQDNKKQKLLKIDIPTSLNSGLE
QESPGADNSYSSLSFSDTDNDSLFGDLTDSSTTSHCEDDTVASRTAPSIMGMFFIPTIRL
PPELADTVMTFCMNTYFRRPEDNQAMLFGRLCSPDPSEPLDVNKGLEVLDDLLESVR
VRLEPVLPTDKYELLFPKCPTMARQAIINLYQPEGITPHVDLLSRYGDGIIGVSLGSGC
AMRFDKTATTDNSFPSEGERHDLYLPERSVIVLSEDARYQWTHGIDKKTRDFVANDTD
STQGAWVERSVRLSVTYRWLLPGADVVGETAD

Codon-optimized NHI 187 sequence:

ATGTTTTTCTCCACGACGTACCAAGCACTGCGCGTTTATACTACAACCGGCCTATAC
AACCGCCTTCGTCTGGAACAATGATCCGTTACCGCTGAGCAAGGGGGTTTCCCGA
AAGAACCACCAGCCACAGACCCAGAGCAGGCGATGACCCCTACCGACAATCCAAC
TATTTCTGCTCTGACTACTGAAGAACTGGCTTTCTTTAAACAGTTCGGATACCTTATC
AAACCTAAAGTGCTGAATGCGGCTACCTGTAATGAACTCGTCGACTTGATGTGGCA
GAGCGCCCCAACTGCTCTGAACAGATTCGATCCCCGTACCTATCAGACGATTCCTA
AGTCGGAATGGTCAAGCGATTCCCTTCGCTTTGTTCAAGGATGATAAATGGCAATTC
GTGCCCTCGGAACTGACCAGCCCCTGATCGATGCGATGACACATCCGCAGCTGTG
GCAATGGGGAGAACAGCTGTTGGGAAAAGGAAATCTACGTGCACCAACACTCGGG
GGCCAGCCCATGGGGTCACGCGGTCCGGCCTGGCCTGGGGGTCCGGTTGATCCG
CAGGACACAGAAGGCGTTAGAGGAATCTATGCGACACTGCCGAATCCCAGTTTAGA
AGGGGCGCAAGACTCTTTACATACAGACGGCCACCCATTTTCATGTTGGATGGGTGT
GTTTATTGGAGGACTGTCCACCGGAGTCAGGCGCTTTTAAAGTTTGGCCGGGTCT
CACCAGCGTTTTTATCCCTTATTTCCCATGCCTTACGATCAAACAAGAATCCCCTTTT
ATGATCATCTTCCTAGCCATAAAGGCATTTTGCATCCTGCGCCGTATTTAGCAGAAGT

CAAAGCTGTCGAAACCGATACGGACCCGGTTGATGTTTGGGGGTCACAGGGCGAT
GTGATACTTTGGCACCATCGTCTTGGACATATGGCGGGTCACCACAAAGGCCATCA
AGCATCTATTAGACAGGCTCTCTTATTTGATTACAACCATATAGCGCTAGATGCGATG
CGTTGCGAGCCGACTCCTTGCGATATGTGGCAGCACTGGTCGCCGGAGCTGCGCG
CAGCGCAGGTTACTACCTTCTCGGACGCACTGATGAGTGCGCAACGTCTGGCTGC
CGAAGGTCAGAATCGCCCGTAA

NHI 187 protein sequence:

MFFSTTYQALRVYTTTGLYNRLRRNNDPFTAEQGGFPKEPPATDPEQAMTPTDNPTIS
ALTTEELAFFKQFGYLIKPKVLNAATCNELVDLMWQSAPTALNRFDPRTYQTIPKSEWS
SDSLRFVQDDKWQFRALGTDQPLIDAMTHPQLWQWGEQLLGKGNLRAPTLGGQPMG
SRGPAWPGGPVDPQDTEGVRGIYATLPNPSLEGAQDSLHTDGHPFHVGWVCLLEDCP
PESGAFKVWPGSHQRFYPLFPMFYDQTRIPFYDHLPSHKGILHPAPYLAEVKAVETDT
DPVDVWGSQGDVILWHLHRLGHMAGHHKHGHQASIRQALLFDYNHIALDAMRCEPTCD
MWQHWSPELRAAQVTTTFSDALMSAQRLLAAEGQNR

Codon-optimized NHI 188 sequence:

ATGACAGATCATCAGAATAGCGCTTTTCAGCCGCTGAATGAAGTGTTTGAACCTCCT
CTGCGCCCAAGTTAAATTCCCCCTTATCTTACCGGCACACGCTCCACATCTTAGCAAT
CAGGGCTGGAGTACATTAACATTCTCCCCGACGGATCCGATTCATATTTCTAGCCAA
AAATTACTTCAGGCTAGCCGTAGCTTTTTTGTCTTCCGTTGGAATACAAAACGAGAT
TTCTTACAGGCAAGGGGACGGAGGAAGGGTGGAATTGCGTAGAAGGTGAAAAGGA
GTTTATTACATTAAGGGAAGTGAATAGCACCCCGCCGGAAGTATTAGATGCAACAAA
AGAGTTCTGGGAAATTACCGGAGATGTACTTAATAAAATTCTTGGCGAAGTAGCTTC
GAGCTTAGGGATGCGGAAAGAATCTCTGACTCAATATTCGGAACCCTGTAGCAAATT
ACTTGGAGAACGGACTGCCTCAATGATTCGCCTGTTTAGGTATGAAATGGATGGAGT
CGGAAATAAAGTGGTATGTGAGCCGCACCGTGACCTTGGCTTACTGAGCCTTAGCA
TTTCAGACGTTCCGGGTCTGGAAGTATTTGATAAGGATTTTAAACGGAGCTTTCCAA
TTGAACGCAGCTACGAAGGGCGGGCGGCAGGCACCGTGCTGGTCGGTCGTGAAC
TTGAGTTTCTGTCAAATGGCCGATATCTAGCAGGTGGACATAGCGTACATATGCATC
CAGAAACAGTAACGCGAAATGAACCGGATGGGGAGGAGAACGGTAATGAACGCCA
GATGCCGGCCGGCAAACATTATAGATATTCAATTGTGTTTGTCTGCGCGGCCATGA
GGAAGTTCGCAATCGACACCGATAAGCTACGTACCTCAATTACGGGTCGTTGGAAAA
AACCCATGAGATCCCTAAAAATGGGGAATCTGTACCGGCGTTTTATGTCCAAACAG
TGAATATTAATGCCGAAATTGAAGAACGCCAGGAACAGAGAGAACGCCTGCATGAG
AAGAAGGTAAAAAGTCTGACAGGCGATTAA

NHI 188 protein sequence:

MTDHQNSAFQPLNEVFELPLRPVKFPLILPAHAPHLNQGWSTLTFSPDPIHISSQKLL
QASRSFFDLPLEYKTRFLTGKGTEEGWNCVEGEKEFITLRELNSTPPELLDATKEFWEI
TGDVLNKLGEVASSLGMRKESLTQYSEPCSKLLGERTASMRILFRYEMDGVGNKVVC
EPHRDLGLLSLSISDVPGLEVFDKDFKRSFPIERSYEGRAAGTVLVGRELEFLSNGRYL

AGGHSVHMHPE TVTRNEPDGEENGNERQMPAGKHYRYSIVFVLRGHEELAIDTDKLR
TSITGRWKKPMRSLKMGNLYRRFMSKHVNINAEIEERQEQRERLHEKKVKS LTGD

Codon-optimized NHI 189 sequence:

ATGGCCACAGAAAATAGTAAGGTGGTCGATCTAAATATCAAAAAAGAAGATCGAATTT
TGGACTTCAGTGATTTCCAAAAACAGGAAATCAAATTTGACATTGAGAACTGCAGG
AAGCTTATCATCAGATTGTGAAAATTAAGAAATTTGAAGATGCCGGGGTTACACATTT
CGGGGCTATCTCTCTTACCCAAATTCCTGGAGATCCGGATTCTATTAAGGGCAACAA
AGCCCGAGGCGTTTACTGGACGAAACCTGATAAAAGCGGTAAAGAAGTCTCTCGCG
ATGAAATGATCGATGAAGCTTCATATTCGGAATTTATAAAAGACTATGAGAACACATAT
TTTAAAGAGGTATACGATGTTCTCAGTAAAAAATATAAGTTAGGCCGCGTACGAATTC
TGCTGAAGGAACCGAGATCAACGTTGTCGTGGCATCGTGATCCTGAACCACGGCT
GCATATTCCAATCATTACAAACCCGGGATGCTTAATGGTTATTGACAATGTGGCAAAA
CATATGCCAGCGGACGGTAGCGTTTGGGTCACAAACAATACGAAATATCATAATGCC
TTTAACGGAGGTGAGGAAAATCGTATTCACCTGGTTGCTTGCGTTCTGGACTATAAA
TTTAATTAA

NHI 189 protein sequence:

MATENSKVVDLNIKKEDRILDFSD FQKQEIKFDIEKLQEAYHQIVKIKKFEDAGVTHFGAI
SLTQIPGDPDSIKGNKARGVYWTKPDKSGKEVSRDEMIDEASYSEFIKDYENTYFKEVY
DVLSKKYKLGRVRILLKEPRSTLSWHRDPEPRLHIPITNPGCLMVIDNVAKHMPADGSV
WVTNNTKYHNAFNGGEENRIHLVACVLDYKFN

Codon-optimized NHI 190 sequence:

ATGTGGCGCAAGAGTGTGCGCTCCCATATCGTGGGAAAAGTTGAGCTGGATAAAGC
ACGCCTGCAGCATGATTTGGAGGCTCTGGCTGCGGTACCGAGGGTGGAGGAAGA
GTATGATGAATTTTCTTCGGGTTACTGGAAGAACTTGAGCCTTTGGAATTCATCCGG
AGAAGCCGATGATACGATGTATCGTGATATTCAGGGGTCAGCAAAACCAACTGCACA
CGCCGCGAAAGCCCCCTATTTAGACGAACTGATACACACTGTGTTTGATAGCAAAAT
TGTTAAAATGGCCCGTGACGCAATTTGGTAGACGCAATGGTGATCCCGCATCGTG
ATTTTATTGAACTAAATAAGGATAATGATCAATACTTCCGTACCTTTATGGTTCTTGAG
GATAATGGATCCGCTTTCCATTTCGGACGATGACACGGTTATTCACATGCGTCCAGGT
GAAATTTGGTATCTGGACGCAGCAGCTCCACACAGCGCTGTAAATTTTTCATCTGAT
TCACGCCAGGCCCTGTGCGTCGATTTTGCATTTGAAGGTAGTTTTACCGAAGATCAA
ATATTTGCCGATAAGAGTTTCTATCGCCCAGATCTGGAACCGGAAATTGTAGTGCGT
AAACCGTTTACTACGGATCACCGTGACAACTGGTACAGTTGGGCGGGATAATCGA
TAATGATAACTTTAAAGATATCCTCTTCTTGATGTGCGCGCTTCACTATCGTTACGATG
TTCCTGCAGCAGAAACCTACGAATGGTTCGTTGAGGCGTGTCGCTTGGCAGATGAC
ACTACACTAACCGCAAAGGCAGAGGGTGTGCGGGATTACATGATAGGCCAGCGTG
CTCTGGGAGAACGTCTTGTGGCGGCAGCCTAA

NHI 190 protein sequence:

MWRKSVRSHIVGKVELDKARLQHDLEALAAVPRVEEEYDEFSSGYWKNLSLWNSSGE
ADDTMYRDIQGSAPKPTAHAAKAPYLDELIHTVFDSKIVKMARARNLVDAMVIPHRDFIEL
NKDNDQYFRTFMVLEDNGSAFHSDDDTVIHMRPGEIWYLDAAAPHSAVNFSSDSRQA
LCVDFAFEGSFTEDQIFADKSFYRPDLEPEIVVRKPFTTDHRDKLVQLGGIIDNDFKDL
FLMSRVHYRYDVPAAEYEWVFVEACRLADDTTLTAKAEGVRDYMIGQRALGERLVAAA

Codon-optimized NHI 191 sequence:

ATGAGTAAAGTTGAAAATCATCACCTGCTGAGCGCCGTCCAGATGGCGCAATTTGC
GGTCAATGGTTTTCTGCAGCTAGAGAACTTAGTCCCCCAAGAACTGAATGAGGCCG
TTCACGCAGAACAGCAGGCATGGAAGGGCCCGGGTTCTTATTGGAGCCAGTCGAA
GCTGATTAAAGATGTCTTTGAACTGCCGCAAGTAAAGGGTGTCTTGCAGAGCCTGG
TAGGTGCTGCACCTGTTTATGATCACAGTGCATTACACATAGTATCCCCACAGCACC
ATCATGCTCAAACTATCACGCGGATTCCGTAATCGATACGCGTCCGTTTGCTTTTG
ATATCCAGGCATTTTACTTTTACATGACACTCCAGAAATTATGGGGCCAACACTTAT
CCTGCCAGGATCACATCTGCACAATAAAGTAAATACAGGATCCATCTCCCACTATAAA
AATATCCTGGGACAACGTAAATTAGTGAGCAAAGCCGGCACAATTGGTTTTCTCCAT
CATGCGATCTGGCATTGTGCACAGCCAAATTATACCGATGAGGCACGTTATGTTTTTA
AACTCCGTCTGCGTCCAGGCCAAGAACAACGGTCGCTGTTTAACATTGCTGATTATA
ATGGACCCGAAGTGGCTCAGGTTCTTAACAATTATCACGACTGGGCTGGTAATGATG
CGCGGCTGAATCAGGTAGCCGCAGCAAATTCTATCGGTATTTGTGTGGTGATGAC
CGTGTTGACTTTTCATTCGAAGGCATCTTCACTCGCATGGGCATATGA

NHI 191 protein sequence:

MSKVENHHLLSAVQMAQFAVNGFLQLENLVPQELNEAVHAEQQAWKGPGSYWSQSK
LIKDV FELPQVKGV LQSLVGAAPVYDHSALHIVSPQHHAQNYHADSVIDTRPFAFDIQ
AFYFSHDTPEIMGPTLILPGSHLHNKVNTGSISHYKNILGQRKLVSKAGTIGFLHHAIWH
CAQPNYTDEARYVFKLRLRPGQEQRSLFNIADYNGPEVAQVLNNYHDWAGNDARLNQ
VAAAKFYRYLCGDDRVD FSGIFTRMGI

Codon-optimized NHI 192 sequence:

ATGATCGATCCCATTTTACGCAACTATTATGCCGAACAGATCTTGAACAAACTGGCC
CAGTTTAAATTTGATTTACAGCAGGAATTTGCCACCGACAACCGTATTAATACCTGCT
ATTTAGATGATCTGCTGAATGTAGATGATGTGATTAAGATTTATAATGCCTTTCCTGAT
AAAAAATATCTCCTGCAACTGAAGGACCTTCGAGAATATAAATATGTGGGTATACAGC
TCAACAAATTCAACCCAATCCTAGAAGAAATTGTGTATGCGTTTCAAGACGCCCGTG
TGGTTCAGCTGATTGCGGAAATTACTGGCTTTGAACAACTTATGCCGGACGAATACC
TGTATGCCGGCGGTGTGTCACTGATGGAACACGGCTGTTTTCTGAATCCTCACCTT
GACAATAGTCATGACAAGGATATGATTAACATCGTGTGCTTAACCTGTTGTATTATGT
TACGCCGTCTTGTTGGAGACCTACGGGGGCAATCTAGAACTTTGGGATCAGGATC
TAAACATCCTTG TAGAACCATTCACTCGAAGTTTAATCGCCTGGTCATCATGGTTAC
GAATAAAACCTCTTTACATAGTGTCTCCCCCATCAACCATCACGGACAACGTTGTTG
CGTTAGCAATTATTACTTTAGCCCTAAACCCGCCGATAAGCAGAACTACTATCATGTG
ACGAGTTTCCGTGGTCGGCCTGAACAGAACTGCGTGACATTGTACTGCGAGGAG

ACGCGCGCCTGCGCAAGCTGGTACGCAAAATATTTAAAAATCGCATAAAAAAACCGC
ATTTTTATCGTAAATAA

NHI 192 protein sequence:

MIDPISRNYAEQILNKLAQFKFDLQQEFATDNRINTCYLDDLNVDDVIKIYNAPDKKY
LLQLKDLREYKYVGIQLNKFNPIL EEIVYAFQDARVVQLIAEITGFEQLMPDEYLYAGGVS
LMEHGCFLNPHLDNSHDKDMINYRVLNLLYYVTPSWLETYGGNLELWDQDLKHPCRTI
HSKFNRLVIMVTNKTSLSHSVSPINHHGQRCCVSNYYFSPKPADKQNYHVTSFRGRPE
QKL RDIVLRGDARLRKLRKIFKNRIKKPHFYRK

Codon-optimized NHI 193 sequence:

ATGACAGATAGAGAACAGCTGCACCGTGACGGGTACGCACTTTTACGCCAAGCTGT
GCCAGCAGAATGGCTGGACGAACTACGTACCTTATTCGATGCGCGCATCACCCCTT
CGCATCAGTGGCCTGTGCCACGTGGTGCAGACTGGCACCATAGTCGTCTTGACGA
TCATGCTATGATCCAGGCTGTATGTCGCCTTCCACAGCTGCTGGCCGTAGTAGGAG
AACTGATCGGAGAACGTTTTTTTTTGGCCCAGGTGGAGGGTTCGTGAGCCACTGGC
GGGGGGGGGACATCAACAACTGCATCGTGACCTTTTCGATCCAGCGACCGGGTGAT
ATAGCAAATGCGATGGTCTTCTTTGACGATTACGGGCCCCGATAACGGGGGCTACTCG
CATTGTGCCTGGAAGCCATCGCCCGTATGCCGACGAGCCGCCTTTTGA CTTTAATG
ATGAAAGTAGAAGCGTCCAGTTAGCCGGTGCTGCGGGAGACATTCTGGTGTT CGA
CGTGGACCTGATTCACGCTGGTAGTTTAAATCGCCTGGGTACCAGGCGCCGCTCAA
TTCTGATTTCTTATTTTGC GGAACCTTTGTATGCAAGTTATCTTGAAACGCGATCATT
GCGTGGTATTCGTATGGATGCTTCAGAACGATTTGAACCGGCGAATTTTTCACTGTC
ACCTGCTTAA

NHI 193 protein sequence:

MTDREQLHRDGYALLRQAVPAEWLDELRTLFDARITPSHQWPVPRGADWHHSRLDDH
AMIQAVCRLPQLLAVVGELIGERFFLAQVEGREPLAGGGHQQLHRDLSIQRPGDIANA
MVFFDDYGPDNGATRIVPGSHRPYADEPPFD FNDESRSVQLAGAAGDILVFDVDLIHA
GSLNRLGTRRRSILISYFAEPLYASYLETRSLRGIRMDASERFEPANFSLSPA

Codon-optimized NHI 194 sequence:

ATGACTATCACGACACCTCCTCCAACAGGGAGACGCGAATGGATACGGGATAGCGA
TTGTGACTTGGACGCATTCAGAGCCTCGGTTGATCGATCTACCGATCTGGCGGACT
ACCCCTTCGCACATTCAGTAGAACAGACGTCTTTTATATCGGTCCGATAGCTTAAT
TCCTCACCTCGATGATGCACAATCTCGTCGTGCGGTCCAGACGGA ACTGGTTCGG
GCGTTAACGGATGGACCGGGAATTGTCGTTTTT CAGAATGCATTTAGCGATCTGGCA
GTGCTTGATCGTGCTACGGAACAGTTCGACTTACTTATTGCGCAGCAGCATGCTAGT
GGTGGGGTTGTCGGTGACCACTTTGGTAAAGCAGGAGCTAACGATCGCATT TGGAA
CGCTGCTCAGAACTGGCCTTAGGGGCTCCTGAGGTTTTTGCAGAATATTATGCAAA
CGATATACTTGCGCTTATTTGTCAAAGCTGGTTAGGCCCCCTATATCAGGTTACCTCT
CAAGTGAACGTGGTCAATCCAGGGGGAGCGGCTCAGGTGCCGCACCGTGATTATC
ATCTTGGAATCGTTCCTCGTGAAGAATTAACGGAATACCCCGCACATCTTCACAGAA

TGTCGGCGGCGTTAACTACAAGGTGCGGTCGCCCATTGCGATATGCCGGTCTGA
AAGCGGTCCTACCCTGTATCTGCCTTATTCTCAACTGTTTGAGGCCGGTTATCTGGC
ATTTTCCCGCCCAGAGTTCGTAGAATTTTCCGCGATAGTCATGTACAACTGCCGTT
GCGAAAAGGAGATGCGGTCTTTTTCAACCCTGCCCTTTATCATGGAGCTGGACGTA
ATCGGTCTGACGCAATCCGCCGTATGGCAAATCTGCTGCAAATCTCGTCACCGTTC
GGTCGGGCGATGGAAGTCCTTGACCGTGAAGCCATGGCTAATGCTGTCTACCCCG
CTTTGTTAGCACGTCACTCTGCGGGGGCCTCCGAGCGGACGCTGCGCAACGCAG
TCGTTGCCACAGCAGAGTGTTATCCCTTCCCCACGAATCTCGATCATGATCAACCG
GTACATTCTCTGGCGCCTGCGACGCAAGTTGATACCGTTTGGGAAGCATTGAGAGA
GAATGCGGACCCCGCGGTGCTCGCCGAACGTTTAGCAGCTCAGACTACACGTCGG
AGACCGTAA

NHI 194 protein sequence:

MTITPPPTGRREWIRSDCDLDAFRASVDRSTDLADYPFAHSVEQNVLLYRSDSLIPH
LDDAQSRRAVQTELVRALTDGPGIVVFQNAFSDLAVLDRATEQFDLLIAQQHASGGVVG
DHFGKAGANDRIWNAQKLALGAPEVFAEYYANDILALICQSWLGPLYQVTSQVNVVN
PGGAAQVPHRDYHLGIVPREELTEYPAHLHRMSAALTQGAHAHCDMPVESGPTLYLP
YSQLFEAGYLAFSRPEFVEFFRDSHVQLPLRKGDVFFNPALYHGAGRNRSDAIRMA
NLLQISSPFGAMEVL DREAMANAVYPALLARQSAGASERTLRNAVVAECYPFPTNL
DHDQPVHSLAPATQVDTVWEALRENADPAVLAERLAAQTTRRP

Codon-optimized NHI 195 sequence:

ATGCAGTCAGATCTGAGCAAACATGCCTCGACAAATACCGCGAGTTCGACTCGTATT
AATGACCAGATAGATTTTCCCGAATGTGCGCGTGAGCTGATTCATTCAGGGTGGTCA
ATTAACAAGCTGCCGATAGCATTACAGGCTCACTTCTGGGATGCACTGGATCTGTAC
GCCAATGTGGGTTCCGATGTAAATCGGAATTTAGCTTCCCTGACCATACAGATGGT
TTCCTGCCATTTGGCATGGAATACGCCAAGGCTACTCATCAACCGGATCTGTGCGA
GCGTTATTGCTACTGGCACAAAGATCGTCGTCGGCACGAGGCCCATGCGTTTTCTG
GTAGCGAGTTTTATCGCGCTATCTGCGGGTATGAATCAGGAATCTCTGAAGTCGCAC
AGGGCGTTCTGGATAGTATATGTGGCTTTTTCAATGTTTCTGATCCAGTGGATATACG
TAATAGTTTCATATATTCAATTATGTGCTTATCATAAGCCGCCCGCCGCTGATCGG
GAATACCTTCAGAACCGTCACGAAGATGGACACTTACTGAGCATTATTAAGGGAAAT
CGGGATGGCCTGGTTATTTTTCCCCGGGGCGAGAGACTGGTTGTGCGACTGGCCG
ACGATGAAATGATCGTAATTACAGGTTCTGCTGCTGACGCTCTTATCAGATAATAGGAT
CCCGTATATGGATCATGCTGTGACTAATCCATGCGTTCCGGTTGCACGCTTAAGCAT
AGTTTATTTTCGTCATCCAGATCTGACACGTAAATACACATCGTTTATGAACTGCAAG
CAACTGGACTTAGTGGCCGTGGCGAATGAGAGCCATCAGGCGTTCGGTAACAATC
CTTTCTCAGCAGCCGCGTAG

NHI 195 protein sequence:

MQSDLSKHASTNTASSTRINDQIDFPECARELIHSGWSINKLPIALQAHFWDALDLYANV
GSDVKSEFSFPDHTDGFLPFGMEYAKATHQPDLCERYCYWHKDRRRHEAHAFSGSE
FYRAICGYESGISEVAQGVLD SICGFFNVSDPVDIRNSSYIQLCAYHKPPRRPDREYLQ

NRHEDGHLLSIIKGNRDGLVIFPRGERLVVRLADDEMIVITGSLLTLLSDNRIPYMDHAVT
NPCVPVARLSIVYFVIPDLTRKYTSFMNCKQLDLVAVANESHQAFGNPFSA

Codon-optimized NHI 196 sequence:

ATGAAACACTTGACGAAAATTACAGATTACAAAGTCGATATCGAGAAATTGAAATCTG
TTTACTTTAACACAATAGAAAAAATCATCGAGTGGAGCAAAGATGACCCGACTCTCTA
TGACTTTAATGCAATTTGCGTAAACCAGATTCCGAGCGATCCACTATCTATCCGTGGT
GGCAACATCCGCGGTCTATATTGGACGTATCCAAAAGATAGCTGGGAAGAGGAAAA
ACGTTTGAACAAGATCGATGAAAACCAGTACACGGAAGTTTGCAAACCCTTTAAAAA
TACTTATGTAGAAGAGATCTACAATGAAATCTCTCAGCGTTGGAATATCGGTTCGATTC
AGGTTTCTGATGAAGCCCCCAAGAAGTTGTCTTTCATGGCACCGGGATCCAGAAAA
AAGATTACATATTCCACTGATCACCAATCCGGGATGTCCGATGGTGATTGAATCTGAA
AGCTTCCACATGCCAGCCGATGGATCAGTTTATATTACGGATAATACAAAATATCACA
ATTTTTTTAACGGCGGCGAAATAGACCGAATCCATCTGGTCGCGACTCTGCTTAAC
TAA

NHI 196 protein sequence:

MKHLTKITDYKVDIEKLKSVYFNTIEKIIIEWSKDDPTLYDFNAICVNQIPSDPLSIRGGNIR
GLYWTPKDSWEEEEKRLNKIDENQYTEVCKPFKNTYVEEIYNEISQRWNIGRFRFLMK
PPRSCLSWHRDPEKRLHIPLITNPGCRMVIESESFHMPADGSVYITDNTKYHNFFNGG
EIDRIHLVATLLN

Codon-optimized NHI 197 sequence:

ATGACCGCTGCGAAAACGCGGACCGATTATGCTGAGTACACGGACTGTACCCCCCT
GTTAGACGACCGCAAGGAAGTGGATCGTTTTTATGACGAGCACGGTTATCTGTTTTT
ACGTGGTGTCTCGATCGCGCACTTGTGAGAACCGCTGCTGAACAGATGCTTCATG
GACTGGTGTGCTGGGAGCTGCGGATCCGCACGCTACTCTGGATGAATTTACGATC
GATTCCTTTGAGGCCGTGGACGAAGTTGCTATGCATGGCCACATAAAATACGATGAA
TTTTGGAATTCAGCCCCGACGACCCGACTGTTTGAACAGATATTCGGCGAGGAGGT
TTTCGTTTTTTCGAAGTACCACCATCCGATATTATCCATCGCAGGCTGGATCAGAGGA
TCCAGCCTTTTCGTATCTTACTCCGCTACACCAGGATGGGTCTATATTGGACCTAAT
AAAGACTTCCGTACGGTCTGGGTCCCGTTAATTGGAACAACAGAAGGAACAGGGG
GAGTTGCGTTAGCTGATGGGTCTCATCGTCATGGACCGCGGGATCATGTTGTGAAT
GACTCCTTTAGACGGTTCGGGCACCCGGTGCGGGGGATTCCGCCACATTTTGTAG
GTGAAGACGAGCCACTGCTGTTTTCAGCGATGGAACCTGGTGATGTTCTTGATTT
CATGCACACATGTGTCATAAAAGCATTCCGAATCTGGCTACAGCTCCGCCGCTAATG
CGGATGTCCATGGATACACGTGTCCAGCCGGCAGCATCTCACAGAGGATACAATGC
ACTCACTCCATGGCCAGAATCCGCAAAGGACCCGTCTAAAGGCATAATGTCAAAAAT
TACAGGTACCCCTACTGAGATAGAGTGA

NHI 197 protein sequence:

MTAAKTRTDYAEYTDCTPLLDDRKELDRFYDEHGYLFLRGVLDRLVVRTAAEQMLHGL
VSLGAADPHATLDEFTIDSFEAVDEVAMHGHIKYDEFWNSAPTTRLFEQIFGEEVFVFR

STTIRYYPSQAGSEDPAFSYLTPLHQDGFYIGPNKDFRTVWVPLIGTTEGTGGVALADG
SHRHGPRDHVVNDSFRRFGHPVRGIPPHFVGEDEPLLFSAMEPGDVLVFHAHMCHKS
IPNLATAPPLMRMSMDTRVQPAASHRGYNALTPWPESAKDPSKGIMSKITGTPTEIE

Codon-optimized NHI 198 sequence:

ATGTTTAAATGGTTCATCAAGATACTCAATTTACTGCGTGGTATAGTCGAAGTGCGTA
GAGGTTTCGTAGAATTCGCCAAGACGGGTAACACCCCTCCGGAGGCTTATATTCAA
ATGATTAATCTTTTCTGTGCGACAGGTGGCCTCAGTAATGATTTGATTCATCAGATGA
TTAAGAGTCCGAAGCGTAAGTTTCCTGTGACCTCAGGTATTCTGGGCAATTTTACAG
AAAGCGAACTATCGAAGGTTACTATGACGATTAAAGAGAAAGGTTATTTTGTATTCTGA
AAACGTTCTCTCTGACGAAATAGGGAATTACCTGTATGAACTTGGGTGTAATACGGA
ACTTTATCAACGCCCGTTAGATTCCGAATCGTACAAAAACAAACCAATTTATAGCAAA
TTCAATCCCTCTCTGGCACCAGCGAGCATACGGTACGATTTTGACGAAGGAATGTTA
ATAAACGATCCCATTATTAGTCAATAATGACAGACTTTTCGGTATTGTCCATAGCGC
AGAGTTATTTAGGCTGCTTGCCGAAAGCTGATGTTACGGGTATGTGGTGGCATACTG
ATTTACAGCAAAGAGCCAAATGCCGAAGCGGCAACTATGTGGCATTTTGATATGGACC
GCGTTAAATGGCTGAAAATTTTCTTCTATCTTACAGACGTCACTCCGGAAACAGGCC
CACATTGTTTTATTGAAGGTTTCGCATAGAACCCGTGGGATTCCCAAACATTTTTTAAA
ATATGGTTACTCTAGGCTGTCAGATCGCGATGTTGAAGACTACTATCCAAAAGAGAA
ATATATAGAGTATAACGCCAAGAAATTCACGGTGATTGTAGAAGACACACGTGGTTTA
CACAAAGGCAGAGCAGTTACGAATGGCTCCAGACTTATATTCCAGACACAGTTTTTCC
GATCACCTGTTTCGGTGGTACGTACTCGAGTAAGAAAATAAACCGCTTTGCGAACGC
TAAACAGAAGAATTTATTAAGTCAACAAAGATATTTATGAACGCTTTCTTTAA

NHI 198 protein sequence:

MFKWFILNLLRGIVEVRRGFVEFAKTGNTPPEAYIQMINLFCATGGLSNDLIHQMIKSP
KRKFPVTSGILGNFTESELSKVTMTIKEKGYFVFENVLSDEIGNYLYELGCNTELYQRPL
DSESYKNKPIYSKFNP SLAPASIRYDFDEGMLINDPIIQSIMTDFSVLSIAQSYLGCLPKA
DVTGMWWHTDFSKEPNAEAATMWHFDMDRVKWLKIFFYLTDTVTPETGPHCFIEGSHR
TRGIPKHFLKYGY SRLSDRDVEDYYPKEKYIEYNAKKFTVIVEDTRGLHKGRAVTNGSR
LIFQTQFSDHLFGGTYSKKINRFANAKTEEFIKLNKDIYERFL

Codon-optimized NHI 199 sequence:

ATGAATATAGCTCAGTCCACAATAGCACCCGCTGACCTTTCTAGTGTCTCACTGCAG
CGCAACCCTGTTTTGGTAGTGGAGAACTTCTGGTCGTCGGATGAACGACGCTTCTT
CCGGGAAAAAATGAACGGTGCAAGTTGGAAGTCACTTTTACAGACCTTCCCCACGTAA
GAGCCGACTTCCCAAATTCCGGTAATTGGGCGAAAGCGGAGCTGGGGTCTGAGGA
AGGCCGGCGGTTTTTGTGCGGCCTCGAGCTGCCGTGTATTCGTGATCATATTGAAA
GTTTTCCAAATATTATAGGTCGTCATGTCGGTTTTTAATTACTACTCTTACGCCGCTGG
CGATTGCCTGTTAACCCACGACGACACCCGACCAGGGTCGTCTGGTGGAGGGAAGA
CGCGCGCCTAAGAGACGGATTGCCGTTGTGACATATGTACACGAGGAGTGGCAGC
CCGACTGGGGCGGTGAACTGATTATCTACGAACGGCGCGCTGATCGTGCCGGGGA
GCCCGTAAACTTAATACCCACCCATTGCATTGAACCCAGGCCGGGTAGTTTAGTTAT

GTTTACAGTTCCAAGATTTTCATCGGGTGTGTGCGCTAGATCCGACGGCTGGAGAAC
ATCGCCGTCTGTCTATTGCAGGATGGTTCATGACCGAACACTCCTAA

NHI 199 protein sequence:

MNIAQSTIAPADLSSVSLQRNPVLVVENFWSSDERRRFFREKMNGASWKSLSLDPHVRA
DFPNSGNWAKAELGSEEGRRFLSRLELPCIRDHIESFPNIIGRHVGFNYYSYAAGDCLL
THDDTDQGRLVEGRRAPKRRIAVVTYVHEEWQPDWGGELIYERRADRAGEPVNLIPT
HCIEPRPGSLVMFTVPRFHRVCRVDPTAGEHRRLSIAGWFMTEHS

Codon-optimized NHI 200 sequence:

ATGTTGGATATCTCGCGCATTGGTCGACAGGCACTTGCTAGTGAGCCTTTTGGGTG
GGCCTTTGTGGATCGCCTCTTCAGTTTCGGGGATGCAGCCGCACTGGCCGCCTCA
TTTCCGAGAGATCATTTCAGACAGTTGCAGGCTATGATGGTGAGAAATCGTACGAA
TACGTGGCCCGCAGTCTCGTGCATATGGGGGCACATGTCCCCAGCTTCGCAGAAC
ACCTCAGTCCTGCCTGGAACATTTAGCAAGTGAAGTGTAAAGCTCTGCATATCGCG
CGGCAATGACCCGTCTAACCGGTGCGGATTTAGCGACGGCCCCGATGGAGATCAAT
GTCGTGCATTATGGGACGGGTGCGTGGCTCGGACCGCATCTTGATTTGAAAGAAAA
AATTGTTACGCATGTTCTGTATTTTAATGAGTCCTGGAAAAAAGAAAACGGCGGCTAT
CTGAGCATCTTACGGAGCGCTGACCCTAGCGATATTGTGTCTGAAATTATGCCGATA
GTGGGGAATTCGGTAGTTTTAGCCCGTTCAGACGCGTCGTGGCACGCCGTGAGCC
GTGTAAGTGAAGGTTGTCGGCGCAGCCGCCGGTCGATGAACGTCATATTTTCATCTA
CCCGGTAGTGTCAGCACTATTTGGCCACCTGGCCGCACCGCAGAACTCCATGATTA
TACAGGTATTTAG

NHI 200 protein sequence:

MLDISRIGRQALASEPFGWAFVDRLFSFGDAAALAASFPRDHFKTIVAGYDGEKSYEYV
ARSLVHMGAAHVPSFAEHLSPA WKHLASDLLSSAYRAAMTRLTGRDLATAPMEINVVHY
GTGAWLGPLDLKEKIVTHVLYFNESWKKENGGYLSILRSADPSDIVSEIMPIVGNVVL
ARSDASWHAVSRVSEGCRRSRRSMNVIFHLPGSVSTIWPPGRTAELHDYGTI

Codon-optimized NHI 201 sequence:

ATGAACGCTCAAGATCATGACGAAGATACGGATCGAGATGCCTGCAGCTCACCTGA
TAGCCTGTTCGATGGGGAAGTGGACGCGTGTAGTCTGGACGGAGCCATTACACTG
CTGCCCCGCGCGCCGTACAGCACCGCCAGTTCCGGGTCTGCTATTTTACCCACATAT
CCTAATAACAAGAAAGCCTGGCCAGAGATGTATATGCGTTTTGTGCAAGAACATATTTT
ACGGCCGCTGGTGCGGACCAGGTTATGCTGTTTGAACGTGCGCCTGCCGCCAG
GGTCGCAGCGAGCGGCAAGAAGTCGATCAGAGGCCCGCCGTCAGGTTTACCAGCG
ATTTTGCTATCACTTCTTCGTGCCATTTTACAATCACTGGCTTTGGTCCTGCCTCCT
GATGTTTCATGCCCTTTTGTTCCTTCGCGAGCGCCCAAGCGCGCACGCCAAATTATCGTA
AATCTTTACGCCCCAGGAGAGGGTATATCAGCGCATGTAGATCTTTTGAAACGCTAT
GGCGATGGTATCGTGGGGGTTTCCTTTGGGTGAGGCTGCGTGATGCGCTTTAGTG
ACGTGGGTGATGACGCGGCAGCAAATGAATCTTACAATCCTGGCACAAAGAAAGAT
GATGGTGGTGAACGGGAATTTTACGACGTGTATCTACCGGAGCGCTCCTTCTATGT

GATGATGGGGGATGCACGTTATAGATGGACTCATGGTATCGAACGCCGCCGTGAAG
ATTATGTGGAACAGAGGACGGAGATAAAGCGGAATGGATCGGACGCGGAGAACG
TCTCTCAGTGACCTTTCGGTGGATGCTGGAGGGTGGGGATGTTCTCTGA

NHI 201 protein sequence:

MNAQDHDEDTDRDACSSPDSLFDGELDACSLDGAITLLPARRTAPPVPGLLFYPHILIQ
ESLARDVYAFCRRTYFTAAGADQVMLFERAPAAQGRSERQEVDQRPPSGLPAILLSLL
RAISESLALVLPDPVHALLFSAAPKRARQIIVNLYAPGEGISAHVDLLKRYGDGIVGVSFG
SGCVMRFSDVGDDAAANESYNPGTKKDDGGEREFYDVYLPERSFYVMMGDARYRW
THGIERRREDYVETEDGDKAEWIGRGERLSVTFRWMLEGGDVL

Codon-optimized NHI 202 sequence:

ATGCTTCTTGATACCTATTTCTATCACCTGCCTCGCGCTATCCCATCCCATACCTGTG
AAGACATCATTGAATTTGGGAAATCTCTTAATCCAACCTCAGAGCCGTACCATGTCCA
CAAAAGATATGCTTTCTGACGAGGAAAAAACAGGAAATTAAAGATACCCGCAACA
GTAAAGTCGCATGGGTGTATCATGACAATCCTTGGCTGTTTCGGCAGTTAACCCCGA
TTCTAGAGTACGCGAACAAAAGTTGGGGGTTTAATATTTACGTTACGAGAATATCCA
GTTCACTGAATACGAGCCGCGTGGCCACTATACATGGCATAATGATGCTATCCAGAA
CCCGATGAACCTAAAAACATGATGAGAAAATTGAGCATGGTGGTACAATTGTCTAA
ACCGGAAAAATATGAAGGTGGCGGATTTCTGTTTTAACCTACGTGGTCTGGACTCAC
AGGTCGACGACAACATTATGAGCCCGCCTCCAGAATTTAAACAACAAGGCTCAATAA
TAGTGTTTCCTTCCTTTCTGTGGCATAAAGTGGAACCTATTACGAAAGGCATCCGTC
ATTCCTTGTTTATGTGGGCTATTGGCGAAAAATGGAAATAA

NHI 202 protein sequence:

MLLDTYFYHLPRAIPSHTCEDIIEFGKSLNPTQSRTMSTKDMLSDEEKKQEIKDTRNSKV
AWVYHDNPWLFRQLTPILEYANKSWGFFNISRYENIQFTEYEPRGHYTWHNDAIQNPMN
LKNMMRKLMSMVVQLSKPEKYEGGGFRFNLRLGLDSQVDDNIMSPPEFKQQGSIIVFPS
FLWHKVEPITKGIRHSLVMWAIGEKWK

Codon-optimized NHI 203 sequence:

ATGTCGGATCAGGCGCGTGAGGAGGCACTGGCGCAGGCCGCTCTGGTTGCTTTAC
GAGGAGGTAATGCTGCTAAAGCCAGGGCCACTCTGGCGCAAATGGCCGCACCTCC
TCCGATGTTATTGGCCCAGGTTTGTAACCGTTTAGGTGACCGGGATGGCGAAGTCG
AGGCACTGCGAGCCATATTGGCTGGGGACCCTCGAAATTTACCTGCTCTCTTGGCT
ATGGGACACAATGCTCAAGCACGAGGCGACGATCGTGGCGCGATTTCATGGCGTC
GCACCGCACTGGCGCAAGCAGCGGCCCATGGAGCCCCACCTCAGCTGCAGCCTT
TATTACAACAGGCACAGGGAAGCATTGCACAGAGCAGCCAACAGTTTGAGGATCAC
CTGACGGCAGCCATCAGAGATCTGCCTCCCCTTCCACGTATTGCGCTTGCAACGGA
TCTGCTTCTGGGAAAGCGTGAGATTTATCTTCAGCAACCGTCTATGTTTTATTTCCC
GGCCTCCCCCAGCGAGCATTTTACGAAAGGTCCGATTTTGATTGGGTGGCTCCTAT
GGAAGCGATGACTGATGCCATCGTCGCAGAACTAGCTGCAGTGCGGGCCGGTGCG
CCGGATTTTGCCCCGTATGTGCAGACAAGTAAAGAGCGCCCCGCCCGAACAATAA

GCTGCGTGACGATCCGTCCTGGGGCGCATACTATTTCTGGCAGTCAGGAGCTCCG
GTTGCAGATCACGCGACCCGCTGTCCAGCTGTGATGGCCGCGCTTGATCATGCTC
CAATGCCTGTTATCGAAGGTCGCTCACCCACAGCACTGTGGTCTCTGTTGAAACCA
GGAACATCATATCGCGCCGCGACACGGGTTATTGAACACTCGGCTGATTTGTCACCT
GCCATTACTGACGCCACCGGATTGCGCCTTACGTGTCGGTGCTGAGACCCGTCCG
TGGCATCGCGGTGAGATGTTGATTTTCGATGATTCCATCGAACATGAAGCTTGGAAT
AGATCGGCCGATACTCGCGTCATTTTACTGTTTGAGGTTTGGCGCCCTGAGATCGTA
GCGGAAGAGCGGGCCGCGTTGACACTGCTGTTCCAGGCTATCGACCAGTACGGC
CCGGCGCAGATCGACACCGGCGGGTGA

NHI 203 protein sequence:

MSDQAREEALAQAALVALRGGNAAKARATLAQMAAPPPMLLAQVCNRLGDRDGEVEA
LRAILAGDPRNLPALLAMGHNAQARGDDRGAISWRRTALAQAAAHGAPPQLQPLLQQ
AQGSIAQSSQQFEDHLTAIRDLPLPRIALATDLLLGKREIYLQQPSMFYFPGLPQRAF
YERSDFDWVAPMEAMTDAIVAELAAVRAGAPDFAPYVQTSKERPAPNNKLRRDDPSWG
AYYFWQSGAPVADHATRCPAVMAALDHAPMPVIEGRSPTALWSLLKPGTHIAPHHGLL
NTRLICHLPLLTTPDCALRVGAETRPWHRGEMLIFDDSIHEAWNRSADTRVILLFEVW
RPEIVAEERAALTLLFQAIDQYGPAQIDTGG

Codon-optimized NHI 204 sequence:

ATGGGTGCGGAGCATTCTAATGCCCTTCCTCTGGGTACGAACTAGTTAAGGGGTG
CCTCCCACCAGCTCAGTGCCATGCCATATGCGAGGAATTATCATCGCATCACGTCTG
TGCGGGTACGGGGGGGATTAGAAAACGCTGAAAAAAATTCACGGGGGATCCGCGAT
CTGGCACATTGTGCACACCTGCTTTCTTTGGCAGCTCAGTACTTAAATGGCCCCGGC
GCAATTAGTTCGGGCTATTGTTTTCAATAAAACCGTTGACAACAATTGGTTAGTGACT
TGGCATCAGGACAAAACGGTATGTGTATCAGCCAAATTCGACGCCCCGGATTGGGG
ACCGTGCTGAGAAGGACCACACTCTGCACGTGCAAGCGCCGCTCGCAGTTCTT
AATAACATGGTTACGCTGCGGATCCATCTGGACCCAGCGGATATTAGCAGTGGTTGT
CTTAAAGTCATCCCGTCAAGTCATCAGGCCGGTCTGCTGAGTCAGAAAGATATAGAA
CAAATTGTCCGACAGCAGCCTGAGGTGGTATGCGAGGCCGGCGTTGGAGATGTCC
TTGTTATGCGTCCTCACCTATTACATGCGTCATCAAAAGCTCAAAATCCGTCCCAAC
GCCGGGTTGTCCATCTAGAGTATAGTGGGTATTCTGTTACCAGACCATGTGTCATGGG
CTTAG

NHI 204 protein sequence:

MGAEHSNALPLGYELVKGCLPPAQCHAICEELSSHHVCAGTGGIRNAEKKFTGIRDLA
HCAHLLSLAAQYLNGPAQLVRAIVFNKTVDDNNWLVTWHQDKTVCVSAKFDAPDWGPW
SEKDHTLHVQAPLAVLNNMVTMRIHLDPADISSGCLKVIPSSHQAGLLSQKDIEQIVRQQ
PEVVCEAGVGDVLMRPHLLHASSKAQNPSQRRVVHLEYSGYSLPDHVSWA

Codon-optimized NHI 205 sequence:

ATGAGCGTACAGCAAGTTAAGGAACAGGGTTTCTGCATCCTGGAACATGTTTATAGC
GGTGAGGAAGTTCAACGTATAATTGAAAAAATTGAAGGAGTTGATCCCGTAGGCC

GGCGTTTCGAAAACTAGCGATCTATTTGCAATACGTCGCTTTCTGCACGTAGTGGC
CGATGTCGGCCCCCTGATTTTCACTCCGGCGCTGCAGACAATAAAGCTCCTTCG
GTCCAGATTTCTTTCCAGTCAAGAGCATTACTTTGATAAGCCGGGTGGAAGCAATT
GGTTTGTAGCGTATCATCGTGACTTAACAATTCTGGTTGATCGTAGAGTTGAGTATGA
AGGATACAGTGGCTGGACTGTGAAAGAGGACCGTTTTGGCGTACAGCCACCGGTT
GATGTTCTCGAAAATATCTTTACTATCCGTATCCACCTGGATGATACCGACGAATCGA
ACGGTGCACTTCGGGTGGCTCCGGGCTCACATCTCGAAGGTGTAGTTCGTCCAGA
GCTGACAGATCTCAGTGAAGAACAGGTGTGTGCGTGCCGAGAGGAGGCGTGATG
ATTATGAAGCCGCTCCTTCTGCATGCTTCCGGACGGACAGTTTCGGAGCGTAATAG
ACGCGTCATTCACATCGAATTCAGTAATCGTGCGCTGCCAGGTGGTTTAGAATGGG
GAGAAAAAAAAGAAACAGTATAG

NHI 205 protein sequence:

MSVQQVKEQGFCILEHVYSGEEVQRIIEKIEGVDPVGPAPFRKTSDLFAIRFLHVADVG
PLIFTPALQTISSFGPDFFPVKSIYFDKPGGSNWFVAYHRDLTILVDRRVEYEGYSGWT
VKEDRFGVQPPVDVLENIFTIRIHLDDTDESNGALRVAPGSHLEGVVRPELTDLSEEQV
CRVPRGGVMIMKPLLLHASGRTVSENRNRRVIHIEFSNRALPGGLEWGEKKETV

Codon-optimized NHI 206 sequence:

ATGCTGAAAGAAAACTTGCACGTGGACTGAAAGTTCTTACTGATAAACGCTACAGG
CAATTCTATGCTCAGAGACAGATTTTAGGTGTTCTTCTCGGGAACGCAAGGCCAAT
TACTAGCATCCCGCCTTCCCGTTTTTGGAGGCACGTGGTACAGGCGAGACCTCTGC
ACAGACAGCTATTCTTAAGGATCGCGGTTATCTGTTCTCAATAACTTGATTAAGCCG
GACCAGCTTCAGGCAATGCATGCCTTTTTCAAGTGTAAATGAGTGCGCGGATCCATAC
AGGCCAGAATTTGGCAAATTTGTTGCACCGGATAACGCGCCGCCGCGAGACACATGT
GGCGTTTTTTTCCAATGAAGCGATCATTCGTGCCCCGCACGTTCTGGATATAGCAA
TGATCCTAATGTGCTTTCAATCGTATCAGGGTTTCTGGGAGCGAAACCCACTATCTC
ACTGATCACTGCTTGGTGGTCTATGCCTCATAAAGATGGTAAAGCAGAGGAGGCGG
AATTGTTCCACCGCGATGTGGACGATATTCGGTTTGTAAATATTCTGCTACCTGAC
AGACGTTGACGAAGTGAGTGGCCCTCACATGTTTGTACCCGGATCTCAAAGTATCG
ATAGGTTGACGAAGATCAGAAGGTATCAGGACAGTGAAATCCTTGATGCGTTTGGT
GAAACCAGCGTTACGACATTTACGGGTAAAGCAGGGACAGCTTTTTTAGAAAACACT
TATGGATTCCATCGCGGGGTGCCTCCAACGCGCAAACACGGCTCCTATTTAGGT
ACTATACTCTTTGAGGAACTCGATATACGGACCAGATAAACCTGTGCGTCCGATTGG
GGCAAACGGGGTCCCCGCCGGCATTGACCCCTTTATCAACCGGGTGTACCTATCCC
CGCGCTAA

NHI 206 protein sequence:

MLKEKLARGLKVLTDKRYRQFYAQRQILGVPSRERKANSLASRLPVFEARGTGETSAQ
TAILKDRGYLFLNNLIKPDQLQAMHAFFSVNECADPYRPEFGKFVAPDNAPPQTHVAFF
SNEAIRAPHVLDIANDPNVLSIVSGFLGAKPTISLITAWWSMPHKDGKAEAEELFHRDV
DDIRFVKLFCYLTDVDEVSGPHMFVPGSQSIDRLTKIRRYQDSEILDAFGETSVTTFTGK

AGTAFLENTYGFHRGVPPTKPRLLFQVLYSLRNSIYGDPKPVGRIGANGVPAGIDPFIN
RVYLSPR

Codon-optimized NHI 207 sequence:

ATGGAGATTCGTAAACTGATCCCGGAAGTTTATGGGGAACAAGGGTATTCGCTCCAT
CTTAAACTTTCTGAGGAGGAACTCGGTGCAATCAGAAAAATGATCCGCATGCAGTG
GCTGTACCGCCTACAGCTTATTGCCCCGAAATATACATATCAGTTCGATGAGCTTGG
AATAGAACGATATCACGAACTGGCTCATACAATTGATCACAGCCAGTCGTGGCCGAA
AACGAGCAGAGTCCTGCCTAGAGAGGCAGTGGATGTGATCCGAGCGATGGAATTTT
TTCAGAAAATGGAAAAGGAGTTTGGTGATATTAATTTGCGGACGAGGAAAAGCTG
GGGTGGGAGAACATTTATTGGCGGTTGGTGCGCCCCGGAATTCTGACTTCGGAA
CCTTACATACAGAAAATTGGTTTGTAACTCTCGGGTATTATGGCGATGAAGTGCACGA
TAAAGACTGGGAGAAAATTAATCTGGATCAGCATTTCATGCGACTCCGGGGAAAAA
TGGTTTGATTGTTGTACCGCATTCTCACCAAAAAAAGATTGGAATGGCATGCAGA
GGATAAATATGGGCAGAAAAAGCCTATAATTGATGAAGACGTTTCCCAGTTGAACCA
GCTGCTGCTGGCTACAGAGCCCCGGGAGAGCTGTGGTATTTCACTATGACCTCCTTC
ATGGAGGGGCTGAGAATAAAGCCGATACTACACGCATCTCTACTGAGTTCACCTTCC
TTATTAAGAAAAATAAGATCGCCGGTTTAGTATGTAA

NHI 207 protein sequence:

MEIRKLIPEVYGEQGYSLHLKLSEEELGAIRKMIRMQWLYRLQLIAPKYTYQFDELGIER
YHELAHTIDHSQSWPKTSRVLPREAVDVIRAMEFFQKMEKEFGDIKIADEEKLGWENIY
WRLVRPGNSDFGTLHTENWVFNLGYGDEVHDKDWEKIKIWISIHATPGKNGLIVPH
SHQKKDWKWAEDKYGQKKPIIDEDVSQLNQLLLATEPGRAVVFHVDLLHGGGAENKA
DTTRISTEFTFLIKKNKIAGLVC

Codon-optimized NHI 208 sequence:

ATGCCTTCGACATACCTACAGTCCCTTCAAAAAGAAGGTTTCGTTGTGATCCCTAGC
CTGATTCCGCCAAGTAGTAAAGAATTTGCGACATTAAACAGGCCGCTACTACTGCG
ACGACCAACACCCGAACCGGAACGTGGCCATACTTCCGTACCGTCCCTAAGCAGTT
TCCACCCTGGCCCACAAACGTTCCACCTCCAGCAAGCGAAGGCGGTATATGGGGA
GTGCAACATCTGTTGCACCCGGACATGCCAGGTGCGGAAGAATTGCTAAGTTTTA
TTTCGATGAGAGAATTTTATCTGTAGTTGAAGAACTGCTAGGGCTCGACCTTGCCGA
AAATGATGAAAACTGGTCATGGAACGTGTTCAACCTGTTGGTGGCCCCTGAACTC
ATGATTTTGCCTTACGCTGGCATCGTGATGATATTCCAGAGACCGCTTCCCCGGATG
AAGAAGAAAAGTTGTTAGCATCAAAATCTCCGGGTGGTAGACAGTCTCACGCCAG
TACAATCTGGCATTGTGTGACGATGCATCTCTGATTGTTGTGCCGGGTTCCCACCGT
CGTGTTGCGACGGAAACAGAACGTAACGCAGACCCGTACGAAAATGAACTGCCGG
GACAGCTAGTCGTTGAATTAAGGCTGGTGATGCAGTTTTCTATGATTCAAATATACT
CCACCGCGGCGTTTATAAAGCCAAAGAGGAGGGGGGAGAAGAATCTAGATTGACC
TTACATGGGTGCGTGGGTATAAAGAAACGGATGGTGGCGAAGGTGATGGTAGTAA
AAAAGTTCGCGCCACTGCAGTTCTACAGCATGGAGTGGGCGCATGGGTGCATCGG

GGTGATGCTGCTTTTGGGGTTGGAGGTCGGGCGGAAAAAATGCGTGCGAATCTCG
TGGCAATGGGAACGGGGGAAGGTGTTGGTTACTCGTTACAGGGGTGA

NHI 208 protein sequence:

MPSTYLQSLQKEGFVVIPSLIPPSSKEFATLKQAATTATTNTRTGTWPYFRTVVPKQFPP
WPTNVPPPASEGGIWGVQHLLHPDMPGREEFAKFYFDERILSVVEELLGLDLAENDEK
LVMELFNLLVAPETHDFALRWHRDDIPETASPDEEEKLLASKSPGGRQSHAQYNLALC
DDASLIVVPGSHRRVRTETERNADPYENELPGQLVVELKAGDAVFYDSNILHRGVYKA
KEEGGEESRLTLHGSGVIKETDGGEGDGSKKVRATAVLQHGVGAWVHRGDAAFVVG
GRAEKMRANLVAMGTGEGVGYSLQG

Codon-optimized NHI 209 sequence:

ATGACCGATCCGAACACTTGTCTGCTGATGAACAGATGCGTCATTATATCACCAGG
GGCTACGTCGTGCTGAAGACCGGACTGCCTAAAAGCTTCCACGAACAAATTTACCA
GCGCACATCGGAGGTAATGGAGAAGGAAGGCAACCCGGGTAACAACATTCTTCCG
CGCGTGCCAGAACTGCATCGTGTGTTGCGCCGACCCTACCTTGCGAGGCGCGCTTA
GCAGCGTATTGGGCCCGGATCACATCATGCACAGCCACCGTCATTGCCATGTTAATA
AGGGCCCCGGGCGATGGCGGCGGCGTGCATAAGGATAGTTACTGGGGTTATAGAAG
GGTCCGCTCGCATAGAACTAGGTGGGCCATGCTCCTCTATTACCCACAGGATTGTC
CAGAGGAAATCGGTCCGACAGGCGTAGTGCCGGGCGAGTCAAACCTATGAGCATAA
GGAGAGTTGGGAGCCGCTGGAGCAGATCCTACCATGTGTGGGCGAAGCGGGCAC
CGCTGTGCTGATCCATTTTGACATTTGGCACCGCGCTTGCCCTAATCTCACAGAGA
AGACCCGCTTCATGATGAAATTCAGTTCACGCGGATGACAGAACCTACTGCTCCG
ACCTGGAATAATCGATCGGCCGTTTTTTCCTGGAGGCAACGGGTCAGGCGATAGACA
CCGTGGTTTGTGGACGCGGATGTGGAATTGGCACAGAGGTGACGCGCTAGCTACG
GTTGATGGAACCGGCGCAAGTGGTGTGATGCGGTTGCTCAGTTACGCGATGGGG
ACTTAGAGGAACGTATGGCAGCAGCGGATCACTCGGTGCCGCCGGCGCAGGCGA
TGGTACTTTGGATGCCTTAAGAGCCGCCCTTGCGAGACGAAGCGGAACCGGTGAGA
TTGAATGCTGCGTATTCTCTTGCCGCGATTGGTACCCCGGCAGTAGCTGGCCTCGT
TGATTGCTTGACCGACGATATGGAGCCCGCTCGTCAGGATGCGGGCTCAGGGTCTG
GCCGCTATGGGCCCTGCCGCGGAAGTTGCGTTAGTGAACGCCTGTTCTGACGCAG
ATGAGGCGCGACGGGGATACGCGGCATTTGCATTGGGTGAAATGGGCGCTCACGT
GGTTAGTGATGCTGCTGCAGCTGTTTCGGGCCTTGCTACCGATCCGGCTGAATTTG
TACGACGTAACACAGCGGACGCCCTTGCTACTATGACTGGACACGCCGACCACGC
AGTACCTGCATTGACTACCCTGTAAAGTGATGATTCAGATCAGGTTAGATTTGACGC
GGCATATGGTTTAGCGCATCACGGTGTGGCCGCAGCTCCGGCGACGGACGGTCTG
GTTGGGGCTCTGGATGATGAAAACCGTTACGTCCGCAACCACTCAGCAGAAGCCC
TGAAGCGTATTGGGACCCCGACCGCCACCGCGGCGTTGTTGACTTTCTTGAAAC
ATCACGTTGGTGCCCCATTACCAGCAAAGAAAGCACGTTCTGA

NHI 209 protein sequence:

MTDPNYLLSDEQMRHYITRGYVVLKTGLPKSFHEQIQRTSEVMEKEGNPNNILPRV
PELHRVFADPTLRGALSSVLGPDHIMHSHRHCHVNKGPGDGGGVHKDSYWGYYRRVR

SHRTRWAMLLYYPQDCPEEIGPTGVVPGSQTYEHKESWEPLEQILPCVGEAGTAVLIH
FDIWHRACPNLTEKTRFMMKFQFTRMTEPTAPTWNNRSAVFPGGNGSGDRHRGLWT
RMWNWHRGDALATVDGTGASGVDAVAQLRDGDLEERMAAADSLGAAGAGDGTLDALRAALRDEAE
PVRNLNAAYSLAAIGTPAVAGLVDCLTDDMEPARQDAAQGLAAMGPAAE
VALVNACSDADEARRGYAAAFALGEMGAHVVSDDAAAVSGLATDPAEFVRRNTADALAT
MTGHADHAVPALTTLLSDDSDQVRFDAAYGLAHHGVAAAPATDGLVGALDDENRYVRN
HSAEALKRIGTPTATAALFDFLETSRWCPITSKESTF

Codon-optimized NHI 210 sequence:

ATGAATGAACATTTATTAAATTCGAAACAGATGGCCAGTTTTGTTGCGTCAGGATATC
TGCGCCTGGATAACATGGTACCTAAAGATCTGTGCAATGCATGCCTGGACGAAATGA
AGCACCATAATGGGTATTTTGAGGTTGGTGCGCCTTTTGAAGAGATCTGGCCTAAGA
ATACAGCATTAGGTGACACCTTTCGTCTTCCTCAAATCAAAGGCCTGATCCAGTCAT
TAGTCGGACCCGACCCTCTGTATGACCATCACTTTCGCGATTTAACAAAAGCGAATA
ATACTCGAGGACCTGATATGCACCAGGATTCTGTTATCGATTTCCGTGAAAACATTTT
TGATATCCAGCTGTCTTTTTTTCGCAGCAGACACGCCAGATAAAATGGGCGGTACTTT
TCTAATTCCGGGGACTCACTACAGAAATGTTTCGCACCTCTGAGATTACGATTTATCAA
CACATGAAAGGAAAAGTTTGGGCAGCCTGTCCTGCGGGTACAATATATGTTTGGAAT
ACGCGTGTGTGGCATGGCGCTCGGAGCAATCGGTCTGACAAAGATCGGTATATGTA
TAAATTGCGGCTCAACCCTACACAACCGCAGATCGGACTGTTTGATACCAGTGACC
TGAACGACCCATCTATCGATTGATACTGAATACAAATCATGGATGGGAAGGTAACG
AGCATCGTTATGAGCTTATGCAACGGATTAACTGTGGCGTTACGTTTCCAATCAAC
CAGACTACGATGTAGGTGAACGGTTTATGCGCCGTATTGAATACAACCCCCAGATTG
AAGCGGCGTAA

NHI 210 protein sequence:

MNEHLLNSKQMASFVASGYLRLDNMVPKDLCNACLDEMKGHHNGYFEV
GAPFEEIWPKNTALGDTFRLPQIKGLIQSLVGPDP
LYDHHFPHLTKANNTRGPD
MHQDSVIDFREN
YFDIQLSFFAADTPDKMGGTFLIPGTHYRNV
RTSEITIYQHMKGKVWAACPAGTIYVWNTRV
WHGARSNRSDKDRYMYKLRLNPTQPQIGLFD
TSDLNDPSIDSILNTNHGWEGNEHRYELM
QRIKLWRYVSNQPDYDVGERFMRRIEYNPQIEAA

Codon-optimized NHI 211 sequence:

ATGACAGCAAGTGAGTCACCTAAATGGAAATCATCTCTGTTACAGAAACGGTTAGTC
ACTCGCAGCGCATTTTGAACGAAGCATCGCAGTGTGCGGCATCCATATTGCGGTC
TGAGGCTGCGCGCGTTGATGGTCAGCTGTGTGATGATGTTTCGAGCGGTCCTGGCA
GGTAATGCTGTTTACATTCGCCGCTTCGTGTGCGATGAAGCCGATATGCGGCTCTAT
GAACGCCTCAAGGCAGAACTGGTAGCCGCAACCGGAGCGAACATGTCTCATGATG
GTGGTCGCATAGATTGGTCCCGCCATGAGATTTTTGAAAATCCAACTGGCATCAGC
GCGATGTTTAATGAGATTATTGATATGTTAGCTGAGTATTTTCGATTTGGACGTGTATGC
GACCAGGCTCAATTACTATCGGGATGGGCATCAATGGAAGCCGCAACATCATGATTC
TCATGCGTATGGGGGCACATCAACACACAGCGGCGAACGATTACGTGAAGATTTCA
CCGTGGGCATAACTCTGGGCGCGACCCGTTCACTGCTGTTTGTCCACGAAGCGAG

TGCCACCAATTTGACTTCCCACAGCGTAACGGTGACTGTTTCGCCTTCACTAGTG
AAGTTAATAACGCCTTTACTCATGGGGTCCCCCGGGTTAATACTCCTGTAGGCGACC
GCTTCAGCATTATTGCGTGGGGTTCGCAGACGCGCTATGAACGAACGTAATTCCGAT
CGGGCGGTGCCCTCATTGACGGTAGTGACCTTCAATCTGTAGATGATGCAATTCA
GGCCGCCCATCAGCTCGTCAGCCGTGCCTCAGCATCGACCCGCCCAACCGTCGC
ATCAACGCCGTCTACACGCATGGAACGGAATGATGGTAATAGTCATCGCAATGAATC
TAGCGCAGGTAGTTCGGTGCCGGCTAAAAAAGAAAAAGAATCGCCTGCAGTAA

NHI 211 protein sequence:

MTASESPKWKSSLLQKRLVTRSAFCNEASQCAASILRSEAARVDGQLCDDVRAVLGN
AVYIRRFVCDEADMRLYERLKAELVAATGANMSHDGGRIDWSRHEIFENPTGISAMFNE
IIDMLAEYFDLDVYATRLNYYRDGHQWKPQHHD SHAYGGTSTHSGERLREDFTVGITL
GATRSLLFVHEASGHQFDFPQRNGDCFAFTSEVNNAFTHGVPRVNTVPVGD RFSIIAWG
RRRAMNERNSDRAVPSFDGSDLQSVDDAIQAAHQLVSRASASTRPTVASTPSTRMER
NDGNSHRNESSAGSSVPAKKKKKNRLQ

Codon-optimized NHI 212 sequence:

ATGAACCAGGACACGACGACATTACTGAACCAACCGCAGCATCGGCAGCTTTCTTT
AGCGGATGAAGTTGCCGGAGCGCCTGGGTATAGCACTCGCCTGTCGCTGAGCGAG
CCGGAGCTGACGGTTATTTCGTCAATTAGTCAAGGATCAATACCTGCAGCAGATTGAA
GCCCAACCATCCTGATAAATTAGAACTGTTTCGTCATGGTTCAATGGAGGATTATCACC
TGATATGCCACCACGTTGACCACGAGAAATTATGGCCGAAGGTGCGTCGTATTTTAC
CGCAATTCAGCGTCGAAATGATTGAGACTTTACCATTCATGAAGGAACTGGAAAAAG
CGTATGGTGCGTTCACGCTGAGCGATGAAGAGGAGATGGGATATGGAGAGATGTAT
TGGAGAATTGTTTCGTCCTCGATTCTCCTACAGATGTTGGCCCAATTCATGCGGACTAC
ATGTTTTGGGAACTAGGCCACGGCAAGATGCCAGCAAACCTGTTTCAGGGTGAAAGT
TTGGCTCGCATTATTCGCAGAACGGGGAAAGTCCGGCTTCCAATTCGTGGCTGGTT
CTCATTTAGAAGATATTCCTTATATAGGTCAGAAAAGAGGGTTAATTCTGAAACCTAA
GCTTCTGGTGGATGTTAATAGCCTTGCCCTGGTGCCCTTTGTCGCCTCTGCCGGCG
ATATAATTGTGTTTCACGACCGTCTGTACATGGCGGCCTTGTTGGTGGACAGAAGA
CCCGAATTAGTTTAGAGTTTACTATGATCATTCCACACCGTTCATAG

NHI 212 protein sequence:

MNQDTTLLNQPQHRQLSLADEVAGAPGYSTRLSLSEPELTVIRQLVKDQYLQQIEAHH
PDKLELFRHGSMEDYHLICHVDHEKLWPKVRRILPQFSVEMIQTLPFMKELEKAYGAF
TLDSEEMGYGEMYWRIVRPDSPTDVGP IHADYMFWE LGHGKMPANCFRVK VWLAL
FAERGKSGFQFVAGSHLEDIPYIGQKRGLILKPKLLVDVNSLALVPFVASAGDIIVFHDRL
LHGGLVGGQKTRISLEFTMIIPHS

Codon-optimized NHI 213 sequence:

ATGGCAAGTTCGGCACACAACCTGCAAACGCTGCCCACTTTACATGGCGTGACTGC
AGCGCCGCCGCCAACCCCGGCAACACAGCCAAATAGGTTGAGTACTAGCGATGCA
GCCGATGCATTAAGCCGTCTTCTGCATCGGTTACCGCCGACCTTATCTCTGCCTACT

CGCTTATCTCCGCCAACAAACGAGCCCGCCTTCCGTATCGTTCTCGCAGGATAATCC
AAACTTCCTGAAAGATATTACCAGCTCATCAAGTCAGTTAGGTTTTTTTTCAGTTAACT
AATCATCATCTCCCTAGCCAGCTGGCAGTTTCAGCTGAGATCGAGTCACTCTCCTTG
CTGAACCTGCCCCCGGATCAAAAACAGGAGACTTTTTCTAAAATTTGGCCTCTCAG
CTTCGAGTATGATCACGAGGATGGTAATGGCGACGGAGATGCGGAAAGTTTTTGCT
TCGATTCTAGTTGCAGTACCGAAAGCACCTCTCAAAGCCTGTCCTCACTGAAAGAAT
TTGCTCTGGCTCTGGAAAAAGTTGGTTTAGAGGTGGTTGAGGCACTTAGTTGCGCA
GTTGGATTTGAAAATCCACTGCGTGAAGGGGGGAACAAATTTTGCTCACTGATGTG
GATTTCTGAGGGGGTAGCTGGAAATAAACAGGCTTTTCGGGCCGTTTTTACCCATA
TATTATTGCACTGCACTATCAGATCCGCTGCCGGAAATATCTTTTATTGGCTGATTCT
GGCTGGGTCAGTGTCTCACCGCAAGTTGACTCCGTGCTGGTCACCGTAGGTGATAT
CGCACAGGTATGGTCAAATGGAAAATTAAGAAGGTAAGAGGCAGACCTATGGCGG
TGTCGGTCGATGGTGAATCAACACGCTGCATTAGCATGTCTCTGCTTCTGACATTGC
CAACGGAGAGTACTGTGAGTACACTGTTGCCAGGCACAACCACAACACCTCTAGT
TCATCAACTGACGACCACGGACAAGACAACAATGTGAAAGATTATAATAATGATGATG
ATGATGATGAGAAGGAAGCCGAAGATAATAAGAAAAATAAATTGTTTTCGAGCTTTTC
ATTTGAAGATTACGCTTGGAGAGTTTACCATGAACGCCTGCCCTTTAAAGATCCACT
GGATCGCTTTCGCATCATCTCAAATTTATTAA

NHI 213 protein sequence:

MASSAHNLQTLPTLHGVTAAPPPTPATQPNRLSTSDAADALSRLHRLPPTLSLPTRL
PPTTSPPSVSFSQDNPNFLKDITSSSSQLGFFQLTNHHLPSQLAVSAEIESLSLLNLPPD
QKQETFSKIWPLSFEYDHEDGNGDGAESFCFDSSCSTESTSQSLSSLKEFALALEKV
GLEVVEALSCAVGFENPLREGGNKFCSLMWISEGVAGNKPFGFSGRFYPIIALHYQIRC
RKYLLLADSGWVSVSPQVDSVLVTVGDIAQVWSNGKLKKVRGRPMAVSVDGESTRCI
SMSLLLTLPTTESTVSTLLPGTTTTTSSSSTDDHGQDNNVKDYNDDDDDEKEAEDNKK
NKLFFSSFEDYAWRVYHERLPFKDPLDRFRIISKFI

Codon-optimized NHI 214 sequence:

ATGTCTATGAAGCCGAACCCTCTTCCTGGTGTCCCTAACGTTGAATCACCATTTTTT
GATCAGCTGTTTGCTGAGGGCACTACTCCTCCTGAGTTGCGTAGTATTGCTCTCAG
CCTTGACGCGATGGATATGCAGTGATTGATTTCCCTGACGAGGACATCTCAGGCA
TGGCCGATCGAATACGTGCCGATCTGCATGATCGCTATGATTGGGAGGCAAGGGCA
GCGGGGCGTGGCGCGGATATGCGTCTGCAAGATGCGTGGAATTCAATGATGATGT
ACGACGTATCGCAGTAAATCCGAATATAATCAACCTGCTTAGTGAGCTATATGGTAGG
AAGGCCTGGCCGTTTTAGACCCTAACTTTCTGTGGGTACTCAGCAGCACTTTCA
CACCGATTCTGTGCACTTTTCGTCAATGCCAGAAAGATACATGTGTGGTGTGTGGG
TTGCTCTAGAAGATATAGGACCCGACCAAGGCCCTGTCATCTACTATCCGGGATCGC
ATCGCTGGCCGATTTACACTAATGAACATATTGGTTATCGGCCGACGCGTGCTCGTA
CCACGCAGATGGTATACGAGGGGTATGGTCCCAGTTAGTTGCTGCCTCGGGAATT
CAGCCTAAAACATTAACAATTAAGAAGGGCCAAGCACTTATATGGCTTGCCAATTTAA
TGCATGGAGGAGATATTCATTTAAGTCACGAAAAACAAGATGGAGCCATGTCACTC
ATTATTTTTTCGAGGGTTGTGTCTATTATACGCCGATGAATTCGGACGTGCCGGGAG

GTCTGGTTGATATCAGGAGCCCGTATAACATCATTTTGGACACCCATGCTCAGAATG
CAAGCCTGGGAAGACCGCTGCGCCGAGAATACTTAGACGCCATCGCGGCCACGAG
CTATTTTGCCGCAGAGCCCGAAGATGATTTTGACCCTGCGGCTTACCTTCTGGCTAA
CCCGGATGTTGCTGCAGCGGGAGAGGACGCATACGCGCACTGGATCAAGCATGGC
AGAGATGAAGGCCGTCCGCTGAAGTTGATATAA

NHI 214 protein sequence:

MSMKPNPLPGVPNVESPFDFQLFAEGTTPPELRSIALSLARDGYAVIDFPDEDISGMAD
RIRADLHDYDWEARAAGRGADMRLQDAWKFNDDVRRIAVNPNIINLLSELYGRKAWP
FQTLNFPVGTQQHFHTDSVHFSSMPERYMCGVWVALEDIGPDQGPVIYYPGSHRWPI
YTNEHIGYRPTTRARTTQMVYEGLWSQLVAASGIQPKTLTIKKGQALIWLANLMHGGDIH
LSHEKTRWSHVTHYFFEGCVYYTPMNSDVPGLVDIRSPYNIILDTHAQNASLGRPLR
REYLDIAAATSYFAAEPEDDFDPAAYLLANPDVAAAGEDAYAHWIKHGRDEGRPLKLI

Codon-optimized NHI 215 sequence:

ATGGAAACACCAAAAAAGGGGTGAGGTGGAGTCACCGTCGCAATGAGCGACAAA
AGACCTTTTATCATATTAACAATGGCCTGATCGCGGATAAAAGCGAATTGTCCGCAG
CAGAAATTACGGCTTACGCCGAACACACGCGTAATAAGTGCATAGCCCAGTTTGGT
GTAGTGAAATTAGCCAAATATAATTTAGATGTCTGCTGCGAAACGTATCAAGGTGTTA
TGCTCTTGTTGCTATGCTGCATTACATTCGCCTAATATCGCCCTTGCTCTGGTTTA
CGCTGGCGCAGATCCGTGCTTTGCTAAGAGACACAATGGCGCAGTGGACAGTTTC
CCAGCGGCACGCGAAATACTCATGAAATACCCGCCGCCTTATATGGCTTGGTTGCTT
GTCCTGCTGGCGACAGCCCCTCGTACTGATAGCGCGTGCGGCGATTGCGCTTGCG
TGGGCTCGAACATGGTTTTGACTCCATGTGATCACGCTGTCTGCGAATTATGCTTTT
GGAAACATCTGGCAACAATGCGTGTGGATCAGGACATGACCTGCAGCTGTTGTGAT
GTCCAGCTGGTGTGCCATGTGTTAAATCGCCTCCTGCGCCGCCAAGCCCACCGT
CCCCTATGGAACCGGAAGACATTGCGGTTGCATCCAAAGCCGCATATGCTTTACTC
CCGCCAGAGCCTCGGAAAGGTAGCACTAACGCAAAAGTTAGCAAACCTCGTGAAGC
CTCTTCCACTCCATAATGCCGTTGAGCTGTTTGTGGGAGCATCTCAGTCTCAGAGA
GATGCGAATATGTTTAAAGCCGTCTGCGCAGATAACTGTTCTCGCCTGCGGGCTCT
GGTGGCAGCTGGCGCAGATGTTGATTGCGCGAATGAGTATGGACAGACCCCTCTG
TTCGTAGCTGCTTGGTTAGGCCGTGATAGGGTGGTGGCCTTCCTCATCTCGGTGGG
TGCTTCTGCCGTCGCACAGGATAACGCCGGTACCCTTCCTGTGCAAGCCGCAGCT
GCGCATTGTCATACGGGTGTTGTGTCCCTGTTACAGTCACCAGTCGGCCTGTGTGA
TTTTAATCCAAGATGGTGTCCAGCTGACGAAGTCCCCTCACCCGCCTTCGTTGAAC
TCATTCCCGCGGATGCTTCGCATCCCGGCGCGGGATCCGGTTATTGGGACGACCT
GTTTCCGGAAAGTTTTTTAGCACGACTGGAGCATTTGCATTCAACCTTGCCGGTAGC
ACCAAAAGAAAAAGCCTCATGTTTGAATAGGAGCTATTTTTGTGATTCAATTGGTACC
GTTATGAATGCTGTAGAAGCTCGCTGCGGCCGTAGGCTGTTTCCGAATATGCGATTT
CTGCATTATGAAAGCATCGGCGGGTACTTACCAGCGCACACCGATTAAAGTCGTA
GATGCCGATGGGGTCACCAGTACCACAACTTTTATATTATTTGACCAGCTGTAGC
TCAGGAGGTGAAACAACTTACTGAGCAGCATGGCGCCAGATCGCTCTGTTTTGGC
GCCCGTTAAGCCCAAGCGCGGCCGCTTACTGATCTTTCCGCATATTTGCCCGCAGC

AGGGTGGTCGCGTCGAATCTGTGCCCAAAGCTGCTGCTGCGCGGAGAGATGTATTG
A

NHI 215 protein sequence:

METPKKGSRWSHRRNERQKTFYHINNGLIADKSELSAAEITAYAEHTRNKCIAQFGVVK
LAKYNLDVCCETYQGVMPVRYAALHSPNIALALVYAGADPSFAKRHNGAVDSFPAARE
ILMKYPPPYMAWLLVLLATAPRTDSACGDCACVGSNMVLTPCDHAVCELCFWKHLATM
RVDQDMTCSCCDVQLVLPVKSPPAPPSPSPMEPEDIAVASKAAYALLPPEPRKGST
NAKVSKLVKPLPLHNAVELFVGASQSQRDANMFKAVLADNCSRLRALVAAGADVDCAN
EYGQTPLFVAAWLGRDRVVAFLISVGASAVAQDNAGTLPVEAAAAHCHTGTVSLLQSP
VGLCDFNPRWCPADEVPSPAFVELIPADASHPGAGSGYWDDLFPESFLARLEHLHSTL
PVAPKEKASCSNRSYFCDSIGTVMNAVEARCGRRLFPNMRFLHYESIGGYLPAHTDLS
RTDADGVTSTTTFILYLTSCSSGGETNLLSSMAPDRSVLAPVKPKRGRLLIFPHICPHEG
GRVESVPKLLLRGEMY

Codon-optimized NHI 216 sequence:

ATGAAATATTATCAGCAGCATCCGAATGCCTTTCCGCTGAATTATTTACAGGATCTGT
GGGGTGAAATACAAGCATGCCCCGACTTTGCCATTAATAATTTGAATAGGGATTTCAT
AAATACTAAAGGCTTCTCCGTTGTATTCCAGCGCCAAGGCTTAACCGAGGTTATCCA
GAAATTTCTTACTTTAAACCGTATTTGGATGTTGCTTTATTACCAAAGTGTAAACGCAT
TTTATCTCAACCCTCTGCTGCTGAAAAGCGGGTCCCGTGTTGATCCTCATATCGATC
GTTCTTTGAGATCATATTGCAAGACCGTTGAGCCGCCAGCTTTTGTAGCGTTCTTT
ACGTCAGAGTTCCGGAGGATATGGAAGGTGGCGAACTGGTTCTCAAAGTCCAGAA
ACGGCAAGTAGGCCAGATCAAGCCGCAAACGAATAGCCTTATCTACTTTTCAGGGTG
ATCTGACCCATTCAAGTCAACGCGGTTACGTCACACGGAACCGGCTGAGCTTGGA
TGTGAACAATATTGCTTAAGTGAAGAGGAGCTTGAAGATATTCCGCAGTTTACCTTG
GAGTCAAGGGCAAATCAAGGCAACGGAAAAATAAAGGATCGAAATAA

NHI 216 protein sequence:

MKYYQQHPNAPFLNYLQDLWGEIQACPYFAINNLNRDFINTKGFSVVFQRQGLTEVIQK
FPYFKPYLDVALLPNCNAFYLNPLLLKSGSRVDPHIDRSLSYCKTVEPPAFVSVLYVRV
PEDMEGGELVLKLQKRQVGQIKPQTNSLIYFQGDLSVNAVTSHGNSLSLVCEQYCL
SEEELEDIPQFTLESANQGNGKNKGSK

Codon-optimized NHI 217 sequence:

ATGAGCCTGAGTCACGATTTTAAATCATTACAAAGGACTCTAATATGAATGACAATG
AAAAATATCTGTTTGATCTTAACGGGTACATTATCATTGAGACGTAAGTGGACAGAA
CGAGATAGATGAAGCTAATAAAGCTATCGATCACTATTCGAATGAGATCAAAGAACGT
AAAGGAGAGTTCTCCCTCTCGGGTGGTTCTGGCGAATTATTGGGAGATACCGGGC
GCGGCGACCTTGAGGGTGCTCTTTCTGGCCAAAACCATCGTGATCCGTTTCG
CAAAATGCTGGCACATAGCAAGATTACCCCTTATTTATCAGAATTACTGAACCCTGGC
TTCAGGTTAGACCACAACATGGGGATTATTACGATGCGCAAAGGTGCCGAAGGACA
TATCCTGCATGGGTCCAGTGGCCCTGACTTTGATCCACATCAGTATTACATCTTTAA

AATGGCAAGATGCACAATGGTTTAACCGTGGTAGCGTGGCAACTTTCGGATATTCGT
GACGGCGACGGTGGTTTGGCTGTTGTACCTGGGTCTCATAAAGGTAACATAGCACC
ACCGCGAAGTCTAACGCGCTACGAGGAATATAAAGATGTTGTGAAACAAATTACTTG
CAACGCGGGGCGACGTCATAATCTTCACAGAAGCGCTCACTCACGGTACCCTGCCG
TGGAAGCGGATCATCAGAGACGAACTGCATTGTTTCGGTATTCTCCAGGAAACATT
GCCTACGCTAAAAATTATGATTGGCCGGATTCCATGATGGAGGGCTTGTCCCCTGAA
GAACGAGCCGTGATGGAACCACCGTATCATATCCGTCTGAACCGGCCGATAATTGAT
TCTTGA

NHI 217 protein sequence:

MSLSHDFKIITKDSNMNDNEKYLFDLNGYIIIRDVLDQNEIDEANKAIDHYSNEIKERKGE
FSLSGSGSELLGDTGRGDLEGLSWPKPHRDPFRKMLAHSKITPYLSELLNPGFRLDH
NMGIITMRKGAEGHILHGSSGPDFDPHQYYIFKNGKMHNGLTVVAWQLSDIRDGDGGL
AVVPGSHKGNIAPPRSLTRYEEYKDVVKQITCNAGDVIIFTEALTHGTLPWKADHQRRT
ALFRYSPGNIAYAKNYDWPDSMMEGLSPEERAVMEPPYHIRLNRPIIDS

Codon-optimized NHI 218 sequence:

ATGTCCGACAAACAGCAAAAGTCAGATTGGGATGTATCAAAATTTTCGTACATATTGTG
CGGAAGGGGAGACCAAAGCCCTGAATCTGGGCAATCGGGGACCAATACGTTTTAC
ACCGGATGGTCTCCTAGACCCGAATATTCTTAACGCATATGAGCAATTCGGATTTTAT
GTCTTTGAAAATGTCATAGATCAAGATGAACTGGAAGAATGCCGTGGTGAAGTTGAC
CAACTATTGGAAAGAGCGCCATGGCCTCATAGAGATAGCAAAGTCGATAGGAACGG
TAAACCGGCTTTGGGCCCAGAATTTGCGGTTAATCCTTGGGTATGACCAAACCGCT
GTCCGATCCTTTGGGGGGCACGCGGGAAAACAATGGGCGTCACCCGGTGAAGAT
GTTCAACCGGAACCGGGTGCTGACAGCCCGGCCACGTGCCGTATCTTGTAGCA
GGATGTCTACAACTTCTGCCTAGTTTTCTTCGGATCTTCGGCCATCCGGATATTCTC
AGAGTAGCAGAAGCTATTAACGGACCTGACTTCACCCCATTTACCGAAGTGCTTTTT
GTAAAAGAACCGGAGATTGGTCCATCTGTGAGTTGGCACCAGGATGGACAGATTCA
TTGGGATCATCCCCAATGGGATTCTGAATATTCACGGCTATAATGTCCAAGTACAGCT
GTATGGTTCCACCCCAGCCAACGGGGTATGGGTTTTGCCTGGTACCCACCGCGAA
GGGAAGGTCGATATTGCCGAACGTGTGTCTAAAAACGATGACAGTGATTGGTTTGA
GGACGCGGTACCTCTGGTCTGTAAACCGGGTGACGCTTTCATCACGAACCGTCAG
ACTGTGCATGGGAGTTTCCCTAATACCTCCAAGGACCGTCGTATCACGGTAAACTTC
GGGTTCCACCGTTATAGCAGTGTTATAGGACAGACGGGCAAACCTAAGTGGCGAGGG
CAATCAGTATACGGAAGACTATATCCGGGAGCGCTGCCGGTGTATACCTTTGGCTAT
TGATGCCCGTAGCCAAAGATTCCCTAATGAAGAGCGCTATACGTATCAGCCCTTTCG
CGGTTTAGAAGACCAATATCGTTATGGTCCAGATACGATTTCGTGACATTCTCACAGAT
TACAACTTGCGAGATCTGGGCATCTGA

NHI 218 protein sequence:

MSDKQQKSDWDVSKFRITYCAEGETKALNLGNRGPIRFTPDGLLDPNILNAYEQFGFYV
FENVIDQDELEECRGEVDQLLERAPWPHRDSKVDRNGKPALGPEFAVNPWVMTKPLS
DPLGGTRENNGRHPVKMFEPEPGADSPAHPVYLVAGCLQLLPSFLRIFGHPDILRVAEA

INGPDFTFPTEVLVKEPEIGPSVSWHQDGGQIHWDHPQWDSNIHGYNVQVQLYGSTPA
NGVWVLPGTHREGKVDIAERVSKNDDSDWFEDAVPLVCKPGDAFITNRQTVHGSFPN
TSKDRRITVNFGFHRYSSVIGQTGKLSGEGNQYTEDIYIRERCRCIPLAIDARSQRFNE
ERYTYQPFAGLEDQYRYGPDITIRDILTDYNLRDLGI

Codon-optimized NHI 219 sequence:

ATGGCTGTAGTTGATCATGCCCCGCGGCTACCGCAATTCGACAATCTTGACTTTGAC
AATTTCCACGACGAGGAGCTGCCACGCCGTTTGGCCGCTGGCAACGGTGAGCTG
GCTGCAGACATCGCAGCGGGCGCATTAGTTAAAAGCTTTGCTTGGCGCCTGAGTG
ATGGGAGAAGCGTTACGTATGTGGTAGATGATAATGGCATTGTAATACGGCCTGGTG
ATGATGCTGAACTGGTAGTTGAGGTTTCAGAGCAGGATTGGTTAACTATGTGGCAG
AGTTATGGACTTGGGTAGGCCTTATGTACGCGGAAGCGGCAGAGATGGTACGTGGA
GATTCGCGCCTGTTTGAAGATTTGAACCGGTTCTGTTGGCCATGTATCACGGTCG
ACCTCTGTATGAGGGATGGGAACCGACAGTCGATCTTTCTCGCTCTTTTACGCTAGA
TGATGATCACGAAGAAATGAGTCGATTCTTGATGAGGCAGGGTTTCTACATATACG
GGGCGTATTCAGTCCTGAAGAGATTGATGCACTGCGTGCTGAGGTTGAACGCCAAC
GTTCCGAAGCTACCCCCGATGACCACAGGTCATGGTGGGCGACAACTCTGACGG
AGATGAAGTTTGTGTCGGGTGACACATATGGATGACCGTAGTGAATTGATGCTGAG
TCTGACCGATGATCGGCGTCTTGATGCCATTGGTGCCTTGGAGGTGACGATTTCTG
CTGCTTACCCGGAACGGGGGGATGGAGTGTCAGTTGTAATCAAATTACCGTCTATT
GTAGAGGGCTTAGCAGATCTCCCATGGCATCGTGACTGTGGTACCGGCGGGCACT
CCATTACTTGTCTACTGTCTCCATTGGTATCCAGCTGGACGAATGTACGGAAGCAA
ACGGTCAGTTACATTATCTTGCAAGGATCAAATAAATCTAGCAATAGAGCTCGCGAGG
TACGAGATGGTTGGCCGACCGTCGCTATTTCTGCAAGTCCCGGCGACGTGACAGTT
CACTACGGTCATACCATGCATGGCGCCCCGCGCCTACAGGAGCCCCGGTCAGCG
GGGGTCGGCGTACCATTATGTTGGCTACCATAAACCCATCTGGGCGGAGTTTATAC
CGCCGGGCCAAGGTTATAACGACATTCTGTTTAAGGATGGAGGCCGTATCTTAAGC
CCGGAGGAATTTACGGAACAGTCCAGTTGA

NHI 219 protein sequence:

MAVVDHAPRLPQFDNLDFDNFHDDEELPRRLAAGNGELAADIAAGALVKSFAWRLSDGR
SVTYVVDDNGIVIRPGDDAELVVEVSEQDWFNYVAELWTWVGLMYAEAAEMVRGDFG
LFEDFEPVLLAMYHGRPLYEGWEPTVDLSRSFTLDDDHEEMSRFLDEAGFLHIRGVFS
PEEIDALRAEVERQRSEATPDDHRSWWATNSDGDEVCCRVTMDDRSELMMLSLTDDR
RLDAIGALGGDDFAAYPERGDGVSVVIKLPSIVEGLADLPWHRDCGTGGHSITCPTVSI
GIQLDECTEANGQLHYLAGSNKSSNRAREVRDGPVVAISASPGDVTVHYGHTMHGA
PPPTGAPVSGGRRTIYVGYHKPIWAEFIPPGQGYNDILFKDGGRIISPFEFQEQSS

Codon-optimized NHI 220 sequence:

ATGGCTGTAGTGATCTCCCCAACACCTGACCCGATCACATCTTCGTCGCCTTTCTCA
ATCTCTAAATTCATTGCTAAACTGAAGCTTCGCCGCTCTAGCAAGCAAAAAGACCAA
ATCCAGGATGAGATTTTCGCATGATCCGCGTCCTCCTCTGATACCTAGCCCTCTTCCA
TTGGTTCTTCCAGAGCACCAAGCATACTTTGTGTCAATTGGGCTGGACTACCCTGAC

CTTTCCTCGTCCTCGTCCCATCCTAGCCATCATTATCACCGACGAGTAACCGCGA
ACCGGAACCGGAACCTGAACCAGAACCTGGCCCGCATCCCCTGCAATCAGCCTAT
CTTACCCTGTTCTTGGCAGCCCAACAATTCTTTGACCTTGAAGATGAAAAAACAG
CAGTGGAAGCACCGGTTGGGCACAGAGGAAGGTTGGACCAAATACGCGGTGAAA
AAGAATTTATTACACTGCGCACCTTGCCTATACACCAGATATCCTCAAAGAACCGG
CGAAACGTTATTGGGATTTAATGGGTAGTCACTTGTGTAGTACGCTGGGACGGATT
CTATGTCAGTCGGCCTCCCAGGCGAGGGCGAAGGCGATGGTCTGAAGGGCTTCGT
TGGGCCCTGCAATGTCATGGGCGAGAAAGAAGAGGACAAAACCTCAACGATGCTG
CGAATCTTCAGATACGAATCTTGGGAACGAAAAATTGTGGCGGAAGCTCATGCCGA
TCTGGGCCTCCTTTCATGTGTAGTGGGGGATGTTCCAGGTCTCGAAGTATGGAATG
GACAACAGTTCGTAGATATCGAACGACTTTACGAACGTGAGCGTATAAGGAAAGGTA
CCAATGGTGTGGGCGGGATTAACGGCGATCGAGCGGGCTTTGGAGGGAAAGCCA
CACTCCTGTCGGGTCGCCAGCTTGAACGCTTTTCAAATTTTCGCTATCCTGCTGGC
GGTCATCGTGTTGTGTCTACGGGCTTCCCAGCCCGCCATCTTCCAAAAGTCTGTC
GGCTCATACCCCAGTGCCTAACGAGGCCGAACGCAAAGAGAAAAAATATCGGCATA
GCATTGTGTTTGTCTGCGTGCCACGAACCGACGCCGGTAAACAGCAGTACATTG
GAGACAGAAATCACAGGTAAGTGGGCAGAACCAATACGCGATGTAACGGCAGGTG
AACTTTATGAACGTATTCGTAACCAGCACTTTAATGTGAATATAGGTCATGAGGAACG
TGAAGCTCAACGGCGACGCGTCCGTGAAGCTAAAGCAAATGGAAAGTCCGTTGAA
GGAACCTGGCTAA

NHI 220 protein sequence:

MAVVISPTPDITSSSPFSISKFIAKLKLRSSKQKDQIQDEISHDPRPPLIPSPLPLVLPE
HQHTLCQLGWTTLTFPSSSSPHSHSSPTSNREPEPEPEPEPGPHPLQSAYLTLFLAA
QQFFDLEDEKKQQWKHRLGTEEGWTKIRGEKEFITLRTLAYTPDILKEPAKRYWDLMG
SHLCSTLGRISMSVGLPGEGEGLKGFVGPCNVMGEKEEDKTSTMLRIFRYESWER
KIVAEAHADLGLLSCVVGDVPGLEVWNGQQFVDIERLYERERIRKGTNGVGGINGDRA
GFGGKATLLSGRQLERFSNFRYPAGGHRVVSYGLPSPSSKSLSAHTPVPNEAERKE
KKYRHSIVFVLRAHEPTPVNSSTLETEITGKWAEPIRDVTAGELYERIRNQHFNVNIGHE
EREAQRRRVREAKANGKSVEGTG

Codon-optimized NHI 221 sequence:

ATGAATTCGGCAGCCACCATAAAGGCGTGGAATTCCTCACTGAACGTTTCGGCCA
TCTTGCGGACTTCCCTGGGAGTGCTTCGGATACGGCGTTGCTGAAACAACATTTAA
ACGATCACGGTTATGTCTTACTGAGAGATGTTTTAGATGTTGATCTGGTCACTGCCG
CACGGAGCGAAGTATTCGGTAGGCTCGCGGAGCAAGGTGAAATAGCAGATCCACC
TGGCGGAGGTGTTGCAACTGGTACATCTCGCCGTATCGATCCGGCGGATGATGATG
GCGCCTTTTGGACATCGGTTTCTGAAGGTGAAGCACTGCGTCGTCTGACACATGGT
CCAGGCATGATGGCGATTATGCAAGCCACCCTGGAAGAACCAGCGCGTGCCTTTG
ACCTGATTTATTTGCGCCCCACCTGTGTGGGCAAAGCGACCGCGATGCATTATGAC
TATCCTTTTTTTGCAGGAAGCGCCAGCCGATGTTCAACGCATGGATTCCGGTTGG
TGATGTTCCAATTCAGGACGGGCCGTTAGTACTGGTTGAAGGTTCTCACCGGTTCG
ATGATCTTATTGCACCTTTTCTTAGCATCGACTATGCTGCAGATCGCAGCGATAGTGT

AGTTCAGCAGGCCGCGTATGCCGGATCCGGCCAGTCGGATCCAATTCACCTGATTG
AGGATCGAGACACGCGCCTCTTGACAGCGGACTTTTGAACGGGCGACATGTTGAT
TATTAGTATGTTTCATCCTGCATGGCAGTCTAGATAACGTGTCTCCCACTAACCGAGTT
CGTCTCTCGTGTGACGTTCCGGTATCAGCCCGCTTCCCAACCGGCAGATGATGGTC
GTTATTTTGGGGCCATTCTTCAGGCTCAAAAGGTGGTGGATATGGAGATATGCGTG
GGGCCAGCCCCTGAATTAA

NHI 221 protein sequence:

MNSAATIQGVEIPTERFGHLADFPGSASDTALLKQHLNDHGYVLLRDVLDVDLVTAARS
EVFGRLAEQGEIADPPGGGVATGTSRRIDPADDGAFWTSVSEGEALRRLTHGPGMM
AIMQATLEEPARAFDLIYLRPTCVGKATAMHYDYPFFAGSAQPMFNAWIPVGDVPIQDG
PLVLVEGSHRFDDLIAPFLSIDYAADRSDSVVQQAAYAGSGQSDPIHLIEDRDTRLTAD
FRTGDMLIISMFILHGSLDNVSPTNRVRLSCDVRYQPASQPADDGRYFGAIPSGSKGG
GYGDMRGAQPLN

Codon-optimized NHI 222 sequence:

ATGACGGCTGCTGCCGTGTTAGACCGCGATGCCATTGCGCGCACCAACCATGGCGC
ACGAGCCTTTTGTGTGGGGGTTAGTAGCGGACAGTCTCCCTGGAACGTCCGCGCG
AAGACGTCTCGCGGCAGAAATCCCCTCAGAAGGTTTTGTAAGTGAACGTGCTG
GCGGGACTACGGCAGGGAAAGGTTACCATACTTGAACTTGACACTGGTGCAGGG
CGGAGTTCCGCTTCCGGAAGGTATGCGTAGGCTCACACCGCTCTGGCAGGGCCTA
GTTACCAGCCTGTTAGATGATGGGTACCGAACCGCAGTGGCTGAAGCGACAGGCC
GTTCACTTGATGGTTGCGCTCAAGAGATTCGCGCTGTCCGGTATGGATCAGGCTCG
TGGATCGATCCTCATAACGGATCGCGCAGATAAAATAGTAACGCAGACCTGGTACTTT
AATGAAGGCTGGCGCCCTGAGTGGTCCGGCGCCCTGCGTATACTGCGTTCGCCAG
CAGTTGACGATGTGGCTGCTGAAGTTCTTCCAGGTTTAAACAGACTCAGTTTTACTGG
TCCCTTCACCGGACTCTTGGCACACCGTGATGCCAGTGGCCGATGGTGCAGAACA
GGATCGGCGAGTTCTGCTCGTGCAATTTGTTTCTCAGGAGAGAACAACCTTGGTAA

NHI 222 protein sequence:

MTAAAVLDRDAIARTTMAHEPFVWGLVADSLPGTSARRRLAAEFPSSEGFVLTERAGGT
TAGKGYHTWNLTLVQGGVPLPEGMRRLTPLWQGLVTSLLDDGYRTAVAEATGRSLDG
CAQEIRAVRYGSGSWIDPHTDRADKIVTQTWYFNEGWRPEWSGALRILRSPAVDDVAA
EVLPLGLTDSVLLVPSPDSWHTVMPVADGAEQDRRVLLVHFVSQERTTW

Codon-optimized NHI 223 sequence:

ATGGACCAGAAGGAATACCTGCTGAACGATCAGCAAATGAAAAAATTTATTCGGGAT
GGTTATGTCACTGTGCAGACAGATCTGAGCGTAGAATTCCATGCAGCTGTTTTTCAG
CAGACCGAAAATGTCTTCGAAATGGAAGGCAATCCTGGAAATAACCTGCTGCCTCG
CATAACCAATGATACAGGAAGTGTTTGATGATCGGTGGGTCAACGGCGCTCTTACGT
CAGTACTTGGGAAAGATTATTACCTTCAACCTCATAGACATTGCCATTACAATCCGCC
CAATTCTAATGGTCAGAATCTACACCAGGATGGCGGCAAACGCTGGAGCCACTACA
CTAGATGGCTGCTTGCACTTTATTATCCACAGGAAACCCCTGAAGAGCTGGGTCCG

ACAGGCATTATACCGAAATCGCACTATTATGGTAATCGCGACTCTATTAATGACGAAG
ATGAAATACCACTATGTGGAAAGGCTGGCACTGTGACTATTACAAATTATGACCTTTG
GCATAGGGCTATGCCAAACAGTTCGAGCAAAAAACGTTATATGATGAAGTTTCTGTT
TGCGCGTATGTCAGAACCTGAGGTGCCCTCCTGGAACAATCAAGACACAGAATGG
GCGTCTGCCGACATTTCCCAGTCAGACGCCGAGAATGACGTCAAAATGTTCCATCA
AATTTGGAAGTGGCATTGCGGCCAAACGAATGGTAACGGTCACTGCCCATCGTCTG
TCACCATGCCCCGACTCGACCAAGATTGCGCGAGCTTGTTGAGGTTCTGGGCCAGGA
TAGCGAACCCTAAATGCATCGATATCGCATACTGCTATTTCCGAATTCGGAGCGAATGC
TGTGCCGGCTCTGGTGAAGCAGCTGGAAAGCGAGAGTGAGGACATTCGTAGATAT
GCTTTCTGTGCTTTGTCGGCGATTGGAAAACCGGCGGTATCCGCGCTCATCACTGC
CACACAGCATCCTGATTGGTGGGTTCGGGCGTCCGGCAGCCGAAACCCTGGGGAA
CATCGGGAATTCGGCACAGGCAGCTGTTCCATCCCTGATTGCGCGCATTACAGGATG
AGAGTGAGCAAGTTCGACAGCTCGCTGTGCGAGGCCCTCGGGACCATTGGACAACA
TGATAGTACGGCTGTTCCGGACCTGGCACTCTCCCTGCAGGATGAAAACGAAAGG
GTTCTGAAGAAATAGTGTACTGGCACTCGCCAGACTGGGTCGCTATGCCACCAAGC
TGTGAATGATCTGCAATGGTTCTGTCTGATGATAATAGATATGTCCGCGGTGATGCA
GTGCACGCGTTGCGTAGAATCGATACGCCGGAAGCTCGCGAAGTTCTGATTCAGTT
CCTCTTAAAGAGCCGTTGGTGTCCCTTAACAAGCAAAGAATCAGGATACTAA

NHI 223 protein sequence:

MDQKEYLLNDQQMKKFIRDGYVTVQTDLSVEFHAAVFQQTENVFEMEGNPGNNLLPR
IPMIQEVFDDRWNVNGALTSVLGKDYYLQPHRHCHYNPPNSNGQNLHQDGGKRWSHY
TRWLLALYYPQETPEELGPTGIIPKSHYYGNRDSINDEDEIPLCGKAGTVTITNYDLWHR
AMPNSSSKKRYMMKFLFARMSEPEVPSWNNQDTEWASADISQSDAENDVKMFHQIW
KWHCGQTNGNGHCPSSVTMPDSTKIRELVEVLGQDSEPKCIDIAYAISEFGANAVPALV
KQLESESEDIRRYAFCALSAIGKPAVSALITATQHPDWWVRASAAETLGNIGNSAQAAV
PSLIRALQDESEQVRQLAVEALGTIGQHDSTAVPDLALSLQDENERVRRNSVLALARLG
RYAHQAVNDLQMVLSDDNRYVRGDAVHALRRIDTPEAREVLIQFLLKSRWCPLTSKES
GY

Codon-optimized NHI 224 sequence:

ATGAACCATCCTGTTGATAATCCCGAAATTCATCGTGAATCCTGGAGGAGAGACGGT
TATGCGCTGCTGCGCGGCGCGGTGCCCGGCGATTGGCTGGATGGTCTTCGTGCG
ACCTTCGATGCGGGCATTCTTTCCTCAGATCAGTGGCCTGTCCCTCGGGGACACCAC
CTGGAATCATGCTCTTGTTGACCTGGATGCACGGGTTCAGGCAGTTTGTAGGCTGC
CTGAAGTATTAGTTGTCGTTGGGGCCTTAATTGGAGAACGTTTCTTTTATCACAGGT
AGAGGGCCGTGAGCCGTTAGGGGGGCCATGGTCATCAAACATTGCACAGAGATTTAT
CAGCACAACGCCCGGGTGATACTGTCAATGCCCTTATTTACCTGGACGATTATGGGC
CGGATAATGGCGCTACGCGTATTGTCTCGGGCACCCACCGGCCGGCGCCGGATAG
CCCTCCGTTTGATGACCTGGATGAGAGTGGTAGCGTTCAGTTGAAGGGTCAGGCC
GGTGATACTGGTTTTTCGATGCGGACTTGGTTCATGCTGGTAGTCTGAATCCGTCT
GGGAGTCGTCGACGTACCCTGTTGTTAGGGTATTTTGCTGAATCCCTGCATGCCCT

GCATCTTCAAACCGCACACTTGCGTAATGTTAGGATGGATACCCGGGAACGTTTCG
ATCCGGTATAA

NHI 224 protein sequence:

MNHPVDNPEIHRESWRRDGYALLRGAVPGDWLDGLRATFDAGILSSDQWPVPRGTT
WNHALVDLDARVQAVCRLPEVLVVVGALIGERFFLSQVEGREPLGGHGHQTLHRDLA
QRPGDTVNALIYLLDDYGPDNGATRIVSGTHRPAPDSPPFDDLDESGSVQLKGQAGDIL
VFDADLVHAGSLNPSGSRRTLLLG YFAESLHALHLQTAHLRNVRMDTRERFDPV

Codon-optimized NHI 225 sequence:

ATGCAGGATTTTATCGAAGTCATCGATAACGCACTGCCTGCTGACTTTTGCGATCAG
TTCATCCAGACATTCGACCAGAGCCCGCATCTCGTTGAAGGCCGCACTGGCGCGG
GTGTCGAACAGGATAAAAAAGTTAGCCATGACTTATATCTGGACCAGCATCAGGAGT
ACCAGGCCGCCCTGCAACACATTCGCGCCACAACCGCACAACACGTCCTGTCTTAT
TTTAAGAAGTATCACTTCGGTCTGATCGCCCCTGTGGCGCTGAAGGTGAAAGATCC
CGGTTCCGGTAAAGTCGTCAATCTGACACATGAAACTTCCACGATGTCGCAAAAG
GTCAAGAAGAATCTTACATGAAACTCCTGTATAGGCTCGGCAGTATCCAGGCCAG
CGGTATCGAGCGGGAATTGGTAACTATAATTACTGGCATTGCGAAGTTTATCCACAG
CCACAGAGCGTGGAGGCCTTACATCGCACTTTATTGTTTATGTACTACCTGAATGAT
GTCTGAAGATGGCGGTACGACGGATTTTACTATCAGGATCGCCAGATTAAACCGAAA
AAAGGTCGTATGGTTATTGCGCCTGCTTATTTACCCATACCCATCGTGGTACTACG
CCGAAAAGCAATGACAAATATATATTAACCAGCTGGGTCTTAATGAATACAGGCTCCA
CGCTGTATAACGCTGA

NHI 225 protein sequence:

MQDFIEVIDNALPADFCQQFIQTFDQSPHLVEGRTGAGVEQDKKVS HDLYLDQHQEYQ
AALQHIRATTAQHVL SYFKKYHFGLIAPVALKV KDPGSGKV VNLTHENFHDVAKGQEE
YMKLLYRLGSIQAQRYRAGIGNYNYWHCEVYPQPQSVEALHRTLLFMYYLNDVEDGG
TTDFYYQDRQIKPKKGRMVIAPAYFTHTHRGTTPKSN DKYILTSWVLMNTGSTLYKR

Codon-optimized NHI 226 sequence:

ATGCTTATCAATGCTAATCAAGACGTAGCACCATATTTGGCTGACCCGCGTGTATTAG
GGACTGCTGAAGCGTTTTTTGGACCGTTTGTCCGTATCGCATCAGCAGGTAGTATTG
TGACCCACCCTGGCAATGCACGCGGGTATTGGCATGCAGATTGGCCATACAATCAG
ACCAACGCAAGTCGGATCCAAGTTCCTTATCCAGATCTGATGATGGTATTAAGCACC
CTGTGGATGTTGACGCCTTTCAATCAAGAAACGGGCGGGACGCTACTGGTACCAC
GGTCACATCGTTCGTCTAATAACCCGTCAGGCTCAAACGGCGTTGATCCGGATGCG
CCACATCCGGATGAAATCCACGCTACTGGCGCGAGTGGATCTGTGCTGGCTTATGA
CGCTCGTCTTTGGCATGCGGTGGCCCCTAATCGTTCGAGTGAGCCACCGAGCAGC
GCAGGGGGCCGCGTTGTGAATACCACGCACTCTCGCAGTTGGAGCGGCTCGGCG
CATGCAACAGTGCCGGTCGCGCCAACCTGGCCAAATGTAAGACTGGGAGCCCGTA
GCCCGAAATAA

NHI 226 protein sequence:

MLINANQDVAPYLADPRVLGTAEAFFGPFVRIASAGSIVTHPGNARGYWHADWPYNQT
NASRIQVPYPDLMMVLSTLWMLTPFNQETGGTLLVPRSHRSSNNPSGSNGVDPDAPH
PDEIHATGASGSVLAYDARLWHAVAPNRSSEPPSSAGGRVVNTTHSRSWSGSAHATV
PVAPTWPNVRLGARSPK

Codon-optimized NHI 227 sequence:

ATGGACGATGATGATGCCCCAAGTGTCGAAATCGCAGCACGTGCAACCCATTGGGT
AGTCGAGAGAGACTTTCTACCCGGAGCGCTGGCGCGGGCTGTGCGCGGACGGTA
TGATGACGTGTTTCAAGATGCGACGCGCGCGCGTGGTGAAAGGTTTTGCTGGGAT
TTGTGGCACGTTCTAAACAGTATACACTTCTACGGACACCCGCAGAAGACTTTTTT
GGCGAAGAATTACATGGTGCCTGGAGGAAATGCTTATGACATATGCTCGGGAGCG
TCTCGGATGTGCAAGCATGACCCCAATGTGGATGTCGTGCTACGTACATGGCATGC
GACAGGAAGTCCATGCGGATGTGCCACATGGCCCGTTTGCATTCGTACTGAGCCTG
ACTCGCGAATCAGGGGCAGACGGTGGTTTACCTTTAGCGGTGGCGAAACGCAAA
TCATGCGCCCAGAGAGACTTAACTATTGGCGCAATTTGATAGTAGTGAAGTTGTAG
AGCGTGCTCAGATCATGGAACTATTAGTCCACGCTTTAACCAGCTCGTTGTATTCTG
ATCCCCGTTTGCCCCATGGGGTCACGGAGGTTTTTGGCACACAAGATATTCGGGAT
GGCCGGCTGGTACTGCACGGGTGGTTCAAGGACCCGGAACCTTCATTCAGCGGTG
CATTATCTGAAGAGGATGCAGCGGAAACCCTTGAGACCGCCCTTGCGGAATTATAT
GCCCCGTTGGTTGAGCTCCCGCGTGCGTCTGGTATGGTGTGCGCCCGTATCACGA
TAACCCCTAGTGGGGATGTTGATGATCTGACTTGGACATGTGATACGTTGACGCCG
ATCCCTCTGCCGGAAGTCCAAAGCGAGACAGATATCCGCGACGCAATTATGCTTGA
TATCGCTAGTGCTTTACTCGAATTAAAGTTCCCTCATGAAACCGATCACCCATCGTAA

NHI 227 protein sequence:

MDDDDAPSVEIAARATHWVVERDFLPGALARAVRGRYDDVFEDATRARGERFCWDL
WHVPKQYTLLRTPAEDFFGEELHGALEEMLMTYARERLGCASMPMWMSCYVHGMR
QELHADVPHGPFVLSLTRESGADGGFTFSGGETQIMRPERLNYWRNFDSSSEVER
AQIMETISPRFNQLVFDPRLPHGVTVEVFGTQDIRDGRVLHGWFKDPEPSFSGALSEE
DAAETLETALAELYARLVLPASGMVCARITITPSGDVDDLTWTCDTLTPIPLPELPSET
DIRDAIMLDIASALLELKFPHETDHPS

Codon-optimized NHI 228 sequence:

ATGCCGGGAAGCCCTGGTATAAGTATACATACAGATGGCCAGCCGCATGGTTCAGAT
TTATTTGGGTATGAGGGGAGCTGCGCTCGATTGCTCCGCGTATTGTATTACTTAGAT
GAATTGACCCCGGATAGGGCCCCCTTTTCGCTTGATCCCGCGTACCCATCTGAGTTT
CCACGCTGATGCAAGTCCTTACGTGCGTTATAAATCTCATCCGGAAGAAATCACCCCT
GGTGGCGCCTGCGGGGACAGCGGTTATAGTTTCCTCGATGCTGCTGCATGGAAGT
CACGCAAATCGCGACGAACAGCCACGAGAATTGGTTCAATTAGGTTATCGGCCGAG
CTGGGCCGGCCCTATCCAGCCGGTTGATGAGTGGGCACCTGGCCTGGTCGAAGG
TGCGCCGGAAATAGCAAAACCGTATCTGAAACCGCTGAATACCACGCATGCTCAGT

GGGTTCAGCCGCATAAACCTGAGGGCATGCGAACAGAAGCAAAGGGAATAAACCC
GTCGCGGTGGGGGGACTAA

NHI 228 protein sequence:

MPGSPGISIHTDGQPHGSDLFGYEGSCARLLRVLYYLDLTPDRAPFRLIPRTHLSFHA
DASPYVRYKSHPEEITLVAPAGTAVIVSSMLLHGSHANRDEQPRELVQLGYRPSWAGPI
QPVDEWAPGLVEGAPEIAKPYLKPLNTTHAQWVQPHKPEGMRTEAKGINPSRWGD

Codon-optimized NHI 229 sequence:

ATGGCATCGCAGGACGCAGGTCCGCATGTCTGGCTGAGCGGAGAAGCTGACTGTG
ATTTGGACGCTTTTCGCGCACTGGTGGCGGAGAGGACCGATCCTCGCGATTACCC
GTATGCGGCGGAAGTCGCCCATGATGTACCACTGTATGATAGCGATCGCGTAAGAG
CAGCCGCGGATCCACGCGGTTTCGTACAGAATGGGTGCGTGCCCTTTTGGATGG
CCCAGGGATCGTCGTCTTGCGCGGAGCGTTTGCCGACCCTGATGTGGTGGATCGA
GCCTCTGGAGTATTTGGCGCCCTGATTGAACGGGAACGCGCGGAAGGGACTGCTC
GCGGTGACCACTTTGCAAAGCCCGGCGCAAACGATCGTCTGTGGTTCGGCACTTGA
TAAACCGCCCTTCATGCGCCTGAAGTGTTGCGCGATTATTATGCTAATGACATGTT
GGCCCTGGTGTGCGAAGCTTGGCTGGGTCCGGGTACCAGGTCACCTCACAAATT
AACGTCGTGAACCCAGGTGGGGCTGCGCAATCTGTCCATCGCGACTATCATTTAGG
TTTTCTGAGTGACGAACGCGCAGCTAGATATCCCGCCCATGTGCATAGGCTGTCTC
CCGTGTTGACCCTGCAAGGTGCCGTGCGCCATTGCGACATGCCTGTAGAGAGCGG
ACCGACACTTTATCTGCCTCATTCTCAGAAATACGAGCCCGGATACCTGGCCTGGC
GCCGCCCGGAATTTGCTGCTTATTTTGAAGAACACCACGTTCAATTACCACTTGCAA
AAGGGGATGCAGTTTTTTTTTAATCCTGCACTGTTTCATGCGGCAGGACACAATCGTA
CCACCGATGTGCGCAGGATGGCCAACTTACTTCAGATTAGCTCGGCATTTGGGAGA
GCTATGGAAACCGTAGATAGGGAGGCGATGGTAAATGCGTTATTCCCGGTCCTGCT
CCGGCGTCGCTCCGAGGGTGTGGGTGAAGCGTGGTTACGCCGTGTAGTCGCGGC
ATGCGCAGAGGGGTACCCGTTTCCAACGACCTGGATCTGGACCCGCCGCTCGCC
GGGTTAGCCCCTCCAAGTCAAGCAGACACTGTGTGGAGAGCTTTGACGGAGGAAT
GGAGTCCGGGGCGGCTCCGGCGTGAGCTGCGAGCAGGTGCCGAACGGCGTCGG
GCATAA

NHI 229 protein sequence:

MASQDAGPHVWLSGEADCDLDAFRALVAERTDPRDYPYAAEVAHDVPLYDSDRVRAA
ADPTAVRTEWVRALLDGPGIVVLRGAFADPDVVDRA SGVFGALIERERAEGTARGDHF
AKPGANDRLWSALDKTALHAPEVFADYYANDMLALVCEAWLPGYQVTSQINVVNPG
GAAQSVHRDYHLGFLSDERAARYPAHVHRLSPVLT LQGAVAHCDMPVESGPTLYLPHS
QKYEPGYLAWRRPEFAAYFEEHHVQLPLAKGDAVFFNPALFHAAGHNRTTDVRRMAN
LLQISSAFGRAMETVDREAMVNALFPVLLRRRSEGVGEAWLRRVVAACAEGYPFPTDL
DLDPPLAGLAPPSQADTVWRALTEEWSPGRLRREL RAGAERRRA

Codon-optimized NHI 230 sequence:

ATGTTGCGCATGGTACGCGATCGCCCGGGTCAGATCGGATGCGAACGTATCAGAAA
TATGACAAACCATCCGACCGATGCGGAACGCTTCCTGTTTGATAACCAACGGGTACC
TCGTATTAGAGGATTTTTTACCCCTGCGCTGGTGGATAGTCTGAACGCCGCCCTG
GAACGGACGATCGTACGCCGTCGTGACCCCGGTTTTTCATCGCACACATGAACCCG
CGTTCGCTGATAAACTTGATTCCGCAAATACTCGGGTAATGCATCTTCTTGCGGAGG
ATCCGCTGTTCCCTTGACATGCTGGACTACGACCCAATGATGGAGTATATCCATGGGT
TGTTCAATGAGATGCCTCACTTACATTCGACAGATGCTATCTATGAGATCGAACCGG
AGGAACATCACGGTAAGGGGTGGCACACTGACGGTATCCAAGATGGGTTCGGGA
TTTTCGCCCGCATATCCCACTGCTCCAGTTTAAAGTCGGATATTATCTGAGTGATATG
ACCGAACCTGGACAGGCAAACCTGACCCTCGTTCCGGGAAGTCATAAATCCCTCGT
GGAACCCGATTCTAAGAACGGCATGCCGGGCGCGGTACAGGTCTGTGGGCGTCC
GGGAACGGCTGTCCTGTTTCACAATGGAGTTTGGCATTCCGCTGGTCCGTTCACTC
GCCGCGGGGGCAAGAGGGTCATCCTGTACTATGGTTACGAACACCCTTGATGTTG
GCTTGCGCGGAACAGTGGCGTTATGACAGAGGATTTCTCAACAGCCTGACACCCC
GTCACCGCCGGTTTTTCCACGGCTTTGTTTTTGATCCCCCTGAATATCGTTGGGGCT
GA

NHI 230 protein sequence:

MLRMVRDRPGQIGCERIRNMTNHPTDAERFLFDTNGYLVLEDFLPPALVDSLNAALERT
IVRRRDPGFHRTHEPAFADKLD SANTRVMHLLAEDPLFLDMLDYDPMMEYIHGLFNEM
PHLHSTD AIYEIEPEEHHGKGWHTDGIQDGRDFRPHIPLLQFKVGYLSDMTEPGQA
NLTLVPGSHKSLVEPDSKNGMPGAVQVCGRPGTAVLFHNGVWHSAGPFTRRGKRVI
LYYGYEHPWMLACAEQWRYDRGFLNSLTPRHRRFFHGFVFDPEYRWG

Codon-optimized NHI 231 sequence:

ATGGATGAATATGATGGTTTAACTTCATTTACTACAGCTTCTAGCACTTTAGAGGAGAT
CTTGAACGAAAACACCATCACGCTGAACCAGCCTCTTTATGTAATGGGCCAGAGTA
CCCTATTATACGCAGCAAGTATGTATGATGAGGAACTTGCCCTTAAAATAATTAAAGC
GAAGAAACGAATTCCTGTTAACATTAACGCAATCGATAATCACGGTTATTCAGCATT
CACTACGCAGCCGATGCTGATATGTATCAGCTGGTCAAAACCTTACTGCAATGCGAT
GATATAGACGTCATGTTAAAAACCGAAGACCTAAATGTAGTAGGGCAGATTCAGTCA
GGCGGTGCAACTGCATTACATCTGGCGGTCAATTCGGGGAAATCTTAAACTAGTTAAG
GACTTAGTGGAGTATGAAAAGAAGCTGGTCATGGTGCCGGATCTCGATGGAAATTTA
CCTGTGCACCTGGGCCAACTGTACTCTGTGAATAGCGATCTGATTCAGTATCTGTTG
AGTATACATAACACGAAGCTGTACGAAGAGTCGAATCTGATTACCCTGCAGAAGACA
CAGCAGTTACAGGCCAATAAGCGATATCAGGACAGCCTAAAAGTACCTTCCAATCTG
TATACACCTCATACTTTCCCAAACGTATGGACCAGTGCCGAGTGTAATAAGATACTGG
AACAGGTCCTGAGCAAGACGAGTCAGACTGGCTGGAATACGTCGCGGCACACCTC
CTACCCGACCACAGATCTGCCGAGCTACAGACTGCCTTTTCTGGATCAGTGGATAC
AAGAATCACTACAAATTAGACTCTTTCCTAAGATCATTGATGAATTTAGCTGCACCC
CTCCACTGTTTTATCTTTCCGTGACCTGTTCTATGTAAATACGAAGCAGCAGGGCC
TCATGTGCAGAACGACTTAGCACTGCATTGCGACGGTTCATACTCAGCTTTAACAT
TTTACTGAATCGTAAAGAGGAATTTGAGGGAGGAGGCACATTCTTCGCCCCCACAA

AAACTACCATACATATCCAGCAGGGCGATGCACTTGTTTCATTTCAGGTAGGGTGGTGC
ACGGAGCTGCGTCAGTCGTGAGTGGTAAACGCCTTATAATAGTAGCCTTCTTAAATG
TTCGCCCATCACCTCCGTTATGA

NHI 231 protein sequence:

MDEYDGLTSFTTASSTLEEILNENTITLNQPLYVMGQSTLLYAASMYDEELALKIHKAKKRI
PVNINAIDNHGYSALHYAADADMYQLVKTLQCDIDVMLKTEDLNVVGGIIQSGGRTAL
HLAVIRGNLKLKDLVEYEKKLVMVPDLGPNLPHLGQLYSVNSDLIQYLLSIHNTKLYEE
SNLITLQKTQQLQANKRYQDSLKVPNLYTPHTFPNVWTSAECKILEQVLSKTSQTG
WNTSRHTSYPTTDLPSYRLPFLDQWQESLQIRLFPKIIDEFQLHPSTVLSFRDLFYVKY
EAAGPHVQNDLALHCDGSILSFNILLNRKEEFEGGGTFFAPTCTTIHIQQGDALVHSGR
VVHGAASVVSGKRLIIVAFNVRPSPPL

Codon-optimized NHI 232 sequence:

ATGCAAATAATACGTTACGCCCGTTTGCCCATTTGGCTATGATGCTCACGCGATCCAA
AATGAACTGACTGGAGGTAATCAGCAGTGGCTGCCTCATCTTAATACGTATCATTATT
CAGGCGGATGGCAGGTGTTACCTCTGAGGTCACCGGGAGGTAATCATCTGAACCC
TGTACCTGATCTCATGGATAATGCCAGCTTTCTTGACACGGATTATATGCAGTTATTT
CCTACTGTAAAAAAGCTGGTCAGTCTGCTTCAGTGTCCAGTCATGAGTGTACGTTTT
CTGAATTTACGTGCCGGTTCGGTCATTAAGGAGCACAGAGATCATCAACTTAGCTTC
GAAAAAGGGGAAGCCCGACTGCACTTTCCCGTTATCACGCATGCTGACGTGGAATT
TTATTCCGAAGATCAAAGAATCATTCTGAATGAAGGAGAATGCTGGTATCTGAATGC
GAACCTGCCTCATCGGGTGAGTAATAATTCTAGTATTGATCGTATTCATCTGGTGATC
GACTGCAAAGTAAACCCCTGGCTGACCGCGATGGTGCAGGGGGGCCCCCAGATC
GCAATCAAGGAAGAAGAGGAGGATCTGTCTCATTACCCGGCCATGATCGAGAATTT
GCGCCAGCAGGATAATGAGGCGGCCCGTCAGTTAATAATTAACTGGAACAAAAAC
TAAAAGATGCAGAACGTTACGTCCAGCAGCCGCGCAAGGTAA

NHI 232 protein sequence:

MQIIRYARLPIGYDAHAIQNELTGGNQQWLPHLNTYHYSGGWQVLPLRSPGGNHLNPV
PDLMDNASFLDTDYMLFPTVKKLVSLLQCPVMSVRFLNLRAGSVIKEHRDHQLSFEK
GEARLHFPVITHADVEFYSEDQRIILNEGECWYLNANLPHRVSNSSIDRIHLVIDCKVN
PWLTAMVQGAPQIAIKEEEEDLSHYPMIENLRQQDNEAARQLIKLEQKLKDAERSRPA
AAQG

Codon-optimized NHI 233 sequence:

ATGGAAGAGCGTTGGGCCGTAGCGGGTGCGGTGGATGAAGATGGGTGCTTAAAGG
CCGATCAAATCAAAAGTTGGAGAGAACAGGGGTTTGCTTTTGTGTCAGGCCTGTTA
CCGAAAGCCCTTCTGTCAGAACTGCGCAGCGACGCTCAATCGTTCTATCCAAATCC
AGAAGATGCAAACGTAGCTGATTATAACCACTTCGGGTCTAACGGACACTTTGTGTT
TCCATCCGATTATAGTTCTGCAAACAGTGTGTCTCTCCATCCAAACCTGCTGCGAGC
CGTTGCAGATCTCTTGGGCGTCGAAATAGCCGAATTGCGTCTGACACAAAGCGATC
TTTGGCCCAAATATGGAGGCAATGGACTGGAAGATAGGAGCGAAAACAGAGATCAG

CGCATCCACTGCGATTATCCAAACCATACTTTGACGCATCCGCCAGAATGGGACCG
GCCTGAGGCAGTTGAAATCATTATTTACTTGTGAGATGAAAGCGAGTGTCAAGGTAG
CACCGCTGTTGTGCCAAAACCTGGGATCTGACGATCCGGCCTATCGGTGGCCGATAG
TAAACACCCCCGGGGGTAGCAGGTAGCAATTACGTGATGACAAGGCCAAAAGCCGA
GCGTTATATGAGAGAGCTCAACCCAGATGCAGCTGATTTGAGACAAAAGGAATTATA
TGAACGCGAAGTGAAAACCCGCTATCAGATGGGTGATGTTATTTTCTATCGACACGA
TACCTGGCATCGCGGCACGCCTGTAGCCCAGGGTGCCCTTCGGCTGGTCCAGAAC
ATGACATTTAAAAAGGCTACATCGGATTGGGTAAAGTGTTTTGCATTCGGGTTGGGCC
TGGAGTCTGTACCGTGCCGGCCACGGCCCAGAAAACTGATTGCGGAACTGAGTC
CAGATCAGCGTACCGTTCTGGGCTTTCCGCCACCAGGCCATTTTTACTGGACCGAA
CAAACCTCTGGAAGCTGTAGAGGCTCGATACAAACCATTCTGGGATTGATATGACTCCA
TACCATCAGGCCGTTAATCAGACGTAA

NHI 233 protein sequence:

MEERWAVAGAVDEDGCLKADQIKSWREQGXAFVSGLLPKALLSELRSDAQSFYPNPE
DANVADYNHFGSNGHFVFPSPDYSSANSVSLHPNLLRAVADLLGVEIAELRLTQSDLWP
KYGGNGLEDSENDRDQRIHCDYPNHTLTHPPEWDRPEAVEIIIIYLSDESECQGSTAVVP
KLGSDDPAYRWPIVNTPGVAGSNYVDDKAKAERYMRELNPDAAFRQKELYEREVKT
RYQMGDVIFYRHDTWHRGTPVAQGALRLVQNMTEFKKATSDWVSVLHSGWAWSLYRA
GHGPEKLI AELSPDQRTVLGFPPPGHXYWTEQTLEAVEARYKPFIDMTPYHQAVNQ

Codon-optimized NHI 234 sequence:

ATGGATCAGCTGATCAGCGATAGTCTCAAGGAAACTGATGAGGCTAGCCTGAGAGC
AAAATACTCAGAGCAGGATGAGTTCGTTATCGTTGATGAATTCCTGCCGTTGGAAGT
GCTGAGCCGCTGGGAACTTGAGCTTGCACGTCTGAAGCCACATGTGCATCGTAACT
ATTTGCCGGGCCACAAAAAAGGCGGTAGCGTCGCATATCAGACAATGAATGCCCTC
GCGCCATCTATTACTCGCGTTTACCATTCTGCACCCCTGCTGGGCTTTCTGCGACG
GGTTGTAGGTGCGGAGGTAAATGAATGTCCCTCAAGTGATCCCCATCGCTGTGCAC
TTTATGCATATACTGAGGAAGGGGATCACATGGGGTGGCACTATGATACGAGCTATTA
TAAAGATCGTAGATGGACCGTTCTGGTTCGGTTTTAAAGACGAAAGTAGTAGCCACCT
GCTTTGTCTCTTGCATACCCGTAACCTCCGACCGTGATGTAGTAAATTTAGAGCTCCA
GATCAAACCAGGCACTCTGGTCATTTTCAATGGCGATAAAGTGTATCATGCAGTTAG
CCCGCTGAAGAAAGGGGAAAGCCGTTATATTGTCAGCATGCAGTACGTTACCACGG
GGAATGAATCCTTTTATGCGTTTTATCAGCAACATGAAAGACGCCGTTGCCTACTT
TGGCCTGAGAGGCGTTTTTCGGCAAAGGTCGACTGCGTTGA

NHI 234 protein sequence:

MDQLISDSLKETDEASLRAKYSEQDEFVIVDEFLEPLVLSRWELELARLKPHVHRNYLP
GHKKGGSVAYQTMNALAPSITRVYHSAPLLGFLRRVGAEVNECPSSDPHRCALYAYT
EEGDHMGWHYDTSYYKDRRWTVLVGFKDESSSHLLCLLHTRNSDRDVVNLELQIKPG
TLVIFNGDKVYHAVSPLKKGESRYIVSMQYVTTGKMNPFMRFISNMKDAVAYFGLRGVF
GKGRLR

Codon-optimized NHI 235 sequence:

ATGTTTCAGGATATTACCCAGAGTATTGAAGCGTGTGCAGCCTACTATGGGGATGAG
GCAGAGCAAATGAAGCAGTATTTAATTCAGGGTCAGGAAAAAGCGCTGGCACTTCC
AAACCGCGGCCCAATACGCTTCGACGAGAACGGCGACATTCATCCTGATATCCTGA
AACTTATTCAAATATGGCTTCTATATTTTTGAAAACGCATTGGGCGAAGACGAGCT
TAATGATTTAAAAAATGATCTTGAACAGATGCGCGAGAATTTTCCGACCGAGATGGG
CGCAACTTTAGATGCACGCGGTCGTCCGGCATTGGGTTCCGGATCGTGAGGGCTTC
GCGCTGCAATGGGCTAAGCCTCTGTCAGATCCATTGGGCGGCACAGCAATGCTGA
ATGGTCGACATCAAGTGAATTTGTTTGAACCGAAGCCAGGTGAAGGCGCCCCCA
GGCCACGCCTCTGTATTTGGGCGGTAGCTTGCAATATAGCGACGCGTGTCTGCGGA
CATATGCCACCCAGGGCTGCTCAAAATGGCGGAAGCAGTCAACGGTAAAGACTTT
GTACCGTTCCACGAGGGACTGTTTATAAAGGATGCAGGTTTAGGTGCGAGTGTTTC
GTGGCATCAGGATGGTGATACCCACTGGGATTCTCCGGTTTTTGATGAAAATATCCA
TGGCTTTAATTTTATGGGACAGGTGTATGGGAGTACCGCAGTTAACGGGGTGTGGG
TGGTCCCCGGTAGTCACAAAATTTCGCGGGTGAATATTGTAGACCTTGTTAGCAAG
GCGGGTACAGAACGCCTGCCTGAGGCTATCCCTATTATTTGCAACCCGGGTGATGT
TGTCATTAAACAGCAGACAGATCCTCCATGGCAGCTTTGCAAACACTGGGTTGGAAA
AACGCGTCACTGTAAATTTTGGTTTCCATCCAATGGCATCAGTGTTAAACGTCCGGG
GAGCAGGTCTCCATGCCGAGGCAGCAACATTCGATGAGGCATTTATAAAACACCGG
AGTCGATTGATTGGCATAGCTATCGCAGCGCGTAAACAACGGTTTCCAGACGAACT
GCCCTATGCGTATCTGCCTATTGCAAATGATCCTGAAAAACCAGTGTGGAACCAAGA
AGTGAAAGAGAGCCTGAAAAATTACAATCTGATGGACCTGTCAATTTGA

NHI 235 protein sequence:

MFQDITQSIEACAAYYGDEAEQMKQYLIQGQEKALALPNRGPIRFDENGDIHPDILKTY
KYGFYIFENALGEDELNDLKNLEQMRENFPTEMGATLDARGRPALGSDREGFALQW
AKPLSDPLGGTAMLNGRHQVKLFEPKPGEGAPQATPLYLGGSLQYSDACLRTYAHPGL
LKMAEAVNGKDFVPFHEGLFIKDAGLGASVSWHQDGDTHWDSPVFDENIHGFNFMG
QVYGSTAVNGVWVVPGSHKNSRVNIVDLVSKAGTERLPEAPIICNPGDVVINSRQILHG
SFANTGLEKRVTVNFGFHPMASVLNVRGAGLHAEAATFDEAFIKHRSRLIGIAIAARKQR
FPDELPYAYLPIANDPEKPVWNQEVKESLKNYNLMDLSI

Codon-optimized NHI 236 sequence:

ATGTCGGCCGCTACCCTCGAGGAAGGCCCTCTGAGAGGTCCTTCGTTCTTTCTATC
TCGTACGGCCCTTCGCGCCCTGGCCTTGCTCATAGAGACCGCTACGGACCTGCG
AAGCCATATCCTCACGTTGTAATCGATGGATTTCTGGGACAGCCTCTGGCAACTGAA
CTGGCCGAACGGTTCCCGGGGGCGACCGAAGCAAGTTGGAAAAGGCGAGATCAC
GTAGAGCAGGCGGCACGTCTGGGTCAGCTTCAACGCAAAGCTTTTGAAGATGTTTC
CGGGTGTACTACGTCATCTGCTGTCAGAACTGAGTAGCATGTCGTTTATAGATTTCTT
AGAACTCTGACTGGTGTTTCAAGGTCTGATAGCCGACCCGCACTTTCGTGGGGCG
GGCTTGATCTGACGCTGAAAGGGGGCCATCTAGCCCTGCACGCCGATTTCAATC
GCGACAGGTTCCGTGCCTTGACAAGACGTCTGACCGTGCTCTATTATTTAAATCCG

GGTTGGGAAGCTGCCTGGGGCGGTGACTTGGAAGTGTGGTCGGCCGATCTGTCC
CGCTGTGAGGCCCCGTATAGCCCCACTGTTGGACCGCCTTGTAGTAATGGCCCATGG
AGATGATTTTTGGCATGGTCACCCGTCGCCTTTAGAATGCCCGGAAGGGAGGGGC
CGCGCTGCCGTCGCCGCCTATTTCTATACAGCAGAGGAGTCACCTCATGCACCCGA
GCCACATTCCGCACTGTGGGCGCCGCCACGTAGTTCGTAA

NHI 236 protein sequence:

MSAATLEEGPLRGPSFFLSRTALRALALAHDRYGPAKPYPHVVIDGFLGQPLATELAE
RFPGATEASWKRRDHVEQAARLGQLQRKAFEDVPGVLRHLLSELSSMSFIDFLETLTG
VQGLIADPHFRGAGLHLTLKGGHLALHADFNDRDRFRALTRRLTVLYYLNPGWEAAWGG
DLELWSADLSRCEARIAPLLDRLVVMAGDDFWHGHPSPLECPEGRGRAAVAAAYFYTA
EESPHAPEPHSALWAPPRSS

Codon-optimized NHI 237 sequence:

ATGGAACAGGCGACCCAGGTGGTTCGAGAGGCCAGCGGATGGGTTCGAAAACATAA
AGACAATAGACGTGGCCCGTCTGCTGCGTGAAGAACATCTGTTAACCAACGAGTTC
GCCAGCAACGTCCGTTCCGTTATATTGTTATTGATAATTTTCTGCATGCTGATCAAG
CAGAGCGGATATATAAAAGCTATCCTGCAATTGACGAAGCGTGGCAGAACGGCAAT
GATGTGCATACCAAAGGGAAATGGGGTACTCCGGACGTGGAAGGGACCGTAGCCG
GCGAATTTTACCGGGAAGTCAATAGTCCAGAATTCCGTACCCTATTAGGCCGAATAA
CAGGAATACCGCAGCTGCTCCAAGACCCAGACTTACAGGGCGCTGGACTTCATCA
GATCAGGGATGGTGGCTTCTTAAGCGTGCACATCGACTTTAACAGACTGCGCGGAT
CTAATCTAGATCGTCGCCTGAATCTGCTGGTTTATATGAACCCTGGCTGGAAAGAAG
AATGGGGTGGTTGCCTTGAGCTGTGGGACATGGAGAAAAACAAATGGTGTCCAG
TGTA CTGCCGTTGCTAAATCGTTGTGTGATTTTTGAACTAATGAAATTAGCTATCAT
GGACACCCTGTCCCCGTATCTAGCGGTGGCGTGACTCGTAAATCGCTCTCTGTGTA
TTATTATACTGACGGCCGCGACGATGTGGTAGCAGACAATCATTCCACGATATATGTA
AATACACAAGGAGCTGAAGGTT CAGCGAAGCTTCTAAAAAATGGAATAGCACATTCA
GCACGTAAACTGATAAAGAACTGGGTGGCGATGGAAATTTT TAG

NHI 237 protein sequence:

MEQATQVVERPADGFENIKTIDVARLLREEHLLTNEFAQQRPFYRIVIDNFLHADQAERI
YKSYPAIDEAWQNGNDVHTKGKWGTPDVEGT VAGEFYREVNSPEFRTLLGRITGIPQL
LQDPDLQGAGLHQIRDGGFLSVHIDFNRLRGSNLD RRLNLLVYMNP GWKEEWGGCLE
LWDMEKKQMVSSVPLLNRCVIFETNEISYHGHVPVSSGGVTRKSLSVYYYTDGRDD
VVADNHSTIYVNTQGAEGSAKLLKNGIAHSARKLIK KLGGDGNF

Codon-optimized NHI 238 sequence:

ATGACCACTACCGCTCGCTGCTCCCGAAGCACTAATACCGCTCCCGTAGAAACCGA
TCGCGATGGTAGTCGTACGATGACAGATACAGCTCAGAGCCAACAGGCCAGCGA
GGCAAACAGGAAGAAGCCCATCTTATCTTACCTGCAGCAGAAGTCGATGCCGGTAC
CCTGAGATTTGCAACGGCTGACGATACAGAGCGAGCCCTGTGTATTGGTGCCTTTA
ATGTTGCGGTTCCGGAGGACCTTGATGTGCGCGCAGGGTTGGATTTTGTAGGACA

TTTTACAAACCCGGCACCCGGAGATCCTGCGGATCGACATCGAGGTCACCCGGGAAC
ACGGCCATCCGGCAAGTAAACTTGGGTATGAGGATCGCCCAGATCAAGTCGAGCA
GCTGCAGTTAGAATCCTTTTTGTGGCAAGAATACTTGCCTGACGAGGTAAGTACCG
TCTTACGCCGAATGCGCCAGCTCACACTGGATGTGCTGTATGGGGTGTTTGACGCC
GCTTGGGTGCCAGAAGCAGATCGCGATCTGATCACCGGCGGGGGCCCCCTCAGGAC
ACCGGTCTTTGTTATACCACCGTAAATCATTATCGCGCAGGTCTTTCAGATCGCGCA
GGTATAGTAGAGCATTCCGATTGAGGTTTCATTACTACGATTTGTACGGACCAGCCC
GGCTACGAAATCCTGCATGAGGGACATTGGCTCCCGGTTTCAGAGAAAGGCCGGGGC
ATTTACCCGTGAACTTGGGTGATGCATTTAGGGTTCTTACCCGCAAAGTCCACGG
CCCGTGACAGCCGTGTACCATCGTGTGCCCCGAATTGCGACCCGATGCGGGCGTTTC
CGGATCGTTCATCGTTCAGTGTGTATATGGGTCCCCGCTATGACATGCCGTTATATCA
GTATGATACTGAAGGTCGCCTACGTGAGTATCAAAGTTTCAGAGATTTAGTGTTGA
AAAAGCAGGCAAAGTGGGGTATGACTTCCATTCACGCGTGTA

NHI 238 protein sequence:

MTTARCSRSTNTAPVETDRDGSRTMTDTAQSQQAQRGKQEEAHLILPAAEVDAGTL
RFATADDTERALCIGAFNVAVPEDLDVRAGLDFCRFTFYKPGTGDPADRRHRGHRHGH
ASKLGYEDRPDQVEQLQLESFLWQEYLPDEVTAFLRRMRQLTLDVLYGVFDAWVPE
ADRDITGGAPQDTGLCYTTVNHYRAGLSDRAGIVEHSDSGFITTICTDQPGYEILHEG
HWLPVRERPGHFTVNLGDAFRVLTRKLPRPVTAVYHRVPELRPDAGVPDRSSFTVYM
GPRYDMPYQYDTEGRLREYQSFRDFSVEKAGKLGDFHSRV

Codon-optimized NHI 239 sequence:

ATGCTGGTGGGTATCAGAGCGTTACTGCACAGCAGCCCCCATGAACAGTTCGAAAT
GCTGGACCCTGCACGCCGGATACGGACCCTGCTTCTCCATACCTTGTTACATACTA
CATCGTGTGATATCGATAAGGTTTACGCGGCAGCAAGTGTGTACCCATTTGCCATGA
GCAGTCTGCTTGCACTCCTCAGTCAACAGACACCGAACTTACAGACCCAAAAAGGG
CTGATTTTATTAGGTCTGCAGAATGATTTTCTCAGCCCAGATGGCAAATTGCCCGTTT
CCCCGCGTAGCGGTTATCTGGAACGTCTCAAGGAATTAGTGCCGGCCTTTTCGCGA
GTTTGGCTCTGTCATATGGGTCAGAAGTGAGTTTGAAGCTAACCGCGTGGCGAATC
CTGATGACGGCGATGGTGACACCGTTATTGCTGGTTCAACCGTAGACGGGAGCTTA
GGCCTGAAACATGTAAGCAAACGGAAAGCATCAGAGGACGCCACTTTACCGCTGAA
GAGGGCTAAACCCGCTGATGCGGAGGAGGAAGAAGAAGATCCGGAAGTGTCTG
AACCGTACATCGAATCGGGAACCATGTTGCGTCAAGGCGAGCTGGGGCGCCGAAT
ACCCGGAAGATATTCAGAACCTGATAGCCGATAGCGACATTCAAGTGACAAAACTT
ACTATAGCGCTTTTGGCTCTACTTCTTTATTACTAACGTTGAGAAGCAAAGTTATCAC
CGAGCTTTTTGTATGCGGATGCAATACAAATTTGTGAGTCTTTGCTACTGCGATGGAT
GCCGCCCGGTACGGGATACAGATAACGTTAATTGAGGATTGTCTGGGGTACCGCCG
AAGGGAGAGGCATGACGAGGCAATTCGTAACTAGTTGATATTATGGAATCGGATGT
TATGTCGAGTTCTAAAACAATAGATATGCTGCGTTCGCCGCCAGACGATGACAAACG
TGAAGATCATTGTTCCGATGAAGACGAAGAGAGCGCGGCAGCAACAGATGAACTG
GAAGCGGACTCTAGCGAAGAGGAAGAAGAAGAGGATTTTCATCCCCGACCCTGACG
CACATATTGAAAAGCAGCATTGGCCGCCTTCAGCTCGGCTCCCACTTAATGTGGAG

AATTTACGTTTCATTGAGTCTGCGTCGTAGCACAGCCGTTGCAGCTCGCAGTACTGT
GACATCCCAAGCTGTGGATCTTGAAGAAAAGACGGCCGAGACCACGGTGGATGAA
TACGAAAAAGCCATGCACGAAGCGAGAGATCGCGTTGCCCGTTATGATGCTCTTGC
CCAGCGCTCGACCCCCCTCTTTGGTATGAATAAGGAGAAAGAGTCAGCGGGTTCG
CGGATACTGTATGACCTGCTTCCGGCAGAATTAGCCGAAACGGTTTTTTGAGAACT
GAAACCAGAAGTCCATTGGCAGCCTGCTGAGGAAGCCGCCGGTCACGCCCTCAAC
CATGTTTTGATCCAGTTGTATAGGGGGGGGACCGATTTTATCTCAGAACACAGCGAT
AAGACTCTGGATATTGCCAAAGGGTCATCAATCGTAAATGTTAGCTTTGGCGCACAA
CGTACTATGCGTCTGCGATCTAAGCGTGGTGCAACCAGTCATCTGGCTGCACCTAG
TAAACATGGGGCCGCAACCTCTCCTTTAACGCGTACGACCCACCGCATCCCCTGC
CTCACAATAGCATGATCACAATGTCACTGGAACTAATGCTGAATATCTCCATGGTAT
TAACGCCGATAAACGTCCAGCGGTTGAATTTCCGCTGCAGAAAAAGAGTATGATG
GACAACGCATATCGTTAACATTCCGTCAGATCGAGACTTTTCTGTCAGCCGATGAGC
GACTTATTTGGGGTCAAGGCGCTACCGGCAAATCACGCGCCCTTGCCCGTGAAGT
GGTGAATGGGGACGGGAAAGAATCAGAAGAACTGGTCCGCGCGTTCCGGTGCTGA
GAATAAAGCAAGCCGTATAGAGTGGGAAGTGGTTTACGGACGCGGTTCCGGATGTCC
TGCATCTCAAATAA

NHI 239 protein sequence:

MLVGIRALLHSSPHEQFEMLDPARRIRTL LLHTLLHTTSCDIDKVYAAASVYPFAMSSLL
ALLSQQTPNLQTQKGLILLGLQNDFLSPDGKLPVSPRSGYLERL KELVPAFREFGSVIW
VRSEFEANRVANPDDGDGDTV IAGSTVDGSLGLKHVSKRKASEDATLPLKRAKPADAE
EEEEDEPELFLNRTSNREPCCVKASWGA EYPEDIQNLIADSDIQVTKTYYSAFGSTSLLL
TLRSKLITELFVCGCNTNLSVFATAMDAARYGIQITLIEDCLGYRRRERHDEAIRKLVDIM
ESDVMSSSKTIDMLRSPDDDKREDHCSDEDEESAAATDELEADSSEEEEEEDFIPDP
DAHIEKQHWPPSARLPLNVENLRSLSLRRSTAVAARSTVTSQAVDLEEKTAETTVDEYE
KAMHEARDRVARYDALAQRSTPLFGMNKEKESAGSRILYDLLPAELAETVFEKLKPEVH
WQPAEEAAGHALNHVLIQLYRGGTDFISEHSDKTLDIAGSSIVNVSF GAQRTMRLRSK
RGATSHLAAPSKHGAATSPLTRTTHRIPLPHNSMITMSLETNAEYLHGINADKRPAVEFS
AAEKEYDGQRISLTFRQIETFLSADERLIWGQGATGKSRLAREVVNGDGK ESEELVRA
FGAENKASRIEWEVVYGRGSDVLHLK

Codon-optimized NHI 240 sequence:

ATGAACCTTAAGCAAATACGAAGTTCATATTGTCAGCACGGTTTTTGCGAAGTTCACA
GGATTTTCTGGAGATAAGGAAATTGCGCAGCTTACCGAAACGCTGATGCGTTTTTCAT
CGAAATTGGTGCGTCGCCAATTTAGAACTTTATCAACAGGGTGCCGTTAACTCCGCA
TATCTGACAGGTACCCAACATCTAACACTCGCAGAACGGTCAGTTTTATTTGATTTTA
TTGGTAGCCATAAAGTTATGGAATTAGTGCAGGCTCTGCCTCTGTAAACCGGCAT
TTATGAACACGCAACTGTTTTTCGACCCAGTTAACGGAGCGCAATCGAATTATTGGC
ACCGCGATCCTCAGTACCACCTCAACCTGCAACAGCAACGTCAAGCTCTGTCTGGA
CCTGAGGTTATTCATCTTCGTCTGGCGCTGAAAGACGAACTGGGCATTGAACTGGT
CCCGGGAAGCCACCGCAACTGGGATAATGATGAGGAGCTCGCCGTACGTCAGCAA
GTTGGTGGTCGTTCAAACAGTGAACCGCTGTCAACGGGGCTCTCGCTGCCGCTTC

AACGGGGTGACCTGCTTATCTTTTCGGCAAACATGATTCACCGTGGCTTGTATGGTA
ATCAACGCCTTGGGTTTGATATGTTGTTTTGCGAAGCCCTTCCCGAATTACTGAAGT
TTGCGGACGCCAACTGCTTGCCGTCCGAGGCACACCTGCAACAACTTCAGGCACC
TATGGCATTTTTAAACACTTTAAATGTTAAAGGTCGCGGAGACTGTAAACCTGATTAA

NHI 240 protein sequence:

MNLKQIRSSYCQHGFQFTGFSGDKEIAQLTETLMRFHRNWCVANLELYQQGAVNSAY
LTGTQHLTLAERSVLDFIGSHKVMELVQALPLLKPAFMNTQLFFDPVNGAQSNIYWHR
DPQYHLNLQQQRQALSGPEVIHLRLALKDELGIELVPGSHRNWDNDEELAVRQQVGG
RSNSEPLSTGLSLPLQRGDLIFSANMIHRGLYGNQRLGFDMLFCEALPELLKFADANC
LPSEAHLLQQLQAPMAFLNTLNVKGRGDCKPD

Codon-optimized NHI 241 sequence:

GCTGAAAACGGGGGTAGACCACCGGTTTTTCAGATTCCCATGGAGGGGGGCAGAG
CCTCACTTGCTTTCGCCGCACATCAATTGGTAGGGGTCATTACTGACCCAGCGGTT
CTTATGCACGGCGATGCGCCGGTTCGGAAGAAGTTACCACACCGGTTGCAAGCG
CCGCCGTAGCAGCCGCTGCAGCCGCGAGTGCGGCACAGGGTCATGGCGACCTAG
TACGTCACCCGGTTGCCAGCTTGACGGCTTTCTGAGCGGCGCGGAAATAGCCTG
GCTGATGGAATTAGCCTTTAGCGCAGAAAATCGTTTTCTTTCGTGCTCGTGTTAGTGA
TGAACGCGATGATTACCGGCATTCTCTGGTTCTTAGTGCTCCAGATGCTTTGCGGCA
ATTGTTTGTGGCAAAAATAACTGCCGCTATGCCGGAGGTTATGCCACAGCTCCGCTT
GGGTCAGTTCGCGTTAGGTAAAATTGACTGCCAAGTGACTGCCTCTGTTGATGGTT
CCTACTTTAAGATCCATACTGACGCGGGCGCAAATGAGACTTATAAGCGACAGTTTA
CATACGTTTATTATTTAATCGGGAACCGAAAGGGTTCCTGCGGAGAACTGCGTA
TTTACGATGACATGGTCCGCAATGGTAAGCTAGCCTCTACGGAAAGCTTCCAGGTTA
TTGAACCCCGGCACAATTCAATCGTGTTCTTTCAGGCGGCGGTGCGCCACGAAGTT
ATGCCTGTGCACGTCCCTTCGAAACAGTTCAGAGATGCACGCTTTACTGTTAACGG
TTGGATCGAACGTGCTTAG

NHI 241 protein sequence:

AENGGRPPVFQIPMEGGRASLAFAAHQLVGVITDPAVLMHGDAPVPEEVTPVASAAV
AAAAAASAAQGHGDLVRHPVAQLDGFLSGAEIAWLMELAFSAENRFFRARSDDERDD
YRHSLVLSAPDALRQLFVAKITAAMPEVMPQLRLGQFALGKIDCQVTASVDGSYFKIHT
DAGANETYKRQFTYVYYFNREPKGFTGGELRIYDDMVRNGKLASTESFQVIEPRHNSI
VFFQAAVGHEVMPVHVPSKQFRDARFTVNGWIERA

Codon-optimized NHI 242 sequence:

ATGCATCTGTGGAACATATTCAAACCCAAAAAGCGGAACAATTGGGAACCTGTACCG
CACCCTTCGTAAATCTGCCCCGCTGCGTGCAACCTTGCATGAAGAGGGCCTGG
CCGTGCAGCCATTCTCTCTCCGGACCAACTCCTGGCTTTGCGCCGACTGTATGAA
CAGGAACATCAGCTCGAAAATTCTGACCAGGGTGGCATGTTTTATTCCGTCTACTCT
CAGGATACACCGTACCGTCGTGTTTCATGAAGCCATTCAGGCCATACTGCAGCC
CGCGCTGGACGCATTATTGACAGATTACAGAACGGTGCTTAATTCTTTTATCATAAAA

ATGAATGGTCCTGCATCAGAGTTCGCCTTGCATCAAGACACTACGGGTCTGGACGA
GACACAGTATTCCAGTCTCTCGCTGTGGATCCCGCTGGAAGATATTACCGAACAGA
ATGGAGCATTGTGCGTTGTTAAGCGATCACATCGTCTGTTTAGTCCCTATAGAAGCA
TCAGTTTCCCTCCGCCATATGCAGGGATAAAACACACCATCAAGGATTACTTAGAAC
CTGTTTTTATGAAAGCCGGTGAAGCGTTGTTATTTGATAATCGTCTATTACATTATAGC
CTGCCGAATCGCTCAGATCACAACCGCGTTATCGTAATGTGTGGAATTTTTCCGCAG
GAAGCACCGGTCGTTACGTGTTTTAAATCTCCAGGTGACCCGCGCATCGAGCTGAT
CCAGCATGAAGACGATTTTCATGATTAACCTATCCTAACTTCTTCGTAGATTGTCATCAA
AAACCTACTGTTGGCCAAACGATCGGTTATGTAGCAGATACATACGGCCCAATGACA
GAAAAAGCCTTTCTGAGAGCCTGCCGTGAGCATGGAATCCAGAAAACAAATTATCT
GCGGAATGCGCTAAGCCAGGGTTACCTTCAGGGCGAACCTGTTTTAGAAAAAACCG
CACGCTAA

NHI 242 protein sequence:

MHLWNIFKPKKRNNWEPVPHHFVKSAPLRATLHEEGLAVQPFLSPDQLLALRRLYEQE
HQLNSDQGGMFYSVYSQDTPYRRRVHEAIQAILQPALDALLTDYRTVLNSFIIKMNGP
ASEFALHQD TTGLDETQYSSLSLWIPLEDITEQNGALCVVKRSHRLFSPYRSISFPPPYA
GIKHTIKDYLEPVFMKAGEALLFDNRLLHYSLPNRS DHNRVIVMCGIFPQEAPVVTCKFS
PGDPRIELIQHEDDFMINYPNFFVDCHQKPTVGQTIGYVADTYGPMTEKAFLRACREH
GIQKTNLYLRNALSQGYLQGEPVLEKTAR

Codon-optimized NHI 243 sequence:

ATGCTTGCACGTAGAGTTGTTACCCGGCACCTAGCGCGGGTGGGCGCCCGAACTT
TTTCATCAGCACCCGCTTTTTGATGAAAACACACTGCTGCCATTATTTCCACAGGCGA
CACAGGAAGCCATTCAACACGGCCTGAATACCCAGGGTTTCTGCATCTTGCGGGAT
TTCGCTTCTTCTGCAATTGCGCAGGCCATGCGTACTGAAGCTGAAAGGCTCTATGC
GGAAGGTCACATGTTTCAATCGAACTCGGTAGACGCCGAAGGAAGATCGTACCCGA
AGCCGAACGTTTTTCGCCAGCGAACTTAACGGTAATGAATGGGATTTGGCACCGACC
ATTCTGCACTATACCCGTTCCGTAATGTTGCAGGCTCCGGCCATGTAAACAAATTG
TTTCCAGGGTTACAAATGAGCAGCCGTGCGTACGCCTCTAAATTGGCTGTGTCCTT
GGGCGATGGCGCCAGCTACCCGAAACATTGTGATACGGCTGGTCTTCCTGACAAG
CGTAAAGTTACCATGGTATATTATCTGAATCCAACTGGGCACCGGGGCCACGGCGG
AGAACTACGCGTTTTTCGCACCGAATGGTACCCATGTCGTCGAGCCACGTGCAGATA
CCCTGGCGGTTTTTTGGTTCGGATCTAGTCGCCCACGACGTACTGCCGTGCGAAAC
GCCTGTAGAAGATGTCGCGGCACGCAGATATGCACTGACACTATGGCTTGTCACCG
ATAATCCCAGACAAATTAATGATAAGAACCATCACTTATACCCATTGCGCCAGCACCA
TTTCCCGGATGCGGCTTAA

NHI 243 protein sequence:

MLARRVVTRHLARVGARTFSSAPAFDENTLLPLFPQATQEAIQHGLNTQGFCILRDFAS
SAIAQAMRTEAERLYAEGHMFQSNSVDAEGRSYPKPNVFASELNGNEWDLAPTILHYT
RSVMLQAPAMLNKLFPGLQMSSRAYASKLAVSLGDGASYPKHCDTAGLPDKRKVTMV

YYLNPNWAPGHGGELRVFAPNGTHVVEPRADTLAVFWSDLVAHDVLPCEPVEDVAA
RRYALTLWLVTDNPRQINDKNHHLYPLRQHHFPDAA

Codon-optimized NHI 244 sequence:

ATGGCCACTGAGCTGGATAAAGAACTGCCTTTATTTTCAGCCGTTTGCAACCGCCGC
GGGATTTCTTAGTGACGCGGAAATAGACCGTCTTATTGCCGAGCACGTACCACTCTT
ATCAGAGGGCCGCCTTGGATTTGGGGGGACCAACGCTACCATTGCGCGCAGCCAA
GTAGTATTCTTAGGAACTGAGCCGAAATACCGGTGGCTTTACGATAGACTGTGGACC
GCAGCGCAGGAGTGCAACCGTTTATTTTTTTGTGTTGACATCGCCGGTGTGGAAGC
TAACGTGCAGTTAGCGCGATATGATTGTGGCAATCGGGGCTTCTATGATTGGCACAC
CGATTTGCGCGGGATCGCGCCGCGCCGTAAATATCTCTGTCTATCCAACCTTTCACG
TCCGGAAGATTATGATGGCGGCGACCTCGAGCTGCTGTATGGGAATGCCCCCGC
CGGCTAGATCGTACCCGCGGCACCTTAATTGCGTTTCCAAGTTTTATGCTGCATCGG
GTGACGCCCCGTTACACGGGGTGCCCGCTGGAGCCTGGTTGCATGGATTCTAGGCA
ATCGATGGCGGTAG

NHI 244 protein sequence:

MATELDKELPLFQPFATAAGFLSDAEIDRLIAEHVPLLSEGR LGFGGTNATIRRSQVVFL
GTEPKYRWLYDRLWTAAQECNRLFFCVDIAGVEANVQLARYDCGNRGFYDWHTDFA
GIAPRRKISLSIQLSRPEDYDGGDLELLYGNAPRRRLDRTRGTLIAFPSFMLHRVTPVTRG
ARWSLVAWILGNRWR

Codon-optimized NHI 245 sequence:

ATGGCCAGCAGTACCCAGGAGCACCATTACACCATTTACCAACACCTTATGGCACG
ACAACCGCACCCCCCTCCAACACCGTCTGCACATAGCAATCACAATCATAACCTCCT
GTCTACGAGCGAGGCTGCAGACTCCCTGTCTCGGCTGCTGCATCGCATCCCTCCT
ACCCTGTCATTAAGTTTCCCGGGCCGCCGAGGCCCTCCGTCAACGACGATAA
GTCCGCCATTGATTTCCCTGTCTGACGCTAAAAGTATTCTCCATAGTAACCTGTAAAG
CGCCGCAAAACAACATGGTTTTTTCCAGCTTACCCAGCATAGTATTAGCTCGCATTTA
GCGCAGTCTGCGGAATCGGAAAGTGCCAGTATCTTCAATGACGAAAAACAGTTGTG
TTTTCCGAAAAATTGGCCGCTGGGGTTTGATAACGATGATGACGATGATGACTT
TGTATGTGGACGCTCACTGTGCCTGGATGAGAGCAGTTCTATGGAGTTAGGCTTTT
CGTCAATTCATGAATTTACCAGTCAGATGGAGAACTGGGCCTGGAAGTATTGAA
GAGTTAGCTTGTGCAGTTGGTTTTGAGAATCCTGTCGGTGAGGACCGGACACGTTT
TTGCTCCTTAATGTGGATCAGCGAATCCGAAAGCGGTTTCATCAAATATGCCGCTCTC
TCCAGGCAGGGTCTATCCTTATGTGGTGGGGCTGAATTATCAAATTCGTTGCCGCAA
ATATAGCATGTTAGCAGACAGCGGCTGGATCTCTGTGGCGCCTCAGGTGGACTCTG
TTATGGTTACTTTAGGCGATATCGCACAAAGTATGGAGCAACGGTAAAGTAAAGAAAG
TGCGTGGCCGGCCGGTCCCAATAACAAACGGAGAAATTAATTCATCCAGTAATAATA
GCCATTATGTTTCCCTGACCCTGCTGGTTACATTGGCCACGAAAAATACCGTAAGCC
CTCTTCTTCCAATTAAGCTGCCGTCTGTTAGGAGCAGTGTTGAATAAAGAAGATGATG
ACGGTTTAGGTGATAATGACTCTACAACCATAGCGGTGAGGAAGAATCCCCTATGT

TTAACTCATTTTCATTTCGAGGATTACGCCTGGCGTGTGTATCATGCACGCCTGAATAT
CAAAGATCCTTTGGTACGTTACCGTGTGTTGA

NHI 245 protein sequence:

MASSTQEHLHLPTPYGTTTAPPPTPSAHSNHNHLLSTSEAADSLSRLLHRIPPTLS
LSFPGRRQSPSTTISPPLISLSDAKSILHSNLLSAAKQHGGFFQLTQHSISSHLAQSAES
ESASIFNDEKQLCFPNWPLGFDNDDDDDDDFVCGRSLCLDESSSMELGFSSIHEFTS
QMEKLGLELIEELACAVGFENPVGEDRTRFCSLMWISESESGSSNMPLSPGRVYPYVV
GLNYQIRCRKYSMLADSGWISVAPQVDSVMVTLGDIAQVWSNGKVKKVRGRPVITN
GEINSSSNNSHYVSLTLLVTLAHTVSPLLPIKLPSLGAVLNKEDDDGLGDNDSTTIAVE
EESPMFNSFSFEDYAWRVYHARLNKIDPLVRYRV

Codon-optimized NHI 246 sequence:

ATGTTATCTAGATTGCAACCGGACCTGGCTGGACCGATGACGAAGCGTACAATGCA
GCAATCTAGTCAGGCCCCGGCTGAACCACCCCTCGTAGCGGACAACCGTACATTAG
CAGGTTTCAGTTCATGAACCTGGTTGGGCCAACGCTGATGCAGATGCCTTCGCCGAG
CAGAGAGAATATCTGGCTACTCACAATGGCATGAGAGGTCTAGAAAGGCTCCAACC
AACCGATATAGAAGATGCCGTGCGCATATTCTACCGAGACGGATTTGTGGTAATAGA
AAATGCATTAACCGAAAAACAGCTGGCCGAATTAAAGGAAGCATCGGAAAGAGTCAT
TCGTGAGGTACTTGCTATGGACAAAAATCGGCAGGGCAACAGGGGTAGTCATCGTT
ATTCATTTGGCAGCGCATCTTTGACAGGCCAGCTGCTGCATGAACCAGCATGGGCC
ATGTTGGTAGATCTACCCACGGTCACCCCGATTATAACCGCAATTTTTGGCTCACGA
GACTATATTTTACGTGGTGCGGGAGGTGATTTCTGTCTTCCAGGTGCCTCAACCTAT
CAGCCTCTTCACTCGGATATGAATGACAGGATTGAATATCAAGGTAAGACATTCCGT
AGCTTCCACGACCCACGGGGCAAATTGAACTATCGTGACCTGCCCTGCCCTTATGT
GTGCTGCAACTTTCTAACATCCGACTTTACCGCCATAAATGGTCCCACCCGACAGAT
CCCAGGTACGCAGCACAGCCAACAACGCATTCCGAAATTAGCCCAAGAACCCGAG
TGGATGAAATTATCAACGGTGTGCCCCGGCCCCAGCAGGTTTCAGTCCTAATTCGGGA
CACCCGCGCATGGCATGGAGGAACGCCAACTTATCGGATCAGGTCCGCGCCATC
CCAAATACGGAATATCACGCGCCATGGTACCGAGAACCAATGCGTAAGTCAATGCCA
CGCGAGATTTATCAAAGTCTGAGCGAACACGCACAATATATAGCTCGGTACGTAGTC
GCTGAGGCGGGCGAAGCGCTCATTACGGGATTTCTGTAAGATCTGGGCACCACCC
CTCCAGGGCTCCGGGACGATGCTTAG

NHI 246 protein sequence:

MLSRLQPDLAGPMTKRTMQQSSQAPAEPLVADNRTLGSVHELGWANADADAFAEQ
REYLATHNGMRGLERLQPTDIEDAVRIFYRDGFVVIENALTEKQLAELKEASERVIREVL
AMDKNRQGNRGSRYSGSASLTGQLLHEPAWAMLVDLPTVTPITAIIFGSRDYILRGA
GGDFCLPGASTYQPLHSDMNDRIEYQGKTFGSFHDPRGKLNYRDLPCPYVCCNFLT
DFTAINGPTRQIPGTQHSQQRIPLAQEPEWMKLSTVCPAPAGSVLIRDTRAHWGGTP
NLSDQVRAIPNTEYHAPWYREPMRKSMPREIYQSLSEHAQYIARYVVAEAGEALITGFR
EDLGTTTPGLRDDA

Codon-optimized NHI 247 sequence:

ATGCATGCATCCCGAGCAACCGACCCTATTTTTACAGTACCCGATCTTCTTAGTCCT
GAAGAATGTGCAGCGTTTATCCGGAGAGCGAGTGGTACCGGCTTCGGTGAGGCCC
CCCTTACCGTTGGGGTTAATCGGTTTATGATGGAGAAATCCGTGCGTAATAATGAGC
GGGTGATCGAAGATGATCCCGCCTTAGCTGCACGGCTATGGAAGCGTCTTGAGCCT
CACGTCCCCGCGGTGTGGGGGTCAATGTGTGCGATTGGTCTGAACGAACGCTTCC
GTTACTATCGGTATACTCCTGGTCAGTATTTCCGCTGGCACCGAGATGGTAGCTTTG
TCCGTAGTCCAACCGAAAGATCTCTGTTACCGTTATATATTATTTGAACGACGATTT
TCTCGGTGGTACCACTGACTTTTACGGGAGAAGGCGAACTCCGTATCACACCGCGG
CGTGGGATGGCATTAAATATTTGCACATCCGATATATCATCAGGGGGCCGAGGTTGTG
AGAGGTACGAAGTATGTGTTGAGGTCCGATGTTATGTACACCCGTTTGTAG

NHI 247 protein sequence:

MHASRATDPIFTVPDLLSPEECAAFIRRASGTGFGEAPLTVGVNRFMMEKSVRNNERVI
EDDPALAARLWKRLEPHVPAVWGSMAIGLNERFRYYRYTPGQYFRWHRDGSFVRSP
TERSLFTVIYYLNDDFLGGTTDFHGEGERITPRRGMALIFAHPIYHQGAEVVRGTTYVL
RSDVMYTRL

Codon-optimized NHI 248 sequence:

ATGTCGGACGCCCCCTTTGTGCGATCTCAAACGAACCGTATATGCTTTCCCCAGAGCAA
ATTACTTCCTTTGAGCAGCAGGGTTTCCTGATTCTGAGAGACTTTCTGGCCGACACA
TCATCTGTGCAAACACTGCGGCAATGGACCCAGCAGGTACATGACTGGCCTCTGGT
GAAAGGTAAATGGATGCCTTATCAGGAGAAAATCAGTGATGGTAGAATGGTTCTTAC
CCGCACGGAGAATTATTCACCATTCCATCCCGGTTTTAATGCTCTGTTTCAGAGGTGA
CCGTTTAATGGGCGTTTTAAAGCAGTTATCAGGCGAAGAAATGATCCTGTTTAAAGA
AAAAATCAATTATAAACTCCCGAAAGCAGGCGGTTTTCGATGCACACCAGGATGCGC
CTGCCTATACGCATGCTGGTGCTCTTAAACACCTTACAGTAAACATTGCAGTAGACA
AAGCCACACTGGAAAATGGTTGTCTGGAAGTCGTCCCGGGAAGTCATCGCATGAA
CATCCCAGTGGGTAGTGATAATTGTATCCCAAAGGACTGGGAAGATGCTAACGAGT
GGATACTGGTGCCGTTGAAACCAGGGGATGTTCTGATTTTTTGGCAGTTATCTCGCA
CACCGTTCGGGGCCAAATTCTTCGAGCAACCCGAGAGCAGCAATATACGCAACCTT
TAATGCCGTCTCCGATGGTGGCGATCGTTTCGCTGAATATTACATAAACCGCCGCGC
AGAATGGCCGCCCATGCATGAACGCGAGGAAGGCAAAGATTACAGCCTGGGAGCC
CTGCGTTACGGTTTTTGGCAGCCCTATGACGGGAGCGACCCAGCACAGCCGCATGC
AAACGGAGTTGTTTGGTGCGACTGTCAGCCAGGCGGGTGTTAGCGCGAAAATTGC
CTAA

NHI 248 protein sequence:

MSDAPLSISNEPYMLSPEQITSFEQQGFLILRDFLADTSSVQTLRQWTQQVHDWPLVK
GKWMPYQEKISDGRMVLTRTENYSPFHPGFNALFRGDRLMGVLKQLSGEEMILFKEKI
NYKLPAKGGFDAHQDAPAYTHAGALKHLTVNIAVDKATLENGCLEVVPVPGSHRMNIPVG
SDNCIPKDWEDANEWILVPLKPGDVLIFGSYLAHRSGPNSSSNPRAAIYATFNAVSDGG

DRFAEYYINRRAEWPPMHEREEGKDYSLGALRYGFGSPMTGATQH SRMQTELFGATV
SQAGVSAKIA

Codon-optimized NHI 249 sequence:

ATGACGGGCAAAGTGAGCCTTAACACCGATATTGGAGTTTTCCGTGCCAATCAGCT
CGATGAACTGGATGCTTTTTTCCGCGCGCATGGGTTTGCCATACTGAGGGGATTATA
CGATCCTGCCGACTTAGAAGCCTTAACCAAAGACTGCGTGGCTGCGCAGACCGGG
GTTATCGAAGGTTCTCTGGATGAACGCTACGGCAATTCTGTTCTGCTTGATGATGCT
GTTGATGGCAAACTCGGTTTACTAACTACGTGCAGTATATTACGGAAATTTCTTCGA
AAACTCGTAACGCTATCGTCCATCCTGCGATAGTGGATTTAATGGCACGGTGGGTTC
CAGAAGGATATCTGTTGGAGCATAGTCGCTTTGGTGTGTGTATCAAGATGCCAGAG
GGGGTCGTAATAGCGGTTACCGCAGAATCGGATGGCACAGTGACTGGCAGAGCGG
TCCGCATCGGGATGTGTGGCCTTCGGTCGCGTTTACTATTAATATTGATGCGACATC
GCCCCGAGAACGGTTTCCTTCGCGTTGTTCTGGCTCGCATAAATGGGCTACCCCCG
CACCGTTTAAAAATGTTAATAACGTTGCCGTCCCAGCAGGAGCCCCGCCGACCGGC
GGCTACACAGACGAAGCACCGCCGTTTGAATGCCGTTAGCATTCCAGAAAGCTCC
GGGAGAAATTGGTATATATTGTGATGCCGGCGATGTTCTCTTCCATGACGCATACCT
GTGGCACTCCGCCGCGACTGCGACGAACGACCGCGCCATCCGTCGTCATGTTTCG
CGGTAGTTGGTACTCTGGTGTACAAGATGCCGTTAACGATGATTTTATCGAAGATTTC
GTAAAAAATGCGGCACGATGA

NHI 249 protein sequence:

MTGKVS LNTDIGVFRANQLDELDAFFRAHGFALIRGLYDPADLEALTKDCVAAQTGVIE
GSLDERYGNSVLLDDAVDGKTRFTNYVQYITEISSKTRNAIVHPAIVDLMARWVPEGYL
LEHSRFGVVYQDARGGRNSGYRRIGWHS DWQSGPHRDVWPSVAFTINIDATSPENG
FLRVVPGSHKWATPAPFKNVNNVAVPAGARPTGGYTDEAPPFEMPLAFQKAPGEIGIY
CDAGDVL FHDAYLWHS AATATNDRAIRRHVRG SWYSGVQDAVNDDFIEDFVKNAAR

Codon-optimized NHI 250 sequence:

ATGAAAGTTCTTTACAAGGATTTTAACCAGGAAGCTTTCAGTTTCCAGAATGACCCG
GTTCTTGTTCTTGAGGAGTTCTGGAGCCCGGAAGACCAGGCGTTTTTTTTTGGATGC
AATGAAAAAGACTGCGTGGAAGAACTCTTAAAGATATGCCGGCCGTTAGCCACGCCT
TCCCGAACAGCGGTAATTGGCATCGGGGTGACGTGGGTCAGAGAGAACGCGAGG
CGTTTCTTAGTAAAATAACCCTGGACTGCATCGCCGACCATATTGATTCTTTTCCTAA
CATTAAACGTCGCCATCTGAATCTTAATTATTATTTCGTACGGCGTAGGGGACTGCCTA
CCTACTCATGACGATACGGATGAAGCCTACGTACCAAGTAACTTGCCTCGCCCACC
ATTGCGGCGTCTCGCCATGGTAGCATATTTTCAACGAATGGGAGCCTTCATGGG
GAGGGGAGCTAATCGTTTATGGGGAACGCAAAGATAAAACGGGAAAGAAGACATTA
AATATCGTGGGTTGTGTAGCTCCAAAACACGCTCTCTCGCCATGTTTACCGTACCC
AGGCATCATCGCGTGTGTGCGCATAGATCCGATCGCTGGCGATAATAGACGCAAAAC
GATTGTCACGTGGTTCATGACGGAACATTGA

NHI 250 protein sequence:

MKVLYKDFNQEAQSFQNDPVLVLEEFWSPEDQAFFLDAMKKTAWKTLKDMPAVSHAF
PNSGNWHRGDVGREREAFLSKITLDCIADHIDSFPNIKRRHLNLYYSYGVGDCLPT
HDDTDEAYVPSNLPRLRLAMVAYFHNEWPSWGGELIVYGERKDKTGKKTLNIVG
CVAPKPRSLAMFTVPRHHRVCRIDPIAGDNRRTIVTWFMTEH

Codon-optimized NHI 251 sequence:

ATGAGTGCGGAAGCCAGAGGTTTAGTCCCGGAAATAAGGCGGGGTAAGATCTATAA
AGATGTATATGCAAAACGGAGCCCGGGGCGTCCTGCGTTGGGCGAAGCACGGCTG
GAACGTGCCCCGCGCCGTGGATGGTGTACTCGAGTTTGACGGTAGTGGTGCACGTG
ATCGCGCATTGCGGATGGAGTATTTCTGCTAGAAATTCCTTCCGGTCTGGATGTAA
GCGCGGGCGACGCTTTTGCACGTCAAGTCTATGAAGGTCCTGCCAAGGCCCTTAT
GGCCGCTTCCGCGATATCGGGTCAGACAGGTTTGGTGACCCGCTGCTTGGTTTTTC
ATCAGCGTATTAATCAAATAGAACAATTCTTGCTTGAAAGACGCTTCTGGGCAACGG
CGTACCCGCCCCGAAATAGCCGCACTGGGTGAATCACTGACGAGGCTGTCACGTGA
AGTGCTGTGTTCAAGTCCTGAGTGCTGTGGGAATTCCCGAACAGGACCGTCGTCGT
GCAACCGGTGGATGTTCCGAATGTGCCGGCTCCTATCACTTAACATTTAACCATTATA
GACCGGCCAGGCAGATATTGGCCTCTCCAGCCACAAAGATGATGGATTCTTAAC
ATCCTGCGTACAACCGCCCCGGGGCTTGAAGTGTCCCGCGAGGACCATTTGGGAAA
CCGTTCCGGCCGATCCGGATTGCTTTATTGTTAATTTTGGACTCTCTATGGAAATCTT
GACAGCCGGCTCCCGCATGCCAGTTGCCGCTATTATGCATCGAGTAGCCCGTCAGC
AGGACGATAGATCATCATTCCGGCATTTCATCGTCCGGTTGTCTCCCGGGTGAAG
ATGAAGGAATATATCGGTACGCTACAGGATCCGGGCTTGAACGGGTATGCGGTTCA
CGCGAATTAATTGACGAAAACGATTACGAAATTTACGCCGGCACAGACGCGCCGGA
TGGTCCTGACACTGCTACTGCCCCAGGTTCTCCGATGAGCATCCACCGTAAAGG
GCGCGGGTCGATAA

NHI 251 protein sequence:

MSAEARGLVPEIRRGKIYKDVIYAKRSPGRPALGEARLERARAVDGVLEFDGSGARDRA
FGDGVFLLEIPSGLDVSAGDAFARQFYEGPAKAPYGRFRDIGSDRFGDPLLGFHQRIN
QIEQFLLERRFWATAYPPEIAALGESLTRLRSREVLCSVLSAVGIPEQDRRRATGGCSECA
GSYHLTFNHYRPAQADIGLSSHKDDGFLTILRTTAPGLEVSREDHWETVPADPDCFIVN
FGLSMEILTAGSRMPVAAIMHRVARQQDRSSFHGFSSSGCLPGEDEGIYRYATGSGL
ERVCGSRELIDENDYEIYAGTDAPDGPDTATAPGSSDEHPTVKGAGR

Codon-optimized NHI 252 sequence:

ATGGCGCTTTCTACAGAGAATTTATCGCAACAAATCCTTGAATTCAAAGCCGCCCTG
AAACAACAGAGTCCGGATTATAAGGCCCGGTTGAAAGTGTCCTGATGCGATGAA
AGCCGAAATTGCACGGATTAAACAGGAAGAGTCCGAGGGTTCAGCCATTCCTCAGT
TCGAATATTCGGATATTGCAGCCAGTGGCTTTAACGATGAACAACGTGCGAGAGTCC
ATCGGGCCGGTTGCTGTATCGTTTCATAATGCTTTACCTCGAGATGAAGTTAGTGCAC
ATAATGAGAATCTTTGAAATACATCATAGATAATGGTTATTATGACCACAAGAGTACC
GTAGAAGATAATTATTTTCACTGAAATCCGACAAGCCACAAATATTTGGCATTTA
CTGGTCACAGGCACAGGTCTGGGCGCGCCAGCACCAAAATATGGCCACCCTGCGT

CGGCATCTTAATCACCTCTGGCAGTGGCAGGATCCGCGTAGTGGTAAGGCATACTT
CAATCCCGATTTGGAGTCGAGTTATGCCGATCGTGTGAGACGACGAGAACCTGGAG
ATGCAACCTTAGGTCTATCTCCGCACGTAGATTCAGGCAGCATAGAGCGCTGGATC
GAGCCACACTATCGCGATGTGTATCATGAAGTATTCCATGGGGATTGGACCAAATAC
CAAGCATTTCGACGGTCGTAGTCGAGTGGATGTTGAGGAATTTCCATCTCCCGCGGT
ATGCTCGGTCTTTAGGACCTTTCAGGGATGGGTGCGGTTAACGCAACAAGGCAAAG
GTGATGGCACGCTACAGACCGTTCCGTCTACTCTGGCGATGCCTTTCATGTTACTG
CGCGCTATTCAGGACGACGTGGAAGAAGAAGATCTGTGTGGTGCCGCACCTGGCC
GTGCCTTAAGTGTGTTACAAGGAATGGCATCCCTTACTGCTGGAAGGCCTTGTCAGTA
TACCGAAAATGCAACCAGGGGACACAGTTTGGTGGCATCCGGACACGATTCACGC
CGTAGAAGATAAGCACAAATGGTGATGGATACTCTAACGTACTTTTTATCGGGGCCGC
ACCTGATTGCGAAAAGAATCGCGCGTTCCTGGCTAAACAGCGACCAACATTCTTGA
ACGGGGACTCTTGTCCAGATTTTGCGGCCGAAAATCATGAACGCAACTATGAAGGC
CGCGCGACCGAGGCCGACCTCAGTCCCCTGGGTAGACAGCAAATGGGTTTTGAAT
AG

NHI 252 protein sequence:

MALSTENLSQQILEFKAALKQQSPDYKARFESVTDAMKAEIARIKQEESEGSaipQFEY
SDIAASGFNDEQRARVHRAGCCIVHNALPRDEVSAHNENLSKYIIDNGYYDHKSTVED
NYFSQLKSDKPQIFGIYWSQAQVWARQHQNMATLRRHLNHLWQWQDPRSGKAYFNP
DLESSYADRVRRREP DATLGLSPHVDSGSIERWIEPHYRDVYHEVFHGDWTKYQAF
DGRSRVDVEEFPSPAVCSVFRTFQGWVALTQQGKGDTLQTVPSTLAMPFMLLRAIQ
DDVEEEDLCGAAPGRALTVYKEWHPLLLLEGLVSIPKMQPGDTVWWHPDTIHAVEDKH
NGDGYSNVLFIGAAPDCEKNRAFLAKQRPTFLNGDSCPFAAENHERNYEGRATEAD
LSPLGRQQMGFE

Codon-optimized NHI 253 sequence:

ATGTCTCCACGCAGATAGATCATTTTCGTACCCATGGCTACCTGCACCTGGAGGGT
TTCCACCCCAAACCTCGCGTGGCTCCGCTGCGCCAGAAGCTGCGTGACGAATTAA
AACGTCTCGCCGGGGCGAAAGGTGGTTTGTGCGGCTGTTCAAGCGGCTTCCTCTATT
CCAACAGATCGGTAAACTGTCGGCACTTGTTAACCTCCCCGGAGTGACGAAGCCT
TGGTGACCCCGGCTTTGCTGGACGTCATTGCAAACTGGGGCACCGGGGCCCTCC
TAGTGACGCGGCTGGCCATCCTGGAACGCAACTGCTCTTATCACCGCCCCAGCAG
GGTGATTGGACACTTCAAGGTCTTAAGTGGCACGTGGATCTGGCCGCCCGACCTC
AGGATCCTTTGCCTGGAGTTCAAGCGTTTTTCCTGATAGATGATGTCGCACCGCATG
GTGGTGCAACGTTAGCACTGGCTGGTAGCCACAGAATCGACACTCGTCGACCAGC
ATCGGATAGCCAGCCTGCGCTGCGGGAAGTCTGAAGGATCCGCTTGATCTTGAAC
GTCGTCTTGGTGAACAAGGGCTGGCCATTGTTGAGATGTGTGGGAGAGCTGGTGA
CGTTTTTTTTGATGGATATGCGTGTTTTGCATACGCCAAGCGTCAATGCCACCAAGCA
ACCTCGCATGATGGCAACAAGTAGATGCATCTTCGGAACCTAA

NHI 253 protein sequence:

MSPTQIDHFRTHGYLHLEGFHPKPRVAPLRQKLRDELKRLAGAKGGLSAVQRLPLFQQ
IGKLSALVNLPGVHEALVTPALLDVIKLGHRAPPSAAAGHPGTQLLLSPPQQGDWTLQ
GLNWHVDLAARPQDPLPGVQAFFLIDDVAPHGGATLALAGSHRIDTRRPASDSQPALR
ELLKDPLDLERRLGEQGLAIVEMCGRAGDVFLMDMRVLHTPSVNA TKQPRMMATTRCI
FGT

Codon-optimized NHI 254 sequence:

ATGACGGTGAGCCCAGAAGCATGGGCCGGTCTGGCGACTTTATCTACTGCGGTGG
AAACCCTTGAACGTATAGTCAGCAGTGCAGATGTGCATATGGATCGTCCACTGCAAG
TTATGGGCCAAGACGCTCTGCTCTTTGCCGCGAGCATGTATGATGAAGACCTGGCA
CTGGAAGTTCTGGCAGCTCGTGCTAGAATACCGGTTGATGTCAATGCCATAGACAAT
CATGGTTACACTGCGTTGCATTATGCCGTTGATGCCGATATGCACGCTTTGGCCCAG
ACACTGCTGGCGACTGAAGCGATTTGCGTGCGTTTACAGACGGAAGACCTCAGTG
TCGTGGGCCACATCCAGTCGGGCGGCCGTACCCCTCTGCATCTGGCAGTATTGAG
AGGTAGCATGGATTTGATGCAGCTTTTACTGGAAGCGGACCCTCTGCTTGCAACTG
TGGCGGATCTGGACGGCCATTTACCGGTGCATCTGGCTCAGTTACACGGCGGGAG
CATGGATCTTATTGGATGCCTACAGGCATTTTCATGCGGCCACTCTCTATGATGTCGC
TTCACCTCCGGAGAGAAAGTTGGCGACTCAGGCTATTGCTGCCGCCCGCTACCAG
GCGAGCCTGTCCGTACCCGATGCGGTGGCGACTCCTCATATATTTCCAGGTATCTG
GACGCAAGCAGAATGTGAAAGAGTCTTAGGTCAATTGGAAGCGACTACACACGTTA
CCGGATGGCATAAGACACGCCACACCTCATATCCGACAACAGATCTTCCATCTTACT
GCTTACCGTTGGGTGAACACTATTGGGTTCAAGACACGCTCTCTGCGAGATTATTC
CTTCTATTCTGAGCCGTTTTCAACTTCCTGCTTCCACGCAACTGTCATTTCCGGGATC
TGTTTTATGTCAAATACGAAGCAGCCCAAGACGCCAGCGCGATCTAGCTCTGCAC
TGTGATGGGTCTGTTCTGTCTTTAATGTACTGCTCAACAAACGGACTGATTTTCGTA
GGTGGAGGTACTTTTTTCGCGCCCACACAGACAACGGTTCATATTGAACAGGGTGA
TGTAGTCGTTCACTCGGGCCGGGTAGTTCACGGTGCGGCACCCGTGACGGAAGGT
AAGCGTTTAATTCTAGTCGCGTTTCTGAATGTCCGCCACCGGCATGCGCGCCTTG
A

NHI 254 protein sequence:

MTVSPEAWAGLATLSTAVETLERIVSSADVHMDRPLQVMGQDALLFAASMYDEDLALE
VLAARARIPVDVNAIDNHGYTALHYAVDADMHALAQTLLEATEAICVRLQTEDLSVVGHIQ
SGGRTPHLAVLRGSMDLMQLLLEADPLLATVADLDGHLVPVHLAQLHGGSMDLIGCLQ
AFHAATLYDVASLPERKLATQAIAAARYQASLSVPDAVATPHIFPGIWTQAECERVLGQL
EATTHVTGWHKTRHTSYPTTDLPSYCLPLGEHYWVQDTLSARLFPSILSRFQLPASTQL
SFRDLFYVKYEAAQDAQRDALHCDGSVLSFNVLLNKRTDFVGGGTFFAPTQTTVHIE
QGDVVVHSGRVVHGAAPVTEGKRLILVAFLNVRPPACAP

Codon-optimized NHI 255 sequence:

ATGCCAGTAGTCATTCCGACACCGGAATTACTGGCACAATTTAATCGTGATGGATTTT
TGGTTTTACCGGATGTTCTTTCTCCGGCTGAGGTGACTACTCTTCGCAACGGCCTG
GAACGGGTCTTTAACCAGTTTAGTCCAGAAGCCGATATATACCATATGCAGGACATTT

GGCGTCCAAAAATGTTTGAACATGGCGAGGAATTCGAACAACCTCATTGACCATCCA
GGAATCGCCGATTTTCGCTGAGGCCGTTCTCGGGAAAGATTGCCACATGATCGCAGA
AACTGGTCTCCGCACGTCACCGGGAAAGACTATAACCTTCTGGCATTTAGACGACA
CTGTACGGTTTTCCTTTAGCAGATGGTCAGAACTCGATCCGGCGATTGTTATGCCAA
CATTTGTAATAAATATGAATTACTATCTTTGTGATGTAGATGAGGAACTAGGACCTACA
CAGTTTATACCAGGATCACACCGCGCAGGTCGATCGATGCGCCCGGAAGATAATGA
TGCGTCGGGGAAACCCCACTTGGGAAGGGCGGTCGATGGTGTCTTCGACTGGAAA
GGCAGGCACTCTCGTTCTGTGGAATGACCAAGTTTGGCACCGCGGTGGACCAAAT
GTCAGCAACGGGCGTATTCGCTGGGTTATTCAAACACCGTTTCGCACAGAGGTGGG
TTAGCCTGAGATACTGGCCACATGTTAATTATCATATGCCAGAAGAAGTCATCGCACG
GGCGAATCCCCGACGTAAACGTCTCTTAGGTTTCCATCCGATAGGGGCATATGGTTA
A

NHI 255 protein sequence:

MPVVIPTPELLAQFNRDGFLVLPDVLSPA EVTTLRNGLERVFNQFSPEADIYHMQDIWR
PKMFEHGEEFEQLIDHPGIADFAEAVLGKDCHMIAETGLRTSPGKTITFWHLDDTVRFP
LADGQKLDPAIVMPTFVINMNYLDCVDEELGPTQFIPGSHRAGRSMRPEDNDASGNP
TWEGRSMVSSTGKAGTLVLWNDQVWHRGGPNVSNRIRWVIQTPFAQRWVSLRYW
PHVNYHMPEEVIARANPRRKRL LGFHPIGAYG

Codon-optimized NHI 256 sequence:

ATGCGTTACTTTAAAAATGAGCAGGATTTTACTTTCCTGAGTAGTTTCAAAGATAATTA
TAAGGTGATTACAGAGGAACTGATCAACTTTTTGAACAATCCCAATGTGGACTTTCA
AAAATTTGTTT CAGACAAGTTCTGCGCAGAATTTCGATTAACAAAGTTGATTTTTGGAAA
ATCATCGCTTTATATATTCATCTGAAAACCCCTCTGGAAATATTCCAGGAGTACAGCG
ACTTTTATCAGGTGAATATCGAGCAGGTTAAGCAAGCCTATGACTATATTTACAGCCA
GTTTCCGCAGACGTGGAAGATCGTTTCGCGAACTCACAGCGTCGCAGGACAATGGG
ATCATAAATGTCTTTTTTTCATACTTTGAGCCCGGCGCAAAACTCAACTTACACGTCA
ACGATGATCCTTACATGTACCGTGCTCATTTAGGACTGATAGTCCCGGAAGGCAACA
TTGGCTTTAAAGTCTGTGACGACTTAGTAAATGGCAGGAAGGCGAGGTGCTTGTG
TTTACGCCTACCAACCCCCACACAGCCTGGAACCTGACAAAACACCCGAGAGTAAT
TATGATTATAGATTTTTTTTAAACCCGATGCTGATAGGGCCGAAATGAAACAGTTGGAA
CGGCAACAGTTCGAAAAAATGATGCTGATCAATCCATGGTCATTTGGGATGTCTGGT
GGCTACTCCGATCTGGATCCGGAAACAATAAAAAAATATGCCGTCTTAGATATCTAA

NHI 256 protein sequence:

MRYFKNEQDFTFLSSFKDNYKVITEELINFLNPNVDFQKFVQTSSAQNSINKVDFWKII
ALYIHLKTPLEIFQEYSDFYQVNIEQVKQAYDYIYSQFPQTWKIVRELTA SQDNGIINVFF
SYFEPGAKLNLHVND DPYMYRAHLGLIVPEGNIGFKVCDDLKWKQEGEVLVFTPTNPH
TAWNLTKHPRVIMIIDFFKPDADRAEMKQLERQQFEKMMLINPWSFGMSGGYS DLDPE
TIKKYAVLDI

Codon-optimized NHI 257 sequence:

ATGCCTCCTACTCAAAAATTCGAGGAACGTATTGTACGAGTCGAAAAACCATCAGCC
CATTTTGGCCTGTTGCGTAATGAGAAGGGTGATTTTCTGGGGGTATCAGAGGGGGA
GACGGCGGTATTTGATCATGCTGATGATAAAGCTATATGGAAAGAGGAAGGTGGCAC
CTATCGGCACGTGGTGAGTGATTTGACTCTCGATGTCTCGGCCGCTAATGGTGAGC
ATCTCGCCCTGAACGGTAAAAGTCTGGGCGAAGACGGTTCTGAATCCTCAGTACCG
GGTCGTTTTTCAGCCTGGGCATGGCCCAGCGCATTGCGGTCCGAATACTTACAGCA
GATCCGGGAGAAATGGTTGGGTATGCCTTCCTTCCATTCTTTCTCCTCACGCCATTGA
GGAActACAACGCGTAAGCTGCACGGGAAAATGGAACCATGAGGAACTGGACCGT
TCGTTGCCTACCCTCACTCAATCGGTTGAGGTTTCACGGGCAGCCACAGAGCCCG
TATCCCTCTGGGTGATGCGCGAGTACATGCAATGCCGAGAGATTCGGCTTGACAT
CGACCGGGTTTTGCAGTCGTAGCTCCGGATGATGGTGAACGTCAGGTTGGCGGAT
GGCATAGCGATTACCCCTACCTGTGGGGGATTGCTCGAAATAGTAGTGATGGCAAC
CGTATCGTGCCACATAATGCCTTAAATCTGGTACTCGGAGTTCAACGTAACCTGTGC
GTGACCGAGTTTAAAAGTGAAAATGGTGCCACGTGCTTCAAGCTGGGCTCACACAC
ATTTGGACAGGGTCCGCCACCGGAGTGGGGTTCCGGCAATGCGTTTCGTGAGGA
GGGTGCCAGGGAAGCGAAGGGTTTGCCGTACACGGGTTCTGAAGCGGACGTTGT
GGAAGCCCCGGCAGGGAGTTATGTTATATATGATGCCCGCACCTGGCATCGCGCGG
GTGTAAATCGCACCGATAAGAAACGGGCTGCCATGCTGCAGAGCGTTGTGCCCATG
TACATTATGCCGTTTATGGAGACTTCACGCCCCGTATAAAGAGTTTTTGGCCTCGCCA
CTAGCATCTCAACTTACCGAAAGGGAACATATCGAGCTAGAGTCACTGCTGGTCTCT
CGTATTGATGGCCCGCAGGGCAGATTGGCCATTACCATGGACGAGGAGTTGTCTGC
GCTAA

NHI 257 protein sequence:

MPPTQKFEERIVRVEKPSAHFALLRNEKGDFLGVSEGETAVFDHADDKAIWKEEGGTY
RHVVSDDLTLDSAAANGEHLALNGKSLGEDGSESSVPGRFQPGHGPAPHLPSYLLQIR
ENGWVCLPSILSPHAIIEELQRVSCTGKWNHEELDRSLPTLTQSVEVSRAATEPVSLWV
MREYMQCREIRLAHRPGFAVVAPDDGERQVGGWHSYDYPYLWGIARNSSDGNRIVPH
NALNLVLGVQRNLCVTEFKSENGATCFKLGSHTFGQGPPPEWWSGNAFREEGAREAK
GLPYTGSEADVVEAPAGSYVIYDARTWHRAGVNRDCKKRAAMLQSVVPMYIMPFMET
SRPYKEFLASPLASQLTEREHIELESLLVSRIDGPQGRLAITMDEELSR

Codon-optimized NHI 258 sequence:

ATGCCGCATTACATTGATACTACCGCGAAAACCGGTGACTGGCGGGATGCTCTTTAC
GAGGATGGCTTTGTTGTTGTCAAGGGCGTGCTGAGTCAGGATAAGGCCTCTTCATA
TGTTGACCGTATGTTCCAGTGGCTGGAAAGTTTTCCATTGTTTAAACGGGACGA
TCCTTCGACGTGGAATGCTAACCATCTCCCAGCTCATATGAAAGGTGGAATGTTCCA
CGGCTATGCCGTTTACGCATGAAAAAGTCATGTGGGATGCACGCTGCGAGCCTAAGC
TTATTGAGGCGTTTTCTAAATTATGGGGTACTGATAAACTACTGGTTTCCTTCGATGG
AATGAATCTGACAATACCTGCCACTGATCGTCCTGCGTCTGAACCTGGCCCCATG
TCGATCAGAGTCCACTGCGTAAAGGAATGCAGTGCGTACAGGGCATCCTTAATTTTC
GCACCCAACGGCCCTAAAGACGGTGGTTTGATCGTTGTGAAAGGTAGTTCCAAACT
GGAAGAACTTTTTCTTTGAGGCCACAGAGATGTGCAGGGGCGTGAAACTTGGGGA

CCCGCAGATTGGTTTGGCTTCCAGCAGGACGAAGTTGGATGGTTTCTGGACAGGG
GTTGTGAGGTCTTAAAGGTATGTGCGGATCCGGGCGATCTTATTTTATGGGACAGTC
GCACCGTACATTGGAACGTTATGCCGGAATCAGATGTTACACGAGCTGCGATGTATA
TTTGTTATACACCCGCGAGCTTTGCGTCAGCGGAGGATCTGGAAATCAAATCAGCA
CTCTTTAAACAGCGTTGCGGGACTGTCACCATACACCCCCGCTTTCCTAA

NHI 258 protein sequence:

MPHYIDTTAKTGDWRDALYEDGFVVVKGVLSQDKASSYVDRMFQWLESFPFGFKRDD
PSTWNANHLPAHMKGGMFHGYAVQHEKVMWDARCEPKLIEAFSKLWGTDKLLVSFD
GMNLTIPATDRPASEPWPHVDQSPLRKGMQCVQGILNFAPNGPKDGGLIVVKGSSKLE
ELFFEAHRDVQGRETWGPADWFGFQQDEVGWFLDRGCEVLKVCADPGDLILWDSRT
VHWNVMPESDVTRAAMYICYTPASFASAEDLEIKSALFKQRCGTVTIHPRFP

Codon-optimized NHI 259 sequence:

ATGACAAATGTTGGTCGCGTGACGGCACTGCCAGTAGTTGACTTACGGGCGGACC
CTGCAGCGCTTAGAGAGCTGTAAAGAGAAGCGGCTCACCGTGTAGGCTTTTTTTTAT
CTTACTGGGCATGGTGTGCCGGGTGAACTGGTGGAGAAAGTGCTGGGAGCAGCA
CGGTCTCTGTTTGGTTTGCCTCAGGCCGATAAAGATGCCGTTGCTATGGTACGTTCT
CCGCATTTTCGCGGCTATACGCGACTGGGTGGAGAGATTACCAGAGGCGCCACTG
ATTGGCGCGAGCAGATTGATATTGGACCGGAACGTCCCCCGGTCCGGTGGGCGGGG
AAAACCTGGATATCTTTGGCTCCAGGGTCCGAACCAGTGGCCTCCAGCATTGCCGG
AACTGCCGGGTATTATCGCTGAATGGGAAGCGGCGTTATCTGGTGTGCGGAGAGAA
CTGCTGCGTCACTGGGCGGCCAGTTTAGGCTCTCCGCCAGACGTCTTCGACCCCG
CGTTTGCAGAAACCCCGCGACATTGATTAAAGTTGTAAGATATCCAGCAAGAGCC
GATAGCCCGCAGGGCGTAGGTGCGCATCGTGATTCCGGAGTGTTAACCTTATTACT
GGCAGAGCCGGGTTCTCGTGGCCTGCAGGTTCTGCTCTCCACGTGCCGGGCACGA
TGGATGGATAGATGCACCGCCTGTTGATGGGGCTTTCATAGTAAATATAGGCGAGCT
TCTTGAAGTCGCAACCAGGGGATACCTTCGGGCAACCGAACACCGCGTGAACCTC
GGTGGCGCAGAGAGAATCAGCGTCCCGTTTTTCTTCAATCCGCGGTTGGATGCGC
AGATTCCGGTACTGTCGCTCCCGGCGGACCTGGCAGCGCGTGCCGGTAATGGCTG
GACGCCGGCAAGAGACCCATCAGACCCGATTTTCGCGGTGTATGGACGTAATGCAT
GGAAGAGCAGATTACGTGCACATCCGGACGTAGCTGCTGCGCACGGTTACTCTTAA

NHI 259 protein sequence:

MTNVGRVTALPVVDLRADPAALRELLREAAHRVGFFYLTGHGVPGEKVLGAARSL
FGLPQADKDAVAMVRSPHFRGYTRLGGEITRGATDWREQIDIGPERPPVGGPGKPGY
LWLQGPNQWPPALPELPGIIAEWEAALSGVARELLRHWASLGSPPDVFDPAFAETPA
TLIKVVRYPARADSPQGVGAHRDSGVLTLLEAPGSRGLQVRSPRAGHDGWIDAPPVD
GAFIVNIGELLE VATRGYLRATEHRVNLGGAERISVPFFFNPRLDAQIPVLSLPADLAARA
GNGWTPARDPSDPIFAVYGRNAWKSRLRAHPDVAAAHGYS

Codon-optimized NHI 260 sequence:

ATGAATTTGTCTATAATGATCATAATTGGCCTCATCGTTTACAGCATTGCCGCGATGA
CCTATGTCTATCGCTGGCGTGGGAAAGCACGCTACGACAGCTTATCCCAATACTTAC
GCAAATCTTGGCCAATATTTGCCCCATTCAATTGCGTGCTATATATGACGACCCGGTC
CTTTGCCAAAAAACCGATCCTTGATGCGGATTTTCTGGAGAATATCGAAGTTTTGCG
CGCAAACCTGGACAATTATTCGCGACGAAGCAATAGCACTAGAATCCTCAGGTGTATT
TGAAGTGATAAAAACCCCGGGTGCGGACGGGTATTATGATGTAGGATTCCGCGACCT
TCTATAAGCGCGGTTGGAGCAAGTTTTATCTTAAATGGTATGGGACAACCTCATGTCTC
TGCCCAGCGCCTTTGTCCCAAGACGCTGGCATTACTGAACCAGGTTCCCGCTATCC
GTGGAGCCATGTTTTCCCTGCTGCCGGCGGGCTCAGAACTAAGTCTGCACGCAGA
TCCAATTGGCAGCAGCTTTAGGTACCACCTTGGGCTGCTCACGCCAAATGCTGAGA
ATTGTCAAATTAACATAGACGGCCAGACGAGAACATGGTTTGATGGCAAAGACTTTG
TGTTGACGAGACGTATCCGCATTTTGGCTACAATACTGCAAATTCGGCCCCGTCTGA
TTCTTATGTGTGACGTGGATCGCCCAATGTCTCTGTTTGGACGAATTTTCAATTCTG
CCTATAGGATCCTAGTTAAGGGTACTGTTGTTCCCTAATATGCCAGAAGATAAACGTGG
AGTATATAGTGCGATTTTTAAATTCTTTGCTCCTTTTCGTCAGTACTCAATACAATTTA
GGGAAAAACATTTTAAACATATAAGATACTGAAATTTATATTAAATTTGACGCTGCTTA
GCCTAATTTTTATGCTTCTGTACGGGGTTCGTTTATTTGTTTGAGATGCTTTTTTTAATG
AATTAG

NHI 260 protein sequence:

MNLSIMIIIGLIVYSIAAMTYVYRWRGKARYDSLQYLKSWPIFAPFNCVLYMTTRSF
AKPILDADFLNIEVLRANWTIIRDEAIALESSGVFEVIKTPGADGYDVGFRIFYKRGWS
KFYLKWYGTTTHVSAQRLCPKTLALLNQVPAIRGAMFSLLPAGSEL SLHADPIGSSFRYH
LGLLPNAENCQINIDGQTRTWFDGKDFVFEDETYPHFAYNTANSARLILMCDVDRPMSL
FGRIFNSAYRILVKGTVPNMPEDKRGVYSAIFKFFAPFRQYSIQFREKHFKEYKILKFIL
NLTL LSLIFMLLYGVVYLFEMFLMN

Codon-optimized NHI 261 sequence:

ATGGAAGGCCAGTACCCGGGAGGTAGCGCTCCTCGTCGGCGTCTGCTGAATCAAG
AGGAATTGGCTATTACCAGTCGGTTGGGTAGTGCTGTGCGCCAGATGAATCGTAGT
GGCAACTTCGCATGGGGGGGTGTTTTAAGTTGCAACCTTCGAATTTATTATGAACGG
TTGGGAACCTGATGAGGCTAAAGGTCAGCTTAAACTGACAACACTGCCTTTTCGACGA
AGAAGGTGCGAGGGGAAATATATCTGGCAGGTACTCCGTCACCGTTTGGGCGAGGC
AAACAGACCGTATATGATGAAGAGTACCGCCAGGCTAGAGAAATCAAAGTTCGCGAT
TTTGCGCTGTCTCCAGACCCTTTCCTGCAGTCCGGTCTGGCAGGACTGTTATCGCG
CAAACCTGGATTATGACTCAGCTCTGATTTTCCGCTCTGAACAAATTAAACGCCTATGGT
AAAGGCGGTTTTCTTTAAAGAACATCAGGATACTAGCCAGGGTGTTGGCCATATTGGC
ACGCTTATTCTGTGTTTACCCTCACCATTAAAGGTGGCGATCTCCTTGTCGCCCAT
GGAGGACAGACCACAACCTTCGACTGGTCAGATCAAGTATGTGATGGAGAAATAGC
ATGGGGTTTCCTGTACTCAGATTGTAAGCACGAGGTACTGCCTGTGGAAGATGGGT
TTCGTATTACCCTTCAGTATGATGTATTCGTCAGTGATTGTAGCAGCGATTTCAGACTC
CGATAGTGATAGCGATTTCAGACTGCGCCTATATGTATGATACGCGTTTGAAAACGGTA
AGCTCAGCATTTTCATGAAGTGATCCATGGGCCATTTCTTCCGGAAGGCGGAGTTCT

GGCCTTCGGTCTGCAGTATGAGTATCCTATCGAAGAAAGGTATAAAAACACTACCTT
AGGTAATTTGGAGTTTAGGTTGAAAGATATTGACGCCGTCTTTATGAAATGCTTGAAG
AGCACCGGGCTGAAATGGGAATATTTCCGGTGTATTTGACACTTGGGAGCTGTTCTGA
AGACCGCCAAAGCAGGAGTTCGGATTGTTGGGATGGGTTCGAGTTCGCCTGAATGG
GCCGAAGACCTTGCGAAAAATGCAAATGCTGAAACTGAACACTCTAGCGATGGAGC
CAGCACGGGTGAAGATGACACACCAACGGATTTCGGCTCAGAGTAACCAGCAGAGA
CGCAAGTCTGACCCTACCTATCGGCGCGATTATGGGTGTCTGAACAGCATCCATGG
CGTCGATGGACAGTTCCTTGAAAGCGGGTCCTTAGACGAGCTGGATTTAGAACGAA
AGAGCGAAATCGTTTGGATTACTGAACCTTATTATTATAATGTAAAAAATGATTACCAT
ACTTATGGTAATGAATACCAGAAAGAAACGATTATGCAGCGGTTGCCGTGAAGGTA
TTCTTGCCGGCGGTTGGTGACGCTATCCGCCAACCTCGTCCTACTACCGAACCACT
CAATAAACGAAAACGTA CTGATTAA

NHI 261 protein sequence:

MEGQYPGGSAPRRRLNQEELAIT SRLGSAVAQMNRSGNFAWGGVLSCNLRIYYERL
GTDEAKGQLKLTTLPFDEEGAREIYLAGTSPSPFGRGKQTVYDEEYRQAREIKVPHFALS
PDPFLQSGLAGLLSRKLDYDSALIFRLNKL NAYGKGGFFKEHQDTSQGVGHIGTLILCL
PSPFKGGDLLVRHGGQTTTFDWSDQVCDGEIAWGLYSDCKHEVLPVEDGFRITLQY
DVFVSDCSSDSDSDSDSDSDCAYMYDTRLKTVSSAFHEVIHGPFLPEGGVLAFLGLQYE
YPIEERYKNNTLGNLEFRLKDIDAVFMKCLKSTGLKWEYFGVFDTWELFEDRQSRSSD
CWDGSSSPEWAEDLAKNANAETEHSSDGASTGEDDTPTDSAQSNQRRKSDPTYR
RDLWVSNSIHGVDGQFLESGSLDELDLERKSEIVWITEPYYYNVKNDYHTYGNEYQKE
TIYAAVAVKVFLPAVGDAIRQPRPTTEPLNKRKRTD

Codon-optimized NHI 262 sequence:

ATGACATCGGACTCAGCATTGCAAGAGGGTATGATGAAAACGGTGGTTGAGAGCTT
TGAAACCGTTGCATTACGTACACACCTGGAAGCGAACAACGGTATCAAAGGCCTGG
AACTATTACAACCACATGAAGTGGACCGGGCTGTAAACTCTTCCAGAGGGATGGC
TTTGTAGTAGTAACCAACGTTCTTGACGAAGAACAGACAAGATTCTGGCTAATGGC
TGTTCCGAAGTCGCTGCTCAGATCCTGGCGCTGGACCGCGAGCGCTCTGGAAATC
GTGGTTCACATCGTTACAGTTTCGGAGGCAGTTCGCTCACTCGTTCTCAATTACATA
GACCTGAGTGGCAAATGCTGTTAGAGATTGCTGTAGTGGACGAACTAGTCACGGCT
ATTTTTGGCTCTGGTGACTATTCCCTGCGCGCAGCATCAGGGGACTTTTGTCTTCC
GGGAGCAATTGGGTATCAACCACTGCATAGCGATGTGGGCGATTGGACGTCTGCAT
CTAAATCCCGCTTCAGTTCTTTTCACGATCCTCGTGGTCAGCTGACCATTCTGTACC
TGCCATGTCCGTATGTTTGTGTTAACTTCCTGCCACAAGACGTGACTCGTCTTAATG
GTCCACGCGCCAAATCCCGGAACCCAGCATTGCGGTGTCCCAATACCAACACTT
GAGGACGAACCGGAATGGATGAGGCTTCAACCGTTTGCCCCGCACCGGCAGGC
GCCATTATGATCCGCGACGTTCCGGGCCTGGCATGGTGGTACGCCGAATATAAGCGA
AAAAATGCGGGCCATTCCAACTTAGAATTCTACGCACCATGGTTTCGCGAGCCTAT
AGTCCCTGGCATATCATACTCTGACCATAAAAAATTGTCTAAACGTGCCAGTTTCTG
TCGCGTTTCTGTGTGCGCGATTTCATCAGAACCGCTGAAGACCGAGGCCACCCTATG
GGCGCCCTGA

NHI 262 protein sequence:

MTSDSALQEGMMKTVVESFETVALRTHLEANNGIKGLELLQPHEVDRAVKLFQRDGFV
VVTNVLDEEQTRFLANGCSEVAAQILALDRERSGNRGRSHRYSFGGSSLTRSQLHRPE
WQMLLEIAVVDELVTAIFGSGDYSLRAASGDFCLPGAIGYQPLHSDVDWTSASKSRF
SSFHDPRGQLTIRDLPCPYVCVNFLPQDVTRLNGPTRQIPGTQHRSRVPITLEDEPEW
MRLSTVCPAPAGAIMIRDVRAWHGGTPNISEKMRAIPNLEFYAPWFRPIVPGISYSDH
KKLSKRAQFLSRFCVADSSEPLKTEATLWAP

Codon-optimized NHI 263 sequence:

ATGAATCTGTCTGAAGAACAGAAAAAATATTTCAAGAAACGGGCATACTACATTTAA
AAAAAGCTCTGCGTCCGAATGACGTGAGCATTGTACAACAAGCGATACTGTCTGAG
CTTGAACGGTTAAAGCTGAAAGTTGGCAGCAAGTTAAGCTCCTCCAAGATAAATGAT
CTGCCATTATTTTCAGCAGACAGGTCGGCTCGGTCAGATGGTTAAAGTGGAGAGCCA
GTTAACCACCTCTTTTTCGTCATGATGTAATTGTAGCCGCAGTTCGGAGCTTAGTCTC
CAACAACCTCTCCCGCCCTGTCAATTCACCCGCAGTTATTATTGAGCTTCCCTCACAA
AGCGGAATGGTCCTTAGCGAAGTTAAATTGGCATCTTGATATCACCGTTCCGAAGCA
AAATGTCCTGGTCGGCGTGCAGGCGTTTGTGTTAATTGACGATGTACAGACAAGAG
GGGGCGCTACCCTGGCAATAGCTGGCTCCCACCAGCTGCACTATCTGCCGCAGGC
TCAGGGTGGTAACGCACATAAATTATTGAGGGACGATCCACTTTTCACTCAGTTATAT
TCTGGGGACTCTGTTGACGCGGAAAACTGGCCAAGCCACATAATATCCATGGCGT
TGACGTTCTGGTCATTGAAATGGCAGGTAAGGCGGGCGATGTGTTCTTTATGGATAT
GCGGGTCTTACATGCACCGAGTGTGAATGCTACTAAAAAATACGGATGATGGCCA
CTAATAGGTTTATATAA

NHI 263 protein sequence:

MNLSEEQKKIFQETGILHLKKALRPNDVSIVQQAILSELERLKLKVGSKLSSSKINDLPLF
QQTGRLGQMVKVESQLTTLFRHDVIVA AVRSLSVSNNSPALSIHPQLLLSFPHKAEWSLA
KLNWHLDITVPKQNVLVGVQAFVLIDDVQTRGGATLAIAGSHQLHYLPQAQGGNAHKL
LRDDPLFTQLYSGDSVDAEKLAKPHNIHGVDVLVIEMAGKAGDVFFMDMRVLHAPSVN
ATKKIRMMATNRFI

Codon-optimized NHI 264 sequence:

ATGGCTGTGTCTCCCGAAAGCATGCCGGAAAAACCCTTTGATTTCCGCGCCCCCCC
TCCGAGTCCTATTGCGAGCGGTAGACGTAGCTGCGTGACGAATGATGACGTATTAA
GCGAATTTT TAGAGCATT CATTACGAGTTCCGGATTTGATTCTGCCGGATAAAGTATT
TCCCGTCAGAAATTTATTGAAAATCCGCCGAAAATCGATTTCCAAC TGTCTCATC
CATGGAATCAGATACTCTGCCTAAAATTCTGGACTCAATAGCAACAATCGGTTGCTTT
CAACTCGTAAACTACGGAATCCCAGGCGAAAGCATTCTGTCAGCCCTGGCTGCTGC
GGCCGGTATCTTCCAGCTGCCTCCAGAGAAACGTGCAGCAGTTACACGTTACACCG
GAAAACTGTACGGGTTTGAAGAGATACATGAAGAAGAAGAAGGAGAAATGTCCGA
AGAATTTGTTTGGTGTGCGCGGAGAAGGATTAAACTGGAGATGGAAGGGATCTGGC
CGGTTGGCTATTCAAAC TCACTGAAAAGATGGAAAACCTTCTGTGCGGATATAGAAA

AAGTTGCAGAAAAGATTTTGTGGTTATCACCGAGAGTTATCCGACGAAACCGGGC
TACGAAAATGAGATGATGCAGGGACAAGATGTAGTGGGGAGTGCATGCTACTTATAT
AAGCATTACAAAACGTGAGCGCCGATAAATGGTCGAGCTCATTAAAGATATGACGTT
ATTCGAATGTTAATTCGCGGTATTGACTACTCACATGCTTTGTGTCTGCACATTTGTG
ACGGTAGCTCAGAGTTTCATGTGTATTCTAAGAAAGGCTGGGTCTCCTTTTCGACCG
GATAAAGATGCGCTGGTGATCACGATCGGAGACCAAACACAGGCATTGTCTGGTGG
CCAGTTTAAACATGTAATCGGCAGGCCGATCTATAAAGGAGAAAAGAAGGAATATATC
TCAATGGCATTCCCTGTATACTCCTCCGTCAATCACAAGTCGGCCGGTGGATCCAGAA
AAGGGTAAAACTATTTTCGCTGAGCCAACAGGCAATTGCCGCAATTGTACTTACGCTT
GTGTATCACATTTTGGTATATGTATATAACAAATTTTAA

NHI 264 protein sequence:

MAVSPESMPEKPFDFRAPPPSPIASGRRCVNTDDVLSEFLEHSLRVPDLILPDKVFPR
QKFIENPPKIDFQLLSSMESDTLPKILDSIATIGCFQLVNYGIPGESILSALAAAAGIFQLPP
EKRAAVTRSPKLYGFEEIHEEEEGEMSEEFVWCRGEGLKLEMEGIWPVGYSNFSEK
MENLLSDIEKVAEKILLVITESYPTKPGYENEMMQGQDVVGSACYLYKHSQNVSADKW
SSSLRYDVIRMLIRGIDYSHALCLHICDGSSEFHVYSKKGWVSFRPDKDALVITIGDQTQ
ALSGGQFKHVIGRPIYKGEKKEYISMAFLYTPPSITSRPVDPEKGKTISLSQQAIAAIVLTL
VYHILVYVYNKF

Codon-optimized NHI 265 sequence:

ATGCCCAAAGGACGTCCGAATACGATGAATAATTATGGTGTCTTCTCCATGAACTC
GGCCTGGATGAACCATTAATGACGCCTCTGCGCGAACGTTTCCTGCAACCACTGAT
GGCGCTGTTGTATCCAGACTGTGGGGGTGGTCGCTTGGATTTCGCACAGAGCATTT
GTAGTCAAGTATGCACCTGGCCAAGACTTGGAGCTGGGCTGCCACTACGACAATGC
CGAACTGACATTAAATGTGGCACTGGGTAAAGTATTTACCGGGGGCGCCCTATATTT
TGGCGGGCTGTTCCAAGCACCGACTGCATTAACCGAGCCTCTTGAAGTAGAGCAC
GTTGTAGGGCAGGGCGTATTGCATCGGGGGGGACAATTACATGGGGGCACGCCCGT
TAGGGACCGGCGAGCGGTGGAACCTCGTAGTGTGGTTGCGTGCGTCTGCCGTTAG
GAATTCATTGTGCCCTATGTGCTGTCGCGAACCTGATCTAGTTGATGACGAAGGATT
TGGTGATGGCTTCACTCGTGAAGAACCGGCCACAGTGGACGTGTGTGCTCTTACC
TAA

NHI 265 protein sequence:

MPKGRPNTMNNGVLLHELGLDEPLMTPLRERFLQPLMALLYPDCGGGRLD SHRAFV
VKYAPGQDLELGCHYDNAELTLNVALGKVFTGGALYFGGLFQAPTALTEPLEVEHVVG
QGVLHRGGQLHGARPLGTGERWNLVWVLRASAVRNSLCPMCCREPDLDDEGFGD
GFTREEPATVDVCAALT

Codon-optimized NHI 266 sequence:

ATGGTAGAGTTAGAGCAGGAAGTGGAGAGACGCCAGAGATTGGGACAGGAATCCG
CGGCTCGCAAAGCTCTCATCGCCTCATCTTACCACCCGGCGCGCCCGGAGGTCTA
TGATTCATTGCAAGACGCTGCTCTTGCTCCCGAATTCCTGGCAGTTACGGAATATAG

TGTTTCTCCGGATGCCGATCTCAAAGGACTCCTTCAGCGCCTCGAGACTGTGTCCG
AGGAAAAACGTATTTATCGTGTACCGGTTTTTACCGCTCCATTTTGTGAGGCACTGC
TGGAAGAGTTGGAGCATTTTGAACAGTCAGATATGCCTAAAGGCCGTCCAAATACCA
TGAACAATTATGGCGTCTTACTACACGAGCTGGGTCTGGACGAACCCTTGATGACA
CCACTGCGTGAACGGTTTCTGCAGCCTTTAATGGCTCTCTTATACCCGGAAGTGTGG
CGGAGGTCGTCTGGATAGCCATCGCGCATTTGTAGTGAAGTATGCCCCAGGCCAG
GATCTGGAGCTTGGGTGCCACTACGATAACGCGGAACTTACCCTGAACGTAGCTCT
GGGTAAAGTTTTTACAGGTGGGGCACTGTATTTTGGCGGTTTGTTCAGGCTCCCA
CCGCTCTTACCGAGCCTCTGGAAGTCGAACATGTCGTAGGTCAAGGCGTGTTACAT
CGTGGGGGTCAGCTGCATGGTGCCCGACCGCTTGGGACAGGTGAAAGATGGAAC
CTGGTGGTTTGGCTGAGAGCCAGTGCGGTAAGAACTCACTGTGCCCGATGTGTT
GCCGAGAACCAGATCTTGTGCGACGATGAGGGTTTTGGAGATGGCTTTACTAGAGAA
GAACCAGCTACGGTTGATGTCTGTGCCCTTACATGA

NHI 266 protein sequence:

MVELEQEVERRQRLGQESAARKALIASYHPARPEVYDSLQDAALAPEFLAVTEYSVS
PDADLKGLLQRLETVSEEKRIYRVPVFTAPFCQALLEEHFEQSDMPKGRPNTMNNY
GVLLHELGLDEPLMTPLRERFLQPLMALLYPDCGGGRLD SHRA FVVKYAPGQDLELGC
HYDNAELTLNVALGKVFTGGALYFGGLFQAPTALTEPLEVEHVVGQGV LHRGGQLHGA
RPLGTGERWNLV VWLRASAVRNSLCPMCCREPD LVDDEGFGDGFTREEPATVDVCA
L

Codon-optimized NHI 267 sequence:

ATGTCCCAGAGCACGCCATATAGGCAACAGGGGCTGTGTTATGATCGATACAGATCA
ACCCACCGCGCTAAGCAGATGTTGTCCGATGCCAAAGTGCGTGATGCCGTAGAGC
AATATCGTGAAGAAGGTTATGCCATCGTGCGCCAGTTTGTGCGATGGTGACGAAATGA
TACGTTTGCCTAAGGAGACAGATGCTGTATACGAAGAAGGCCTGAAACACCATGCC
ACGTATCGTCATGGTAATTTGAGCTTTCAGATATTGCCTGAAAGCGACTTCGGTAAG
CGGTATGTCCTGCAAGCATATTGGTTTTCTTGGGTTAATTCATACTTCGAACAATTCC
GTCGCAACCCGGACTATCTGCGATTGTTAGAAGGTCTGATCGGACGTGATATACGTC
AGATTGCGCAGCAACTTCACTGGAACCGCCGGGGGCAAATCTTACGGGCTATCG
CTACCATCAAGATCTGAGGTTTCGCAATCAAGCGGCATTTGATAACGTGGCGGATG
CAACGGTTACTACAGGGTTGGCCGTTGACCGCGCGACCTTAGAAAATGGTTGTCTG
CAGGTGGTTCCGGGCAGTCATAAACTGGGGTATCTTGGCTTGAGTGATGAGGGTAC
CGGGGAAGTATGAAAGGGCTCACAGCGGAAGATGAATTACGTAAAGTTGGGATC
GATCCGGCTACTATCGTCCCGTTGGAACCTCGAACCGGGTGACTTGGCTATGTGGGG
TTTGCTGACTGTACATGGATCTGCACCGAACAAGCCAAACATGATCGAGCATTG
CGTTGTCTTCCTATGTACGCGCCGATACAACCTCAGCGGGGAGAATGGGCTTTTGA
GACGGAGTTTCGGTTCCACTGGGTGATGCGCCAGTCCTTTGTAAGTATGAAAACT
GTTTGAAGAACCAGGCCCCATGTACGACACAACCGCGTGGTGGAAGTGA

NHI 267 protein sequence:

MSQSTPYRQQGLCYDRYRSTHRAKQMLSDAKVRDAVEQYREEGYAIVRQFVDGDEMI
RLRKETDAVYEEGLKHHATYRHGNLSFQILPESDFGKRYVLQAYWFSWVNSYFEQFR
RNP DYLRLLLEGLIGRDIRQIAQQLHWKPPGANLTGYRYHQDLRFRNQAAFDNVADATV
TTGLAVDRATLENGCLQVVP GSHKLG YLGLSDEGTGELMKGLTAEDEL R KVGIDPATIV
PLELEPGDLAMWGLLTVHGSAPNKSQHDRAFALSSYVRADTTQRGEWA FRDGVSVPL
GDAPVLCKYEKLFEEPGPMYDTTAWWK

Codon-optimized NHI 268 sequence:

ATGGATTTGATTGCACGCTATAGTAATGCTGGGTATGGAGCTGTTGCCGATGGGGTC
ATGGCCTTCTTTGATAGACGCCCAGATTTACAACGCCCTGGCGTGGCGTTTGGTCC
AGCAGATGCCGCGGAACCAGCCAAGGTGTGCGACTGACATTAGCCTCGTGGCGATC
GATAGATCGGACCCGGAGGCATTGCGCCCTGGCAGAAAGTGATCGTGCGGGGCGTTT
CGGCCGCATTAGACAGATACCTCCAAGAGCGTCCACTGTTCCGACAGGTCTGCCC
GGATCAGGAATTGTTCTGCTTACCGATTTTCAACCTCCAGAGATACGCCCTGGTGA
AGGATTCCGGCAGTGGCATTGTGATTGGACAATCGGGGATGAAGCCACTGAACCT
GTCCATCGCGTTTTAGCTTGGATTCTTTACTGCAATAGCGTACCTGAAGCCGGTACC
GAGTTCCACTGGCAGCAACATCATGAAGACGCGGAACGCGGTAAGCTGGTCATATT
TCCGGCTGGCCCGTCTCACATCCATCGCGGTCGTGTTAACCATACTCATTCCAAAA
CCATTGCAACGGGTTGGATAAATGCTGGCTCGCGTCAAAGTTTTTTGGAACGTCTC
GCAGGCTTAAGTGTGGAGGCAGCGCCTGGCTGGTACTGA

NHI 268 protein sequence:

MDLIARYSNAGYGAVADGVMAFFDRRPDLQRPGVAFGPADAAEPAKVSTDISLVAIDRS
DPEAFALAEVIVRGVSAALDRYLQERPLFRQVCPDQELFVLPPIFNLQRYAPGEGFRQW
HCDWTIGDEATEPVHRVLAWILYCNVPEAGTEFWQHHEDAERGKLVIFPAGPSHI
HRGRVNHTHSKTIATGWINAGSRQSFLERLAGLSVEAAPGWY

Codon-optimized NHI 269 sequence:

ATGAAATACTATCGCCAACAGTCAAATGCATTCCCGGCAAGTTACCTGAATGACCTG
TTTGGAGAAATTAAGGCGTGTCCATATTTCTGTAAC TAACAACCTGAATCGTGATTTTG
TTGCCACTAAAGGTTTTTCCGTAGTCTTT CAGCAGTCGGGGCTCATGGAAGTTGAG
GAAAAGTTCCCGTTCTTCAAGCGCTACCTGGACGTTGCCCTTCAGTCAAAC TGCAA
TGCATTTTATCTGAATCCGCTGCTGCTGCAAAGTG GATCAAGAGTCGACCCGCATAT
AGACCGCAGTCTTCGTAGTTACTGTAAACTATTGAACCGCCGCTGTTTGTTAGCGT
GCTTTATGTACGTATCCCAGAAAACATGGAAGGTGGCGAACTTATTTTACGTTGTCAA
AAACGTCAGGTAGGCCAAATCAAACCGGAAATAAATAATCTCGTCTATTTTCAAGGT
GATCTGACTCACTCAGTAAATGCGGTGAAAACCA CGGGTAATCGGCTTAGCCTGGT
TTGCGAACAGTACTCGCTGAGCGATT CGGAACTTGAAGAGATCCCGAATTCCTAG
TAGAATCGCGCATCGTACAGCCTGCCAAAAAAAAGATGCGTTAA

NHI 269 protein sequence:

MKYYRQQSNAFPASYLNDLFGEIKACPYFVTNNLNRDFVATKGFSVVFQQSGLMEVEE
KFPFFKRYLDVALQSNCFYLNPLLLQSGSRVDPHIDRSLRSYCKTIEPPLFVSVLYVR

IPENMEGGELILRCQKRQVGQIKPEINNVLVYFQGDLTHSVNAVKTGNRLSLVCEQYSL
SDSELEEIPNFLVESRIVQPAKKKMR

Codon-optimized NHI 270 sequence:

ATGCATTTACATAGTGATCGCATCGCAGCAGCATATGATCAGCTAGACCGTCGTCAG
AAACATGCGTTTTTCGTTTCCCGAACACACCGATGGCTTCCTGCCGTTTGGATTAGA
GTATAGTCGGACTCCTGAACGTCCGGACCTCTGCGATCGCTTTTGTTACTGGGTTG
CGCGTAGAGATGCCCACCACGGATATGCCCTCACAAAAAGCGCTTTCTATCGAGAA
TTAGCTGCATATGAGGCTTGTATAAATGCAATTGCTCAGGACCTTCTGGATGCGGTG
TGTGAAAATCTGGGAGGAACGGCACAGCCGCCGGTACGTCACTCTAGCTATCTGCA
GTTCTGTGCCTATGAACCGCATCACTGCTCACAAGATCGTGCATATCTGCAGGATCC
CCACGAAGATGGACACCTGCTTAGTTTTATTCTGCTCCTGCCGGCGCCCGGCCTGGTC
CTGCTGCCGGATAGCTCACCTGTTAGTGCGTTGCCACGTCCAGATGAAATGATAGT
GCTTGCCGGCTCATTATTGGAAGCGCTGACTGACGGAGAAGTACCGGCGATGCAA
CACGCCGTAGAACGGCCGAGAGCACGCGTGAGACGTACATCACTTATGTACTTTGT
GAATCCAGATCTTAGCGCGGGTTACGCTAGTACAATTGTTCATAGACGGCCTCTGGA
TCTACGCGCATTAGCAGAACAGCGTCACGTAGCTTTTGGAATTCGACCTTAGGAC
GTCCGTGTGCCGCGTAA

NHI 270 protein sequence:

MHLHSDRIAAYDQLDRRQKHAFSFPEHTDGFLPFGLEYSRTPERPDLCDRFCYWVA
RRDAHGHGYALTKSAFYRELAAYEACINAIAQDLLDAVCENLGGTAQPPVRHSSYLQFCA
YEPHHCSDRAYLQDPHEDGHLLSFIRPAAPGLVLLPDSSPVSALPRPDEMIVLAGSL
EALTDGEVPAMQHAVERPRARVRRSLMYFVNPDLASGYASTIVHRRPLDLRALAEQR
HVAFGNSTLGRPCAA

Codon-optimized NHI 271 sequence:

ATGAATCCTTTGCCTGGCGTTCCCTTAGTAGAATCTCCACTATTCCCTAAGCTGAAG
GACTCATTGGGCCTGACAGCGGAAGAAGAACGAATTGCAACATCATTACATGAAAAT
GGTTATGCCGTGCTTGATTTCCAGATGCGGATCTGCATGCTCGCATCGAACGCAT
CAAACATAATCTTGGCCCTCGGTATGGTATTGACTTTAACGATCCTCAGTCCGATAAG
ACCGTTGGGGAGCGCAGAATCCAGGATGCATGGACCTTTGATGAAGACGTGCGTG
CAATAGCTGCAAACGCTGCTGTCATGGATCTGCTCTCCAAGCTTTATGGTCGCAGA
GTGTTCCCATTTTCAGACTTTAACTTCCAGTTGGTACCCAGCAAGATGCACATTCA
GATTCTGTACATTTTAGCTCTCTGCCGGAACGATTATGTGTGGGGTTTGGCTCGCC
ATGGAAGACATCGGCGCCGATGCGGGTCCACTGTTCTATTACCCGGGGTCCCACC
GCTGGCCCATTATGACTAATGCGATGATTGGGCGTCGTGGCTTCGGGAGCGATCTG
CAGTCAGCACAGGATCCGTACGGTCCGGCCTGGGATGCCATGCGAGACGCCTGC
GACGCGCGCCAGGATATTTTTTTAGCGAGAAAAGGCCAGGCTCTGATTTGGTGTGC
TAATCTGCTGCATGGCGGGACGCCGCAAGCCAATCCGCACCTGACAAGATGGTCG
CAAGTTACTCACTATTATTTTACGACTGTATTTACTACACGCCCGCTTTTTTCGGATG
AAGCACTCGGTCAGCTGAGATTACGCGACTTAGTTGGGATCGATGATGGCCAGCCG

CGGTAAACCGCTATCTGGGGGAAGCGCTGCCAAAAACACCAGTAACACCGCGTG
CCAGCTTCCGTCGTAGGCTGATCAAACGTATACGAATTTGA

NHI 271 protein sequence:

MNPLPGVPLVESPLFPKLKDSLGLTAEERIATSLHENGYAVLDFPDADLHARIERIKHNL
GPRYGIDFNDPQSDKTVGERRIQDAWTFDEDVRAIAANAAMVMDLLSKLYGRRVFPFQT
LNFPVGTQQDAHSDSVHFSSLPERFMCGVWLAMEDIGADAGPLFYYPGSHRWPIMTN
AMIGRRGFGSDLQSAQDPYGPAWDAMRDACDARQDIFLARKGQALIWCANLLHGGTP
QANPHLTRWSQVTHYYFDDCIYYTPAFSDEALGQLRLRDLVGIDDGQPRLNRYLGEAL
PKTPVTPRASFRRLIKRIRI

Codon-optimized NHI 272 sequence:

ATGCAGCAGCAGGCTTCATCTGCACCAGCGGCTTCTTCTCGTCCAGATAATCCCTC
ACCACCACCAGCACCTCCGGCCAGCTCGGCGTCAGCAGTTAAACGCAGTCCGGCA
GCCACACCTACCGCCCCGCCATCACTCGCCCCGCTTAGCTCCGCGCCTCATCGAC
GGGACGCACCCGGTGGCCCGGCTCCCCACCTTTACTGGCTGCCGCCCATCCTC
CAGGCGCTGAAGCAGCTGTTACTAGCAGCATGGCTCGCGTTGCGCTTTCCGACATT
GCCCCTTATGACGGCGCCCCAGCGGGTCCTTATTTACGGGCGGTCTGAAGCACTGT
CAGGTTCTCTCACCCGTCATAACGCCGCGGTAATCGAATTGGGCCCGGAAGATGCA
GCGTTGATGCGTTGCGGGCTCGAGGCAGCGAACTTTATTTGCGGACTCGTAGCC
AAGCTCCAGCCGCAGGCAAAGGCTCACCGCGAGGGGTGTACTTATATCGTGCCGG
CCGTCCAATTGAAGATGGCGACTCAAGTCCTCCTTGTATGGGTGAGATCTTCAAATG
CATGGGTAAAGTTGCCCGTGCCGCATTGTGCGCCATCGCACGCCACTTGCGGGCTC
CGATCTGATGCGTTCAACCATTTACTTGATGATATTCGTTACCTGCGAATGAGGTTT
CAAGCTCGGTATTACTGGCCTCGTATAGTCCCAATACTCTCCAGAATGGCAAAATAA
CCATAAGTGGTGGGAACTCTCAGTTAATGCAGAAGTTGAGAAAGGACTGCTGACG
CTGATCAGCAGTGACAACCCGGGTGTCCAAGTGTTTGACCCGAACGGTTCGTTGGT
ATTTAGCAGATGGGGGCTCTTCTTTTCGGAGATCTGCTTCTTGTGACTGGCAAAGCAT
TATCCACATTACAGCCGGTCTTAGACCAGCCGCACTGCATCGCGCTGCACCCGAT
TGTTTTGCAGGTAGTTTTTGAGGGCGGACGAACGTCTCTTTCGCTTTTCGCTTAATGCC
GCAAAGCAATGCTGTGCTGGATTGTTTCGCCGATCGCTGCAGCCGGACACGTTATTC
CCCAGTCATACGTTAGAGTTTCTGTTAGTCAGTTCATGGATGACTTGAGCGCTGAGG
AAGATGCACTGTGCCCCAATCCGGAAAATGTCTGTGTTCCGGGTAACAATAGCAAC
AAGGAGCCAACCTTTAAGGAGTGTTCTGTTCGGATCCAATTAGTGGTGCGTTTTTGG
AGACGCCGTGGTCGTGAGCTGTGGCCACTCCTTTGGAGGTTTCATGTTAAAAAGG
GTAATTGAAACGTCTCGGTGCACATTATGTTCCGCGGAGATCAAAGGTAATACGTTT
ATACCAAATCATGCCCTTCGGGCGCGCAGCCGCGCTGTAAAATTGGAGGATGATCG
TCGTCTGTTCCATAACGCACAACCTTCGTAAACGTCGTAAAGAAATGGGGGATCAGG
CCGATCCGATGCGGCGTTTACACCGTGAAAACGGTGAGTCTGGTGCAGACAATGG
GCTCCATAAAGGTGTTTCAGTACCCATTTATGGTTAACGAAAAAGTGTTAATCAAGGG
GAACAAACGTACTCCGGAAAAATTTGTGGGTAAAGAAGCAATAATTACCTCCCAATG
TCTGAATGGCTGGTATCTGTTAAAAATAATAGAAAGCGGTGAAAATGTACGTCTGCAA

TATCGAAGCCTGGAGAAAATCCCAAATTCGAATTCGATTGATGATCGTTGCCCATCA
CAAATACAGAACGGCTCATCCTAA

NHI 272 protein sequence:

MQQQASSAPAASSRPDNPSPPPAPPASSASAVKRSPAATPTAPPSLAPLSSAPHRRDA
PGGPAPPPLLAHAHPPGAEEAVTSSMARVRLSDIAPYDGAPAGPYLRAVEALSGSLTR
HNAAVIELGPEDAALMRCGLEAAKLYLRTRSQAPAAGKGSPRGVYLYRAGRPIEDGDS
SPPCMGEIFKCMGKVARAALCAIARHLRLRSDAFNHLLDDIPLPANEVSSSVLLASYS
NTLQNGKITISGGKLSVNAEVEKGLLTLISSDNPGVQVFDPNGRWYLADGGSSFGLLL
VTGKALSHITAGLRPAALHRAAPDCFAGSFGGGRTSLAFRLMPQSNVLDSPISGAAAGH
VIPQSYVRVSQSQFMDDLSAEEDALCPNPENVCVPGNNSNKEPTLRVLSDPISGAFL
EDAVVVSCHSFGGFMLKRVIESTRCTLCSAEIKGNTFIPNHALRAAAAANKLEDDRRL
FHNAQLRKRRKEMGDQADPMRRSHRENGESGADNGLHKGVQYPFMVNEKVLKGNK
RTPEKFVGKEAITSQCLNGWYLLKIIESGENVRLQYRSLEKIPNSNSIDDRCPSSIQNG
SS

Codon-optimized NHI 273 sequence:

ATGCCGGCCAAAAGAGATCTTGCGAGGTACGTGCAGTGGTTTGATAACGCACTGGA
TGCTTCGCTCTGTCAGCGCATGATCGACTCTTTTCAGCAACTGAGCCGCTTACAGA
CTCGTAACGAACGGGGCTCGATAAAAGCCCTGACCTCTTCTGCCTGGACCGAACT
GAACGTGAGCAAGGTTGCAGATGCAAGTTTTGAAGGGCTCTTTCGCGAACAGATTC
AGCGACACCTGGATTTATATAATGAGCGCACAGGCCTTACTCTGCCGGTGCCAGCA
CGTGGAAGACTGGAAAATTTGCGAATCAAGCGGTATCTCGTGGAAGCCGCGGATCA
GTTTCAACCGCACTTTGACGCTCTGGATTATACGTGTAATCGATACATGGTGTTCTTG
TGGTATTTGAATGACGTGAGCGAAGGTGGAGAGACGGAATTCTGTGACCTTGATATA
CGTGTGAGCGCCTGTGCAGGTAGACTTCTGATGTTTCCACCATACTGGATGTTTCA
GCATGCTGGTCTGCCGCCGAGATCGAATGATAAATATATAATATCGACATATCTTCTAT
TTAG

NHI 273 protein sequence:

MPAKRDLARYVQWFDNALDASLCQRMIDSFQQLSRLQTRNERGSIKALTSSAWTELVN
SKVADASFEGLFREQIQRHLDLYNERTGLTLPVPARGRLNLRKRYLVEAADQFQPHF
DALDYTCNRYMVFLWYLNVDVSEGGETEFCDLDIRVSACAGRLLMFPPYWMFQHAGLP
PRSNDKYIISTYLLF

Codon-optimized NHI 274 sequence:

ATGCCGGAATCTACAATGCCGCAAATCACCATGATAGAGCTCACCCCGGAAGAACT
GACTTCAAATCGATTGGCTCACACAATTTACAGAAGAGCCTAGAGGCACTCCATCG
AGACGGCATAGTGGTGTTAAAAAATGCAGTCGATCCGGATCACCTGGACCGCTTGA
ATGCACGTATGGTACCGGAGGCCAAACAGCTGTACGCGCGGACCGAGACCCACCG
TAATTTTGGGCCGAATACGGGTAAATATCCAACAGGAGCCAGTTTGTGAGCCAGAGTA
TATCTTTGAAGACGTTATCGCAAACCCATTTGCGACGTGGTGATCGAGGGCATTAT
AGGCCCCCACCACCAGCTGCGGTTTTATAGTGCAAACACGGCGTTTAAAGCGACC

GCGCGTCAGCCCGTACATATTGACATTGACTTTGACATGCCTCGCATGCCGTTTCGC
ATTCTGCGTAAACATAAATCTGGTGGATACCAGTCCGGAAAACGGAGCTACAGAAGT
CTGGCTCGGTTTCGCATATTGATACGGATCAGTCGGTACTCGATCACTCAGTGAAACA
TAAGCAAGTGAAAGCTGATCTGCTGGAAGCTCGTCGTAAGATTTGCCCCGGTATCC
AACCTGGCTTGCCTAAAGGTTCCCTGGTGATTCTGTGATTTCCGGCTTTGGCACGCT
GGGATGCCGAATCAGACCGATGATCCGAGAGTAATGCTCGTGACAATTCAGTTTGC
GAATTGGTATAGATCTGAGCAAAAACCTGTTACTGCCTAAATCCTTACAAGGCAAGGT
CAATTTTGGGCGCTTGGTTCCGTGTGTTGACTGGGTAGATGATGGCTATGACTATCT
GAAAGGGGCCCATACACATGACTTTTCCCTGTTACCTTAA

NHI 274 protein sequence:

MPESTMPQITMIELTPEELTSKISGSHNLQKSLEALHRDGIVVLKNAVDPDHLDRNLNARM
VPEAKQLYARTETHRNFGPNTGNIQQEPVCEPEYIFEDVIANPFATSVIEGIIGPHPQLRF
YSANTAFKATARQPVHIDIDFDMPRMPFAFCVNINLVDTSPENGATEVWLGSHIDTDQS
VLDHSVKKHKQVKADLLEARRKISPGIQPLPKGSLVIRDFRLWHAGMPNQTDPRVML
VTIQFANWYRSEQKLLLPKSLQGKVNFGRLVPCVDWVDDGYDYLKGAHTHDFSLLP

Codon-optimized NHI 275 sequence:

ATGCGTCTTGAACGAGAACTGAGCGACGCCCAACAGCGTCTAGCCCAAGCTACGT
ACAATCCTGAGGATTTTACCGCTAATCTTGTGCAGTCGCGCAGTTACGCTGTCAACC
CGAGCTCCGGAACCTCAGCTAAAGTGGTCGATCAGTGTGTCGATTCAGTGCAGCG
CTATGGTTTTTTCGTCGTTGATAACGCCATACCGTCAAATCAAGTGAGTGTCAATCCG
GCAGGAGGTTATTCAGGCACAAACAACCATTTTCGCAGAATTTAAAAAAATTCGTGA
GCGTGTTGACAACATTGGGCTGAATCCCAGTAAACTGTTGACAGGTACCGATGCAG
AACTGCGTCCGGTCCGCCGTGTGCGGCATTTGCCCAAACCTCCTAATGACATAGTG
TGGATGCCGCAATATGCTCAATATTTGGCACACCCGGTAATTACAGCACTGGCGCG
GCAAGTCCTGGATGACCATCTGCGCATTGCCCAGCTGCATCTGCGCATTATAGAATC
AAAGCAGCCAGATGGTACACCGGGCGGGTTCGGTAAAGTGCAATATCGCAGCCGT
GATAATCGTGGTTGGCATAAGATTGGCCACATGACTTATCGGCTTATGGCGGCGAT
AAACCGGAACACAATGCCGGTTGTATCCGTCAACCGTTTCCTGACGTCACTATGTG
CCTGGTCATGATATGGTACCTGACCGATGTGACGAAAATAGCGGCGGTACGTGGG
TGGTTCCTGGCAGTCATCGCGATAAGAGGAATCCGCGCGGACCAAGTGACAACATT
AACGTAGTCGCTCCAATTCCAGGTGAAATGCAGGTTACAGCGACCGCTGGTTCTGT
GTACATACAGGACAGCCGGAGCTGGCATGCATCGCCAATGCATAACCCGGGTGGG
CAAGACCGTGTTGCCGTTATCAACCGATGGTGTCTTGGTGGCTGTCAGTTGATGA
TTATGCACCTGAAAATATTTTCAACACCATCTGCCAGCCTCTTTTCGAGAGCCGAATA
CGAATCACTGCCACCGAACTGCAACCATTCATGAAACATCTCTGCCCGGAGGTCT
GGGACACGATTCAGCAGCCAGTCCTGGATCGCGCGAAAGCAGCGGCCTTGCGGA
CGAAGTGGGGTTTTTGATCAGCTGGAAAACCTATCCGGGAATCTTAGCCAGGCCAAC
GCCCATATTCGAGTTCGGTCTCTTCCGGATAGTCATTGA

NHI 275 protein sequence:

MRLERELSDAQQLAQATYNPEDFTANLVQSRSYAVNPSSGTPAKVVDQCVDLSLQRY
GFCVVDNAIPSNQVSVIRQEVIAQTTSQNLKKIRERVDNIGLNPSKLLTGDAELRPV
RRVGHLPKPPNDIVWMPQYLAHPVITALARQVLDHLRIAQLHLRIIESKQPDGTP
GGFGKVQYRSRDNRGWHTDWPDL SAYGGDKPEHNAGCIRQFPDVTMCLVMIWYL
TDVDENSGGTWVVP GSHRDKRNPRGPSDNINVVAIPGEMQVTATAGSVYIQDSRSW
HASPMHNPGGQDRVAVINRWCPWWLSVDDYAPENIFNTICQPLSRAEYESLPTELQPF
MKHLCPEVWDTIQQPVLDRAKAAALRTKWGFDQLENLSGNLSQANAHIRVPSLPDSH

Codon-optimized NHI 276 sequence:

ATGAGATTGATTGCGTCCTTTGAGGACCCCGGCTACGGTGCCTTAGCAGATGCGGC
CATGGCATT TTTT GATCGTCGCACTGATCTTCAGCGGAGGGGGGTTGCATTCTCCT
TTGGCGGTGAACCCGGAGATGGCGCAGAAGCGGATCCTGGGAAAATCTCGACTGA
TATTCGTTAGTTTGGATTGATCGGTCAGACCCGGAGGCGCAGGGCCTTGCTGATG
TCATTATGCGTGGCGTGAGTGCCGGTTTAAAACGTTACCTGGCCGAACGTCCCCTG
TTCCTTTCATGTTGTCCAGAAAGGTCCTTTTTGTAAATCCAATTTTCAATCTACAGC
GTTATGCCCCCTGGCGAAGGCTTTTATAGTTGGCACTGTGATTGGAACACAAGTGATG
AAGCTACTGAACCTATTCGACGTGTCCTGGCATGGATACTGTACCTGAATACTGTAC
CTGACGGCGGCACTGAGTTTCATTGGCAGGATCACCATGAAGAGGCTGTAAAGGT
AAACTGGCGATTTTTCCTGCTGGCCTTAGTCATCTGCACCGTGGCAGAGTTAGCCA
AACCATACTAAAACAATTGCGACTGGCTGGATTAATGCAGGCACGTTAGATGCCTA
TGTCACAAGACTGGCAGCGGGCTAA

NHI 276 protein sequence:

MRLIASFEDPGYGALADAAMAFFDRRTDLQRRGVAFSFGGEPGDGAADPGKISTDIS
LVWIDRS DPEAQGLADVIMRGVSAGLKRYLAERPLFLSCCPERSL FVNPIFNLQRYAPG
EGFYSWHCDWNTSDEATEPIRRVLAWILYLNTVDPDGGTEFWQDHHEEAVKGKLAIFP
AGLSHLHRGRVSQTHTKTIATGWINAGTLDAYVTRLAAG

Codon-optimized NHI 277 sequence:

ATGTCTCAATTTATCGAAGTTATTGATAATGCGCTACCGGAAGAACTCTGCGAGAGT
CTAATCCGCCAGATTGACACCTCACCTAATCGTCAACCTGGCCGGACAGGTGGTGG
CGTTGATACATCTAAAAAGGTCTCTGAGGATGTTTCAATCTGTACGCATGATGAATTC
GCATCGCTTTTGCCTAAGGTGTTAGAATGCACCACATCTCATCTCGTTGAATATTTTCG
CTAAATACTATTTGCACTGATAGGCCCAATTGGTCTGACACTCAAACATCCGAAAA
CTGGTGAACCTGTAAAGCTGACCCAGGATAATTTGATGAAGTTGCTCGTCCGAATC
TGCAGTCTCTCGTTCAGTATCTTTTCCGTATGGGTGATATCAACGCACAAAAATATCC
CGCATCCAAAGGTGGTTATCCTTACTGGCATAGTGAAGTGTATCCGCAGACTGGAC
ACAATGAAGCTCTTCATCGGATTCTGCTGTTTATGTATTATTTAAACGATGTTGAGGA
AGGAGGAGAGACTGACTTCTATTATCAAGATCTGAGCATTAGACCTAAGAGAGGAAC
TATGGTTATCGCACCGCCTATTTACGCATACCCACCGTGGCAATGTCCCTGTTAG
TAATGATAAATATATTCTTACATCGTGGGTGTTGTTTAATCGTGCGGAGAATCTGTATC
GGGGCTGA

NHI 277 protein sequence:

MSQFIEVIDNALPEELCESLIRQIDTSPNRQPGRTGGGVDTSKKVSEDVSICTHDEFASL
LPKVLECTTSHLVEYFAKYFALIGPIGLTLKHPKTGEPVKLTQDNFDEVARNLQSLVQ
YLFRMGDINAQKYPASKGGYPYWHSEVYPQTGHNEALHRILLFMYYLNDVEEGGETD
FYYQDLSIRPKRGTMVIAPAYFTHTHRGNVPVSNDKYILTSWVLFNRAENLYRG

Codon-optimized NHI 278 sequence:

ATGTCAGCGCCTACTGCGCCGGTAACCGGAGCGTTGCCGGATTACCGCATTGGG
AAGAAACCCCGGCGGATTTGCCGGCCGCCATTCCGGAGGTGAAAGCAGCACTGC
GCGCTCGAATAACTGCGAGTGGAAGAACAGTGGAGGAAGTTTTTTCGGTTGTCTGA
AAGACGCGTTCGTACTGCCGTTGATGAGATCACAAACGGCCCGTGAACGCGGCGAG
AGTGTATGGCCCGTTGTGGATTACGCAGATATAGCCGCTGGTACTGTGCCGGCGGG
CCAACTGGCCCTATTACGCAGGCGGGGCTGCCTGGTTGTTAGAGGCCACTTTGATC
GAGATCAGGCTCTGCGTTGGGATCGGGAAATCGTTGATTACGTCGATGGCAACCGG
TTTTTTGAAAATTACCGTGGACCAGGAGACGACTTTTTTGGTTTCAGTCGGTAGTAGA
CCGGAGATCTATCCAATCTATTGGAGCCCAGCGCAGATGCAGGCGAGACAATCCGA
CAGGATGGCCCGTGTGCAGGCATTTCTGAATCGTCAGTGGAAACACGAGAGCGAA
GGCGTTCGGTGGTTTGACCCCGAACGTGATAGCCTGTATCCTGACCGTATTAGGAG
ACGTCCTGAGGGTGCTGATTCTGCGGGCTTGGGTGCACACTTGGATCCGGGCACA
CTGGATCTTTGGATGACTGAGAGTTATCAACGCGCGTTCCGCCATCTTTTTGATGGG
ACCGTGGAGCAGTATGACCCGTGGGATGCGGCCCATCGAACGGCAGGGCCCGCAA
TATCCTGGCAGTACCATGTGCAGTGCCTTCCGGACCTTTCAGGGTTGGACCGCGCT
GTCCGACATGGACCATGACCAGGGTGTGCTCCATACGGTGCCGATACCTGAAGCA
ATGGCATACTGATGTTACGCCCACTGCTGGCGGATGTCCCGGATGACGATATGTG
TGGTGTACCGTAAATCAGGTCTTTCCGGCGAATGAACGTTGGCACCCCTTTACTAAT
GAGGGCGTTGTCTGGGATCCCTGATGTCCGCGCTGGCGACTCAGTGTGGTGGCAT
TGTGATATGATTCAGTAGCCCTGTACGGGATCAAAAAGGTTGGGGTAATGTT
ATGTATATCCCGGCTGCACCGTGGTGCCCGCGTAATGAACGGTATGCTGCAACGGT
TCGCGAGGCGCTGCGAACCGGGAGCTCACCTGATGATTTTCCTGCGGAACATTAC
GAACGGAGTTGGCCAGATCGCTTCCGCCCTGAACAATTGAATGATACCGGCCCGGC
GCGGATTGGGTTTAGATTAA

NHI 278 protein sequence:

MSAPTAPVTGALPDLPHWEETPADLPAAIREVKAALRARITASGRTVEEVFAVVERRVR
TAVDEITTARERGESVWPVVDYADIAAGTVPAGQLALLRRRGCLVVRGHFDRDQALRW
DREIVDYVDGNRFFENYRGP GDDFFGSVGSRPEIYPIYWSPAQMQRQSDRMARVQA
FLNRQWKHESEGVRWFDPERDSLYPDRIRRRPEGADSAGLGAHLDPGTLDLWMTES
YQRAFRHLFDGTVEQYDPWDAAHRTAGPQYPGSTMCSAFRTFQGW TALSDMDHDQ
GVLHTVPIPEAMAYLMLRPLLADVPDDDMCGVTVNQVFPANERWHPLLMRALSGIPDV
RAGDSVWWHCDMIHSVAPVRDQKGWGNVMIYIPAPWCPRNERYAATVREALRTGSS
PDDFPAEHYERSWPDRFRPEQLNDTGRRGLGLD

Codon-optimized NHI 279 sequence:

ATGGCCGGTAACGGGCTGCCGACTCTGGGACGTGTCAAACCTCACTGACCTGGTCTG
CTAAGGAGGGGTTACCTTCCGAATCATATAAAGTCTCAGTCAGCACGCTGTGCCAG
TCACTAGCTCAGTATTCTGCGGCAATTGTGCAGTTACCGTCGGGCGACGGTGCCT
GCTGCGTAGCTGCCTTGACAGTGCACGTTTGTATTTTCATCATCGTCCTTCATATCC
CAGCGAGGACATGATCCATACAAACGAATCGAGGGATTGGTGTAAAACGTCCGGAT
ATTATGCCGATCCCCAATTATGGCAAGAGAGTATGATTTCCGTCCGGGTTTGACGC
CCGCGGAGTTAGAAAATGAAATTCACTTTTCGCCTGCTGGCCTCGTTGATATCTACG
CCCTTTTAGGAAAGGCGGCAAGAGAAATATTGGATGCTATTTCAATCAACCTATCACT
CAGGTCCAGCCCGTTTACGGAAATACTGGATAATGTACCGCTCCGTGGTTCGTGAAA
TCTCGTCCTCGGTATTGTCCGTATGTTGTCAAGCGCGACCGACGTTCCATGGCGCC
CAGCACCACAACTTAACGAGTCAAGAAGATGGTAGCCTGATTATGTTTAGCGAGCAT
GAGCACCCCGCGGACAAAAGCCTGATCAGCATTGTCAAATCAGATAAATCAGGCTT
ACATATTAGGGATTTTCAGGGTCATTGGGTACTGGTGGATGGCGATCTGGGCCCGC
AAGAAGCAGTGATTTATCCAGGGCTGGCTCTTTACCAGGCGACTGCAGGCTATATTA
ATCCAGCCTTGCATCGTATCGAAATAACAAACCCGCAATCTAATATTCACGGCCGTT
GCTCCGTGACGTTTAACTTATGCCTAAGAGCATGACCCATCTTAGTTGTTCTGAAAT
GCGCGCGGCGGGCCATGGGGTTGAAGCGCAGTTTCAGCTGCCGGTTCCAGTAGA
TGACTTTATGCAGCGTCCACATTGACAGATAATCCGATGAATCGTCAGAGCGTCCC
GTCCTTTAATTTCCCTAATAGTCATGACGGTCCCATGAAACCGCTGATAAAGCGCCG
AAAGAACAATTCTAAATGTAAACCGCTTCCGCCTTCTAAACGGCTGCGACTGGAGG
CCCAGCGCGTTTTTAAAAGAACGCGTTCAGGATATTGCAGATAAAAAGGGTATAAAGC
TGCGTTTCTGCTCACTTAAAGAGTGCGAAGCTCATATTCATTCTCTGGATTCTCCGT
GTGCGAATATTCGTATGGAAATTGGTTGGCCACCGGGTGTGCCGTTTCGTGCATCCG
CATGATTTGCCAAACAAGGCAAAGATTGGGTTTTTGGAAAGCGTATGAGCCGGGTTG
GACCATTACCCACGACATGGAGCTTTCTCTGATTGACCCGGGGCAATCAACAGAAC
ATTCCGGGCATTATAGTCAGGAAATTTAA

NHI 279 protein sequence:

MAGNGLPTLGRVKLTDLVAKEGLPSESYKVSVSTLCQSLAQYSAAIVQLPSGDGALLRS
CLDSARLYFHHRPSYPSEDMIHTNESRDWCKTSGYYADPQLWQESYDFRPGLTPAEL
ENEIHFSPAGLVDIYALLGKAAREILDAISFNLSLRSSPFTEILDNVPLRGREISSSVLSVC
CQARPTFHGAQHNLTSQEDGSLIMFSEHEHPADKSLISIVKSDKSGLHIRDFQGHWWL
VDGDLGPQEAVIYPGLALYQATAGYINPALHRIETNPQSNIHGRCSVTFKLMPKSMTHL
SCSEMRAAGHGVEAQFQLPVPVDDFMQRPHSTDNPMNRQSVPSFNFPNSHDGPMK
PLIKRRKNNSKCKPLPPSKRLRLEAQRVLKERVQDIADKKGIKLRFCSLKECEAHIHSLD
SPCANIRMEIGWPPGVFPVHPHDLPNKAKIGFLEAYEPGWTITHDMELSLIDPGQSTEH
SGHYSQEI

Codon-optimized NHI 280 sequence:

ATGGATGCAAAATTGAAAGGTGTGCAACAGATACTGTCTTTTGCTAAGTTACCGTTAA
GCTGTGATTTTGAAGCAATTAATACCGAAATAAATAACCTCGATGACTCGCAGTGGT
GTATGCATGTCAATCAGGCGTGTTATAATGGTGCCTGGGATGTGCTGCCCTTTATG
CACAGACGGCTCACTTGCAGAGTCATCCTATACTCCAATGCTTCAGCATTGAACATA

CAAACGTGACTTTCTCTCCACTGCCTATAATGGAACGACTGCCTGTTCTTGCTCAGT
TCCTTGCTCAGTTTGTGTGCCCTCTCAAAGCGTTAGACTTATGAGATTACACCCGC
AGTCTGAGATTTTGCCACATAATGATCAGCAGGTTGCTCTGGAGTACGGCGAAGCC
CGCATTGATGTCCCTCTTACGGGCATAGACGAGGTAGACTTTGTCATAAACGGCAG
CACAGTTCCCATGATGCGCGGGGAGGTTTGGTATATGAATGCGGACTTAGAACACT
ATGTTTCTAATCGTGGGGCAGAGCCGCGTATTAATTTAGTCATCGATTGCAAAGTGG
AGGATTGGTTAAAAAAGATCATTAAATCTAGCGAGCAGAAATTCTAA

NHI 280 protein sequence:

MDAKLKGVEQILSFAKLPLSCDFEAINTEINNLLDSQWCMHVNQACYNGAWDVLPLYA
QTAHLQSHPIQCFSIEHTNVTFSPIMERLPVLAQFLAQFVCPLKSVRLMRLHPQSEI
LPHNDQQVALEYGEARIHVPLTGIDEVDFVINGSTVPMMRGEVWYMNADLEHYVSNR
GAEPRINLVIDCKVEDWLKKIISSEQKF

Codon-optimized NHI 281 sequence:

ATGCCTATTGCCGTTGATACAGCACCGGTAGAACAAACCAGACAGTAGTTTTAAGAAA
CGCTTCACGGAGCGTACCACACCGCCTCCAGAATTGCCTCCGTTTTTCTATGCCGA
CCGGTTTGATAAAGCGCATTTTAATGATTGGCGGGATGAGCTTGCAACCAAAGGCTA
TACGGTGGTAAAAGCTATACCTACAGAAAAGGCGCTGGAATATCGGGAGAGGGCCT
TTCAGTGGATGGAAAGCTTTCCTCTGGGCTTTGATCGGAACGATGTCAAACGTGG
AAGAACGAACACTTGCCAGTTCATATTAAAGGCGGTATGTTTCATGGCTATGGATTCT
ATGACATCTCCCAGTTGTGGGATATCCGCTGCGAGGATGGTGTCATTGATGCCTTC
GCCAAAGTGTGGGGCACAGACGAGTTAATAACATCATTTGACGGCGGATCTATCAT
GCTCCCGAATCGTGTTGATGTGATGGACAGCGGTAAATGGGAACACATGGACCAGT
CACACTACCGACGCGGATTTTACTGTTGTCAAGGCTTGGTTAACTTAAACAACAATG
GTCCTGATGATGGCGGTCTGATGGTACTGGAAGGAAGCAGTACGCTTGTTGAAGAA
TACTTCGACACTCATGGCCGAAATAGCTATAAAAGCTGGGGGCCATTGATTGGTTT
GGATTTACCGAAGAGCAGCAGCAATGGTTCTTTGACAGAGGCTGTAAATGGGTAA
GGTTTGTGCCGAACCCGGGGATTTGATTCTGTGGGATTCTCGTACCATGCACTATAA
CGTACGCCCCGAAAGGTGATAGAGACAGGGTGTGCACCTATATTTGTATGGCGCCGG
CTCACTTACTTACAGAGGAAGATAGGCAGCTCCGCGTGGAGTGCTTTGAGCAGAGT
AAGGGTACGACCCATGTTCCGTTTGCTGCTATCTACGCGAGGGATTTGAACCAGT
GAAAAGAAGTGACGGGTCCCCATGCCACATGATACTGGCAAGCCAAAAAATCCGC
CGGTACTGAATGACCGTATGGCCAAGCTGGTCCGTATTAAACCGTATTAA

NHI 281 protein sequence:

MPIAVDTAPVEQPDSSFKKRFTERTTPPELPPFFYADRFDKAHFNDWRDELATKGYT
VVKAIPTEKALEYRERAFQWMESFPLGFDRNDVKTWKNEHLPVHIKGGMFHGYGFYDI
SQLWDIRCEDGVIDAFKVVWGTDELITSFDGGSIMLPNRVDVMDSGKWEHMDQSHYR
RGFYCCQGLVNLNNNGPDDGGLMVLEGSSTLVEEYFDTHGRNSYKSWGPFDFWFGFT
EEQQQWFFDRGCKWVKVCAEPGDLILWDSRTMHYNVRPKGDRDRVCTYICMAPAHL
LTEEDRQLRVECFEQSKGTTHVPFAAIYARDFEPVKRSDGSPCPHDTGKPKNPVLND
RMAKLVGIPY

Codon-optimized NHI 282 sequence:

ATGCGCAGTCTGACTGAATCCGAGAAAACAAGCTTTCAAGAGAATGGCTTTCTGAA
ACTGACTGACCTGCTTACGAAAGAAGAAGTTGATTGCTGATTAGATGGACCTCAG
AAGTTAAGAACTTACCGAACGTGGTGGGTGAGCACATGCCATACGAAGAGATTGAC
GCTACAGGTAAAAGAGTTCTGTGCCGAACAGAGAACTATTGTAATTATCATCCGGGT
TTTAACAGTATTCTGCGTGGCGAAAACGCCCTGTCCGTGCTGCAGCAACTGTCAGG
CGAAAAAATGCATCTGTTTAAGGAGAAAATCAATTATAAACAACCGCACGCGGGGGG
CTTTGATGCACATTTTATGATGCGCCGGCATAACAATCATATAAAAGCCACTAACATCTC
ACTATCTTAATGGCGGTTGAACCTGCTACGCTCGAAAACGGATGTCTTGAAGTGGTT
CCGGGGAGTCAACAAGGAGCCTATCCCCCTGGGTGAGGATCGCTGCATTACCAAAG
AATGGTGCGAAATGCACGATTGGATACCTGTTCCACTGGATACGGGCGAGCTCCTG
ATATTTGGTAGTTACCTTGCACATCGCAGTGGAGCTAATCATTATCATTAGGTCGCG
CGGCAATATACGCAACCTATAATTGCGATTGAGATGGTGGGGATTACATGATGAGT
ATTATGTTTCATCGCCGGAAATTGTGGCCGCCTACACACGAACGCCTGGTAGGTGAA
AAGTACGAAGAAGGTGCAGTGCTTTACGGGTTCGGTTCGCCTATGCTGACAATTCA
AGAGAAGGCCTGA

NHI 282 protein sequence:

MRSLTESEKTSFQENGFLKLTDLTKEEVDSLIRWTSEVKNLPNVVGEHMPYEEIDATG
KRVLCRTENYCNYHPGFNSILRGENALSVLQQLSGEKMHLFKEKINYKQPHAGGFDH
FDAPAYNHIKATKHLTILMAVEPATLENGCLEVVPGSHKEPIPLGEDRCITKEWCEMHD
WIPVPLDTGELLIFGSYLAHRSGANHSSLGRAAIYATYNCDSDGGDSHDEYYVHRRKL
WPPOTHERLVGEKYEEGAVLYGFGSPMLTIQEKA

Codon-optimized NHI 283 sequence:

ATGACACTTAATTATGTTCTGATCCATAATGCAGACGGCTATCTAGTAAGCTGTGATG
GTCAAGCACTTCGCACTGCCGCTCATGGGGATGATGCCTGCTACTGGCAGACCAC
TGATGATGGATATCGTCACGCCCTGACAGGCCGTCTGCTCACCGGTGTGGAAACG
GAAACGGGAACCGACTTACAGGATCGCAATGGCAATAAAGTGTGCGGGCGGACCGT
TTCGTCTGACGGATGCACCGCCGCGCCTGGCTTCCGAGATACTCGCCACTCTTAAT
GATCAGGGCTTTGCAATTATCGAGAACTGATGTCTGATCGGGAAATTTCCCAGCTT
AAAGAAGAGACTGCAGTTGCACGTGCGAAAGATCATGCTGATGAAACCCCAAAGA
TGGAAACTTCTGGATAACAGACTCACTTATTTGGTCTGTAGATGTCGCCCCGGGCTGT
CACGCACCCGGTAGCATTATGGGTGATGCAGCAATATATGAGTTCTGATGAGATCCA
TTTCTGTCATCAACCGATAGTGAACACTCTGAAGCCCGCTGACAACTGAAAGGTC
AGTTTCCGGAAGGAGGATGGCATAAGTACTATCCGTATCACGGAAATGTGTTCCCC
AATAATGATTGGCCTGAGCAGCCGGTGTGTTGGCGTTCAATTCAATATTTGCATCGAT
GAATTTGCGCGAGATAATGCAGCTACACAGTACGTACCGGACAGTTATAAACTAAGA
AAAATTCCACCTCAAGAAATAAATACCGGAGGAACACGTATGGGTGAGGCCATACAT
AAGGATGTGAAACAATGGATCGCTCCTGCCGGCTCAGCTGTAATCTATGATTCACGA
ATGTGGCATAGAGCTTGTACGAACTGAATGTGAGTGGTCTGGATCGGATTGCAGT
TCTTAATGCTGTTAGCCCGGCTTATGTCCAACCCATGATGAATAAATCTAACTTAGGA

CGTCAGTATCCGAAGTCGCCGGTGGCAGCGATGTTGACTGAAAGGGAGAATATCG
ACATCGACCGGTTATGTTGCCGCCCAATGCTGCCGCTACCCCAGGGAATGCCAGCT
ATGCATGGCCGCACGCCCCGAGTAG

NHI 283 protein sequence:

MTLNYVLIHNADGYLVSCDGQALRTAAHGDDACYWQTTDDGYRHALTGRLLTGVETET
GTDLQDRNGNKVSGGPFRLTDAPPRLASEILATLNDQGFAIEKLMSDREISQLKEETAV
ARAKDHADETPKDGNFWITDSLIWSVDVARAVTHPVALWVMQQYMSSDEIHFCHQPIV
NTLKPADKLKGQFPEGGWHSYPYHGNVFPNNDWPEQPVFGVQFNICIDEFRADNAA
TQYVPDSYKLRKIPPQEINTGGTRMGEAIHKDVKQWIAPAGSAVIYDSRMWHRACHEL
NVSGLDRIAVLNAVSPAYVQPMMNKSNLGRQYPKSPVAAMLTERENIDIDRLCCRPMLP
LPQGMPAMHGRTPE

Codon-optimized NHI 284 sequence:

ATGGTGTCATCAACGCCATATCTCGACGCTCTTCGTAGAGACGGATTCGTTAAATT
CCCCATCTGCTTCCGTACACGTAAGTGTGGCCCTGCGCGGCTGCTGTAAATACAC
TACCGAAAGGGCCAGGAACGGAAATGGCCCTATGTTAGGACAGTACCCAAACAAT
ATCCGCCCTGGCAAGAATGGACGCCAGGTGATAATATTTGGGGGGTGCAGCACTTG
CTCCACCCGGACATGAACGTTAGAGATACATTTGCTGAAGTCTATTTACGCGACGAA
GTTCTGGATGTTGTTAAAGAACTCGTAGGGCTTAAGGGCGATCCTACCGGGGATGA
CAAAGTGGTTATGGAAGTGTGTAACCTCCTCGTCTCACCGGCGGATAGGGTTGATTT
CGAAGTTTGTGTCACCGGGATGATATTAGACCAGACGTAACCCCGGAAGAGGAG
GGCCGGCAACTCAAAGAGAAATCACCAGAAGGACAGCAGTTGCATGCTCAGTATAA
TATTGCACTTTTTGAAGATACCTCATTAGTTGTCTGTCCTCCGGGCAGCCACCGCCGGG
TGCGAACAGAACTGAACGTCAGGCCGCACCATACGAACCTAATCTTCCTGGGCAA
GTAAGTGGTGAAGTCCACCCCGGTGATGCAGTGTCTATGATAGTAATATCCTGCAC
AGAGGCGTGTATAAAGGGATAGATACATCATCCGAAGTGGTTCGCATGACGCTGCA
CGGATCAGTTGGATTAGCAGGACACGGAAGTGAACGTGCGAGACAGGTGCTGCAG
CATGCGGTTGGTGAATGGGTGGATCGGGCACCGTTTAGCATAGAAGGCCCGAAAG
GTGAACGCGCAGAGGCCATGAGAAAACGACTGGTTGACATGGGATCAGGGAAGGA
TCTGGGTTATAGTCTTGTTGGTTAG

NHI 284 protein sequence:

MVSSTPYLDALRRDGFVKIPHLLPSHVLLALRGCKYTTERARNGKWPYVRTVPKQYP
PWQEWTPGDNIWGVQHLLHPDMNVRDTFAEVYFSDEVLDVVKELVGLKGDPTGDDK
LVMELFNLLVSPADRVDFFELCWHRDDIRPDVTPEEEGRQLKEKSPEGQQLHAQYNIAL
FEDTSLVVVPGSHRRVRTETERQAAPYEPNLPQGVLVELHPGDAVFYDSNILHRGVYK
GIDTSSELGRMTLHGSVGLAGHGTERRARQVLQHAVGEWVDRAPFSIEGPKGERAEAM
RKRLVDMGSGKDLGYSLVG

Codon-optimized NHI 285 sequence:

ATGGCAGCAATGACACTGTCCAAAGCTTTGTCTCGTTCCAAAAGCCAACTGCTAAC
ACGCACGGTGCCCATTCGATGCATGTCCACATCAACGAGTACTTCGACACTGCCAA

TCACAATCAAACCCACAACCTAAAGAACTTTCTGGCTGGCTTCTTGGGTGAGAGAAATT
TAGAAAAGGCGATGAGAGGCTTACATGAAGATGGTTTGGTTGTTGTGAGTGACGTC
ATCCGCGATGATCATCTGGATCACTTAAACAATAAAATGGTGCAGGATGCACGCACC
CTCCAGGCCAAAGGGAAAGATATGCCTTATAATTACAACGTAGGAAACATTCAGCAG
GATCCTCCACCTGTAAAAGAACATTTTGAATCTAGTATATTCCTGAACCCGATTGCCA
ATCAAATTACTACCGGAATGCTTGGGCCACGCCCTAAATGGACCTTTTGTTCAGGCA
ACAGTGCCATGCCACCAGAGGCTGGGGCGCAGCCCCAGCGACAACCCGTCCATG
CGGACAGTGATTTTGGCGATCCGAGTCATCCCTTCGCCCTGGTCGTGAATGTCCCG
CTGATTCGTTTTACTCCCGAGAACGGTTCCACTGAGACCTGGCTGGGTACTCACAC
GGGTGACTTGAGCGGCCTGCAGGTTCAACAGGGTGCAAAAGGTACCCGAGCATCC
GGCCGTATATGTGAAGATCTGCTTGAACAGCGTCGGCAAGTGCCTGGACCGAGCC
AACCGGTAATTGAAAAAGGTAGTATCGTGATCCGGGATTAAAGATTGTGGCATGCTG
GAATGCCTAACATGACGGATGAAGTGAGAGTGATGCTGGCAATGATCCATTTGCA
CCCTGGTATCGCAACCCGATGAAATTAGAATTATCTGAAGATTTAATGCACAAGATTA
GATCACAAAATGATCTTCAGATCCCAGTTGATTGGGTGTCTCGAGATGAAGCCCTCA
AGAGATATCTGAATAGGGGCTTTGGCAATTCATATGACTTTGGGCAGGCGGAATAA

NHI 285 protein sequence:

MAAMTLKALSRKSQLLTRTVPIRCMSTSTSTSLPITIKPTTKELSAGFLGERNLEKA
MRGLHEDGLVVVSDVIPHDHLDHLNKMVQDARTLQAKGKDMYPYNYNVGNIQQDPPP
VKEHFESSIFLNPIANQITTGMLGPRPKWTFCSGNSAMPPEAGAQPQRQPVHADSDFA
HPSHPFALVVNVPLIRFTPENGSTETWLGTHTGDLGLQVQQGAKGTRASGRICEDLL
EQRRQVRGPSQPVIEKGSIVIRDLRLWHAGMPNMTDEV RVMLAMIHFA PWYRNPMKL
ELSEDLMHKIRSQNDLQIPVDWVSRDEALKRYLNRGFGNSYDFGQAE

Codon-optimized NHI 286 sequence:

ATGCCCCGGGATGGGAATGATGGGGTTAGGCGCTGGCGCACCTAGACAGGCCCGC
AGCGAGGGCGATATTAGTTCTGTATTTGCTAGCCTGTCTGGTGAAGAAGCGAAACC
GTTGCCTCCTCGCTTTATCGACCTAAAACGTCAAATTATTGGCGACGAAGCCAACCA
ACGTGCTTTGACTGACTCATGGGTGAGGCTCTGTGATCGTCTGGGCCAGGCCGGC
GAGGAAATTGAACGTAAAAACAGGACTGCATTCCACAGATTGCCTACGAAGAACT
GACATCACCCTACCGAGCACAAGTACCCTGTCTGCCCTGAAACGCGCGGGCTCT
TTCGTCGTCAGAGGCGTGGTGGATGAGGAGACCGCCACTGGATGGCTGAAAGGCT
GTGAAAAGTATATTGGGGACAATCCACAAGTTCGGGGGTTCCCAAAAGATGATAAG
CAAGTTTTTGAACCTTTATTGGAGTAAACTCAATTAGCTGCACGGTCTCATCCACGG
AGTCTGGCTGTGCAGCGGAGCCTCCTGACCCTCTTTACACATCCGCCGGATTTTCC
GATCTCATTAACGACTCCCTTAACGTATGCGGATAGGCTGCGTATAAGACATCCGGG
GGATGCCCAATTTGCTCTTGGTCCGCATATGGATGGCGGGTCTATCGAGCGCTGGG
AGGATCCCATGTATCGGGCTGTTTACAAAGACATCCTGAAAGGTAAGTGGGAAGCG
TTTGATGCGTATACGATCGGCGAGCGTGGTGTAGCTAAACAAAATCTGTACGACGG
CCCGGGTAGCTGTGGCGTTTTTATAGACATTCCAGGGCTGGACCTCTATGAGTGAAA
CGGGCCCAGGAGAGGGGACGCTGCGGGTCTATCCTTTTATAAGAGAACTTACGGC
CTATACGATATTACGTCCGTTATTTCTGTGAAAAAGAGTCCCGAGCTAGCCTGACTTAT

GAACAGTATCTGGATCCGTCGAATTGGGTTCTGGACCCGACCACCAGTGCCTTTCC
TGGCTCGCCTCTGGCACGGAGCCAAGAATTCAATGATGGTACCCATCCGCATCTGG
CTTTGAATCGTAGCATGGTTAGCATGCCGACTGTAAACCTGGCGATCAGGCGTGG
TGGCATTGCGAAAGCGTGCACAACGGTACGGGTCCTAGTGCGGTTCTGTACATACC
GGCCGTGCCACTTACGCCTTCAAACATTGAATATATTAAGGATCAACGGAGCACGTT
TTTTGAAGGGCGTCCAGGCCCTGATTTCCCTGGAGGTGAGGGTGAGGCCCATTTTC
GTCGGTAAAGGGGTAGAGGGTGATATACTGAGTAAAGAGGGTCGCCGCGCAATGG
GTTTGGAGCCATTTGAGATCACAGCCGATCTCTCCTCTGGTGAGAAGTCTATACGTC
AGAAAGCGTGA

NHI 286 protein sequence:

MPGMGMMGLGAGAPRQARSEGDISSVFASLSGEEAKPLPPRFIDLRQIIGDEANQRA
LTDSWVRLCDRLGQAGEEIERKKQDCIPQIAYEELTSPLPSTSTLSALKRAGSFVVRGV
VDEETATGWLKGCEKYIGDNPQVRGFPKDDKQVFELYWSKTQLAARSHPRSLAVQRS
LLTLFTHPPDFPISLTTPLTYADRLRIRHPGDAQFALGPHMDGGSIERWEDPMYRAVYK
DILKGNWEAFDAYTIGERGVAKQNLVDGPGSCGVFRAFQGWTSMSSETGPGEGLRVY
PFIRELTAYTILRPLFREKESRASLTIEQYLDPSNWVLDPTTSAFP GSPLARSQEFNDGT
HPhLALNRSMVSMPTVKPGDQAWWHCESVHNGTGPSAVLYIPAVPLTPSNIEYIKDQR
STFFEGRPGPDFPGGEGEAHFVGKGVEGDILSKEGRRAMGLEPFEITADLSSGEKSIR
QKA

Codon-optimized NHI 287 sequence:

ATGTTAAAACTGGATACCCTTCGCGCTACCCGCATTGAGCGTGATCCCTTCGACTAT
GTGATTGTGCGGAACCTTTGTTGAGCGTGAAAAATTGGATCAGGTTCTTTCCGATTAT
CCGGAGGTACCGGGACCCGGGTCTCATCCGCCAGGTGGCCTGAAGATTTTCAGGTC
ACTTTAAAAAACTGATGGACGAACTGCTGGATGCTCCTTTTCGTCAGCTGGTTGAAG
AAAAATTTGACGTTGATCTGACGGGGCCGTCCGACAATGTATACGGTTCGCGGATTCT
GTCGTGCGCGGGACGGAAAAATCCATACCGATAGTAAGACGAAGATTATCACTGTC
CTGCTTTATATGAACGATGAATCGTGGGAGAGTTCCACTGGGCGTCTGCGCATACT
GAGAAATGGTACCGATCTAGAAAACTTTGCGGCTGAGGTCGAGCCTTCTGGCGGAA
CTTTGCTAGTTTTCAAGCGTTCAGATAACTCGTGGCACGGGCACCATCCGTTTGAA
GGGAAACGCCGAGCGGTTTCAGCTGAACTGGGTAACCGATCAGTCAGTCGTGGACA
GAGAACAGGGCCGACATGGATTCTCATCAGCAATTAAACGCCTGATGGGTCTCGGC
GCTCATTA

NHI 287 protein sequence:

MLKLDTLRATRIERDPFDYVIVRNFVEREKLDQVLSDYPEVPGPGSHPPGGLKISGHFK
KLMDLLDAPFRQLVEEKFDVDLTGRPTMYTVRGCRRARDGKIHTDSKTKIITVLLYMN
DESWESSTGRLRILRNGTDLENFAAEVEPSGTTLLVFKRSDNSWHGHHPFEGKRRAV
QLNWVTDQSVVDREQGRHGFSSAIKRLMGLGAH

Codon-optimized NHI 288 sequence:

MASWSHPQFEKGATTRILGVVQLDQRRRLDDLAVLAKSNFSSEYSDFACGRWEFCML
RNQSGKQEEQRVVVHETPALATPLGQSLPYLNELLDNHFD RDSIRYARIIRISENACIIPH
RDYLELEGKFIRVHLVLDTNEKCSNTEENNIFHMG RGEIWFLDASLPHSAGCFSPTPRL
HLVVDIEGTRSLEEVAINVEQPSARNATVDTRKEWTD ETL SVLGFSEIISEANYREIVAI
LAKLHFFHKVHCVD MYGWLKEICRRRGEPALIEKANSLERFY LIDRAAGEVMTY

Codon-optimized NHI 290 sequence:

ATGCCCTGACACATGACCAGCGTGATCATTTTTTAGAACATGGCTGGGTAAAGTT
CCGAATGCCATTAGCAAAGCAATTATAGAGGAATGGACGAAGAATATCTGGGTCCGC
CTTGTTACGACGAGAACGATAAATCAACATGGCTGGAGGAATATCTGAACTGCC
GCGTCACCGTGAAATACCGGTTGAAGAACTGGCCCCTGGTGCTTGGGAGGCTATG
TGTGAAATTGTGGGCGGAGAAGATAAAATAGACCCAGTACGTGAGAGATATCACGG
TGATCAATTTATTATAAACTTCGGTTCGGACTATTGGGAAACAGCTACGCCTCCGCC
GCAGGAACCTTCGCGGATGGCACCATGATAATGATTGGTACCGTCAGTTTTTAGACTC
ATCCGGTAATGCACTTACTATTGTCCACTGCTTTACCGATATACCGCCGCGCGGAGG
TGGGACCTGGTTGGCTGAAGATGGCATTGCAAGCATCGTGAAATACTTGTATGATCA
TCCTGAAGGCCTGGATCCACCGCTGGAAAAAGTGCTGCATACGCATACCAAAGACT
GTAAAAAATTTGTGCAGGTCGAAGCCAAAGCTGGTGACACACTTATTCTTCATGGCT
TACTGCCACACACCAATGGAGTGAACCATCTGCATTATGCCCGCGTTATCACCAATC
CCCACGTTAATATGGTAGAACCTTTCAATCTTAATCGCTCAGATGGTAACTACACACT
GTGTGAGCAGGTCATCTTACGGGCTCTGGGGCGTGAATCAGTGCCGGAATATAAGC
CGACACGAGAGCGAAAACGGTACCATCCTCGCACTGCGCCGTTTAAACGTGTACG
GGTTGAACCAGAATTA AAAAG AATGCTTGAATATGCAAAGCACATGGGCTGCCGCA
GGACTCGGTTGACTCCGTGTACCTCCGCGGTGAAGAAGAAATTCGTCTGCATGAAC
AACGTAATGGCTATGACCAACCGCATGGCCCCGGACCGATCGCGTACTCAAATAAA
GACTATTTTTTCACAAGAGAGGAAGGAATAACCACACAGGACTAA

NHI 290 protein sequence:

MPLTHDQRDHFLEHGWWKVPNAISKAIIEEWTKNIWVRLGYDENDKSTWLEEY LKLPR
HREIPVEELAPGAWEAMCEIVGGEDKIDPVRERYHGDQFIINFGSDYWETATPPPQELR
GWHHDNDWYRQFLDSSGNALTIVHCFTDIPPRGGGTWLAEDG IASIVKYLYDHPEGLD
PPLEKVLHHTKDKCKFVQVEAKAGDTLILHGLLPHTNGVNH LHYARVITNPHVNMVEP
FNLNRSDGNYTLCEQVILRALGRESVPEYKPTRERKRYHPRTAPFKRV RVEPELKRML
EYAKAHGLPQDSVDSVYLRGEEI RLHEQRNGYDQPHGPGPIAYS NKDYFFTREEGIT
TQD

Codon-optimized NHI 291 sequence:

ATGACTACTACCCTGCCCCTGATTCGCATTAATCCATCCCTTGACCCTGCGCCCTTT
GCAGCGGCGTTTTAGCGTGATGGGGTGGTTCAGATTCCGATCTTCCTGGAGCCGG
CAGACGCAGAGCGTGTTGCCGGAGTTTTGGCTGGTCTAACGTGGAATGTCGTGCG
TCCGGACGAGACTTCGGAAACCCTGGTGATAAGCCCACAAGTTATACAAAAGTTTG
GTGAAGCTCAGGTTTCGACAGTTTTTACAGGGCGCATTGAAACGTGCGAGTCGTGG
ATTTAGCTTTATACACTTATCTTACGCGCTTCAGGATGAATATGCAAGAGATCCGACC

GCACCGATTACCAAGCGACCGCTTTCCTGCAGAGCCGCACTTTTCTGGATTTCGG
ACAGGCGGTGATAGGCTCTGACCAGGTAGATGGGGTCCGCGTTTCAGGCGAGTAAC
TATCGCCCGGGTGATTTTCTTACGTTGCATGACGATTCGCATAAACGAGATGATCGC
CTGGCCGCAATTTACGTTGGGGTTTACGCGAGATTGGAGACCAGATTGGGGGGGAC
AGCTTCTGTTTCATGACGTTGAAGGCCAAATTGAGAGGGGCTTCATGCCTGGTTTC
AATGTTCTCACTCTTTTCAGAGTTCCTCGAGCACATTCAGTAGCTCCCGTGGCTGCA
TATGCCGCAACTAAAAGACTGAGTCTGACCGGATGGTTGGTTCGCGACGCCAATCC
TTCAGCGTAA

NHI 291 protein sequence:

MTTTLPVIRINPSLDPAPFAAAFQRDGVVQIPIFLEPADAERVAGVLAGLTWNVVPDET
SETLVISPVQVIQKFGEAQVRQFLQGALKRASRGFSFIHLSYALQDEYARDPTAPIHQATA
FLQSRTFLDFGQAVIGSDQVDGVRVQASNYRPGDFLT LHDDSHKRDDRDLAAFTLGFT
DWRPDWGGQLLFHDVEGQIERGFMPGFNVLT LFRVPRAHSVAPVAAYAATKRLSLTG
WLVRDANPSA

Codon-optimized NHI 292 sequence:

ATGCCCCGGAATCGATCTCCAGACAGCAGGATTGGCTGTCAATGATGGCCCCCTGTT
ACCGGGCTCATTTGATTTCTTGACACAATCCTCACCGAGCGAACCTCTCGATGTGT
TACGAGCGCGCTATGAAGAGGACGGTATGTTACTCTTAAAAGGATTAATCCCGCGTG
AAGATGTGCTTCGGGGCCCGCACAAAGTACTTCGAAATGATGGCACCTTCAGGCGTT
CTGCAGCAAGACACGCAGCCAGTACAGGGTATATTTAATTCAGACGGTAGTCCTGA
GGACTTCCCTGGCATCGGTGCCGGGGCTACTGGAAACGATGGGCGTCCCGGTGG
AGAATCTGCACAGGCATTTGTTGACCTCGCGATACAGGCTCATTATGAAGATTGGTA
CACGCAAGATCTGGCGGGCCATCCAGCCCTGATTGATTTCTGTCGCGAAGCTGACG
GGCTGGAATGAAAACACATTAGAAGTGAAGATCCCTTCTGCGTAACAACGTTCCA
GGAACAAAACCAATTGGTGTCCACTATGATCAAATCTTTTTGCGCTATGGCGATCCTA
CTTCAGTCACGGCGTGGGTGCCCATCGGAGATGTTTCTCTCACTGGAGGAGGACT
GATTTATCTGGAGGATGGGGATGCAGTAGGTACGCAGATTGAGCAGGATTTCACTC
GTAAAGCACTGGAATCCGGCCTTACAGAGGAACAAGCGCGTAATGCCTTCAATCAG
AATATGATGTCTACCGGTCTCCTGAGTGAATATCCGGCAGAATTTGCACGTAAGTATA
ATCGTCGCTGGCTCGTTAGTGCGTACGAAGCAGGAGACGTAAGTGTACATAAACCT
CACGCTATTCATGCGAGCACCGTTAATAATGACCCAGATAATGTTATCCGACTGGCA
ACAGATCTGCGCTTTGTAGATAAAAGTAAACCGTATGATACACGGTGGACAAAATATT
ATGAGTTTAATGATGGTGTGTGA

NHI 292 protein sequence:

MPGIDLQTAGLAVNDGPLLPGSFDFLTQSSPSEPLDVLRARYEEDGMLLLKGLIPREDV
LRARTKYFEMMAPSGVLQQDTQPVQGIFNSDGSPEDFPGIGAGATGNDGRPGGESA
QAFVDLAIQAHYEDWYTQDLAGHPALIDFVAKLTGWNENTLELKRSLRNVPGTKPIG
VHYDQIFLRYGDPTSVTAWVPIGDVSLTGGGLIYLEDGDAVGTQIEQDFTRKALESGLTE
EQARNAFNQNMSTGLLSEYPAEFARKYNRRWLVSAYEAGDVVLHKPHAIHASTVNN
DPDNVIRLATDLRFVDKSKPYDTRWTKYYEFNDGV

Codon-optimized NHI 293 sequence:

ATGGCGGTGGACTGTATCGCACGCATTCCGAGCATGCCGCCGCATCCGCCTAAAG
AGGAACGTCAGTCAGAAAAGAACTTCTGGTATTTGATGCCGCATTAATTAGACACC
AGCCAACTATCCCGCCTCAGTTTATCTGGCCGGACGAAGAAAAACCCTGCAGTAAT
GTTCCAGAACTGGGAGTTCCCCACATTGACTTAGGCGGGTTTCTGTCAGGGGACC
CAGCGTCAACCACAGAAGCTTCCCGGCTGATTGGCGAAGCCTGCCAACGCCATGG
ATTTTTCTCGTGGTAAACCACGGAGTGGATTATAAACTGATAGCAGATGCACATCG
TTACACCAAAGATTTCTTTGAACTGCCGCTTTCAACAAAACAGCGTGCTCTGAGGAA
GGCCGGTGAGCACTGTGGATATGCATCCAGTTTTACGGGGCGGTTTTCAAGTAAAC
TTCCTTGGAAGAAACCCTGAGTTTTCGCTATTCGGCTGACCCGAATTCCTCAAACG
CCGTGCGGGACTATCTGTGCGATAAAATGGGTGAGGATTTTAAACAGTTTGGTCGG
GCATATCAAGGTTACTGCGAAGCGATGTCGACACTGTCTCTTGGTATTATGGAACG
TTAGCGATGTCTCTGGGTGTCTCCCGAGCTCACTTTAAGGAGTTCTTCGAGGAGAA
TGATTCCATCATGCGTCTGAACTATTATCCACCCTGTCAGAAACCTGAGCTAACATTA
GGCACGGGGCCACACTGCGATCCGACCTCCTTGACCATTCTGCACCAAGATCAGG
TGGGAGGACTGGAAGTCTTCGTTGACGATGAGTGGCGATCAATAAGCCCGAATTTT
AACGCTTTTGTATTAAACATAGGCGACACGTTTATGGCATTATCAAATGGGCGCTATA
AATCCTGTCTGCACAGGGCCGTGGTTAATAGCAAGACCCCCCGGAAAAGCCTTGC
GTTCTTTCTGTGTCCACGCAATGACAAAGTCGTAAGTCCGCCTAGCGAGCTTGTGG
ACACTATGTGTCCGCGTCTGTATCCGATTTCACGTGGCCTATGCTTCTCGAGTTCA
CGCAAAAACATTATAGAGCGGACATGAATACTTTAGAGGCCTTCTCAAATTGGCTTC
AGCAGAAGAATAGTTGA

NHI 293 protein sequence:

MAVDCIARIPSMPPHPPKEERQSEKKLLVFDAALIRHQPTIPPQFIWPDEEKPCSNVPEL
GVPHIDLGGFLSGDPASTTEASRLIGEACQRHGFLLVNVHGVVDYKLIADAHRYTKDFFE
LPLSTKQRALRKAGEHCGYASSFTGRFSSKLPWKETLSFRYSADPNSSNAVRDYLCDK
MGEDFKQFGRAYQGYCEAMSTLSLGIMELLAMSLGVSRAHFKEFFEENDSIMRLNYYF
PCQKPELTLGTGPHCDPTSLTILHQDQVGGLEVFVDDEWRSISPNFNAFVINIGDTFMA
LSNGRYKSCLHRAVVNSKTPRKSLAFFLCPRNDKVVSPPSELVDTMCPRLYPDFTWPM
LLEFTQKHYPADMNTLEAFSNWLQQKNS

Codon-optimized NHI 294 sequence:

ATGACCACGCCAAACGCAAGTCCGGCGAGTACAGGGTTGCCTATAATAGACGTAGC
TGCCCTTCGCGCGCCTGATGATGCGGATGGACGTA CTGAGTAGCCCAACAGATTC
GCGCCGCGTGTGAGAACACGGTTTTTTTTTATGTGAATGGCCACGGTGTTTCGCCG
GAACTGATCAGCCGCCTGGAGCGCCTGTCCACGATTTCTTCAATCTGGATGAGAC
GACTAAACTGCAGTGGCGCATGGCCCTTGGTGGACGTGCATGGCGTGGCTATTTTC
CCTCTCGGCGGTGAACTTACATCTGGCCGCCCGGATTGGAAAGAGGGTCTGTATCT
GGGCACTGAATTAGCCGCCGACCACCCGCTGGTCCAGGCCGGTACACCAGTTCAT
GGCGCCAACCTTTTTCTGATGTGCCCGGCTTAAGAGCTACCATTCTGGAGTATATG
GCGGCTACGACCCAGTTAGGGCACCACTGATGGAGGGTATCGCCTTGAGTTTGG

ATCTTCCGGCGTCTTACTTTTCATGATCGCTACACCGCCGATCCATTGATTCTCTTTTCG
GATTTTTTAATTACCCCAAGTCAAGCAGTTCCCGAGGGTCTGGATGTACAGTGGGGGG
TGGGTGAACACACGGATTATGGCCTGTTGACCATTCTGTTACAAGATCACATAGGTG
GTTTAGAAGTGCGCACTCCGGGCGGTTGGATTGATGCACCGCCCCGTTGCGGGTGC
TTTCGTTTGTAAATTGGGGATATGCTGGACCGTATGACTGGTGGGCTGTACAAATC
GACACCGCATCGCGTTAAACGTAACACGAGCGGGCATGACCGCTTAAGTTTTCTC
TGTTTTTTGATCCGAACTTTTTGGCCGCGTGCAGCGCATCGAAGGTTTAGCTGATG
CCCCGGCGCAGGATGATAGCGCTGAGCGCTGGGATAGAGCGAACGTCCATGCGTT
TAATGGTATTTACGGTGACTATCTTCTCGACAAAGTATCGAAAGTGTTCCTGAATTA
CGCCGTGAAGTTTTGTAA

NHI 294 protein sequence:

MTTPNASPASTGLPIIDVAALRAPDDADGRTRVAQQIRAACREHGFFYVNGHGVSP
SRLERLSHDFFNLDDETTKLQWRMALGGRAWRGYFPLGGELTSGRPDWKEGLYLGTEL
AADHPLVQAGTPVHGANLFPDVPGLRATILEYMAATTQLGHQLMEGIALSLDLPASYFH
DRYTADPLILFRIFNYP SQAVPEGLDVQWGVGEHTDYGLLTILLQDHIGGLEVRTPGGW
DAPPVAGAFVCNIGDMLDRMTGGLYKSTPHRVKRNTSGHDRLSFPLFFDPNFFARVQ
RIEGLADAPAQDDSAERWDRANVHAFNGIYGDYLLDKVSKVFPELRREVL

Codon-optimized NHI 295 sequence:

ATGTTGTCTCCACAAGAAGAGAAAAATATGGTTATCACTAAGACCGGCGAGGTTTCAG
GCAATTACAATGCCCCGATTACGATCGCAAAAGTGAATTGAAAGCATTGATGAGTCA
AAAGCAGGTGTTAAGGGTCTGGTTGATAACGGACTGGCCAAAATTCCGCAAATTTTT
GTTTCATGAACAGAACAAATTTTCGAAGAAAATCTGGGCTACGGGAACACTCAGTTCTC
CATACCCGTGATTGATTTTCAGGGCATAGATGAAGATGCGACACTGCGCTCAGAAAT
AATTTTGAAGGTAGGCGACGCTTGTGAAAAATGGGGATTTTTTTCAGGTTGTCAACCA
TGGTATTCCTGAATCTGTTATGAACGAAACCATCGACGGCGTCCGCCGCTTTTACG
AACAGGATACCGACGTCAAAAAACAGTTTTTATAGCAGAGATGTAACCTCGGAAGTTTA
TTTATAATAGCAATTTTGATCTGTTTCAGGCGCCGGCCGCAAATTGGAGAGATACGG
TATACTTTGTCATGGCTCCTCATCCCCGGATCCTCAGGAAGTGGCCCTGGTGTGT
CGTGACATACTCTTCGAATATAGTAAATATGTCATGAAACTGGGTCTGACCTTATTCG
AACTGCTGAGTGAAGCCCTGGGTCTGGACCCCAATCACCTGAAGGACCTGAATTG
TGCGGAAGGCCTGTTCTTTGTCTGGTCACTACTACCCGGCATGCCCCGAACCGGGT
AGTACGTTTCGGGACCAGCAACACAGTGACTCAAGCTTTTTAACGGTTTTTGTACAA
GACCAAATGGGTGGCCTGCAAGTCCTGCACGAAAATCAATGGGTTGAAGTTCCACC
ACTGCCAGGAGCCCTGGTAGTGAACATTGCCGACCTGCTACAGCTGATCACCAAC
GATAAATTTAAATCCGTCAACCACCGAGTTCTTGCAAAAAAGGAAGGACCTAGAATA
TCTGTAGCAGGCTTCTTTTCGAACACATTTACAGAAAGCAAATACAAGTCGAGTATAT
GGTCCACTTAAAGAGCTGCTGTCTCAGGAGAATCCTCCGATTTATCGCGAAACAAC
GATCAAAGACTTTATTACGCACTATTATAAAAAGGGGCTGGACGGTACGTCTGCAATA
AGCCACTTCAAACCTGTGTAAATAA

NHI 295 protein sequence:

MLSPQEEKNMVITKTGEVQAITMPDYDRKSELKAFDESKAGVKGLVDNGLAKIPQIFVH
EQNNFEENLGYGNTQFSIPVIDFQGIDEDATLRSEIISKVGDACEKWGFFQVVNHGIPE
SVMNETIDGVRRFHEQDQDVKKQFYSRDVTRKFIYNSNFDLFQAPAAANWRDTVYFVM
APHPPDPQELPLVCRDILFEYSKYVMKLGLTLFELLSEALGLDPNHLKDLNCAEGLFFV
GHYYPACPEPGSTFGTSNHSOSSFTVLLQDQMGGGLQVLHENQWVEVPPLPGALVVN
IADLLQLITNDKFKSVNHRVLAKKEGPRISVAGFFRTHFTEANTS RVYGPLKELLSQENP
PIYRETTIKDFITHYYKKGLDGTSAISHFKLCK

Codon-optimized NHI 296 sequence:

ATGCATCTTTATGAACCGCCGAAAGCACCACAAAGCATTCCGCTTATCGATTTGTCG
CGGGCTGGTGATGAAGCGGGCGAGAGGGCAGTGGCGTGGGACATTCATAGGGCA
TGCCGTGAAACGGGTTTTTTTTTATATTACCGGACATGGTGTTAGTGC GGCGCTTGTC
GCGCAGCATTTTCAGTTTGCTAAGGCCTTCTTCGATTTGCCGCTGGAGCGTAAGAT
GGCCATCCATATGAAGCAAAGCCCCACAACCTGCGGGATACGAACCTATTGGTGGTC
AGGTACTGGACTCTCAGGATGATAAAGCAGAGGCTGCTCCACCGGATCTGAAGGAA
TCTTTTTACTGCGGTATGGAGTTGCCGGATGATCATCCCTGGGCGATAAAACGGATT
CGTAATTTTCGGTCACAATCAATGGCCTGATGAATTACCAGAATTTCTGCGCCACATG
GTGAGTTATTGGGCCAGATGCGTGCCCTGGGAGACCGCCTGTTGGCGTTATTAGC
TCTGAGCTTGTATCTGCCGGCTGATTGGTTCGCTTCTAGTTTTGATATGCCGTCAGG
AACCGTTCGCCTGATTAAATATCCTCCTCAGCCTGCCCAAGCACTGTTTAATCAGAT
CGGCGCAGGCGCCCATACAGACTGGGGCGGGCGTGACGATTCTCGCACAAGACGC
GACAGGAGGCCTTGAAGTGAGAAATGCCGCAGGCGCATGGATCGAAGCGCCACC
AGTAGAAGGGAGCTTCGTGATAAACCTCGGTGACCTAATGGCGCGATGGACGAATG
GTCTGTACAACAGCAATTTTACCGTGTTAAGAATAACCGTAGCGGTCAAGATCGGT
ACTCTGCGCCTTTTTTCTACACTGCCCGCCAGATGCGCTCATCGATGCCGTCCT
GGTAGCATAACAGAAGCACATCCCCGTCGCTTCGAGGCCTGTACTGCATACGACCA
CATGGCCGAAATGTTCCGACGTAGCTATGGTTATGCCCCGAAGGCTGTTTCTTAA

NHI 296 protein sequence:

MHLYEPPKAPQSIPLIDLSRAGDEAGERAVAWDIHRACRETGFFYITGHGVSAALVAQH
FQFAKAFFDLPLERKMAIHKQSPTTAGYEPGGQVLDSQDDKAEAAPPDLKESFYCG
MELPDDHPWAIKRIRNFGHNQWPDELPEFRAHMSVYWAQMRALGDRLLALLALSLEYL
PADWFASSFDMPSGTVRLIKYPPQPAQALFNQIGAGAHTDWGGVTILAQDATGGLEVR
NAAGAWIEAPPVEGSFVINLGLMARWTNGLYNSNFHRVKNNRSGQDRYSAPFFYTA
RPDALIDAVPGSITEAHPRRFEACTAYDHMAEMFRRSYGYAPKAVS

Codon-optimized NHI 297 sequence:

ATGACGACAGCTTTTTCAACCCTGCCAATTGTTGACGTCGGCGTGCTGTCTGAAGT
TCAGCCCAAGGCCCAGGACGTCCTGGTACTTACCCAGAAGCTTTATGATGTTTTCA
CAACTGTGGGTTTTGCATATCTTGTGAACGCGCCGCTGACATTTACCCATGCTGACA
TCTTTGGATTGGCGTCAGATTTTTTTCGCTTAGAAGAAACGGAGAAACTGCGGCTTG
CTACAAAAACGTCAGTGCCTTCCAACAGCAACACATACCGTGGCTATTTTCTTTCC
AGCCAGGTTGTGATAGTTTGAAAGAGGGTTTTGAGATTGGGCCGGCCAGACGCC

AGGACGTTCTGCATTCGGCAAAACACAGCGCTATGATCTCACTGAGCCGAATGTAT
GGCCGGCCGGAGCAAGTTTTTCATGATGGGCGGGTCAAATTAGAACGACTACATGCT
GAATTACAAAACTCGCCTCAACTCTGCTGCTGATGTTGGGCATTACACTGGGTAAG
CCGGCCGATTATTTTCGCTCCATATCTGCTGGAATCCCTGTCAACGTAAAGACTGCTG
CATTATCCGCCGCAAGAATCTACGGAGACTGAAGTTGCTGCTATTAAATTATGTTGTA
CACCGCACACAGATAGTGGAATTCTGACTTTGCTCCATCAAGATCAGACAGGCGGT
TTAGAGGTGTTGAATGCCGAGGGCCGATGGATTGCAGCACCTTATGTGCCTGGGTC
AATAGTCGTCAATATTGGCGATCTCATGGCCAGAGTGAGTGGCGGGCGCTTCGTTG
CTACTATGCACCGCGTCCGTACAAGCCCGGGTAAGGGCACTAGAAAAGATGGCTTA
GGTCGTTTCTCGGTTCCGTTCTTTTTTTGAACCAGGAGAGGCGTGCCTGATACGCAG
TATAGATGGGGGAGAAGGCGTTGTCTACGGGGAGTATGTGCGCCGGAAGATGGGC
ACATGGTTAGAATATCAGAGTAGTCCGACCCAAGAGCCGGATGCTACAATTGCCGT
CAGTGCTGGTATCGTAAAAGTGGCGTCCTGA

NHI 297 protein sequence:

MTTAFSTLPIVDVGVLSEVQPKAQDVLVLTQKLYDVFTTVGFAYLVNAPLTFTHADIFGLA
SDFFRLEETEKRLRLATKTSVPSNSNTYRGYFPFQPGCDLKEGFEIGPAQTPGRSAFG
KTQRYDLTEPNVWPAGASFHDGRVKLERLHAELQKLASTLLLMLGITLGKPADYFAPYL
LESLSTLRLHYPPQESTETEVAAIKLCCTPHTDSGILTLLHQDQTGGLEVLNAEGRWIA
APYVPGSIVVNIGDLMARVSGGRFVATMHRVRTSPGKGTRKDGLGRFSVPFFFEPEGEA
CLIRSIDGGEGVVYGEYVRRKMGTWLEYQSSPTQEPDATIAVSAGIVKVAS

Codon-optimized NHI 298 (XenC) sequence:

ATGGGCAGTTTGACCGACAATGCCGCGATTCCCACCGTGGATATCTCGGCGTTCTT
AGATCCTAATGCCTCGCAGGAGGCACGTCAGGACGTTGTGAATGCGATGTCGAATG
CCTGCCACGTATATGGTTTTTTCAATCTGGCGGGTTCATGGCATTCCACAAGAGACCT
TGCGCGAAGCATTGCAACTTAATAAGATGTTTTTTCGCCTTGCCAGAGGAATCAAAAA
AAGAAGTGTTGATTAGTAAGTCAATCGGACAATCATTCCGTGGTTACGAGCCGCC
GGCATCCAGACACATCACAAGGGCCTTCTTCCCGATATTAAAGAGACGTTTCATGGTC
GGGCGTGAAGTACCGCTTGACGACCCTGATTGTGGCACATTCTCAACGGGTCCCA
ACCTTTGGCCCTCATCTTTACCGAAAGAGAAATTTACGACCGCATCATGGCTTACC
AGGGTAGCATGTTAGAACTTGTAAGAATATTCTTGCTATTTTGGCACAAGGGCTGC
CAAAGGAATGGGGATGCTCACCGACCGTTTTCAACTCTTTATTGGATAAGCCGAGTA
TTCCAATGCGTTTTCTTACACTACGCCCCCGTCCCCTCACAACCTTGAAGATGTACGCC
AGTTTGGGGTAGCTGACCACACTGACTTTGGTTGTGTAAGTATTTTGTTCAGGAG
CCCGGAACCTTCGGGCTTAGAGGTCTACTATCCCCCGTTCGGAATCGTGGATTCTGT
CCCAGTTATCGACGATGGATTCGTGATCAACATGGGTGATATGATGCAACGTTACAC
CGGGGGATATTACCGCTCAGCTCGCCATCGCGTCCTTACAAACCGCGAGAAACATC
GTCACAGTGTGGCGTTTTTTCTGAATGGGAACCTTGGGATTAAAGGCGAAGGCTTTA
GATGGCAGCGAAACGGAGACAGTAGTTGGAGATTGGATTGCGGGGCGTCTGATTG
ACACTATGGGACAGACGGGGAAAGTTATTGCAGCGTGAGAGTCCGAAACCGGTAGT
ACTGCCG

NHI 298 (XenC) protein sequence:

MGSLTDNAAIPTVDISAFDPNASQEARQDVVNAMSNACHVYGFFNLAGHGIPQETLR
EAFELNKMFFALPEESKKEVLISKSIGQSFRGYEPPGIQTHHKGLLPDIKETFMVGREVP
LDDPDCGTFSTGPNLWPSSLPKEKFQDRIMAYQGSMLELVKNILAILAQGLPKEWGCS
PTVFNSLLDKPSIPMRFLHYAPVPSQLEDVRQFGVADHTDFGCVSILLQEPGTSGLEVY
YPPSDSWIPVPVIDDGFVINMGDMMQRYTGGYYRSARHRVLTNREKHRHSVAFFLNG
NLGLKAKALDGSETETVVGDWIRGRLIDTMGQTGKLLQRESPKPVVLP

Codon-optimized NHI 299 (SwnH2 X. hyp.) sequence:

ATGGAACAGACCGGTAAGTACAAAGAGTCTCCGGTCCCACGGCTGCTGCCGCATTT
TGACGCGAGCTATCAGCCGTCCCGCGAAATTACCAAATCCCCGCCACGTCCAGCT
TGGAAGAAATTCTGACCCTCCTGGATCGTGATGGTGCTGTAATCCTGACCGATTTCA
TCAGTCTTGAAACGCTGGATAAAATTGATGAAGAACTGCAACCTTATACGCATCCCA
TCGTAGAAAATGATTCCTATGACAATTTTATTGGCAAAAAACGCTGGTGTTACCCGG
GTTGGTCGGCAAGAGCGATACGGTTGCGAATATTCTGGACACGAACGAAACCATCG
AAAACTTCTGAAAGTGATCTTAGAAGAACGTCACCCTGCAGTTTTTGAAGAACATA
CCGAAGAGTTAGTTGTTGAACCGTTGCTGAGCATCGCAATGGGTTTTTATGTGGGC
CATGACAGTCCTCGTCAGGCTCTTCATCGCGATGATATGATCTTCAGTTCGAAACAT
CGTGTGGACCTGAAGGTGCATGAAAATGATGGCTTCTCTTGCTTCATTGCCGGTTC
TCGTGTCACCCGGGAAAACGGCGGTACCATGGTTATTTTGGGTAGTCATAAATGGG
AACATGATCGGCGTGGCCGTCTGGATGAAGTTTCGTTCCCTGGAAATGGAGCGCGG
ATCTGCAATGATTTTTCTGGCTAGTGTCGCGCATGCCGCCGGGTATAACACTATTCC
GGGGGAAGTGCGGAAAGTAATCAATTTTGCCTTTGTTGCGGGGCATCCTGCGTCAAG
AGGAGAACCAAAGTCTGTGCAACTCTCGCTCAAAAGTTCTGAAAATGTCACCGAAA
CTGCAAACGCTGCTCGGATTTAAAAAACCGGCGGGGCACGTGGCTCGGTCTGGTGG
AGAATGATGATCCGGCGAAAGATCTGGCCGCCGTTTATGAGCGCGTACTGGCCTAA

NHI 299 (SwnH2 X. hyp.) protein sequence:

MEQTGKYKESPVPRLLPHFDASYQPSREITKIPATSSLEEILTLLDRDGAVILTDFISLET
DKIDEELQPYTHPIVENDSYDNFIGKKTLLVLPGLVGKSDTVANILDTNETIEKLLKVILEER
HPAVFEEHTEELVVEPLLSIAMGFYVGHDSPRQALHRDDMIFSSKHRVDLKVHENDGF
SCFIAGSRVTRENGGTMVILGSHKWEHRRRGRLEVSFLEMERGSAMIFLASVAHAAG
YNTIPGEVRKVINFAFVRGILRQEENQSLCNSRSKVLKMSPKLQTLGFKKPAGTWLGL
VENDDPAKDLAAVYERVLA

Codon-optimized NHI 300 (SwnH2 T. ben.) sequence:

ATGGATAAAAAATATGCTGCAGCGCGTTTGCTTCCTCCAATTGATGCCTCGTATCAG
CCGGCACGCAGCGTTACGAAAATTCGGCAACTTCCAGTCTTGAAGAGATTCTGGA
AGTTCTGGAGCGCGATGGTGGTGTTATTTTAACGGATTTTCGTACCGAAGGAACTAT
GGATAAAATTGACGAAGAATTGGAACCGTATACGAAAACCATCCCGGACAGCGACA
GCTACGACGATTTTATCGGCAAAAAAACCTGGTGATTGCTGGACTCATCAGCAAAA
GCGATACTCTGGCAAATATCTTAGACACCAACGAGACGATTGACAAACTGCTGAAAA

TTATCCTGGAGGAGCGGTATCCGGCGGTATTTGAGGGTCATACGGAAGAGTTGGTA
ATCGACCCGTTGTTGAGTATTAGCATGGCCTTTACGTGGGTCAGGGCAGTCCACG
CCAGGCTTTACATCGTGATGATATGGTTTTTAGCAGTAAACACCACCCGAATATGAAA
ATCAACGAAGTAGATGGTTTTAGTTGCTTCATCGCGGGCAGCAATGTCACACGCGA
AAATGGCGGCACCATGGTTATCCTGGGCAGCCATAAGTGGGAACATGATCGTCGTG
GCCGCCCCGGACGAAGTCTCGTTTTTAGAAATGGAGCGCGGTAGCGCTTTTATCTTC
TTATCGACCATGGCGCATGGTGCGGGGTACAACACCATCCCGGGAGAAAATCCGTAA
GATCATGAATCTGGTCTTCTGTCGCGGGACCCTGCGCCAGGAAGAAAATCAGTTCC
TGTGTAACCCCCGCTCCAAAGTCCTTAAGATGTCCCCAAAAATGCAGACCTTACTTG
GCTTTAAAAAGCCCGCCGGTACCTGGTTGGGCATGGTGGACAACGATGATCCTGC
GAAGGATCTGCCGGCCATTTACGAAAAAATTATGCAGTGA

NHI 300 (SwnH2 T. ben.) protein sequence:

MDKKYAAARLLPPIDASYQPARSVTKIPATSSLEEILEVLERDGGVILTDVFPKETMDKID
EELEPYTKTIPDSDSYDDFIGKKTLVIAGLISKSDTLANILDTNETIDKLLKIILEERYPAVFE
GHTEELVIDPLLSISMAFHVGQGSQRQALHRDDMVFSKHHPNMKINEVDGFSCFIAG
SNVTRENGGTMVILGSHKWEHRRRGRPDEVSFLEMERGSAFIFLSTMAHGAGYNTIP
GEIRKIMNLVFCRGTLRQEENQFLCNPRSKVLKMSPKMQTLLGFKKPAGTWLGMVDN
DDPAKDLPAYEKIMQ

Codon-optimized NHI 301 (SwnH2 A. oxy.) sequence:

ATGGAGCATATGGAAGCTGTTACTAATGCGTCAGCCCCGCGCCTGCTGCCGGCGGAT
CGATGCCTCATATCAGCCATCGCGTAGCGTCACGAAAATTCCGGCGAACTCCAGTC
TGGATGATATCCTTGAGATCCTGGAGCGTGATGGCGGTGTCTGTGCTGACTGATTTT
GTATCGTTTGAAACACTGGACAAAATCGATATGGAAGTGGAGCCGTATACTAAGCCC
CCCGTGAATTCTGACTCATATGACGACTTCATTGGAAAGAAAACCCTGGTCATTCCG
GGCCTGGTGGGAAAAAGCGATACAATTGCGAACATCCTGGATACCAACGAACTCT
GGAGAAACTGTAAAGGTCATTCTTGAAGAAAAATATCCGGCTGTGTTTCAAAAATCA
TACCGAGGAACTGGTTGTGGAGCCACTCTTATCTATTGCGATGGCGTTTAAACGTCG
GCGAAGGCTCTCCGCGTCAGGCGTTGCATCGGGATGATATGATTTTCAGTAGCAGC
CACCACGCTGATATGAAAATCAACGAAGTGGACGGTTTCTCATGCTTCTTAGCGGG
CACCCGTATTAATCGTGAAAACGGCGGGACGATGATTATTCTGGGCAGTCACAAATG
GGAACATGAACGCCGGGGACGTCCGGACGAAGTTTCCTTTCTGGAGATGGAACGC
GGCTCGGCCTTCATCTTCCTGAGCAACTTAGCCCATGGCGCCGGTTATAATACGATT
CCGGGTGAGATCCGCAAAATCATCAATCTGGTTTTCTGCCGTGGTACATTGCGCCA
GGAAGAAAATCAATTCTTGTGTAACCCTCGTTCAAAAGTTCTGAAGATGAGTCCGAA
GTTGCAAACTGCTCGGCTTCAAAAACCGGGCGCGCAATTGGCTGGGCATGGTC
GAAAACGAGGATCCGACGAAAGATCTTGCTGCGGTGTATGAAAAGATGTTGGCGTA
A

NHI 301 (SwnH2 A. oxy.) protein sequence:

MEHMEAVTNASAPRLLPAIDASYQPSRSVTKIPANSSLDILEILERDGGVVLTDVVSFE
TLKDIDMELEPYTKPPVNSDSYDDFIGKKTLVIPGLVGKSDTIANILDTNETLEKLLKVILE

EKYPAVFENHTEELVVEPLLSIAMAFNVGEGSPRQALHRDDMIFSSSHHADMKINEVDG
FSCFLAGTRINRENGGTMIILGSHKWEHERRGRPDEVSFLEMERGSAFIFLSNLAHGA
GYNTIPGEIRKIINLVFCRGTLRQEENQFLCNPRSKVLKMSPKLQTLLGFKKPARNWL
GMVENEDPTKDLAAYEKMMLA

Codon-optimized NHI 302 (SwnH2 *M. rob.*) sequence:

ATGATTAACAGTGATGCCCAGTCCGCACAGAAACAGGTAGAAGTGGAGAAGCCGG
ACGAAAAATACTCAGCACCGCGTCTGCTGCCTCCGATCCCGGACAGCTATCAACCA
GCGAAGGCGATTACCAAAATCCCTGCGACGTCCTCTTTAGAGGACATTCTTGCGAT
CTTAGAGCGTGATGGGGGGGTTATCTTAACAGACTTTGTCAGCCTGCAGGAAGTGG
ATAAAATCGATGAAGAGTTAGAACCCTTACACGAAATCCTCTATTGCCGATGATGATTC
GTATAATAACTTTATTGGCAAAAAAACCCCTGGTGATCCCGGGGGCTGGTTCGGCAAATC
GGACACAATCGCGAACATCCTGGATACGAACGAAACCATTGACAAACTGCTGAAAG
TGATTCTGGAAGAACGCTATCCAGCGGTCTTTGAACAACACACGGAAGAACTGGTG
GTGGATCCGCTGTTGTCTATCTGCATGGGTTTTTCATGTGGGGCATGGCAGCCCTCG
CCAGGCCCTGCACCGTGATGATATGATCTTTAGCAGCAAACACCGTCCTGATATGAA
AATTAACGAGGTGGATGGCTTTTCTGTTTCTTGGCTGGCACCCGGATTACGCGCG
AAAACGGGGGTACGATGGTGATCCTGGGAAGCCACAAATGGGAACATGATCGCCG
CGGGCGCCCCGACGAAGTAAGTTTCTGGAAATGGAACGCGGCTCTGCGTTCATC
TTTCTTTCAACCTTGCGCGACGGCGCGGGTTATAATACCATTCTGGGGAAAGTGCG
TAAATTACCAATCTCGTCTTTTGTGCGCGCACATTGCGCACCGAGGAGAACCAGTT
TTTGTGCGTGCTCGCAGTAAAGTGCTGAAGATGTCGCCGAAAATGCAAACGCTCC
TGGGTTTTTAAAAGCCGGCAGGGAGCTGGCTGGGCATGGTTGAAAACGAGGATCC
GGCAAAAGATTTAGAAGCGATTATGAGAAGATGCTTAAATAA

NHI 302 (SwnH2 *M. rob.*) protein sequence:

MINSDAQSAQKQVEVEKPDEKYSAPRLLPPIPDYQPAKAITKIPATSSLEDILAILERD
GVILTDFVSLQELDKIDEELEPYTKSSIADDDSYNNFIGKKTLPGLVKGSDTIANILDTN
ETIDKLLKVILEERYPAVFEQHTEELVVDPLLSICMGFHVGHGSPRQALHRDDMIFSSKH
RPDMKINEVDGFSCFLAGTRITRENGGTMVILGSHKWEHRRGRPDEVSFLEMERGS
AFIFLSTLAHGAGYNTIPGEVRKITNLVFCRGTLRTEENQFLCVPRSKVLKMSPKMQTLL
GFKKPAGSWLGMVENEDPAKDLEAIYEKMLK

Codon-optimized NHI 303 (SwnH1 *E. spin.*) sequence:

ATGGGCCTGAGTATGTTACTGCCGCCGGAACCAATCGATCCGGCCTATCAGCCGTC
TACCTCTATCGCTAAAATCCCGGCTAACGCCCAATCGAAGATATTCTGGAGATTATC
GAACGCGACGGGGGTGTGATCCTTACGGAGTTTACGTCCGCTGAAGAACTGGCAG
CCATTGACCGCGATGTTGACGCGCACCGCAAGCAGACACGCGCCGAAAAAGGAG
CTCTCCCGCAGATCCCGAAAGAGACACTCAGTGTTGCAGGATTAGTTGGCAAAAGT
CCGACCGTGGCCCGCATGTGCGAATACCCCGTACTGGAAAACTGCGCACCGCTA
TTCTGCAGGAAAAATTCGATGTGATTCTGTAAGACTACGTGGACCGCAATATTATTG
AGCCGTTGCTGTGATTTCTGCTGACCTTCTACATTGGCCCGGGTGCGCCGCGCCA
GCGCCTCCACCGCGACGACAACACCCATGGCACGCGTCATGCGGGGGAAGGAAG

CTTCGACCTCAAGCGCGCTGGTCAATTTGCGTGCCTTGTGGCGGGCTCGAAAAC
ACGCGTGAGAATGGAGCGACCATGTTTCATTCCAGGTAGCCACAAATGGGATGACAC
CCGGGTCCCACGCATCGATGAAGTGTGCTTCGCGGAAATGGAACCGGGCAGCGC
CCTTATCTTCCTGGCTAGCTGTTATCATGGTGGCGGTCATAACAGCACGAAAGATGA
AGTACGCCGCGTCCATGGTCTCTTTTTTGTGCGCGGGACGATGCGTACCGAGGAA
AACCAGTTCCTGGCAGTACCGCGTAGCAAAGTCCTGACCATGAGCGATAAAATGCT
GAGTCTGCTGGGTTATAAGAAACCCACATCTGCCCTGGGCATTGTTGAAAACGAAG
ACCCAGCCAGCAACCTGCGTACGTTCTGGACATGGCCTCGCAATAA

NHI 303 (SwnH1 *E. spin.*) protein sequence:

MGLSMLLPPEPIDPAYQPSTSIKIPANAPIEDILEIIERDGGVILTEFTSAEELAAIDRDVD
AHRKQTRAEGKALPQIPKETLSVAGLVGKSPTVARMCEYPVLEKLRTAILQEKFDVIRED
YVDRNIIPELLSISLTFYIGPGAPRQRLHRDDNTHGTRHAGEGSFDLKRAGQFACLVAG
SKTTRENGATMFIPGSHKWDDTRVPRIDEVCFEMEPGSALIFLASCYHGGGHNSTKD
EVRRVHGLFFVRGTMRTENQFLAVPRSKVLTM SDKMLSLLGYKKPTSALGIVENEDP
ASNLRTFLDMASQ

Codon-optimized NHI 304 (SwnH2 *R. nec.*) sequence:

ATGGAACAGTCATCAAGCCTGCGCCTGCTTCCACCTATTGATGCGTCCTATCAGCC
GGCACGTGAAGTCACGAAGATTCCGGCCACGAGCTCTCTGGAAGAAATCCTGGCG
GTTCTCGAACGCGATGGAGGGGTGGTTCTGACGGATTTTGTAAAGCCTGCATACCAT
GGATAAAATCGATGAAGAACTCGAACCTTACGCCAATCCGATCGCCGACAACGATTC
GTATGACAACTTTATTGGCAAAAAGACGGTGGTGATTCCGGGTCTGGTGGGAAAAA
GCGATACGATCGCAAACATTATTGATACCAACGAGACCATTGAAAACTCCTTAAAGT
TTTACTTGAAGAACGTTATCCGGCCGTTTTTCGAACAACATACAGAGGAACTCGTCGT
GGATCCCCTGTAAAGCATTGCCTTAGGGTTTTATGTTGGCCATGGCAGTCCCCGCC
AGGCCCTCCATAAAGATGATATGATCTTTTCTAGTAAACACCGCCCGGATATGAAAAT
CAACGAGGTGGATGGCTTCAGCTGCTTTCTCGCCGGCACGAACATTACCCGCGAG
AACGGGGGGCACATTGGTCATCTTAGGCTCTCACAATGGGAACACGAACGTCGTG
GCCGTGCGGACGAGGTTTCTTTTTTGGAAATGGAGCGTGGTTCGGCGTTTATTTTT
CTGGGCAACTTAGCTCATGGCTCAGGTTATAACACCATCCCCGGGGAGGTTTCGGAA
AATCATTAAATTTGGTGTTCTGTGCGGGGACACTGCGCCAGGAGGAAAACCAAGTTTT
TGTGCAACCCGCGTAGCAAAGTGCTGAAAATGAGCCCGAAGCTGCAAACCCTGCT
GGGCTTCAAAAAACCGGCCTCCACGTGGCTGGGTATGGTTGAAAACGAAGACCCG
GCCAAAGACCTTGCGGGTGTGTACGAAAAAGTGTTGCGCTAA

NHI 304 (SwnH2 *R. nec.*) protein sequence:

MEQSSSLRLLPPIDASYQPAREVTKIPATSSLEEILAVLERDGGVVLTDVSLHTMDKIDE
ELEPYANPIADNDSYDNFIGKKTVVIPGLVGKSDTIANIIDTNETIEKLLKVLLEERYPAVFE
QHTEELVVDPLLSIALGFYVGHGSPRQALHKDDMIFSSKHRPDMKINEVDGFSCFLAGT
NITRENGGTLVILGSHKWEHERRGRADEVSFLEMERGSFIFLGNLAHSGSYNTIPGEV
RKIINLVFCRGTLRQEENQFLCNPRSKVLKMSPKLQTLGFKKPASTWLGMVENEDPA
KDLAGVYEKVFA

Codon-optimized NHI 305 (SwnH1 X. hyp.) sequence:

ATGCAGGCTTTCCCGCCACCTATTGACCCCTCCTATGTGCCGTCTGTGTCTCTGGC
CGAAATTCGGGCAACAGCGCCAACCGGGCGATATTATCGCCATCCTGGAACGCGACG
GGGCTCTCATTCTGACCGATCTTGTCTCTTCCCAAGATATCGCTGCGGTAAATGAGG
AATTGGAACCGTATATTCAGAAGGCCAAAGCGGAGACCCATGCCGCCTACTCGTTG
ATTCCAGGCCAAACCGTTGTAGTTCCTGGTGTCTGTGGGAAAGTCGCCGACCATGG
CGCGTATTGCCGAATTCGAAGCAATTGATCAGTTACGCACCTTAGTCCTGCAGAAAC
GTTGCACCGCGACGTGGGAAGATCGCATTGAAGAGTTCACGATCGGTCCTCTCCT
GAACAGCAGCCTGACGTACAACATTAGTTACGGTGGCCCACGTCAACGGCTGCAC
CGCGACGACATGATTCATGGTATCTATCACAGTGGCGATTATAGCCTTTCAAATGAGA
CCATGCTTGGTTTTATGATCGCGGGTTCTAAGACAACACGTGAGAACGGCGCAACG
ATGGCCGTACCGGGCAGTCATAAATGGAACCACACCCGTGTGCCCCAGACCGATG
AAGTTTGCTTTGCGGAAATGGAATTGGGTTCGCGCTGGTGTGTTTTGGGCACGGTT
TATCACGGGGCGGGTCATAAGTGTGCCGGACCAAGGTGCGTAAACTGTTCCGGCCT
GTTTTTTATTCCGGGCACTCTGCGCCCCGAAGAAAACCAAGTTTCTTGCCATCCCCC
GCTCGAAAGTGCTGGGCATGTCCGACAAAATGTTGTCCTTGTTAGGCTACAAAAAA
CCTGAGACCTGGTTGGGTATTGTTAATAACGGCGACCCGGCAGAAAATCTCCAGGA
AGTGTTCATCTGGTGAACCTGTAA

NHI 305 (SwnH1 X. hyp.) protein sequence:

MQAFPPPIDPSYVPSVSLAEIPATAPTGDIIAILERD GALILTDLVSSQDIAAVNEELEPYIQ
KAKAETHAAYSLIPGQTVVVPGVVGKSPTMARIAEFEAIDQLRTLVLQKRCTATWEDRIE
EFTIGPLLNSSLTYNISYGGPRQRLHRDDMIHGIYHSGDYSLSNETMLGFMIAGSKTTR
ENGATMAVPGSHKWNHTRVPQTDEVCF AEMELGSALVFLGTVYHGAGHNSVPDQVR
KLFGLFFIPGTLRPEENQFLAIPRSKVLGMSDKMLSLLGYKKPETWLGIVNNGDPAENL
QEVFHLVNL

Codon-optimized NHI 306 (SwnH1 A. oxy.) sequence:

ATGTCCCTCTTTCCGCCGCCCATGATCCGACTTACGTTCCGAGTACGAGCATTGC
GCGTGTGCGCCGCACTGTTCTGTAGAAGAAATGATCTCTATCATTGAGCGCGACG
GTGGCCTTATTCTGAAGTCCCTGGTGGCGGGCAAAGATGTTGCCTCCATCAACAAC
GAACTGGAACCGTATGTCGAAAAAGCCCGCGCCGAAAGCCACGCTGCGTATGATC
TCGTTCCGAAACAAACAATCATGGTCCCTGGCACCGTGGGCAAAAGCCCGACAAT
GGCGAAAATCGCGGAATTAGAAGTCATCGATAAACTGCGCACCCCTCGTTCTGCAGA
AAAAATGTACAGCGACATGGGAAGATAAGACTGAAGAATTTACAATTGGGCCGCTGT
TGAATAGTAGCCTCACCTATCATATTTCCCATGGCGCCCCTCGTCAGCGTTTACATC
GCGATGATATGATTCATGGTATCTACCATT CAGGTGAATACAGCCTGTCAATGAAAC
CATGTTAGGTTTTATGATCGCGGGATCACGTACGACCCGCGAAAACGGGGCTACTAT
GGCAATTCGGGTTCTCATAAATGGAACCACTCGCGCATCCCGCGTGCAAACGAAG
TGTGCTTCGCGGAAATGGAACCGGGCGATGCGTTTGTGTTCTGCGGCACTGTATAT
CATGGCGCTGGCCATAACGCGGCGGTGGATCAGGTTCGTAAACTGTATGGTTTATT
CTTTATCTCGGGAACCCTGCGGCCTGAAGAAAACCAATTTCTGGCTGTCCCACGCA

GTAAAGTCCTGGGCATGAGTGAAAAAATGTTAAGCCTCCTGGGCTATAAAACGCCG
GGCACATGGCTGGGCATTGTAAACAATGGTGATCCCGCTGACGACTTAGTAGAAGT
GCTGCGGTATGCGAACGAATAA

NHI 306 (SwnH1 *A. oxy.*) protein sequence:

MSLFPPPIDPTYVPSTSIARVAATVPVEEMISIIERDGGILKSLVAGKDVASINNELEPYV
EKARAESHAAYDLVPKQTIMVPGTVGKSPTMAKIAELEVIDKLRTLVLQKKCTATWEDKT
EEFTIGPLLNSSLTYPHSHGAPRQRLHRDDMIHGIYHSGEYSLSNETMLGFMIAGSRTTR
ENGATMAIPGSHKWNHSRIPRANEVCFAEMEPGDAFVFLGTVYHGAGHNAAVDQVRK
LYGLFFISGTLRPEENQFLAVPRSKVLGMSEKMLSLLGYKTPGTWLGIVNNGDPADDLV
EVLRYANE

Codon-optimized NHI 307 (SwnH1 *M. rob.*) sequence:

ATGGGGGCCTTTCCACCACCGATCGATCCAAGCTACATTCCATCGGTTTCTCTGAC
CCGCTTGCCGGCGACGGCGCCAACCGCCGACATCATCGCAACGCTGGAACGTGA
CGGAGCTTTGATCCTTGTCGATCTGGTCAGCCCGCAAGATGTTGCGGGCGATCAACG
CCGAAATTGAGCCTTACATCCAGAAAGCTCGTGCAGAGTCTCACGAGGCCTACGAT
CTGATTCCTAAACAAACGATCATGGTGCCTGGTGTGTAGGCAAATCCCCGACTATG
GCGCGGATGGCTGAGCTTGATGTAATTGACACACTGCGTACCCGTGTTTTGCAGCG
TAAATGCACCGCGACTTGGGAGGATCGGACCGAAGATTTCTCCATCGATCCTCTGC
TGAATCCAGTCTTACATATCATATTAGCTACGGGGGGCCCCCGGCAACGCCTCCAC
CGTGACGATATGATCCATGGTATTTATCATCGTGGCGGCGAATATCGCCTGTCAGAT
GAAACCATGCTGGGCTTTATGATTGCGGGCAGTAAACCACTCGTGAAAACGGTGC
GACCATGGCAATTCCGGGCTCTCATAAGTGGGACCATGCGCGCGTGCCCCGCGTG
GATGAGGTCTGCTTCGCGGAGATGGAACCGGGGTGCGCTCTCGTGTTTCTGGGCA
CAGTTTACCACGGTGCGGGCCACAACCTCGGTACCGGATCAGGTCCGTAAAATTTAT
GGTTTGTTCCTTCCAGGTACCTTCGGCCGGAAGAGAATCAGTTCCTGGCAAT
CCCGCGCAGCAAAGTTCTGGGCATGTGCGATAAAATGCTCAGTCTGTAGGCTACA
AGAAGCCGGGTACTTGTTGGGAATTGTGAACAACGGCGATCCGGCCGAGAATCT
GGCAGAAGTGCTGGGAATGGCGAATAGCTGA

NHI 307 (SwnH1 *M. rob.*) protein sequence:

MGAFPPPIDPSYIPSVSLTRLPATAPTADIIATLERDGLILVDLVSPQDVAAINAEIEPYIQ
KARAESHEAYDLIPKQTIMVPGVVGKSPTMARMAELDVIDTLRTRVLQRKCTATWEDRT
EDFSIDPLLNSSLTYPHSHGAPRQRLHRDDMIHGIYHRGGEYRLSDETMLGFMIAGSKT
TRENGATMAIPGSHKWDHARVPRVDEVCFEMEPGSALVFLGTVYHGAGHNSVPDQV
RKIYGLFFIPGTLRPEENQFLAIPRSKVLGMSDKMLSLLGYKKPGTWLGIVNNGDPAEN
LAEVLGMANS

Codon-optimized NHI 307 (SwnH1 *T. ben.*) sequence:

ATGCCGGAGTTCCCGCCGCCTATTGATCCTAGCTACGTGCCATCAGTGTCAGTACTGAC
GCGTATCCCAGCGAACGCGCCGATCGAAGATATCTTGGCCGTGCTGGAACGTGAT
GGCGGTCTTATCTTAACGGACATTATCTTCCCAGGATGTGGCCGCCATTAAACGAC

GAACTGGATCCGTATGTGCAAAAGGCACGTGCCGAATCACACGCGGCATACGACCT
GATCCCGAAACAACTATCATGGTCCCCGGGATCGTGGGCAAATCTCCGACTATGG
CCAAAATCGCAGAATACGAAGTTATTGATAAACTGCGCACTATGGTTCTGCAGAAAA
AATGCACTGCGACCTGGGAGGATCGTACCGAAGAGTTCACCATTTGGACCTCTGCTG
AATTCGTCCCTGACGTATAATGTTTCTTATGGTGGCCCGCGCCAGCGCCTGCATCGT
GATGATATGATTCACGGTATCTATCATCACGATGGTGAGTATTCTCTGTGCAACGAGA
CAATGCTGGGTTTTATGTTTGCTGGCTGCAAGACCACCCGCGAAAACGGCGCGAC
CATGGCTATCCCGGGCTCGCACAAATGGAATCATACCCGTGTTCCGCGTACAGACG
AGGTGTGTTTCGCGGAAATGGAACCGGGTTCGGCCTTCGTGTTTCTGGGAACTGT
GTACCACGGGGCAGGCCATAATTCGGTGCCGGATCAGGTTGCAAGGTTTACGGT
CTTTTTTTTATCAGCGGGACCTTACGTCCTGAAGAAAATCAATTTCTGGCTATTCCCC
GTTCCAAAGTGCTGGGAATGTCTGATAAAATGCTGTGCTGCTGGGATATAAAAAAC
CGGAAACCTGGTTGGGCATCGTTAATAACGGTGACCCGGCAGAGAACCTGAAGGA
GGTTCTCGATATGGCAAACCTCGTAA

NHI 308 (SwnH1 T. ben.) protein sequence:

MPEFPPPIDPSYVPSVSLTRIPANAPIEDILAVLERDGGILTDIISSQDVAAINDELDPYVQ
KARAESHAAYDLIPKQTIMVPGIVGKSPTMAKIAEYEVIDKLRTMVLQKKCTATWEDRTE
EFTIGPLLNSSLTYNVSYGGPRQRLHRDDMIHGIYHHDGEYSLSNETMLGFMFAGCKT
TRENGATMAIPGSHKWNHTRVPRTDEVCFMEMPGSAFVFLGTVYHGAGHNSVPDQ
VRKVYGLFFISGTLRPEENQFLAIPRSKVLGMSDKMLSLLGYKKPETWLGIVNNGDPAE
NLKEVLDMANS

Codon-optimized NHI 309 sequence:

ACGATGGCCACCACAACACTAGCAGCTCGTTTTACATCCCCACCGTTGATATCTCTCCG
TTTCTGAAAGATCCTTCAAGCCCAGCCGCGCAGCAAGTCATCGAGGATGTACGGGC
AGCGTGATGAGCACGGGATTCTTCCAGATCACGGGCCATGGTATCCCTAAAGAGT
TACAACAATCAGTTTTCGAGGCTGCCAAGAGATTCTTCGCCTTGCCATTCGAGGAAA
AGAAGAACTCGACGCAAAGAAGAACGTAGGCCATCGTGGCTACGATGTCTTAGGT
AGCCAGTCATACGAGGATGACGTGTTACCGGATCTGAAAGAGGGCTTCTTTGTGGG
ACAGGATATACCACCTGACGACCCTCGGGTACAGGCGCGTCGGTTCTTCATGGGTC
CAAATGTTTGGCCACCCACGAGTCTTCTCCCGCCCGAGGAGTTTCGGGAACCCGT
AGAAGAATACTACGCGGCGATGTACAAATTATCACTACCGTGCTTGATCTCATTGC
TCGGACACTGCCGTACGGACCTCATGTGTTTCGACGAGTTCAAGTCCAACAACCCTG
CCGCACCGATGCGCCTGCTTCACTACCCGCCGGCCCGCTCGACCGACAAGCGGC
AATTAGGGGCTTCAGCACACACCGACTTTGGAGCAATTACATTGCTGCTTCAGGAC
GAAAATCCCGGATTGCAAGTGCTGGATCAAAGACGGGCGAATGGGTGGACGTTTC
CACCAAACCCTGACGCTTACGTTGTAAACGTTGGGGATATGTTGAGCAAGTGGACA
GGCGGACACTACAAGAGTTCAGTGACACAGAGTGATCAATAAGAACCCTACAGATCG
CTACAGTGTTGTTTTCTTCTTTGATGGGAACCTTGACTGTCCACTGTCTCCGCTGGA
CGGATCTGAGCCGGGCGCTGACAGTTGAGGAACACATGATCGAGCGCATGACAACG
AGCTACGGTAAGAAGAAGTAA

NHI 309 protein sequence:

TMATTTSSSFYIPTVDISPFLKDPSSPAAQQVIEDVRAACMSTGFFQITGHGIPKELQQS
VFEEAAKRFFALPFEEKKKLDAKKNVGHARGYDVLGSQSYEDDVLPLDKEGFFVGGDIPP
DDPRVQARRFFMGPVWPPTSLLPPEEFREPVEEYAAAMYKLSLTVLDLIARTLPYGP
HVFDEFKSNNPAAPMRLHYPPARSTDKRQLGASAHTDFGAILLLQDENPGLQVLDQ
KTGEWVDVPPNPDAYVVNVGDMLSKWTGGHYKSSVHRVINKNPTDRYSVVFVFDGN
LDCPLSPLDGSEPLTVEEHMIERMTTSYGKKK

Codon-optimized NHI 310 sequence:

ACAACGGCGGCCCGAAACCATACCTACGGTGGATATCTCCCCCTGGCTGTCGCCAAA
TGCCAGCGAGGAAGCTAAGGACCAGGTAGTGGAAGCTGTATCTCATGCGTGCTCA
ACATACGGCTTTTTTAACCTGGTAGGCCATGGTATTCCTGTCTGAAGAGCAGCAAAAAG
ATCCTGGATTGTGCCAAAGCTTTTTTTGACTTGCCTCTGGAAGAAAAAATGGAAGTC
AGTATCGACAAGTCCATGGGTAAGTCGTTTCAGAGGGTATGAACCCTCTCGTATTCAA
ACTCACCAAGAAGGGCTGTTACCAGACACCAAAGAGTGTTTTATTATTGGAGCGGA
AGTTCCTGCGGACGACCCTGACGCCGGTACCTTTTCTACCGGGCCTAACCTGTGG
CCGAAGTCCCTGCCGGACGAAGAATTCGAACCCCAATAATGGAGTATAGAGCCCG
TATGCTGGAGTTAGTTAAAGTTTTGTTACGCATTTTAGCAAGAGGATTACCGAAGGAA
TGGAATTGCCCGCCTGACGTCTTTGACGAGTTCTTGATAAACCCATCGATCCCAATG
CGCCTTTTACACTACGCTCCCCAGACTTCTACTGACCCACGCCAATTTGGCGTAGG
AGATCACACCGACTTCGGCTGCATAACGATTCTCCTTCAGCAACCGGGAAGTAAGG
GGCTTGAAGTCTGGTACCCGCCAACAGAGACGTGGATTCCCGTGCCGGTTGTGGA
AAACGCATTCGTGATTAACATGGGCGACATGATGCAGAAATGGACCGGTGGTTACTA
CCGGAGCGCAAGACACCGAGTCATAACATCGGGTGAGAAACATAGATATTCCGTCC
CGTTCTTTCTCAATGGTAACCTGAAATTGAAAATCAAACCGCTGGACGGATCAGGTG
AGGAAACAACAGTGGGCGAACACATTAGACAACGATTGGCAAAGACGTTAGGAGA
GAATGGTAAGGCGCTGAAGTGA

NHI 310 protein sequence:

TTAAETIPTVDISPWLSPNASEEAKDQVVEAVSHACSTYGFFNLVGHGIPVEEQQKILDC
AKAFFDLPLEEKMEVSIDKSMGKSFRGYEPSRIQTHQEGLLPDTKECFIIGAEVPADDP
DAGTFSTGPNLWPKSLPDEEFRTPIMEYRARMLELVKVLLRILARGLPKEWNCPPDVF
DEFLINPSIPMRLHYAPQTSTDPRQFGVGDHTDFGCITILLQQPGTKGLEWYPPTET
WIPVPVVENAFVINMGDMMQKWTGGYYRSARHRVITSGEKHRYSVPPFLNGNLKLKIK
PLDGSGEETTVGEHIRQLAKTLGENGKALK

Codon-optimized NHI 311 sequence:

ATGACAACAACCTGCACCGGAGGCCATCCCGACGGTTGATATTTCTGCATGGCTGTC
ACCAAATGCATCCGAAGAGGCAAAACAGCAGGTGGTGGGAAGCCATGCGCCACGCT
TGTAACACATATGGTTTTTTTTATCTGGTTGGACATGGCGTTTCGCCTGAAGAGCAA
CAAAAAGCTCTGGATTGCACCAAATTGTTCTTTGCCCTTCCAATGGAGGAGCGCAT
GGAAGTTTGGATAGGAAAATCCATGGGGAAGTCTTTTAGAGGTTATGAGCCCCCGG

GTATCCAAACTCACCATGAAGGCTTGTTGCCCGACACCAAAGAAACCTTTATGGTTG
GCGCTGAGGTCCCTGCTGATGATCCTGATGCGGGGACATTTAGCACGGGCCCCAA
TCTGTGGCCAAAATCCCTGCCAGATGAAGAGTTTCGTACACCTGTTATGGAATATCA
GGCGAAAATGGTGGAATTGGTTAAAGTTCTGCTTAAAATCCTGGCGCGGGGTCTGC
CCAAAGAGTGGAATTGTCCGCCAGATGTGTTTGATGAACTGGCGGTGAACCCTTCC
ATACCTATGCGCCTCCTGCATTATGCACCGCAGCCAGTTAGAGATGATCGTCAGTTT
GGGGTGGCCGACCATAACCGACTTTGGCTGCATTACAATTTTGCTGCAGGAGCCGG
GTACCAAAGGGCTGGAGGTTTGGTATCCGCCGACAGAAACGTGGATCCCGGTGCC
TGTAAGGAAGGTGCGTATGTAATTAACATGGGTGATATGATGCAAAAATGGACGGG
AGGGTATTATCGGAGTGCACGACATCGTGTCTAACTAGCAGTGACAAACACCGGT
ACAGCGTGGCTTTTTTTCTGAATGGCAATCTCAAACCTGAAATGTAAAGCCTTAGACG
GGTCCGGTGCCGAAACTGTCGTGGGGGAACATATCCGTCAGCGTCTTATCGAGAC
TATGGGTGAAACAGGTAAAATGCTGAAATAA

NHI 311 protein sequence:

MTTTAPEAIPVTDISAWLSPNASEEAKQQVVEAMRHACTTYGFFYLVGHGVSPEEQQK
ALDCTKLFFALPMEERMEVWIGKSMGKSFRGYEPPGIQTHHEGLLPDTKETFMVGAEV
PADDPDAGTFSTGPNLWPKSLPDEEFRTPVMEYQAKMVELVKVLLKILARGLPKEWNC
PPDVFDELAVNPSIPMRLHYAPQPVRDDRQFGVADHTDFGCITILLQEPGKGLEVWY
PPTETWIPVPVKEGAYVINMGDMMQKWTGGYYRSARHRVITSSDKHRYSAFFLNGN
LKLKCKALDGSGETVVGHIRQRLIETMGETGKMLK

Codon-optimized NHI 312 sequence:

ATGGGCTCGACCACAAGTTCGGCGGCCATTCCCACCGTTGATATATCAGCGTTTTTA
TCTCCAAATGCTTCAGAAGAAGCCCGTCAGCAAGTAGTGGAGGCTATGCGTCACGC
GTGTACTACTTATGGCTTTTTTAATCTGGTAGGGCACGGAATTCCTCAGGAGGAATTA
CAAGAAGCATTGAGTGTTCTAAGCTGTTTTTCGCCCTTCCGGAAGAGAAACGGAT
GGAGGTCTGGATAGGTAAATCTGTTGGCCGGTCATTTAGAGGATATGAACCACCAG
GTATTCAAACCCATCACGAGGGTTTACTGCCAGACACTAAAGAACTTTTATGGTGG
GCCGGGAAGTCCCCGCTGATGACCCTGATTGTGGAAGTTTCTCTACTGGTCCGAAT
TTGTGGCCGAGCAGTTTACCAAAGGAAGAGTTTCAAGATCCCGTAATGGCATACCA
AGCAAGAATGCTGGAGTTGGTGAAGGTCATTCTTAAATTTTGGCCCAAGGGCTTC
CGAAGGAGTGGGGCTGCCCCCGGATGTTTTTGATTCACTGGCTGTGAACCCGTC
CATCCCAATGCGATTCTTCATTATGGACCCGTTCAAGTTAAAGACCCGCGCCAGTT
TGGTGTTGCCGATCATACAGATTTTGGATGCGTTACGATTCTGCTGCAAGAACCTGG
TACGGAGGGCCTGGAGGTATGGTACCCGCCAACAGAGACATGGATTCCTGTTCTCTG
TTGTAGAGGATGGATTTGTCATTAATATGGGTGATATGATGCAAAAATACACTGGGGG
GTACTATCGCTCAGCCCGTCATCGAGTTATCACATCATCTGATAAACACCGTTACTCA
GTCGCATTTTTCTTAAATGGAAATTTAAAGTTAAAGGCAAAAGCCTTAGATGGCTCAG
GAGCAGAGACCGTTGTGGGAGAGCACATACGACAACGTTTAATCGAGACCATGGG
CGAAACTGGTAAGTTACTGAAATAA

NHI 312 protein sequence:

MGSTTSSAAIPTVDISAF LSPNASEEARQQVVEAMRHACTTYGFFNLVGHGIPQEELQE
AFEC SKLFFALPEEK RMEVWIGKSVGRSFRGYEPPGIQTHHEGLLPDTKETFMVGREV
PADDPDCGSFSTGPNLWPSSLPKEEFQDPVMAYQARMLELVKVILKILAQGLPKEWGC
PPDVFD SLAVNPSIPMRFLHYGPVQVKDPRQFGVADHTDFGCVTILLQEPGTEGLEVW
YPPTETWIPVPVVEDGFVINMGDMMQKYTG GYYRSARHRVITSSDKHRYSVAFFLNG
NLKLKAKALD GSGAETVVGEHIRQR LIETMGETGKLLK

Codon-optimized NHI 313 sequence:

ATGGG CACCAACAAGCGCAACCGCAGCTATTCCGACTGTAGATATCTCAGCATTTTTTG
TCCCCGACGGCATCAGAAGCGGCTCGTCAGCAAGTAGTTGAAGCTATGTCACACG
CTTG TACGACCTACGGGTTCTTTAACTTGGTGGGTCATGGTATACCC CAGGAAGCA
CTTCAACAGGCCTTTGAATGTAACAAAATGTTTTTTGCTCTTCCTCAGGAAAAACGC
ATGGAGGTATGGATTGGTAAAAGCCTCGGGAGATCTTTTCGTGGGTACGAGCCCCC
TGGAATACAAACGCATCATGAAGGCCTGCTCCCGGATACGAAAGAGACTTTTATGGT
CGGCCGAGAAGTCCCGGAGGATGACCCAGATGCTGGCAGCTTCAGCACGGGTCC
CAATCTGTGGCCGTCATCGCTGCCGAAAGAGGAATTCAGGACCGTGTGATGGACT
ACCAGGCACGTATGGTAGAGTTAGTAAAAGTGATCCTGAAAATCTTGGCACAGGGA
CTTCCGAAAGAATGGGGATGCCCCCGGATGTATTCGATTCATTGACCGTCGACCC
TTCGATTCCGATGCGTTTCCTTCACTACGGTCCAGTACCAGTCCGGGACCCGCGCC
AATTCGGCGTGGCCGATCACACGGATTTTGGTTGTATTAGCATCCTGTTGCAGGAAC
CAGGCACCAAGGGGCTGGAAGTTTGGTACCCGCCACCGAGACATGGGTTCGGG
TGCCTGTTGTGGAAGATGGTTTTGTGATAAATATGGGCGATATGATGCAGAAATACAC
TGGGGGCTATTACAGATCAGCACGCCATAGAGTTATTACGAATTCCGATAAACATCG
ATATTCAGTTGCGTTCTTTCTGAATGGTAATTTAAAATTGAAATGTAAAGCACTGGAC
GGTTCTGGCGCAGAAACAGTGGTAGGTGAACATATCCGGCAGAGACTCATTGAAAC
TATGGGCGAAACGGGCAAGCTCCTGCAGTAA

NHI 313 protein sequence:

MGTT SATAAIPTVDISAF LSP TASEAARQQVVEAMSHACTTYGFFNLVGHGIPQEALQQ
AFEC NKMF FALPQEKRMEVWIGKSLGRSFRGYEPPGIQTHHEGLLPDTKETFMVGRE
VPEDDPDAGSFSTGPNLWPSSLPKEEFQDRVMDYQARMVELVKVILKILAQGLPKEW
GCPPDVFD SLTVDP SIPMRFLHYGPVPVRDPRQFGVADHTDFGCISILLQEPGTKGLEV
WYPPTETWVPVPVVEDGFVINMGDMMQKYTG GYYRSARHRVITNSDKHRYSVAFFLN
GNLKLKCKALD GSGAETVVGEHIRQR LIETMGETGKLLQ

Codon-optimized NHI 314 sequence:

ATGGG TAGCCAAACAAGTGA CTACGCTATACCGACAGTTGATATTTCCGGCGTTCCTG
AGTCCTACTGCCTCCGAAGCGGAGCGCAGACAGGTGGTGGAGGCAGTTCGGCAT
GCGTGTACCACGTATGGTTTTTTTAACTCTGGTCCGCCATGGTATTCCGCATGAACAG
CTTCAGCAAGCCTTTGAATCTTCTAAGTTGTTCTTTGCCCTGCCGGAAGAAAAGCGT
ATGGATGTGCACGTGGGCAAAGCTTTAGGCCGGAGCTTCCGTGGTTGGGAACCCC
CTCTTATACAGCAACATCAGGAAGGGCTGTTGCCGGACACAAAAGAGACATTCATT
GTAGGGCGCGAAGTGCCCGCGGATGATCCGGAAGCAGGGAGCTTCCTTACTGGG

CCGAATCTGTGGCCAGATCTGCCTAAAGAAGAGTTTCAGGATGTTGTCTTGGCGTA
CCAAGCTCGTATGGTCGAGCTGGCCAAAGTTATTGTCCGTATCTTGGTTCAAGGCC
TGCCCGAAGAATGGGGTTGTCCCCCAGACGCACTTGATGGACTCACAGTGGACCC
CGCAATTCGGATGCGCTTTCTTCATTATGGCCCCGTAAAGAGAAAGATCCGCGCCA
GTTTGGTGTGGCGGCACATACAGACTTCTCAGCGATTACTATTTTATTACAGCAGCC
AGGAACTGAGGGTTTGCAAGTGTGGTACCCACCTACAGAGGATTGGATCCCAGTAC
CCGTAGTTGAGGATGGTTTCGTAATTAACATTGGAGATTTAATGCAAAAATATACCGG
GGGCTATTATCGAAGCGCCCCGACATAGAGTTATAACTTTCTCTGAGAAAGATAAACA
CCGATATTCTATTGCGTTTTTTCTTGATGGTAACTTGCAGTTCAAGACCAAGGCTTTA
GACGGGTCAGGAGAAGAAACGGTGGTAGGTGAATATGTGCGTTCGTGTCTGATGG
GCACCTTAGGAACTCGCGGCCCTTTACTGTGA

NHI 314 protein sequence:

MGSQTSDSAIPTVDISAFLSPTASEAERRQVVEAVRHACTTYGFFNLVGHGIPHEQLQQ
AFESSKLFFALPEEKRMVDVHVGKALGRSFRGWEPPLIQHQEGLLPDTKETFIVGREV
PADDPEAGSFLTGPNLWPDLPKEEFQDVVLAYQARMVELAKVIVRILVQGLPEEWGCP
PDALDGLTVDPAIPMRFLHYGPVKEKDPQRQFGVAAHTDFSAITILLQQPGTEGLQWYYP
PTEDWIPVPVVEDGFVINIGDLMQKYTGGYYRSARHRVITFSEKDKHRYSIAFFLDGNL
QFKTKALDGSGEETVVGEYVRSCLMGTLGTRGPLL